BIO202 - FINALTERM MERGE PPT (LECTURE 113 TO 225) BIOCHEMISTRY-1

Prepared By:

All About Biotech

AARISH ALI & TEAM (+96569002641)

Lipid Chemistry

- **Polyunsaturated Fatty Acids Of Biological Importance** (Cont.)
- Eicosanoids are derived from either omega-3 (ω-3) or omega-6 (ω-6) fatty acids.
- In general, the ω-6 eicosanoids are **pro-inflammatory**.

Lipid Chemistry

- **Imbalance of omega-6 and omega-3** PUFAs in the diet

- is associated with an increased risk of cardiovascular disease.

Lipid Chemistry

- Long chain omega-3 fatty acids such as alpha-linolenic acid and their derivatives such as
- eicosapentaenoic acid (EPA) and
- docahexaenoic acid (DHA) have
- anti-inflammatory effects

Lipid Chemistry

- In addition, DHA is selectively incorporated into
- retinal cell membranes and
- postsynaptic neuronal cell membranes

- suggesting it plays important roles in vision and nervous system function.

Lipid Chemistry

- Current evidence suggests that diet rich in omega 3 fatty acids are beneficial particularly for
- cardiovascular disease and also to some extent for
- Alzheimer's disease,
- Cancer and
- Rheumatoid Arthritis.

Lipid Chemistry

- This disease preventing role is due to the
- anti-inflammatory action of these
- omega-3 derivatives.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family	Common Name	Systematic Name	Occurrence
Monoenoic acids (one double bond)				
16:1;9	ω7	Palmitoleic	*cis*-9-Hexadecenoic	In nearly all fats.
18:1;9	ω9	Oleic	*cis*-9-Octadecenoic	Possibly the most common fatty acid in natural fats; particularly high in olive oil.

Unsaturated Fatty Acids of Physiologic and Nutritional Significance

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family	Common Name	Systematic Name	Occurrence
Monoenoic acids (one double bond)				
18:1;9	ω9	Elaidic	*trans*-9-Octadecenoic	Hydrogenated and ruminant fats.

Number of C Atoms and Common Double Bonds	Family	Common Name	Systematic Name	Occurrence
Dienoic acids				
18:2;9,12	ω6	Linoleic	all-*cis*-9,12-Octadecadienoic	Corn, peanut, cottonseed, soy bean, and many plant oils.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family	Common Name	Systematic Name	Occurrence
Trienoic acids				
18:3;6,9,12	ω6	γ-Linolenic	all-*cis*-6,9,12-Octadecatrienoic	Some plants, eg, oil of evening primrose, borage oil; minor fatty acid in animals.
18:3;9,12,15	ω3	α-Linolenic	all-*cis*-9,12,15-Octadecatrienoic	Frequently found with linoleic acid but particularly in linseed oil.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family Omega	Common Name	Systematic Name	Occurrence
Tetraenoic acids (four double bonds)				
20:4;5,8,11,14	ω 6	Arachidonic	all-*cis*-5,8,11,14-Eicosatetraenoic	Found in animal fats; important component of phospholipids in animals.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family Omega	Common Name	Systematic Name	Occurrence
Pentaenoic acids (five double bonds)				
20:5;5,8,11,14,17	ω 3	Timnodonic	all-*cis*-5,8,11,14,17-Eicosapentaenoic	Important component of fish oils, eg, cod liver, mackerel, menhaden, salmon oils.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family Omega	Common Name	Systematic Name	Occurrence
Hexaenoic acids (six double bonds)				
22:6;4,7,10,13,16,19	ω3	Cervonic	all-*cis*-4,7,10,13,16,19-Docosahexaenoic	Fish oils, phospholipids in brain.

Lipid Chemistry

- **Polyunsaturated Fatty Acids Of Biological Importance**

- Polyunsaturated fatty acids contain more than one double bond

Lipid Chemistry

- These include:
- Linoleic acid (18: $2\Delta_{9, 12}$)
- α-Linolenic acid (18 : $3\Delta_{9, 12, 15}$)
- Arachidonic acid series (20 : $4\Delta_{5, 8, 11, 14}$)

Lipid Chemistry

- Of these
- Linoleic acid and
- α-Linolenic acid are essential fatty acids.

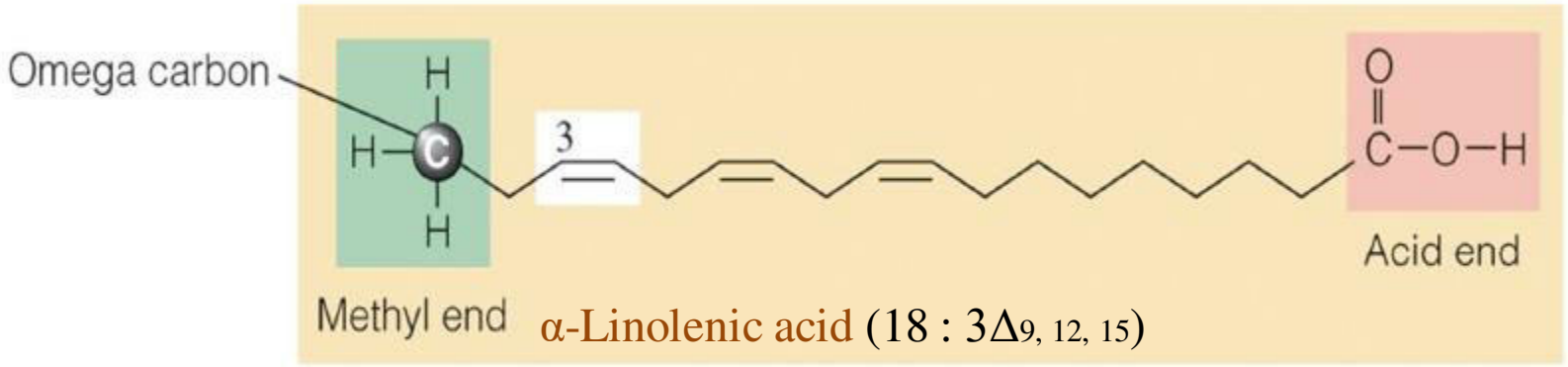
- Whereas, arachidonic acid is synthesized from both Linoleic acid and α-Linolenic acid.

Lipid Chemistry

- Arachidonic acid becomes essential if linoleic acid is deficient in the diet

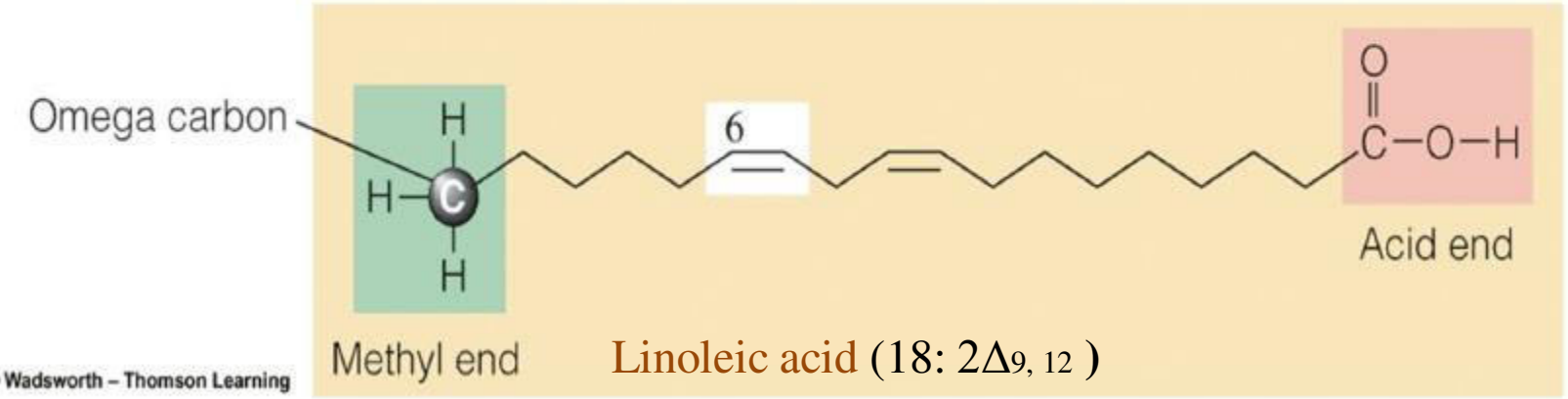
Lipid Chemistry

Linolenic acid, an omega-3 fatty acid



α-Linolenic acid (18 : 3$\Delta_{9, 12, 15}$)

Linoleic acid, an omega-6 fatty acid



Linoleic acid (18: 2$\Delta_{9, 12}$)

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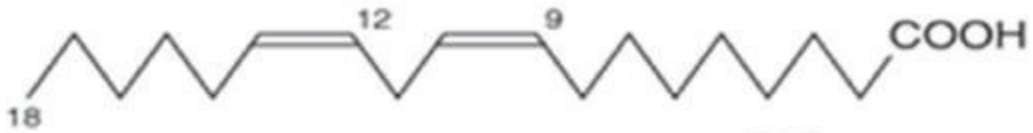
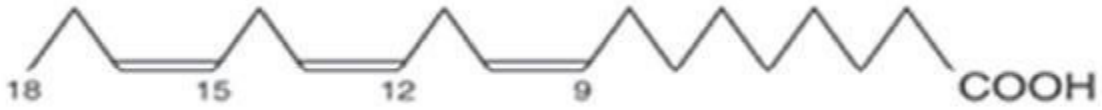
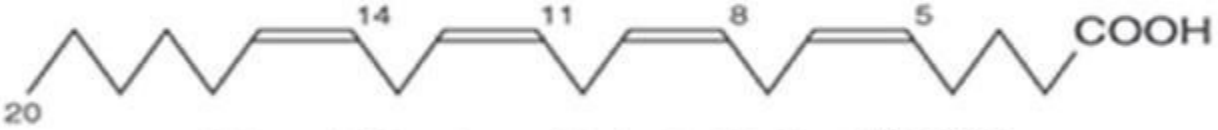
Lipid Chemistry



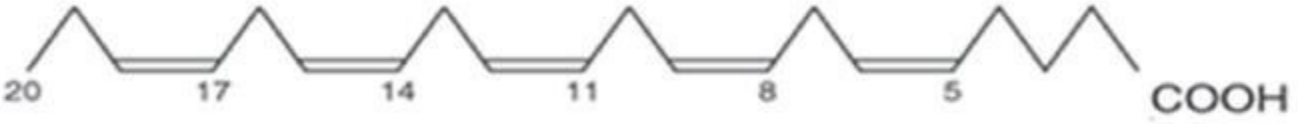
*Linoleic acid (ω6, 18:2, $\Delta^{9,12}$)



*α-Linolenic acid (ω3, 18:3, $\Delta^{9,12,15}$)



*Arachidonic acid (ω6, 20:4, $\Delta^{5,8,11,14}$)



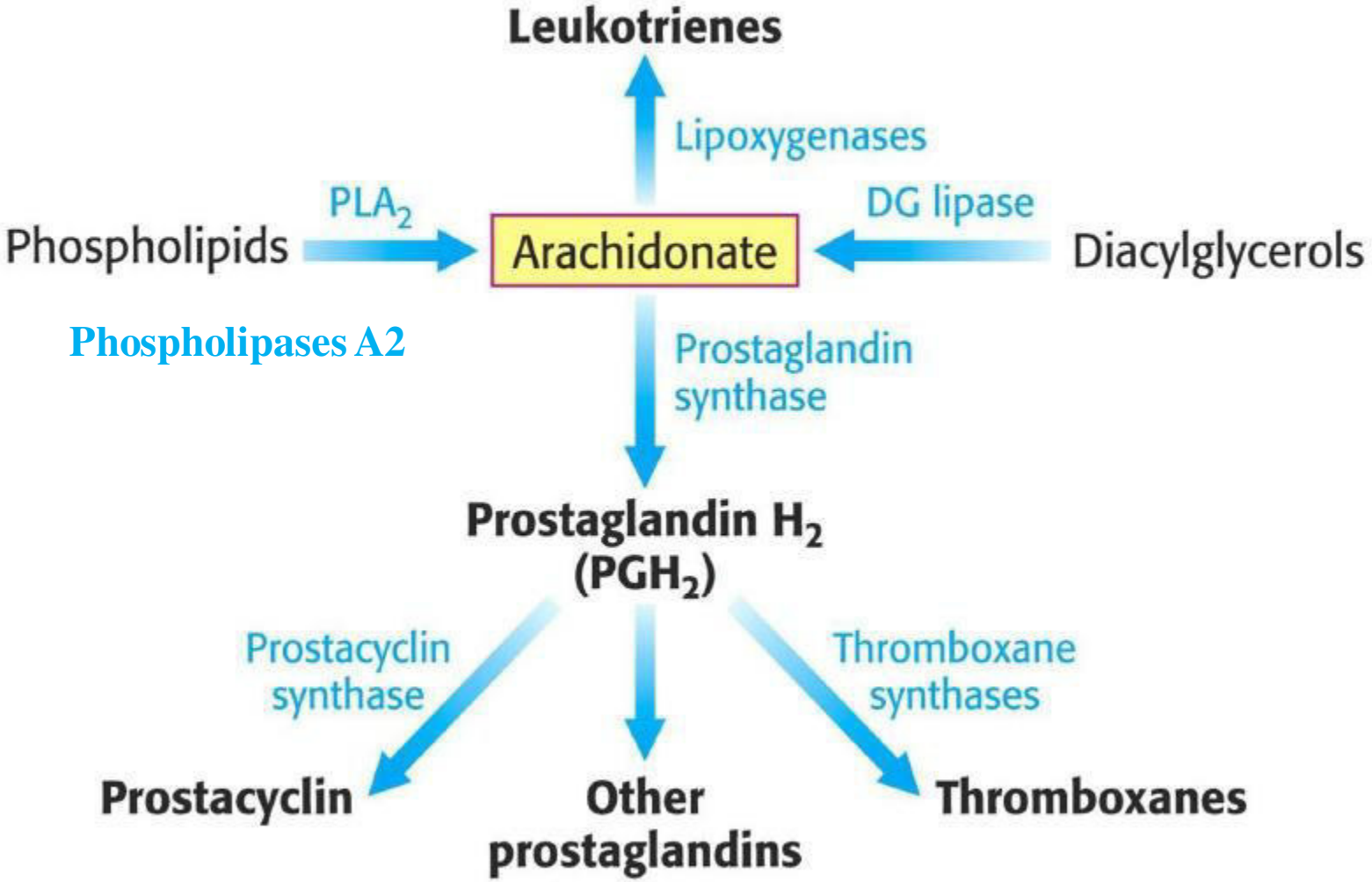
Eicosapentaenoic acid (ω3, 20:5, $\Delta^{5,8,11,14,17}$)

Lipid Chemistry

- Arachidonic acid is the precursor of paracrine hormones called **Eicosanoids.**
- Eicosanoids are chemical messengers which act over a short distance and include:

- Prostaglandins
- Leukotrienes and
- Thromboxanes

Lipid Chemistry

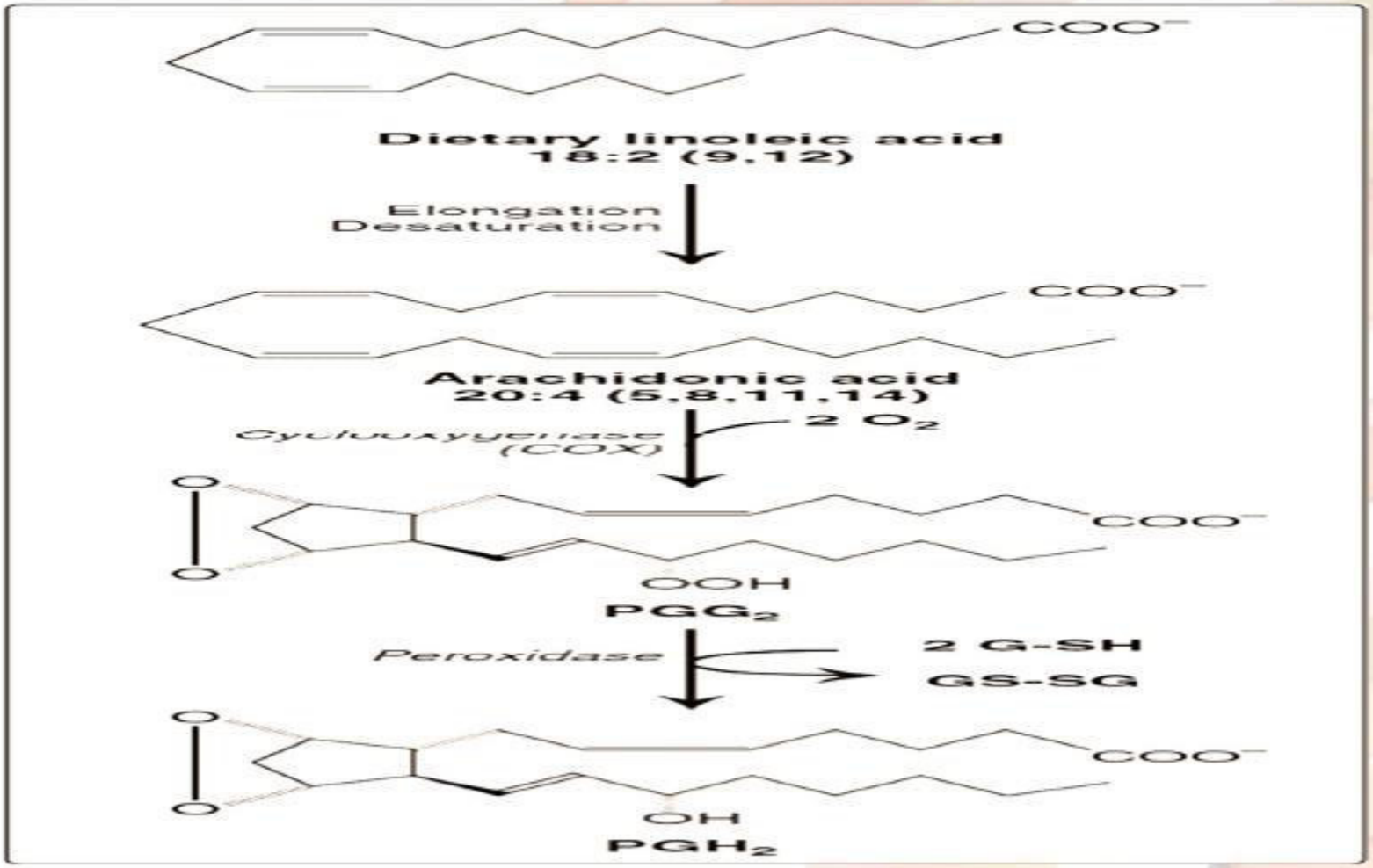


Prostaglandin synthase is more commonly called as *cyclooxgenase*

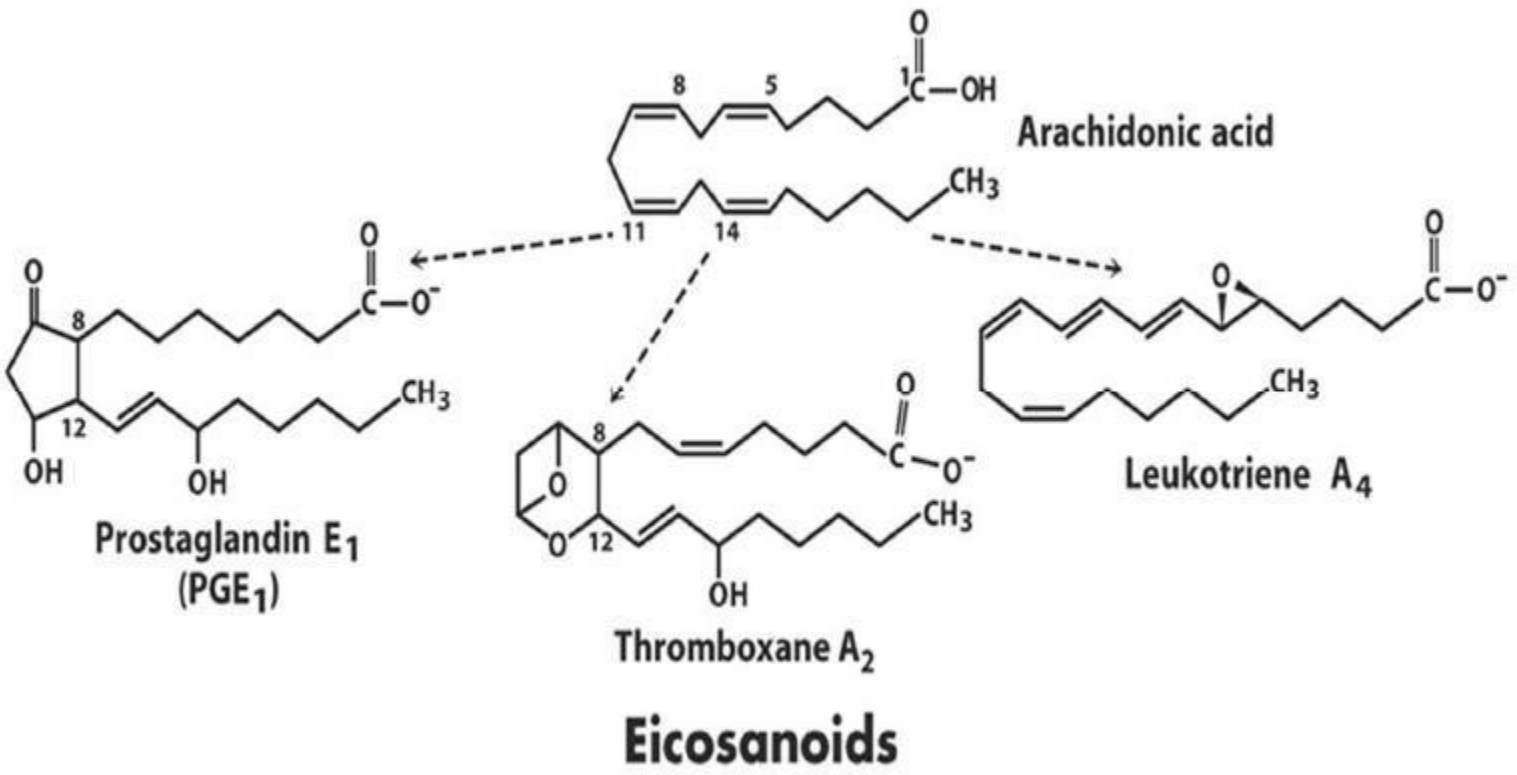
Lipid Chemistry

- These eicosanoids are semi cyclic structures.
- They exert complex control over many bodily systems; in
- inflammation
- immunity, and as
- messengers in the central nervous system.

Lipid Chemistry



Lipid Chemistry

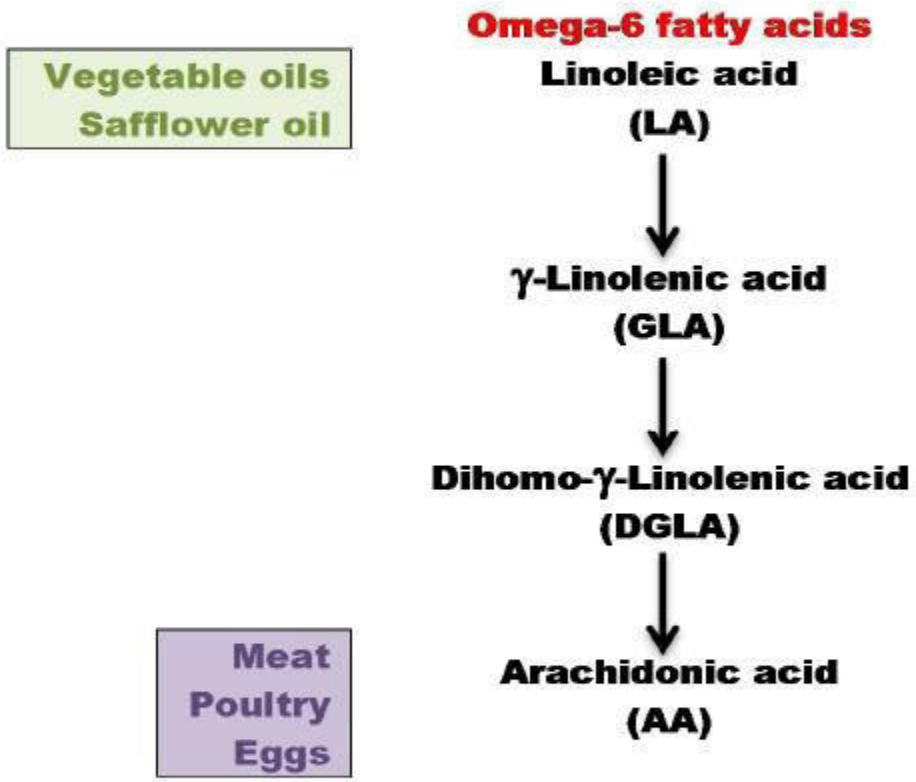


Lipid Chemistry

- **Polyunsaturated Fatty Acids Of Biological Importance** (Cont.)
- Eicosanoids are derived from either omega-3 (ω-3) or omega-6 (ω-6) fatty acids.
- In general, the ω-6 eicosanoids are **pro-inflammatory**.

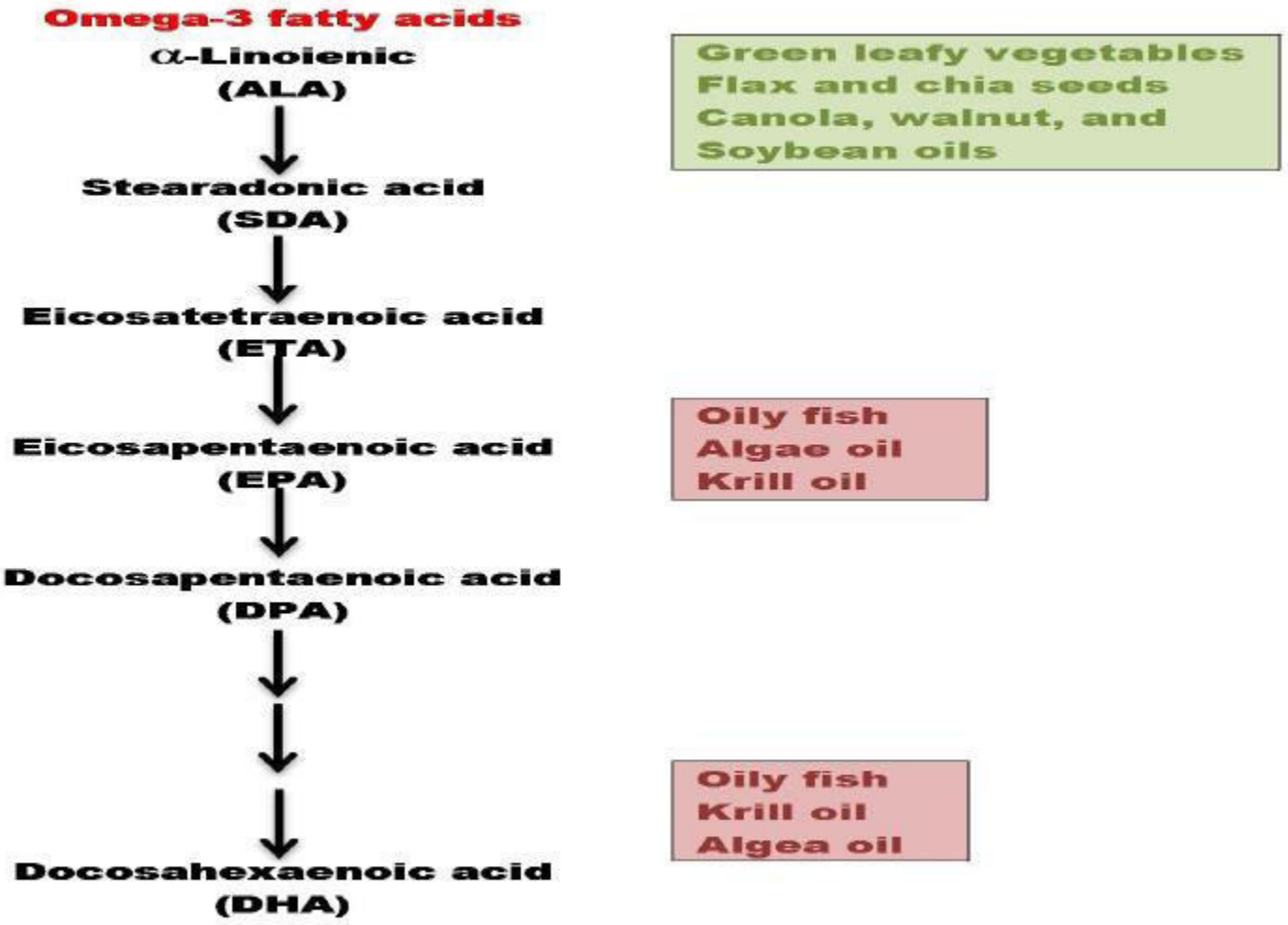
Lipid Chemistry

Figure 2. Classes of Essential Fatty Acids



Lipid Chemistry

Figure 2. Classes of Essential Fatty Acids

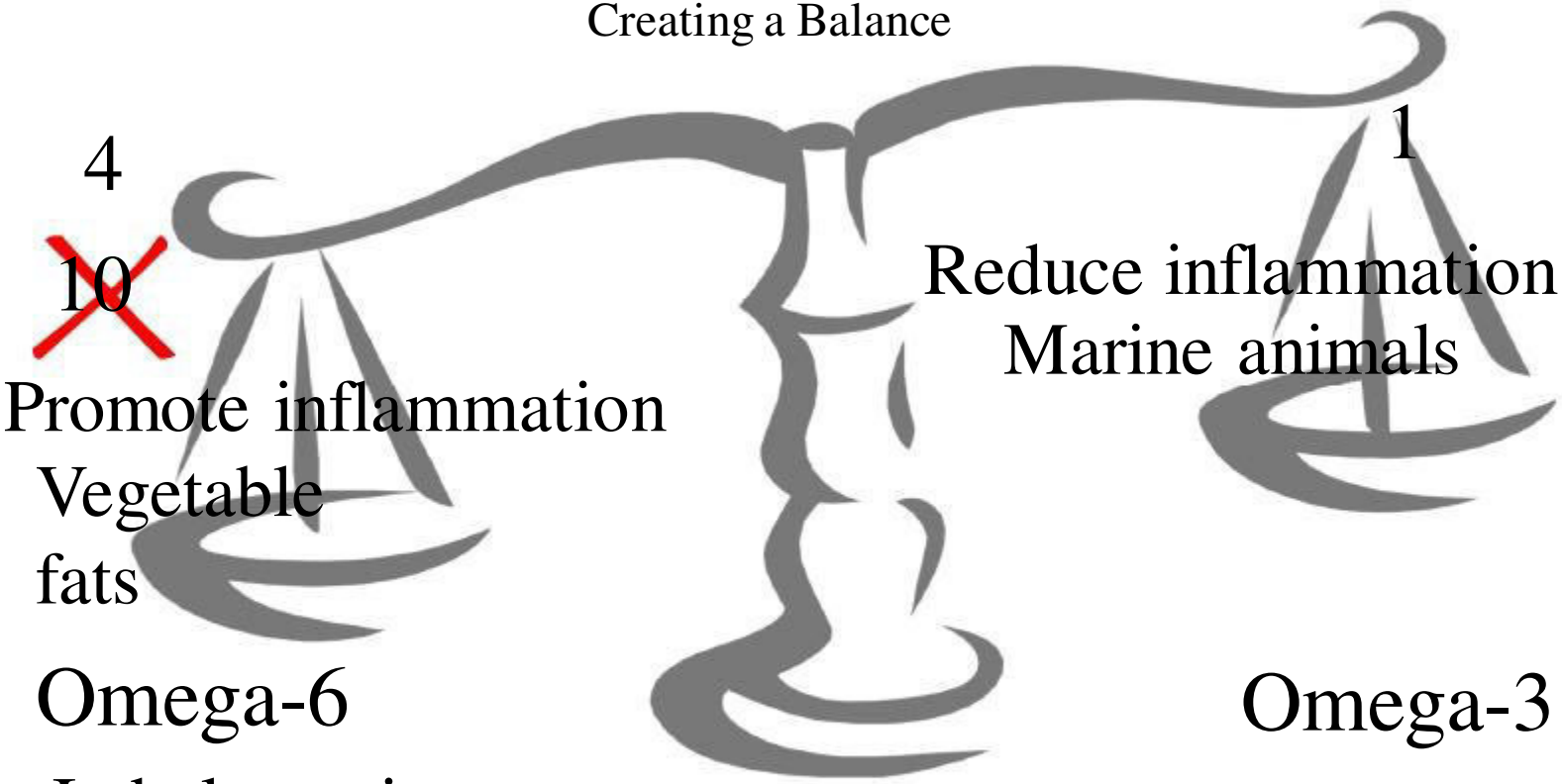


Lipid Chemistry

- **Imbalance of omega-6 and omega-3** PUFAs in the diet is associated with:
  - an increased risk of cardiovascular disease.

Lipid Chemistry

Creating a Balance



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Promote inflammation

Vegetable fats

Omega-6

1

Reduce inflammation

Marine animals

Omega-3

Imbalance in "western" diet

rising rate of inflammatory disease

Lipid Chemistry

- Long chain omega-3 fatty acids such as alpha-linolenic acid and their derivatives such as
- eicosapentaenoic acid (EPA) and
- docosahexaenoic acid (DHA) have
- anti-inflammatory effects

Docosa= 22C **hexa**= 6 double bonds;

eicosa=20C **penta**= 5 doube bonds

Lipid Chemistry

- In addition, docosahexaenoic acid (DHA) is selectively incorporated into
- retinal cell membranes and
- postsynaptic neuronal cell membranes
- suggesting it plays important roles in vision and nervous system function.

Lipid Chemistry

- Current evidence suggests that diet rich in omega 3 fatty acids are beneficial particularly for
- cardiovascular disease and also to some extent for
- Alzheimer's disease,
- Cancer and
- Rheumatoid Arthritis.

Lipid Chemistry

- This disease preventing role is due to the
- anti-inflammatory action of these
- omega-3 derivatives.

Lipid Chemistry

- Classically, ω-3 PUFAs mediate some of these effects by
- antagonizing ω-6 PUFA (arachidonic acid)-induced pro-inflammatory prostaglandin E_2 (PGE_2) formation

END

Lipid Chemistry

- **Unsaturated Fatty Acids of Physiologic and Nutritional Significance**

In this module we will talk about unsaturated fatty acid in a tabulated form.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family	Common Name	Systematic Name	Occurrence
Monoenoic acids (one double bond)				
16:1;9	ω7	Palmitoleic	*cis*-9-Hexadecenoic	In nearly all fats.
18:1;9	ω9	Oleic	*cis*-9-Octadecenoic	Possibly the most common fatty acid in natural fats; particularly high in olive oil.

Lipid Chemistry

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18:1;9	ω9	Elaidic	*trans*-9-Octadecenoic	Hydrogenated and ruminant fats.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family	Common Name	Systematic Name	Occurrence
Dienoic acids				
18:2;9,12	ω6	Linoleic	all-*cis*-9,12-Octadecadienoic	Corn, peanut, cottonseed, soy bean, and many plant oils.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family	Common Name	Systematic Name	Occurrence
Trienoic acids				
18:3;6,9,12	**ω6**	**γ-Linolenic**	**all-*cis*-6,9,12-Octadecatrienoic**	**Some plants, eg, oil of evening primrose, borage oil; minor fatty acid in animals.**
18:3;9,12,15	**ω3**	**α-Linolenic**	**all-*cis*-9,12,15-Octadecatrienoic**	**Frequently found with linoleic acid but particularly in linseed oil.**

Linseed .Alsi **borage oil:** Gao Zaban **primrose:** Gul e Fanjani (used in skin care products)

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family Omega	Common Name	Systematic Name	Occurrence
Tetraenoic acids (four double bonds)				
20:4;5,8,11,14	ω 6	Arachidonic	all-*cis*-5,8,11,14-Eicosatetraenoic	Found in animal fats; important component of phospholipids in animals.

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family Omega	Common Name	Systematic Name	Occurrence
Pentaenoic acids (five double bonds)				
20:5;5,8,11,14, 17	ω 3	Timnodonic	all-*cis*-5,8,11,14,17-Eicosapentaenoic	Important component of fish oils, e.g. cod liver, salmon oils and other fish

Lipid Chemistry

Number of C Atoms and Common Double Bonds	Family Omega	Common Name	Systematic Name	Occurrence
Hexaenoic acids (six double bonds)				
22:6;4,7,10,13,16,19	ω3	Cervonic	all-*cis*-4,7,10,13,16,19-Docosahexaenoic	Fish oils, phospholipids in brain.

Lipid Chemistry

Other Common Fatty Acids

- Other Fatty acids commonly found in body which are worth knowing are:
- Palmitic Acid (16:0)
- Stearic Acid (18:0)

and

- Oleic acid (18:1)

Lipid Chemistry

- **Palmitic Acid**(16:0)
- Palmitic acid, or hexadecanoic acid, is the

- most common saturated Fatty Acid found in

- animals, plants and microorganisms

Lipid Chemistry

- Palmitic acid mainly occurs as its ester in triglycerides (fats), especially
- palm oil.

It is also found in high amounts in

- Butter,
- Cheese,
- milk and
- meat

Lipid Chemistry

- Excess carbohydrates in the body are converted to palmitic acid.
- Palmitic acid is the first fatty acid produced during fatty acid synthesis and the
- precursor to longer fatty acids

Lipid Chemistry

- As a consequence, palmitic acid is a major body component of fats found in the animals.

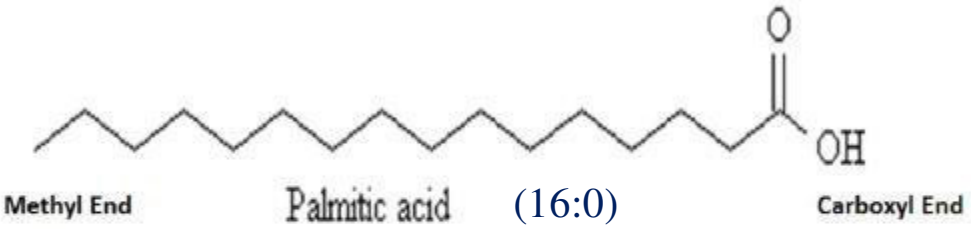
Lipid Chemistry

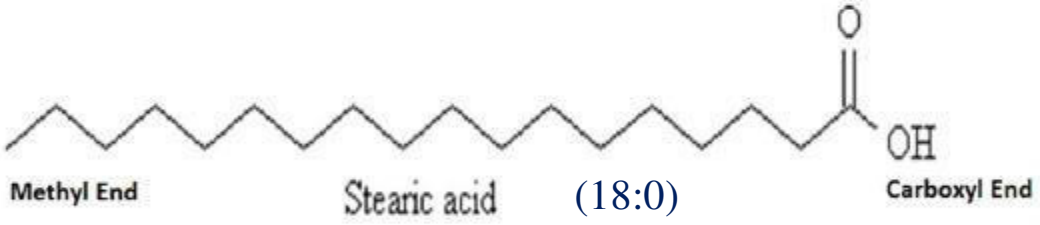
- **Stearic Acid** (18:0)
- As its ester, stearic acid is one of the most common saturated fatty acids found in nature following palmitic acid.

Lipid Chemistry

- Fats and oils rich in stearic acid are
- more abundant in animal fat (up to 30%)
- than in vegetable fat (typically <5%)

Lipid Chemistry



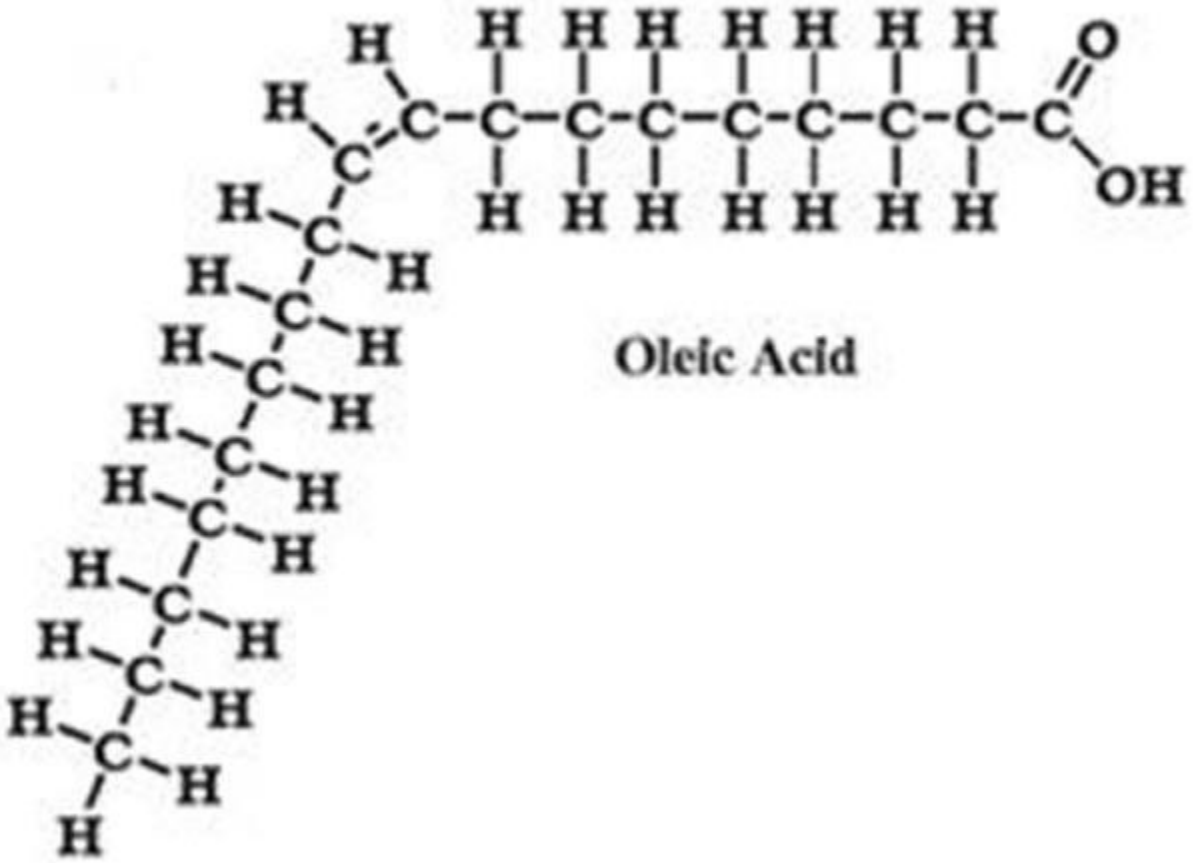


Lipid Chemistry

- **Oleic Acid**(18:1)
- Possibly the most common fatty acid in adipose tissue

- It is particularly high in olive oil
- Oleic acid in olive oil accounts for the blood pressure lowering effect of olive oil.

Lipid Chemistry



Oleic Acid

Lipid Chemistry

- In addition, because of cis configuration and the kink produced due to the cis configuration.
- It is also highly abundant in membrane lipids, providing
- fluidity to membrane.

Saturated Fatty Acids

Common name	Number of carbons	
Acetic	2	Major end product of carbohydrate fermentation by rumen organisms
Butyric	4	In certain fats in small amounts (especially butter). An end product of carbohydrate fermentation by rumen organisms. Also formed in the cecum of herbivores and to a lesser extent in the colon of humans.
Valeric	5	
Caproic	6	

Saturated Fatty Acids

Common name	Number of carbons	
Lauric	12	Spermaceti, cinnamon, palm kernel, coconut oils, laurels, butter
Myristic	14	Nutmeg, palm kernel, coconut oils, myrtles, butter
Palmitic	16	Common in all animal and plant fats
Stearic	18	

Spermaceti by the sperm whale. Palm saeed. evergreen tree *(Laurus nobilis)* of the Mediterranean region valued for its aromatic ovate leaves. Nutmeg Jorenk, Jefal

Lipid Chemistry

- **Glycerol and Sphingosine**

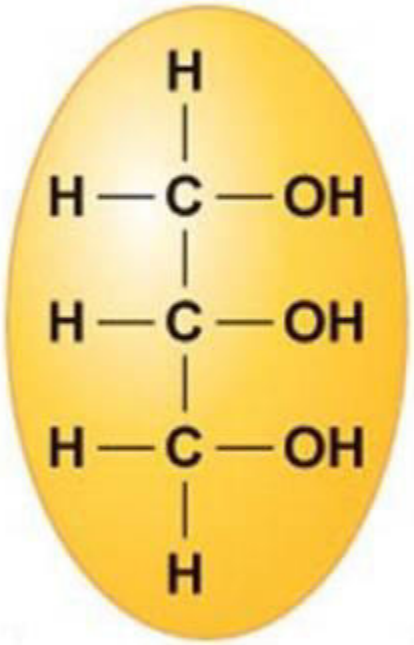
- Glycerol and Sphingosine are the two types of **alcohols** most commonly found in lipids.

Lipid Chemistry

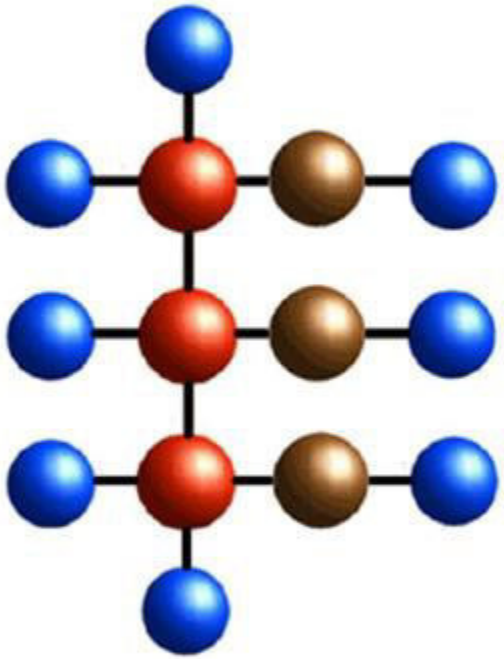
- **Glycerol**
- It is a simple poly hydroxy alcohol (also called polyol or sugar alcohol)
- and part of a class of lipids: glycolipids

- It contains 3 carbons and 3 hydroxyl (OH) groups.

Lipid Chemistry



Glycerol



Lipid Chemistry

- Glycerol is synthesized from

- Dihydroxyacetone Phosphate (an intermediate of the glycolytic pathway)

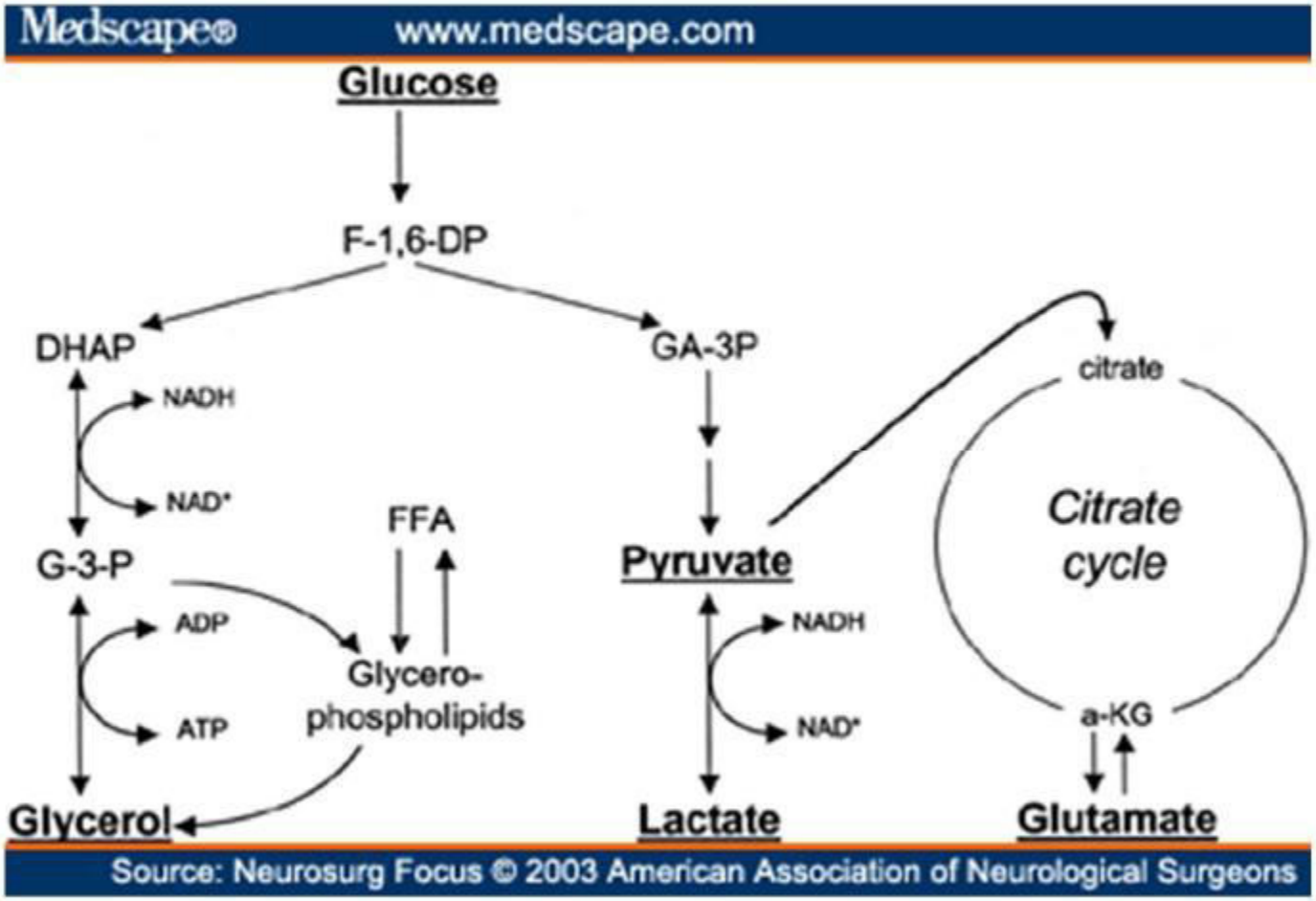
Lipid Chemistry

- Dihydroxyacetone phosphate is acted upon by two enzymes to form glycerol, namely

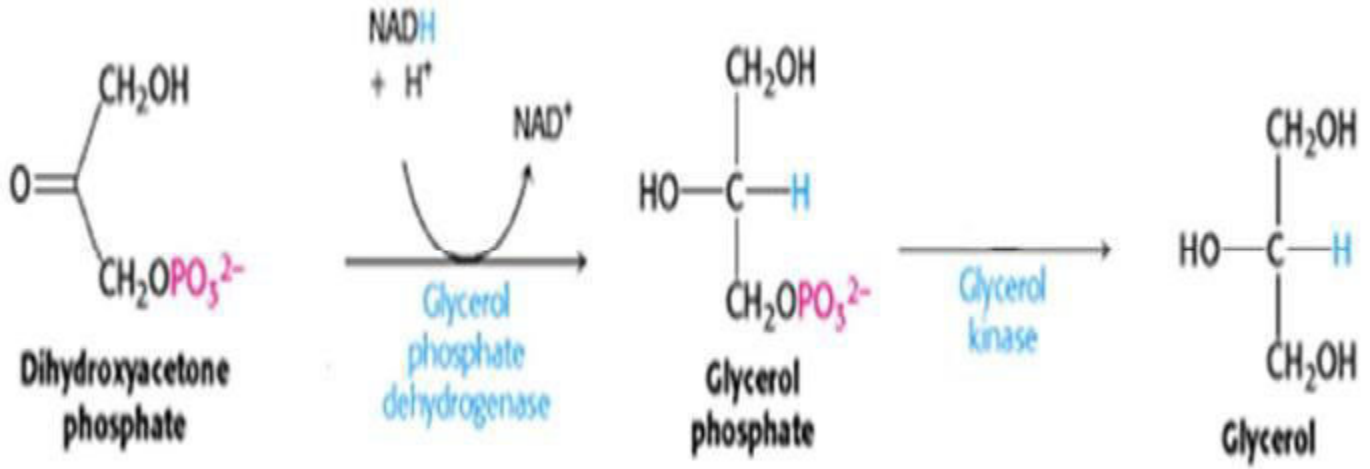
- Glycerol Phosphate Dehydrogenase and

- Glycerol Kinase respectively.

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- Glycerol is a precursor for synthesis of triacylglycerols and of phospholipids in the liver and adipose tissue.

Lipid Chemistry

- When the body uses stored fat as a source of energy, glycerol and fatty acids are released into the bloodstream.
- And enter the glycolysis pathway directly

END

Lipid Chemistry

- Sphingosine is an amino alcohol,
- which is a component of the class of lipids known as sphingolipids

Lipid Chemistry

- Sphingosine is synthesized in the body in the form of ceramide,

- to which different moieties are added to form sphingolipids.

Lipid Chemistry

- Serine and palmitoyl CoA condense to form a product (ketosphinganine) that is reduced.
- A very long-chain fatty acid forms an amide with the amino group.
- a double bond is generated, and ceramide is formed.

Lipid Chemistry

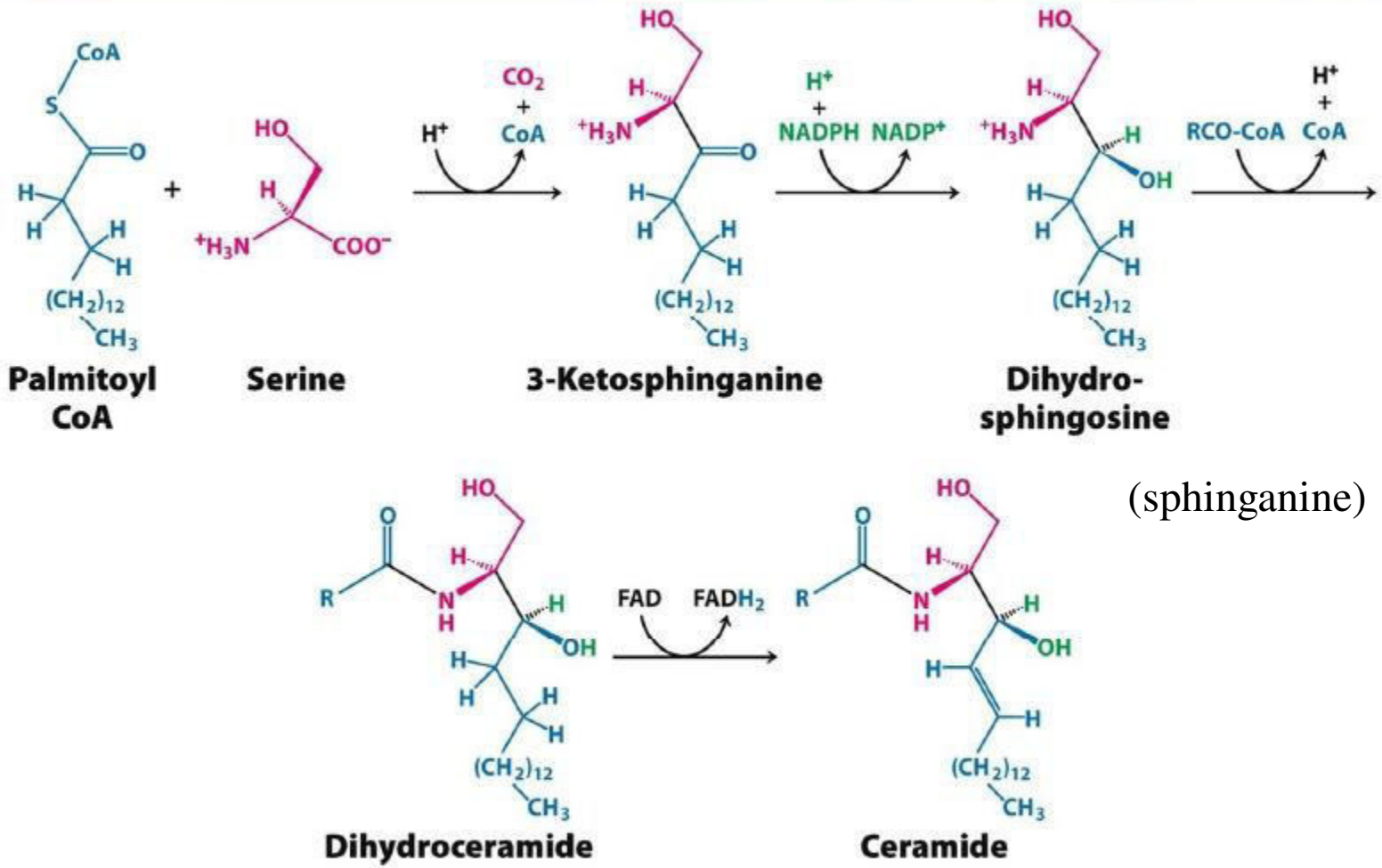


Figure 26.3
Biochemistry, Seventh Edition
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Lipid Chemistry

- There is no direct route of synthesis from sphinganine (dihydrosphigosine) to sphingosine;
- it has to be acylated

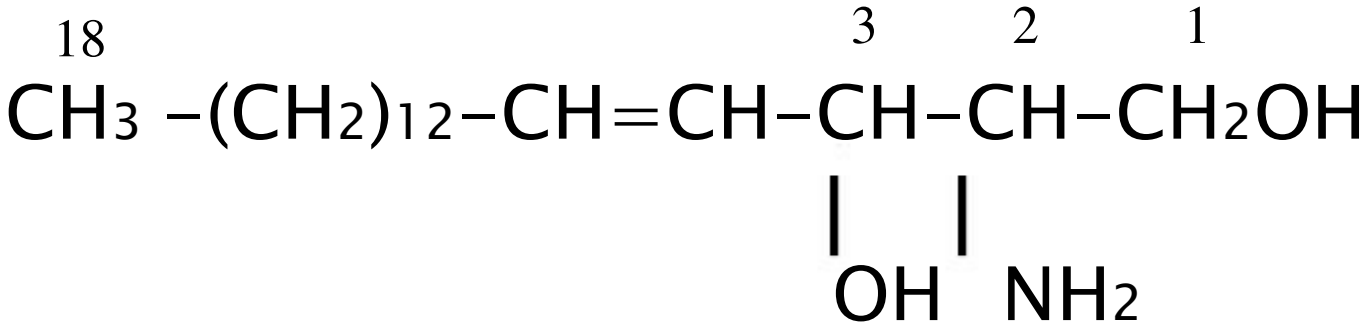
- first to dihydroceramide,
- which is then dehydrogenated to **ceramide.**

Lipid Chemistry

- Sphingosine is formed via degradation of sphingolipid in the lysosome.
- Therefore ceramide is the structural parent of all sphingolipids.

Lipid Chemistry

SPHINGOSINE



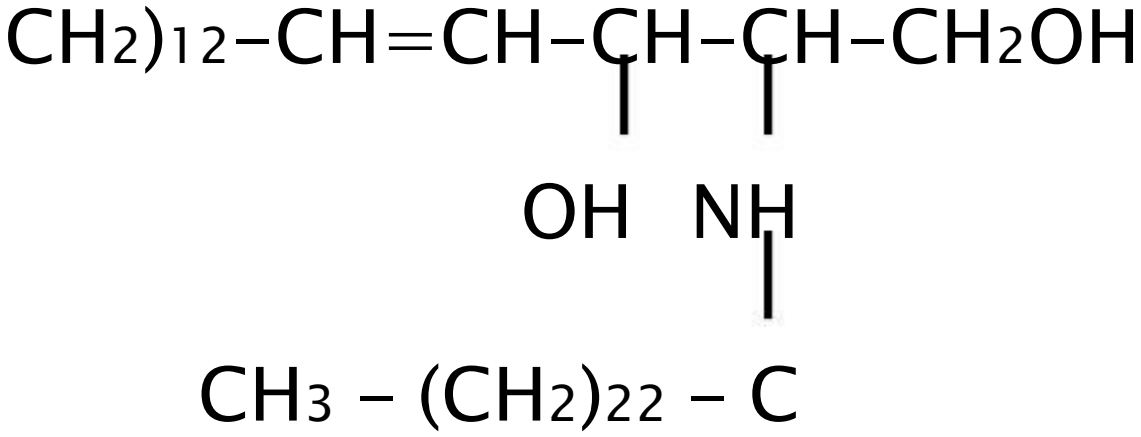
Carbon atoms 1 and 2 arise from the amino acid serine

Carbon atoms 3 to 18 arise from palmitic acid

Lipid Chemistry

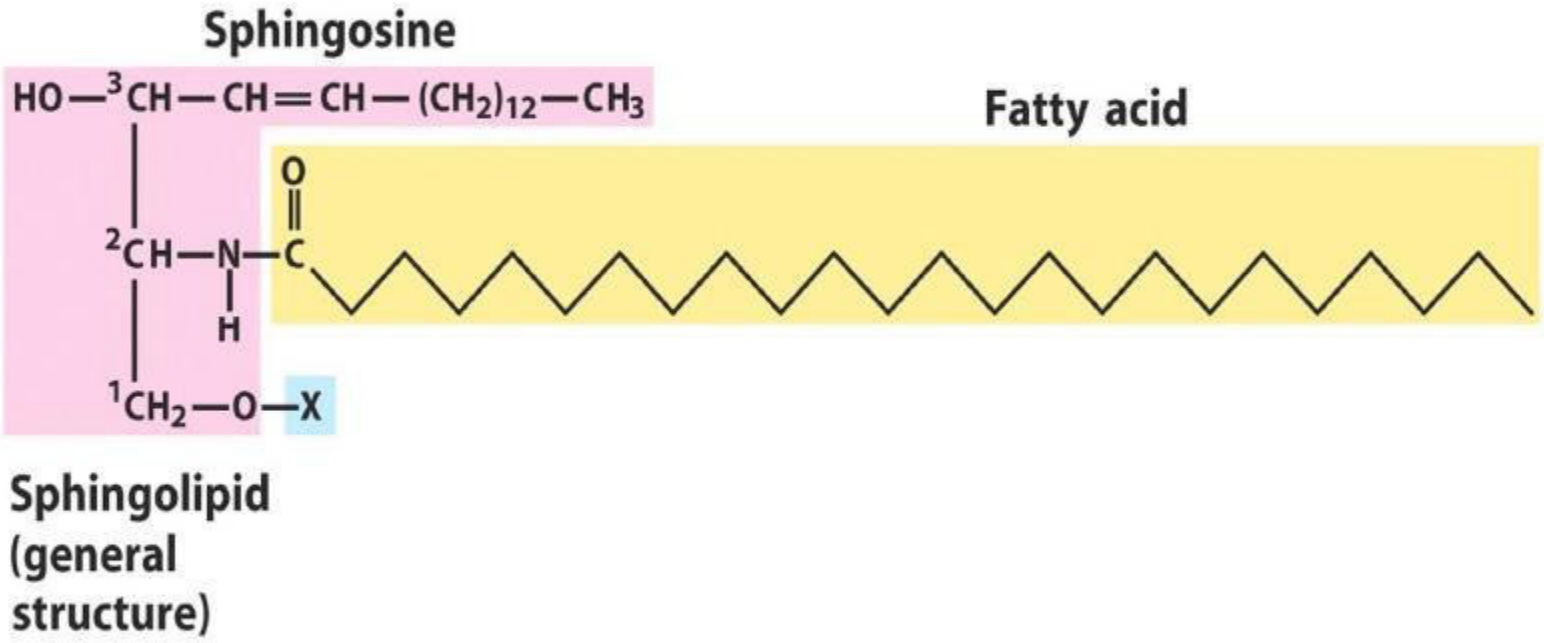
CERAMIDE

Sphingosine



Long Chain Fatty Acid

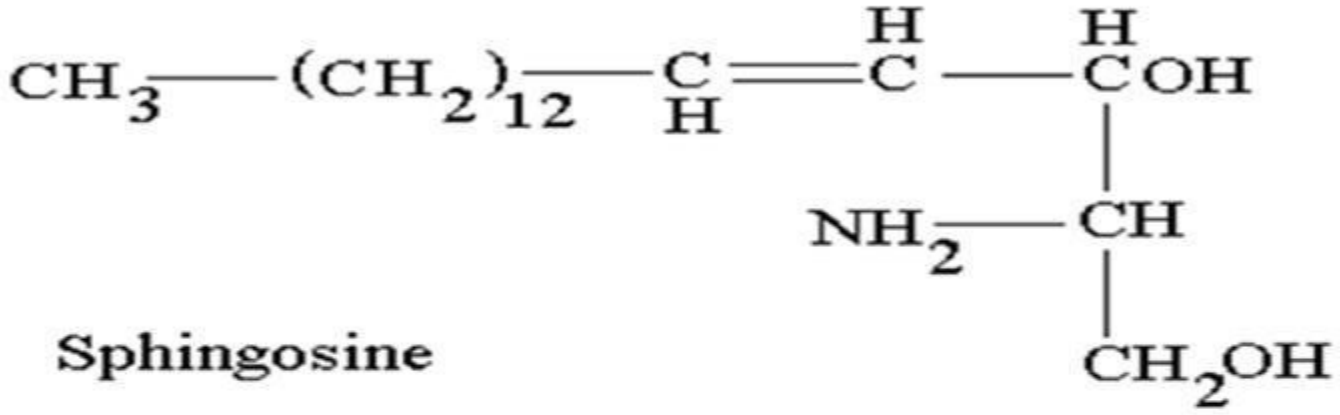
Lipid Chemistry



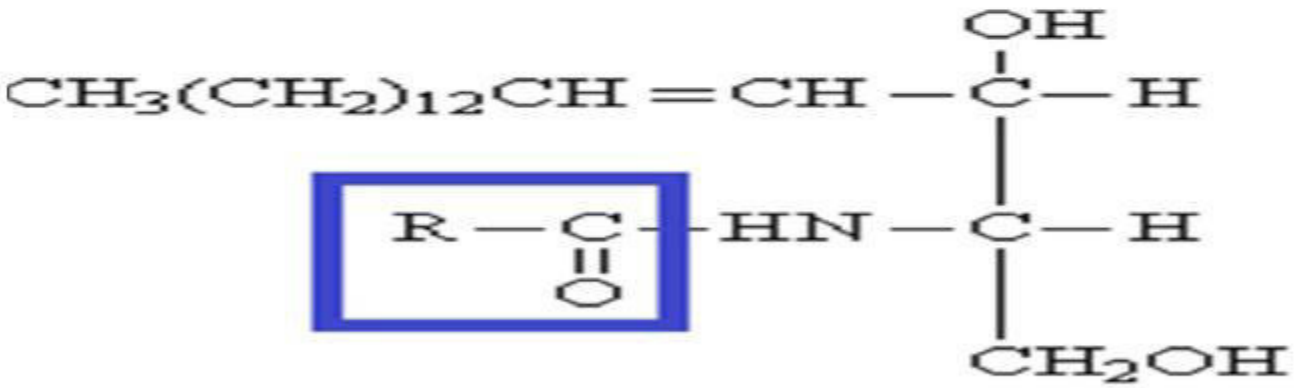
Lipid Chemistry

- In short,
- Serine+ Palmitate
  = Sphingosine
- Sphingosine + FA
  = ceramide

Lipid Chemistry



Sphingosine



Ceramide

Lipid Chemistry

- **Properties of Glycerol**
- Popularly known as glycerin,
- Glycerol is widely used in pharmaceutical and cosmetic preparations.

Lipid Chemistry

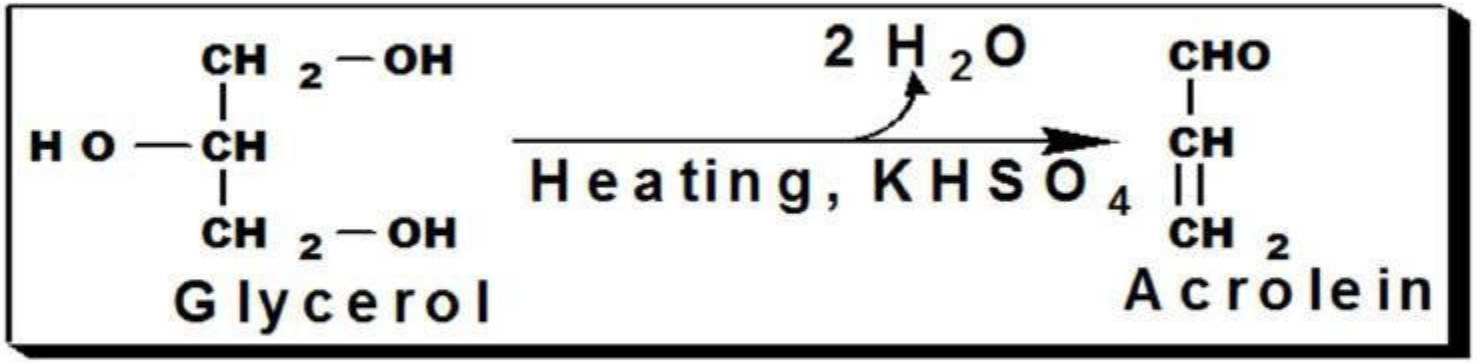
- It has the following properties:
- Colorless
- Viscous oily liquid with
- sweet taste.

Lipid Chemistry

- Acrolein Test
- On heating with sulfuric acid or $KHSO_4$ (dehydration) it
- gives acrolein that
- has a bad odor.
- used for detection of free glycerol or any compound containing glycerol.

Potassium bisulfate is a potassium acid salt of sulfuric acid, with the molecular formula KHSO 4

Lipid Chemistry



Lipid Chemistry

- In contrast to glycerol
- Sphingosine does not show positive acrolein test.
- Therefore glycerolipids and shingolipids can be differentiated on the basis of acrolein test.

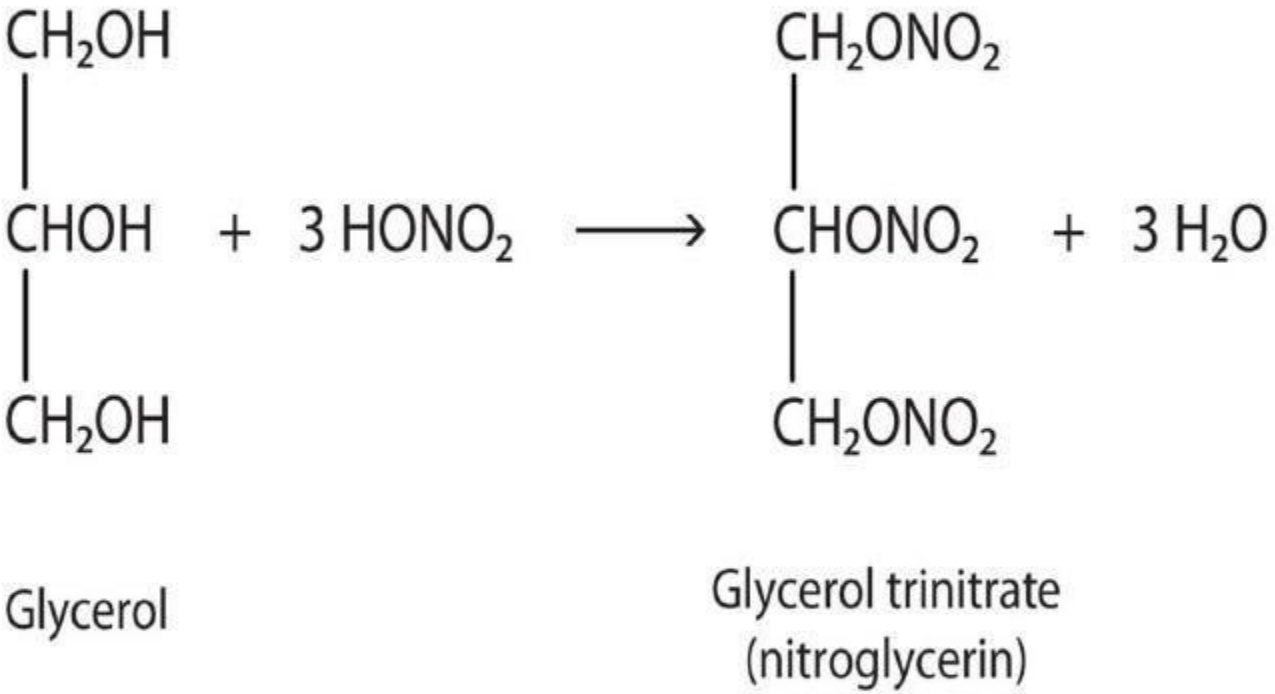
Lipid Chemistry

- Glycerol combines with three molecules of nitric acid to form Glycerol trinitrate

- that is used as explosive and

- vasodilator

Lipid Chemistry



Lipid Chemistry

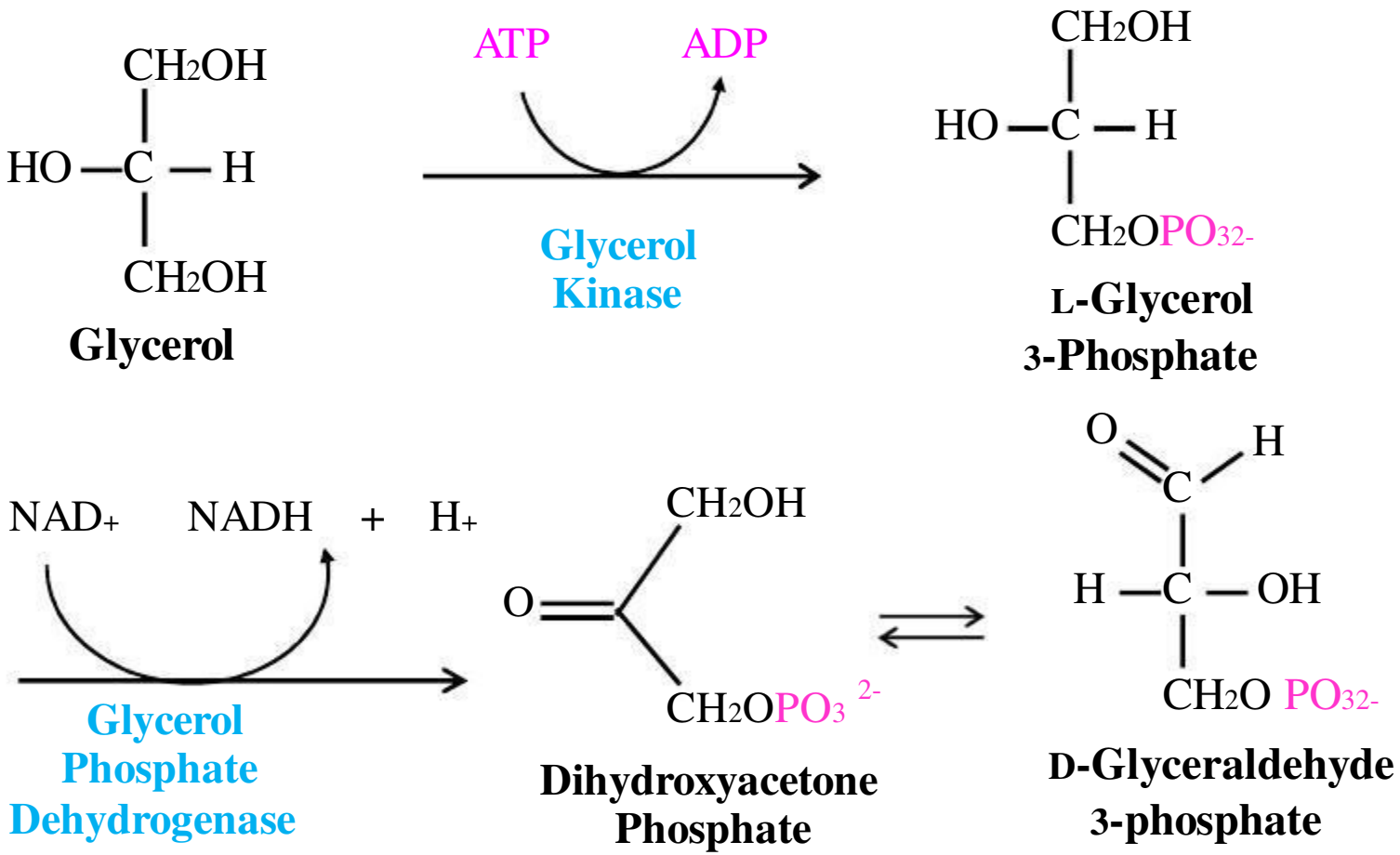
On **esterification** with fatty acids it gives:

- monoacylglycerol: one fatty acid + glycerol.
- diacylglycerol: two fatty acids + glycerol.
- triacyl-glycerol: three fatty acids + glycerol.

Lipid Chemistry

- **Gluconegenic substrate**
- Glycerol can form dihydroxyacetone phosphate
- which can convert into glucose or
- glyceraldehyde phosphate
- to enter into gluconeogenic pathway.

Lipid Chemistry



Lipid Chemistry

Simple lipids:

Esters of fatty acids
with various alcohols

These contain:

a. Fats (and Oils) and

b. Waxes.

Lipid Chemistry

a. **Fats:** Esters of fatty acids with glycerol (**Oils** are fats in the liquid state)

b. **Waxes:** Esters of fatty acids with higher molecular weight monohydric alcohols. (having one OH group)

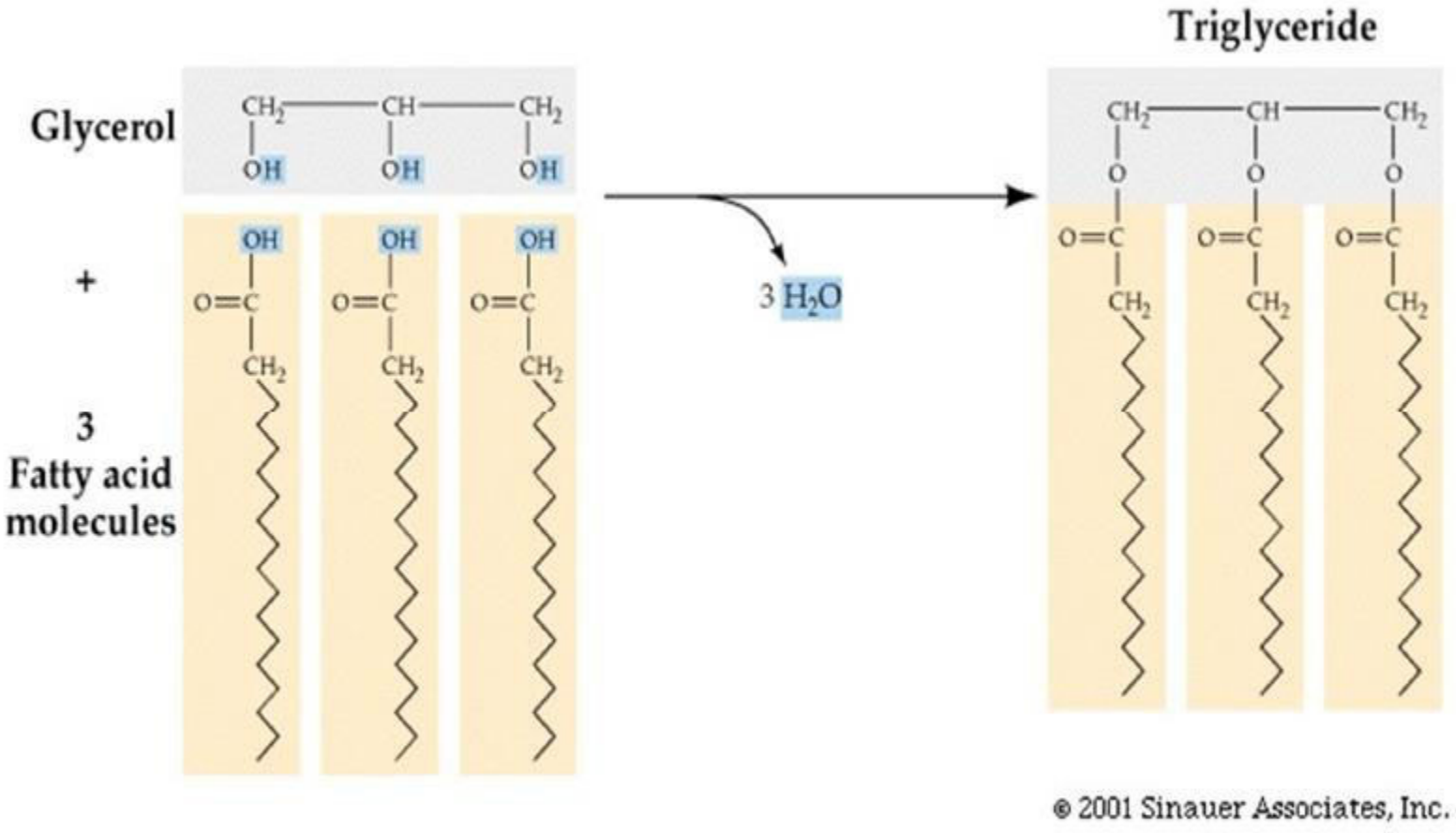
Lipid Chemistry

- **Triacylglycerols (TAGs)**
- The simplest lipids constructed from fatty acids are the triacylglycerols,
- Also referred to as;
- triglycerides,
- fats, or
- neutral fats or
- storage lipids.

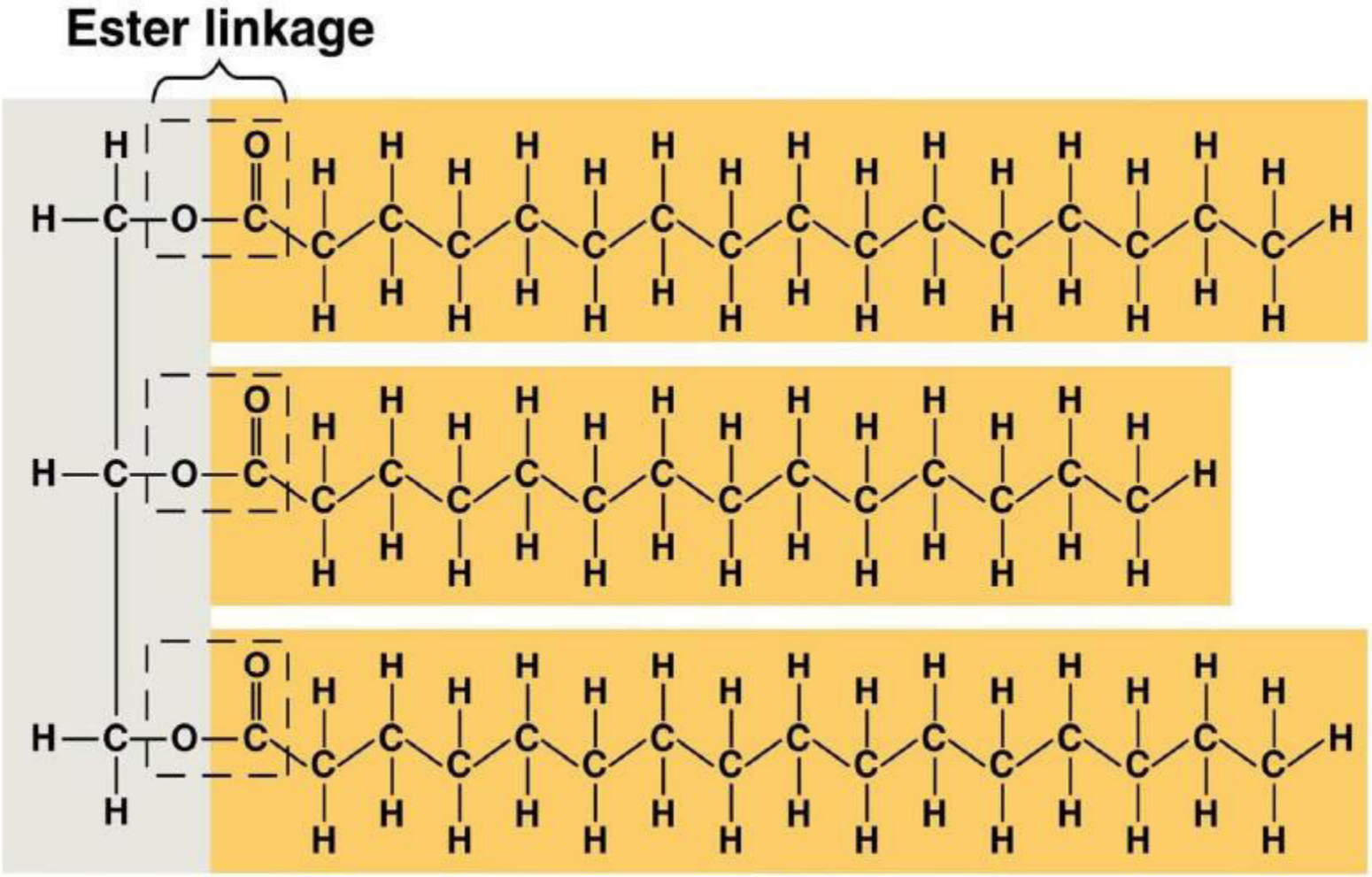
Lipid Chemistry

- Triacylglycerols are composed of
- three fatty acids
- in ester linkage with
- a single glycerol

Lipid Chemistry



Lipid Chemistry



(b) Fat molecule (triacylglycerol)

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Lipid Chemistry

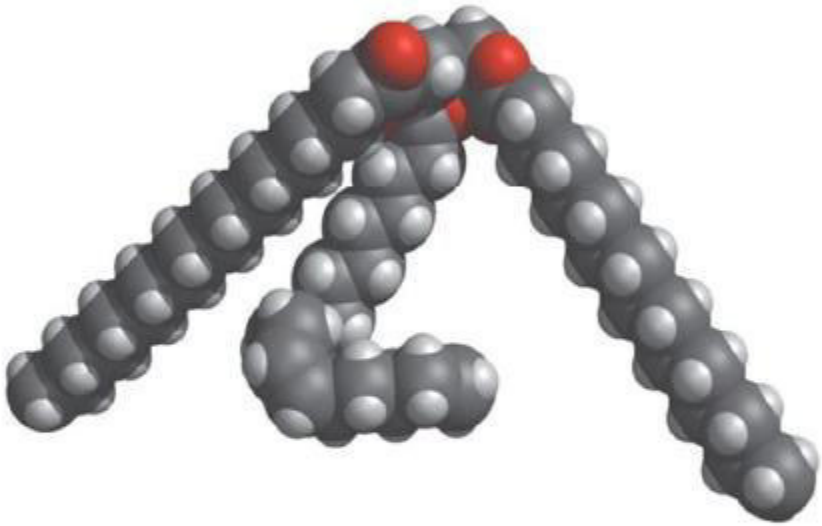
- **Structure of triacylglycerols**

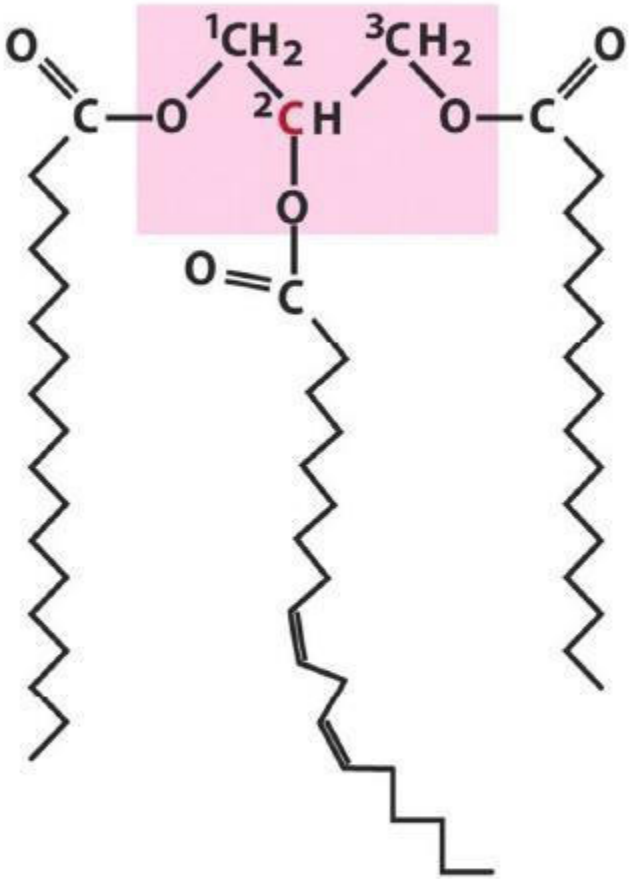
- The three fatty acids esterified to a glycerol molecule are usually **not** of the same type

Lipid Chemistry

- The fatty acid on **carbon 1** is typically saturated,
- Whereas that on **carbon 2** is unsaturated,
- and that on **carbon 3** can be either

Lipid Chemistry





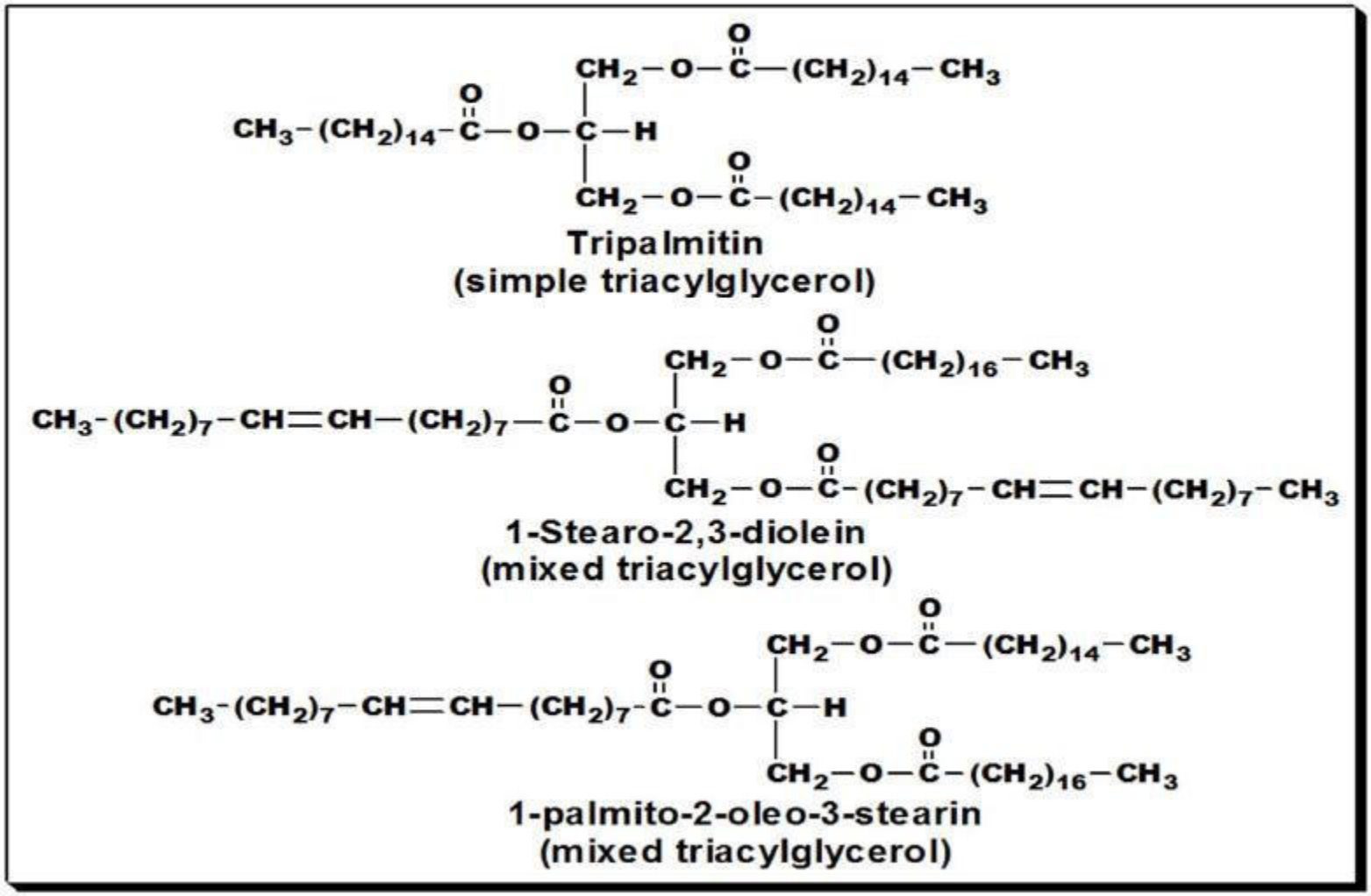
1-Stearoyl, 2-linoleoyl, 3-palmitoyl glycerol, a mixed triacylglycerol

Lipid Chemistry

- Simple triglycerides:
- Fatty acids connected to glycerol are of the same type
- e.g., tripalmitin.
- Mixed triglycerides:
- Fatty acids are of different types, e.g.,
- stearo-diolein and palmito-oleo-stearin.

Stearo-diolein: saturated FA at C1, MUFA at C2, C3.Palmito-Oleo-Stearin: saturated at FA at C1,

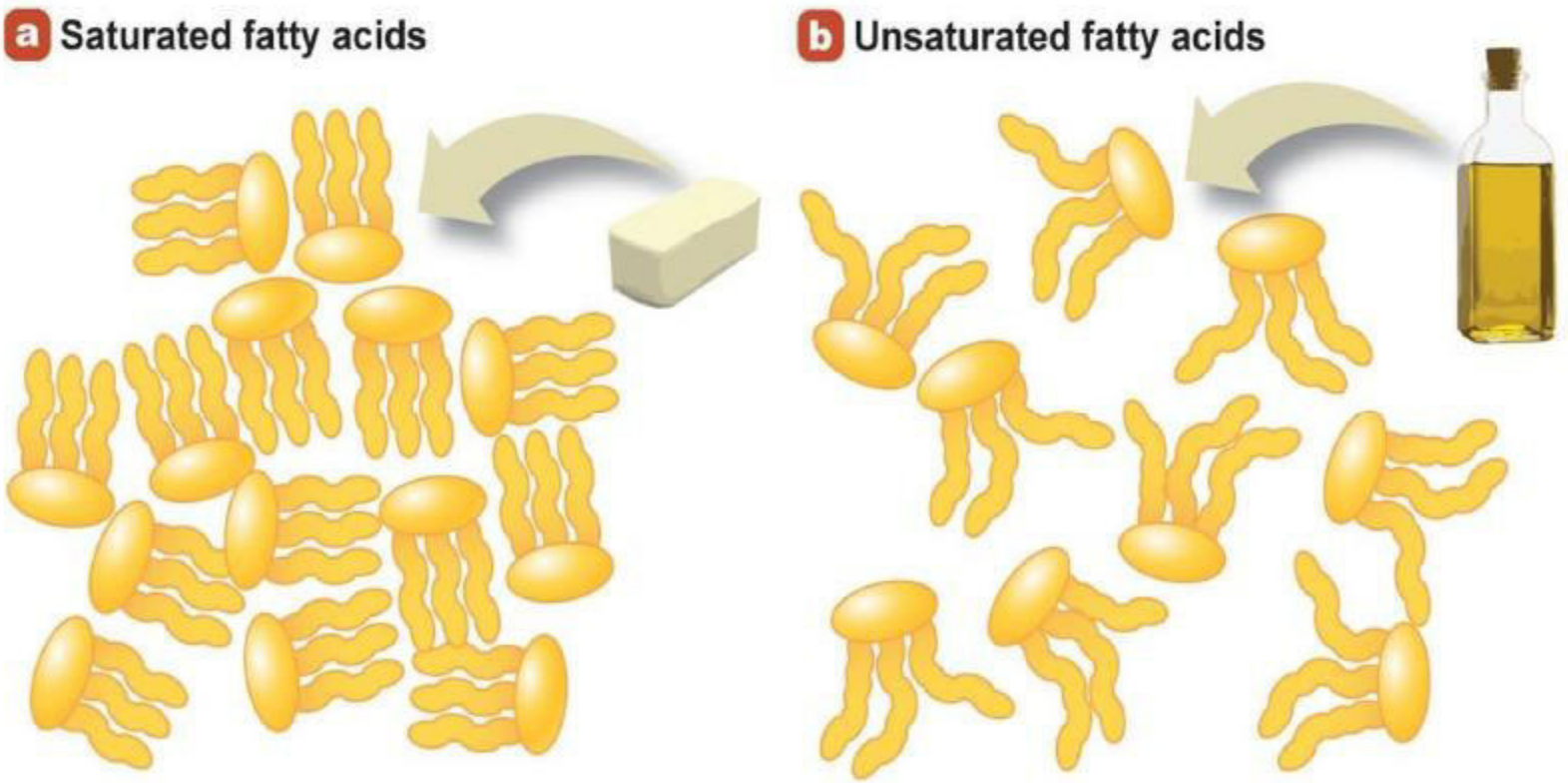
Lipid Chemistry



Lipid Chemistry

- The main difference between fats and oils is for oils being liquid at room temperature, whereas, fats are solids.
- This is mainly due to presence of larger percentage of unsaturated fatty acids in oils than fats that has mostly saturated fatty acids.

Lipid Chemistry



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Lipid Chemistry

- TAGs containing saturated fatty acids are solid at room temperature such as butter whereas
- TAGs containing unsaturated fatty acids are liquid at room temperature such as olive oil.(Olic acid, 18:1,9)

stearic acid, palmitic acid, lauric acid, butyric acid in butter

Lipid Chemistry

- **Triacylglycerols (TAGs)**
- Because the polar hydroxyls of glycerol
- and the polar carboxylates of the fatty acids are
- bound in ester linkages

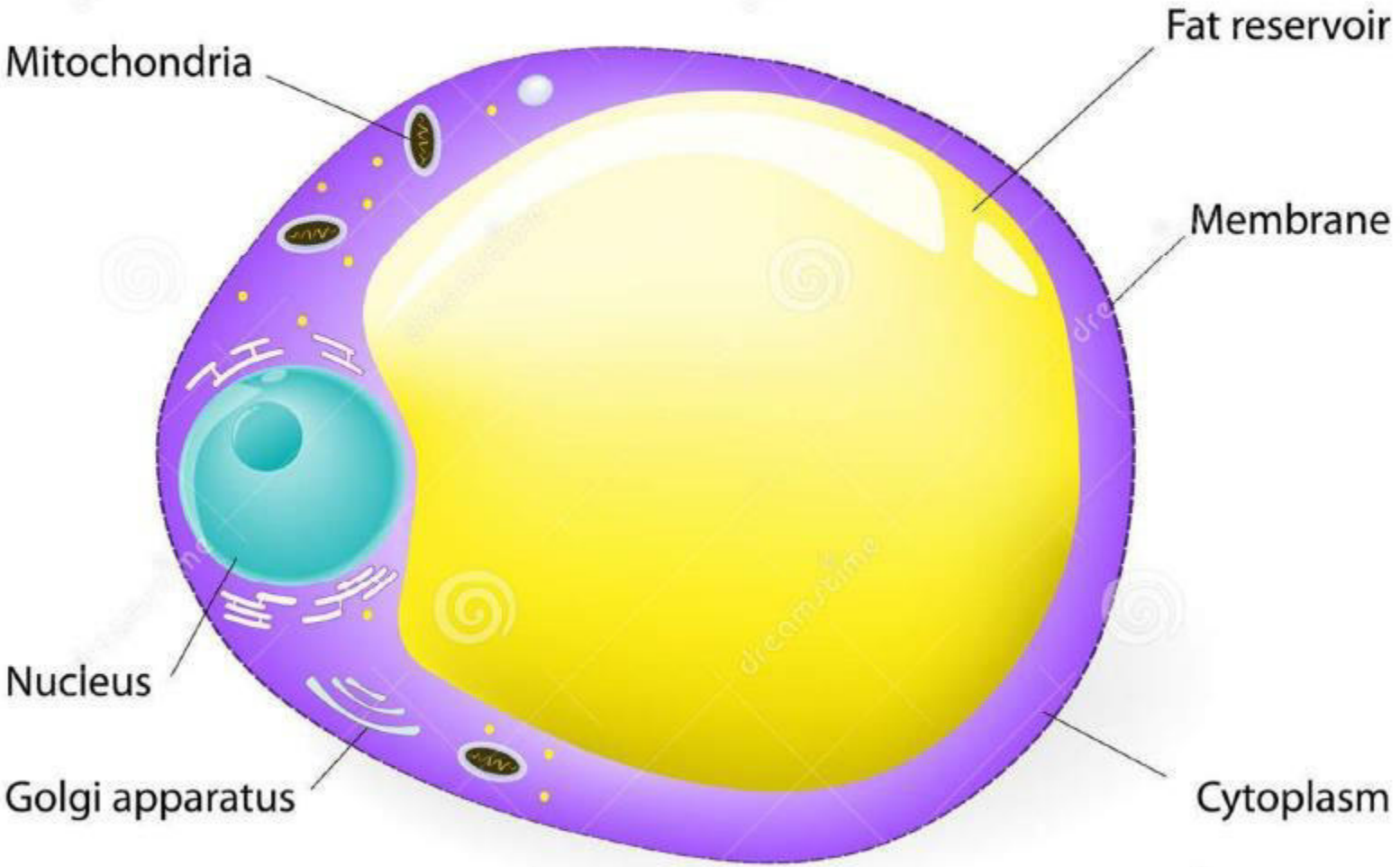
Lipid Chemistry

- Therefore the triacylglycerols are nonpolar, hydrophobic molecules, essentially insoluble in water
- Triacylglycerols provide
- Stored Energy
- and Insulation.

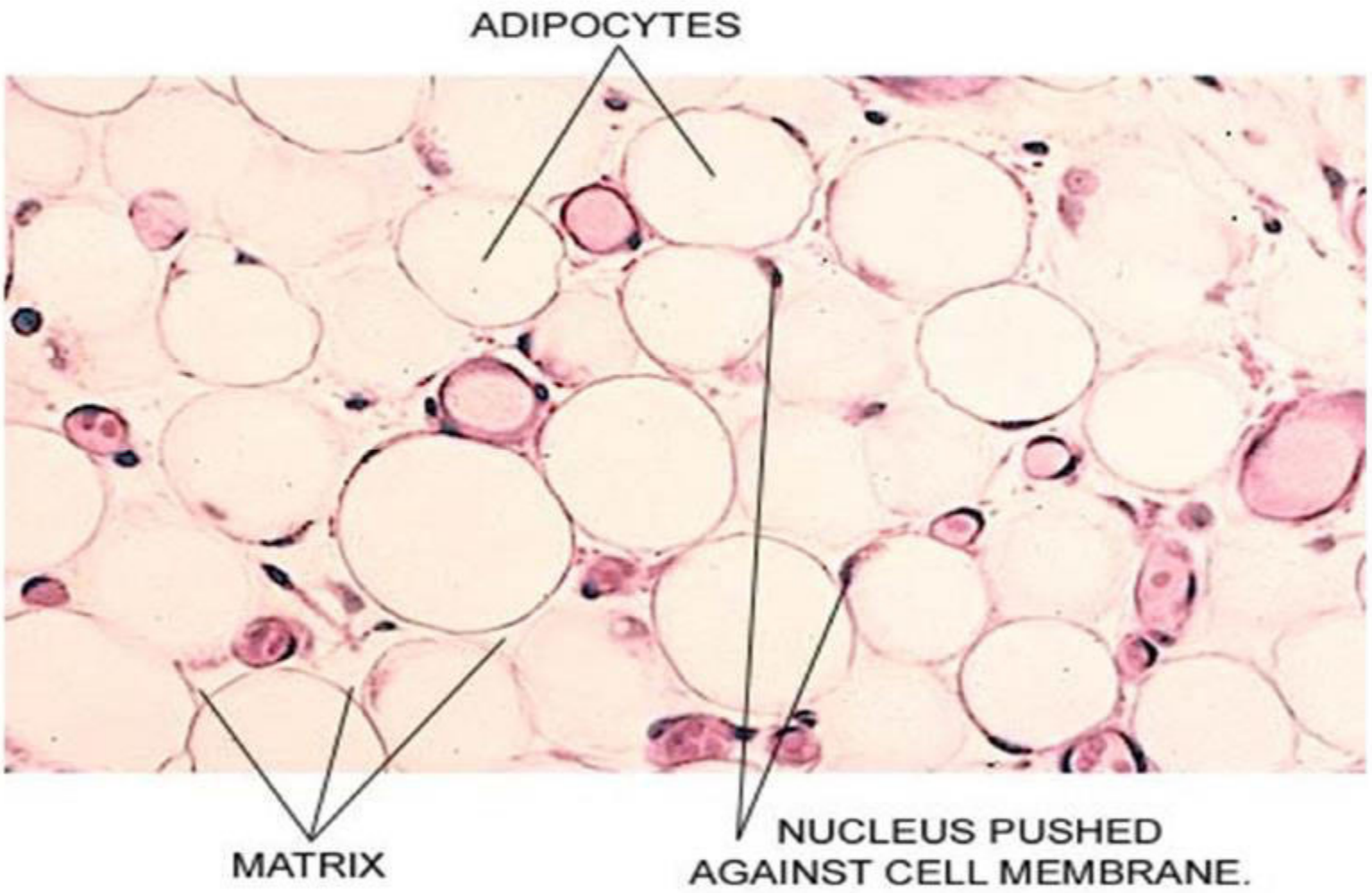
Lipid Chemistry

- In vertebrates, specialized cells called **adipocytes, or fat cells**, store large amounts of triacylglycerols as fat droplets that nearly fill the cell

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- Triacylglycerols are also stored as oils in the **seeds** of many types of plants.
- Thus providing energy and biosynthetic precursors during seed germination

Lipid Chemistry

- Adipocytes and germinating seeds contain lipases, enzymes that catalyze the hydrolysis of stored triacylglycerols.
- Therefore releasing fatty acids for export to sites where they are required

END

Lipid Chemistry

- **Triacylglycerols (TAGs) Cont.**
- There are two significant advantages to using triacylglycerols as stored fuels, rather than polysaccharides such as glycogen and starch.

Lipid Chemistry

- **First,** the carbon atoms of fatty acids are more reduced than those of sugars, and oxidation of triacylglycerols yields more than twice as much energy, as the oxidation of carbohydrates

Lipid Chemistry

- **Second,** because triacylglycerols are hydrophobic and therefore unhydrated.

Lipid Chemistry

- The organism that carries fat as fuel does not have to carry the extra weight of water of hydration that is associated with stored polysaccharides
- (2 g per gram of polysaccharide)

Lipid Chemistry

- Moderately obese people with 15 to 20 kg of triacylglycerols deposited in their adipocytes could meet their energy needs for months by drawing on their fat stores

Lipid Chemistry

- In contrast, the human body can store Iess than a day's energy supply in the form of glycogen (the polymer of glucose)

Lipid Chemistry

- In some animals, triacylglycerols stored under the
- skin serve as insulation against low temperatures.
- Seals, penguins, bears and other warm-blooded polar animals are
- amply padded with triacylglycerols

Lipid Chemistry

Lipid Chemistry

- **Properties of TAGs**
- **Physical properties**
- Neutral fats are

1. colourless,
2. odorless and
3. tasteless substances

Lipid Chemistry

- **4. Solubility**:
- They are insoluble in water but
- soluble in organic fat solvents(e.g., ether, benzene, acetone, chloroform)

Lipid Chemistry

- **5. Specific gravity:**
- The specific gravity of all fats is less than 1.0,
- consequently all fats float in water

Specific gravity of water is **1**

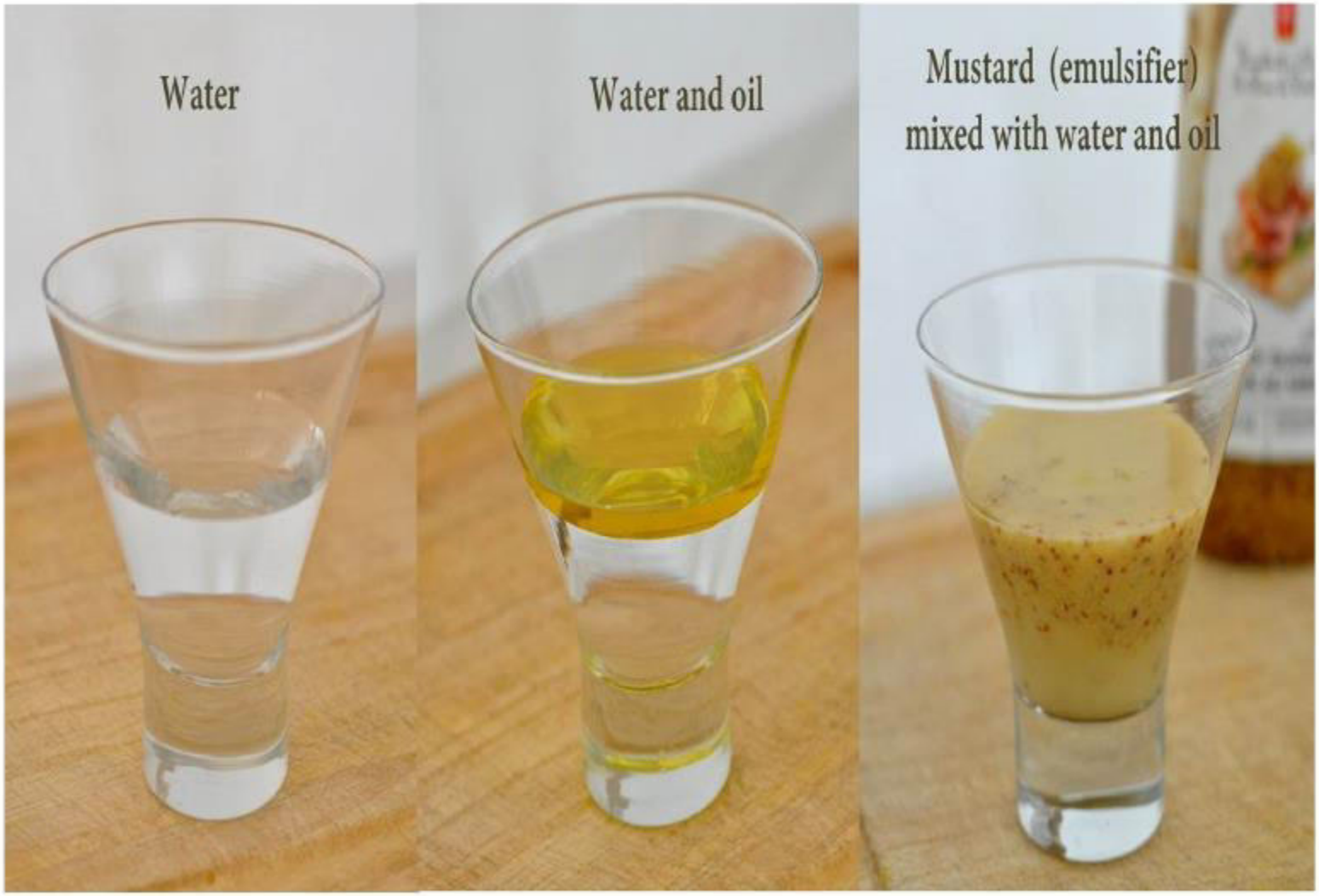
Lipid Chemistry

- **6. Emulsification:**

- Emulsions of fat may be made by shaking vigorously in water and by emulsifying agents such as gums and soaps

These agents have both a hydrophilic and a lipophilic part in their chemical structure

Lipid Chemistry



In Mustard variety of chemicals in the mucilage surrounding the seed hull act as emulsifiers

Lipid Chemistry

- The emulsification of dietary fats in intestinal canal, brought about by bile salts, is
- a prerequisite for digestion and absorption of fats.

Lipid Chemistry

- The **bile salts**, act to break apart the fat globules in the small intestines and allow them to become more "soluble" for absorption.

Lipid Chemistry

- The hydrophobic fat molecules will clump together into globules in the watery mixture in the digestive system.
- The emulsifiers break them down to smaller "globules" and allow them to become more soluble.

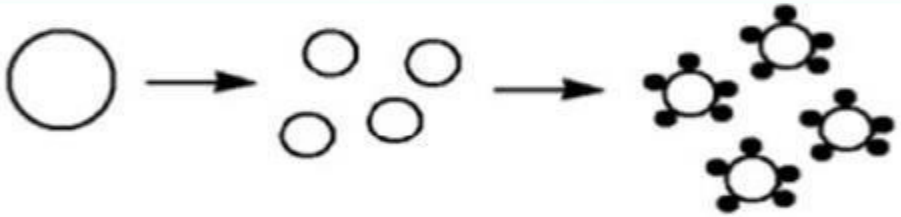
Lipid Chemistry

Bile salts emulsify fats i.e.:

- break large fat globules into smaller globules

What is the benefit of emulsification?

The surface area where lipase can act is increased.

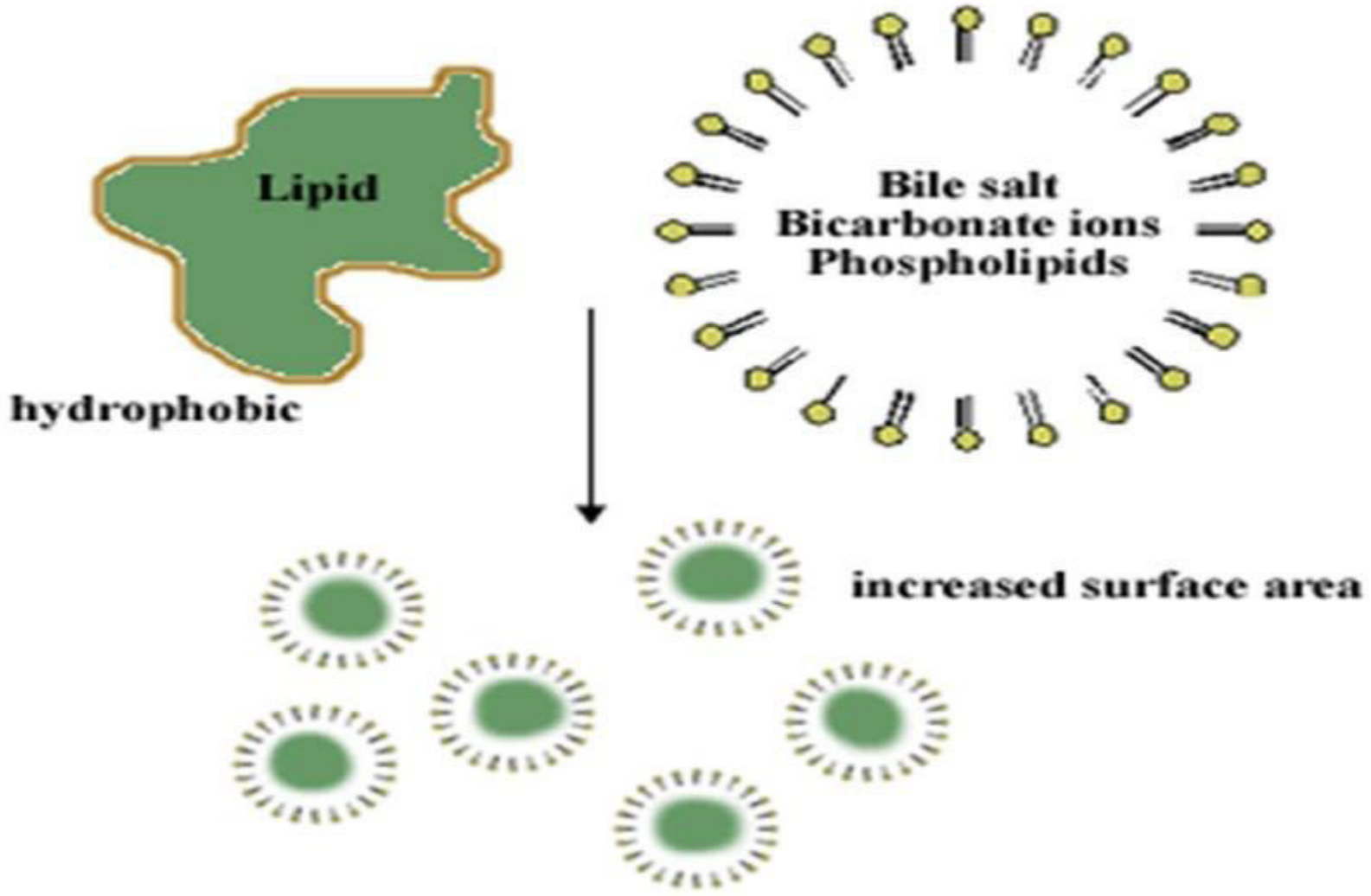


large fat droplet is broken down by muscular actions

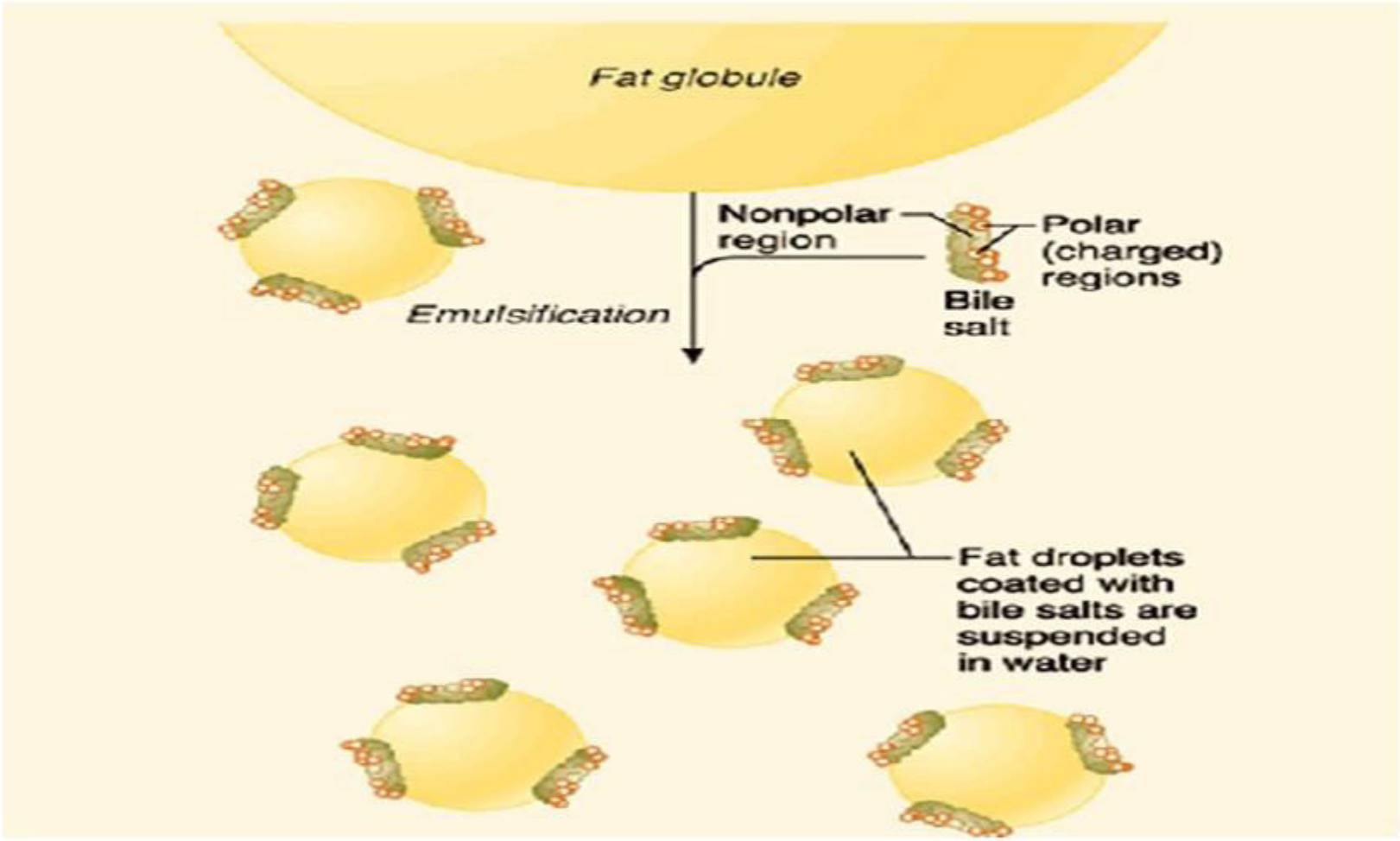
bile salts prevent small droplets from combining

Emulsification of fats

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- **Chemical Properties of TAGs**

1. Hydrolysis

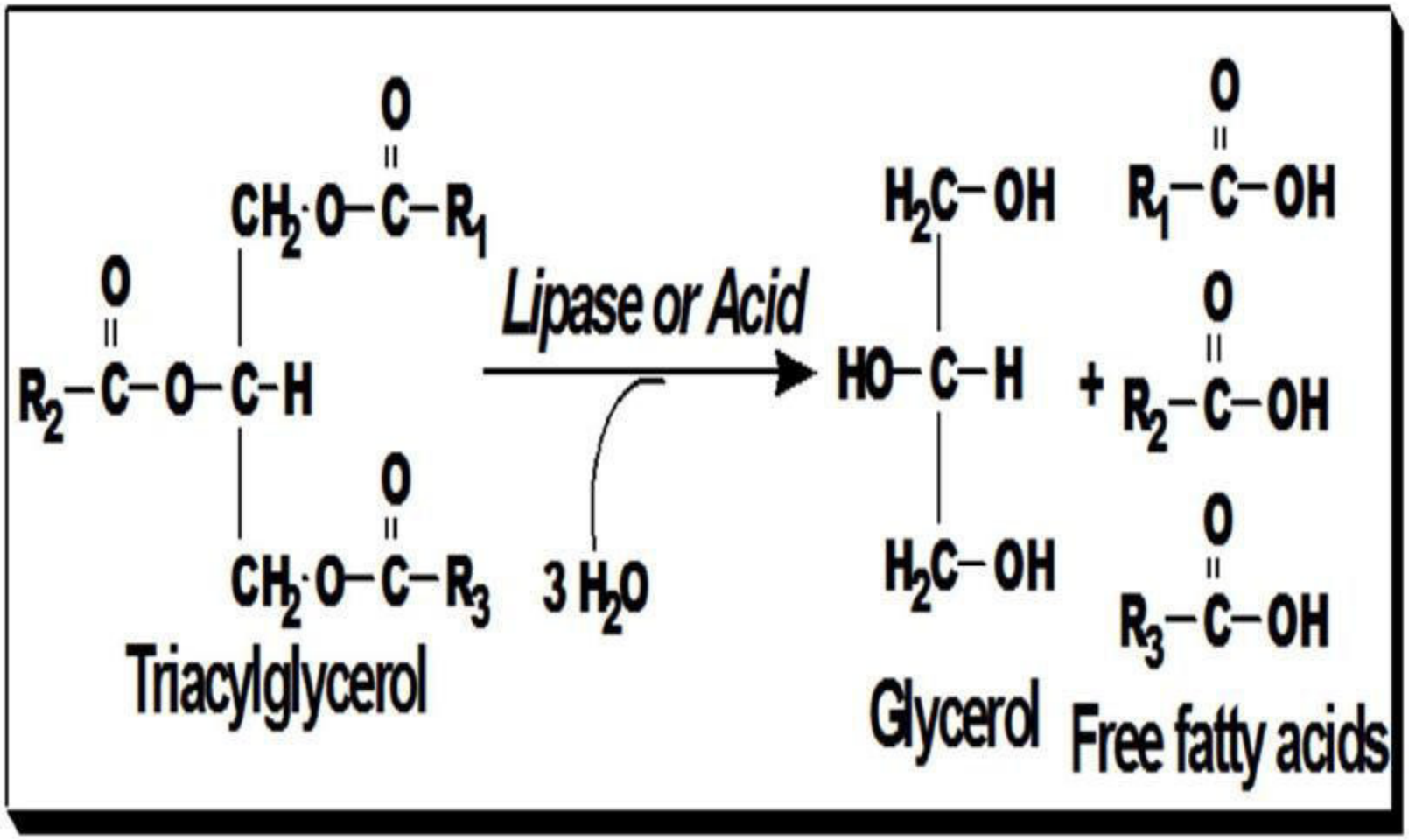
The fats may be hydrolysed with super heated steam, by acids, or alkalies, by the specific fat splitting enzymes lipases to

– free fatty acids

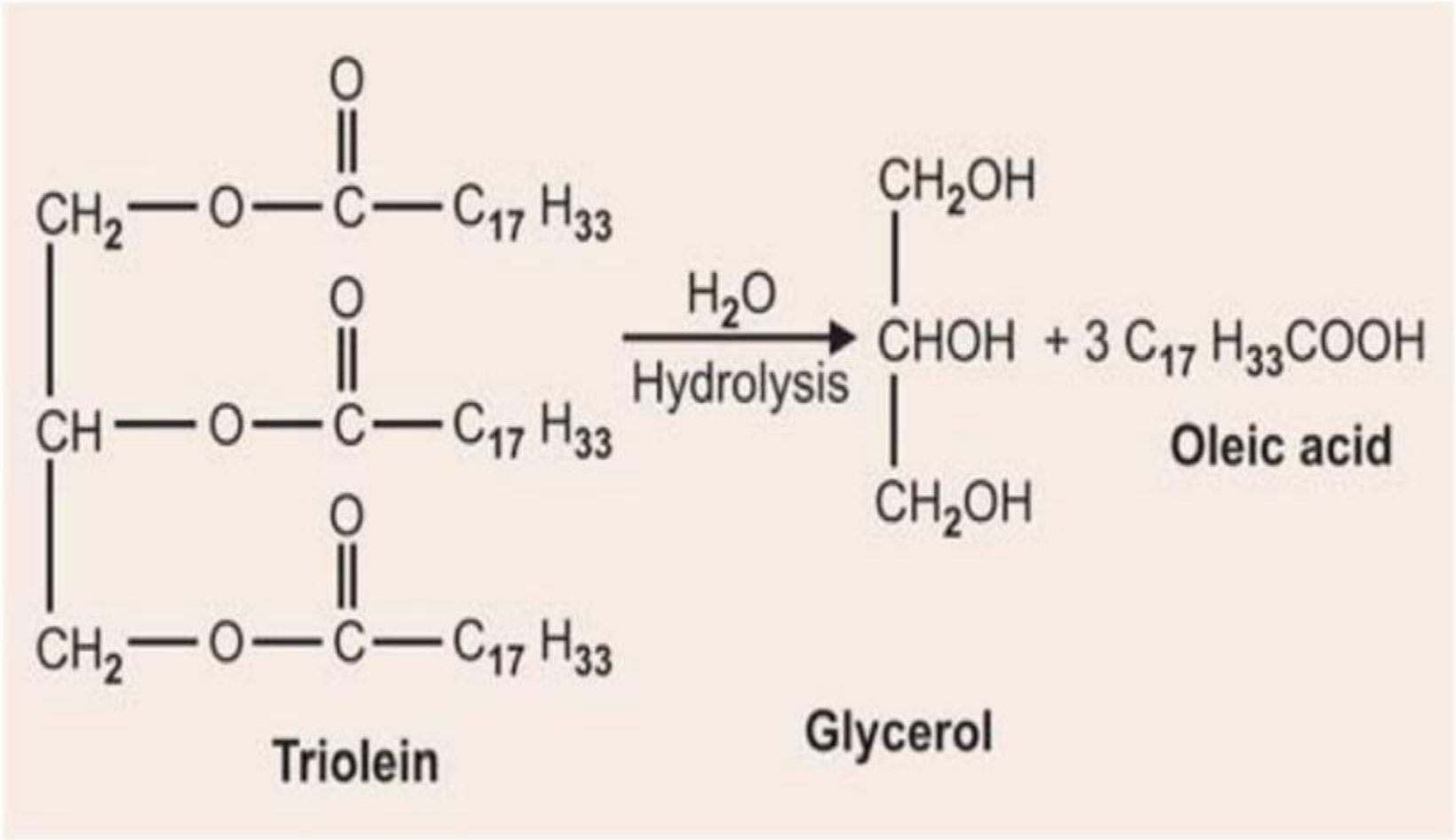
– glycerol

steam at temperature higher than the b.p of water and does not contain any water in the liquid state,

Lipid Chemistry



Lipid Chemistry



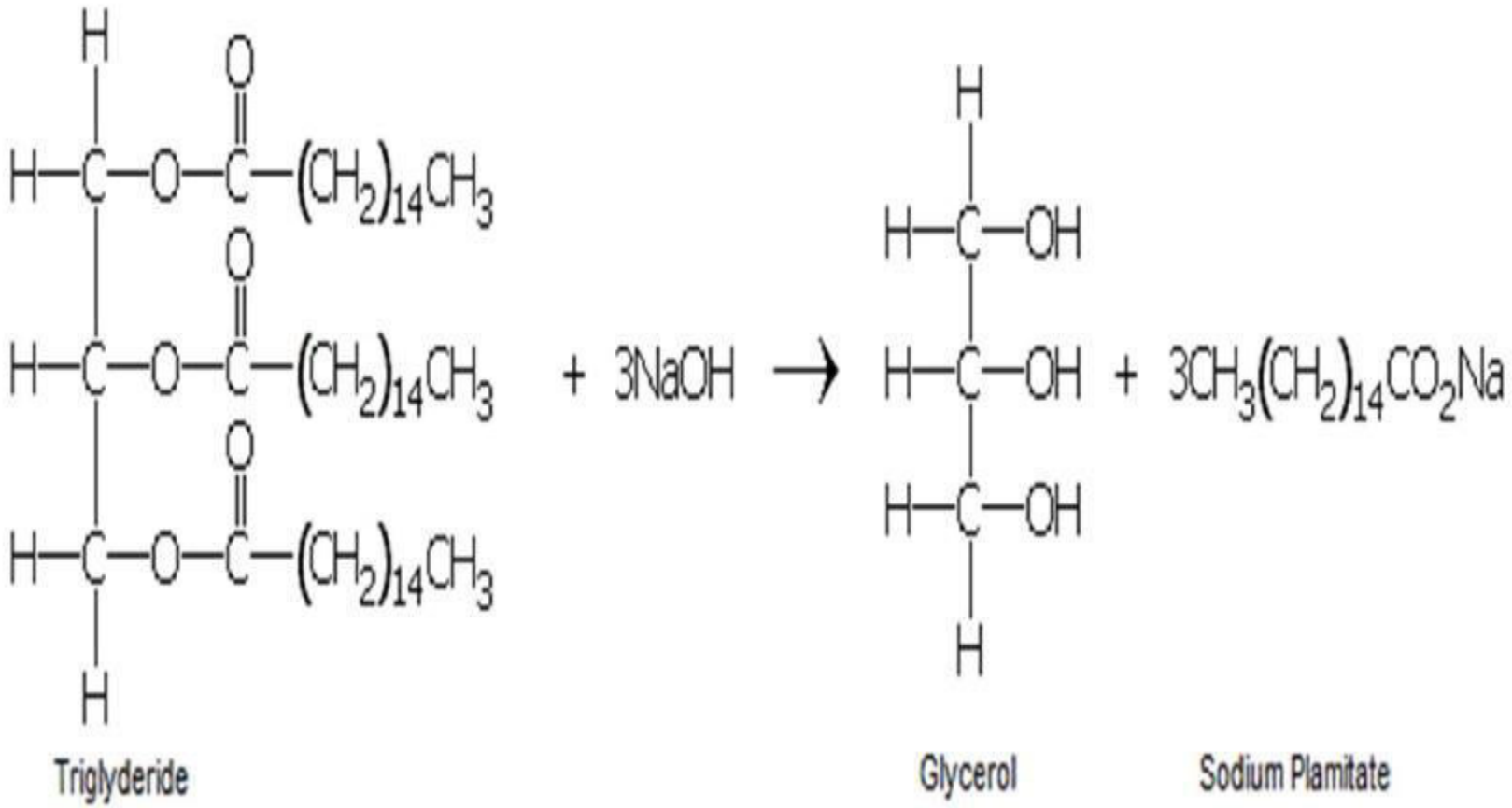
Triolein is another example of simple TAG.

Lipid Chemistry

Saponification

- Hydrolysis of a fat by an alkali is called **saponification**
- The resultant products are;
- glycerol and
- the alkali salts of the fatty acids, which are called “soaps”

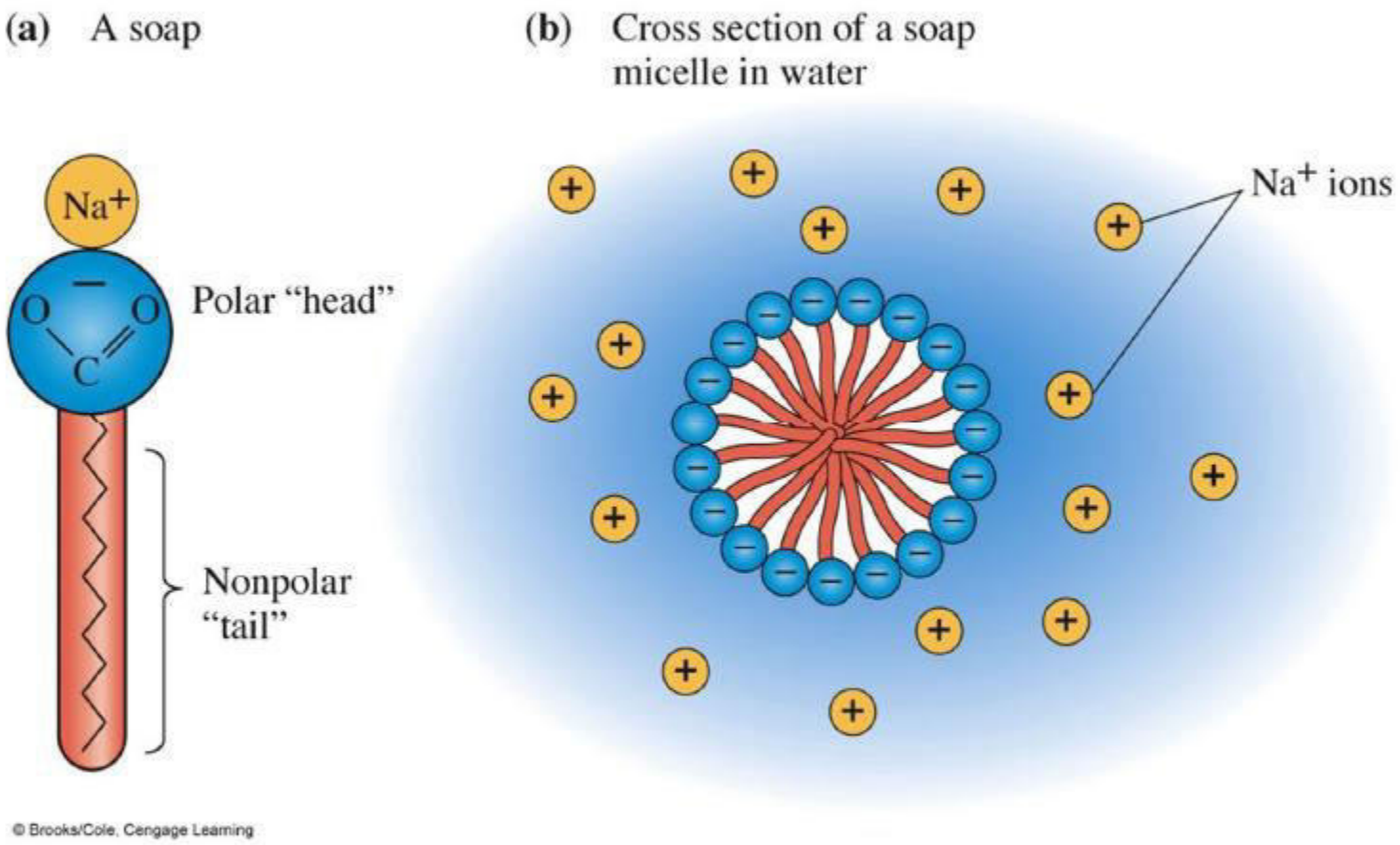
Lipid Chemistry



Lipid Chemistry

- The number of mgs of NaOH/KOH required to saponify the free and combined FA in one gram of a given fat is called its **saponification number**

Lipid Chemistry



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Lipid Chemistry

- The amount of **alkali** needed to saponify a given quantity of fat will depend upon the number of carboxylic (–COOH) group present

Lipid Chemistry

- Thus fats containing short chain fatty acids will have more – COOH groups per gram than long- chain fatty acids and this will take up more alkali
- And hence will have higher saponification number

Lipid Chemistry

Lipid properties

Table 1. Examples saponification and iodine numbers

Fat or oil	Saponification #	Iodine #
Beef tallow	194 – 200	34 – 43
Cocoa butter	192 – 198	32 – 42
Coconut oil	245 – 262	6 – 10
Cottonseed oil	192 – 196	103 – 112
Lard	193 – 200	50 – 80
Milk fat	210 – 233	26 – 35
Peanut oil	186 – 194	89 – 98

Tallow is the fat obtained from cattle used in soap industry.

Lipid Chemistry

- Butter containing a larger proportion of short- chain fatty acids, such as butyric (C4) acid and caproic (C6) acid etc.
- Therefore it has relatively high saponification number from 220 to 230

Lipid Chemistry

- In contrast, Olive Oil (which contain Oleic acid (C18), a longer chain FA),
- has saponification number of 195 or less.

END

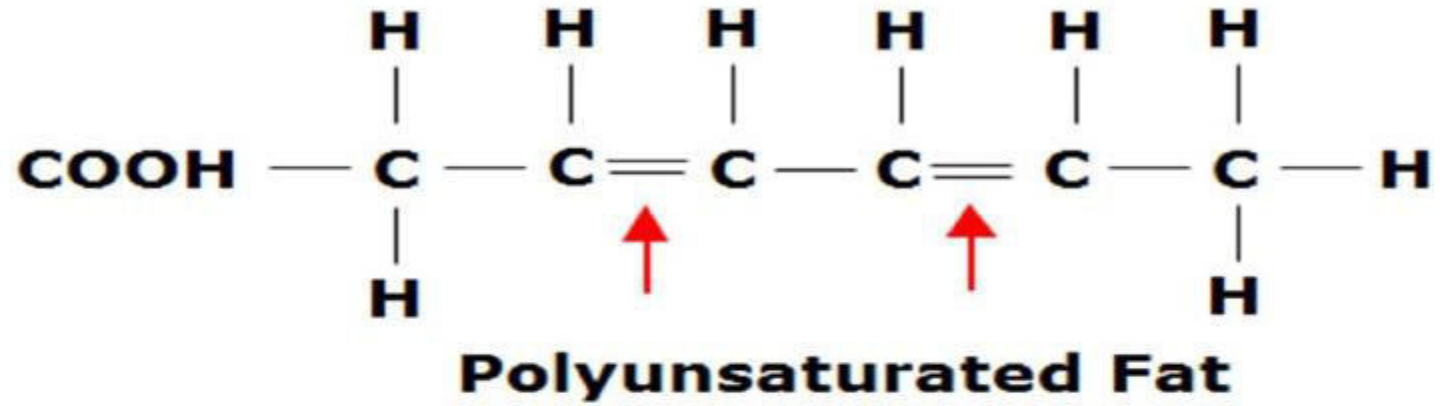
Lipid Chemistry

- **Chemical Properties (Contd.)**
- **2.** Additive Reactions
- The unsaturated fatty acids present in neutral fat exhibits all the additive reactions, i.e.
  - hydrogenation,
  - halogenation.

Lipid Chemistry

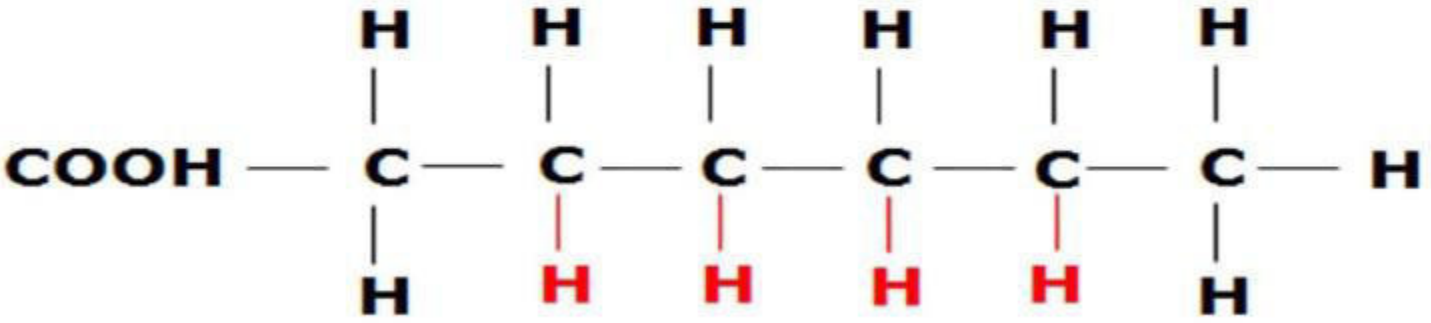
- **Hydrogenation**
- A hydrogenation reaction involves conversion of a carbon-carbon double bond to a carbon-carbon single bond through the addition of hydrogen

Lipid Chemistry



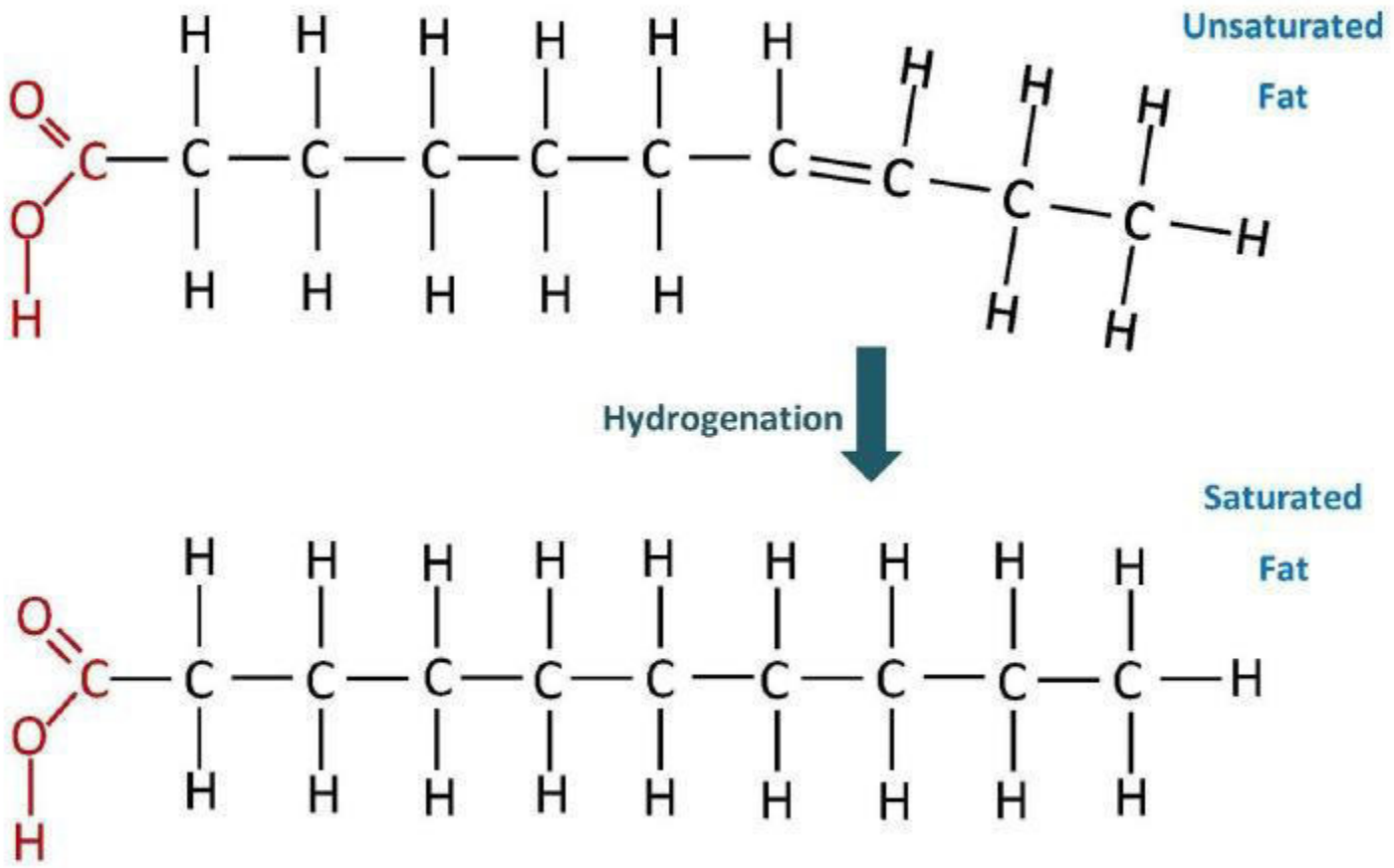
Polyunsaturated Fat

Hydrogenation



Hydrogenated Fat

Lipid Chemistry



unsaturated fats have lower melting points,stearic (SFA) melts at 70$_o$C, oleic (PUFA) at 26$_o$C

Lipid Chemistry

- **Hydrogenation**
  - As you continue to hydrogenate your molecule
    - Melting point increases
    - Fat becomes more solid at room temp

Lipid Chemistry

- **Hydrogenation**

- Oils which are liquid at ordinary room temperature, on **hydrogenation** become solidified

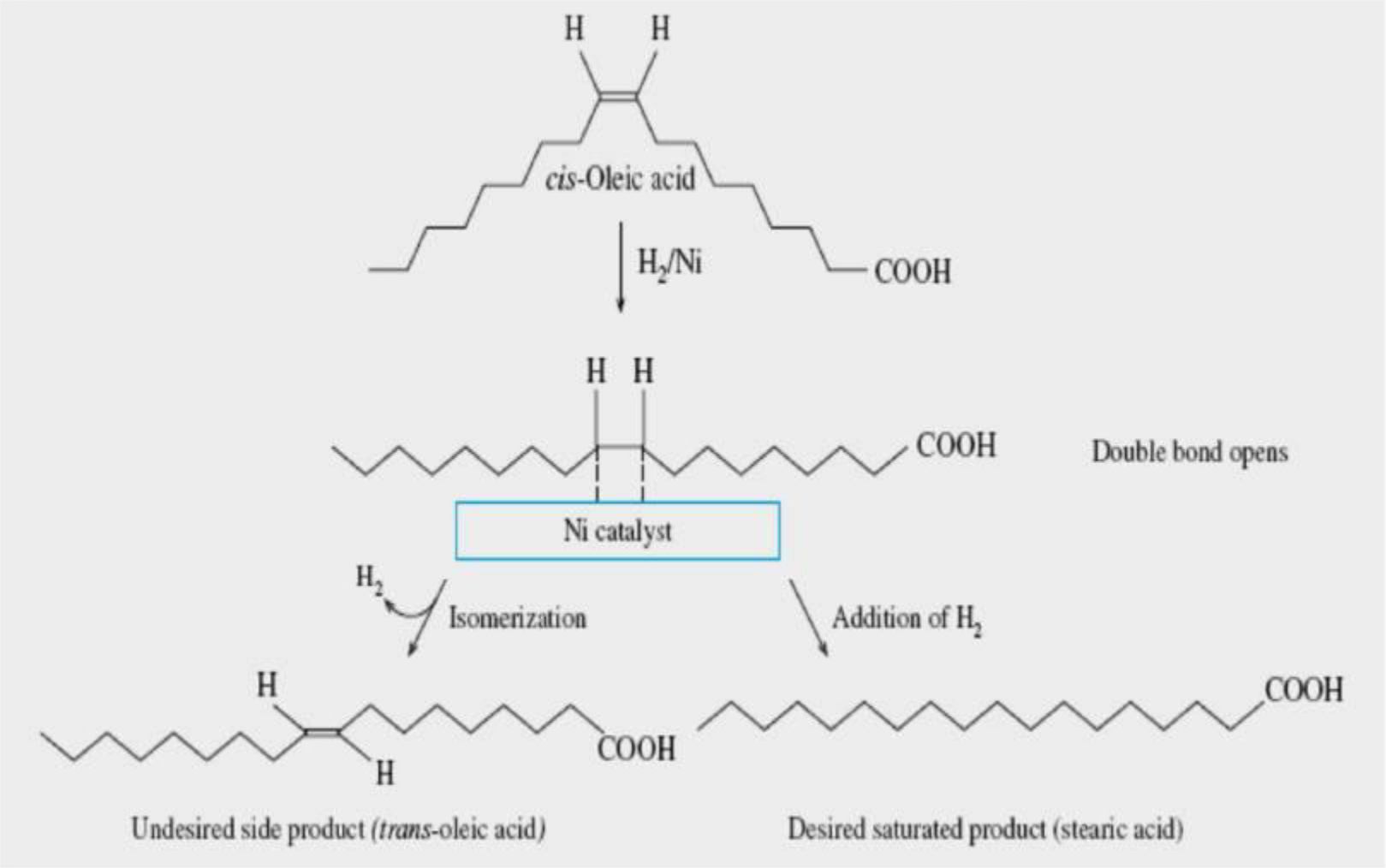
Lipid Chemistry

- This is the basis of Banaspti ghee manufacturing.
- Where inedible and cheap oils like cotton seed oil are hydrogenated and converted to edible solid fats.

Lipid Chemistry

- The hydrogenation is done
- under high pressure of hydrogen and is catalyzed by finely divided nickel or copper and heat.

Lipid Chemistry



Lipid Chemistry

- It is the base of hardening of oils (margarine manufacturing), e.g.
- change of oleic acid of fats (liquid)
- into stearic acid (solid).

END

unsaturated fats have lower melting points stearic (SFA) melts at 70ₒC, oleic (PUFA) at 26ₒC

Lipid Chemistry

- **Chemical Properties (Contd.)**
- **Halogenation**
- Similar to hydrogenation,
- Halogens such as chlorine, bromine and iodine can also be added to double bonds in
- unsaturated fatty acids.

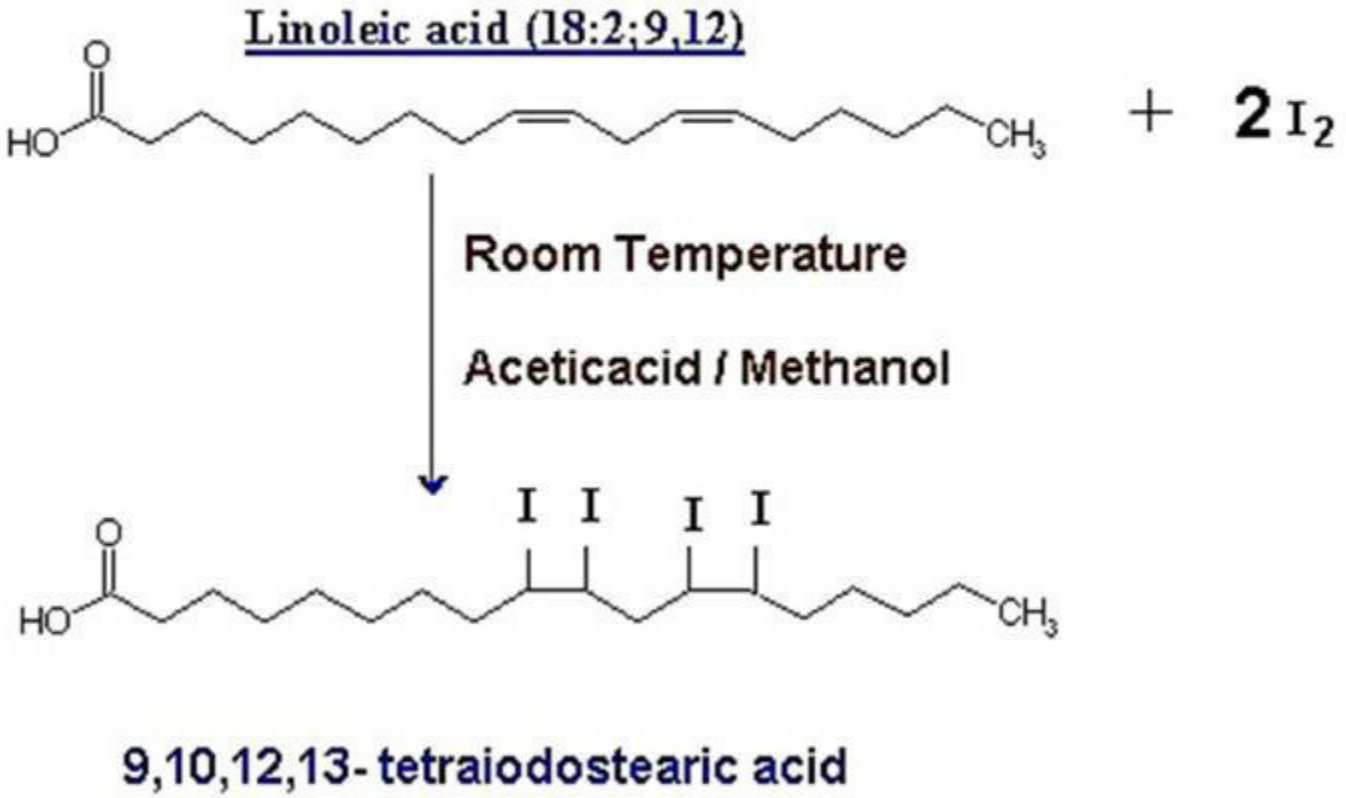
Lipid Chemistry

- It is a very important property to determine the degree of unsaturation of the fat or oil that determines its biological value.
- The degree of unsaturation is reflected by Iodine number.

Lipid Chemistry

- Iodine number is defined as the number of grams of iodine absorbed by 100 gm of fat.

Lipid Chemistry



9,10,12,13- tetraiodostearic acid

Lipid Chemistry

- The more the iodine number, the greater the degree of unsaturation.
- Fats rich in saturated fatty acids have low iodine numbers,
- while fats rich in unsaturated fatty acids have high iodine numbers

Lipid Chemistry

- The determination of iodine number is useful to the chemist in determining the quality of an oil or its freedom from adulteration

Lipid Chemistry

- Iodine number of cotton seed oil varies from 103 to 111.
- That of olive oil from 79 to 88,
- And that of linseed oil from 175 to 202

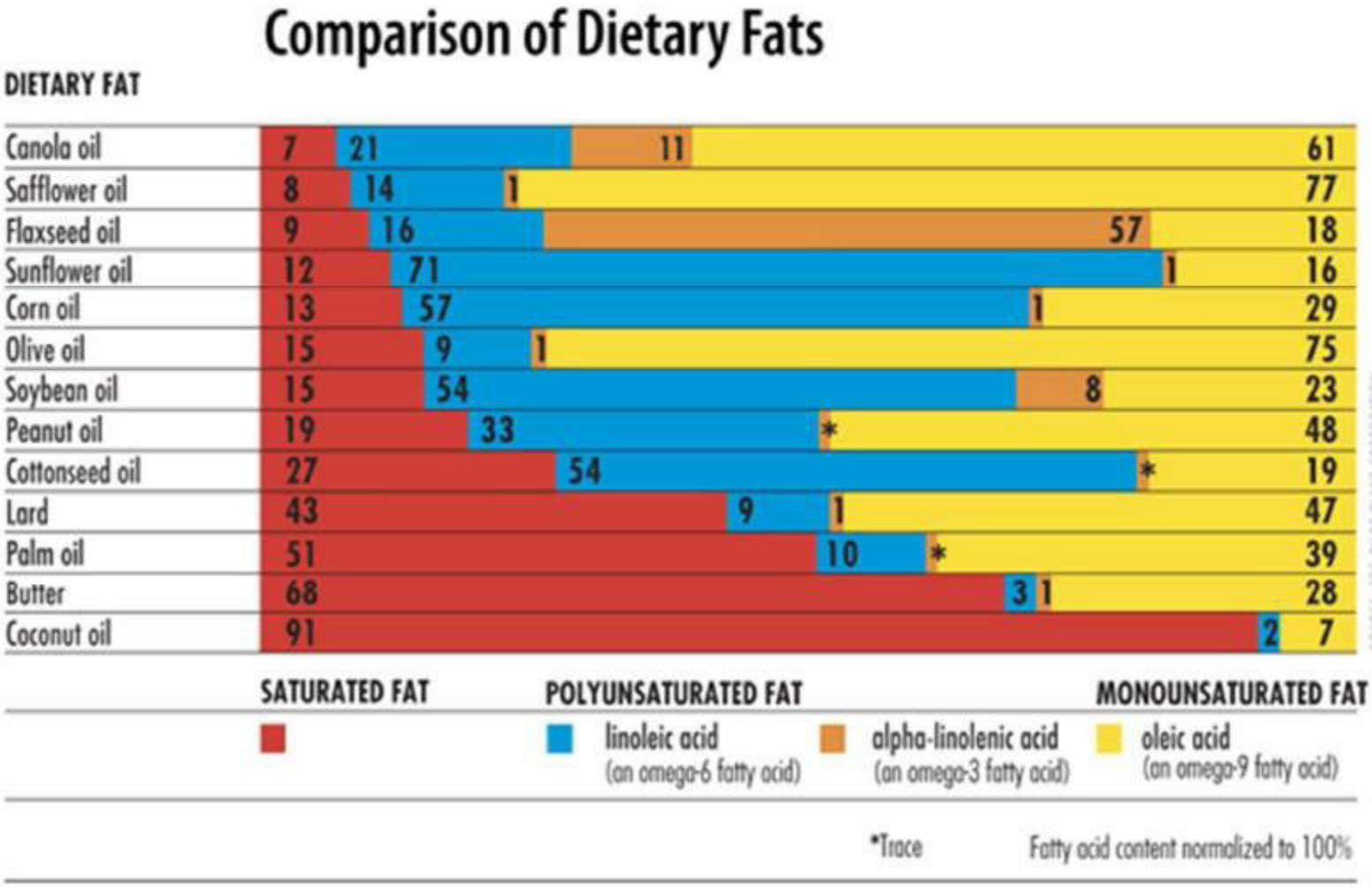
Lipid Chemistry

- A commercial lot of olive oil which has an iodine number higher than 88 might have been adulterated with cotton seed oil

Lipid Chemistry

- The higher is the iodine number,
- the more reactive,
- less stable,
- more susceptible to oxidation and
- rancidification is the oil or fat.

Lipid Chemistry



Lipid Chemistry

Oil/Fat	16:0	16:1	18:0	18:1	18:2	18:3	20:1	22:1	24:0	Average Unsaturation Per Triglyceride
soybean	11	0.1	4	23.4	53.2	7.8				4.6
palm	44.4	0.2	4.1	39.3	10	0.4				1.8
rapeseed	3	0.2	1	13.2	13.2	9	9	49.2	1.2	3.8
sunflower	6		5	20	60					1.4
tallow	27	11	7	48	2					0.6
cottonseed	21.6	0.6	2.6	18.6	54.5	0.7				3.9
olive	13.7	1.2	2.5	71.1	10	0.6				2.8
corn	10.9	0.2	2	25.4	59.6	1.2				4.5
canola	4.1	0.3	1.8	60.9	21	8.8	1	0.7	0.2	3.9
linseed	5.5		3.5	19.1	15.3	56.6				6.6

Lipid Chemistry

- **Chemical Properties (Contd.)**
- Rancidity
- The chemical deterioration of fats.
- When lipid-rich foods are exposed too long to the oxygen in air, they may spoil and become foul smelling.

Lipid Chemistry

- **Rancidity**
- **Definition:**
- It is a physico-chemical change in the natural properties of the fat leading to the development of
- unpleasant odor or
- taste or
- abnormal color

Lipid Chemistry

It occurs particularly

- on aging
- after exposure to atmospheric oxygen,
- light,
- moisture,
- bacterial or fungal contamination
- and/or heat.

Lipid Chemistry

- Saturated fats resist rancidity more than unsaturated fats that have unsaturated double bonds.

- **Rancidity**

is due to

I. Oxidation

II. Hydrolysis

Lipid Chemistry

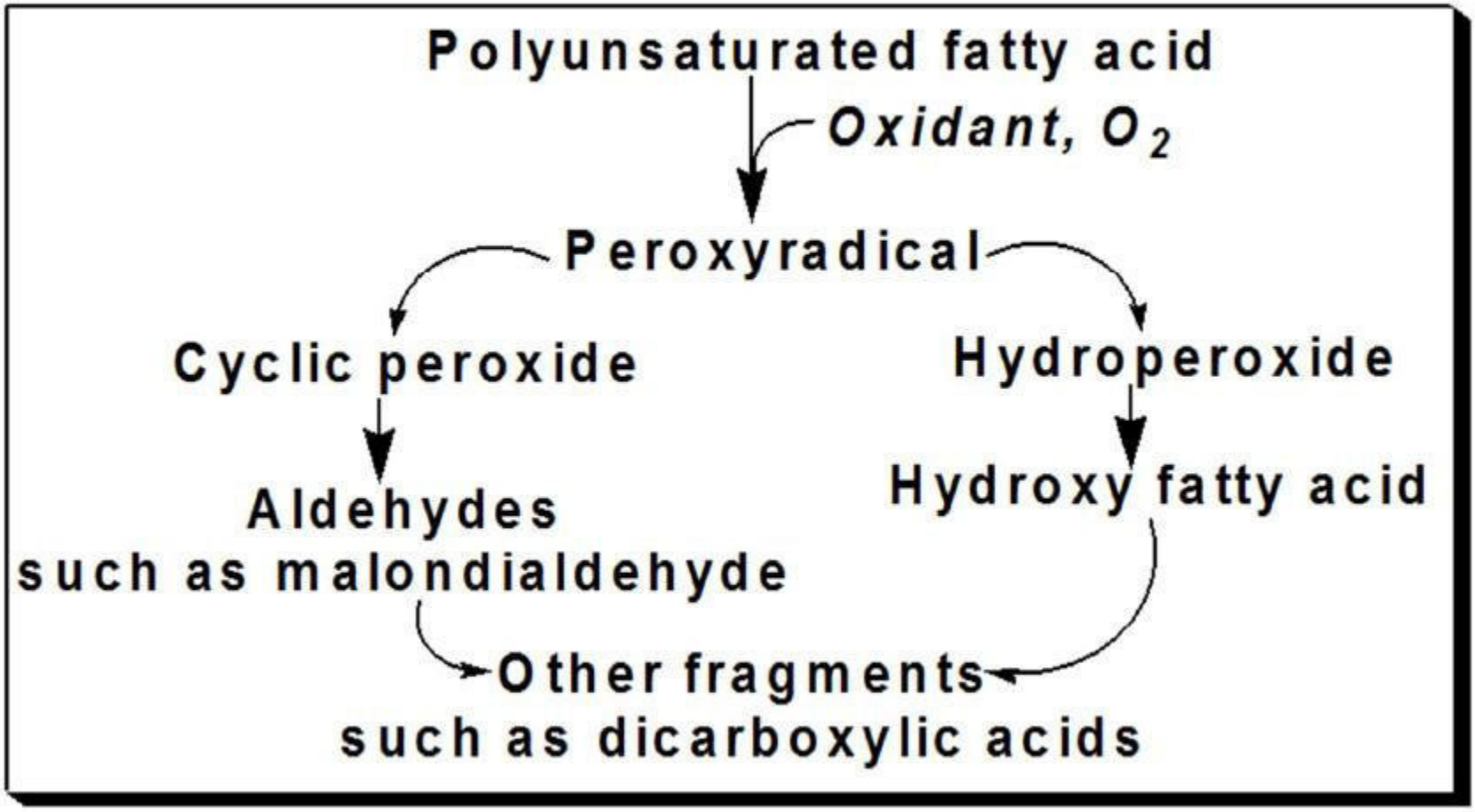
I. Oxidative Rancidity

- Oxidation of the fat molecules give rise to some
- short chain aldehydes, ketones and dicarboxylic acids
- which have objectionable taste and odor.

Lipid Chemistry

- The unpleasant taste and smell associated with rancidity result from the *oxidative cleavage of double bonds in unsaturated fatty acids*

Lipid Chemistry



Malondialdehyde is small aldehyde which is present in rancid foods and it is also produced

Lipid Chemistry

- The oxygen of the air is necessary for this type of rancidity.
- This can be prevented by addition of anti-oxidants such as
- vitamin E to foods.

Lipid Chemistry

II. Hydrolytic Rancidity

- It is due to the slow hydrolysis of fats,
- which in case of fats like butter results in the
- liberation of short chain fatty acids which are volatile and have rancid taste and odor.

END

Lipid Chemistry

- **Waxes**
- A second group of neutral lipids that are of physiological importance.
- Although they are a minor component of biological systems.

Lipid Chemistry

- **Properties of waxes**
- Waxes are insoluble in water, but
- soluble in fat solvents and are
- negative for acrolein test.
- very resistant to rancidity.

Lipid Chemistry

- Waxes are not easily hydrolyzed as the fats
- and are indigestible by lipases (enzymes responsible for fat digestion in body)
- Thus they are of no nutritional value

Lipid Chemistry

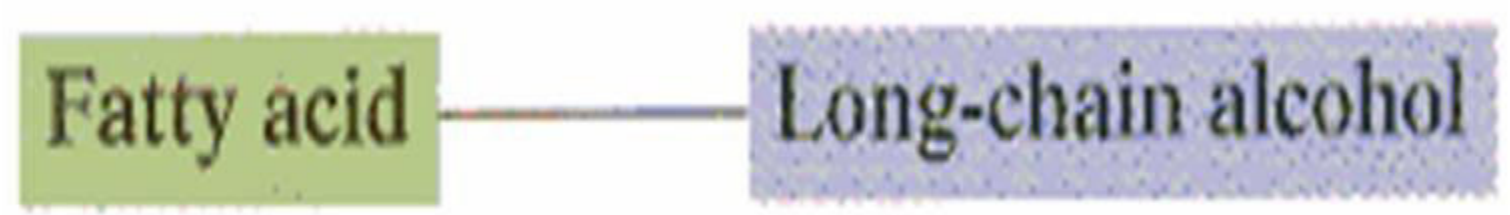
- Waxes are of two types:
- True waxes
- Other Waxes or
  Non true waxes or
  Wax-like compounds

Lipid Chemistry

- **True Waxes**
- Waxes are solid simple lipids containing a monohydric alcohol (with a higher molecular weight than glycerol)
- esterified to long-chain fatty acids.

An alcohol containing one hydroxyl group

Lipid Chemistry



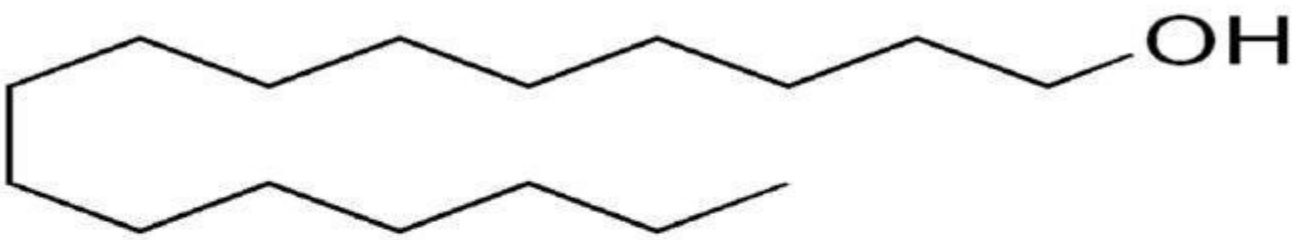
Lipid Chemistry

Myricyl alcohol (30C)

Triacontanol is a fatty alcohol of the general formula $C_{30}H_{62}O$, also known as **melissyl alcohol** or **myricyl alcohol.**

It is found in plant cuticle waxes and in beeswax.

Lipid Chemistry



Cetyl Alcohol (16 C)

The name cetyl derives from the whale oil (Latin: cetus) from which it was first isolated

Cetyl Alcohol can be found in mooisturizer,facial moisturizer,conditioner,anti-aging,hair

Lipid Chemistry

- Waxes are widely distributed in nature such as the secretion of certain insects as;
- **Bees-wax,**
- **Spermaceti** of the sperm whale

Lipid Chemistry

- Waxes also form protective coatings of the skins and furs of animals and
- leaves and fruits of plants.

END

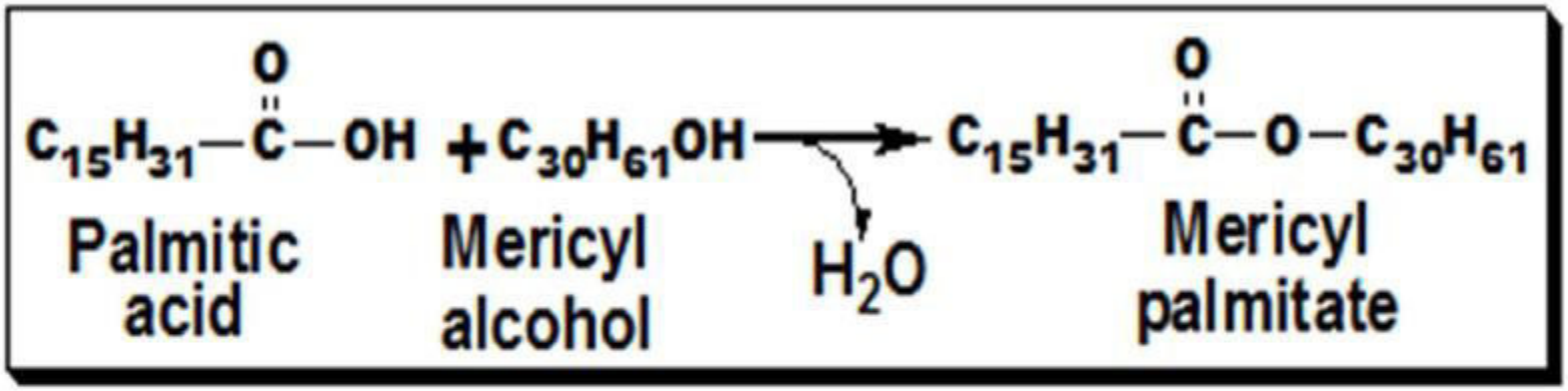
Lipid Chemistry

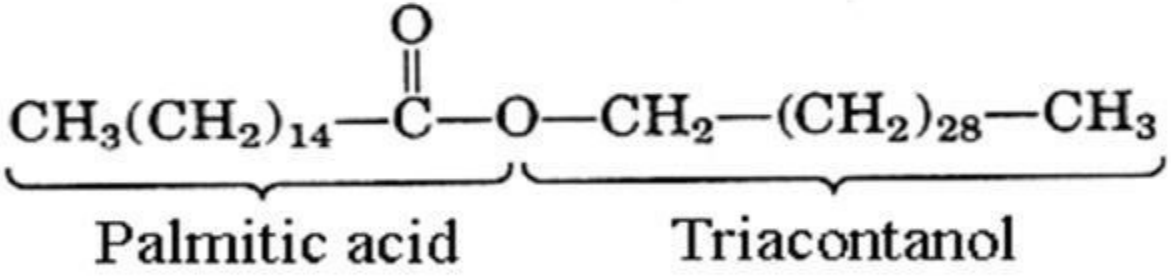
- **1.True Waxes**
- **Bees-wax** is secreted by the honeybees that use it to form the combs.

- It is a mixture of waxes
- chief constituent is myricyl palmitate

(30C) (16C)

Lipid Chemistry





Lipid Chemistry

- **2.Spermaceti**
- is a wax that is most often found in the head cavities of the sperm whale.
- Fatty esters are formed essentially of
- cetyl palmitate and
- cetyl myristate.

Cetyl: 16 C monohydric alcohol **Palmitate**: 16 C FA **Myristate**: 14 C FA

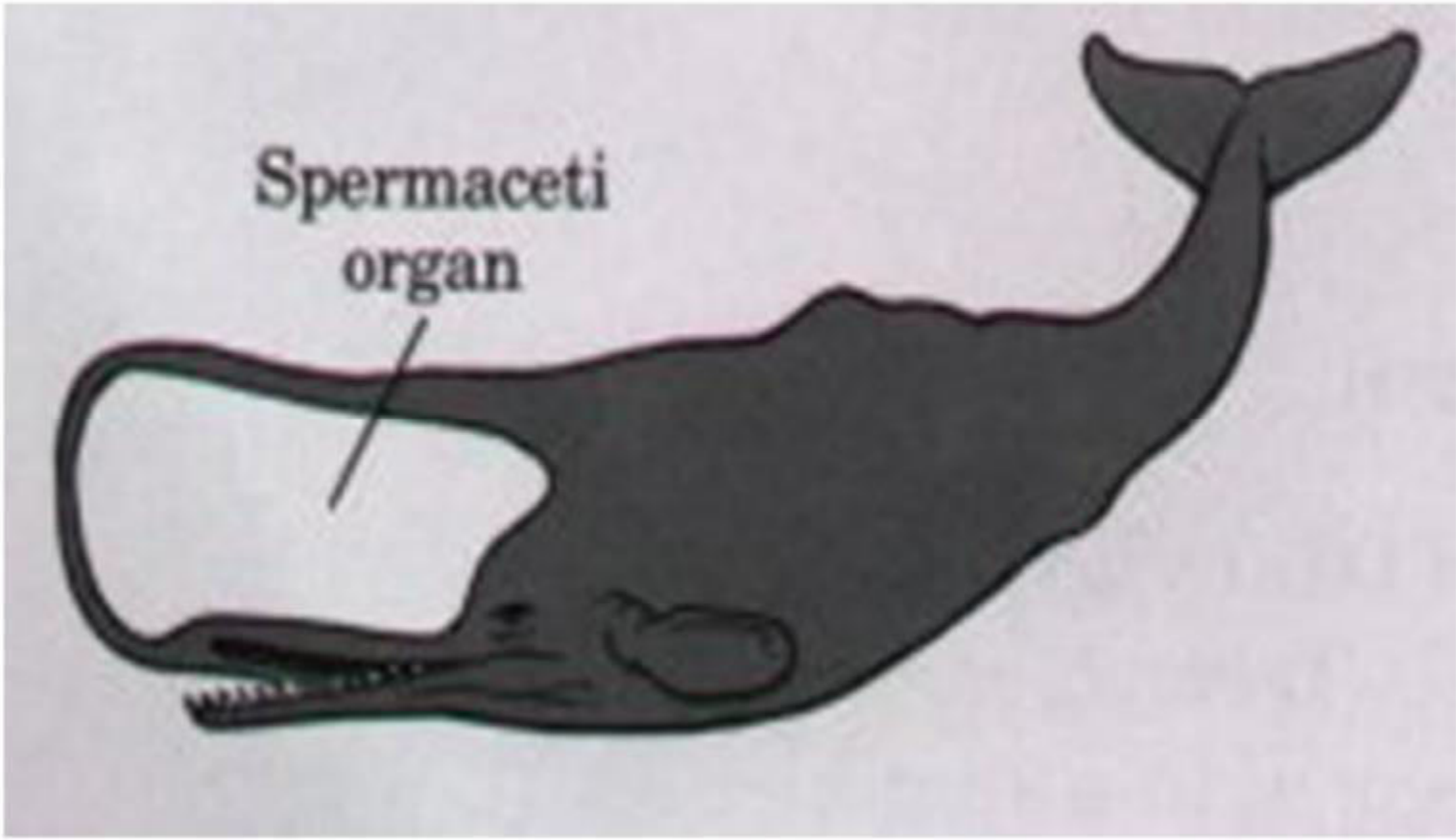
Lipid Chemistry

- It was used in cosmetics, pharmacy and also in candles
- recent international regulation concerning whale captures, has stopped its use.

It is now replaced by synthetic cetyl palmitate.

Lipid Chemistry

Lipid Chemistry



Lipid Chemistry

- **2.Other Waxes or Non true waxes**
- include esters of:
- Cholesterol
- Vitamin A
- Vitamin D

Lipid Chemistry

- **Cholesterol esters**: Lanolin (or wool fat) is secreted by sheep sebaceous glands and

- It contains both free and esterified cholesterol, e.g., cholesterol-palmitate

Lipid Chemistry

- Lanolin secretion helps sheep in reducing water evaporation from the skin.
- It is used as
- industrial lubricant and in
- cosmetics.

Lipid Chemistry

TABLE 15.2 Some Typical Waxes

Type	Structural Formula	Source	Uses
Beeswax	$CH_3(CH_2)_{14}-\overset{\overset{O}{\|}}{C}-O-(CH_2)_{29}CH_3$	Honeycomb	Candles, shoe polish, wax paper
Carnauba wax	$CH_3(CH_2)_{24}-\overset{\overset{O}{\|}}{C}-O-(CH_2)_{29}CH_3$	Brazilian palm tree	Waxes for furniture, cars, floors, shoes
Jojoba wax	$CH_3(CH_2)_{18}-\overset{\overset{O}{\|}}{C}-O-(CH_2)_{19}CH_3$	Jojoba	Candles, soaps, cosmetics

Lipid Chemistry

Lipids and Nutrition

- Dietary fats strongly influence the incidence of coronary heart disease (CHD).

Lipid Chemistry

- In the past, dietary recommendations emphasized decreasing the total amount of fat in the diet.
- Research now indicates that the type of fat is more important than the total amount of fat consumed

Lipid Chemistry

To gauge the effect of these dietary fats on CHD, we measure different biochemical parameters, which include:

- LDL-Cholesterol
- HDL-Cholesterol
- Total Cholesterol
- TAG

Lipid Chemistry

- LDL and HDL stand for
- Low Density Lipoprotein and
- High Density Lipoprotein respectively
- they are included in the class of complex lipids and serve to
- transport lipids in the blood.

Lipid Chemistry

- Serum Cholesterol and TAGs are a risk factor for CHD among others.
- The risk increases progressively with higher values for serum total cholesterol and that of LDL cholesterol.

Lipid Chemistry

- Association of TAGs is weaker than that of LDL cholesterol with CHD.

- A much stronger correlation exists between the levels of blood LDL cholesterol and heart disease.

Lipid Chemistry

- In contrast, high levels of HDL cholesterol have been associated with a decreased risk for heart disease

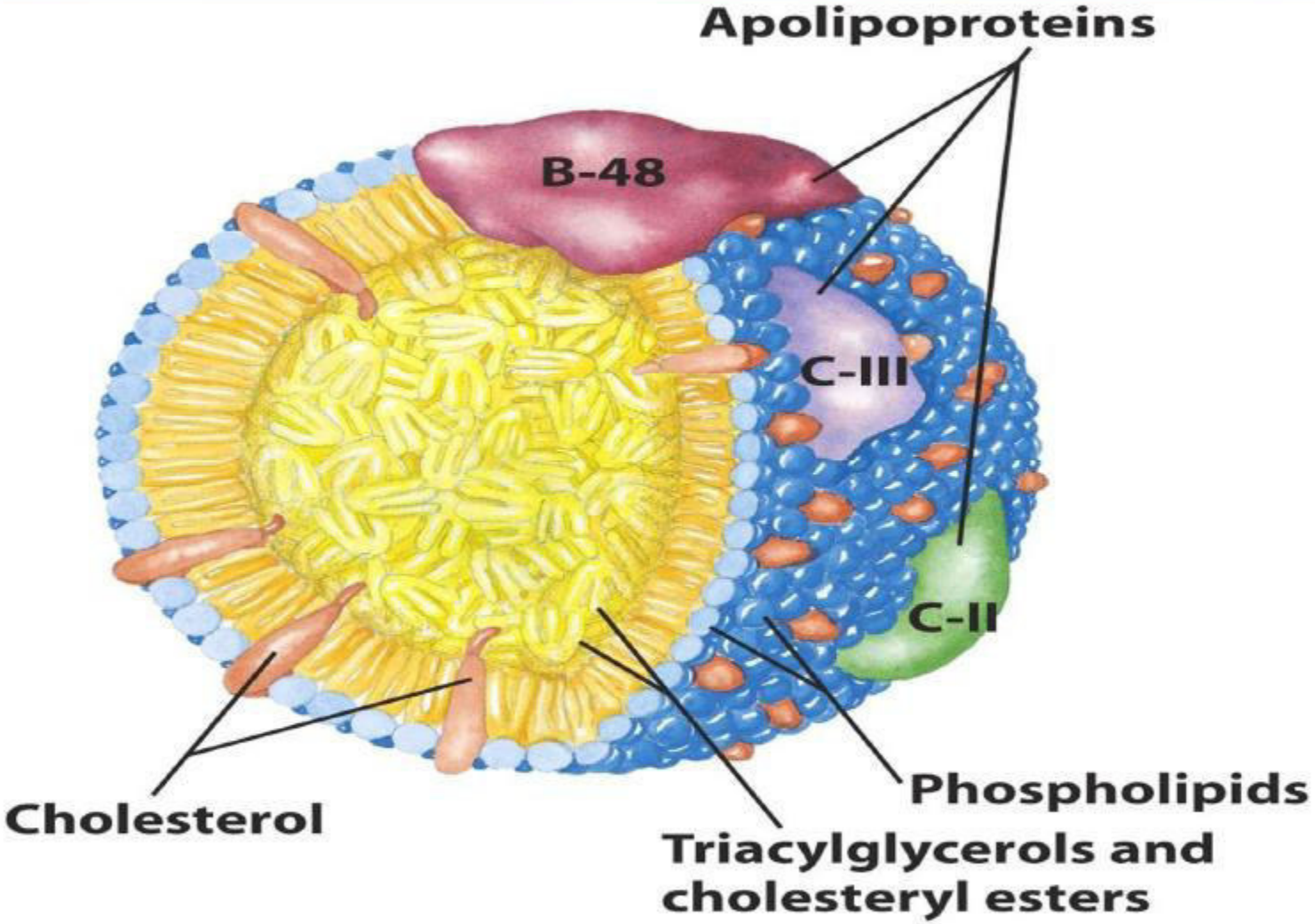
Lipid Chemistry

- In order to understand the influence of these parameters on CHD, we must first understand biochemical role of Lipoproteins:

Lipid Chemistry

- Lipoprotein particles are spherical aggregates with hydrophobic lipids at the core and hydrophilic protein side chains and lipid head groups at the surface.
- These proteins are called apolipoproteins.

Lipid Chemistry



Lipid Chemistry

- Lipoproteins function is to keep their component lipids soluble as they transport them in the aqueous environment of plasma.

Lipid Chemistry

- Due to imbalanced metabolism of these lipoproteins there can be a gradual deposition of lipid—especially **cholesterol**—in tissues, which in arteries of the heart, can lead to CHD.

END

Lipid Chemistry

Lipoproteins

- The lipoprotein particles include
- chylomicrons (CM),
- very-low-density lipoproteins (VLDL),
- low-density lipoproteins (LDL), and
- high-density lipoproteins (HDL).

Lipid Chemistry

- They differ in
- lipid and protein composition,
- size,
- density and
- site of origin

Different lipoproteins contain different *apo*liporoteins, depending on the function of those lipoprotein

Lipid Chemistry

- Different combinations of lipids and proteins produce particles of different densities
- ranging from chylomicrons to high-density lipoproteins
- these particles can be separated by ultracentrifugation and HPLC

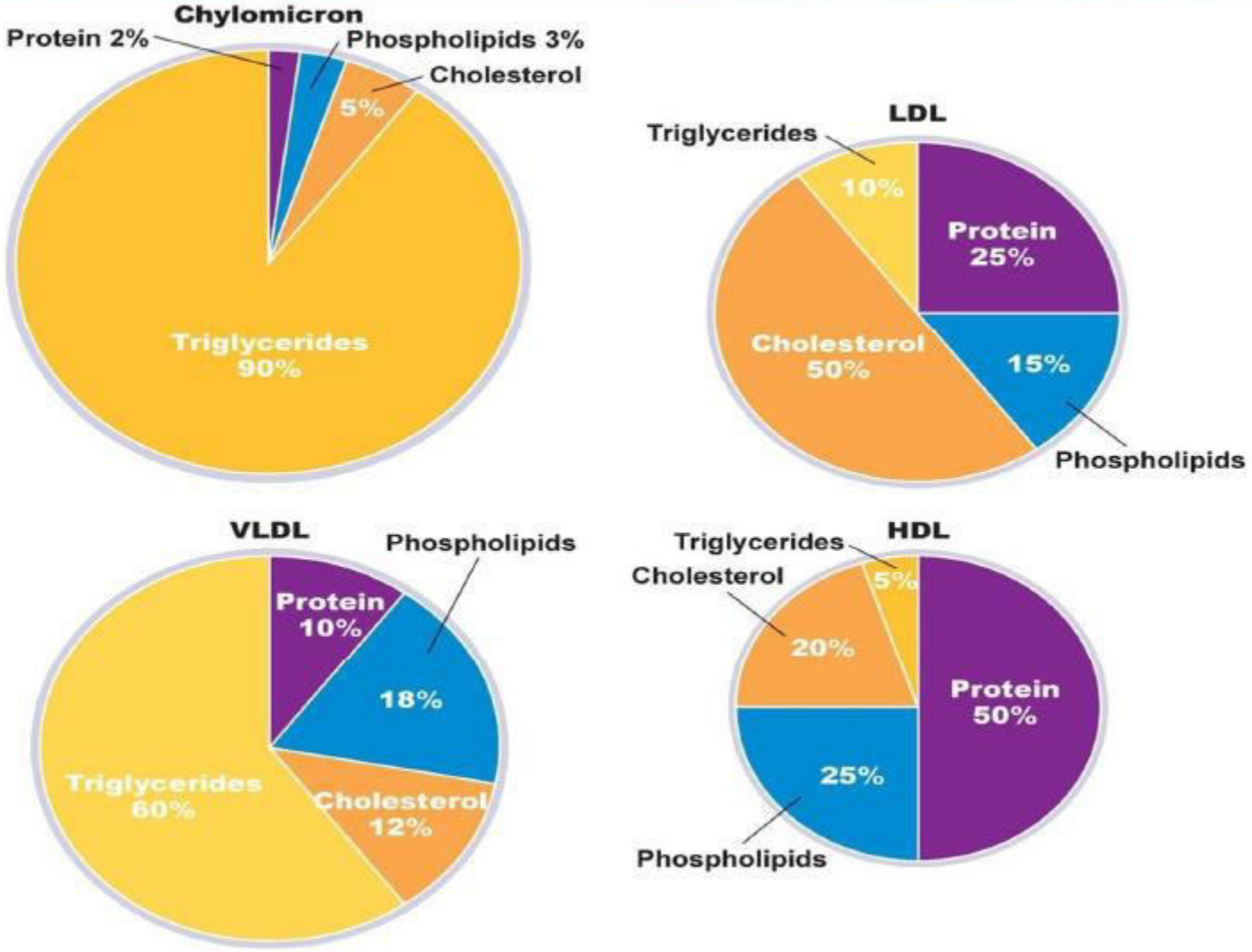
Lipid Chemistry

TABLE 21-2 Major Classes of Human Plasma Lipoproteins: Some Properties

		Composition (wt %)				
Lipoprotein	*Density (g/mL)*	*Protein*	*Phospholipids*	*Free cholesterol*	*Cholesteryl esters*	*Triacylglycerols*
Chylomicrons	<1.006	2	9	1	3	85
VLDL	0.95-1.006	10	18	7	12	50
LDL	1.006-1.063	23	20	8	37	10
HDL	1.063-1.210	55	24	2	15	4

Source: Modified from Kritchevsky, D. (1986) Atherosclerosis and nutrition. *Nutr. Int.* **2,** 290-297.

Lipid Chemistry



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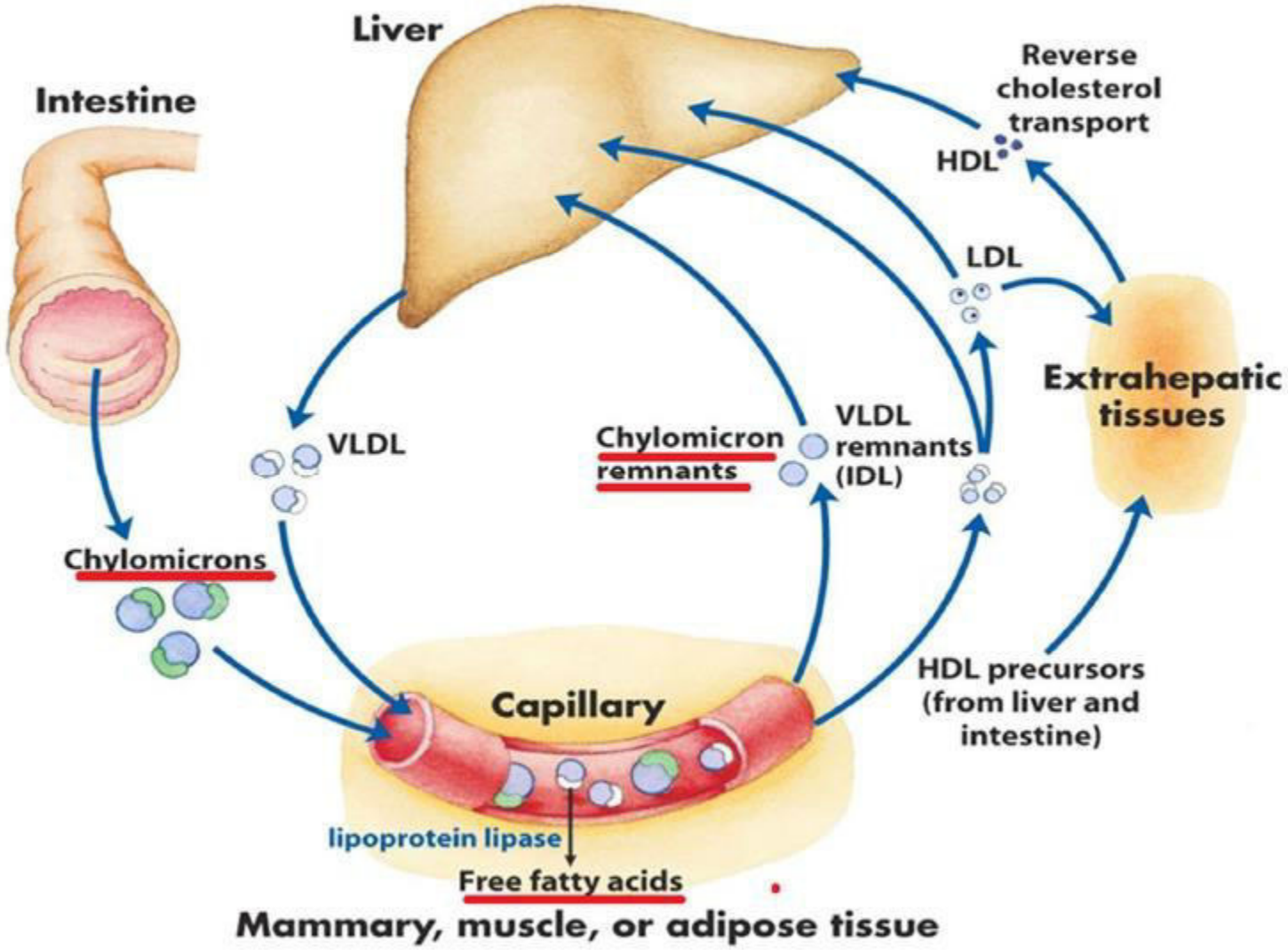
Lipid Chemistry

- Chylomicrons are formed and assembled in intestinal mucosal cells after a fatty meal
- And carry dietary triacylglycerol, cholesterol, and cholesteryl esters to the **peripheral tissues.**

Lipid Chemistry

- As the chylomicron circulates and most of its dietary TAG are degraded and taken up by peripheral tissues in the form of fatty acids,
- the particle size decreases and density increases.

Lipid Chemistry



Lipid Chemistry

- The remaining particle, called Chylomicron Remnant, is removed from the circulation by the liver.

END

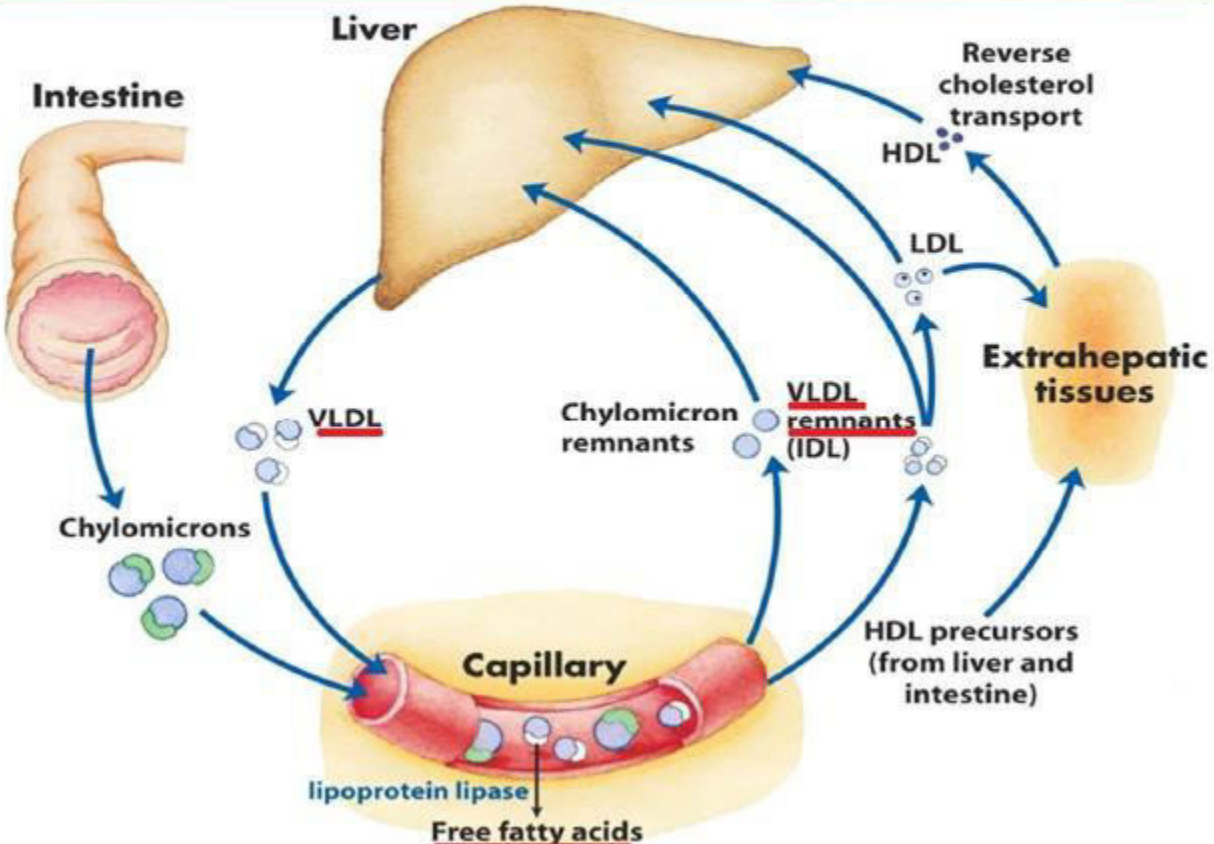
Lipid Chemistry

- **Lipoproteins (Contd.)**
- **VLDLs** are assembled in the liver.
- composed predominantly of TAGs synthesized in liver and
- contain some cholesterol and cholesteryl esters

Lipid Chemistry

- As VLDL pass through the circulation, TAG is degraded and taken up by peripheral tissues in the form of fatty acids,
- causing the VLDL to decrease in size and become denser,
- called VLDL remnant.

Lipid Chemistry



Lipid Chemistry

- further removal of triacylglycerol
- from these remnants produces low-density lipoprotein (LDL)

Lipid Chemistry

- LDL particles contain much less triacylglycerol than their VLDL predecessors,
- and have a high concentration of cholesterol and cholesteryl esters

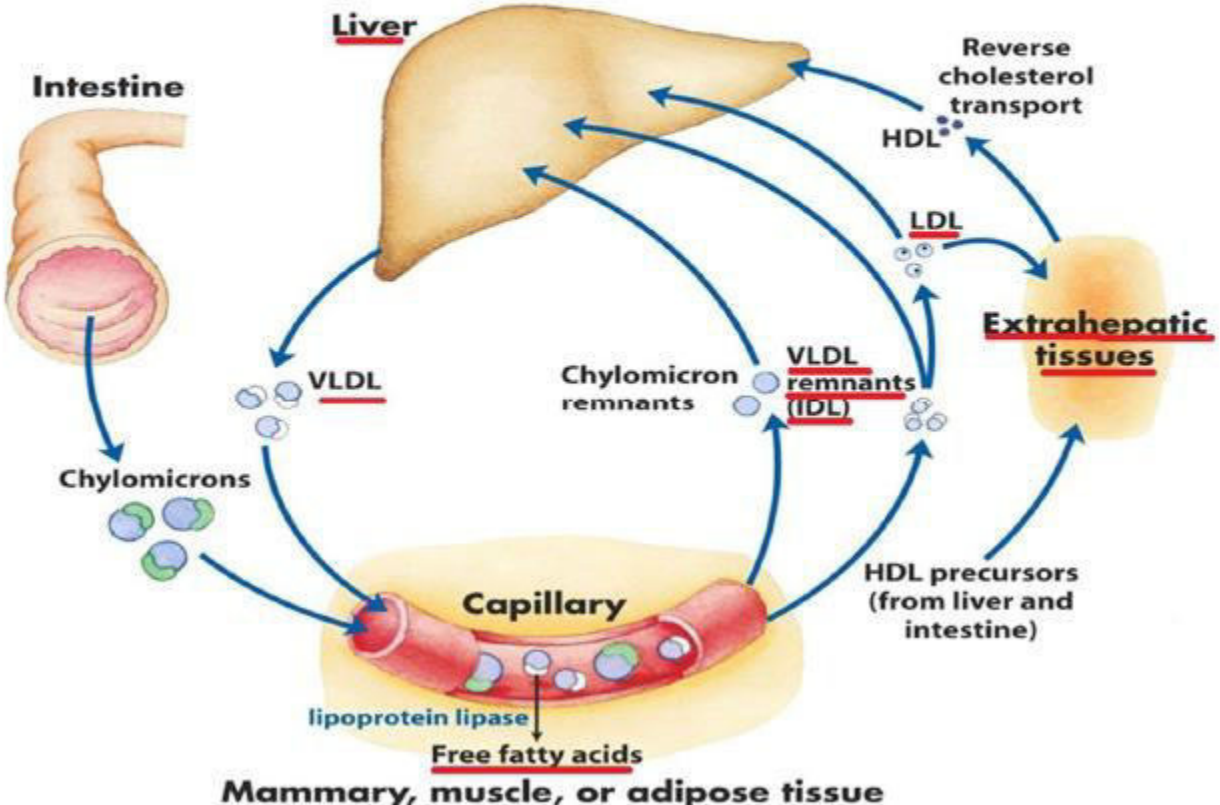
Lipid Chemistry

- LDLs contain apoB-100 as their major apolipoprotein
- they carry cholesterol to extra hepatic tissues that have specific plasma membrane receptors that recognize apoB-100.

Lipid Chemistry

- apoB-100 receptors internalize the LDL into the cell.
- Remaining LDL is endocytosed by Liver cells.

Lipid Chemistry



Lipid Chemistry

- Oxidized LDL can also accumulate in the macrophage cells lining the arteries
- resulting in the formation of atherosclerosis.

END

Lipid Chemistry

Lipoproteins (Contd.)

- **HDL** The fourth major lipoprotein type, high-density lipoprotein,
- originates in the liver and small intestine
- as small, protein-rich particles that contain little cholesterol and no cholesteryl esters

Lipid Chemistry

- They take up cholesterol from non-hepatic (peripheral) tissues and **return** it to the liver as cholesteryl esters

Lipid Chemistry

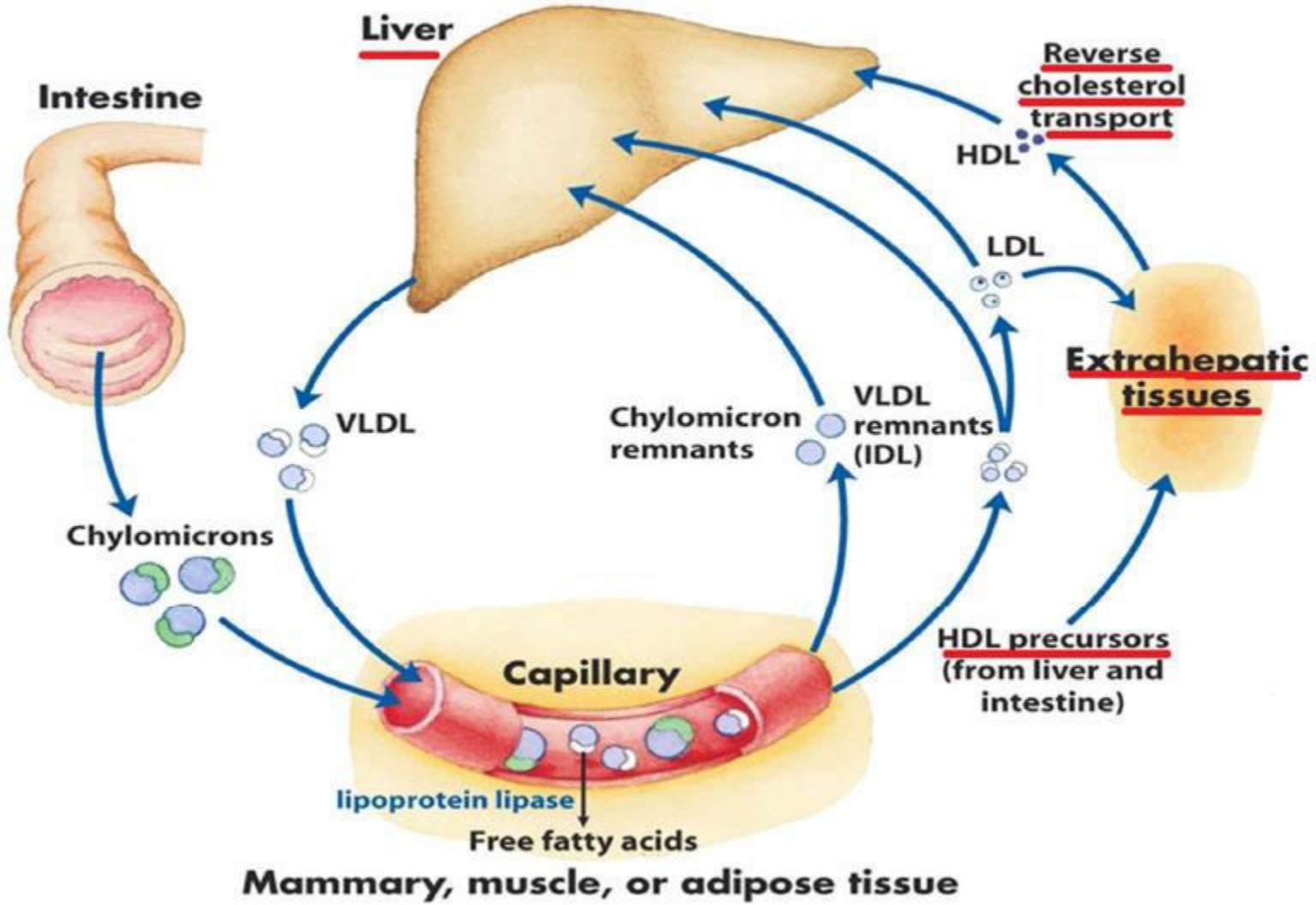
- When cholesterol is taken up by HDL, it is immediately esterified and becomes hydrophobic and
- which is sequestered in the core of the HDL,

Lipid Chemistry

- This transforms the nascent HDL to a mature HDL particle.

- Mature HDL then returns to the liver, where the cholesterol is unloaded in a process called reverse cholesterol transport pathway

Lipid Chemistry



Lipid Chemistry

- This is the basis for the inverse relationship seen between plasma HDL concentration and atherosclerosis, and for HDL's designation as the "good" cholesterol carrier.

Lipid Chemistry

- A much stronger correlation exists between the levels of blood LDL cholesterol and heart disease
- In contrast, high levels of HDL cholesterol have been associated with a decreased risk for heart disease

Lipid Chemistry

- Abnormal levels of plasma lipids (dyslipidemias) act in combination with smoking, obesity, sedentary lifestyle, insulin resistance, and other risk factors to increase the risk of CHD

Lipid Chemistry

- Clinical studies have demonstrated that dietary or drug treatment of hyper - cholesterolemia is effective in decreasing LDL, increasing HDL, and reducing the risk for cardiovascular events.

END

Lipid Chemistry

Role of dietary Lipids (Contd.)

- Unhealthy Fat contains
- Trans fat
- Saturated Fat
- Increased cholesterol content

Lipid Chemistry

Trans fat

- elevate serum LDL (but not HDL),
- Therefore they increase the risk of CHD

Lipid Chemistry

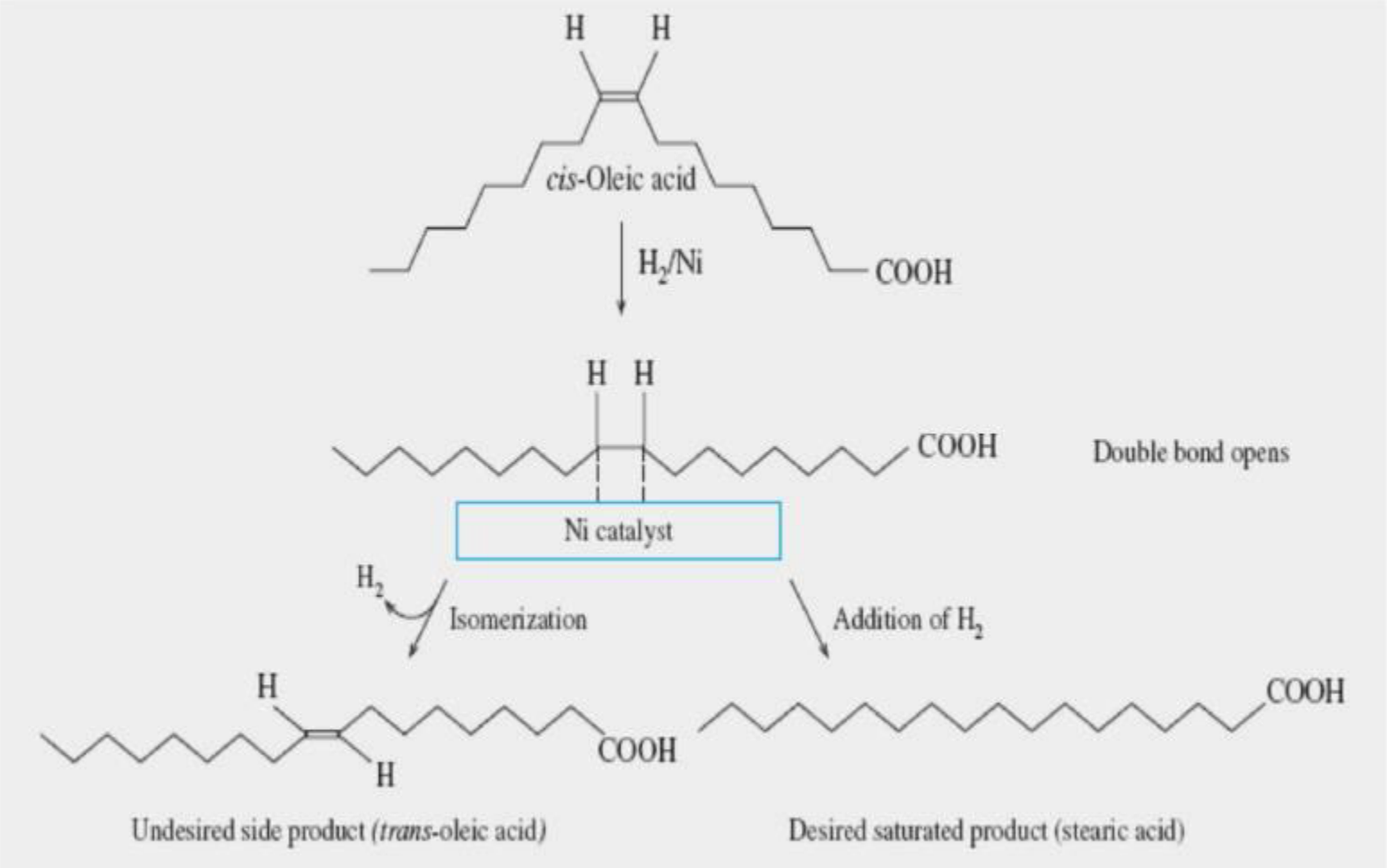
- Fatty acids of *trans* configuration in our food come from two different sources:
- industrially produced partially hydrogenated fat (IP-TFA) and
- ruminant produced Trans Fatty acid (RP-TFA)

Ruminants include cattle, **sheep**, goats, **buffalo**, deer, **giraffes** and **camels**

Lipid Chemistry

- Industrially produced trans fatt (IP-TFA)
- These are made by partial hydrogenation of vegetable fat, and to a lesser extent, of fish oils by heating to about 400° C under high pressure and with the addition of different catalysts such as Nickel

Lipid Chemistry



Lipid Chemistry

- Ruminant produced Trans Fatty acid (RP-TFA)
- are made by bacterial metabolism of polyunsaturated fatty acids in the rumen of ruminants, such as cow & sheep
- And are consequently present in all fats from these animals

word "ruminant" comes from the Latin *ruminare*, which means "to chew over again"

Lipid Chemistry

- The concentration of IP-TFA in partially hydrogenated fat may be as high as 60%.
- Whereas the maximum content of RP-TFA in ruminant fat is about 6%.
- In milk, RP-TFA is 4–6% of the fat.

Lipid Chemistry

- Recent studies suggest that with equal amounts of intake
- IP-TFA is more harmful than RP-TFA when compared on a gram-to-gram basis.

Lipid Chemistry

- The deleterious effects of trans fats occur at intakes of 2 to 7 g/day

- A single serving of French fries in a restaurant may contain this amount of trans fatty acids!

Lipid Chemistry

- Therefore, to reduce the harmful effects of trans fats, one should
- limit intake of foods prepared in IP-TFA containing oils such as baked goods and fast foods etc.
- instead of limiting fat intake from dairy sources.

END

Lipid Chemistry

Role of dietary Lipids (Contd.)

- Saturated fats — Consumption of saturated fats is associated with high levels of
- total plasma cholesterol and
- LDL cholesterol.

Lipid Chemistry

- Saturated fats —

Found in

- butter,
- hard cheeses,
- whole milk,
- animal fats,
- palm oil, and
- coconut oils.

Lipid Chemistry

- Among the SFAs, stearic acid (18:0) appears to have a neutral effect on LDL-C.
- While lauric (12:0), myristic (14:0), and palmitic (16:0) acids are considered to be hypercholesterolemic

Lipid Chemistry

- Saturated FAs increase plasma LDL-C by;
- increasing the formation of LDL in the plasma compartment
- and by decreasing LDL turnover.

Lipid Chemistry

- The lowering of plasma LDL-C observed with PUFAs is likely due to;
- redistribution of cholesterol between plasma and tissue pools
- and up regulation of the LDL receptor

Lipid Chemistry

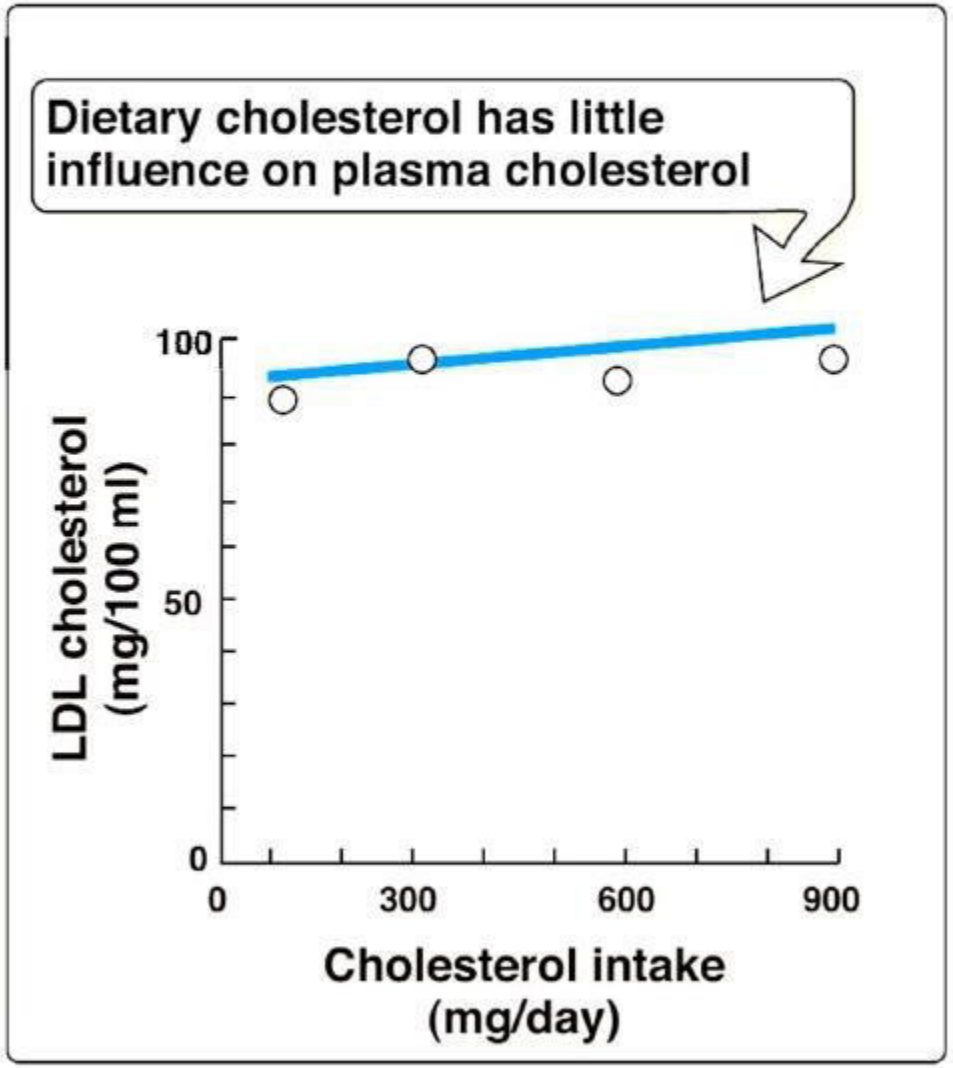
- In contrast, monounsaturated fatty acids (MUFAs) and PUFAs of the (n-6) family have been shown to;
- decrease plasma cholesterol concentrations in clinical studies

Lipid Chemistry

- **Dietary cholesterol:** Cholesterol is found only in animal products.

- dietary cholesterol has little effect on plasma cholesterol.

Lipid Chemistry



Lipid Chemistry

- Therefore, effect of dietary cholesterol on plasma cholesterol is less important than the amount and types of fatty acids consumed.

Lipid Chemistry

- A further reduction in dietary cholesterol seems to be unnecessary in those people;
- who have already reduced their intake of saturated fat
- and increased the ratio of polyunsaturated to saturated fatty acids

Lipid Chemistry

- **The Mediterranean Diet**
- Mediterranean cultures, show a low incidence of coronary heart disease

Spain, France,Italy,Albania, Greece, Turkey, Cyprus, Syria, Lebanon, Israel, Palestine, Egypt, Libya,Tunisia, Algeria, Morocco

Lipid Chemistry

- The Mediterranean diet is an example of a
- diet rich in monounsaturated fatty acids (from olive oil) and ω-3 fatty acids (from fish oils and some nuts),
- but low in saturated fat

Lipid Chemistry

- The Mediterranean diet contains
- seasonally fresh food, with
- an abundance of plant material,
- low amounts of red meat, and
- olive oil as the principal source of fat

Lipid Chemistry

- The Mediterranean diet is associated with decreased serum total cholesterol and LDL cholesterol—but
- little change in HDL cholesterol—when compared with a typical Western diet higher in saturated fats

END

Lipid Chemistry

STRUCTURE OF PHOSPHOLIPIDS

There are two classes of phospholipids:

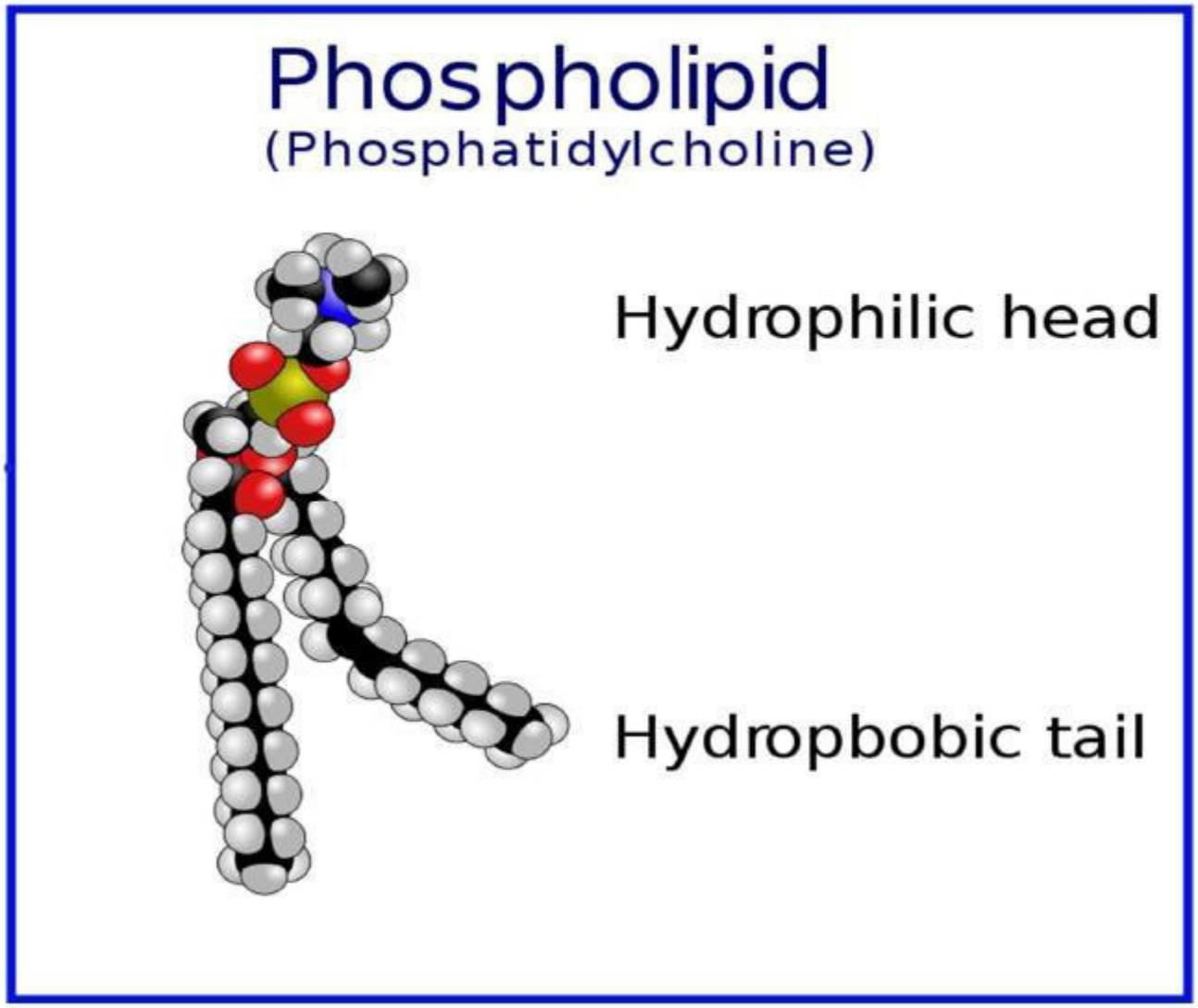
- those that have glycerol as a backbone: glycerophospholipids
- those that contain sphingosine: sphingphospholipids

Compound lipids – FA + alcohol + other groups

Lipid Chemistry

- Phospholipids are the predominant lipids of cell membranes
- Membrane Iipids are **amphipathic** i.e. one end of the molecule is hydrophobic, the other hydrophilic

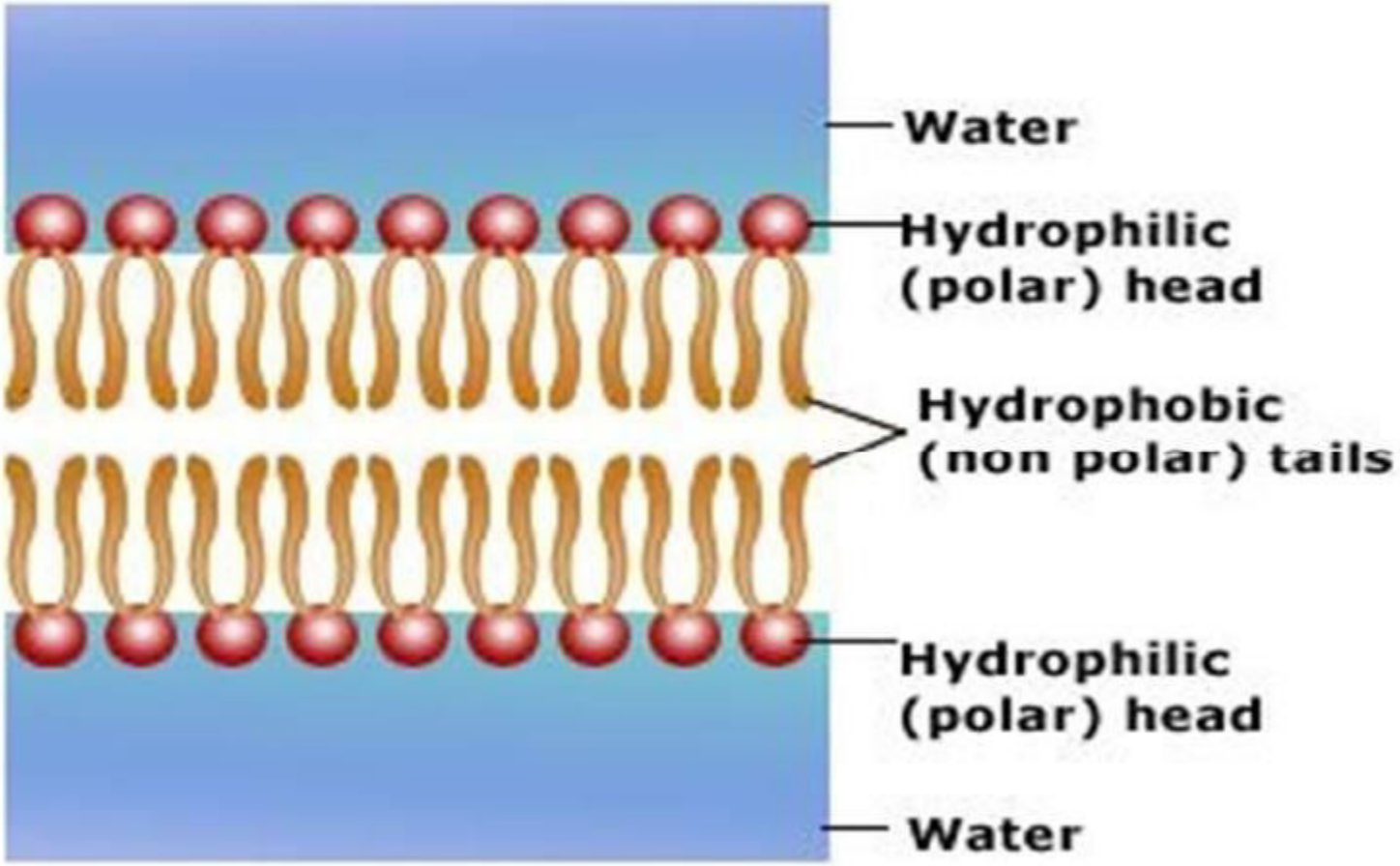
Lipid Chemistry



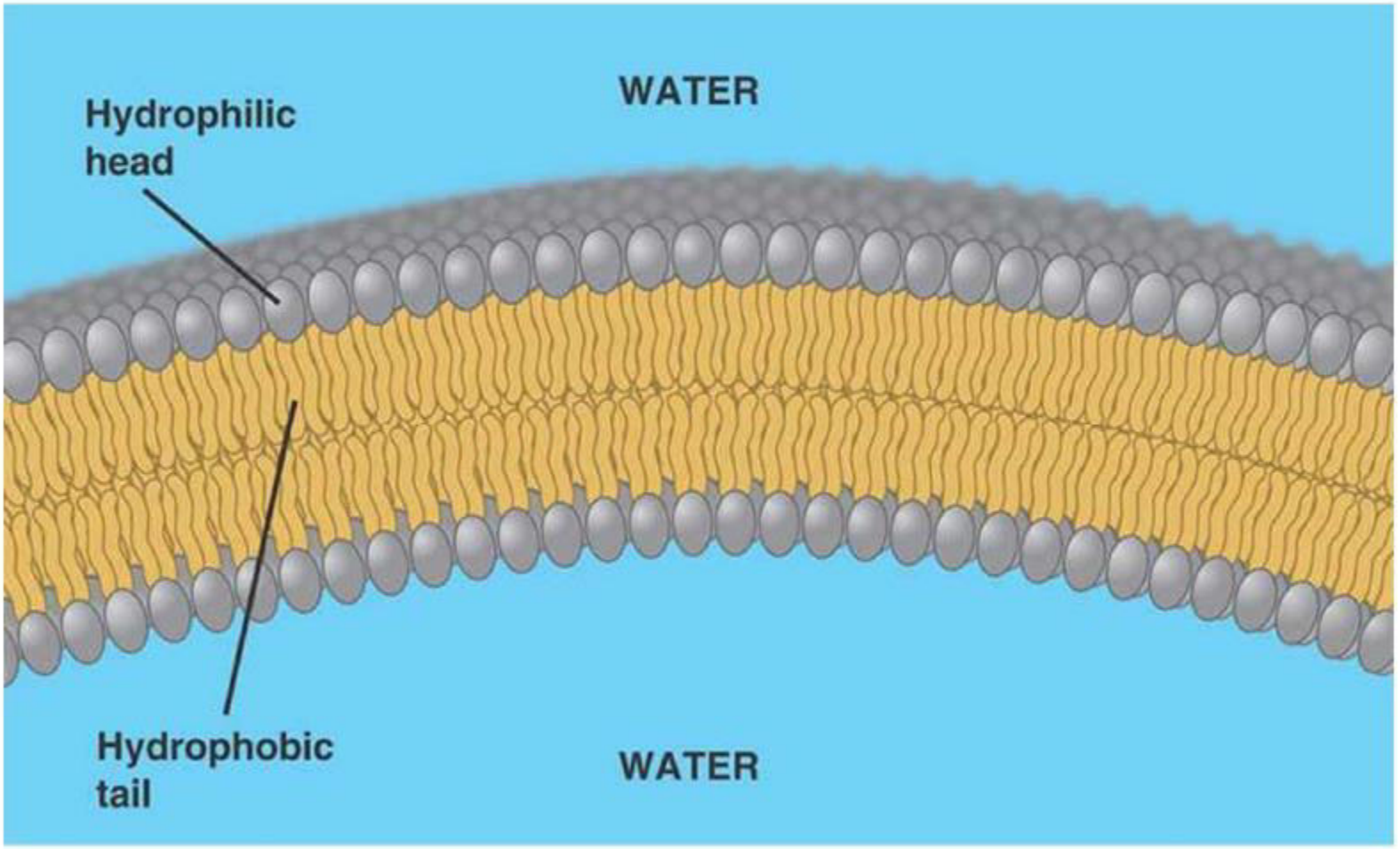
Lipid Chemistry

- Their hydrophobic interactions with each other and their
- hydrophilic interactions with water direct their
- packing into sheets called membrane bilayers

Lipid Chemistry



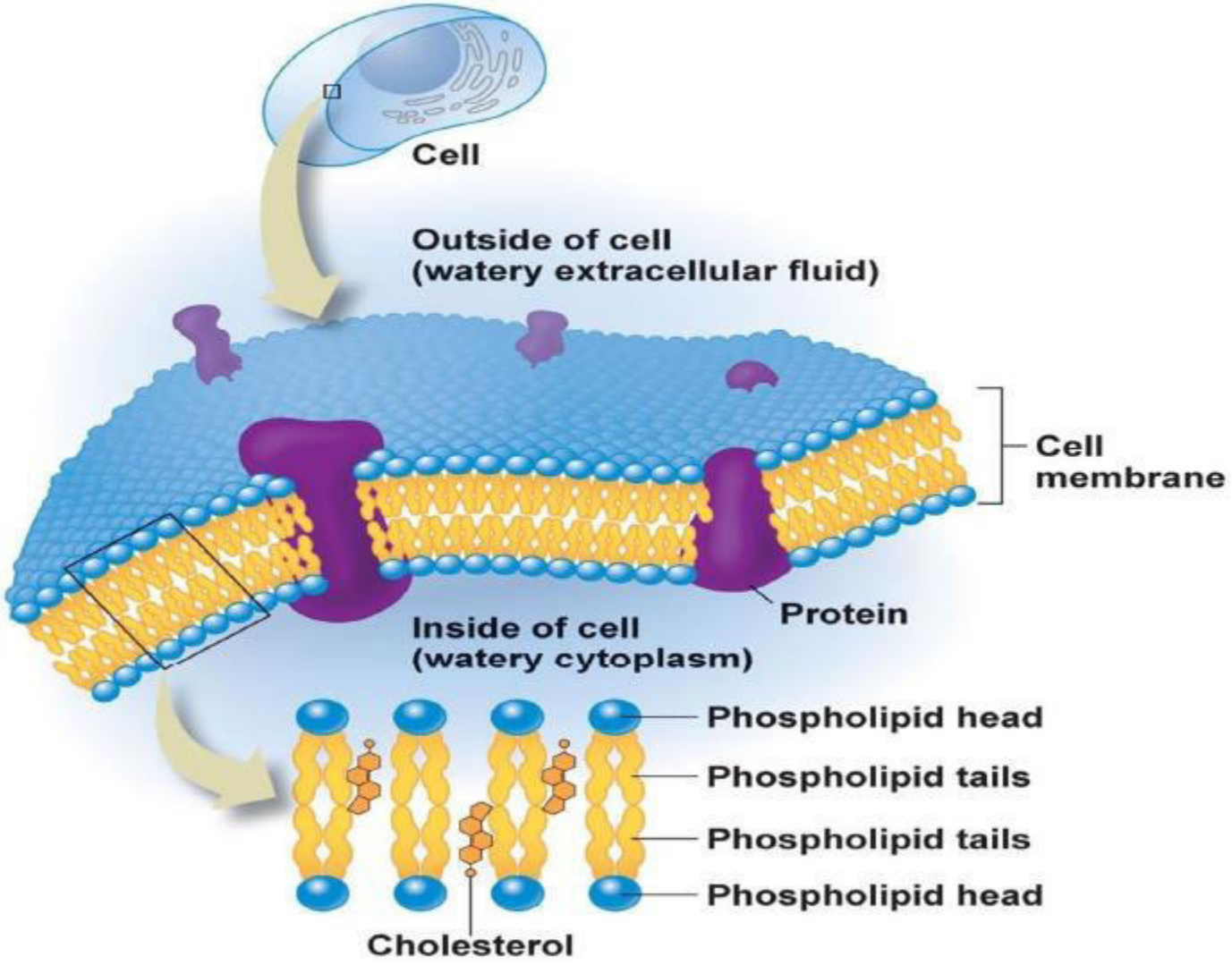
Lipid Chemistry



Lipid Chemistry

- The formation of membrane bilayers
- help in partitioning the cellular environment from extracellular environment.

Lipid Chemistry



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Lipid Chemistry

- Non-membrane-bound phospholipids serve additional functions in the body, for example,
  - as components of lung surfactant
  - essential components of bile.

Lipid Chemistry

- In contrast to triacylglycerol which is essentially synthesized only in liver, adipose tissue, lactating mammary glands, and intestinal mucosal cells

- essentially all cells except mature erythrocytes can synthesize phospholipids.

END

Lipid Chemistry

Glycerophospholipids

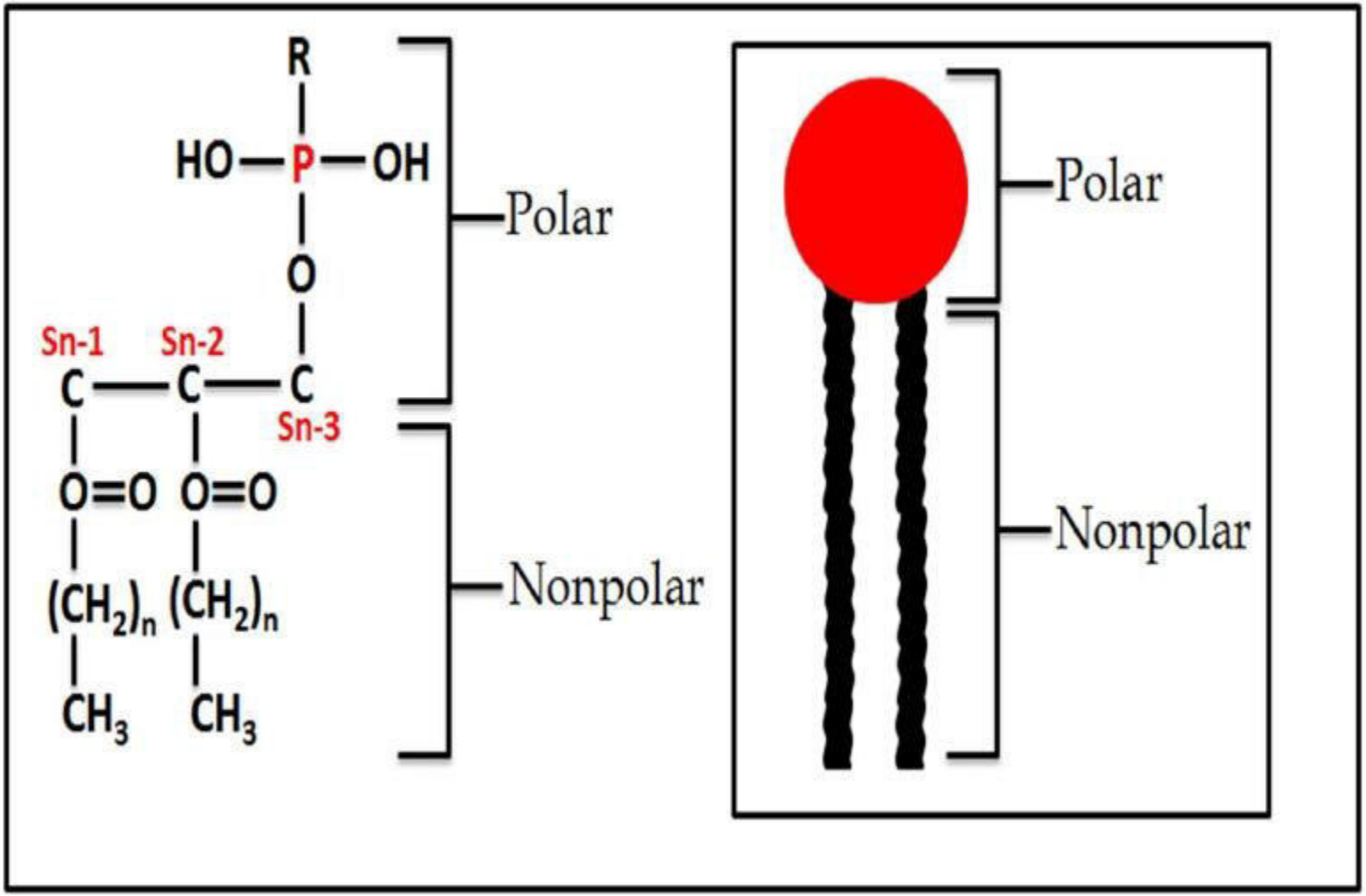
❑Phosphatidic acid

- It is the simplest phosphoglyceride,

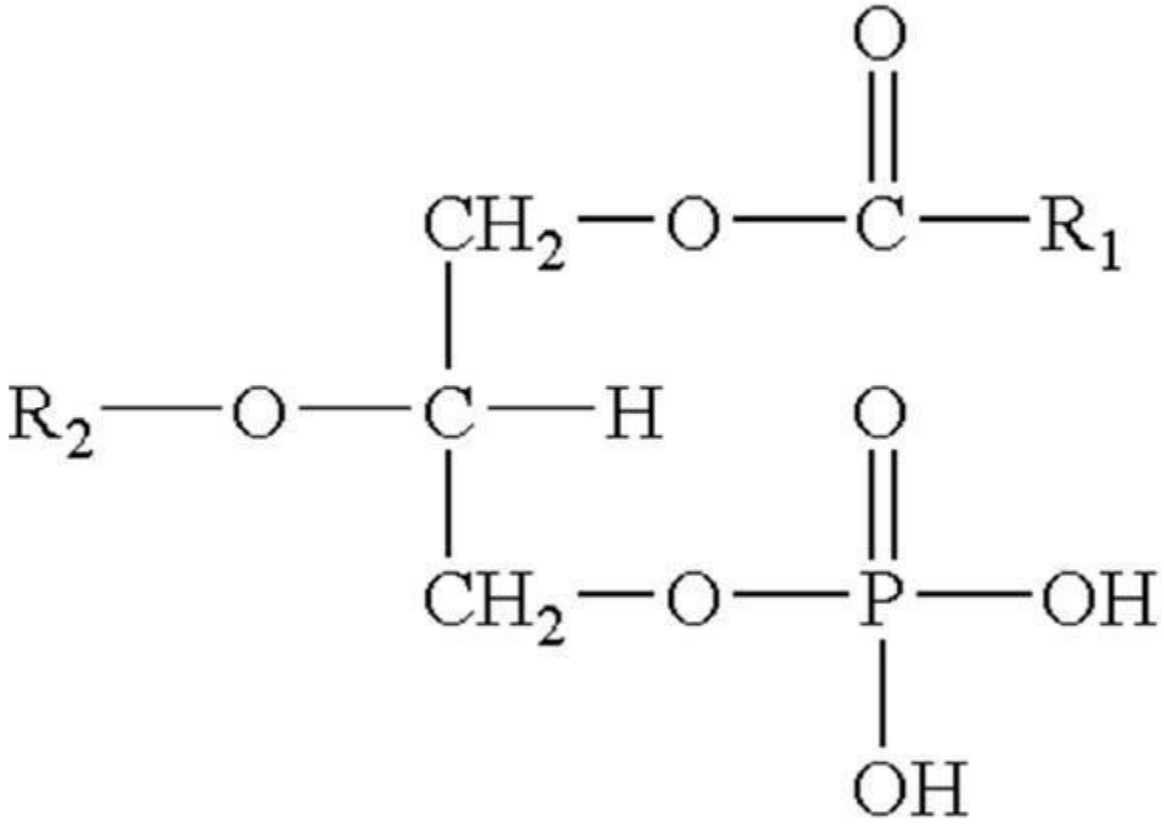
- It is a diacylglycerol with a phosphate group on the third carbon of glycerol.

Compound lipids – FA + alcohol + other groups

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- Phosphatidic acid is the precursor of the other members of this group.

- Further esterification with a low-molecular weight alcohol gives a glycerophospholipid

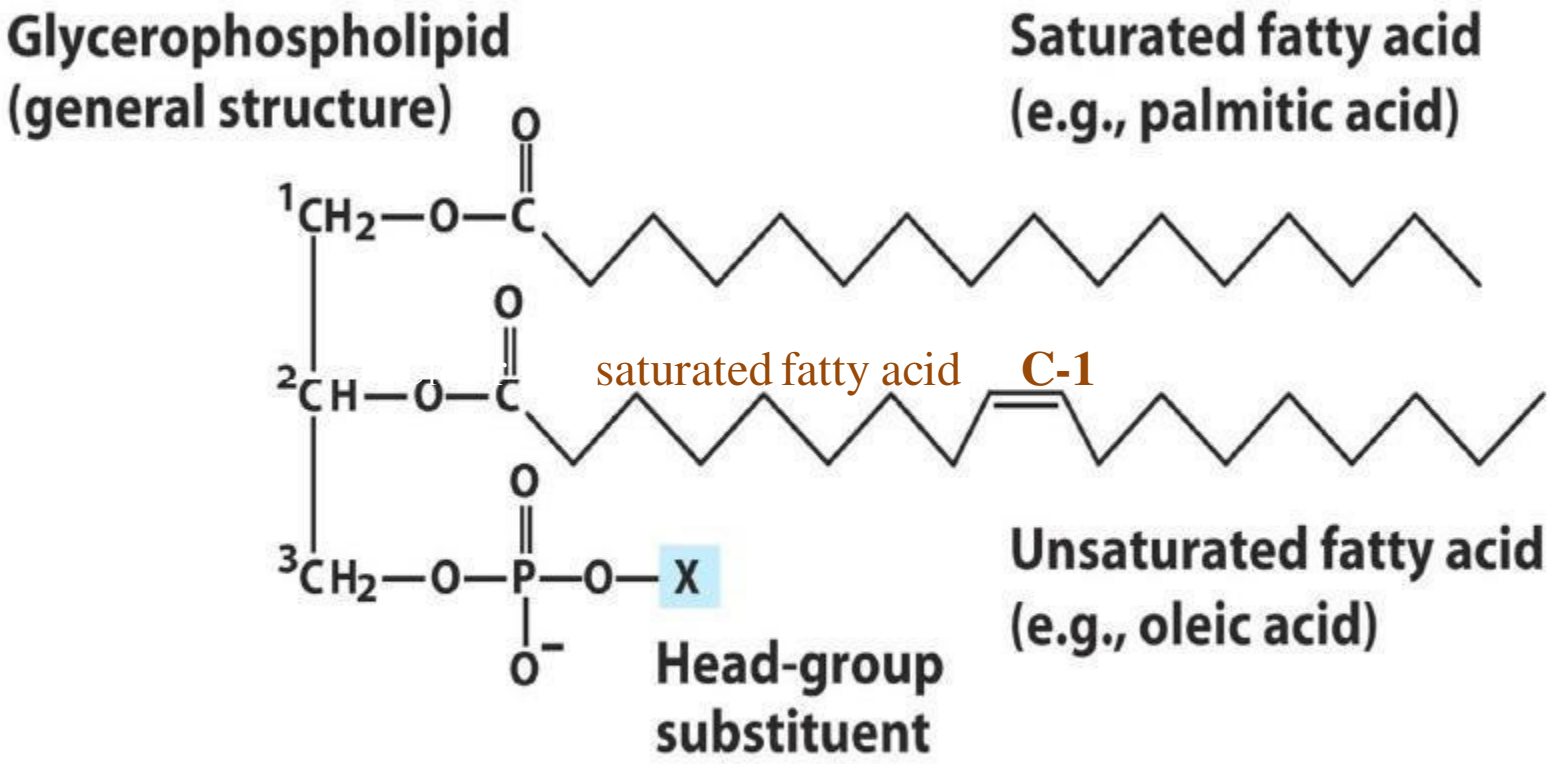
All glycerophospholipids are derivatives of phosphatidic acid and are named for their polar head

Lipid Chemistry

In general, glycerophospholipids contain

- a C_{16} or C_{18} saturated fatty acid at **C-1** and
- a C_{18} or C_{20} unsaturated fatty acid at **C-2** in addition to
- a phosphate group on **C-3**

Lipid Chemistry



C_{16} or C_{18} saturated fatty acid at C-1,C_{18} or C_{20}unsaturated fatty acid at C-2

Lipid Chemistry

- The phosphate group on phosphatidic acid (PA) can be esterified to another compound containing a hydroxyl group
  - Serine + PA → phosphatidylserine
  - Ethanolamine + PA → phosphatidylethanolamine
  - Choline + PA → phosphatidylcholine (lecithin)
  - Inositol + PA → phosphatidylinositol

- The molecule is both a primary amine and a primary alcohol

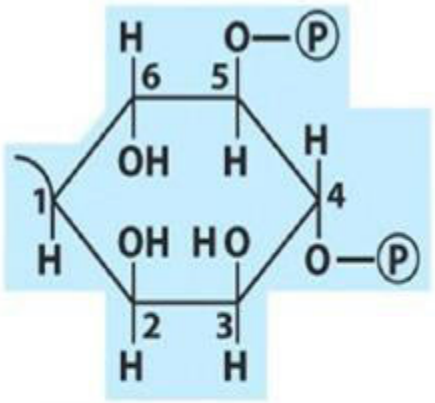
Lipid Chemistry

Name of glycerophospholipid	Name of X	Formula of X
Phosphatidic acid	—	—H
Phosphatidylethanolamine	Ethanolamine	—CH_2—CH_2—$\overset{+}{N}H_3$
Phosphatidylcholine	Choline	—CH_2—CH_2—$\overset{+}{N}(CH_3)_3$
Phosphatidylserine	Serine	—CH_2—$\underset{\displaystyle COO^-}{\underset{\displaystyle |}{CH}}$—$\overset{+}{N}H_3$
Phosphatidylglycerol	Glycerol	—CH_2—$\underset{\displaystyle OH}{\underset{\displaystyle |}{CH}}$—$CH_2$—OH

Lipid Chemistry

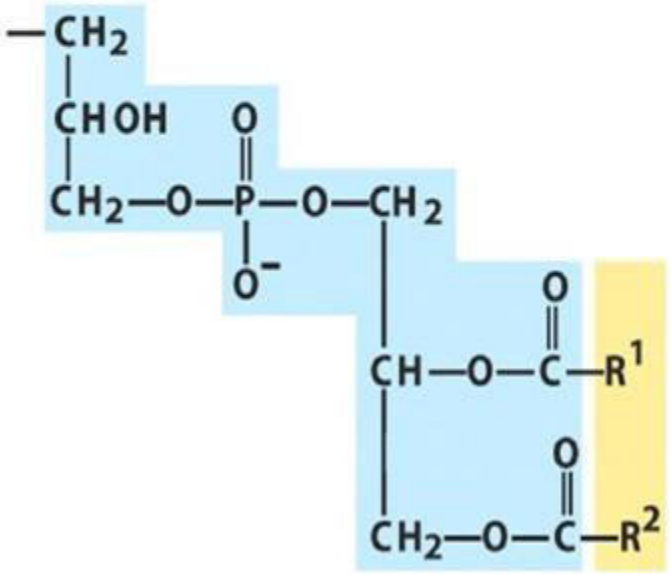
Phosphatidylinositol 4,5-bisphosphate

myo-Inositol 4,5-bisphosphate



Cardiolipin

Phosphatidyl-glycerol



Lipid Chemistry

- The fatty acids in glycerophospholipids can be any of a wide variety, so a given phospholipid (phosphatidylcholine, for example) may consist of several *molecular species,* each with its unique complement of fatty acids.

Lipid Chemistry

For instance,

- Dipalmatoyl phosphatidyl choline

and

- Palmitoyl-oleyl-phosphatidylcholine

are two different examples of Lecithins.

END

Lipid Chemistry

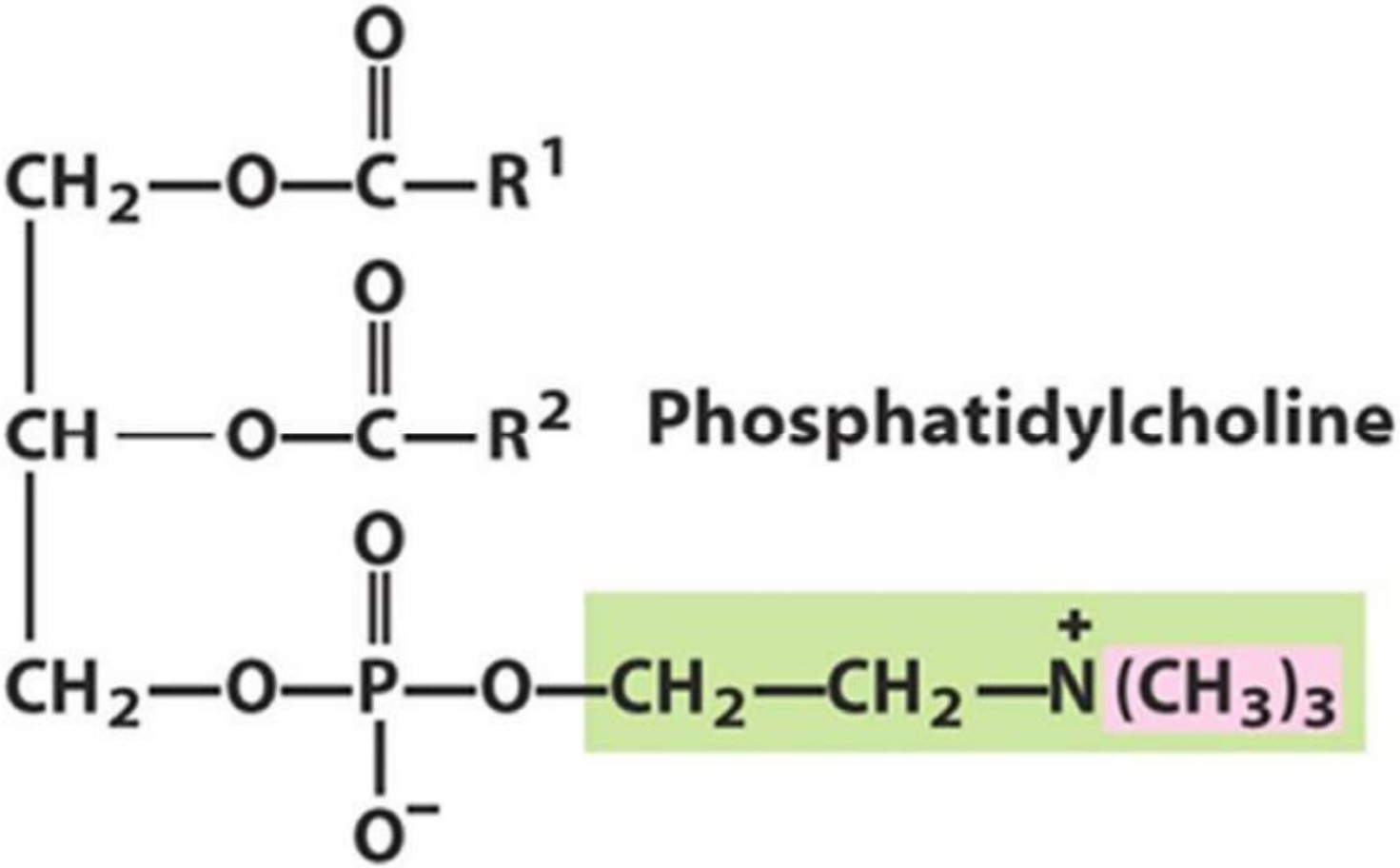
Glycerophospholipids (Contd.)

Lipid Chemistry

Phosphatidylcholines (Lecithins)

- the most abundant phospholipids of the cell membrane
- represent a large proportion of the body's store of choline- important in nervous transmission, as acetylcholine.

Lipid Chemistry



Choline is a quaternary ammonium: tri methyl ethanol ammonium

Lipid Chemistry

- Dipalmitoyl phosphatidyl choline (DPPC or dipalmitoylecithin),

is also the major lipid component of lung surfactant

Lipid Chemistry

- Alveoli are the structural and functional unit of respiratory system in which gaseous exchange takes place.
- A thin fluid layer lines the alveoli for efficient gas exchange.

Lipid Chemistry

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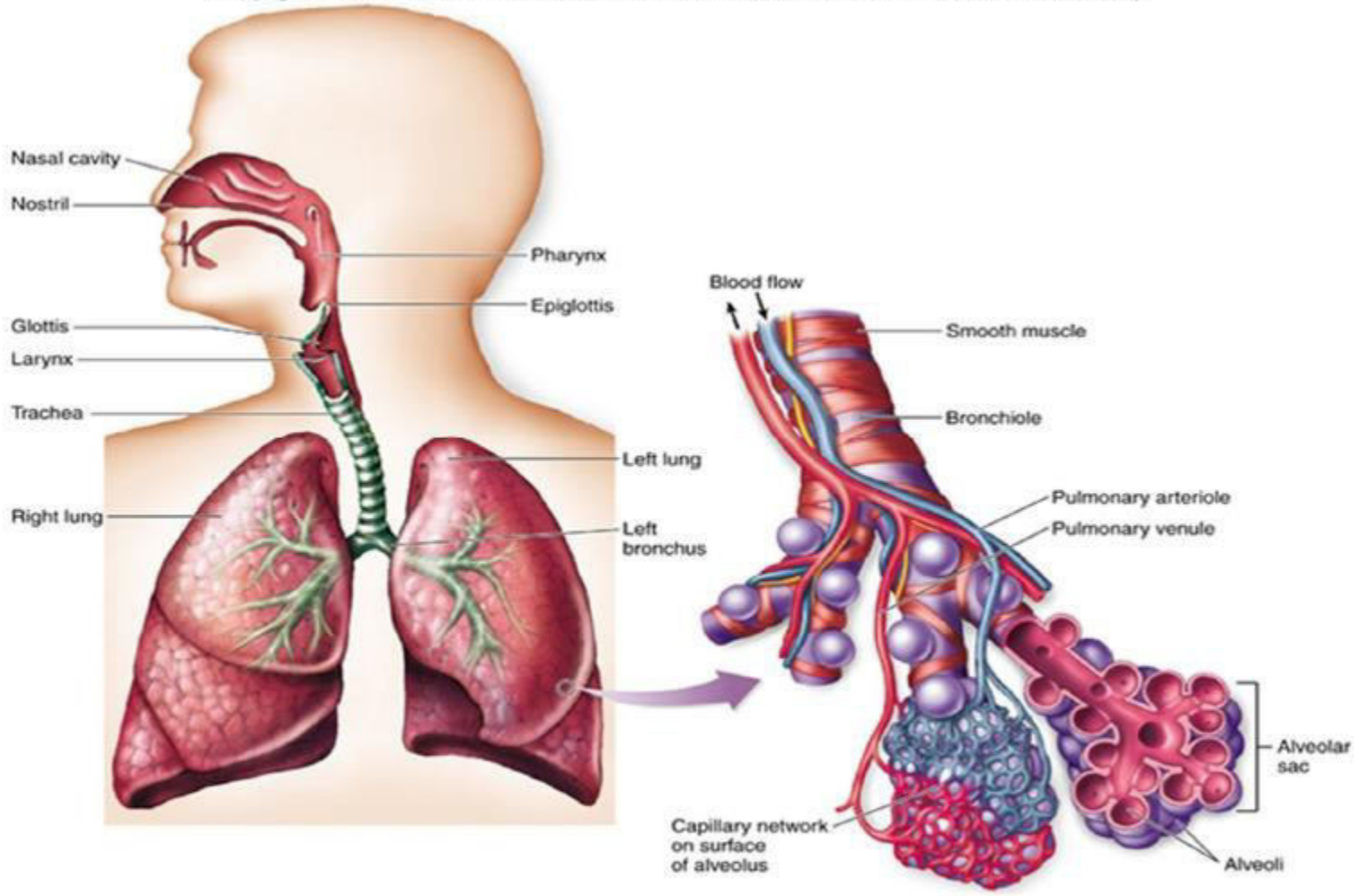
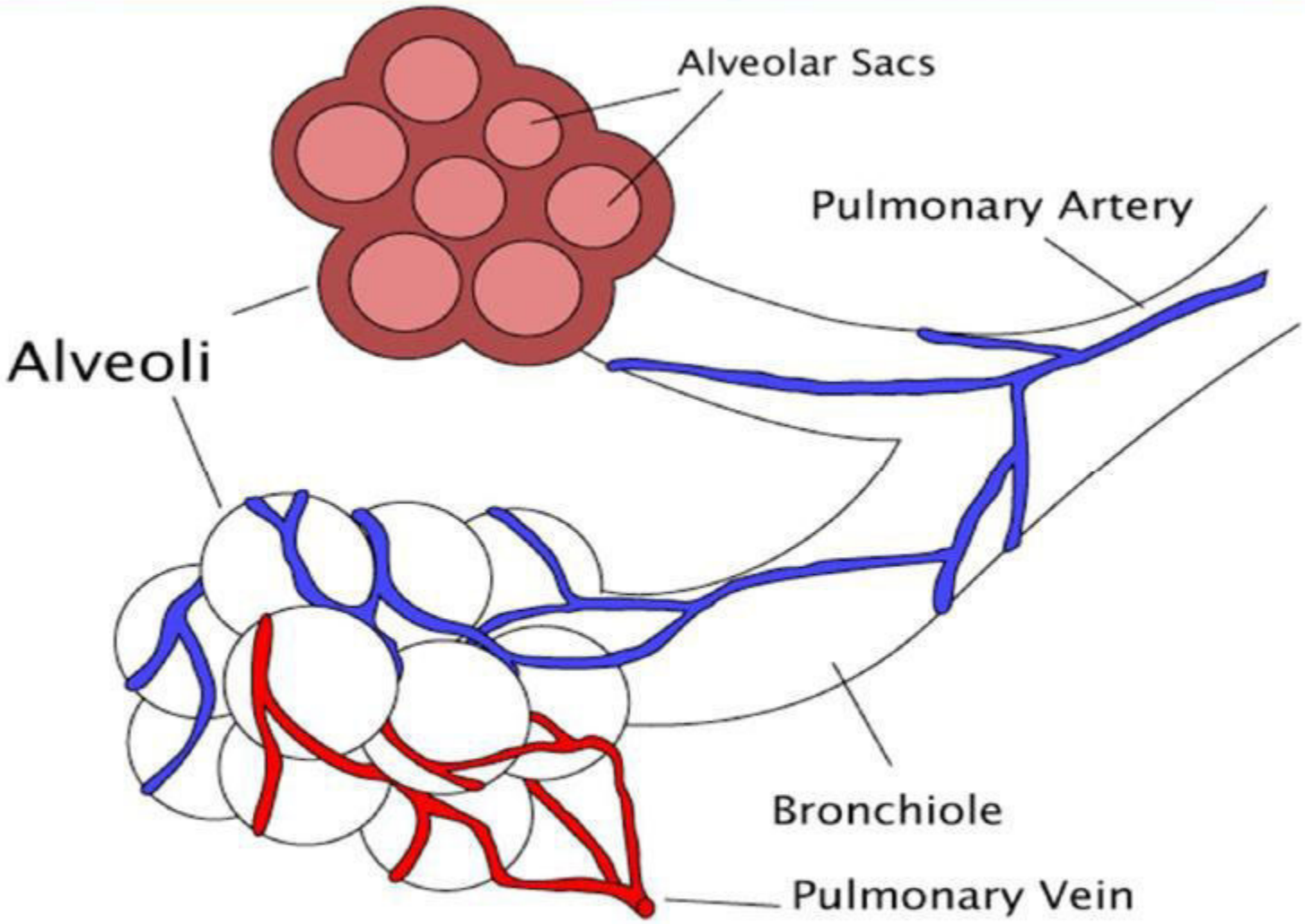


Lipid Chemistry



Alternate diagram

Lipid Chemistry

- When the water forms a surface with air, the water molecules on the surface (in contact of air) are strongly attracted to each other trying to reduce the surface area of contact. This is called **"surface tension"**.

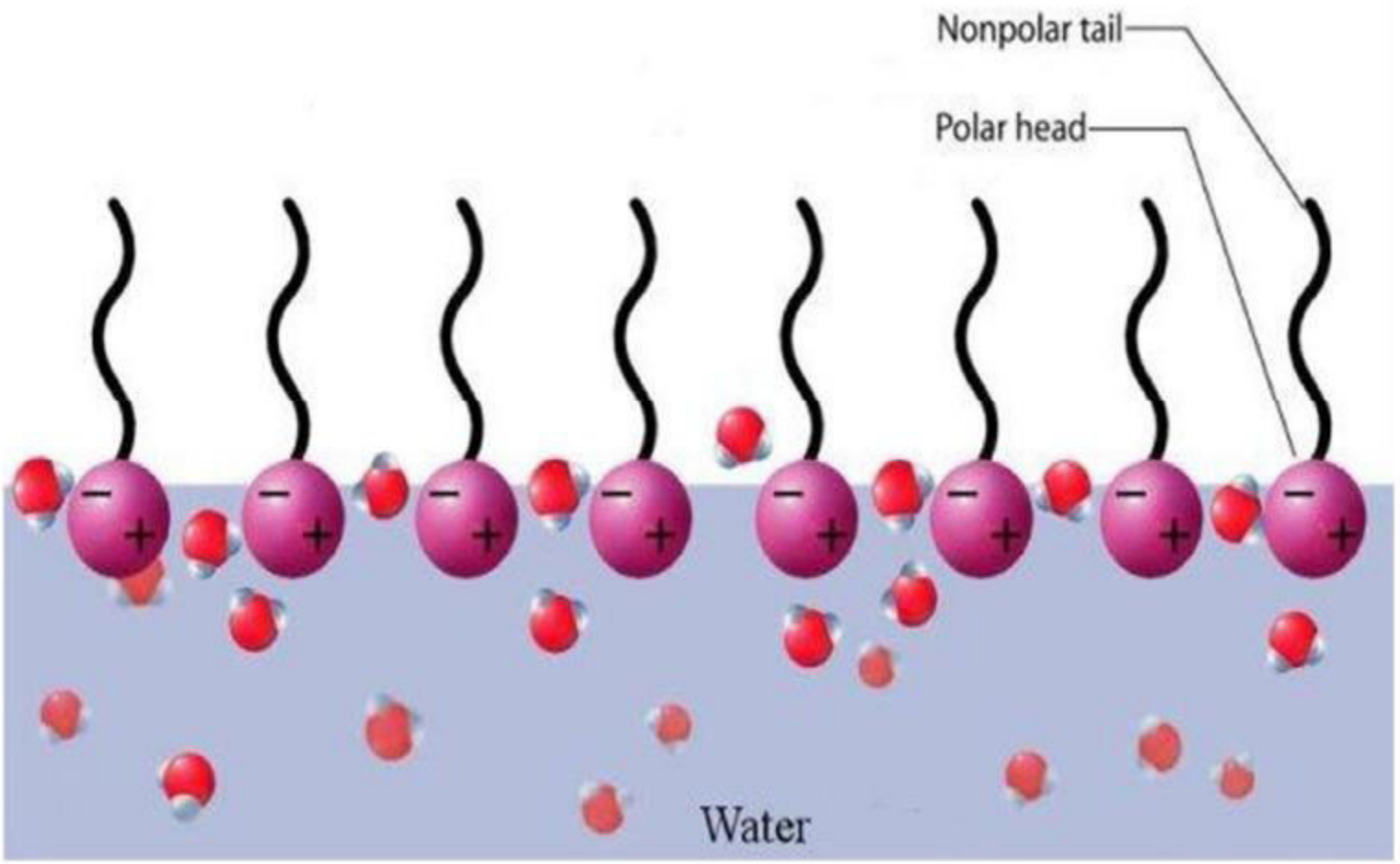
Lipid Chemistry

- **surface tension** might result in alveolar collapse
- it requires a certain inflation pressure to maintain expanded alveoli
- the higher the surface tension, the more pressure required to inflate the bubble, especially in small alveoli.

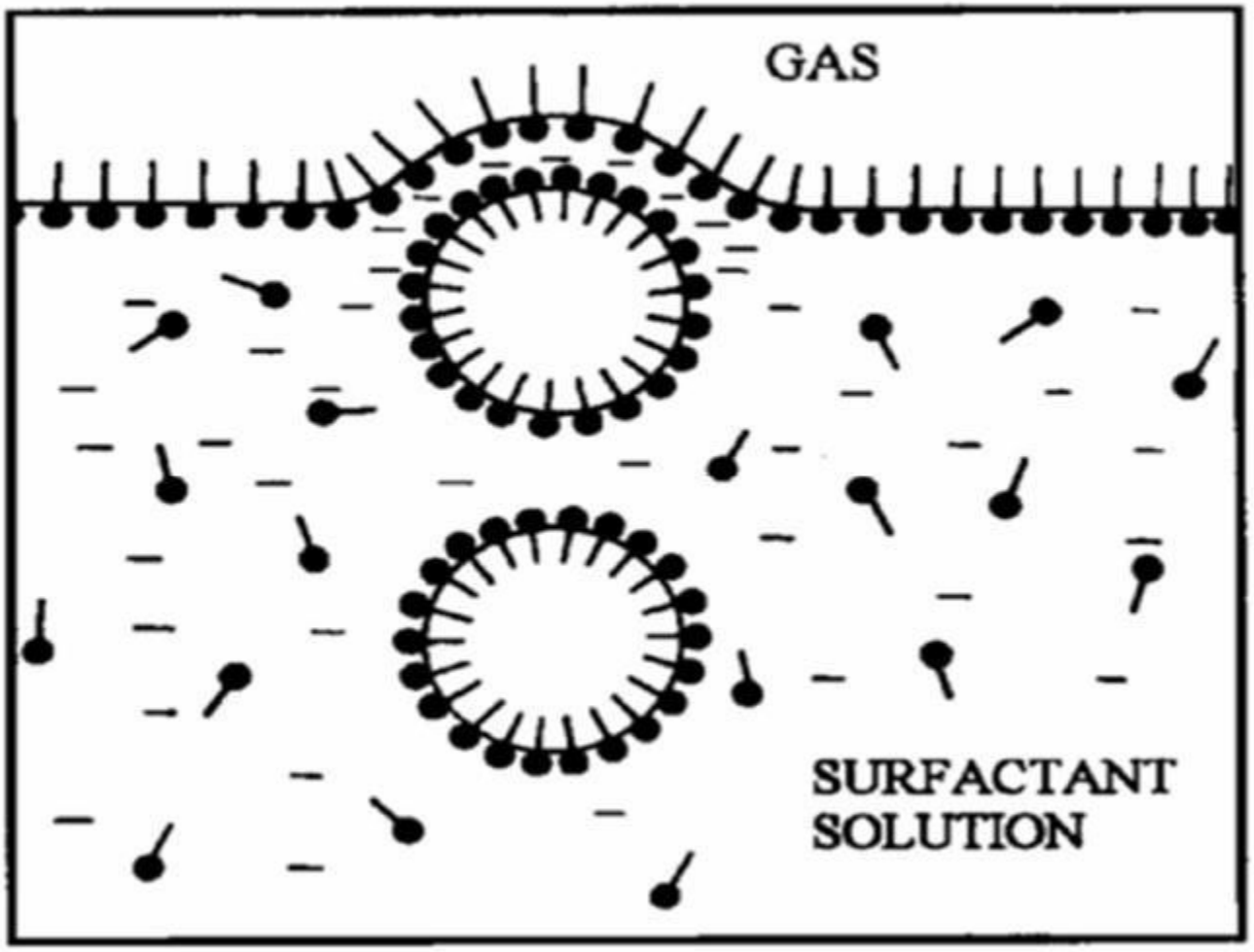
Lipid Chemistry

- made and secreted by lung cells, surfactant serves to decrease the surface tension of this fluid layer.
- It scatters among the fluid molecule decreasing the attraction between them.

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- surfactant have both hydrophilic and hydrophobic regions
- by adsorbing to the air-water interface of alveoli hydrophilic head groups in the fluid and the hydrophobic tails facing towards the air (DPPC) reduces surface tension.

END

Dipalmitoyl phosphatidyl choline

Lipid Chemistry

Glycerophospholipids (Contd.)

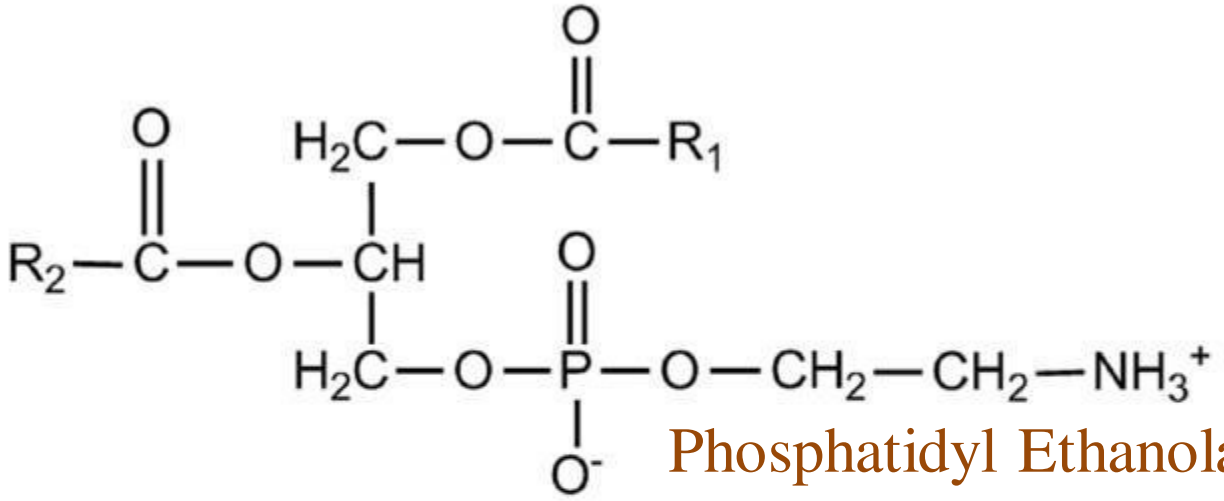
Lipid Chemistry

- **Phosphatidyl ethanol amine** and **phosphatidylserine**
- are also found in cell membranes and
- differ from phosphatidylcholine only in that ethanolamine or serine, respectively, replaces choline

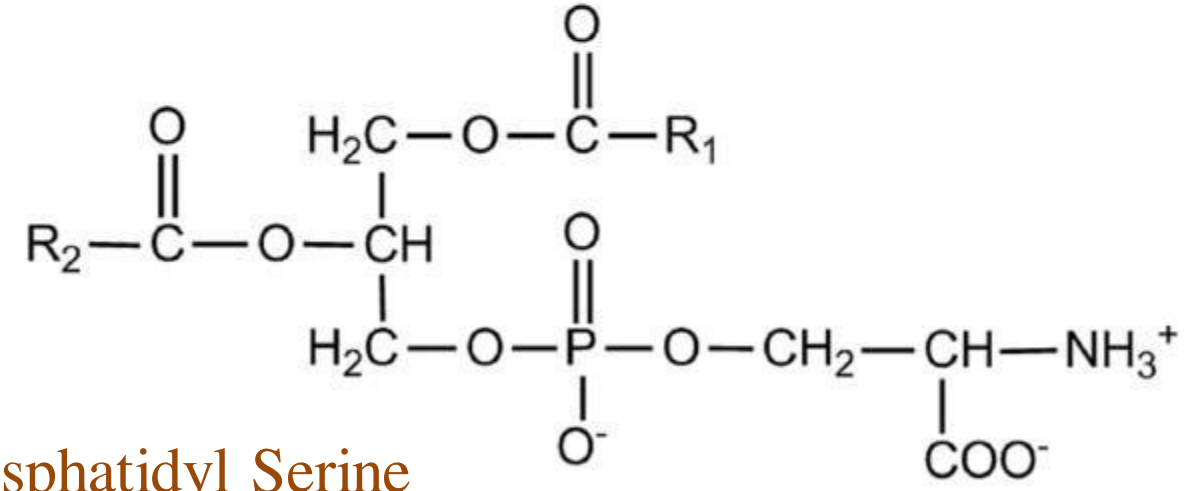
Lipid Chemistry

- Serine is a standard amino acid
- which on decarboxylation produces ethanolamine:
- a primary alcohol and a primary amine.

Lipid Chemistry

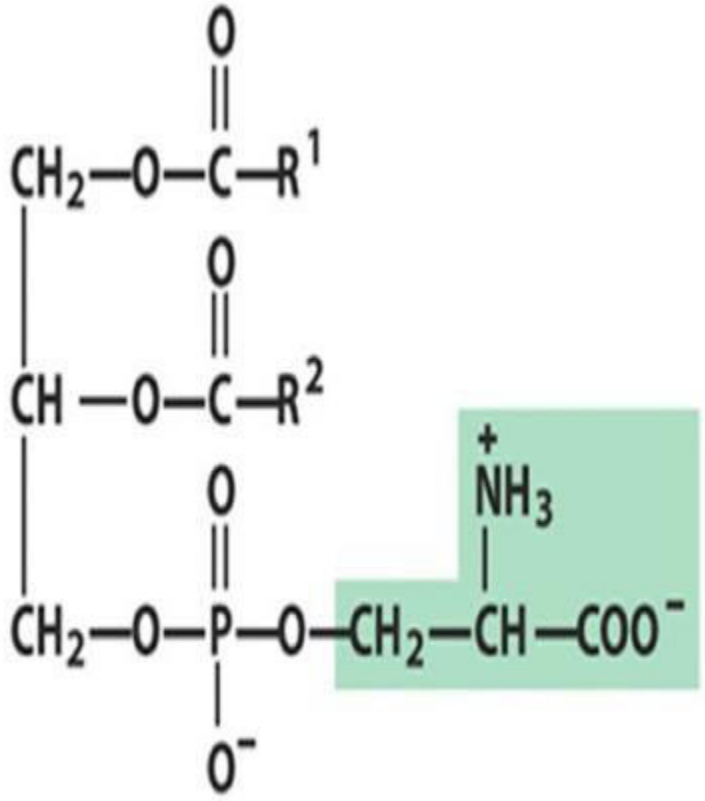


Phosphatidyl Ethanolamine

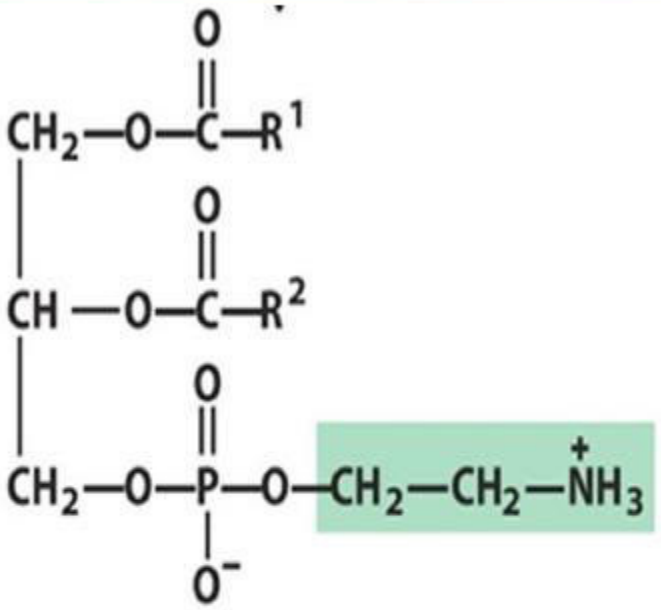


Phosphatidyl Serine

Lipid Chemistry



Phosphatidylserine



Phosphatidylethanolamine

Alternate diagram

Lipid Chemistry

- Phosphatidylserine also plays a role in apoptosis (programmed cell death)

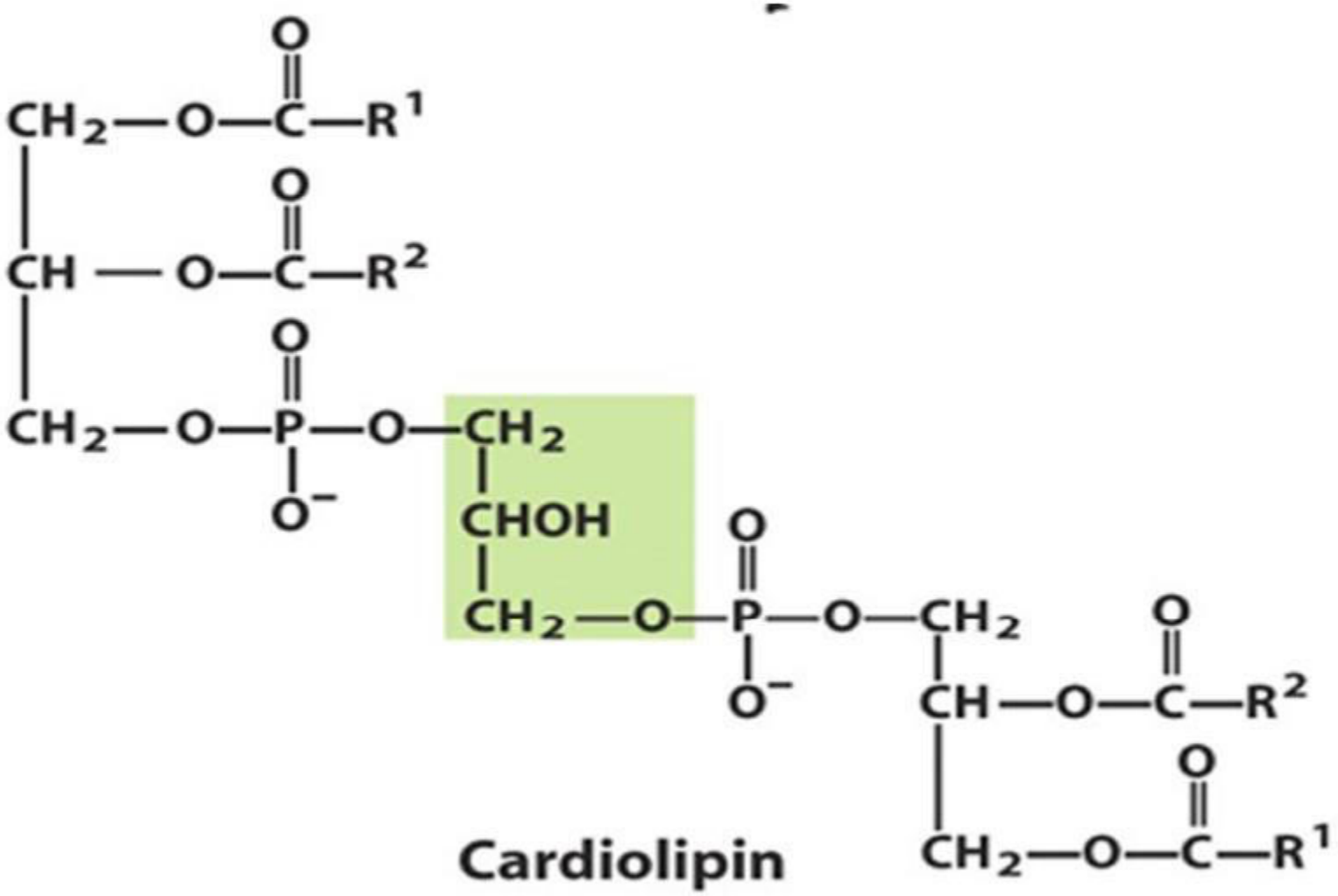
Lipid Chemistry

Cardiolipin

diphosphatidylglycerol

- Two molecules of phosphatidic acid esterified through their phosphate groups to an additional molecule of glycerol

Lipid Chemistry



Cardiolipin

Lipid Chemistry

- In eukaryotes, cardiolipin is virtually exclusive to **the inner mitochondrial membrane**, where it appears to be required for the maintenance of certain respiratory complexes of the electron transport chain.

Lipid Chemistry

- Decreased cardiolipin levels or alterations in its structure or metabolism cause mitochondrial dysfunction in **aging** and in **pathological conditions** including **heart failure**

Lipid Chemistry

- Cardiolipin is antigenic, and is recognized by antibodies raised against Treponema pallidum, the bacterium that causes syphilis

Antigen : usually protein, is a molecule capable of generating immune respons

Lipid Chemistry

- The VDRL test measures immunoglobulin G (IgG) and IgM antibodies to lipoidal material released from damaged host cells as well as to lipoprotein-like material
- and possibly cardiolipin released from the treponemes

.

Syphilis is a bacterial infection usually spread by sexual contact.

Lipid Chemistry

- Anti-cardiolipin antibodies may also be found non-specifically in certain auto-immune diseases.

END

Such as SLE and rheumatoid arthritis

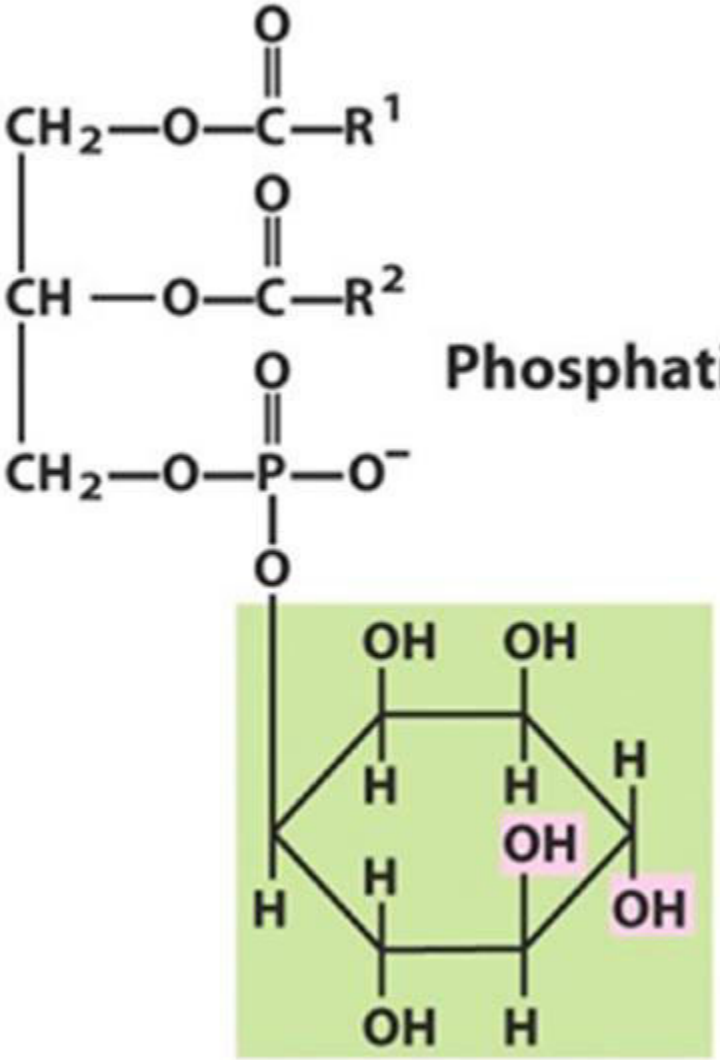
Lipid Chemistry

Glycerophospholipids (Contd.)

Phosphatidylinositol

- consists of phosphatidic acid and
- Inositol in an
- ester linkage

Lipid Chemistry



Phosphatidylinositol

These —OH groups can also be esterified with $—PO_3^{2-}$.

Lipid Chemistry

- Inositol is a polyol synthesized in the body from glucose
- and have the same chemical formula as glucose

Lipid Chemistry

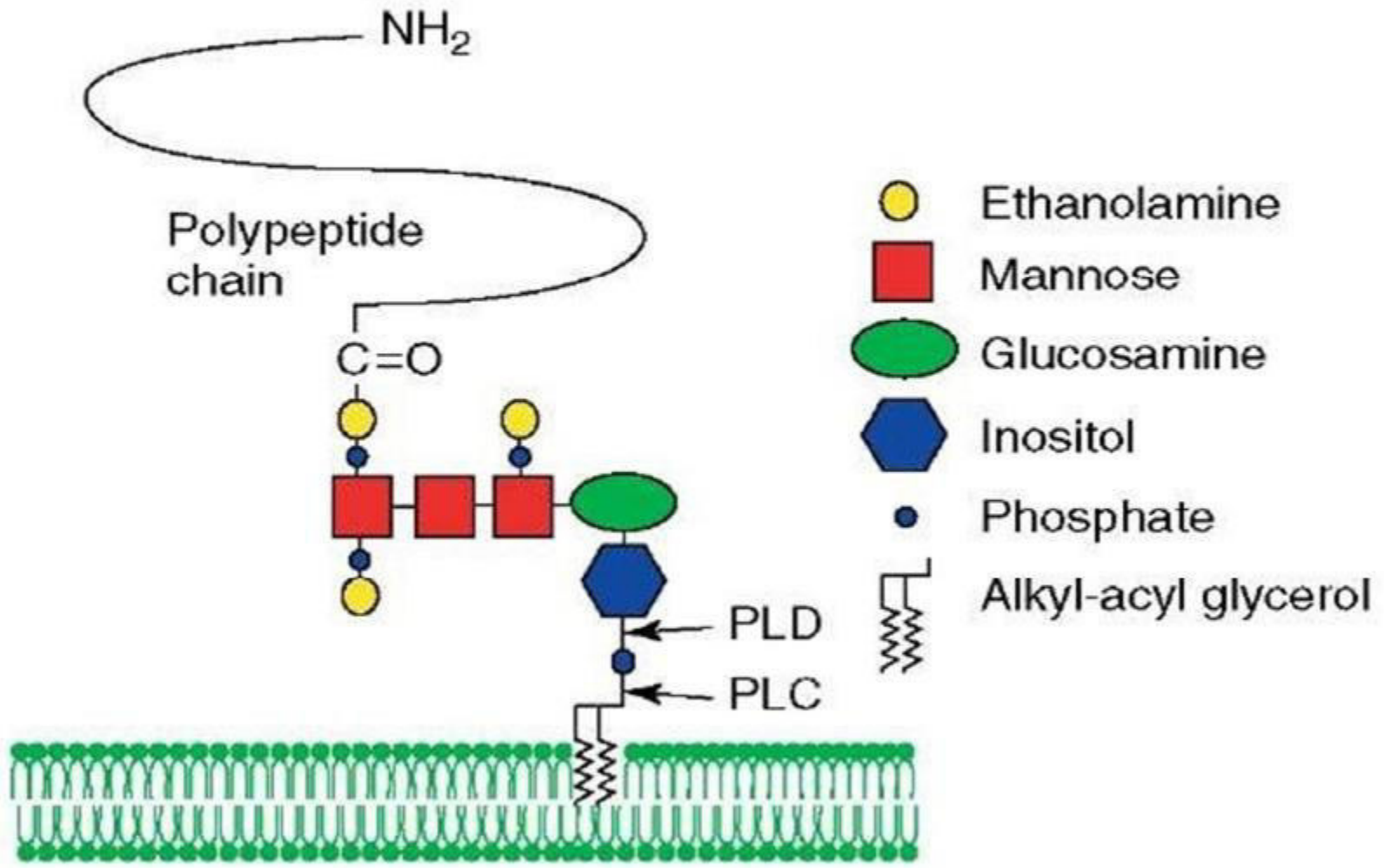
Phosphatidylinositol (PI) is an unusual phospholipid in that it often contains

- stearic acid on carbon 1 and
- arachidonic acid on carbon 2 of the glycerol

Lipid Chemistry

- PI, therefore, as a reservoir of arachidonic acid, serves as precursor for prostaglandin synthesis
- Specific proteins can be covalently attached via a carbohydrate bridge to membrane-bound PI.

Lipid Chemistry



Lipid Chemistry

- Phosphatidylinositol is a precursor of second messengers
- The phosphorylation of membrane-bound phosphatidylinositol produces phosphatidylinositol 4,5-bisphosphate (PIP2)

Lipid Chemistry

- The degradation of PIP_2 by phospholipase C occurs in response to the binding of a variety of neurotransmitters, hormones, and growth factors to receptors on the cell membrane

Lipid Chemistry

- The products of this degradation are
- inositol 1,4,5 trisphosphate (IP_3) and
- diacylglycerol (DAG)

Lipid Chemistry

- These products mediate the
- mobilization of intracellular calcium and
- the activation of protein kinase C, respectively, which act synergistically to evoke specific cellular responses

END

Lipid Chemistry

Glycerophospholipids (Contd.)

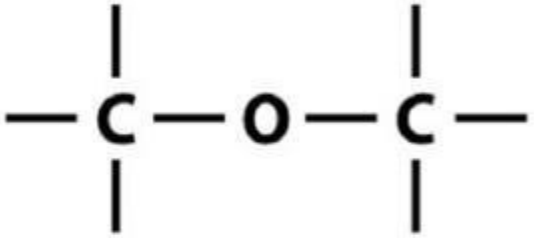
- **Ether lipids**

They are type of Glycerophospholipids, in which one acyl chains is attached to glycerol in ether linkage, rather than ester linkage

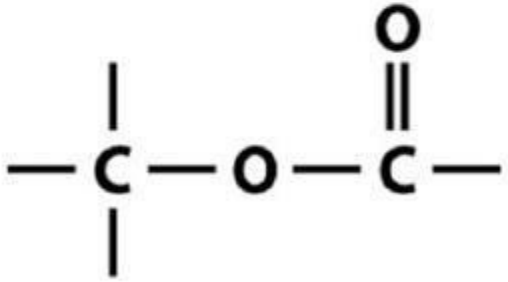
Lipid Chemistry

- Ether lipids with an unsaturated group (alkenyl) at the 1st position on the glycerol chain are called Plasmalogens

Lipid Chemistry



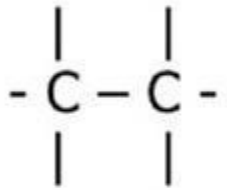
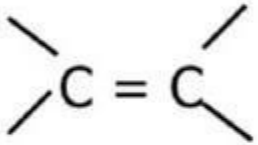
Ether Linkage.



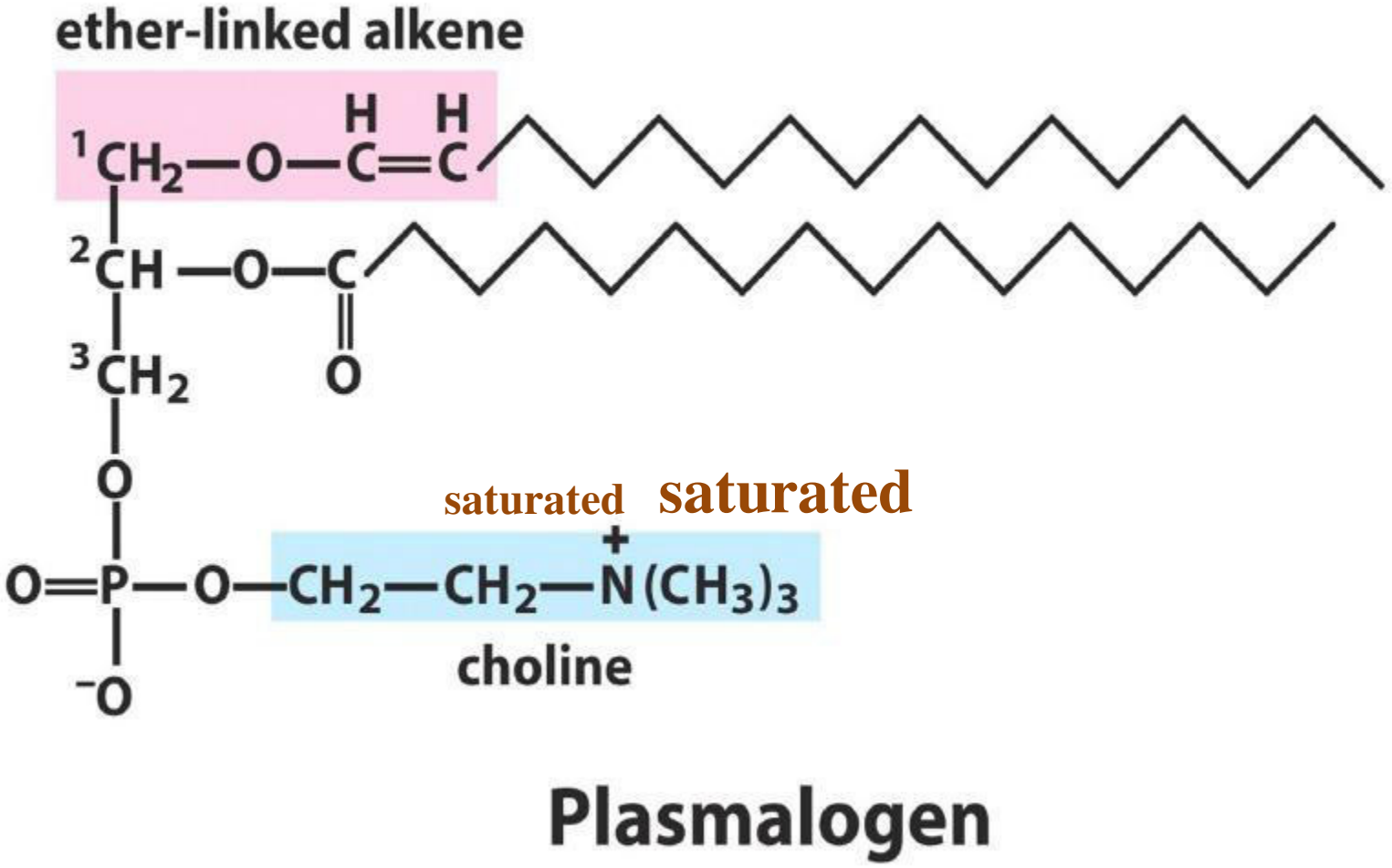
Ester

Lipid Chemistry

Alkane, Alkene and Alkyne

Functional Group			
	Alkane	Alkene	Alkyne
Example	$CH_3CH_2CH_3$	$CH_3CH{=}CH_3$	$CH_3C{\equiv}CH$
	Propane	Propene	Propyne

Lipid Chemistry



Unsaturated ether linkage

Lipid Chemistry

- In some instances serine, or inositol may be substituted for ethanolamine or choline
- These compounds constitute as much as 10% of the phospholipids of brain and muscle

Lipid Chemistry

- Vertebrate heart tissue is uniquely enriched in ether lipids

- About half of the heart phospholipids are plasmalogens

Lipid Chemistry

- The membranes of halophilic bacteria, ciliated protists, and certain invertebrates also contain high proportions of ether lipids.

"salt-loving. **Halophiles** are organisms that thrive in high salt concentrations

Lipid Chemistry

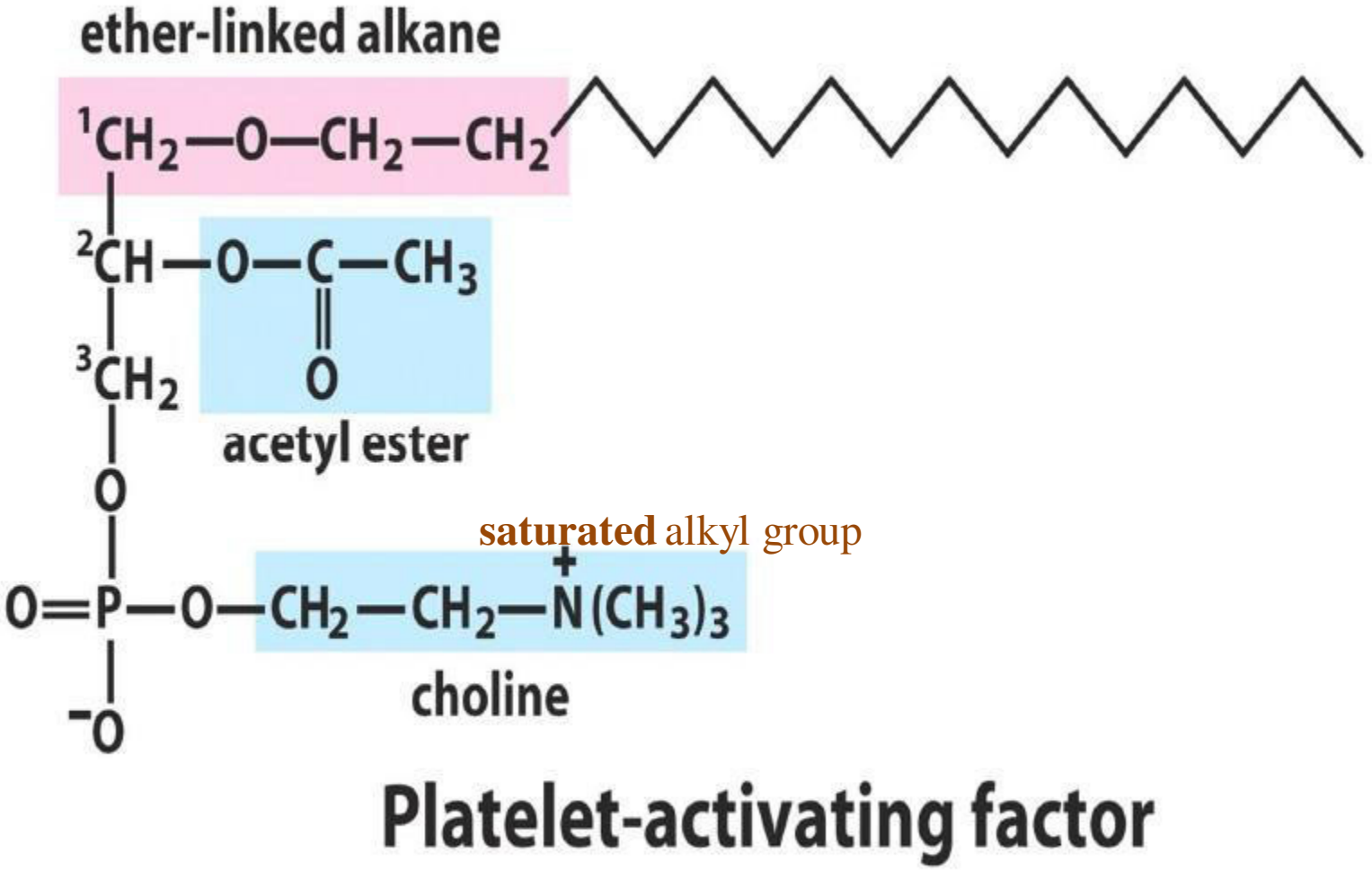
- The functional significance of ether lipids in these membranes is unknown;

- perhaps their resistance to the phospholipases that cleave ester-linked fatty acids from membrane lipids is important in some roles

Lipid Chemistry

- **Platelet-activating factor (PAF)**
- This is an unusual ether glycero phospholipid, with a **saturated** alkyl group in an ether link to carbon 1 and an acetyl residue (rather than a fatty acid) at carbon 2 of the glycerol backbone

Lipid Chemistry



Lipid Chemistry

- PAF is synthesized and released by a variety of cell types
- It binds to surface receptors, triggering potent thrombotic and acute inflammatory events

Lipid Chemistry

- It causes platelets to aggregate and degranulate (required for clotting), and neutrophils and alveolar macrophages to generate superoxide radicals (required for microbial killing)

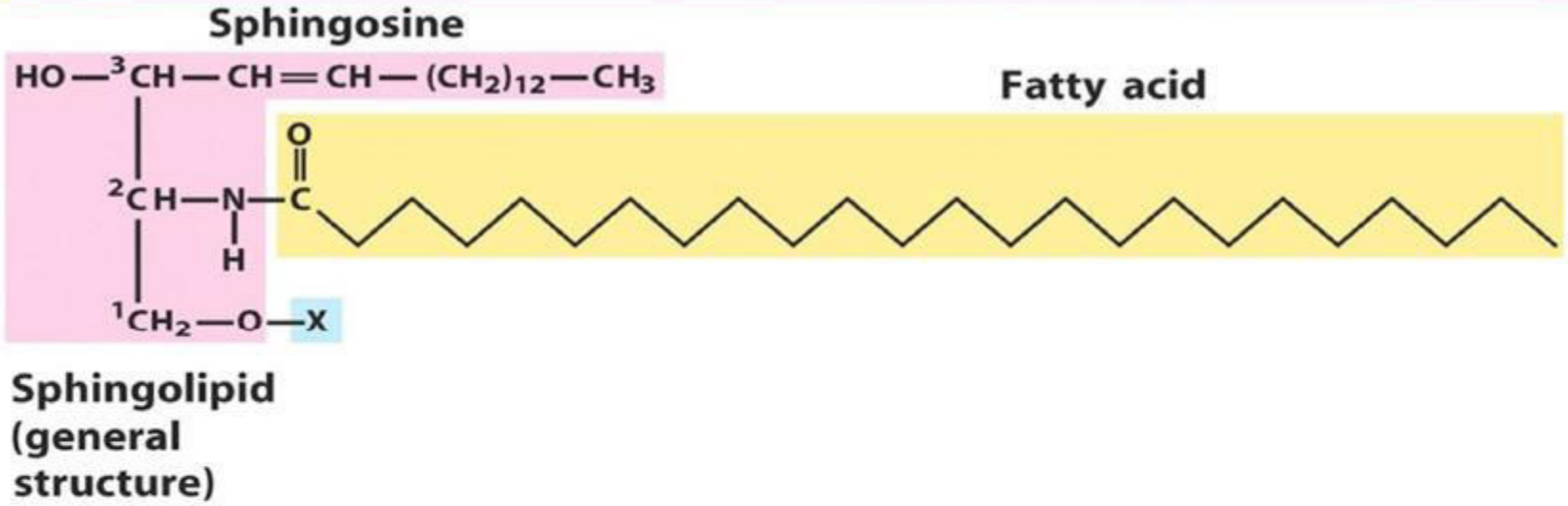
END

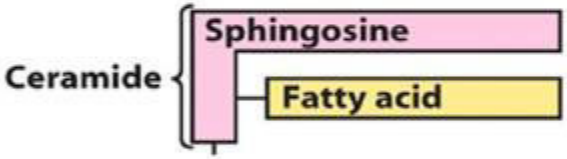
Lipid Chemistry

SPHINGOLIPIDS

- Sphingolipids, like other membrane lipids, are composed of a
- hydrophobic portion, (ceramide) and
- a polar head group

Lipid Chemistry





Lipid Chemistry

- The first three carbons at the polar end of sphingosine are analogous to the three carbons of glycerol in glycerophospholipids

- The amino group at C-2 bears a fatty acid in amide linkage

Lipid Chemistry

- The fatty acid is usually saturated or monounsaturated, with
- 16, 18, 22, or 24 carbon atoms.
- Ceramide is the parent compound for this group

Lipid Chemistry

SPHINGOLIPIDS

- There are two subclasses of sphingolipids:
  - Sphingomyelins
  - Sphingoglycolipids

Lipid Chemistry

- **Sphingomyelins**

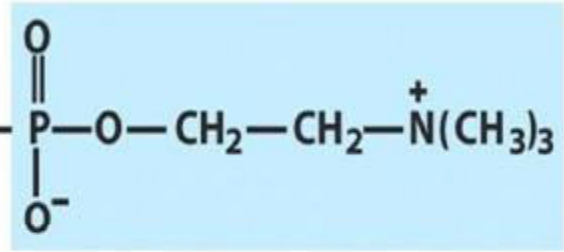
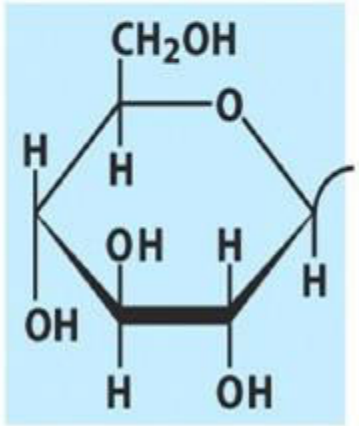
contain phosphocholine or phosphoethanolamines as their polar head group and are therefore classified along with glycerophospholipids as phospholipids

Lipid Chemistry

Sphingoglycolipids are molecules that contain both

- carbohydrate and
- lipid (in the form of ceramide) components

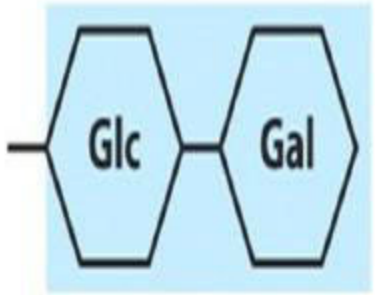
Lipid Chemistry

Name of sphingolipid	Name of X	Formula of X
Ceramide	—	— H
Sphingomyelin	Phosphocholine	
Neutral glycolipids Glucosylcerebroside	Glucose	

Lipid Chemistry

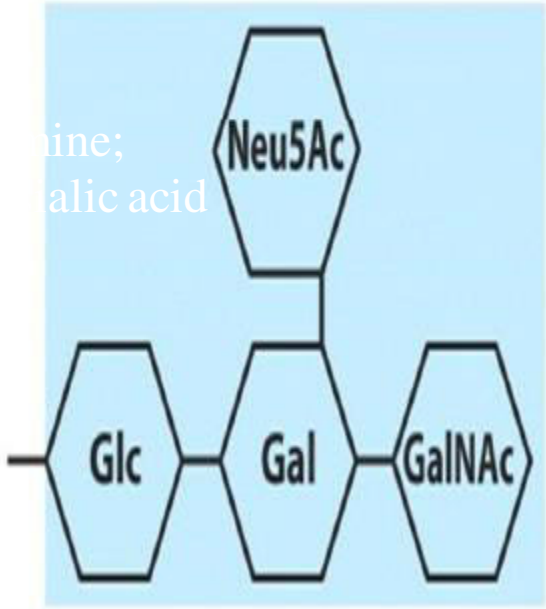
Lactosylceramide
(a globoside)

Di-, tri-, or
tetrasaccharide



Ganglioside GM2

Complex
oligosaccharide

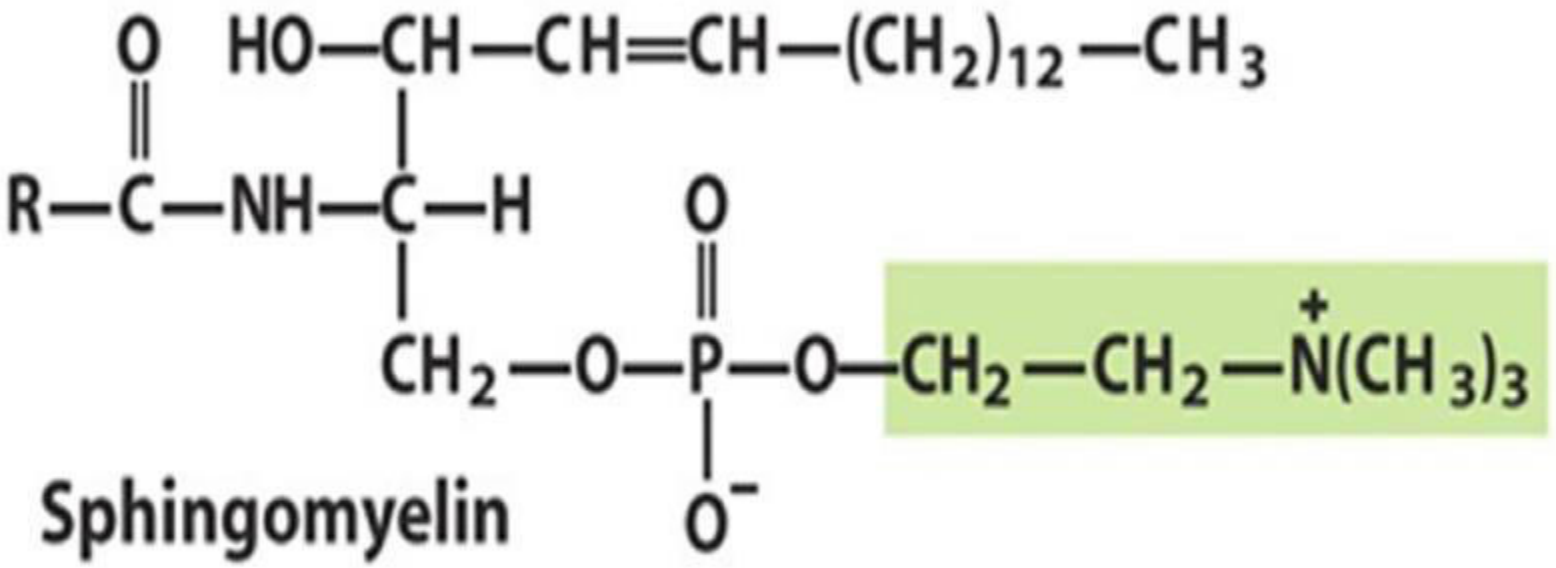


GalNAc, N-acetyl-D-galactosamine; Neu5Ac, N-acetylneuraminic acid (sialic acid

Lipid Chemistry

- When the alcohol group at carbon 1 of sphingosine is, esterified to **phosphorylcholine**, **sphingomyelin**, the only significant sphingophospholipid in humans, is produced

Lipid Chemistry



Sphingomyelin

Lipid Chemistry

Sphingomyelins are

- present in the plasma membranes of animal cells and are especially prominent in nerve tissue including myelin, -thus the name "sphingomyelins

Lipid Chemistry

- Sphingomyelin of the **myelin sheath** contains predominant longer-chain fatty acids such as lignoceric acid and nervonic acid (24 carbon)

Lipid Chemistry

- whereas **gray matter of the brain** has sphingomyelin that contains primarily stearic acid(18 carbon)

END

Lipid Chemistry

Glycosphingolipids are molecules that contain both

- carbohydrate and
- lipid components

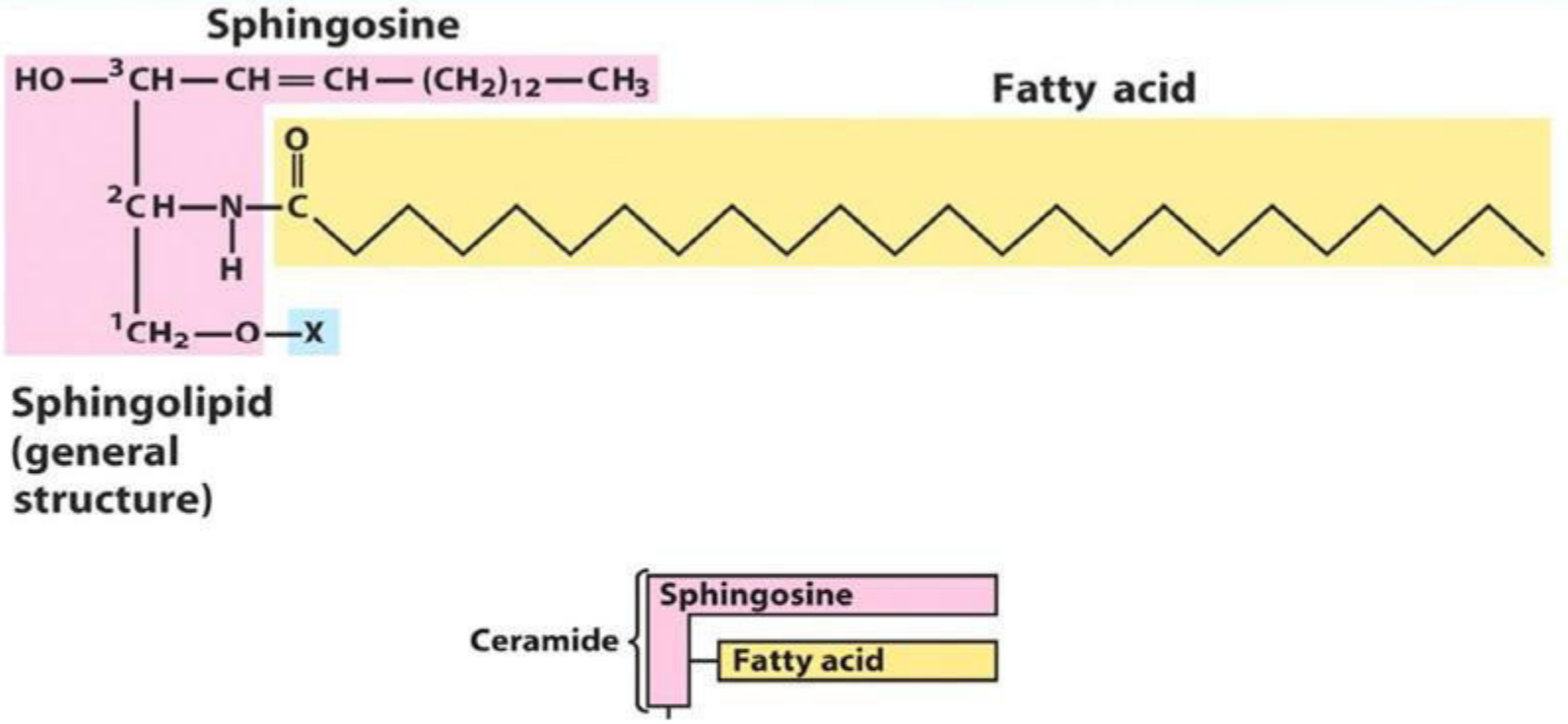
Lipid Chemistry

- Like the phospholipid sphingomyelin, they are derivatives of ceramide
- They are also an important component of membrane bilayers.

Lipid Chemistry

- They occur largely in the outer face of plasma membranes have polar head groups with
- one or more sugars connected directly to the -OH at
- C-1 of the ceramide moiety by an O-glycosidic bond

Lipid Chemistry



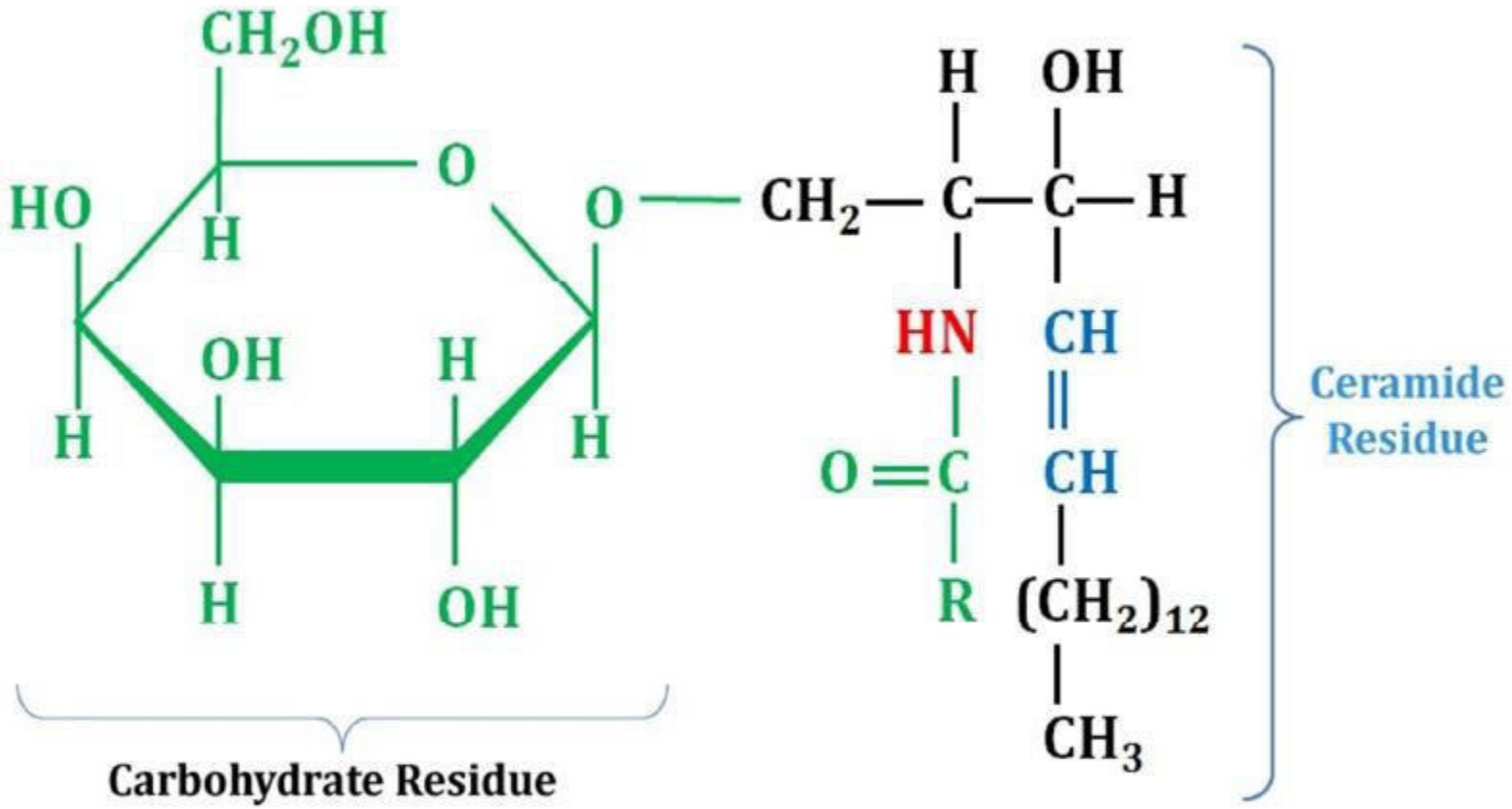
X= one or more sugar molecules.

Lipid Chemistry

Cerebrosides

- have a single sugar linked to ceramide;
- galactocerebrosides are found in the plasma membranes of **neural cells**,
- glucocerebrosides in the plasma membranes of **non neural cells**

Lipid Chemistry



Cerebroside

Lipid Chemistry

- **Galactosylceramide** is a major glycosphingolipid of brain and other nervous tissue,
- It contains a number of characteristic C24 fatty acids, eg, cerebronic acid

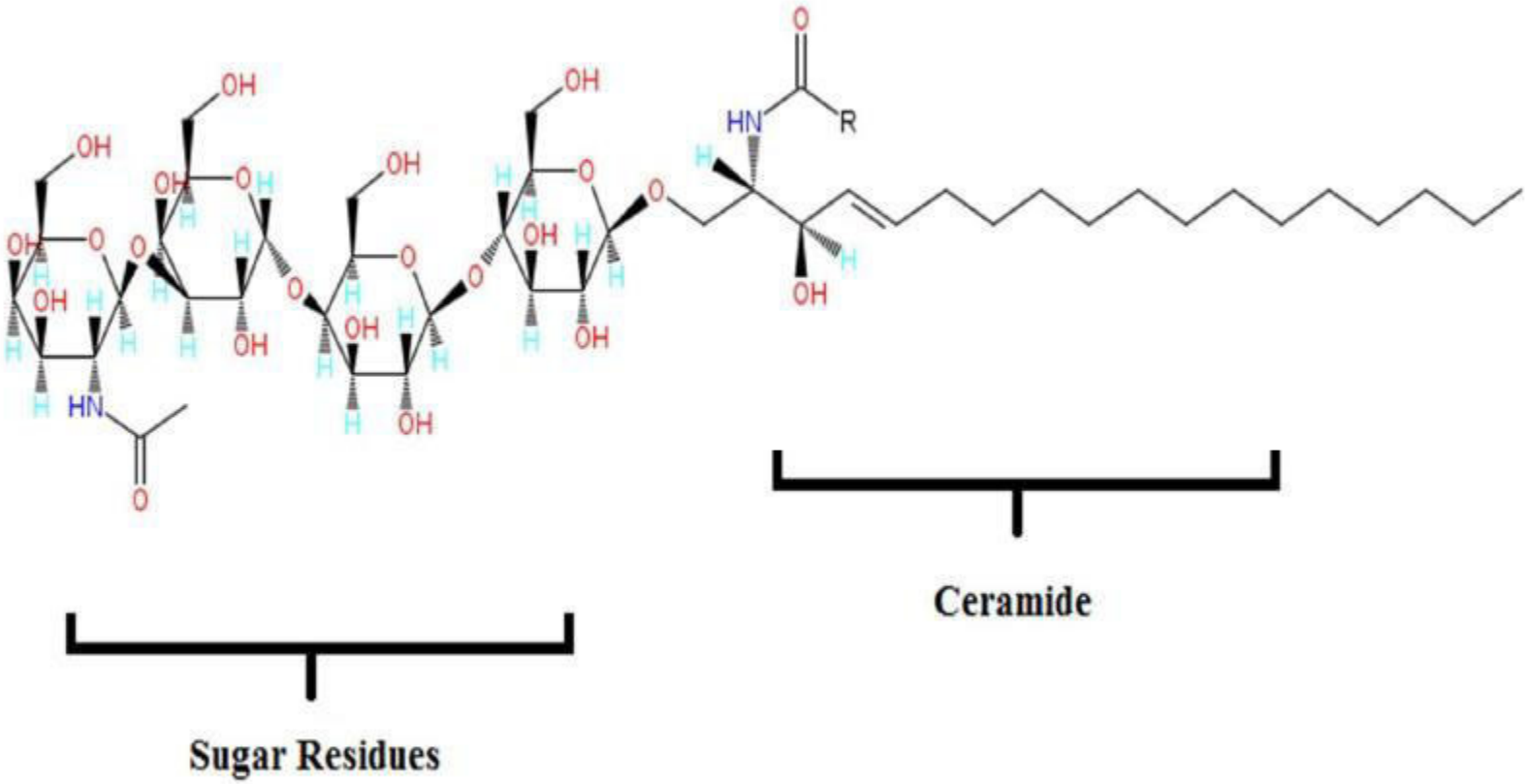
Lipid Chemistry

- **Globosides**

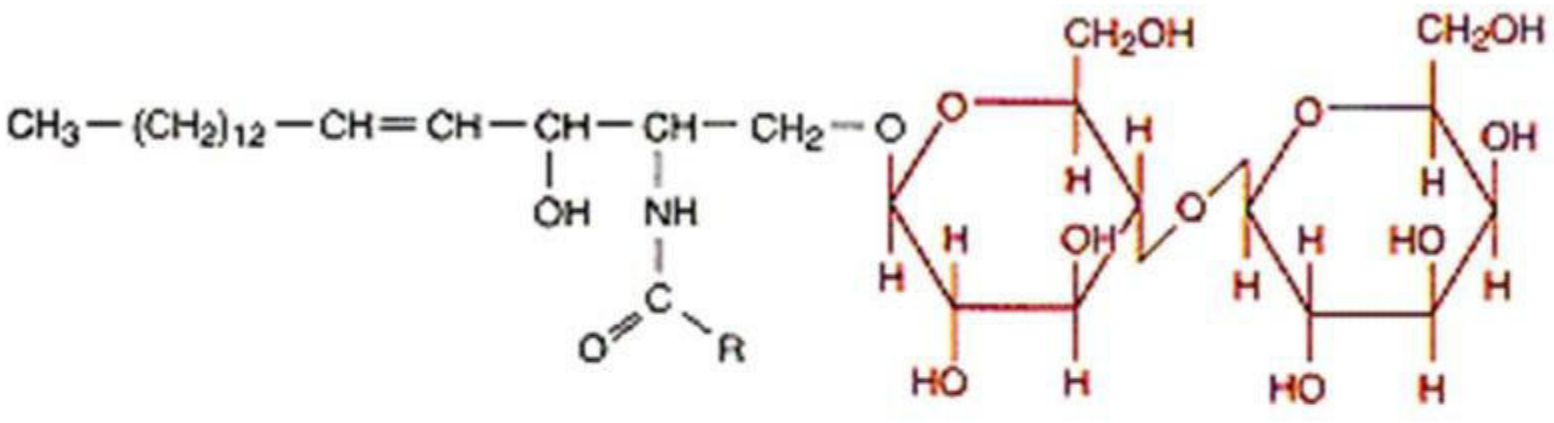
are glycosphingolipids

- with two or more sugars, usually
- D -glucose,
- D- galactose, or
- N-acetyl-D-galactosamine.

Lipid Chemistry



Lipid Chemistry



Ceramide **Sugar Residues**

Lipid Chemistry

- A glycosphingolipid that has only one sugar as the side chain is called a cerebroside
- Globosides are highly abundant in RBCs.
- Globosides are also found in human serum, spleen and liver.

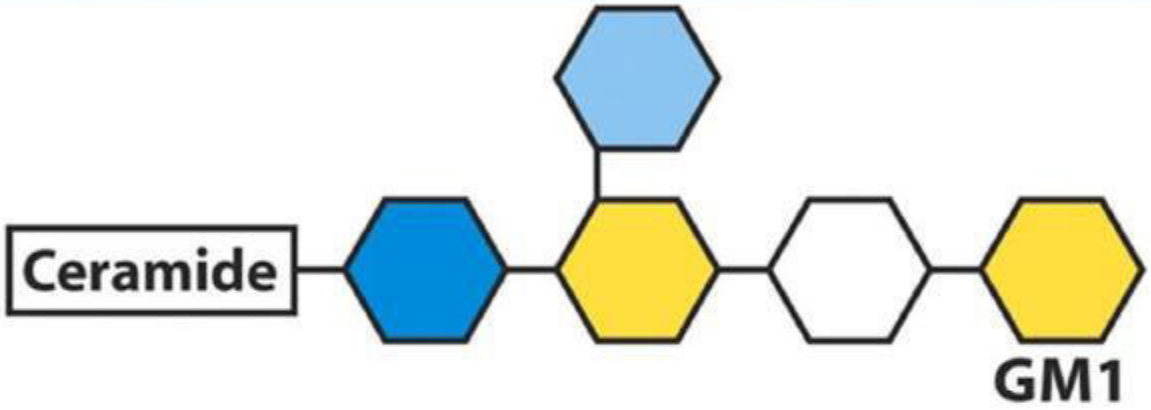
END

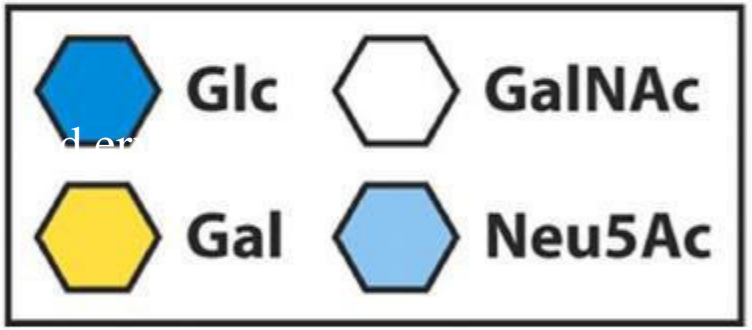
Lipid Chemistry

Glycosphingolipids (Contd.)

- **Gangliosides**
- They have oligosaccharides as their polar head groups and
- one or more residues of N-acetylneuraminic acid (a sialic acid), at the termini.

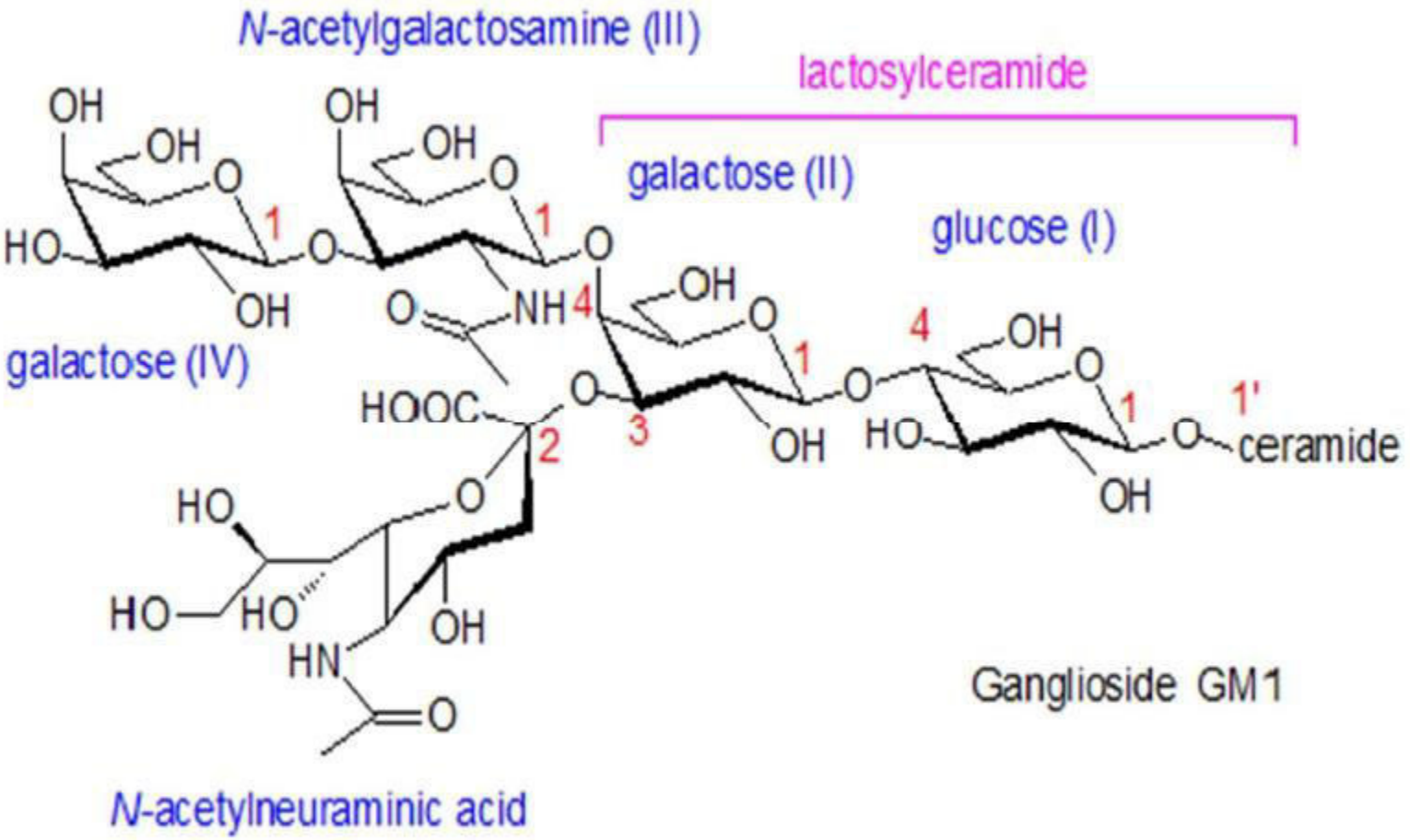
Lipid Chemistry





- D -glucose,
- D- galactose, or
- N-acetyl-D-galactosamine.
- *N*-acetylneuraminic acid

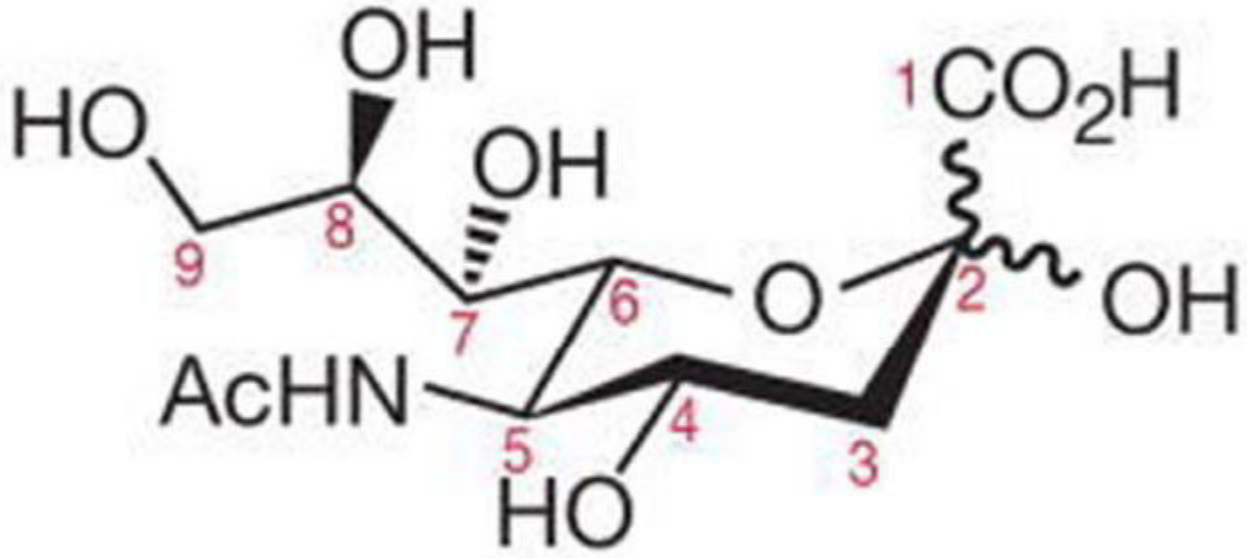
Lipid Chemistry



Lipid Chemistry

- Sialic acids are acidic sugars with a nine-carbon backbone, of which the most common is
- *N*-acetylneuraminic acid
- Sialic acid gives gangliosides the negative charge at pH 7.

Lipid Chemistry



N-Acetylneuraminic acid
(Neu5Ac)

Lipid Chemistry

- Gangliosides with one sialic acid residue are in the GM (M for mono-) series,
- those with two are in the GD (D for di-) series,
- and so on (GT three sialic acid residues; GQ, four)

Lipid Chemistry

- About 6% of **brain** lipids are gangliosides and were first isolated from the ganglion of brain cells
- They act as specific receptors for glycoprotein hormones in the cells.

Lipid Chemistry

- Some gangliosides also serve as receptors for some bacterial protein toxins e.g.
- Cholera toxin binds to the GM1 gangliosides on the surface of target cells

Lipid Chemistry

- Gangliosides also help in
- cell-cell recognition and thus have a significant role in
- growth and
- differentiation of tissues and also in
- carcinogenesis

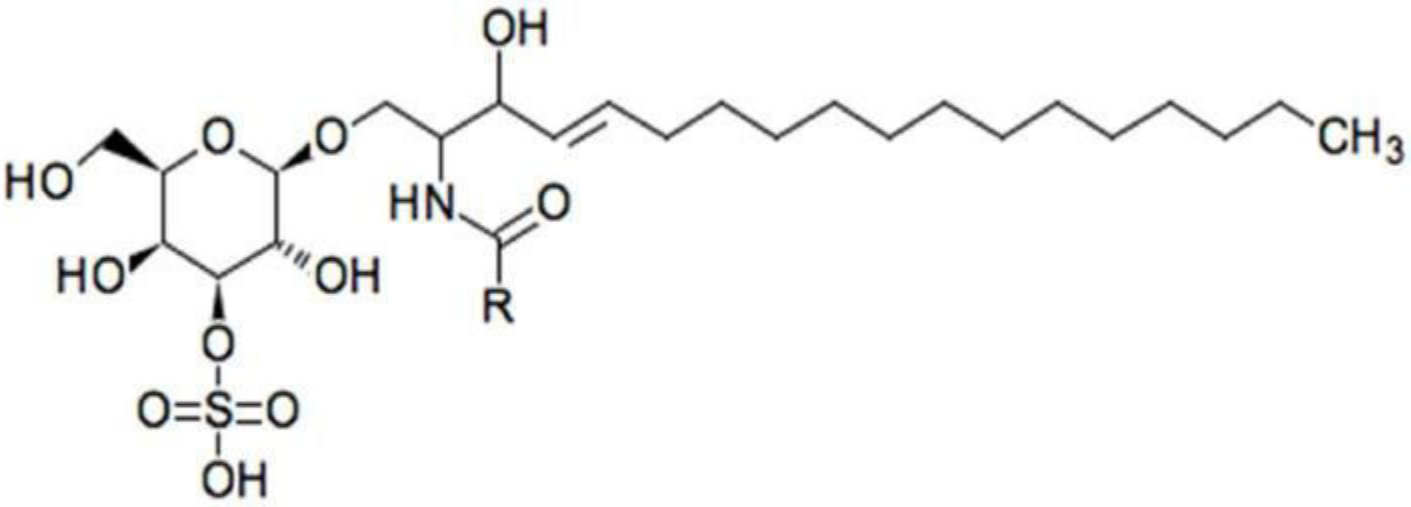
Lipid Chemistry

Sulfatides

Sulfoglycosphingolipids (sulfatides) are cerebrosides that contain sulfated galactosyl residues, and are therefore negatively charged at physiologic pH

Cerebrosides have a single sugar linked to ceramide;

Lipid Chemistry



Lipid Chemistry

- Sulfatides are found predominantly in nerve tissue and kidney

END

Lipid Chemistry

- **Glyceroglycolipids**
- These are the predominant membrane lipids of the plants, such as those of chloroplast.

- They include
  - Galactolipids and
  - Sulfolipids

Lipid Chemistry

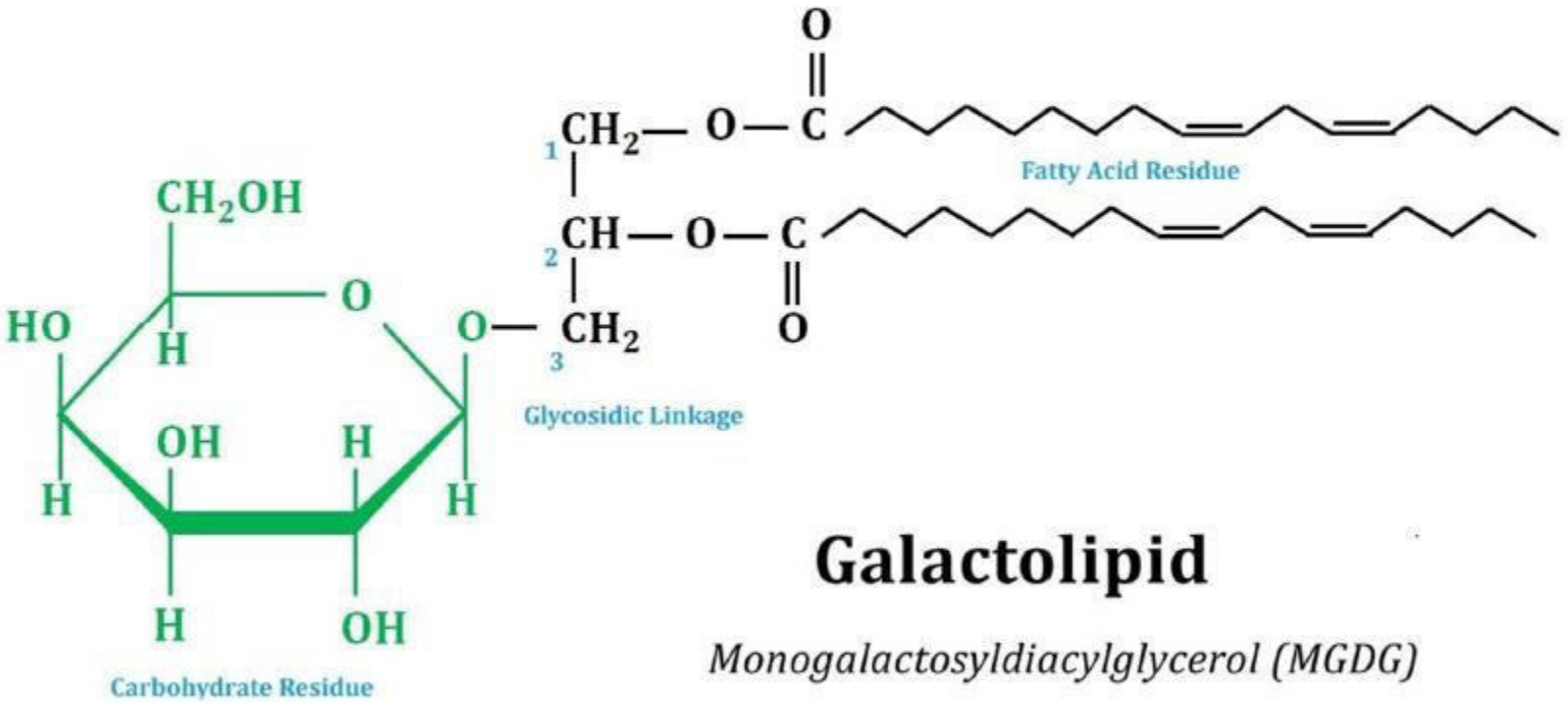
- As the name indicates these glycolipids contain glycerol instead of sphingosine as their backbone.

Lipid Chemistry

Galactolipids

- In galactolipids, the C3 of the glycerol moiety is connected to one or more galactose residues by glycosidic linkages.

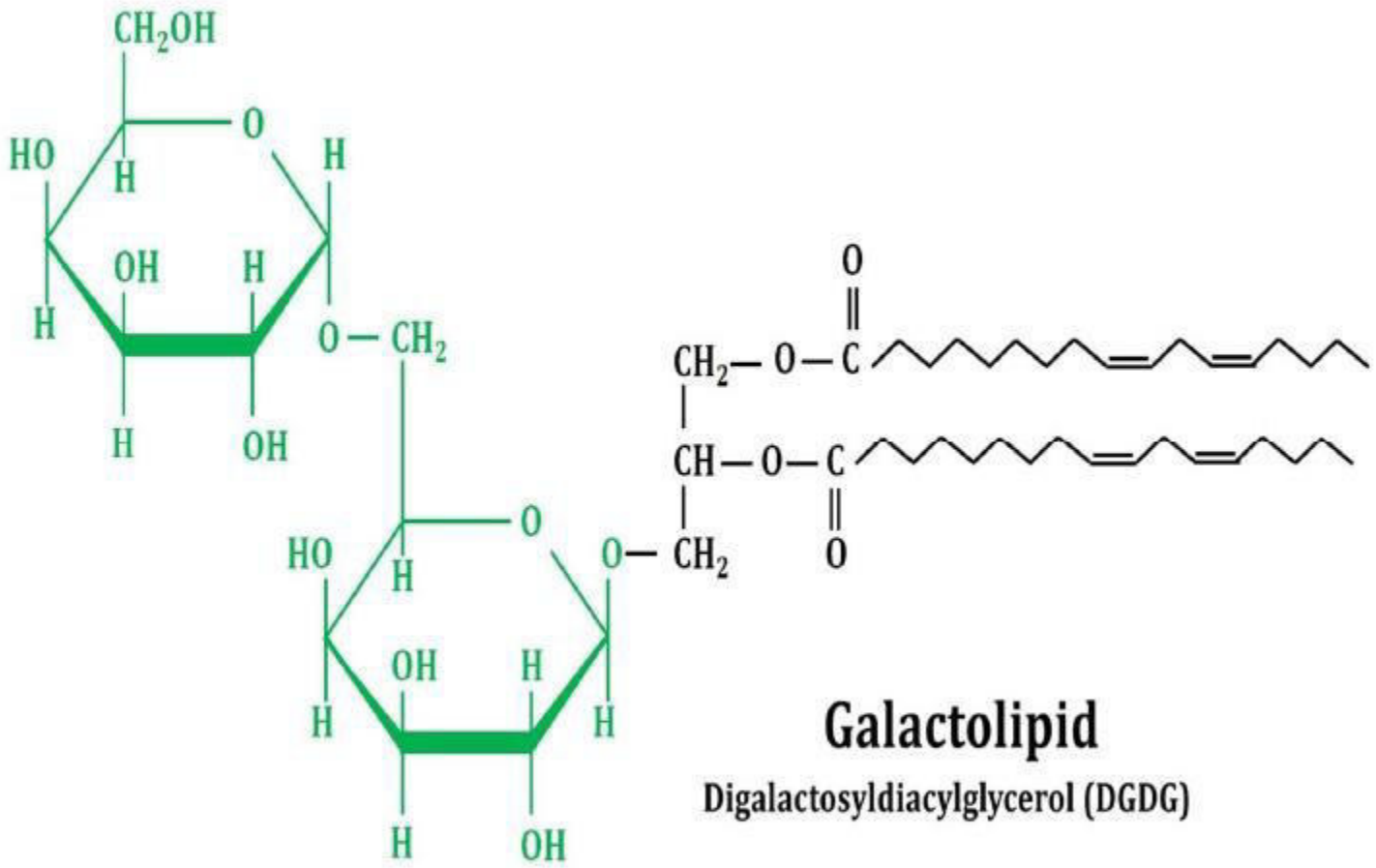
Lipid Chemistry



Galactolipid

Monogalactosyldiacylglycerol (MGDG)

Lipid Chemistry



Galactolipid

Digalactosyldiacylglycerol (DGDG)

Lipid Chemistry

- The head groups of these galactolipids are uncharged but polar.

Lipid Chemistry

- Galactolipids are localized in the internal membranes of chloroplast

- They constitute about 70% to 80 % of plant membrane lipids and
- thus are the most abundant lipids in the biosphere.

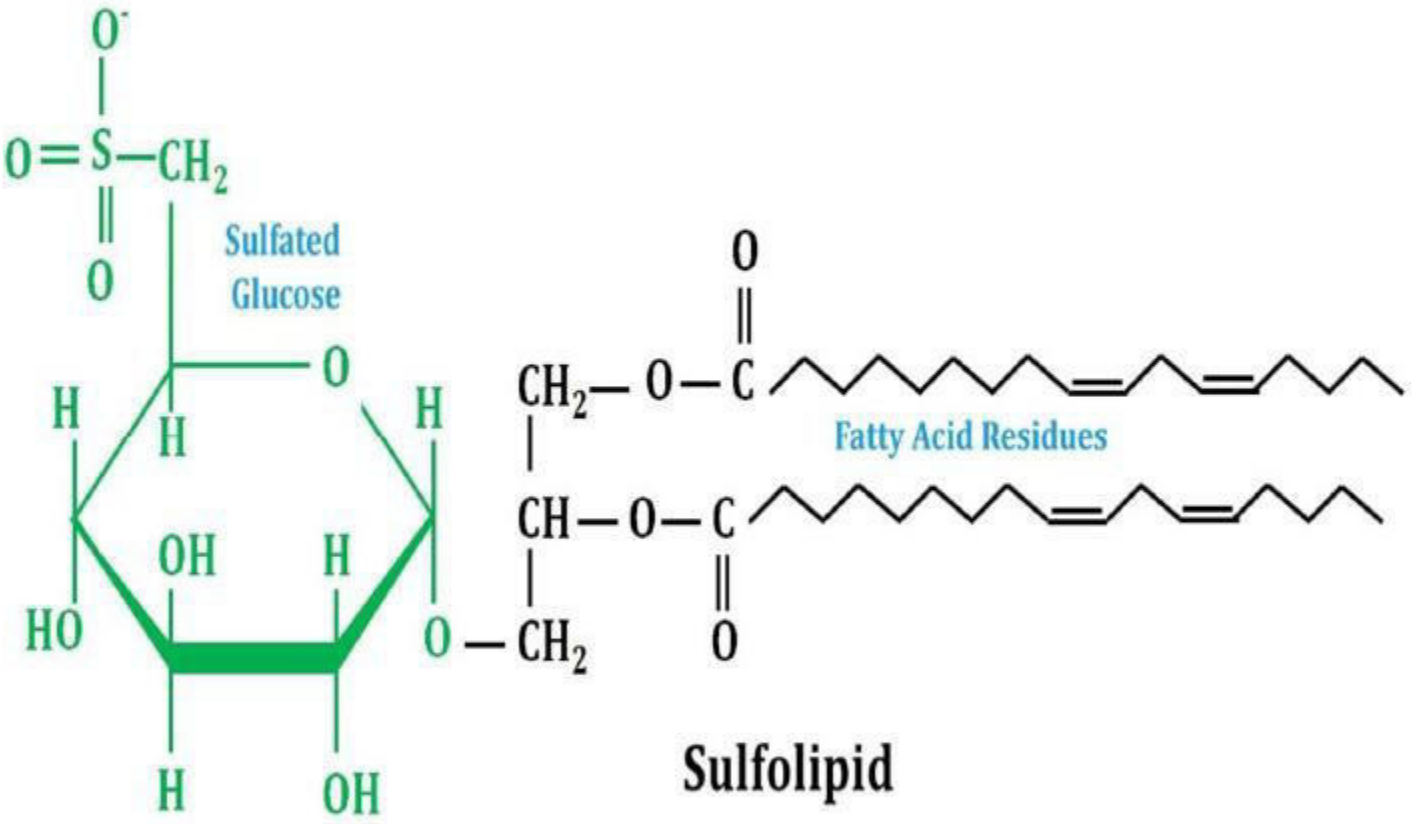
Lipid Chemistry

Sulfolipids

- They are membrane glycolipids with **sulfur** containing functional groups

- Sulfonated glucose is joined to the **C3** of diacylglycerol in
- glycosidic linkage

Lipid Chemistry



Sulfolipid

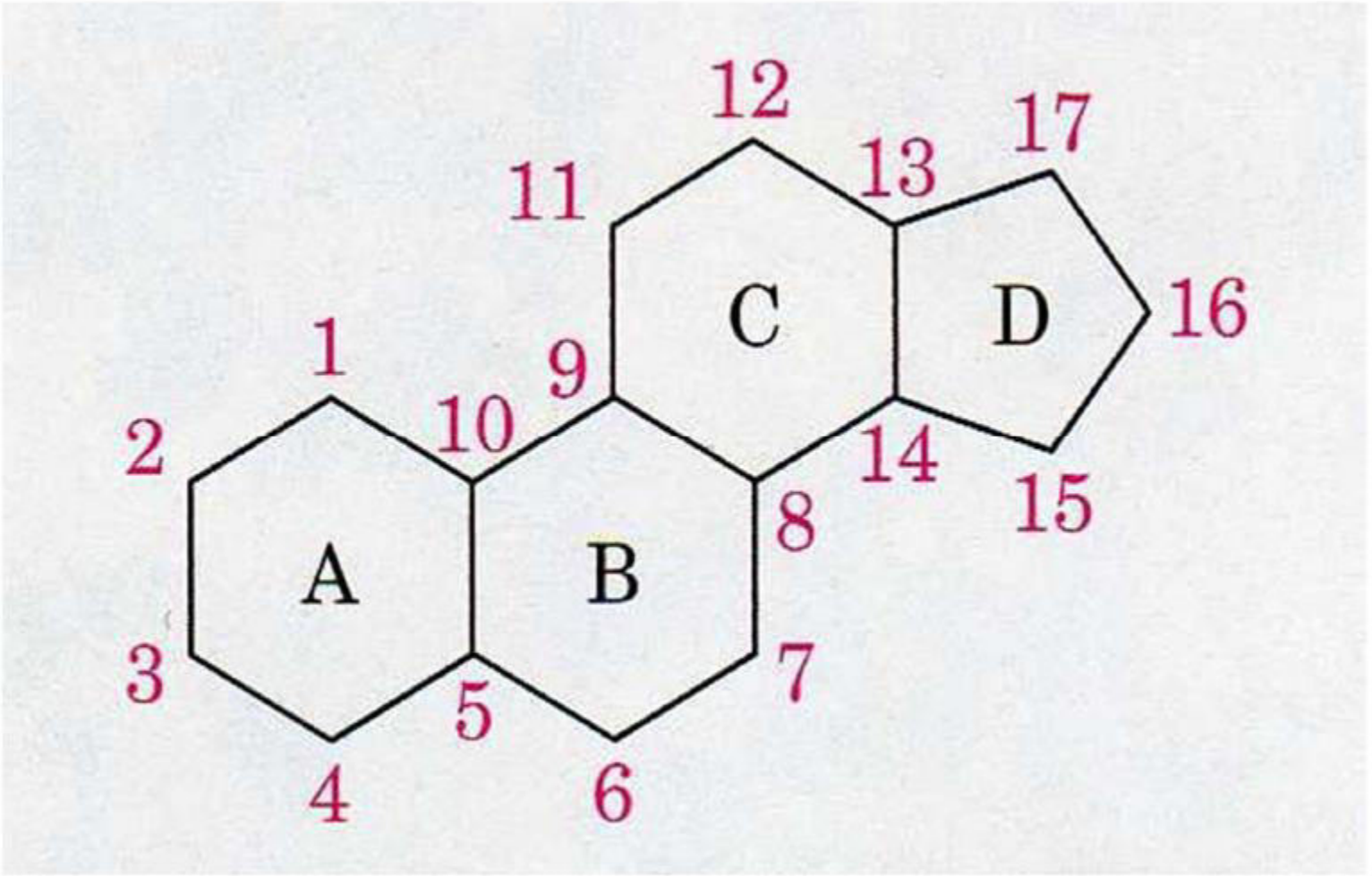
Lipid Chemistry

- **Steroids and Cholesterol**
- A **steroid** is a lipid whose structure is based on the tetracyclic (four-ring) structure

consists of

- 3 cyclohexane rings.
- 1 cyclopentane ring.

Lipid Chemistry



steroid nucleus or the **cyclopentanophenanthrene** ring

Lipid Chemistry

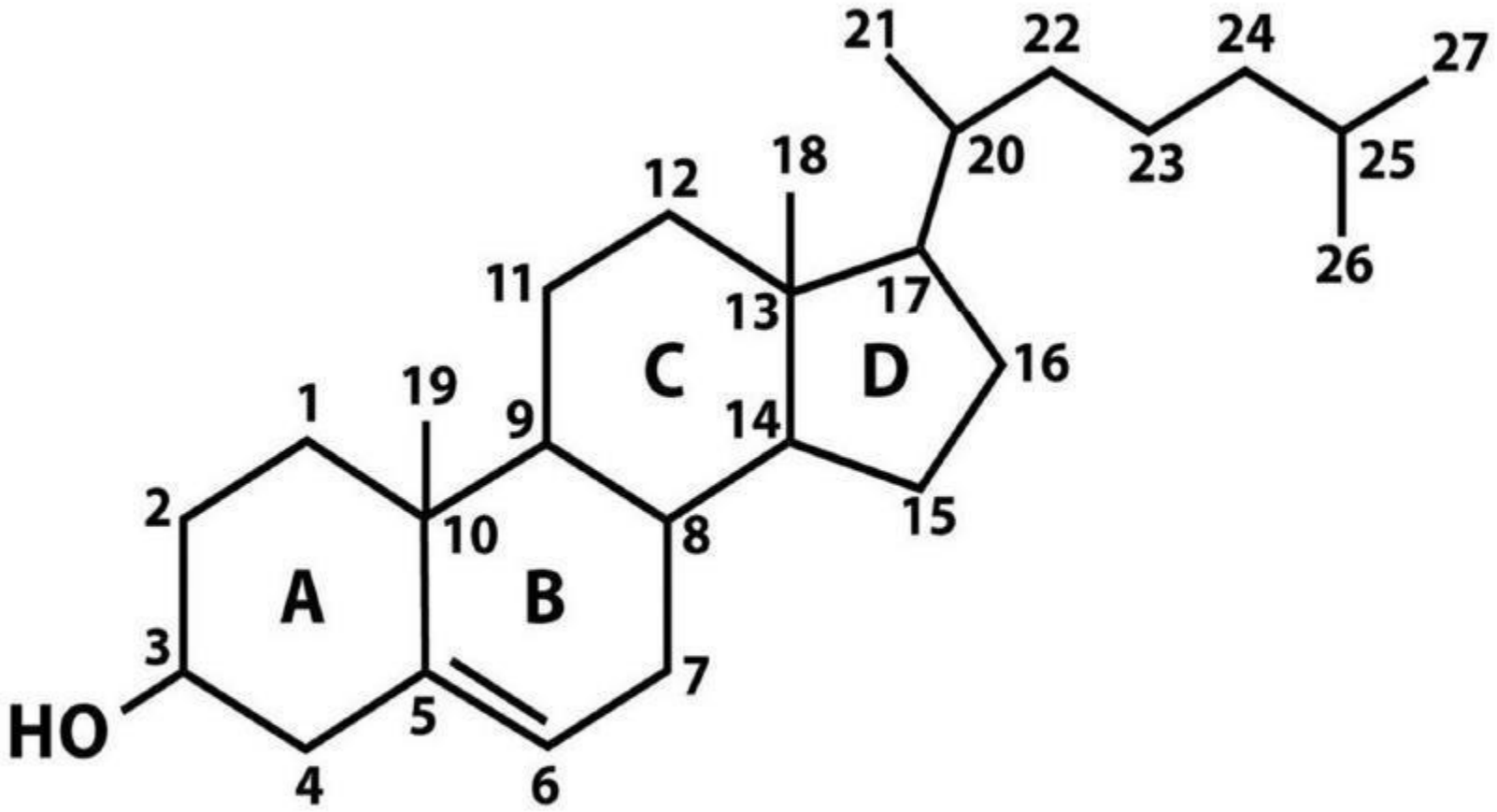
- Steroids with eight to ten carbon atoms in the side chain at C-17 and a hydroxyl group at C-3 are classified as **sterols**

- Cholesterol is the major sterol in animal tissues

Lipid Chemistry

- Cholesterol has an eight carbon branched hydrocarbon chain attached to **C-17** of the D ring
- Ring A has a hydroxyl group at C-3, and ring B has a double bond between C-5 and C-6

Lipid Chemistry



Cholesterol

hydrocarbon chain at **C-17** of the D ring.RingA has a hydroxyl group at C-3, ring B double bond between C-5 and C-6

Lipid Chemistry

- Cholesterol is an amphipathic lipid

Lipid Chemistry

- Cholesterol is a structural component of all cell membranes, modulating their fluidity
- Cholesterol is a precursor of
  - bile acids
  - steroid hormones
  - vitamin D

Lipid Chemistry

- All tissues containing nucleated cells are capable of cholesterol synthesis, which occurs in the
- endoplasmic reticulum and the
- cytosol

Lipid Chemistry

- **Cholesterol** is present in tissues and in plasma either as free cholesterol or combined with a long-chain fatty acid as cholesteryl ester
- In plasma, both forms are transported in lipoproteins

Lipid Chemistry

- in esterified form, with a FA attached at C-3,
- the structure becomes more hydrophobic than free cholesterol

Lipid Chemistry

- Cholesteryl esters are not found in membranes, and are normally present only in low levels in most cells

Lipid Chemistry

- The ring structure of cholesterol cannot be metabolized to CO_2 and H_2O in humans
- Therefore humans cannot utilize cholesterol for energy.

Lipid Chemistry

- Cholesterol is excreted from the body via the bile either in the

- unesterified form or

- after conversion into bile acids in the liver

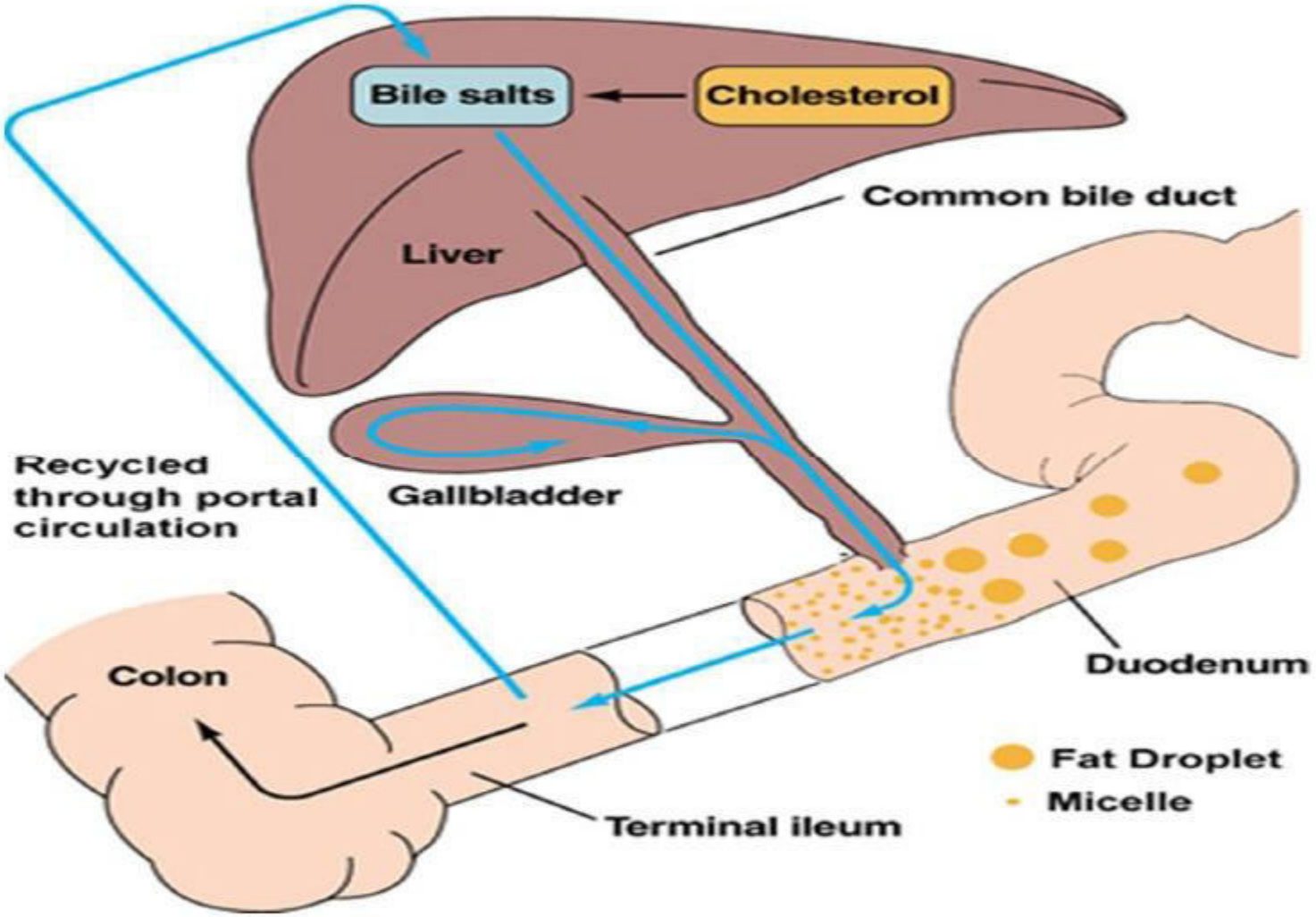
END

Lipid Chemistry

BILE ACIDS AND BILE SALTS

- Bile is a fluid that is made and released by the liver and stored in the gallbladder
- Bile helps with lipid digestion

Lipid Chemistry



Lipid Chemistry

- Bile can either pass directly from the liver into the duodenum, or
- be stored in the gallbladder when not immediately needed for digestion

Lipid Chemistry

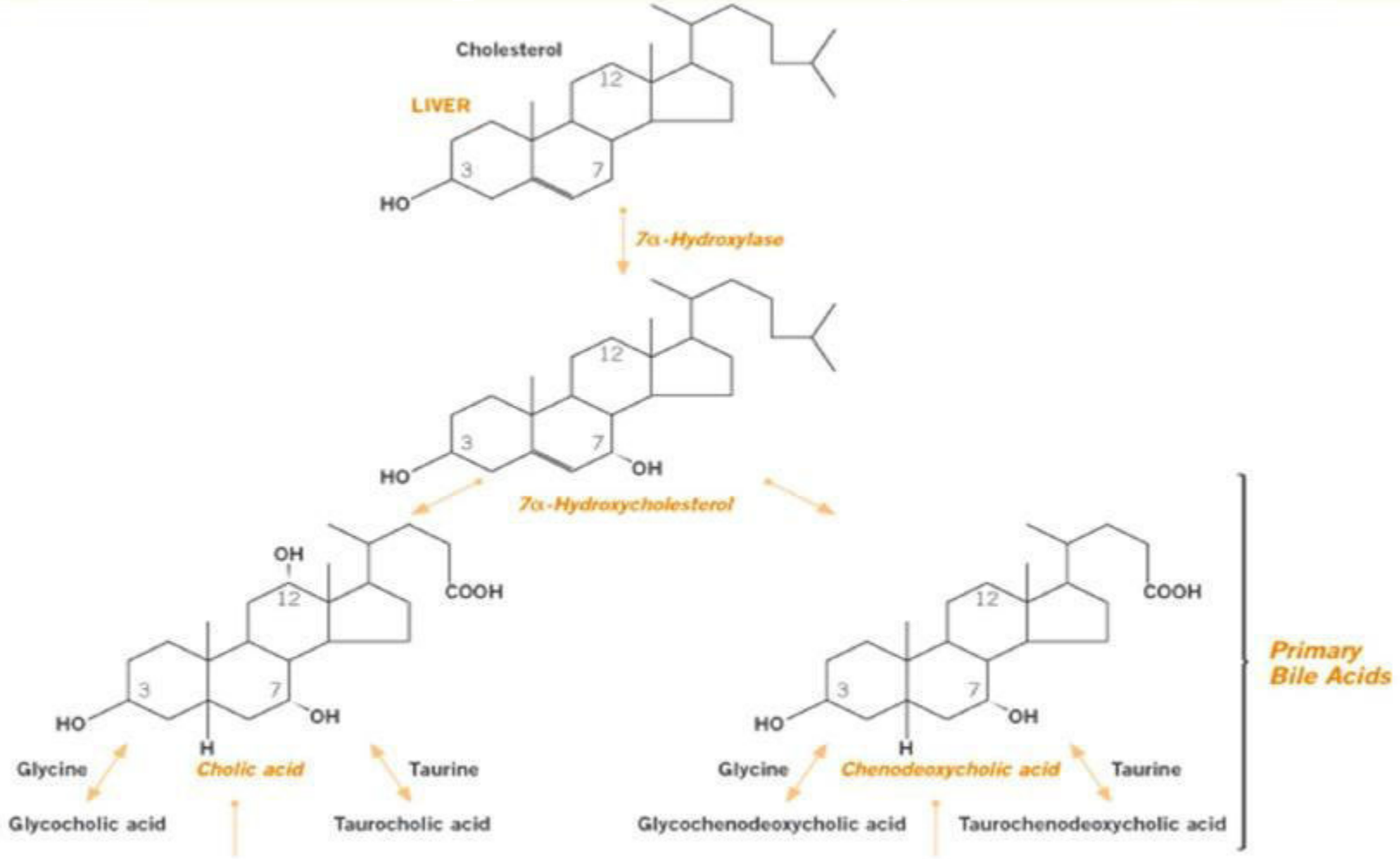
- Bile consists of a watery mixture of organic and inorganic compounds
- Phosphatidylcholine (lecithin) and
- bile salts (conjugated bile acids) are quantitatively the most important organic components of bile

Lipid Chemistry

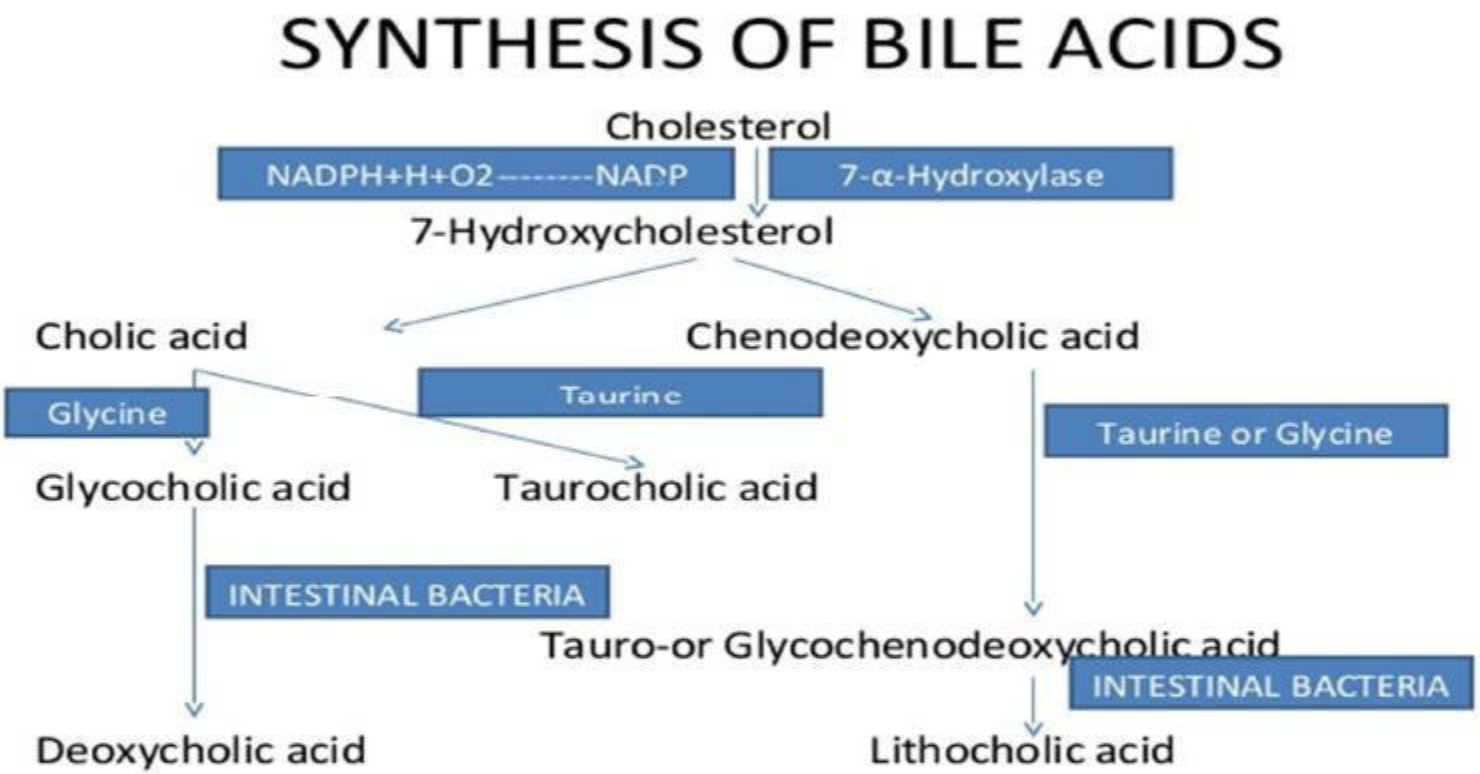
- The primary bile acids are synthesized in the liver from cholesterol

- These are cholic acid and
- chenodeoxycholic acid

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

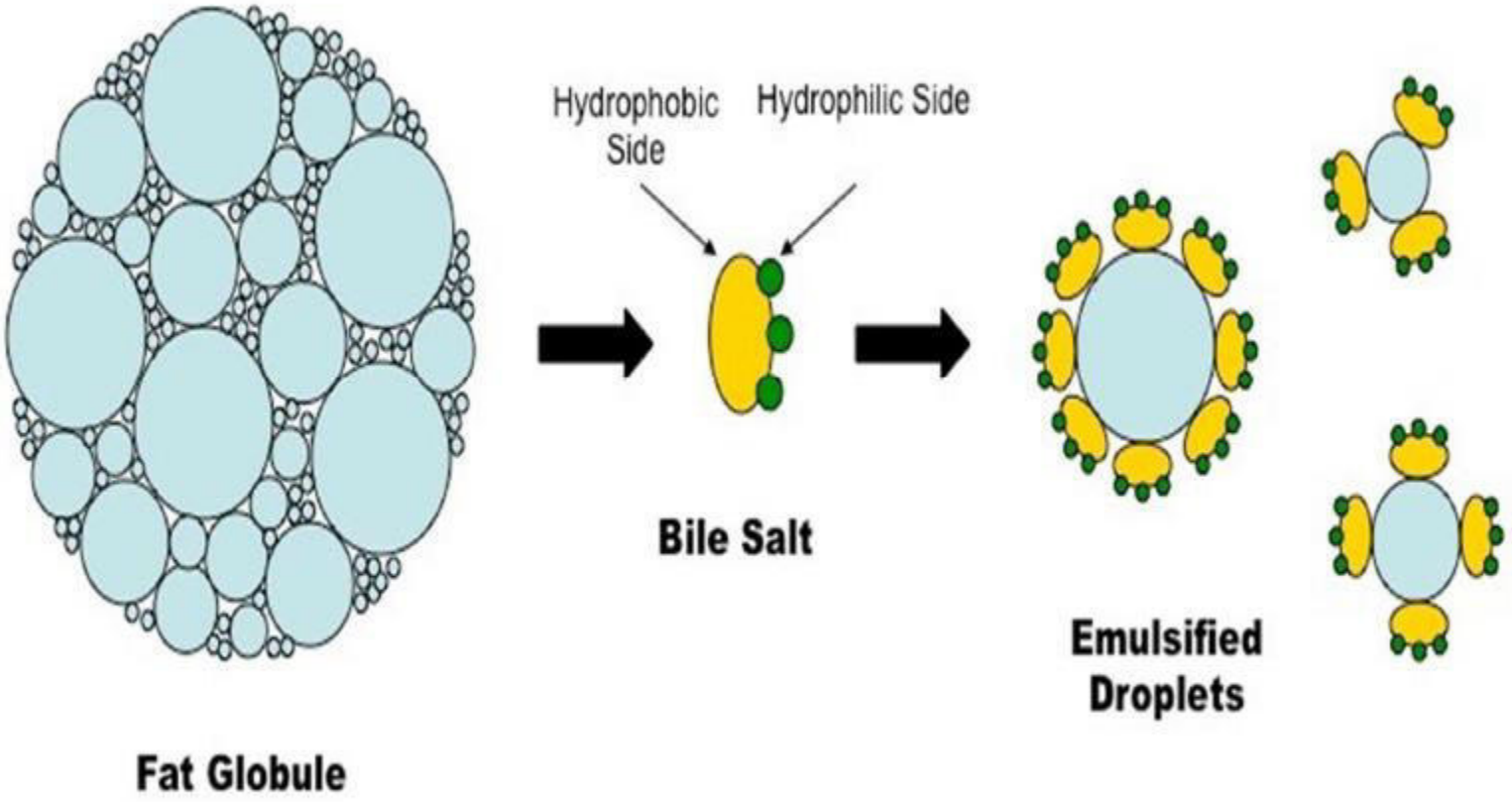
- The bile acids contain 24 carbons, with two or three hydroxyl groups and a side chain that terminates in a carboxyl group

The carboxyl group has a pKa of about 6 and, is not fully ionized at physiologic pH-hence the term "bile acid"

Lipid Chemistry

- The bile acids are amphipathic molecules
- They therefore can act as emulsifying agents in the intestine

Lipid Chemistry



Lipid Chemistry

- They emulsify
- dietary triacylglycerol and other complex lipids
- for degradation by pancreatic digestive enzymes.

END

Lipid Chemistry

EICOSANOIDS

- Eicosanoids are a large group of lipid messengers with potent effects on every tissue in the body

In biochemistry, ecosanoids are signaling moleules made up of oxidation of 20 C FA

Lipid Chemistry

- Eicosanoids are derived from metabolism of 20-carbon, polyunsaturated fatty acids (**eicosanoic acids**)

Lipid Chemistry

- **Eicosanoids** include (but not limited to)
  - ❑**Prostanoids** consisting of
    - Prostaglandins
    - Prostacyclins
    - Thromboxanes
  - ❑**Leukotrienes**
  - ❑**Lipoxins**
  - ❑ **Epoxides**

Lipid Chemistry

- These extremely potent compounds acting through their specific receptors
- elicit a wide range of physiologic and pathologic responses.

Lipid Chemistry

- particularly important in eliciting inflammatory response that occurs after infection or injury and
- produce symptoms such as pain, swelling, and fever.
- they also control bleeding through forming blood clots

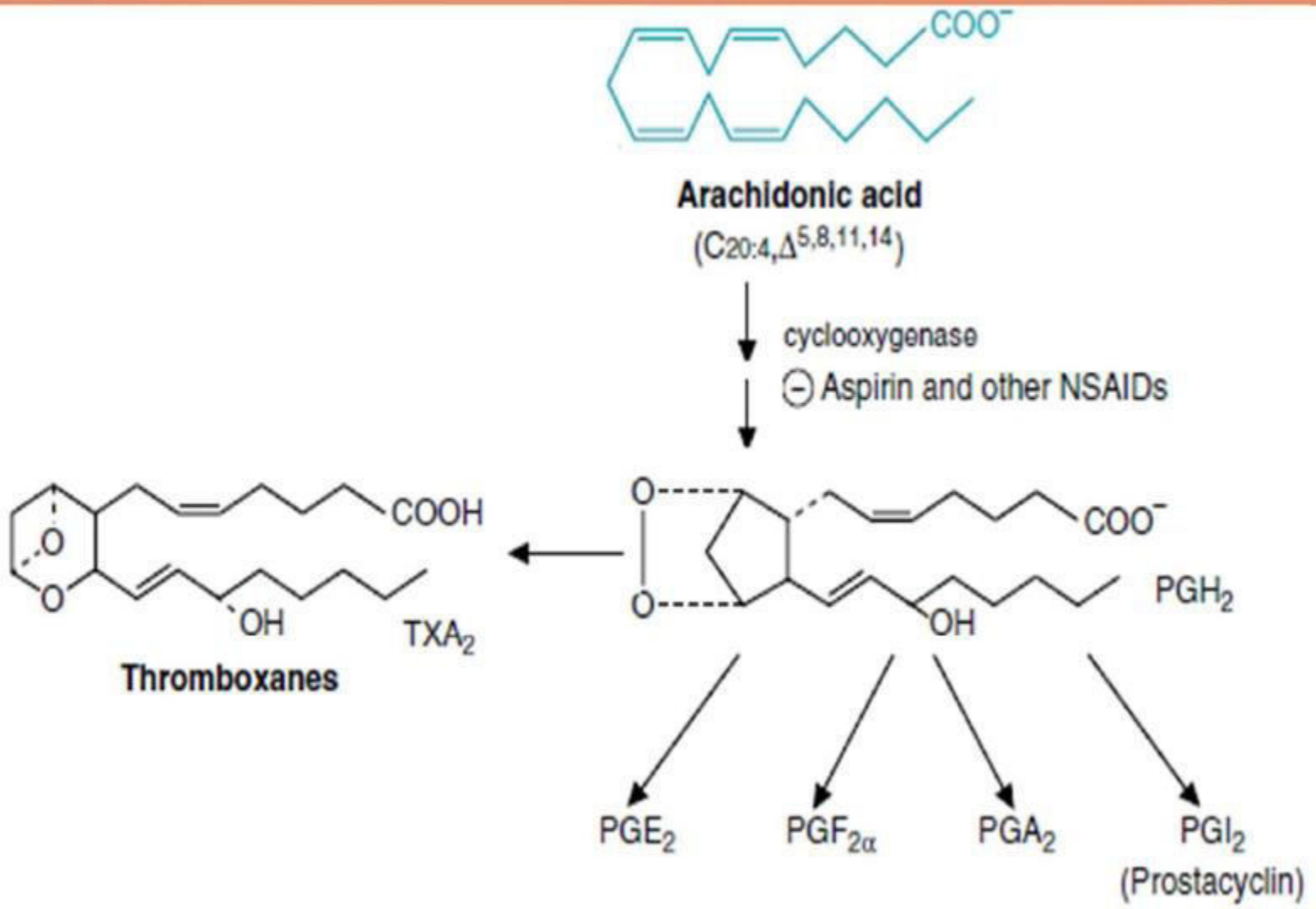
Lipid Chemistry

- Eicosanoids are derived from either omega-3 (ω-3) or omega-6 (ω-6) fatty acids.
- Arachidonic acid is the most common precursor of the eicosanoids.

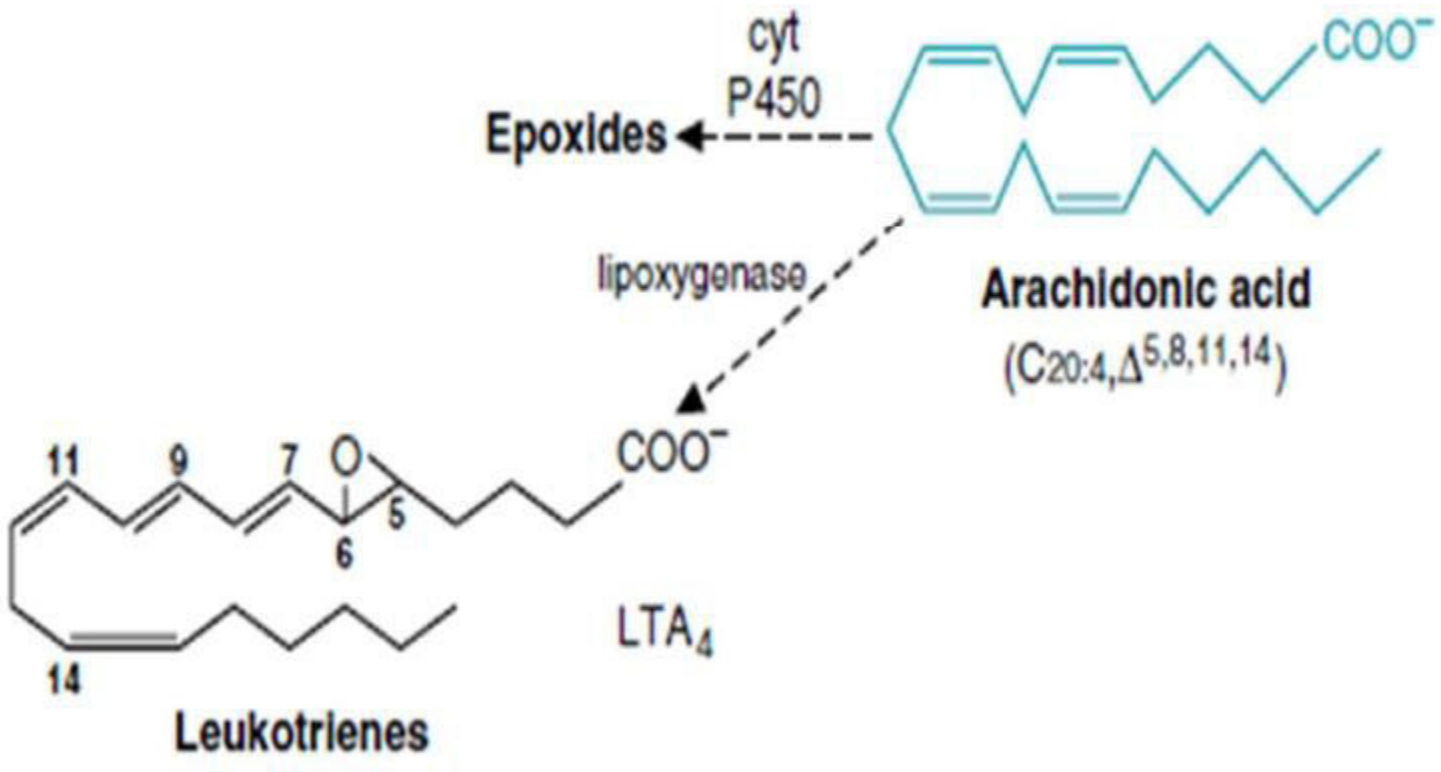
Lipid Chemistry

- Three major pathways for the metabolism of arachidonic have been discovered so far.

Lipid Chemistry



Lipid Chemistry

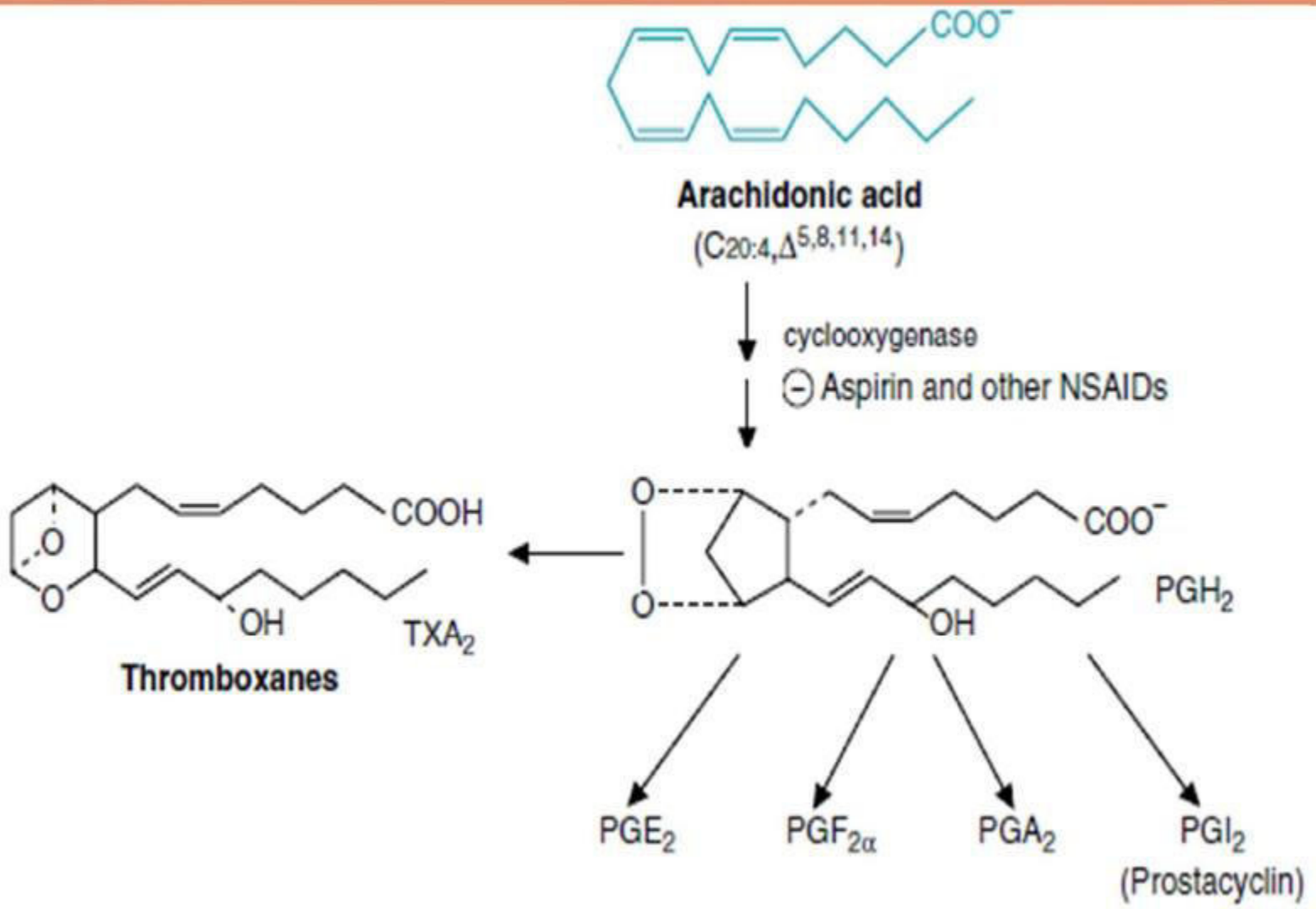


Lipid Chemistry

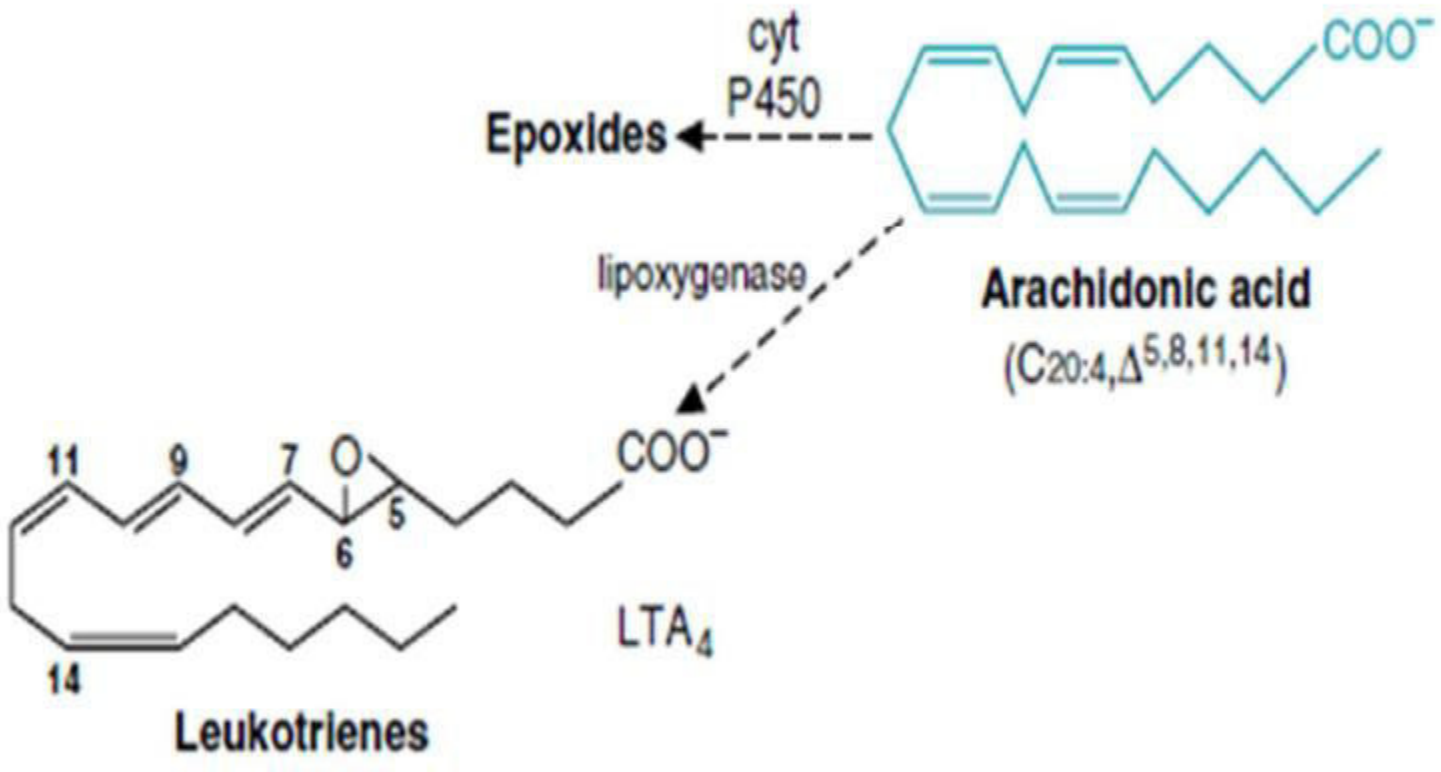
- Cyclooxygenase pathway; prostaglandins and thromboxanes.
- lipoxygenase pathway; leukotrienes, HETEs, and lipoxins.
- cytochrome P450 pathway epoxides and HETEs

HETEs: Hydroxy-eicosa-tetra-enoic acid

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- There are different series of eicosanoids
- depending on the precursor 20 C FA. derived from the essential fatty acids **linoleic acid** and **linolenic**, OR
- directly from dietary **arachidonic acid** and **eicosapentaenoic acid**

Lipid Chemistry

- Depending on the precursor different numbers of double bonds are present in these eicosanoids reflecting the parent 20C FA.

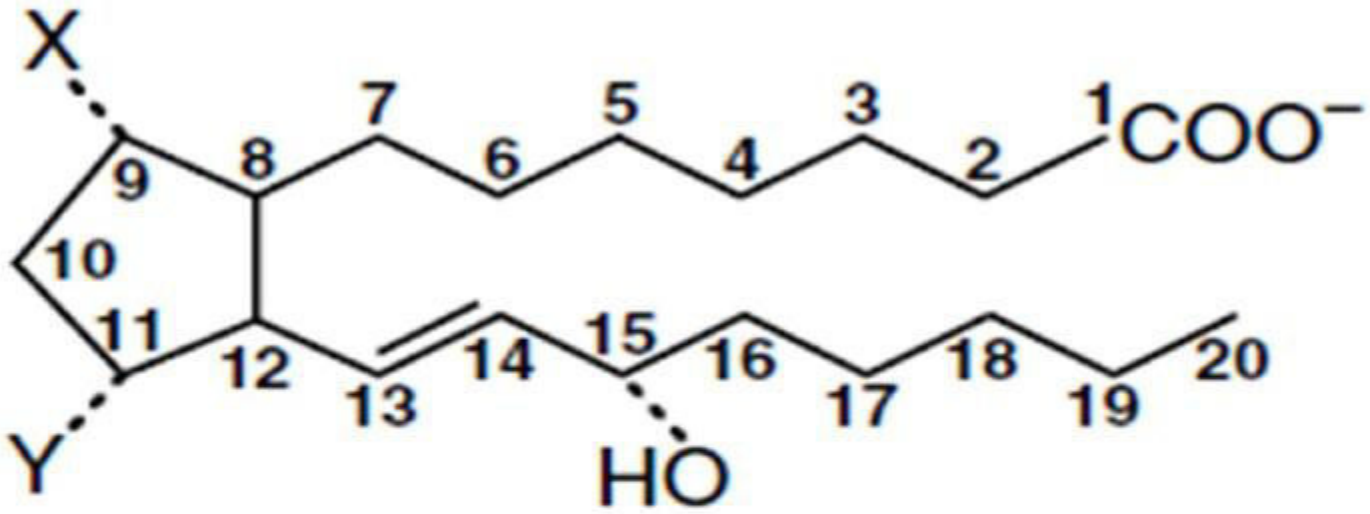
END

Lipid Chemistry

CYCLOOXYGENASE PATHWAY

- **Prostaglandins**
- Prostaglandins are fatty acids containing 20 carbon atoms, including
- an internal 5-carbon ring.

Lipid Chemistry



Prostaglandins have
a hydroxyl group at C 15,
a double bond between C 13 and C 14, and
various substituents on 5 membered ring at C9
and C11

Double bonds also may be present between carbons 5 and 6 and between carbons 17 and 18 in case of series 2 and 3

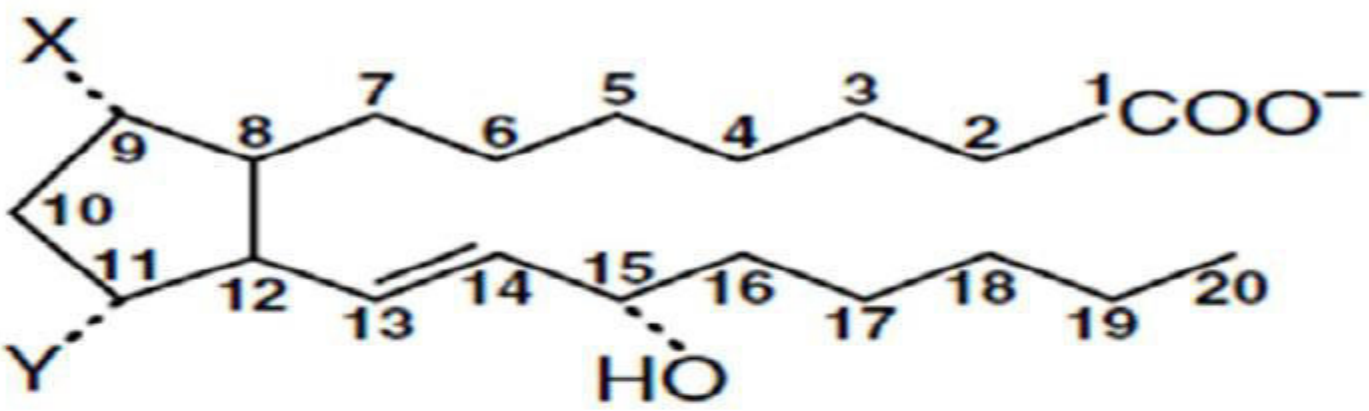
Lipid Chemistry

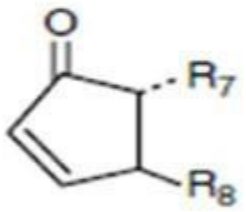
- **Nomenclature** of prostaglandins (PGs) involves the
- assignment of a capital letter (PGE),
- a numeral subscript (PGE_1), and
- for the PGF family, a Greek letter subscript ($PGF_{2\alpha}$).

Lipid Chemistry

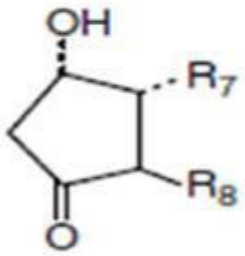
- The capital letter refers to the ring substituents at positions X and Y.

Lipid Chemistry

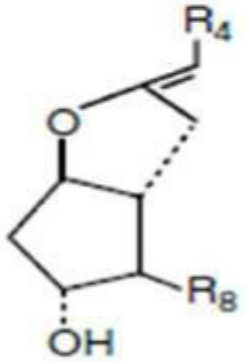




PGA



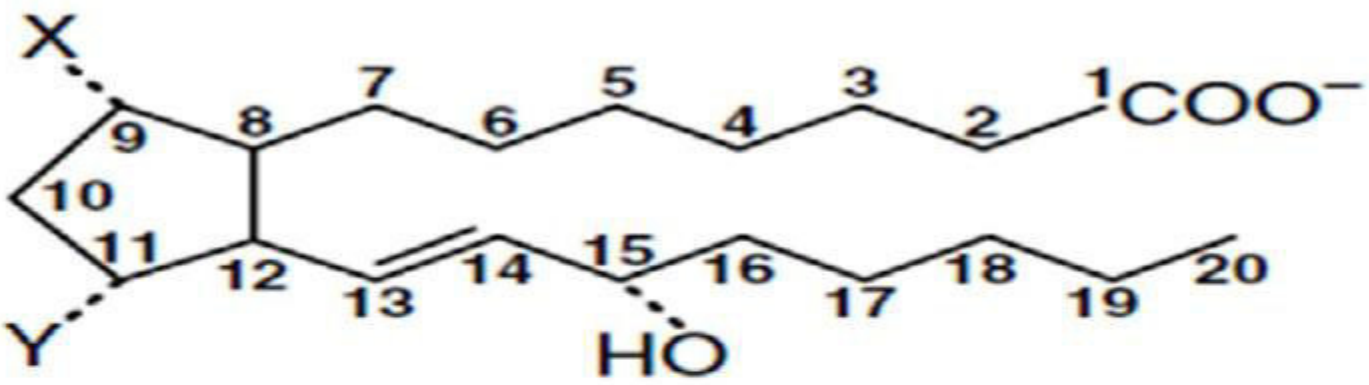
PGD

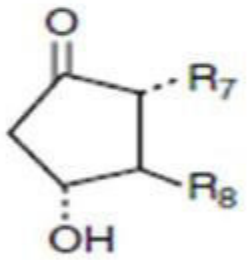


PGI

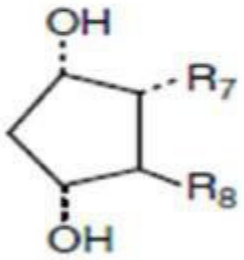
R4 contains four carbons. R7 and R8 contain seven and eight carbons, respectively. prostacyclins (PGI) contain two rings.

Lipid Chemistry





PGE



PGF$_\alpha$

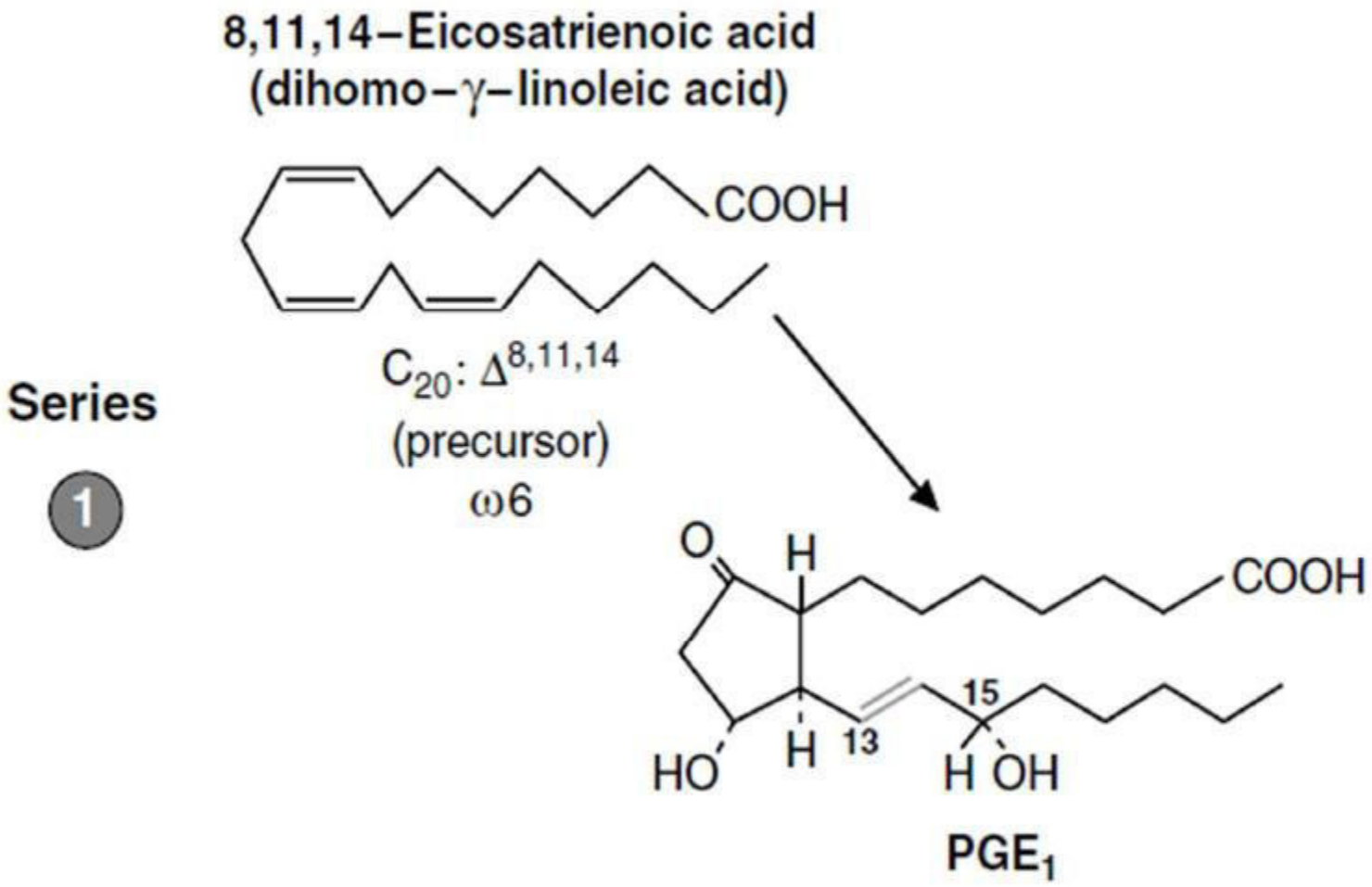
PGG or PGH

R7 and R8 contain seven and eight carbons, respectively.

Lipid Chemistry

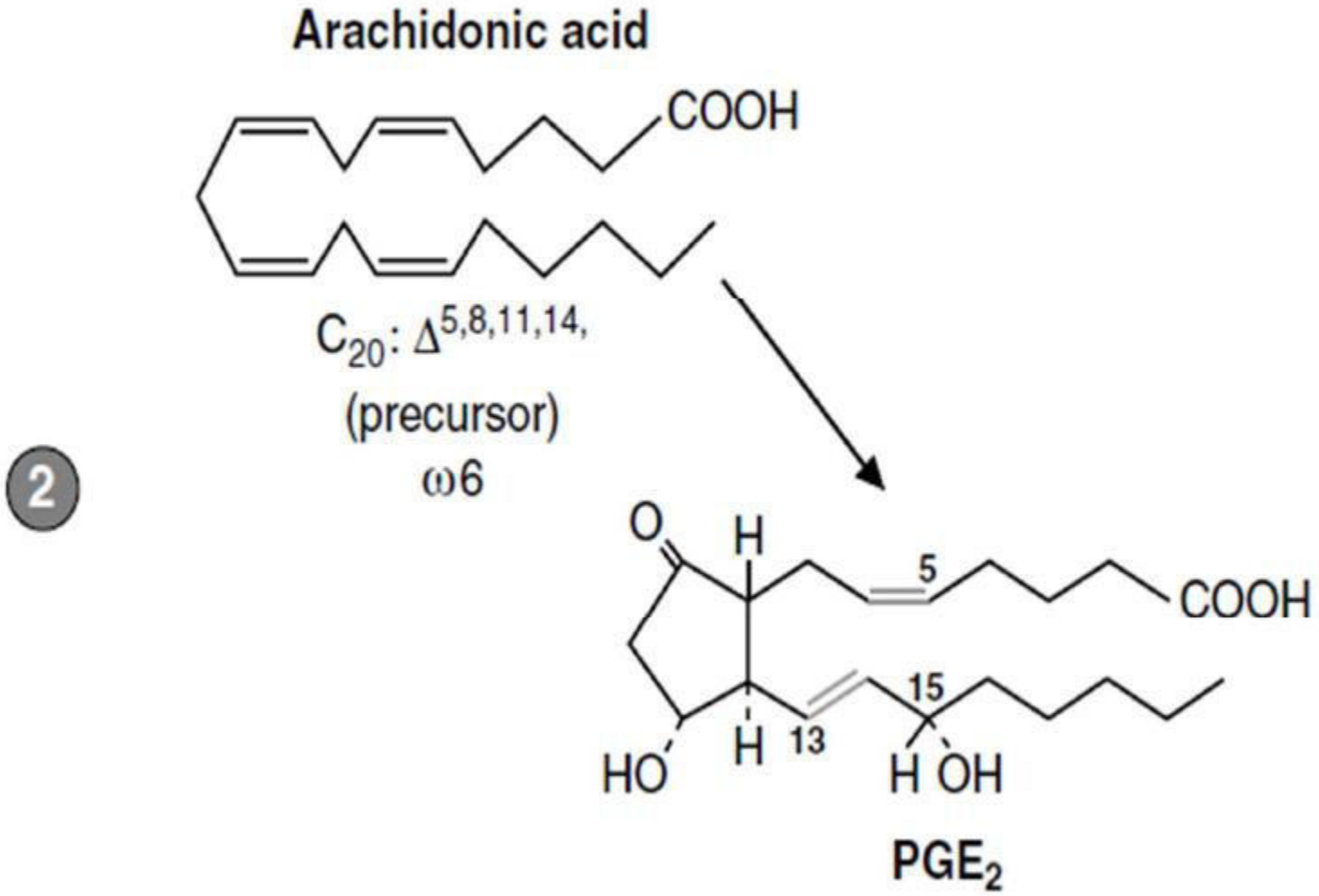
- The numeral subscript that follows the capital letter(e.g. PGE_1) refers to the PG series 1, 2, or 3, determined by the number of unsaturated bonds present in the linear portion of the hydrocarbon chain.

Lipid Chemistry



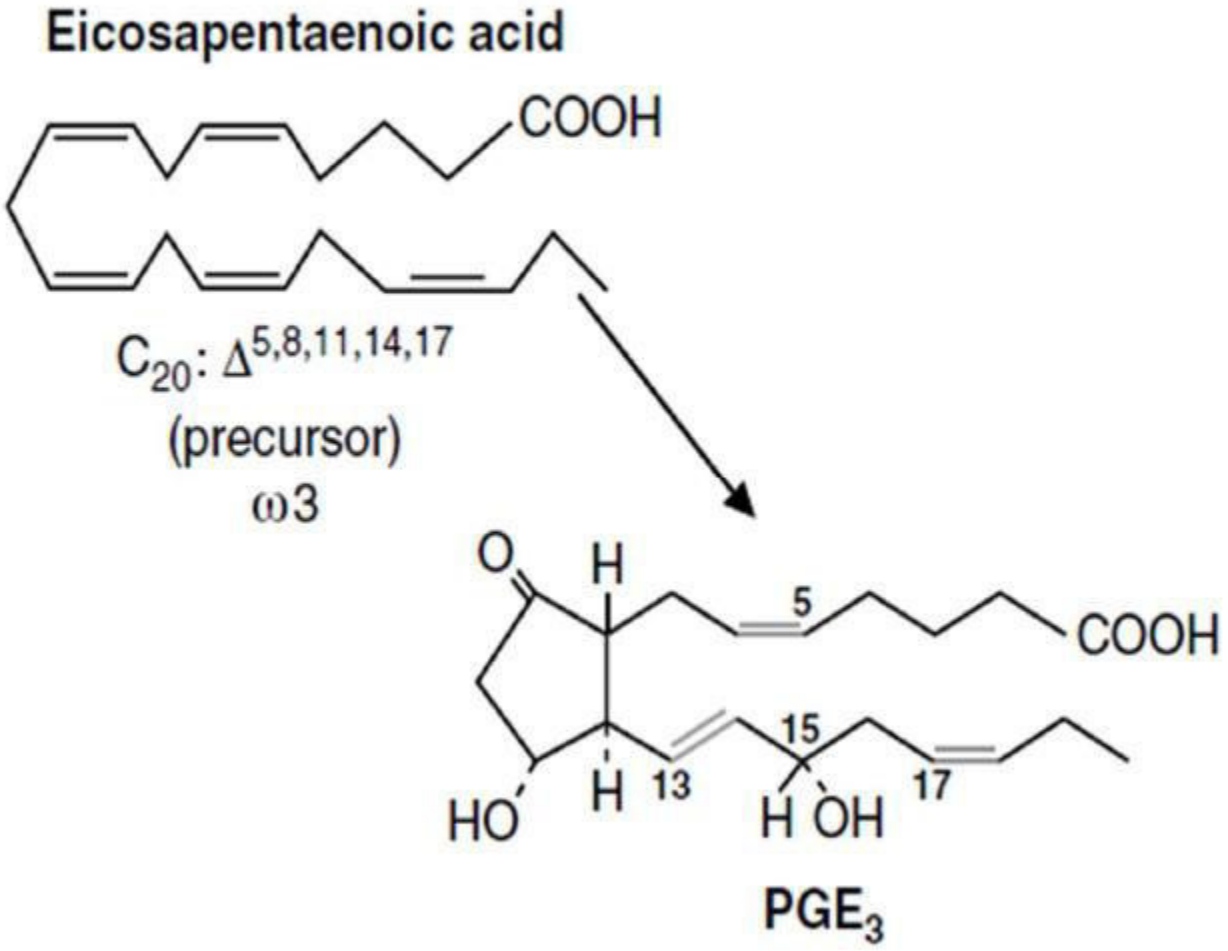
Only 1 double bond in the form of **trans** double between **C13** and **C14**

Lipid Chemistry



2 double bonds: a trans double between C13 and C14 and a cis double bond between C5 and C6

Lipid Chemistry



3 double bonds: trans double between C13 and C14 and cis two double bonds between C5 and C6

Lipid Chemistry

- It does not include double bonds in the internal ring.
- The double bonds between carbons 13 and 14 are *trans*; the others are *cis*

Lipid Chemistry

- The Greek letter subscript, found only in the F series, refers to the position of the hydroxyl group at carbon 9.
- This hydroxyl group primarily exists in the α position, i.e. it lies below the plane of the ring.

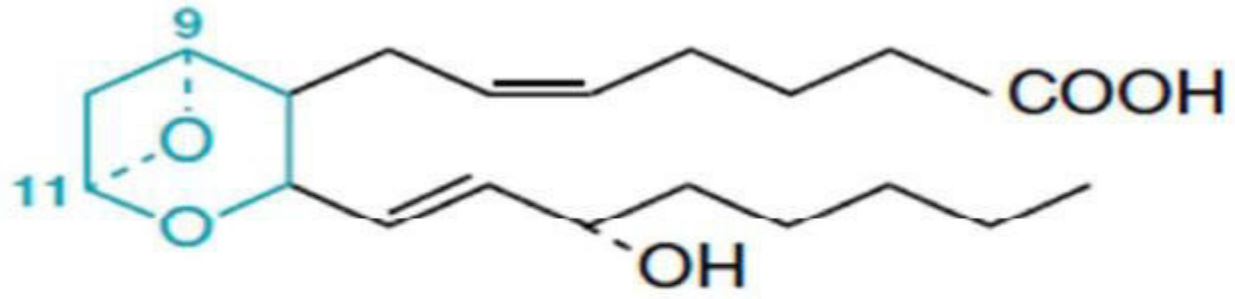
END

Lipid Chemistry

CYCLOOXYGENASE PATHWAY (Contd.)

- Thromboxanes (TX)
- also formed via the cyclooxygenase pathway,
- differ from PGs in that they contain a 6-membered ring that includes an oxygen atom

Lipid Chemistry



Thromboxane A_2
(TXA_2)

Thromboxanes have a six membered ring containing an oxygen atom.

Substituents are attached to the ring at carbons 9 and 11.

In the case of TXA_2(shown above), an oxygen atom connects carbons 9 and 11.

Lipid Chemistry

Biosynthesis of the PGS and TX

- Those derived from arachidonic acid, the 2-series, (such as PGE_2,TXA_2), are described here
- because the 1-series and the 3-series are present in very small amounts in humans.

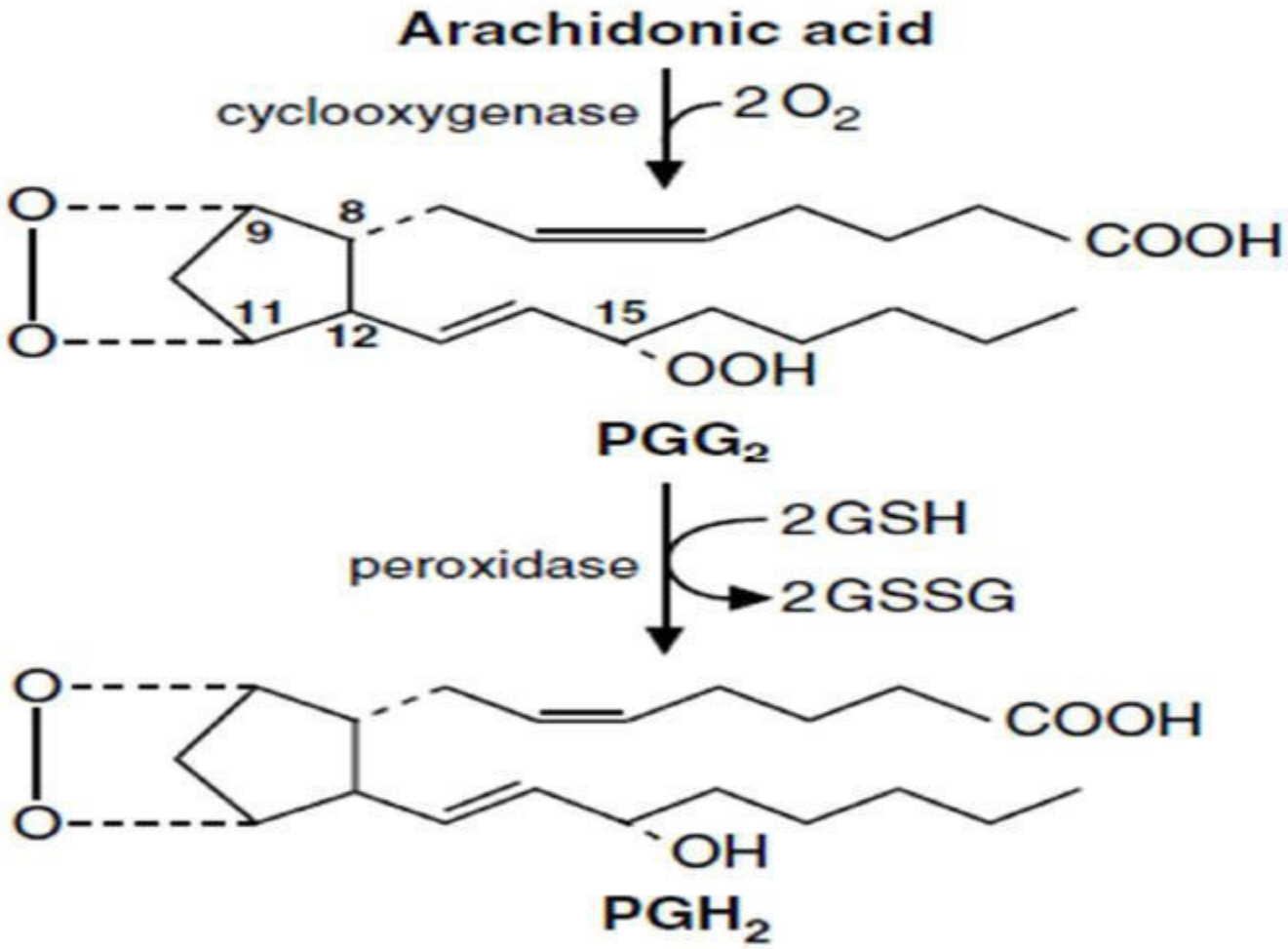
Lipid Chemistry

- The initial step, which is catalyzed by a cyclooxygenase (COX), forms the five-membered ring and
- adds four atoms of oxygen (two between C 9 and C11, and two at C 15) to
- form the unstable PGG_2.

Lipid Chemistry

- The hydro-peroxy group at carbon 15 is
- quickly reduced to a hydroxyl group by a peroxidase to form PGH_2
- PGH_2 is the precursor of all other PGs and TXAs

Lipid Chemistry



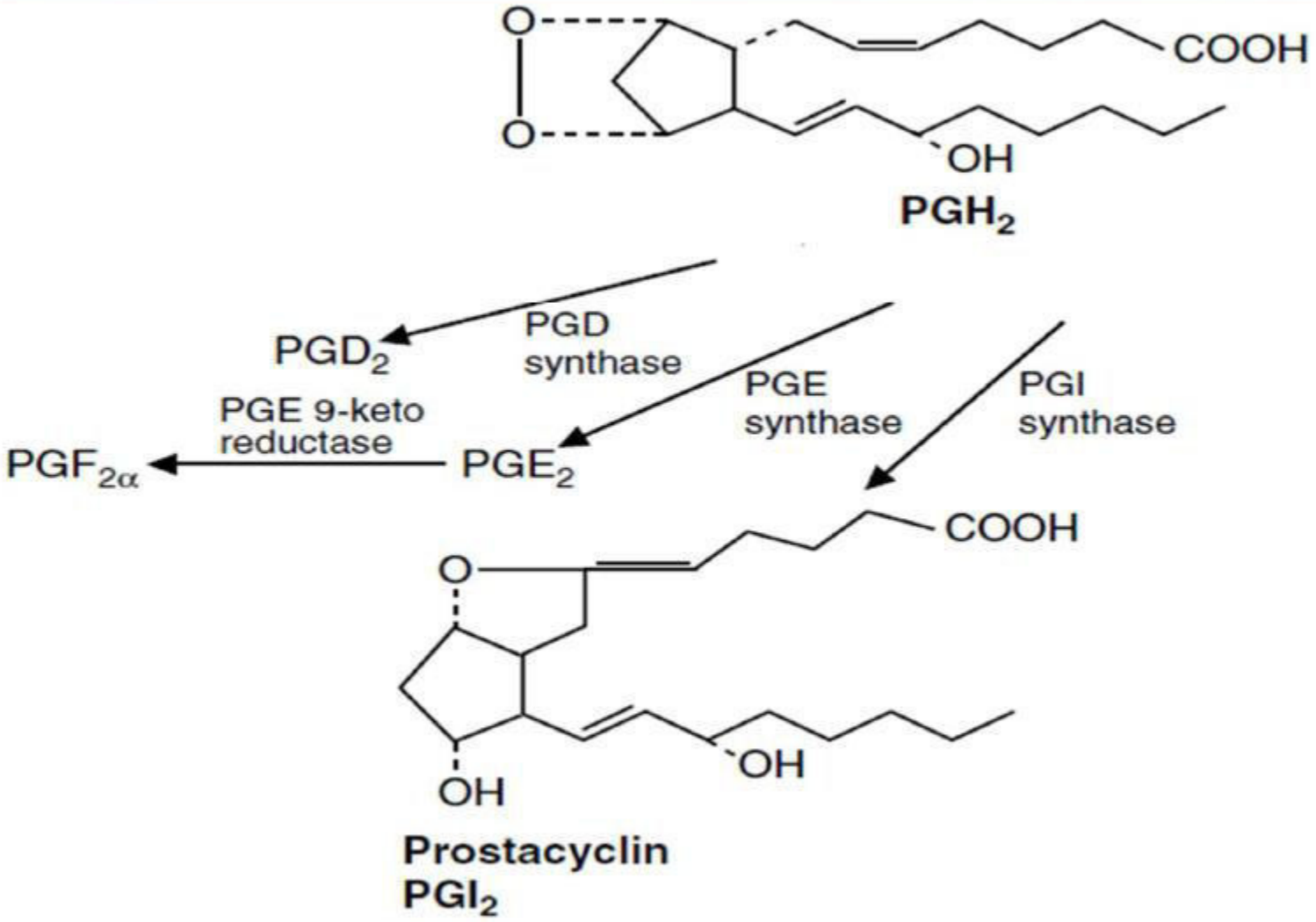
PGG is reduced and as a result GSH is oxidized.

Lipid Chemistry

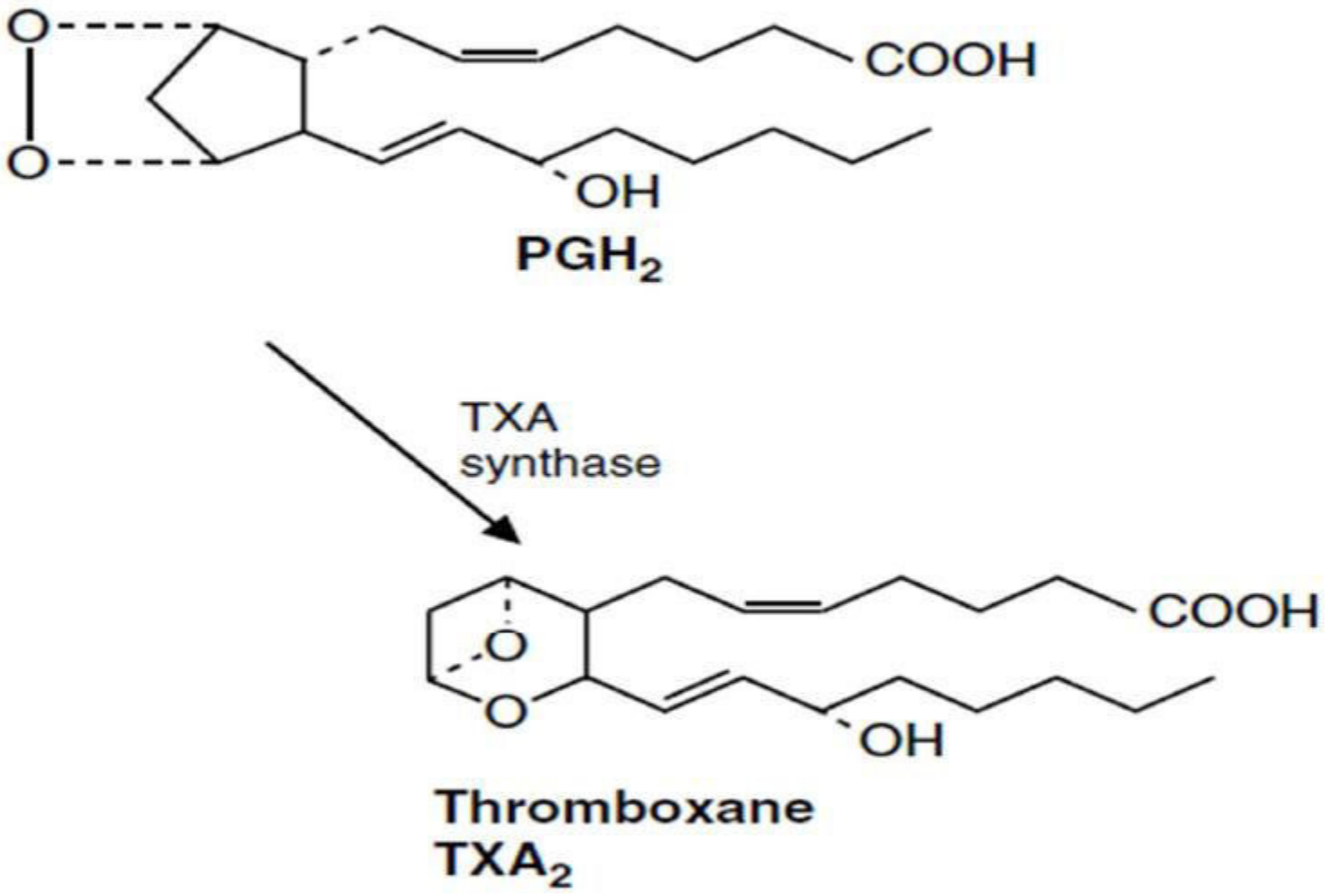
- The next step is tissue specific

- For example, PGH_2 may be reduced to PGE_2 or PGD_2 by specific isomerases (PGE synthase or PGD synthase respectively).

Lipid Chemistry



Lipid Chemistry



Lipid Chemistry

- For example, TXA Synthase is present in high concentration in platelets and forms TXA_2.
- In the vascular endothelium, however, PGH_2 is converted to the PGI_2 by action of Prostacyclin Synthase.

Lipid Chemistry

- The beneficial effect of cold water fish (e.g., salmon), with a high content of eicosa-pentaenoic(TX3) acid (EPA), and docosa-hexaenoic acid (DHA) comes from the fact that they lead to formation of more TXA_3 relative to TXA_2.

As discussed earlier, membrane fatty acid composition is dependent on the dietary intake of these FAs. The more healthy FAs in

Lipid Chemistry

- TXA_3 is less effective in stimulating platelet aggregation than its counterpart in the 2-series, TXA_2.
- Platelet aggregation is the culminating step in the cardiovascular diseases due to atherosclerosis.

END

Lipid Chemistry

Lipoxygenase Pathway:

Synthesis of the Leukotrienes, and Lipoxins (Count.)

Lipid Chemistry

- In addition to serving as a substrate for the cyclooxegenase pathway,
- arachidonic acid and other 20 C FAs also act as substrate for the lipoxygenase pathway.

Lipid Chemistry

- In contrast to products of the cyclooxegenase pathway which are cyclical.
- Products of lipoxygenase pathway are linear.

Lipid Chemistry

- Similar nomenclature rules are followed for Leukotrienes and Lipoxins,
- except that there are no series of 1 and 2.
- The series starts from 3.

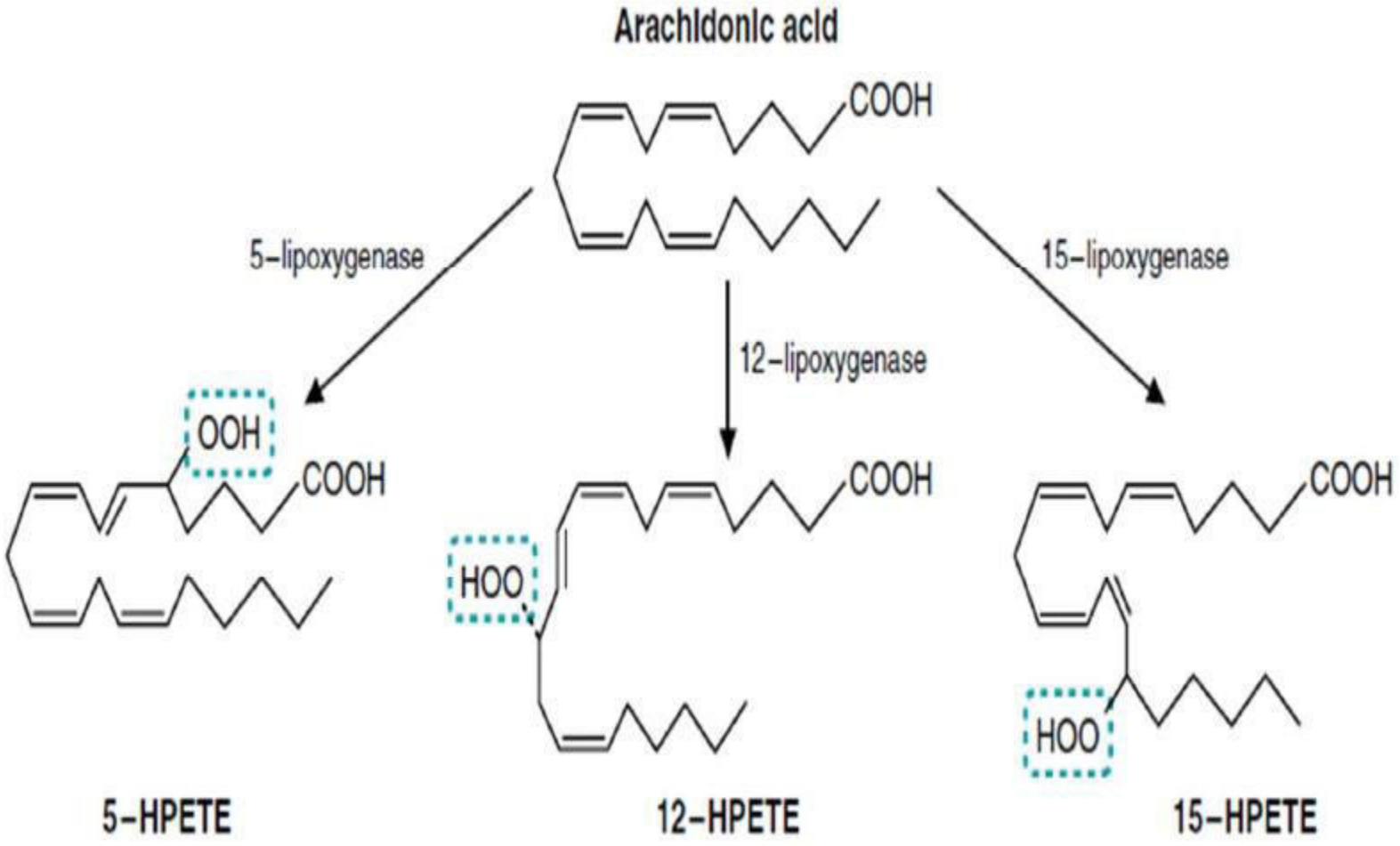
Lipid Chemistry

- the eicosanoids synthesized in lipoxygenase pathway from;
- Eicosa-tri-enoic acid (ETA) have 3 double bonds
- from; Arachidonic acid have 4 double bonds
- From; Eicosa-pentaenoic acid (EPA) have 5 double bonds .

Lipid Chemistry

- The lipoxygenase enzymes catalyze the incorporation of an oxygen molecule on to a carbon of one of several double bonds of arachidonic acid, forming a hydro-peroxy (–OOH) group at these positions

Lipid Chemistry



Lipid Chemistry

- The double bonds at which oxygen is added are between C5 & C6, between C11 and C12 and between C14 and C15.

END

Lipid Chemistry

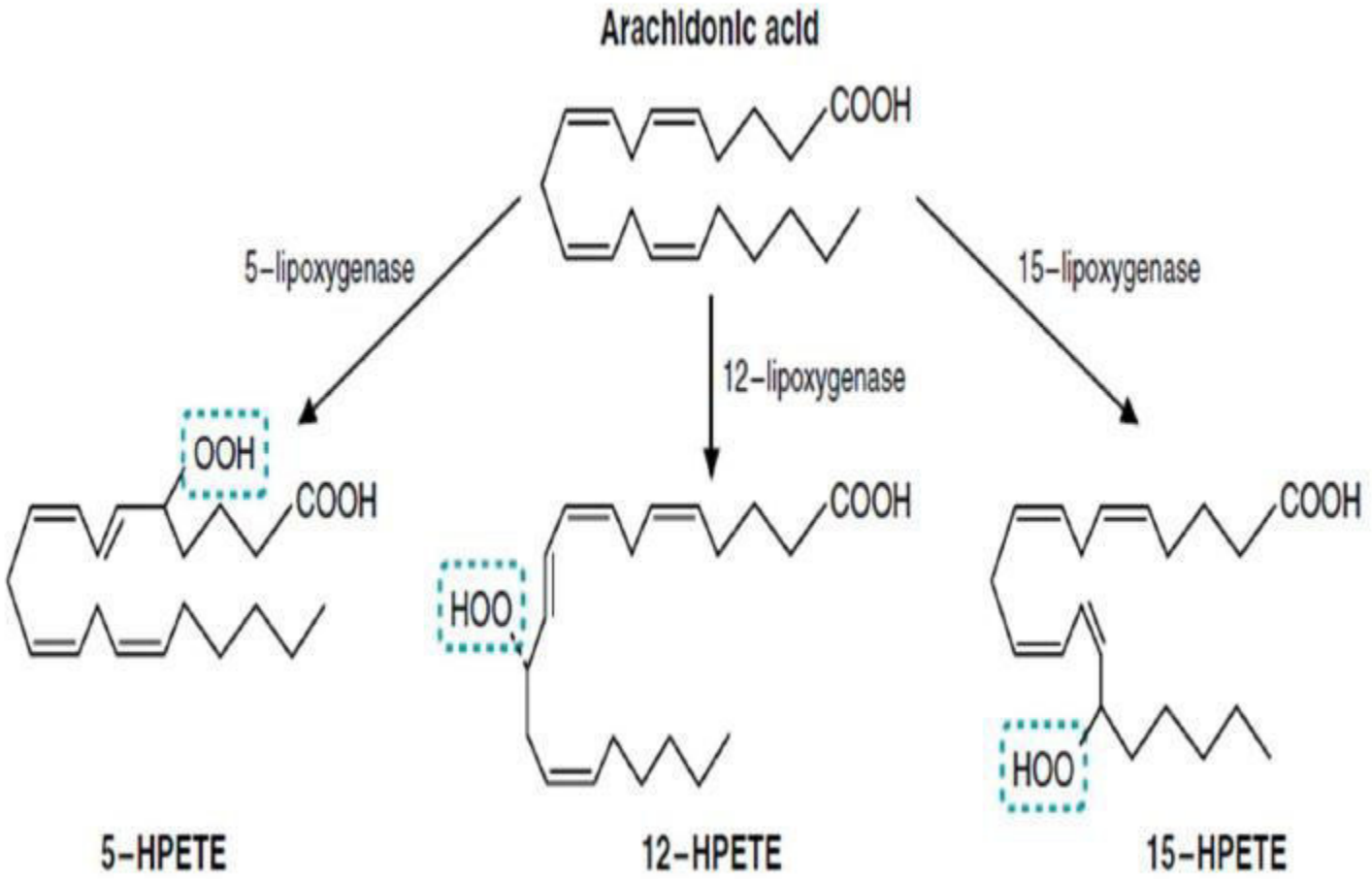
Lipoxygenase Pathway:

Synthesis of the Leukotrienes, and Lipoxins (Contd.)

Lipid Chemistry

- When the oxygen is added, to the double bond it isomerizes to a position one carbon away from the hydro-peroxy group and
- is transformed from the *cis* to the *trans* configuration

Lipid Chemistry



HPETEs: Hydro-peroxy-eicosa-tetra-enoic acid | 5-further.12,14 before

Lipid Chemistry

- As a result of this isomerization, double bond
- between C5 & C6 moves to C6 & C7,
- between C11 & C12 moves to C10 & C11
- between C14 & C15 moves to C13 & C14

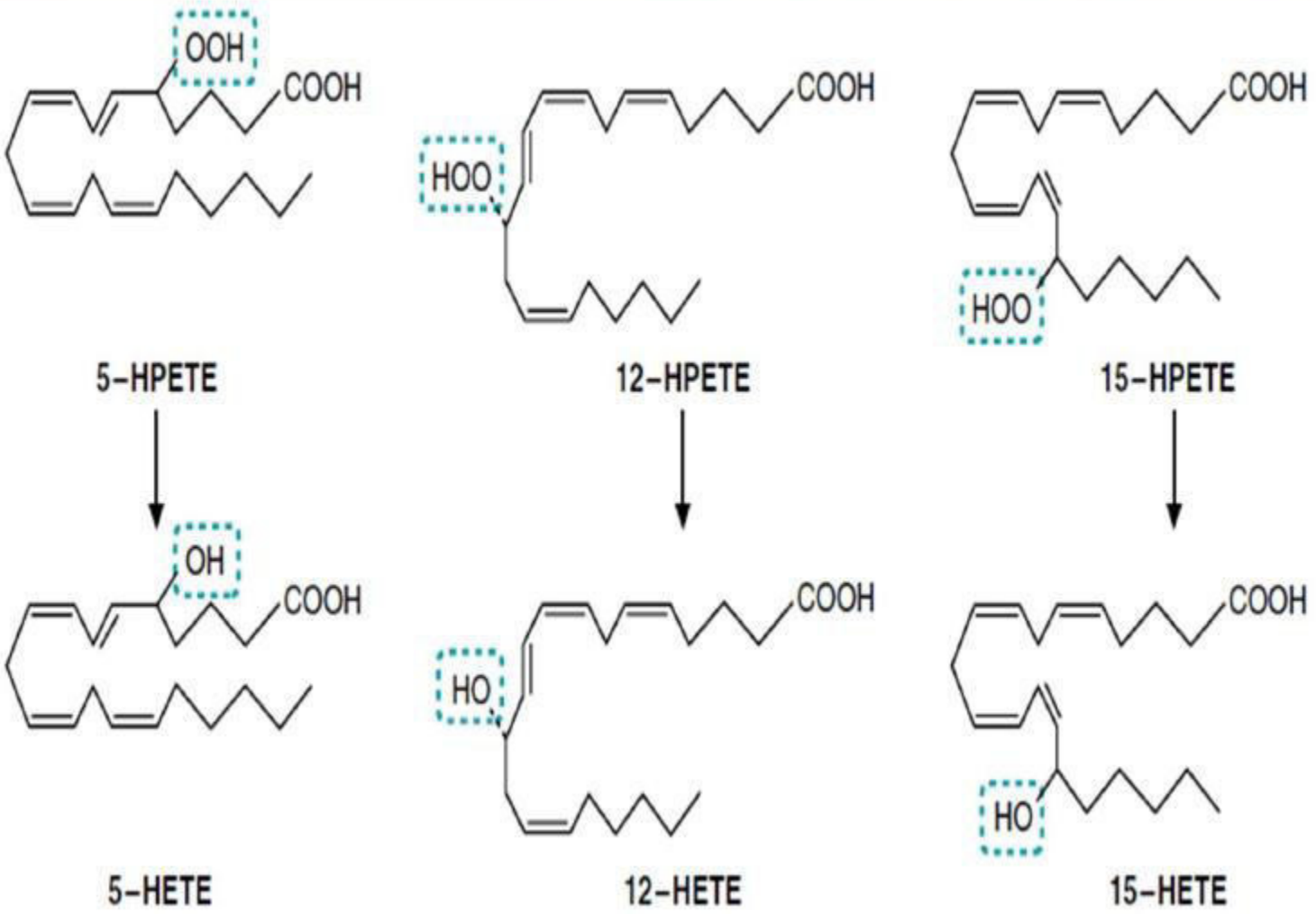
Arachidonic acid; 5,8,11,14

Lipid Chemistry

- The hydro-peroxy group is unstable and
- can be converted to the more stable hydroxy group to form HETEs
- HETEs themselves act as messenger molecules

5-Hydroxy-eicosa-tetra-enoic acid is an eicosanoid

Lipid Chemistry



HETEs: Hydr-peroxy-eicosa-tetra-enoic acid

Lipid Chemistry

- The other alternate is the conversion of HPETEs to leukotrienes(LTs) and Lipoxins(LXs),

- Which are more potent and have more defined physiological roles.

Lipid Chemistry

- The leukotrienes(LTs) and Lipoxins(LXs),

- Have 3 to 5 double bonds as described earlier, in comparison to
- prostaglandins which have 1 to 3 double bonds.

END

Lipid Chemistry

Lipoxygenase Pathway:

Synthesis of the Leukotrienes, and Lipoxins (Contd.)

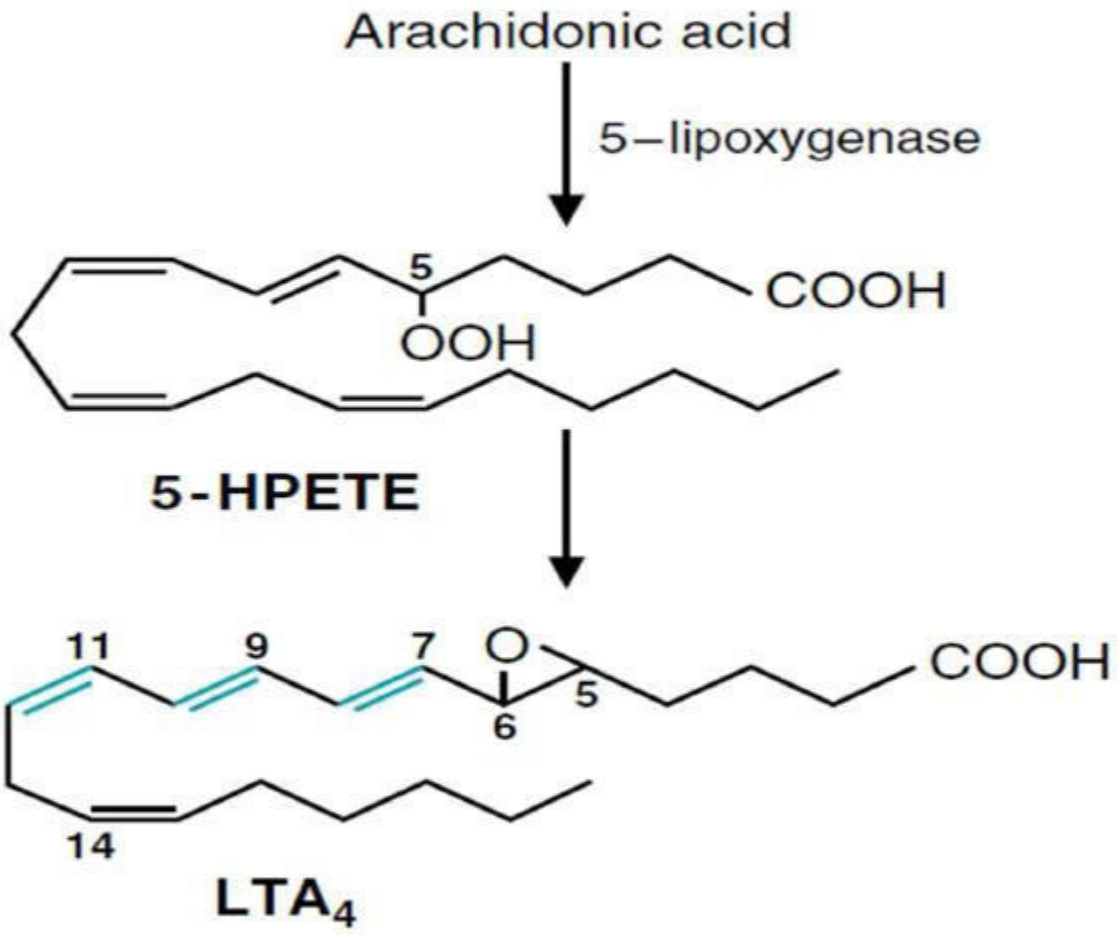
Lipid Chemistry

- HPETEs (Hydro-per-oxy-eicosa-tetra-enoic acids) are the precursors of leukotrienes and lipoxins

Lipid Chemistry

- The major leukotrienes are produced by 5-lipoxygenase
- 5-HPETE is converted to leukotriene A_4 (LTA_4).

Lipid Chemistry

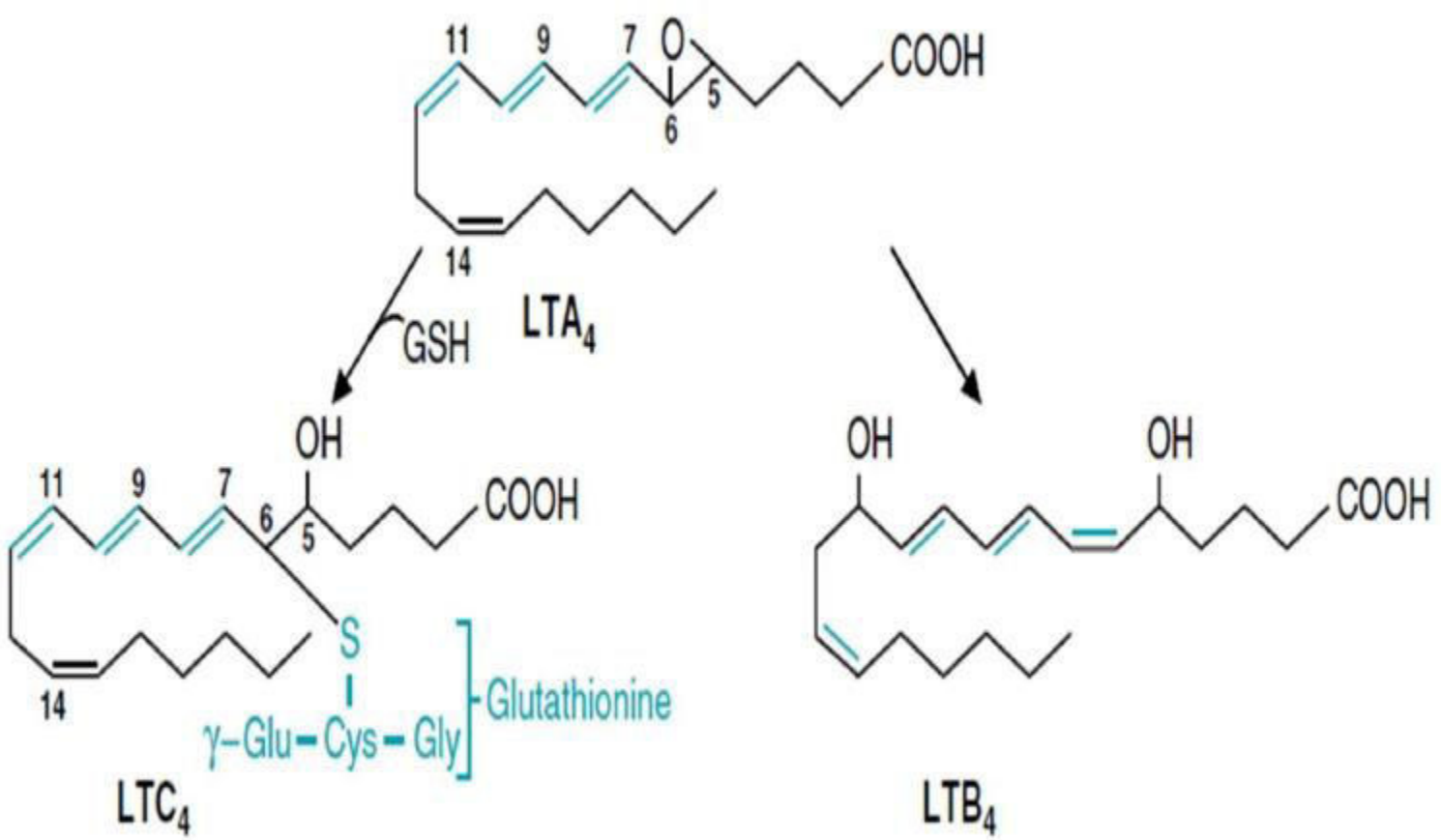


The subscript number 4 refers to the presence of four double bonds in the leukotriene.

Lipid Chemistry

- Other functional leukotrienes are formed from LTA_4 for example,
- LTA_4 is converted to LTB_4, as a 5,12-dihydroxy derivative.
- the addition of reduced glutathione to carbon 6 forms LTC_4

Lipid Chemistry



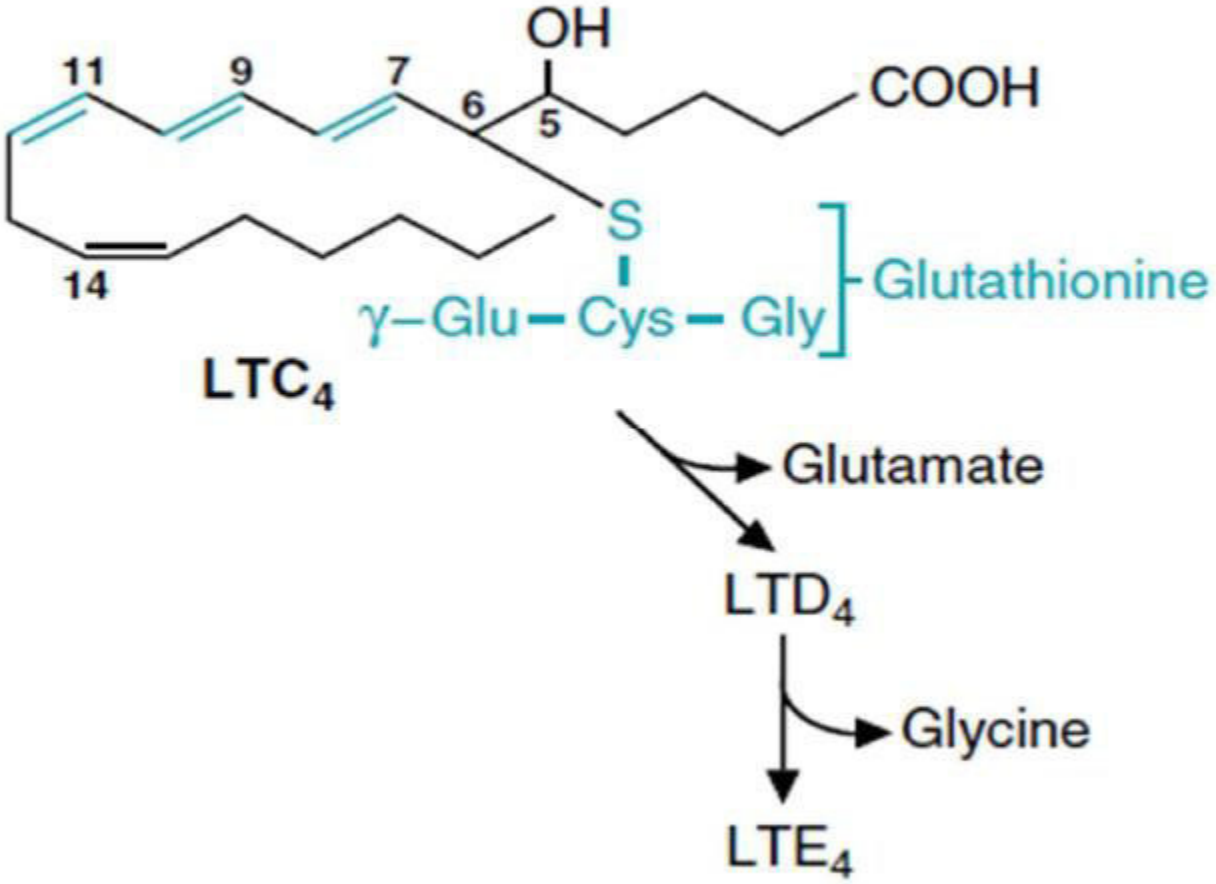
LTB , as a 5,12-dihydroxy derivative

Lipid Chemistry

- Removal of glutamate residue from LTC_4 forms LTD_4

- LTD_4 on removal of glycine becomes LTE_4

Lipid Chemistry



Lipid Chemistry

- Leukotrienes were so named because they were first discovered in leukocytes(white blood cells)

END

Lipid Chemistry

Lipoxygenase Pathway:

Synthesis of the Leukotrienes, and Lipoxins (Contd.)

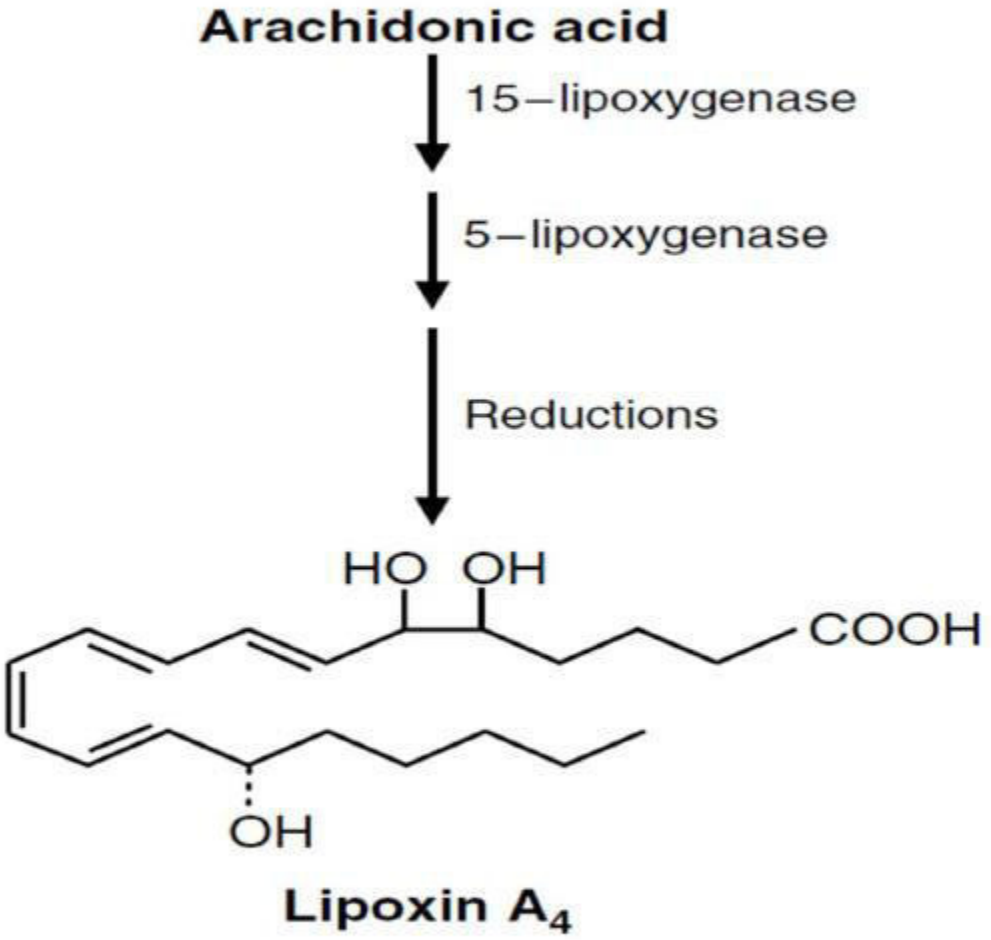
Lipid Chemistry

- The lipoxins are formed through the action of 15-lipoxygenase followed by the action of 5-lipoxygenase on arachidonic acid.

Lipid Chemistry

- A series of reductions of the resultant hydro-per-oxy groups leads to the formation of tri-hydroxy derivatives of arachidonic acid known as the lipoxins.

Lipid Chemistry



Lipid Chemistry

- Lipoxins induce chemotaxis and
- stimulate superoxide radicals for killing of microorganisms

Chemotaxis is migration of WBCs to the tissue site of injury or inflammation

Lipid Chemistry

- Prostaglandins, thromboxanes, leukotrienes and lipoxins have very short half lives and rapidly degraded in the body.

Lipid Chemistry

- In summary, Eicosanoids are derived from C20 (eicosanoic) fatty acids synthesized from the essential fatty acids and make up important groups of physiologically active compounds.

END

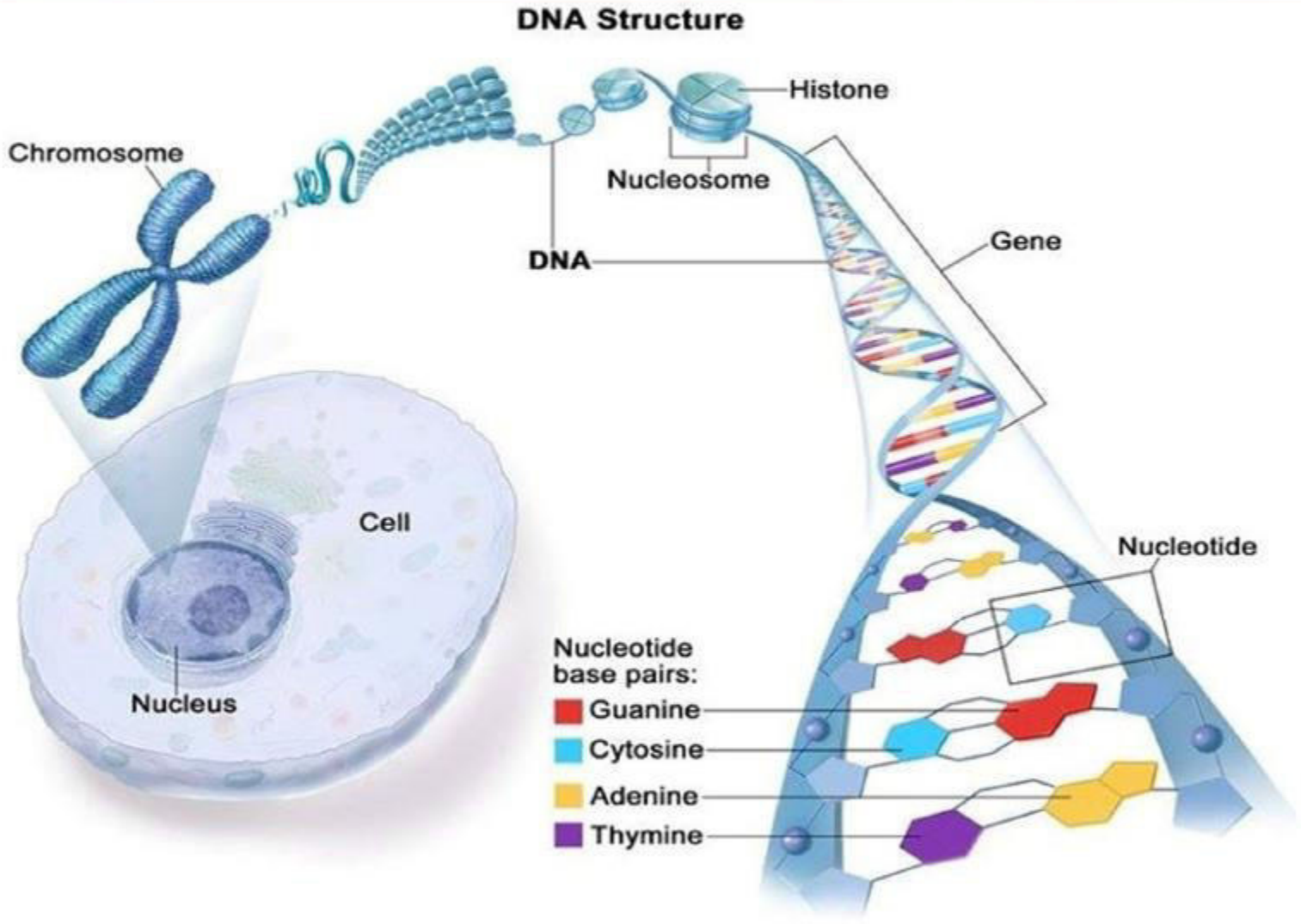
Nucleotides and Nucleic Acids

Nucleotides and Nucleic Acids

Biomedical importance of nucleotides

- **Precursors of nucleic acids:** Nucleotides are the building blocks of nucleic acids.
- Without them, DNA or RNA can not be produced.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- **Transmission of genetic information;**
- This gives them the ability to store and transmit genetic information from;
- one generation to the next which is a fundamental condition for life

Nucleotides and Nucleic Acids

- **Protein synthesis,**
- the ultimate expression of this information, is therefore dependent on nucleotides.

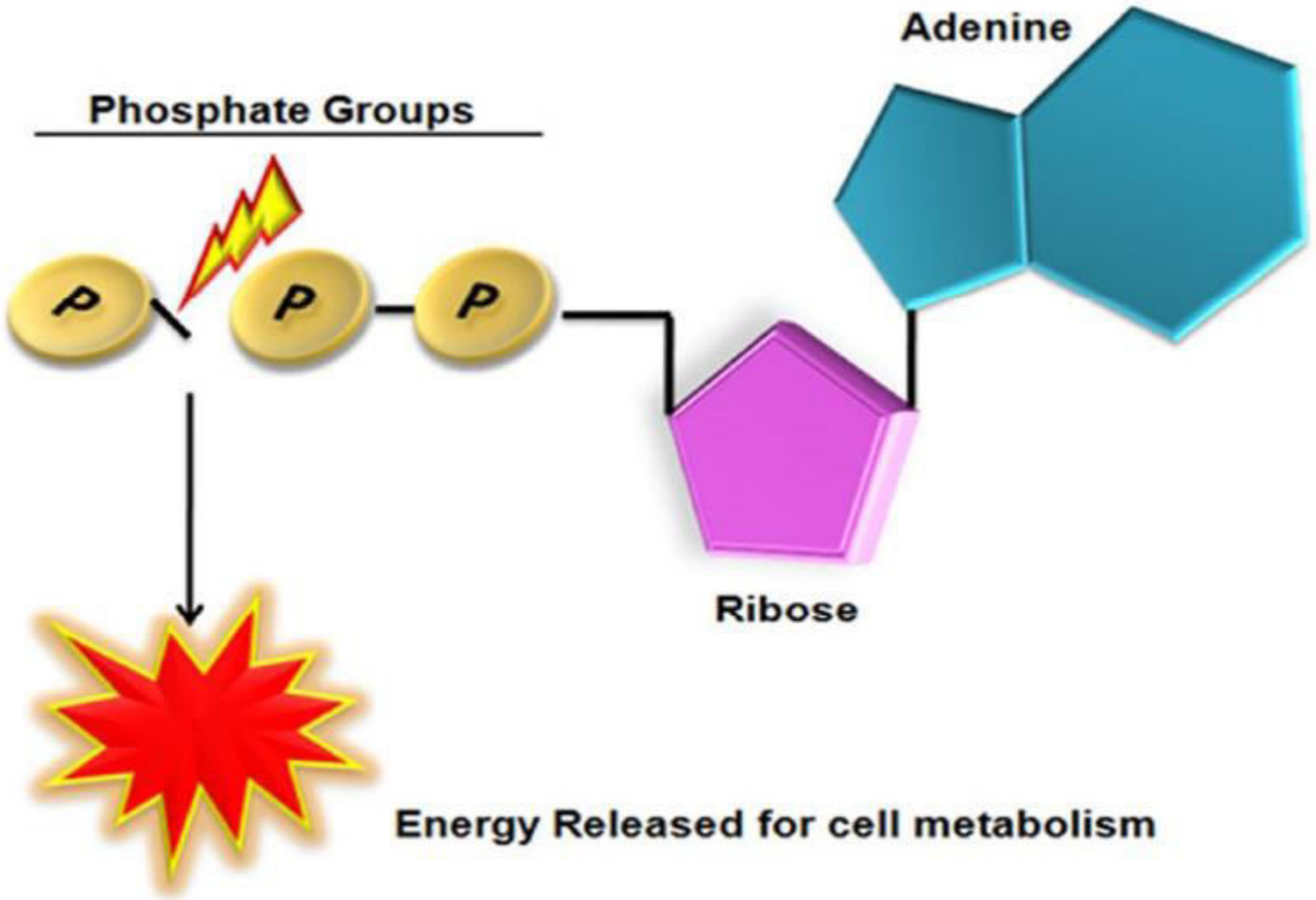
Nucleotides and Nucleic Acids

- **Energy currency:** Nucleotides play an important role as "energy currency" in the cell.

Nucleotides and Nucleic Acids

- Nucleoside tri- and diphosphates such as **ATP and ADP** are the principal donors and acceptors of phosphoryl group in metabolism.
- By doing this, they play a key role in the energy transduction.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- This energy is used in almost every energy requiring process in the body, such as;
- Muscle contraction,
- Transmission of nerve impulse,

Nucleotides and Nucleic Acids

- Transports of nutrients across cell membrane

- Motility of spermatozoa
- And many more energy dependent processes

END

Nucleotides and Nucleic Acids

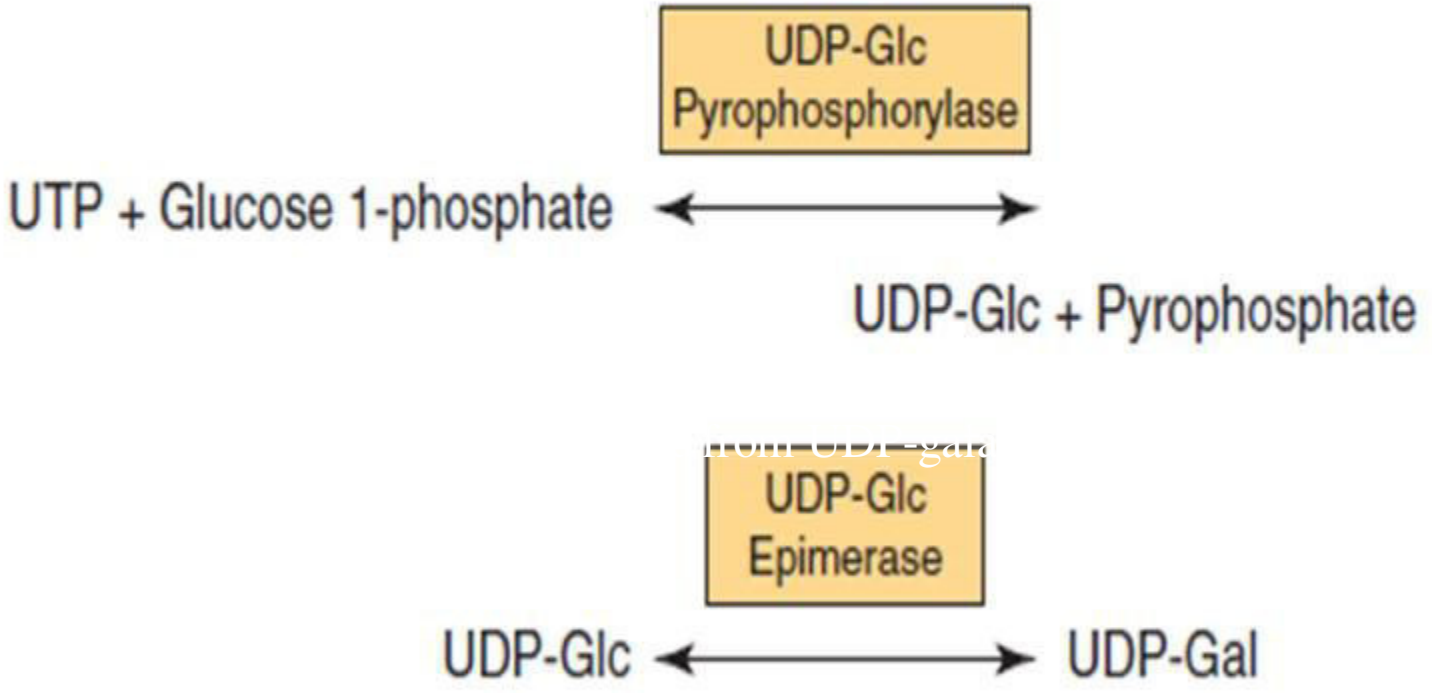
Biomedical importance of nucleotides (Contd.)

- **Carriers of intermediates:**
  Nucleotides also serve as carriers of activated intermediates in the synthesis of some carbohydrates, lipids, and proteins.

Nucleotides and Nucleic Acids

- The sugar derivatives UDP-glucose and UDP-galactose participate in sugar inter conversions
- And in the biosynthesis of starch and glycogen
- Similarly, nucleoside-lipid derivatives such as CDP acylglycerol are intermediates in lipid biosynthesis.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- **Co-enzymes:**
- When linked to vitamins nucleotides are structural components of several essential coenzymes, for example,
- coenzyme A,
- FAD, (Flavin Adenine Dinucleotide)

Nucleotides and Nucleic Acids

- **Co-enzymes:**

- NAD^+ (Nicotinamide adenine Dinucleotide)
  and
- $NADP^+$ (Nicotinamide adenine Dinucleotide Phosphate)

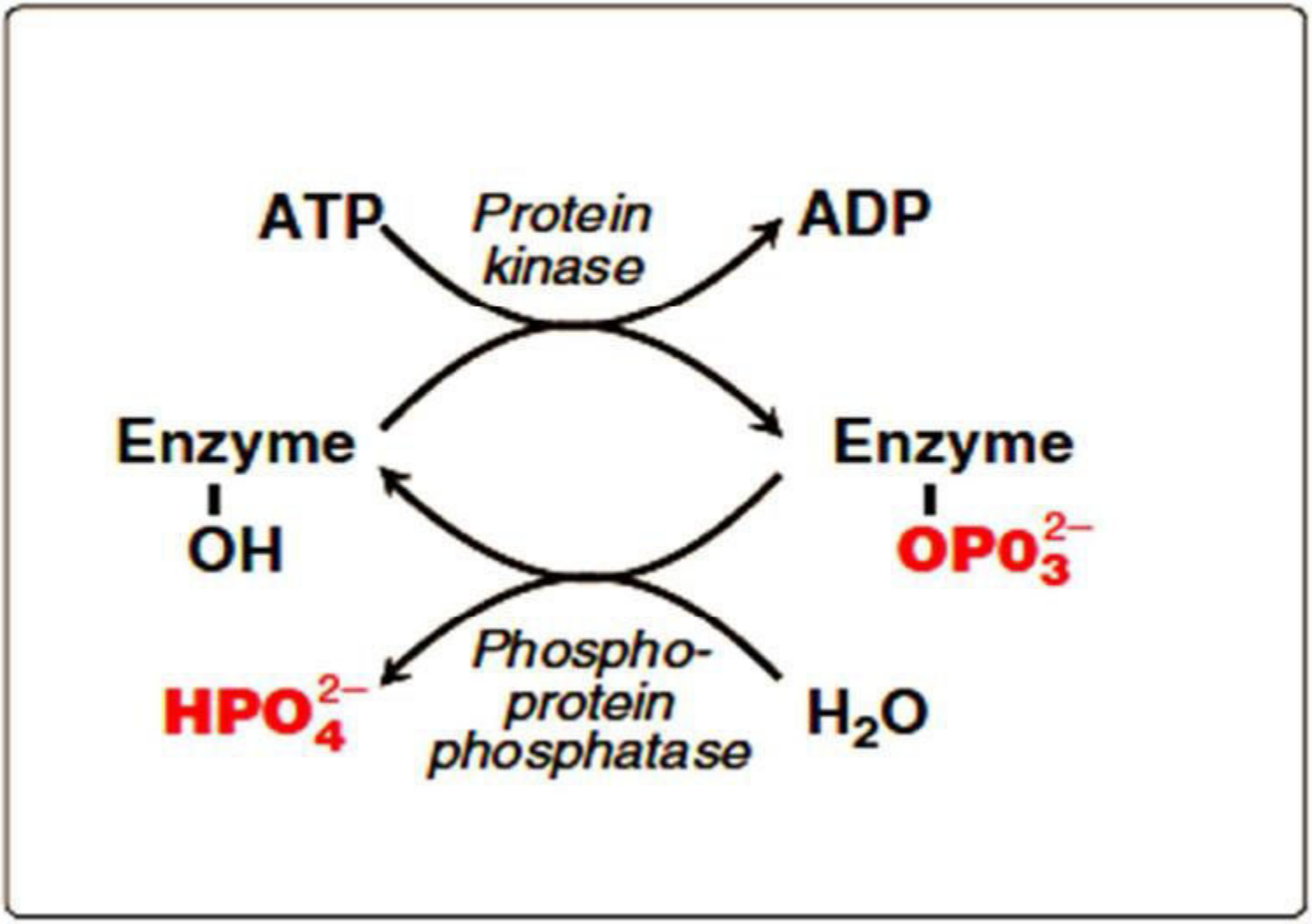
Nucleotides and Nucleic Acids

- **Regulatory compounds:** Nucleotides are important regulatory compounds for many of the pathways of intermediary metabolism, inhibiting or activating key enzymes.

Nucleotides and Nucleic Acids

- Roles that nucleotides perform in metabolic regulation include:
- ATP-dependent enzyme phosphorylation in key metabolic reactions.
- Allosteric regulation of enzymes by ATP, AMP, and CTP

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Phosphorylation can either activate or inhibit ATP- dependent enzyme depending on the type of enzyme.

- Similar is the case with dephosphorylation.

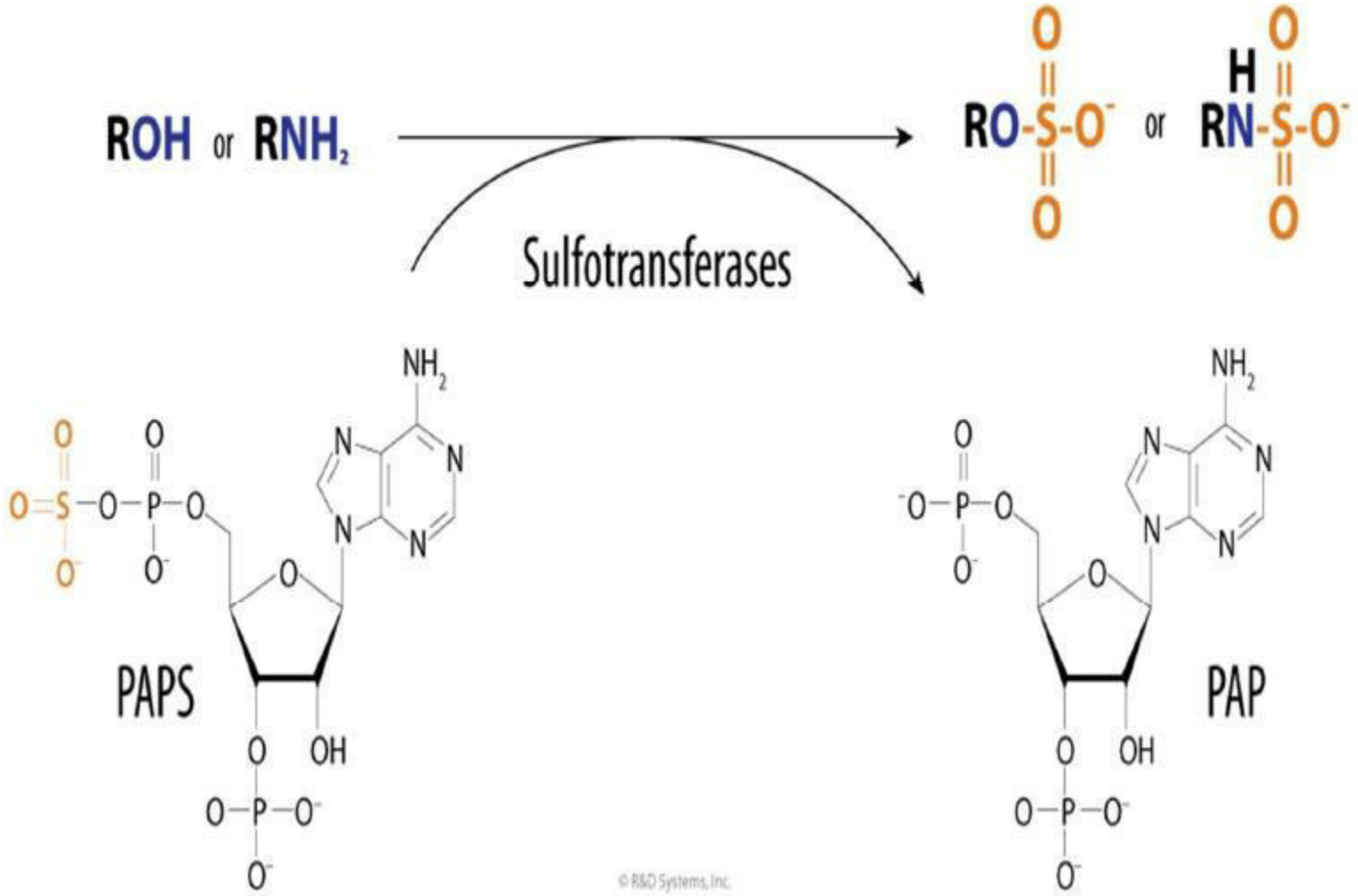
END

Nucleotides and Nucleic Acids

Biomedical importance of nucleotides (Contd.)

- **Sulfate group donor:** Adenosine 3'-phosphate-5'-phosphosulfate is the sulfate donor for
  - sulfated proteoglycans
  - sulfate conjugates of drugs.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- **Methyl group donor:** *S*-adenosylmethionine is a methyl group donor

- e.g Nor-adrenaline → Adrenaline by methylation.

These are neurotransmitter of the sympathetic nerves in the cardiovascular system

Nucleotides and Nucleic Acids

- **Second messengers:** Nucleotides, such as *cyclic AMP (cAMP)* and *cyclic GMP (cGMP)*,

- serve as second messengers in signal transduction pathways.

Nucleotides and Nucleic Acids

- **Signal Transduction**: GTP and GDP play key
- roles in activating or inhibiting proteins in
- various cellular signaling cascades.

Nucleotides and Nucleic Acids

- **Medical applications** Specifically medical applications include the
- use of synthetic purine and pyrimidine analogs
- that contain halogens, thiols, or additional nitrogen atoms;

Nucleotides and Nucleic Acids

- There use include chemotherapy for cancer
- as suppressors of the immune response during organ transplantation.
- as anti-viral drugs such as in the treatment of AIDS

END

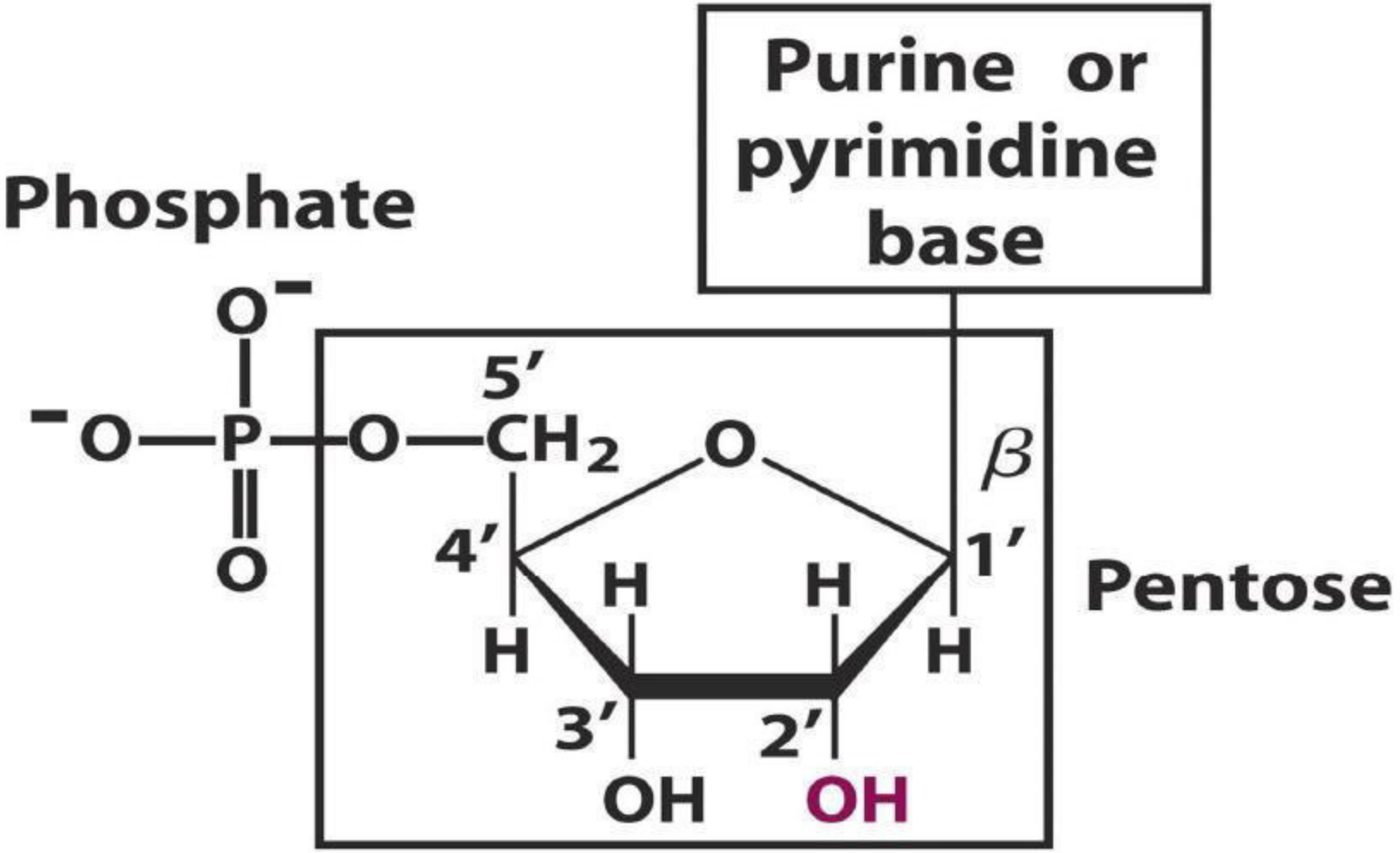
Nucleotides and Nucleic Acids

Composition of Nucleotides

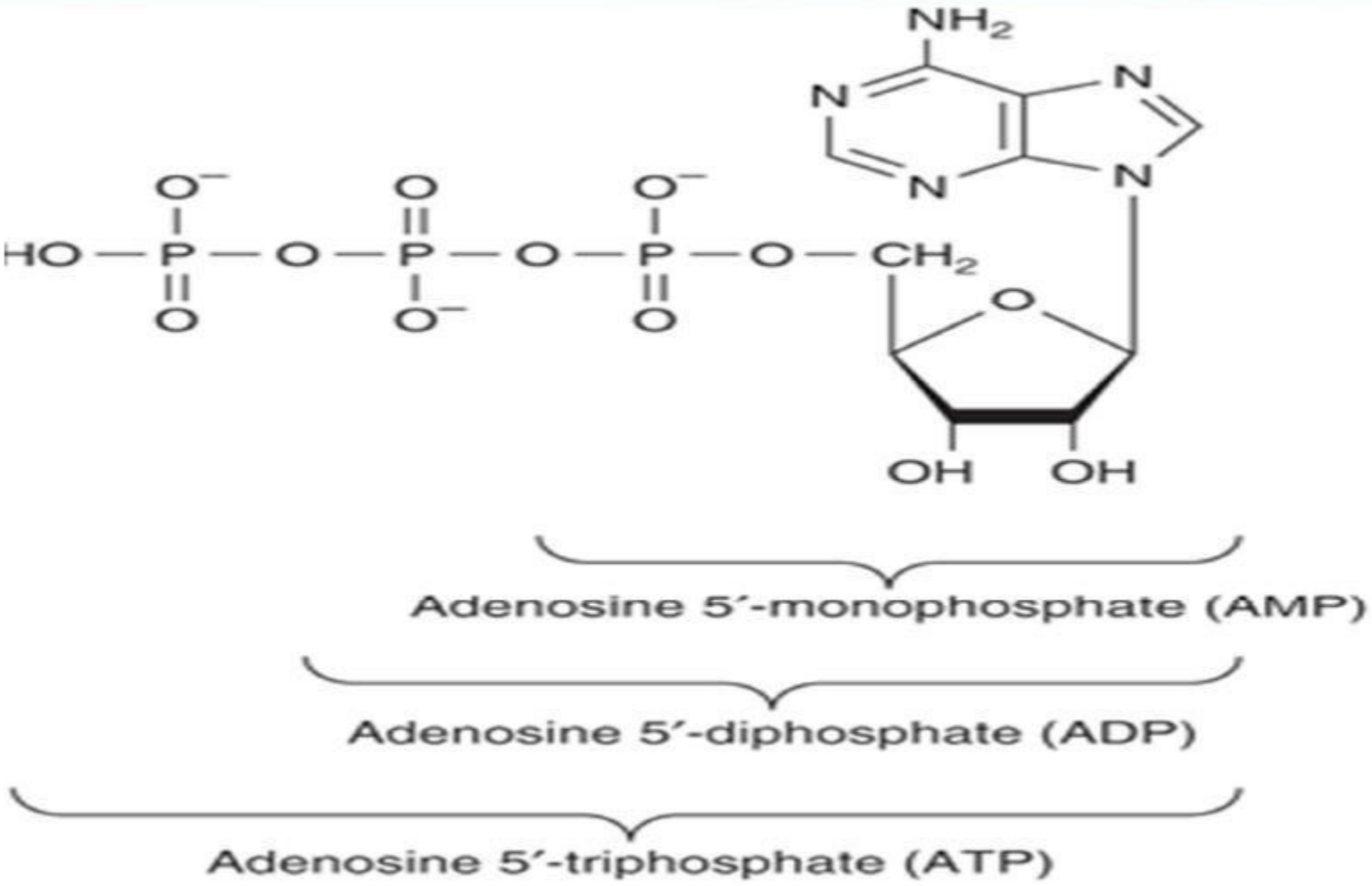
Nucleotides are composed of

- A nitrogenous base (purine or pyrimidine)
- A pentose monosaccharide
- One, two, or three phosphate groups

Nucleotides and Nucleic Acids



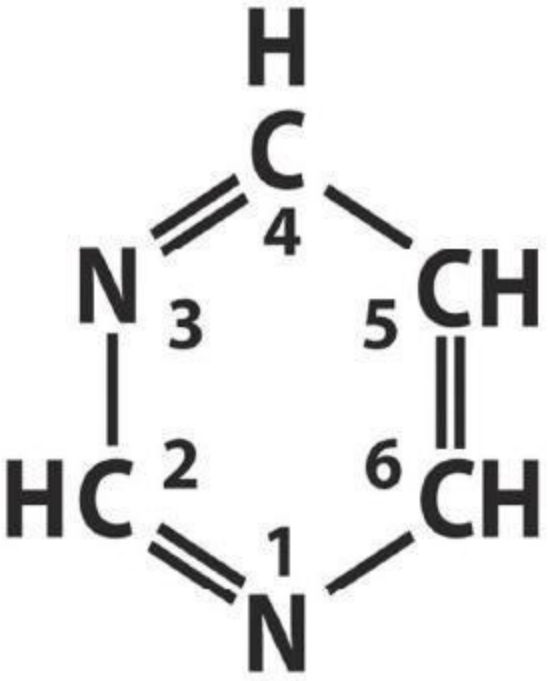
Nucleotides and Nucleic Acids



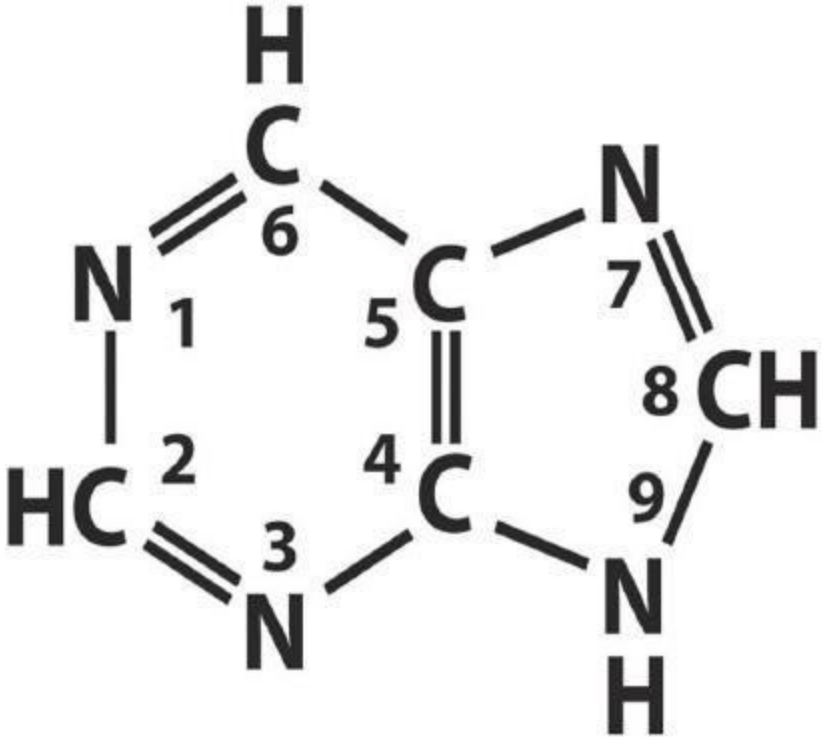
Nucleotides and Nucleic Acids

- **Nitrogenous Bases**
- The nitrogen-containing bases belong to two families of compounds:
- Purines
- Pyrimidines

Nucleotides and Nucleic Acids



Pyrimidine

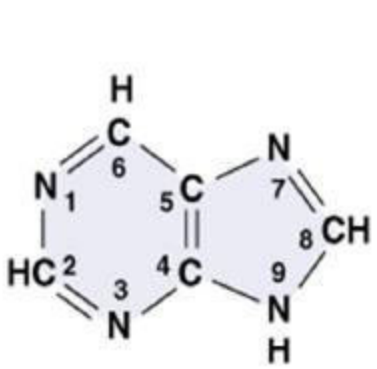


Purine

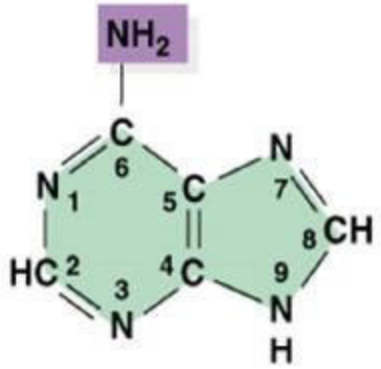
Nucleotides and Nucleic Acids

- By the attachment of different groups to the rings, different types of pyrimidine and purine are generated.

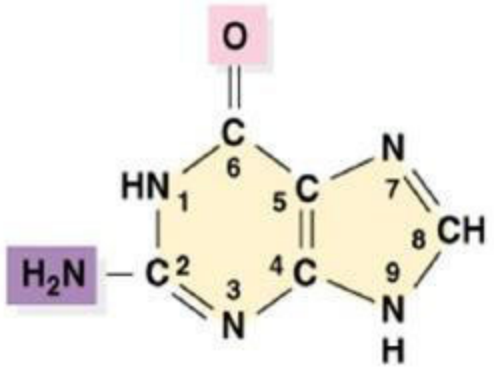
Nucleotides and Nucleic Acids



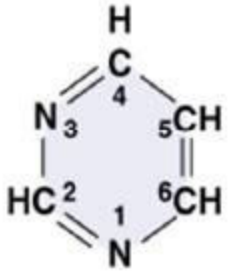
Purine
(parent compound)



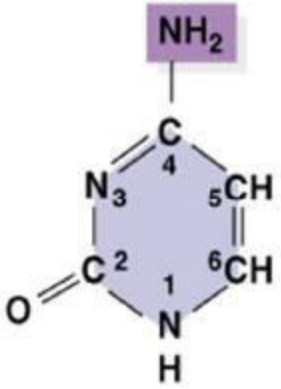
Adenine (A)



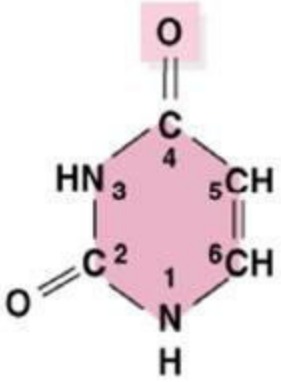
Guanine (G)



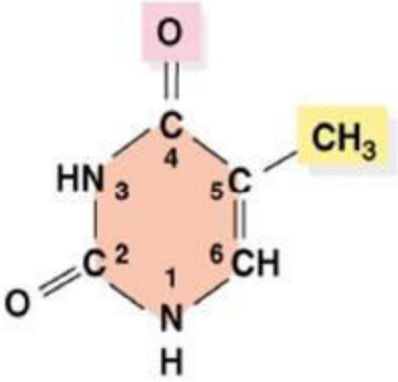
Pyrimidine
(parent
compound)



Cytosine (C)



Uracil (U)
(found in RNA)



Thymine (T)
(found in DNA)

Nucleotides and Nucleic Acids

- The suffix "ine" in these bases denotes the presence of nitrogen (amine) in the ring.

- However, there are some exceptions such as naming of uracil

Nucleotides and Nucleic Acids

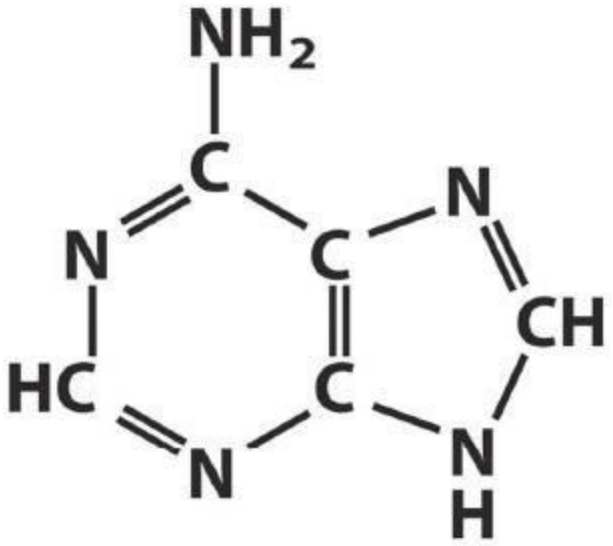
- The utility of these nitrogen-containing ring structures lies in the ability of the nitrogen to form hydrogen bonds and to
- accept and donate electrons while still part of the ring.

Nucleotides and Nucleic Acids

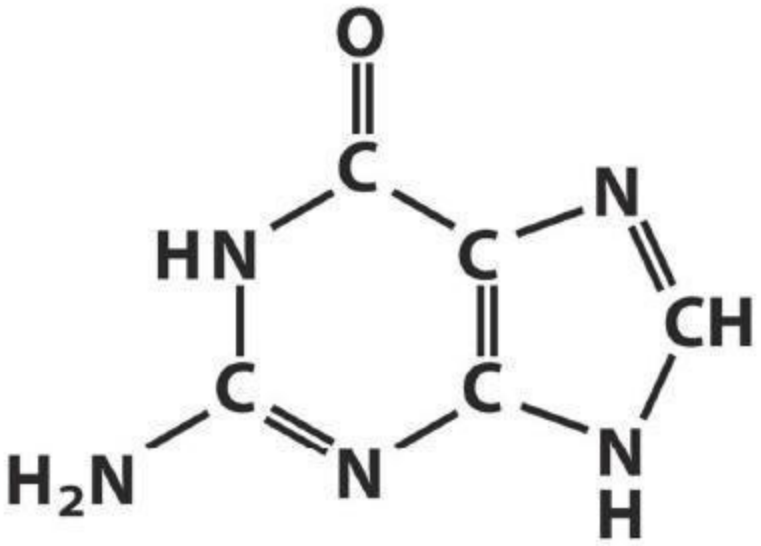
Composition of Nucleotides (Contd.)

- **Purines**
- Both DNA and RNA contain the same purine bases:
  - Adenine (A)
  - Guanine (G)

Nucleotides and Nucleic Acids



Adenine



Guanine

Purines

Nucleotides and Nucleic Acids

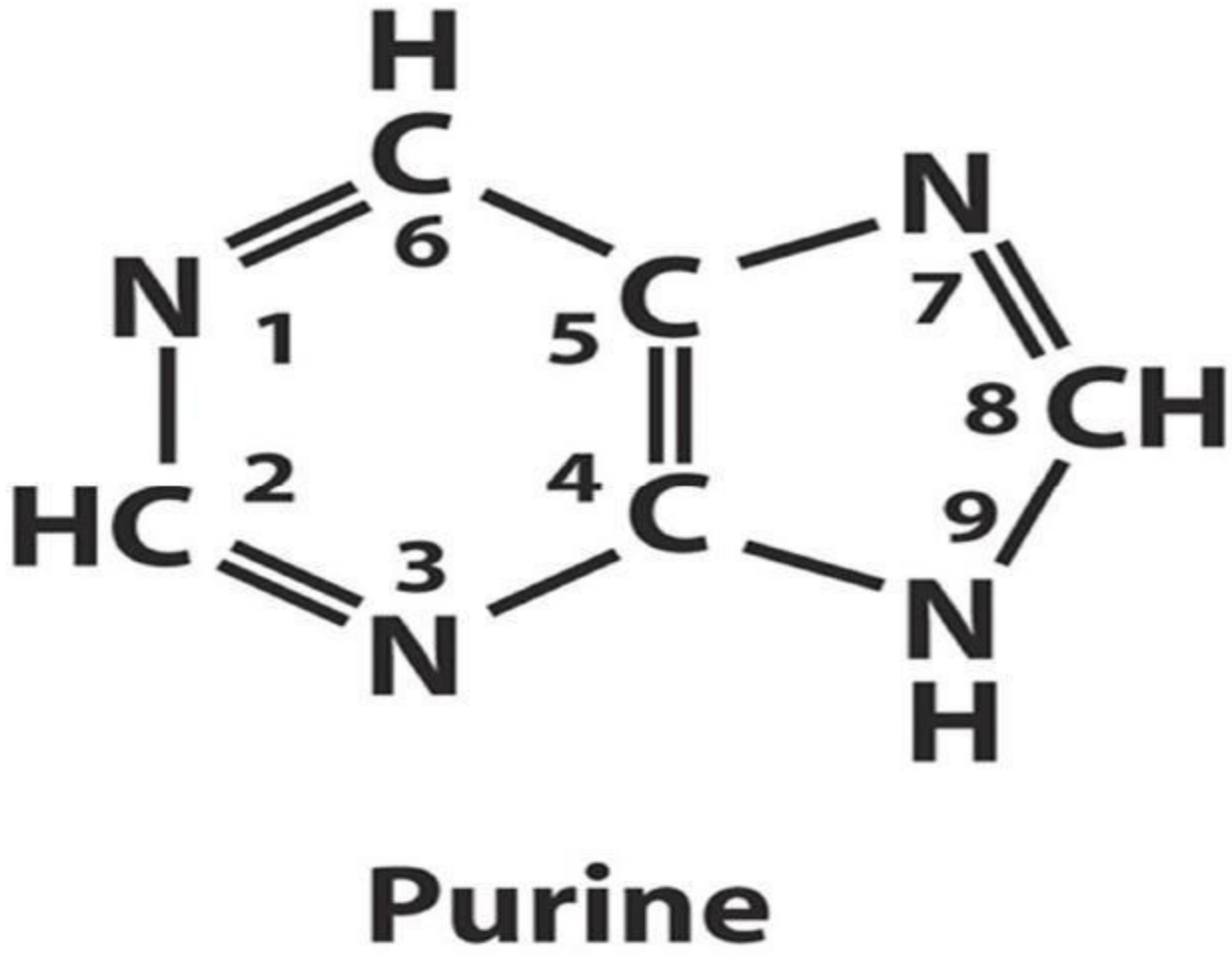
- Adenine when combines with pentose the structure is known as Adenosine or deoxyadenosine
- Guanine when combines with pentose the structure is known as Guanosine or deoxyguanosine

Nucleotides and Nucleic Acids

- Adenine is 6-aminopurine
- Guanine is 2-amino,6-hydroxypurine

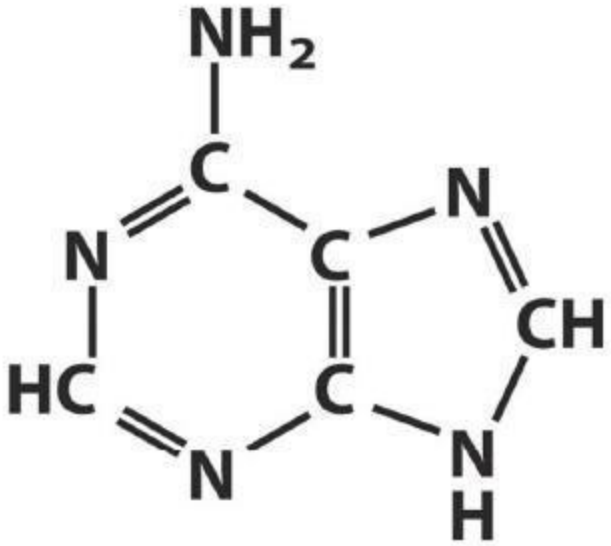
.

Nucleotides and Nucleic Acids

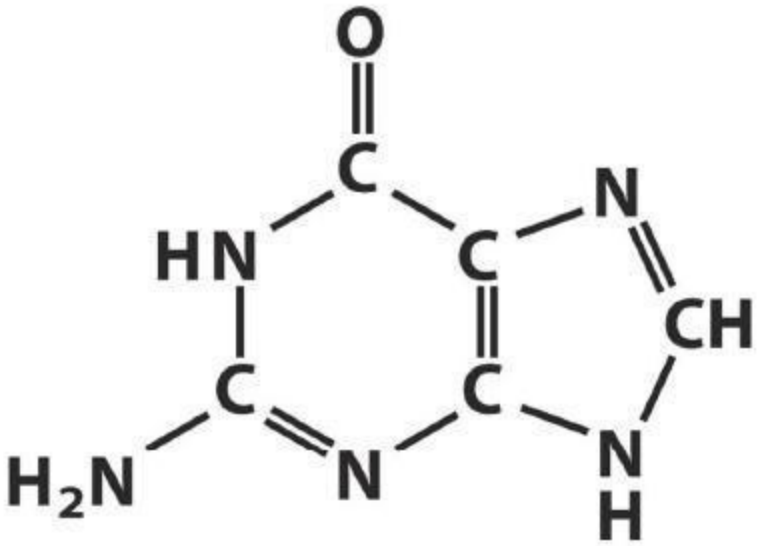


Purine

Nucleotides and Nucleic Acids



Adenine



Guanine

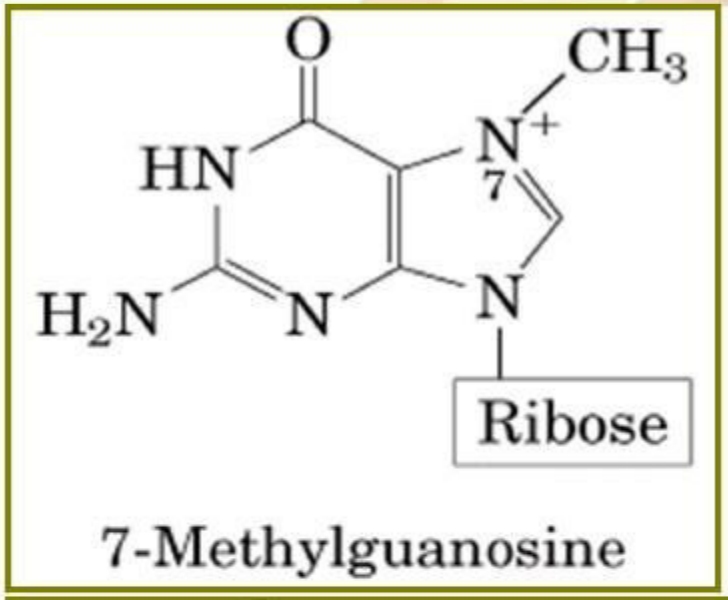
Purines

Adenine is 6-aminopurine.Guanine is 2-amino,6-hydroxypurine

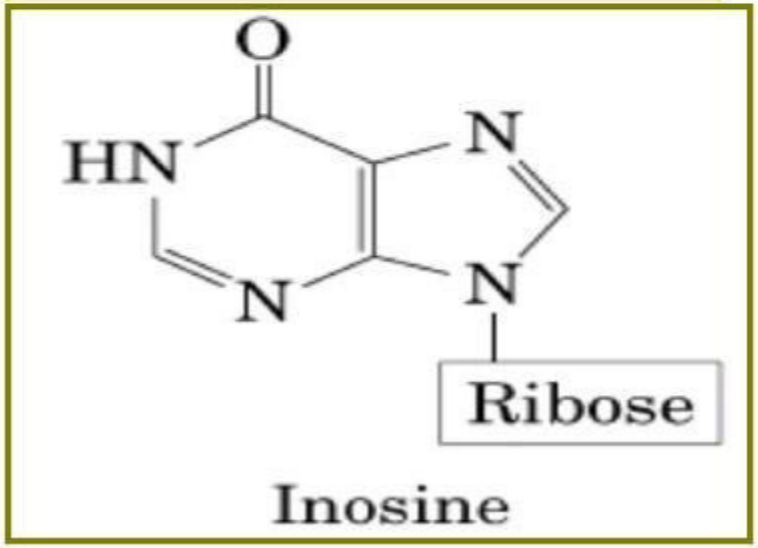
Nucleotides and Nucleic Acids

- Minor Purine Bases: Inosine (I) & methyl guanine (7mG)
- Unnatural: Mercaptopurine, Allopurinol & 8-Azaguanine
- Other purines include: hypoxanthine and xanthine

Nucleotides and Nucleic Acids



7-Methylguanosine



Inosine

Nucleotides and Nucleic Acids

- Uric acid is the catabolic end product of purines in human beings.

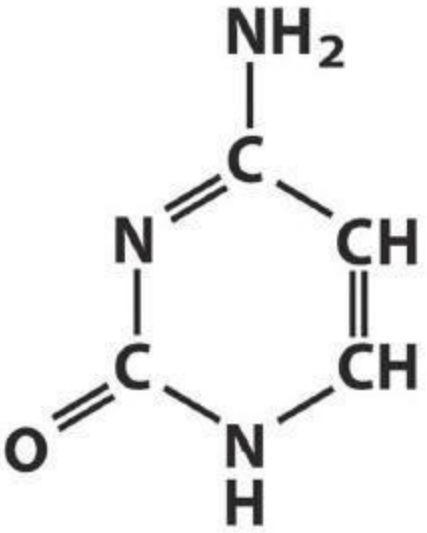
END

Nucleotides and Nucleic Acids

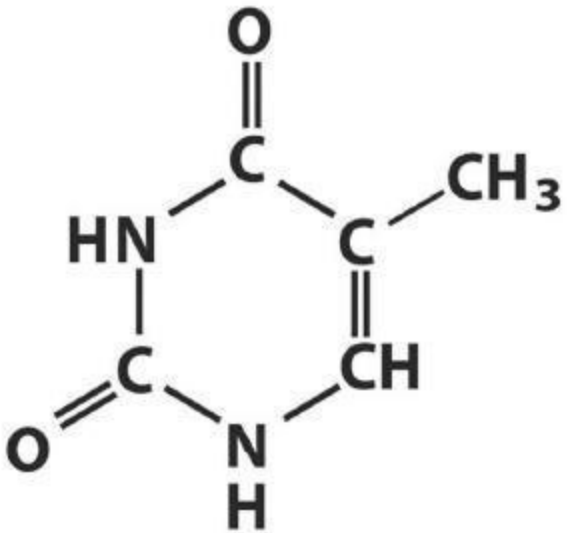
Composition of Nucleotides (Contd.)

- **Pyrimidines:**
- Pyrimidines include:
- Cytosine (C)—in both DNA and RNA
- Thymine (T)—only in DNA
- Uracil (U) —only in RNA

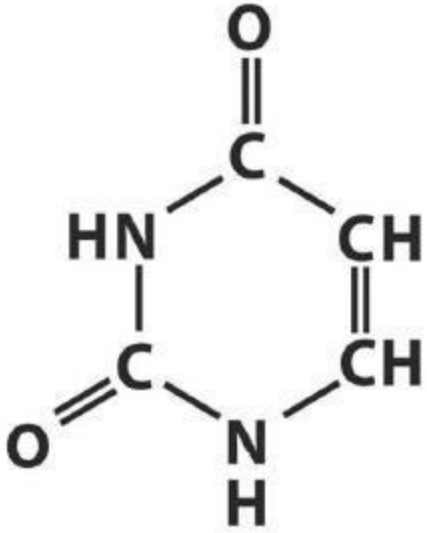
Nucleotides and Nucleic Acids



Cytosine



Thymine
(DNA)



Uracil
(RNA)

Pyrimidines

Nucleotides and Nucleic Acids

- Cytosine when combines with

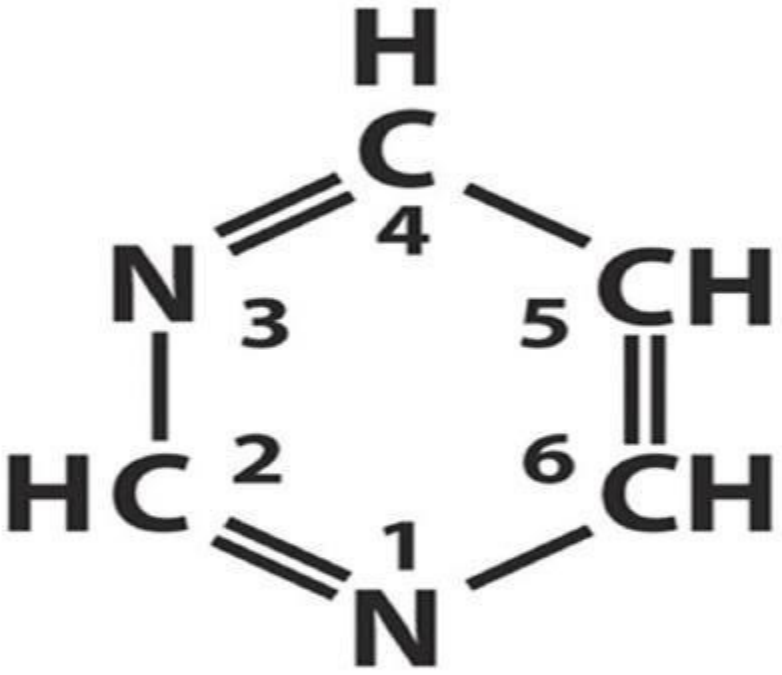
pentose it becomes deoxycytidine and cytidine

- Thymine becomes thymidine and deoxythymidine
- Uracil (U) becomes uridine and deoxyuridine
- depending on the type of sugar.

Nucleotides and Nucleic Acids

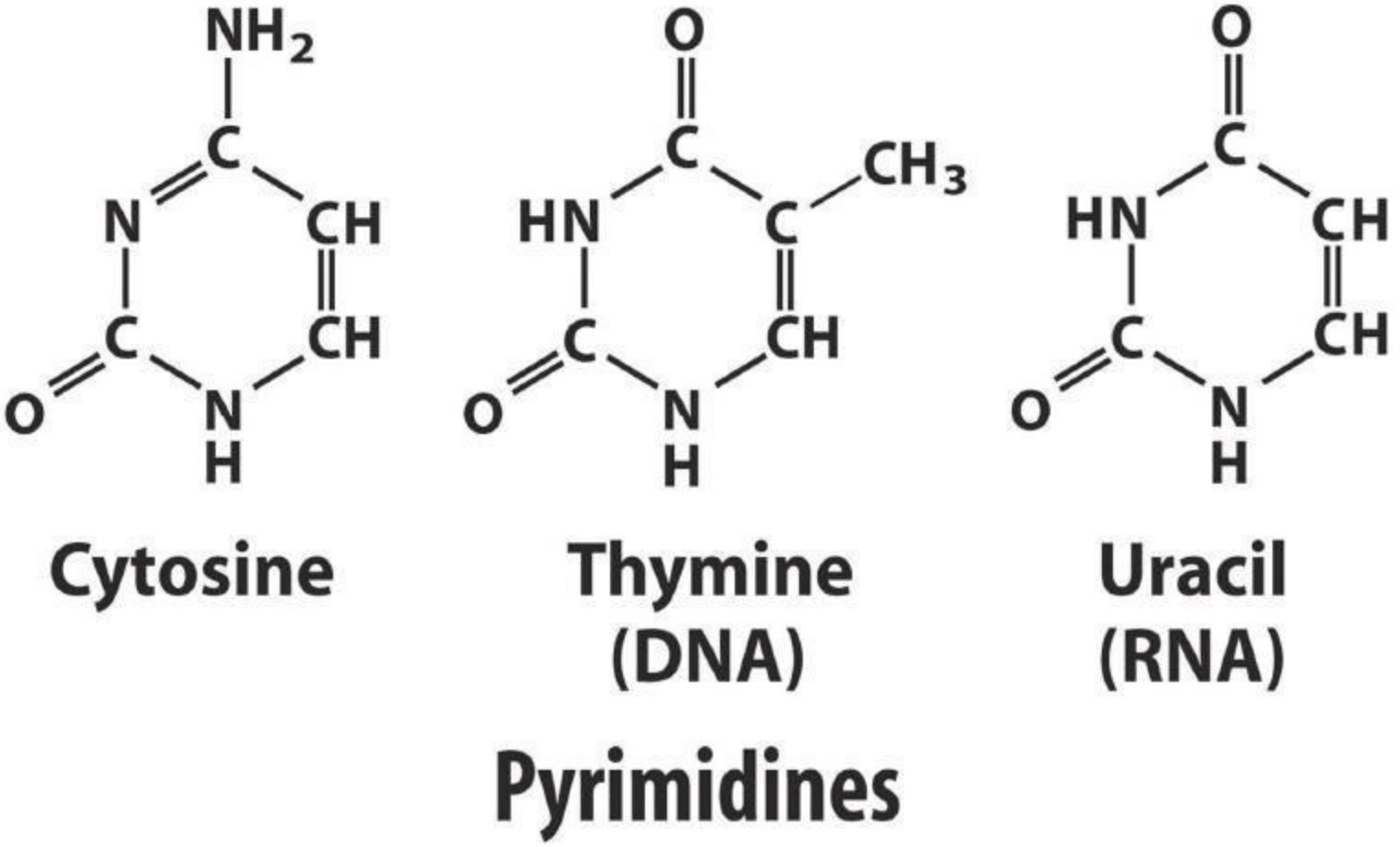
- Cytosine is 2-oxy-4-amino-pyrimidine
- Thymine is 2,4-dioxy-5-methyl-pyrimidine
- Uracil is 2,4-dioxy-pyrimidine

Nucleotides and Nucleic Acids



Pyrimidine

Nucleotides and Nucleic Acids



Cytosine; 2-oxy-4-amino-pyrimidine.Thymine; 2,4-dioxy-5-methyl-pyrimidine

Nucleotides and Nucleic Acids

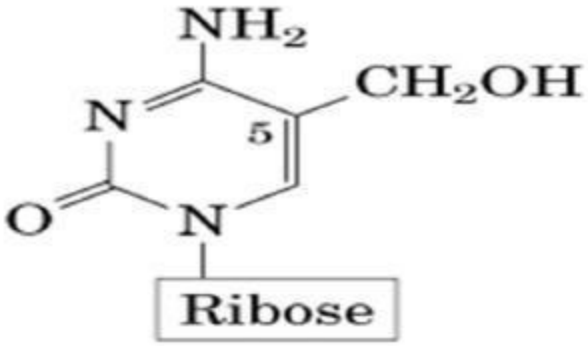
- T and U differ by only one methyl group, which is present on T but absent on U

Nucleotides and Nucleic Acids

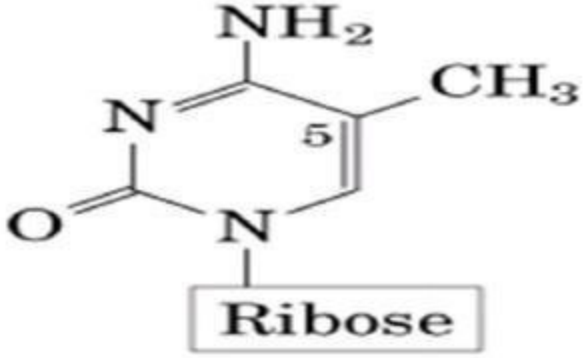
- Minor Pyramidine Bases:
- Dihydrouridine (DHU) ,
- 5-Methyl Cytadine &
- 5-Hydroxy-Methyl Cytadine

- Unnatural Pyramidine Bases:
- Fluorouracil (5FU) &
- 6-Aza Cytosine (AZC)

Nucleotides and Nucleic Acids



5-Hydroxymethylcytidine



5-Methylcytidine

Nucleotides and Nucleic Acids

- Unlike the purine ring, which is not cleaved in human cells, the pyrimidine ring is opened and degraded
- to highly soluble products, β-alanine and β-amino-iso-butyrate,
- with the production of NH_3 and CO_2.

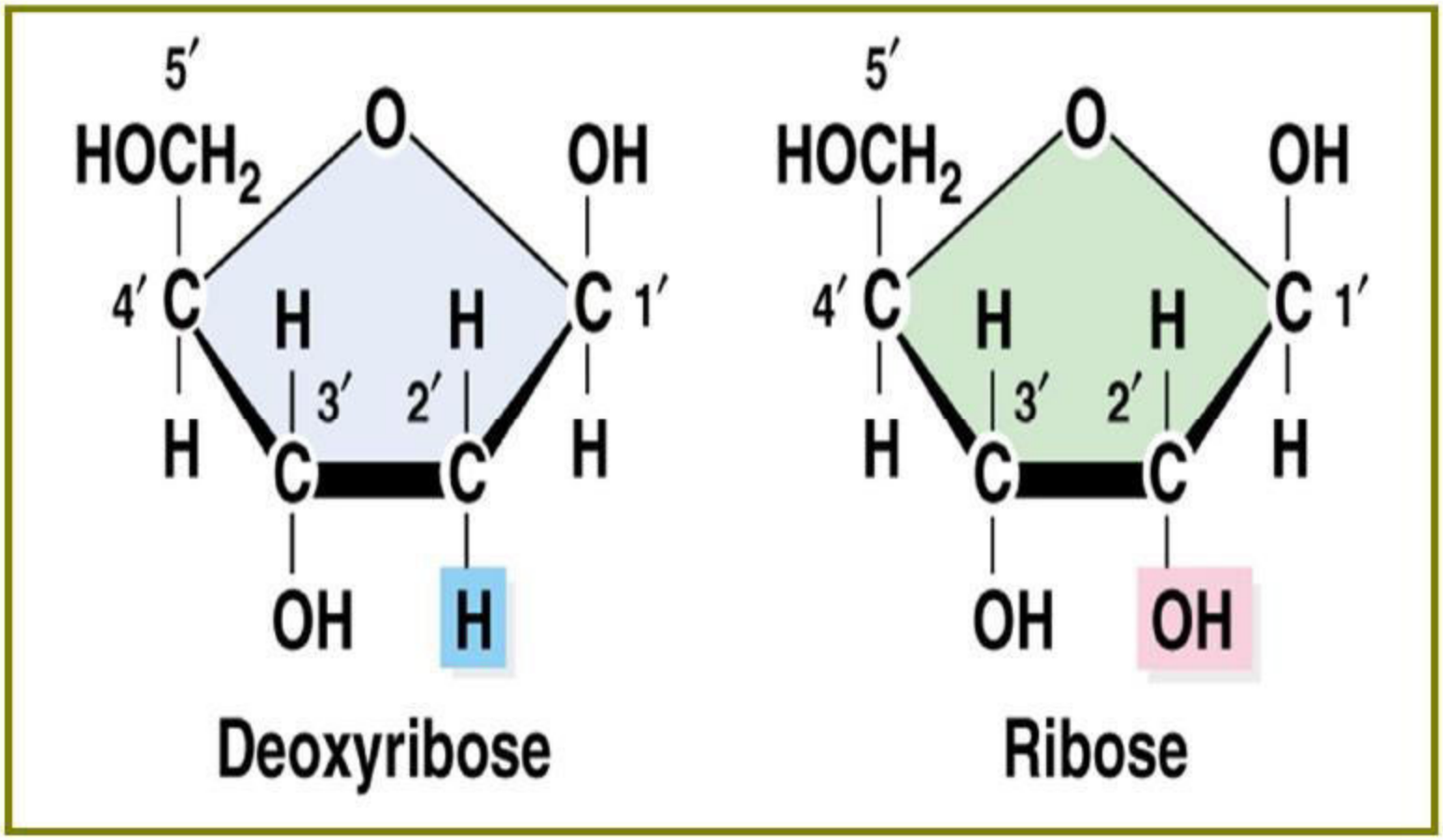
END

Nucleotides and Nucleic Acids

Composition of Nucleotides (Contd.)

- **Pentose Sugar**
- D-ribose and 2-deoxy D-ribose are the only sugars so far found in the nucleic acids.

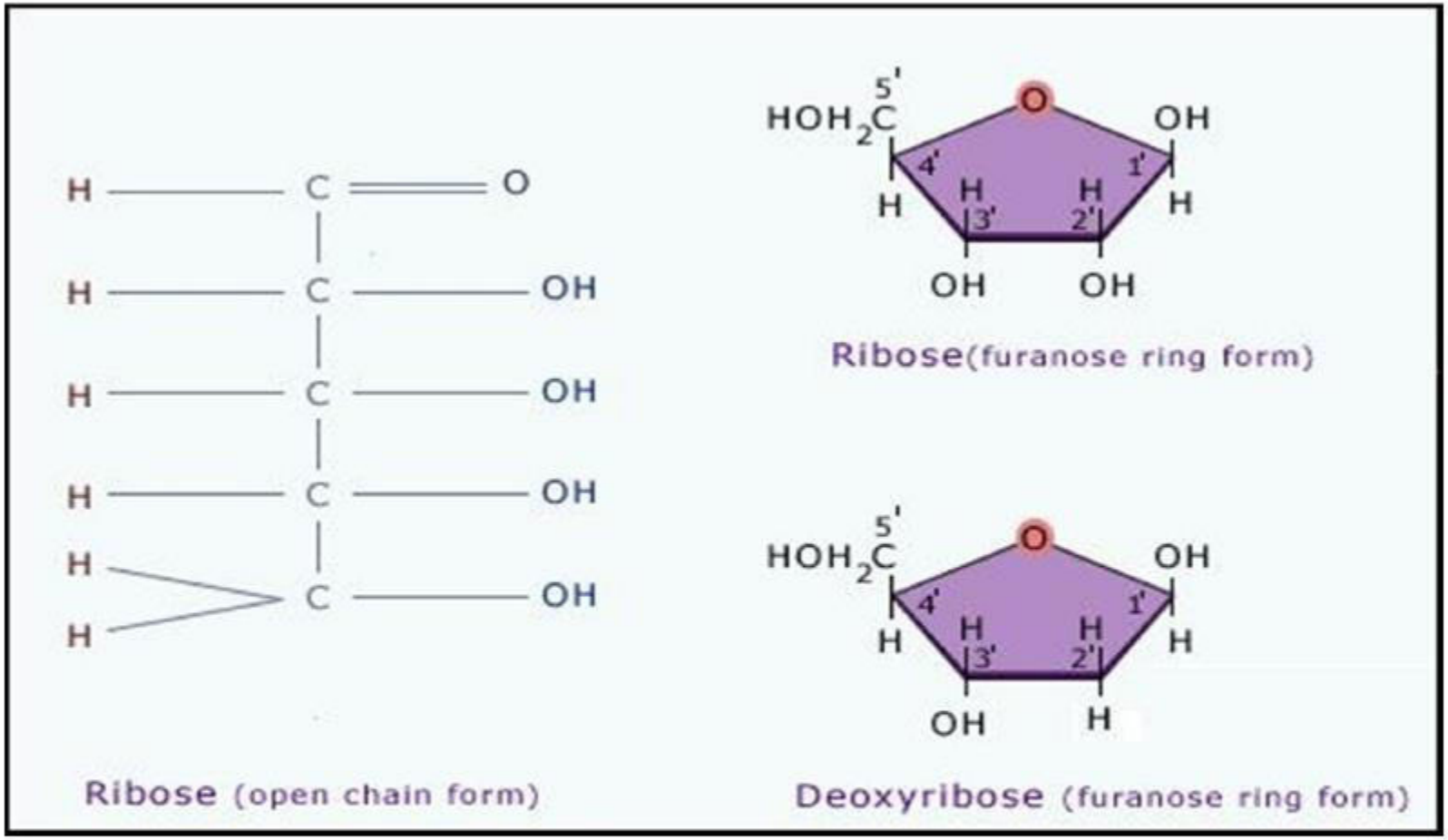
Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- These aldo pentoses belong to D-family
- They are present as Furanose (ring) in the form of β-Anomer

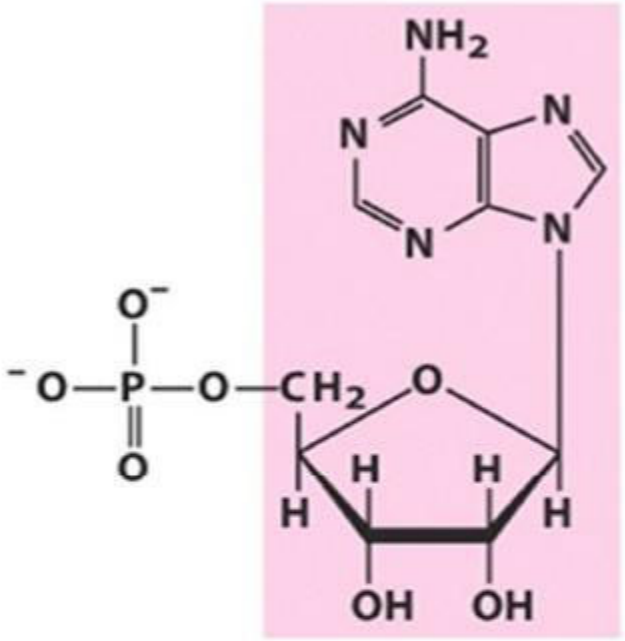
Nucleotides and Nucleic Acids

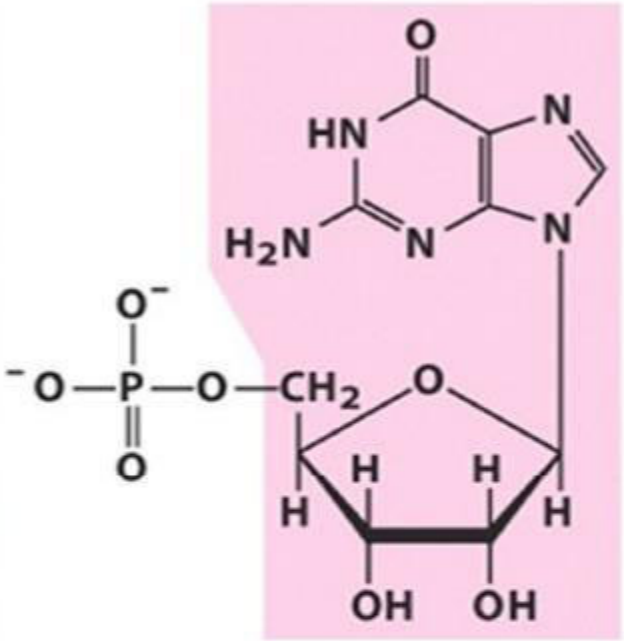


Nucleotides and Nucleic Acids

- The addition of a pentose sugar to a base produces a **nucleoside**
  - If the sugar is D-ribose, a ribonucleoside is produced
  - If the sugar is 2-deoxy D- ribose, a deoxyribonucleoside is produced

Nucleotides and Nucleic Acids



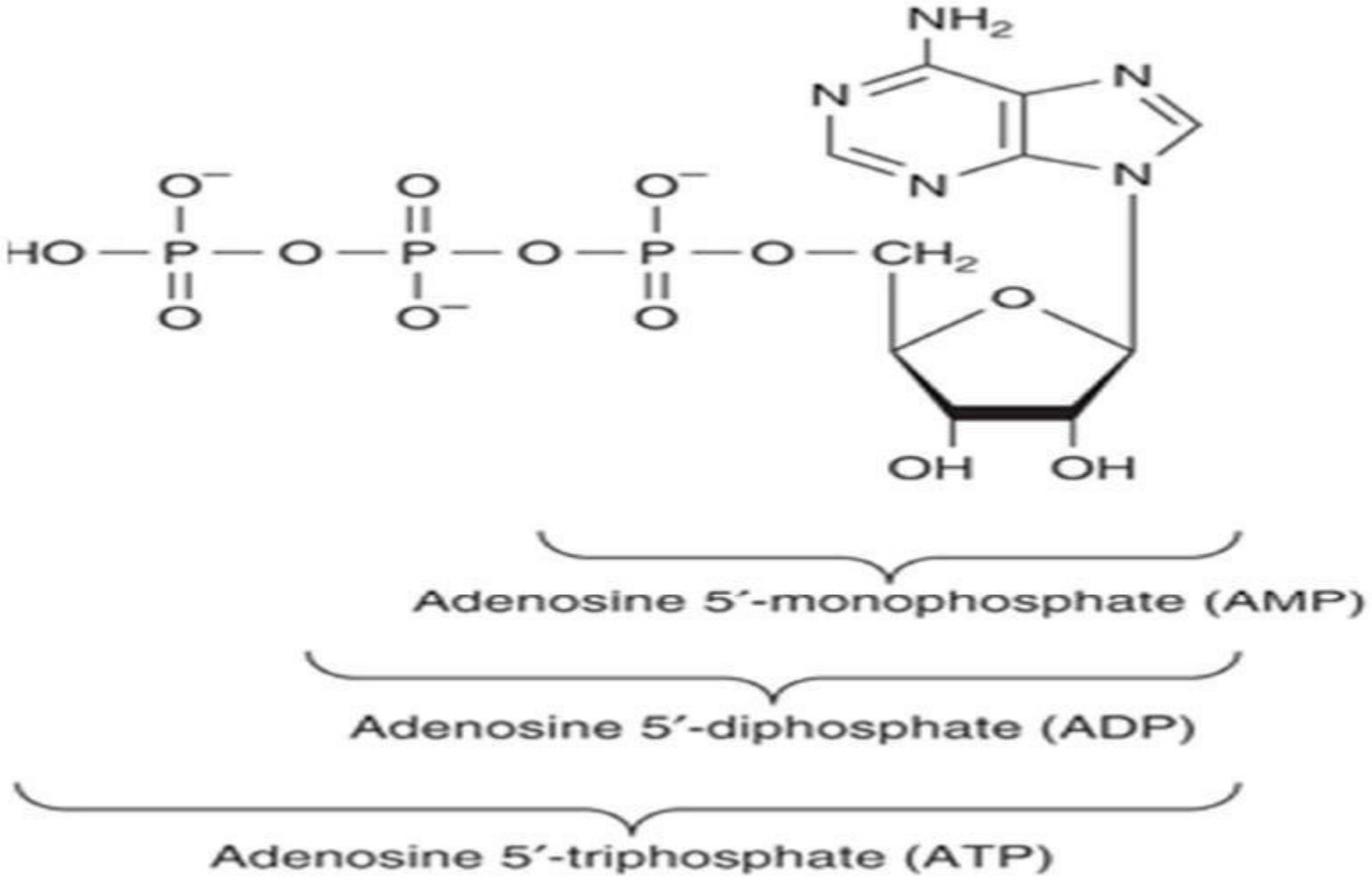


Nucleotide:	Adenylate (adenosine 5′-monophosphate)	Guanylate (guanosine 5′-monophosphate)
Symbols:	A, AMP	G, GMP
Nucleoside:	Adenosine	Guanosine

Nucleotides and Nucleic Acids

- **Phosphate group**
- There may be one, two, or three phosphate groups present in nucleotides.
- **Nucleotides** are monophosphate, diphosphate, or triphosphate esters of nucleosides

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- These phosphate groups give an over all negative charge to the nucleotides.

END

Nucleotides and Nucleic Acids

Properties of Nitrogenous Bases

- **Aromatic:** The Nitrogen containing bases are aromatic i.e. they have alternate double bonds

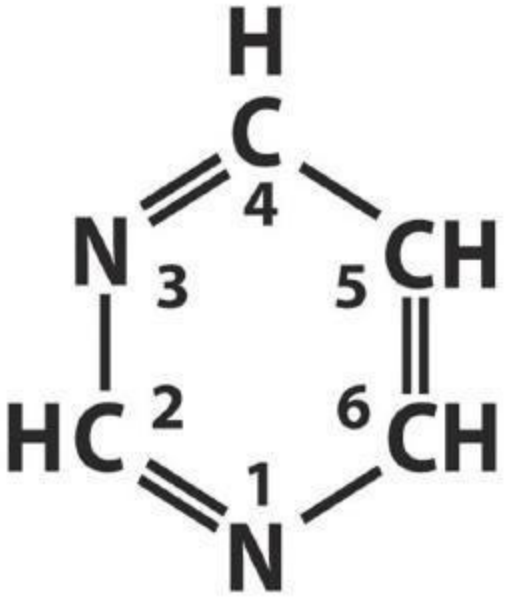
Nucleotides and Nucleic Acids

- **Heterocyclic:**
- They are heterocyclic i.e. structures that contain other atoms in addition to carbon, such as nitrogen in the ring structure

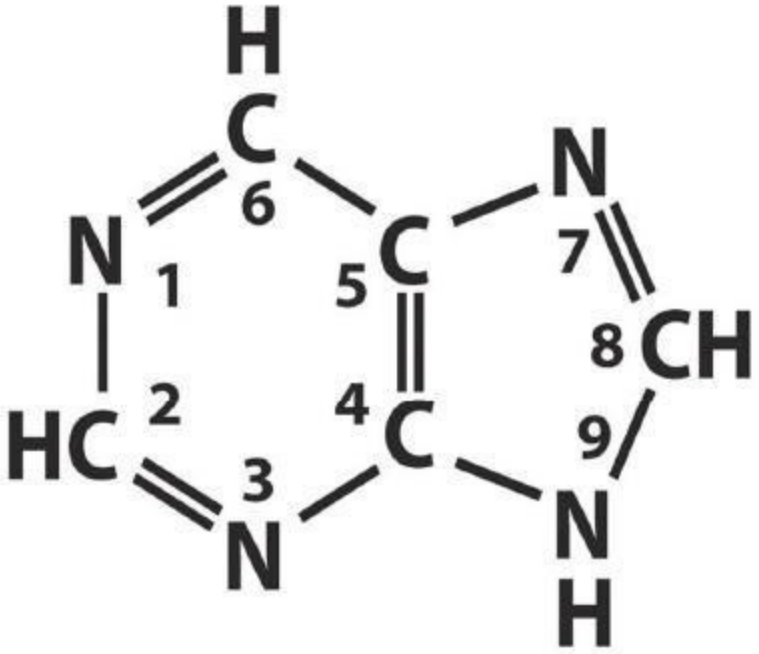
Nucleotides and Nucleic Acids

- The six-atom rings of purines and pyrimidines are numbered in opposite directions.

Nucleotides and Nucleic Acids



Pyrimidine



Purine

Nucleotides and Nucleic Acids

- **Weak Bases:** Purines or pyrimidines with an –NH_2 group are weak bases

Nucleotides and Nucleic Acids

- **Functional Groups:** The most important functional groups of pyrimidines and purines are
- ring nitrogens
- carbonyl groups
- exocyclic amino groups

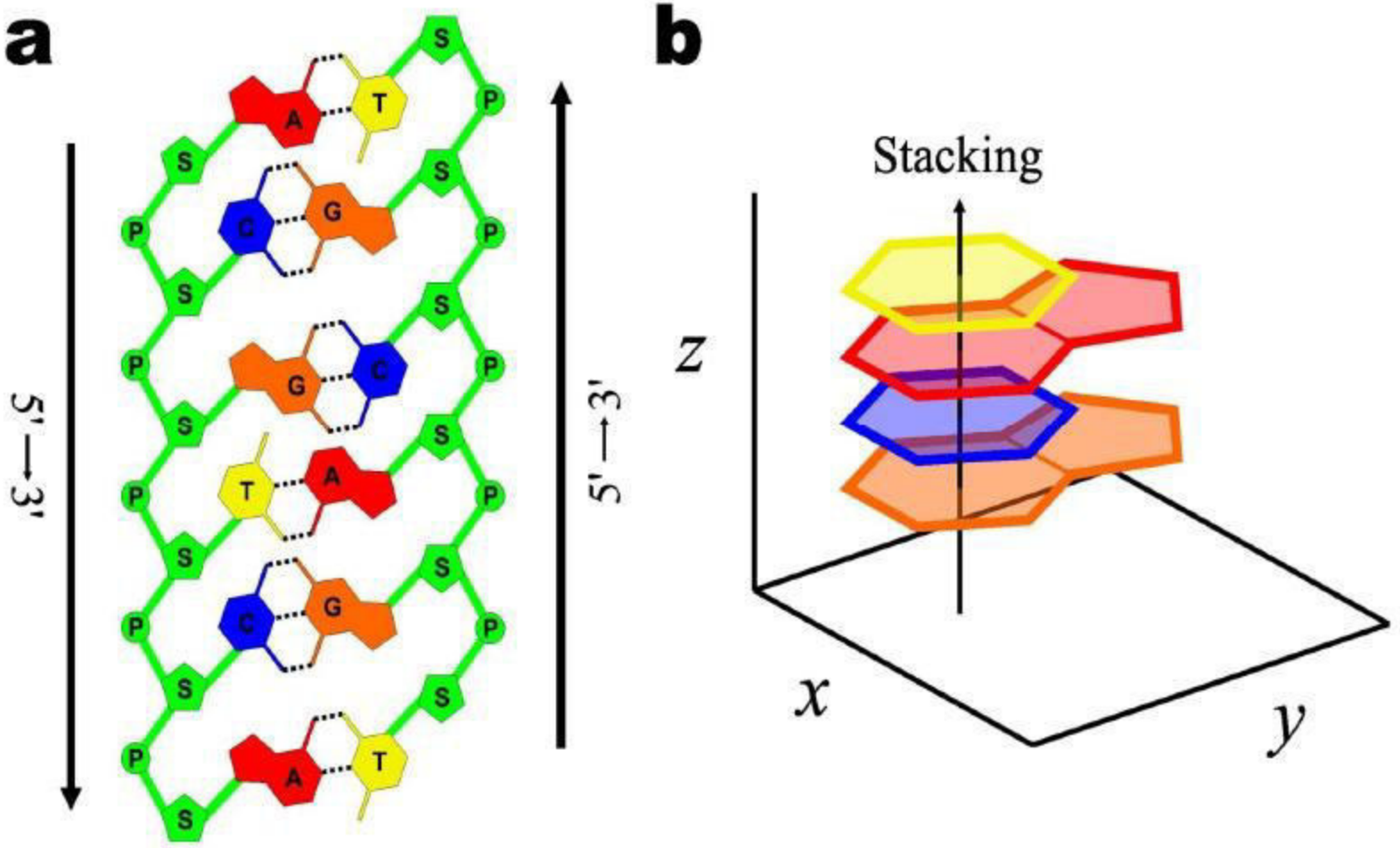
Nucleotides and Nucleic Acids

- **Hydrophobicity:**
- The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral cell pH

Nucleotides and Nucleic Acids

- **Stacking Interaction:** Hydrophobic stacking interactions in which two or more bases are positioned with the planes of their rings parallel (like a stack of coins) are one of two important modes of interaction between bases in nucleic acids.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Base stacking helps to minimize contact of the bases with water, and these interactions are very important in stabilizing the three-dimensional structure of nucleic acids.

END

Nucleotides and Nucleic Acids

Properties of Nitrogenous Bases (Contd.)

- **UV light absorbance:** The conjugated double bonds of purine and pyrimidine derivatives absorb ultraviolet light.

Nucleotides and Nucleic Acids

- Nucleic acids are characterized by a strong absorption at wavelengths near 260 nm.
- The mutagenic effect of ultraviolet light is due to its absorption by nucleotides that results in chemical modifications in DNA.

Nucleotides and Nucleic Acids

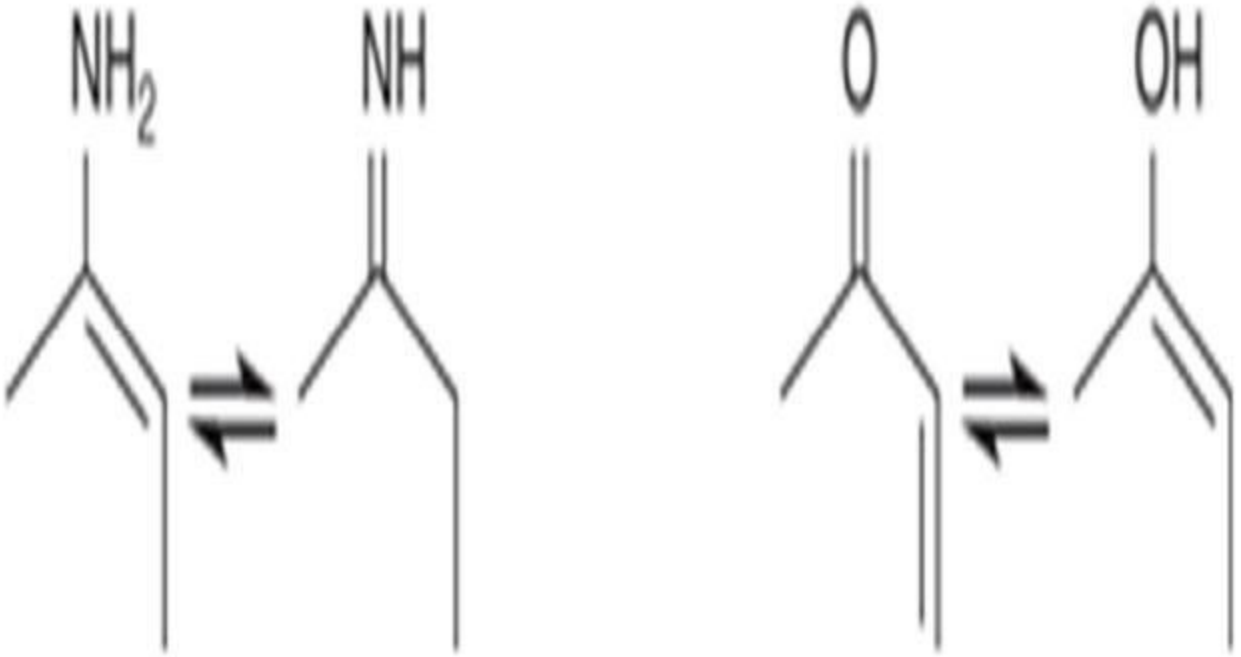
- This property is also utilized in quantitative and qualitative analysis of nucleotides and nucleic acids.

Nucleotides and Nucleic Acids

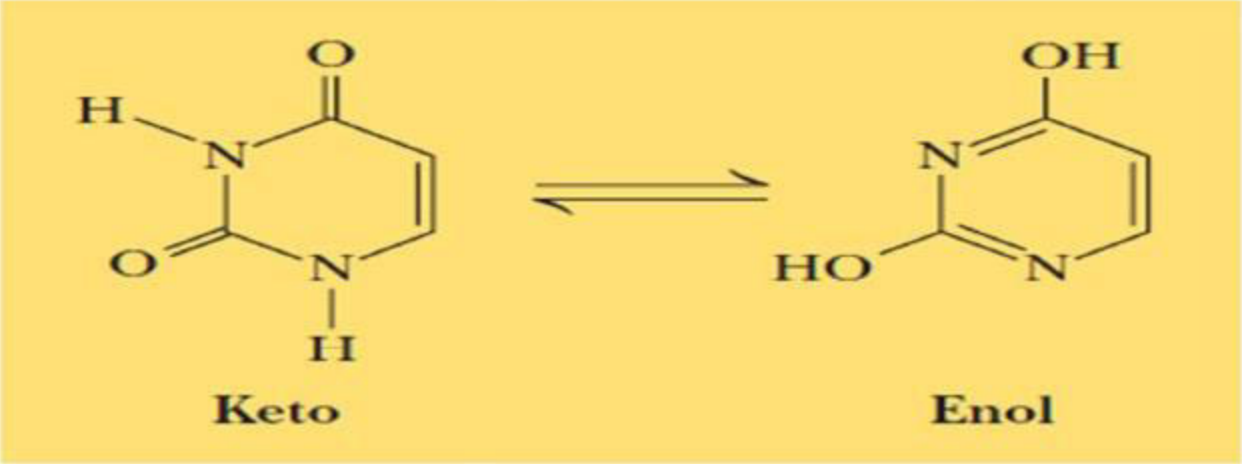
- **Tautomerism:**
- All these bases can exist in keto-enol or amine-imine form.

- At physiologic pH keto and amine form is predominant.

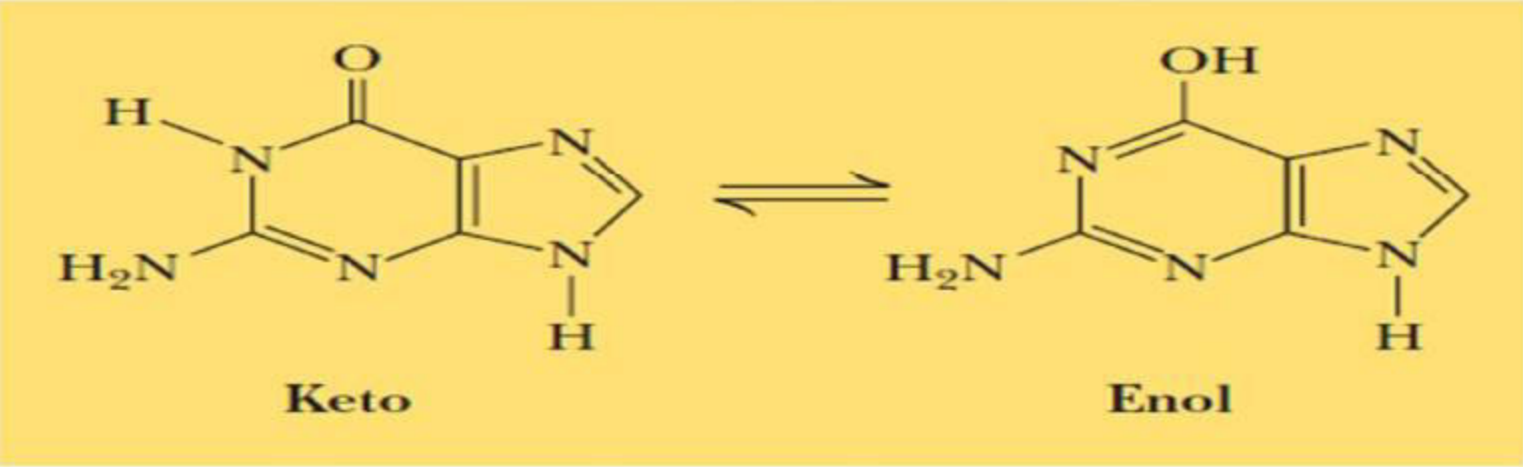
Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids



The keto–enol tautomerization of uracil.



The tautomerization of the purine guanine.

Nucleotides and Nucleic Acids

- Note that the smaller pyrimidine molecule has the *longer* name and

- the larger purine molecule the *shorter* name.

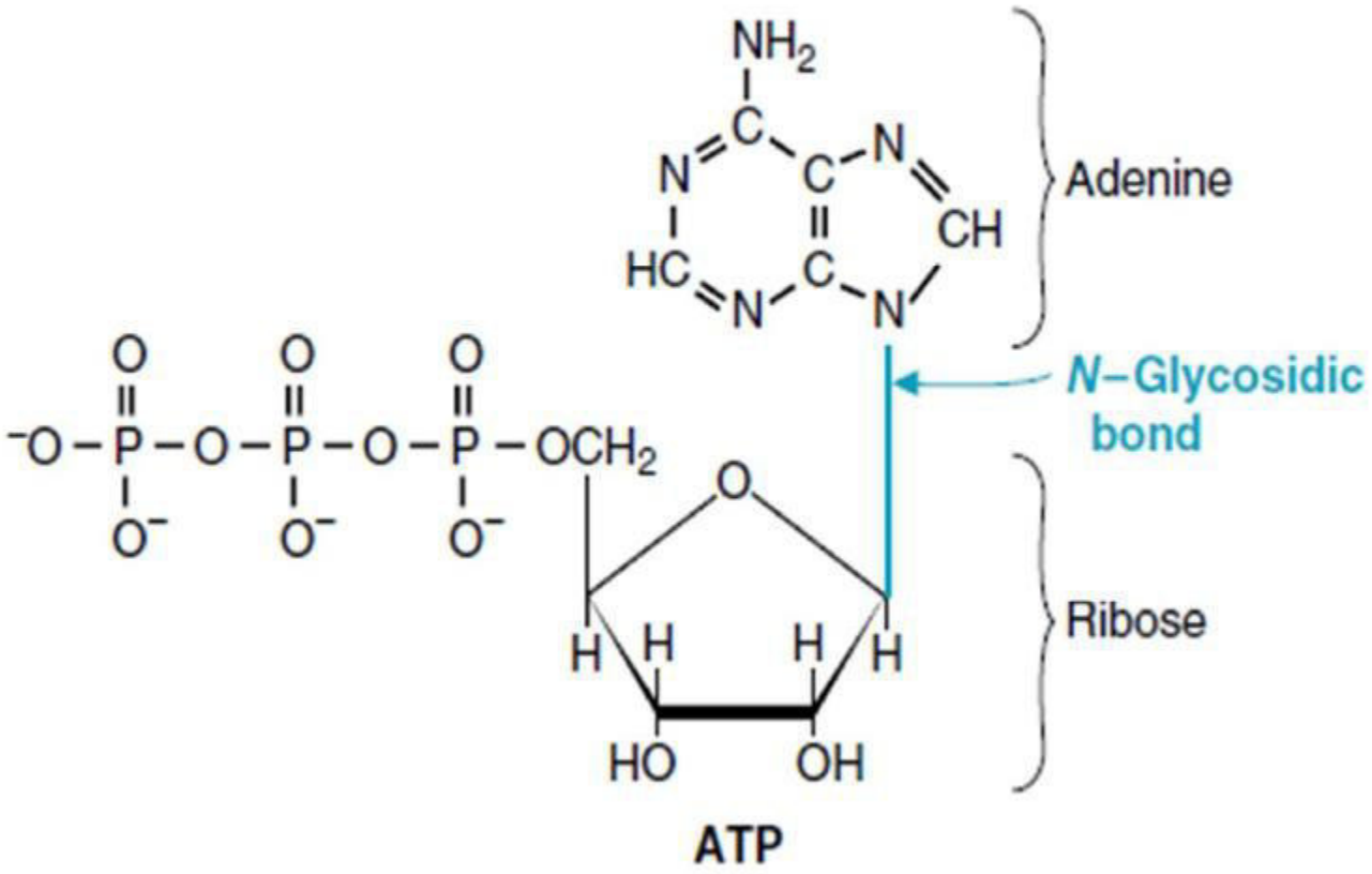
END

Nucleotides and Nucleic Acids

N-glycosidic bond

- Sugars are linked to the heterocycle by a β -N-glycosidic bond, almost always to the
  - N-1 of a pyrimidine
  - N-9 of a purine

Nucleotides and Nucleic Acids

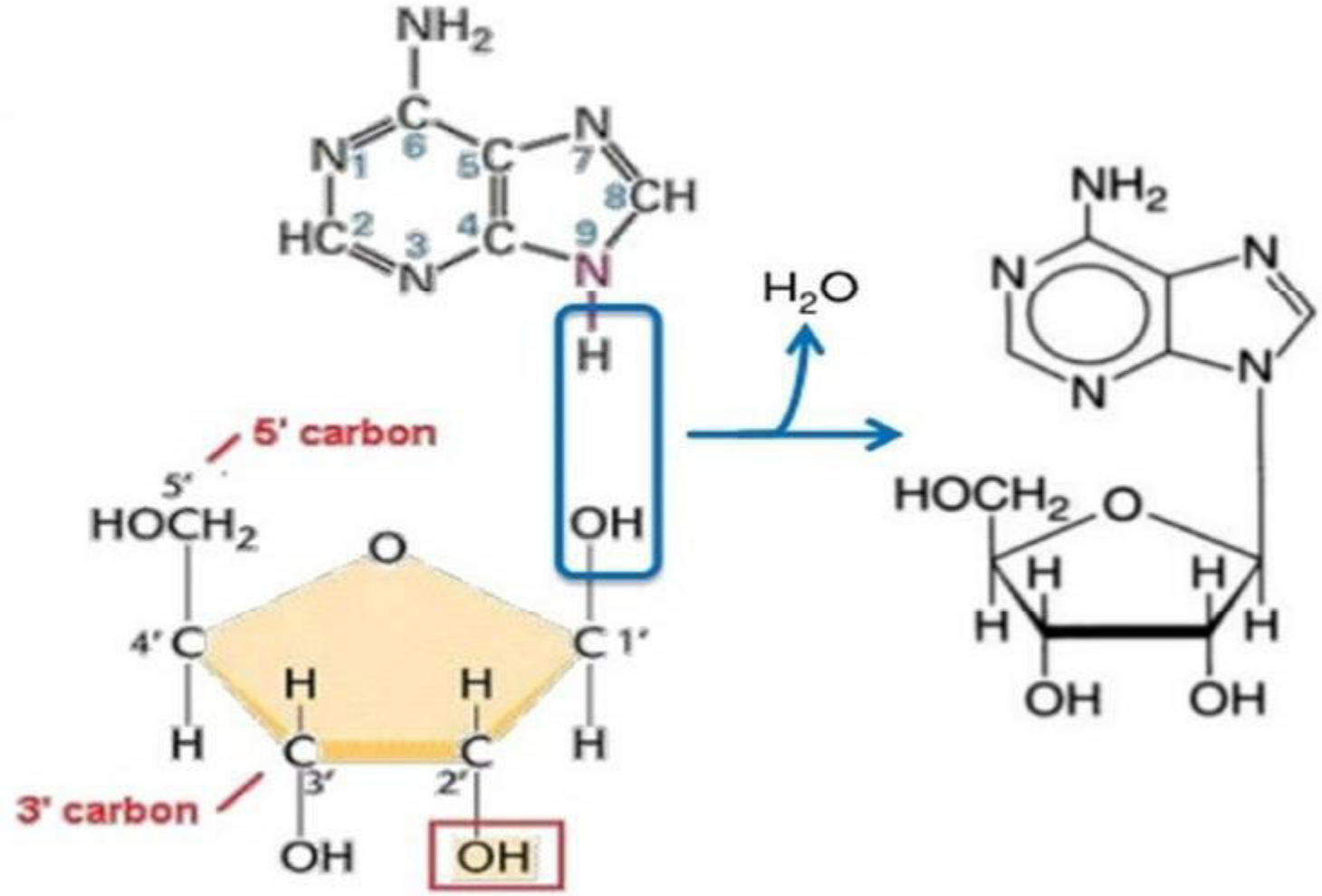


ATP

Nucleotides and Nucleic Acids

- The N-glycosyl bond is formed by removal of the elements of water
- a hydroxyl group from the pentose and
- hydrogen from the base

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Thus it is a condensation reaction.
- Similar to O-glycosidic bond formation in carbohydrates

Nucleotides and Nucleic Acids

- However, N-glycosidic bonds, have Nitrogen atom instead of oxygen linking the two residues.

Nucleotides and Nucleic Acids

- the addition of the glycosidic bond to nitorgenous base is indicated by the name change
- such as from adenine to aden**osine** for the glyc**osidic** bond

END

Nucleotides and Nucleic Acids

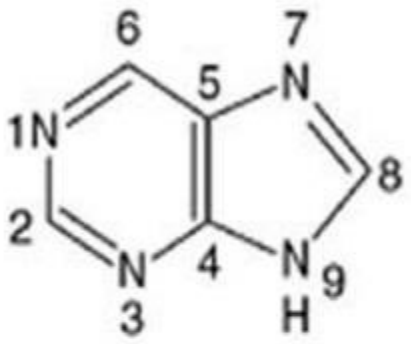
Numbering of Carbon and Nitrogen Atoms

- The carbon and nitrogen atoms in the rings of the base and the sugar are numbered separately.

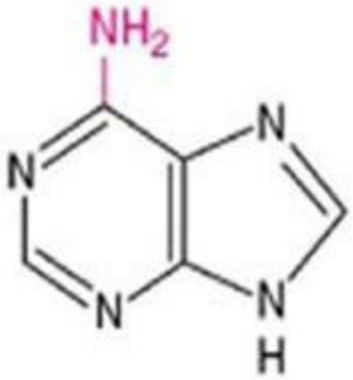
Nucleotides and Nucleic Acids

- The atoms in the rings of the **bases** are numbered
  - 1 to 6 in pyrimidines &
  - 1 to 9 in purines

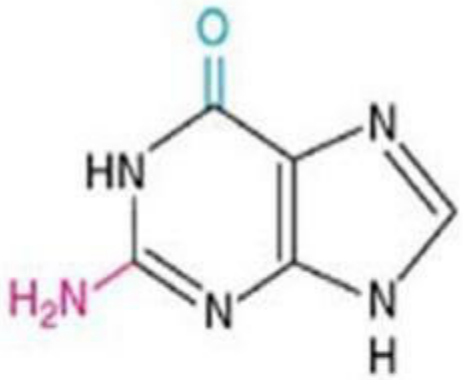
Nucleotides and Nucleic Acids



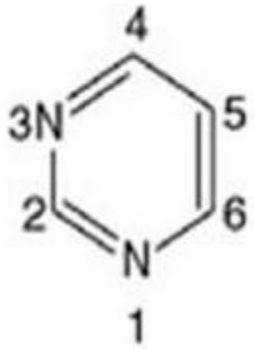
Purine



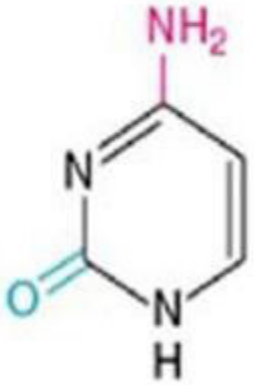
Adenine



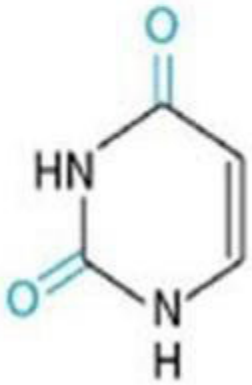
Guanine



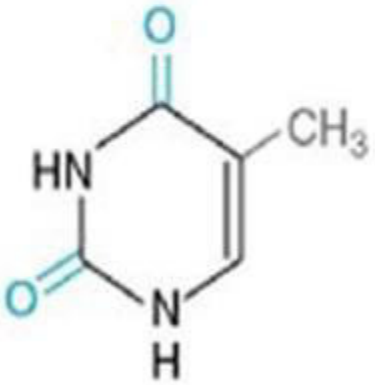
Pyrimidine



Cytosine



Uracil



Thymine

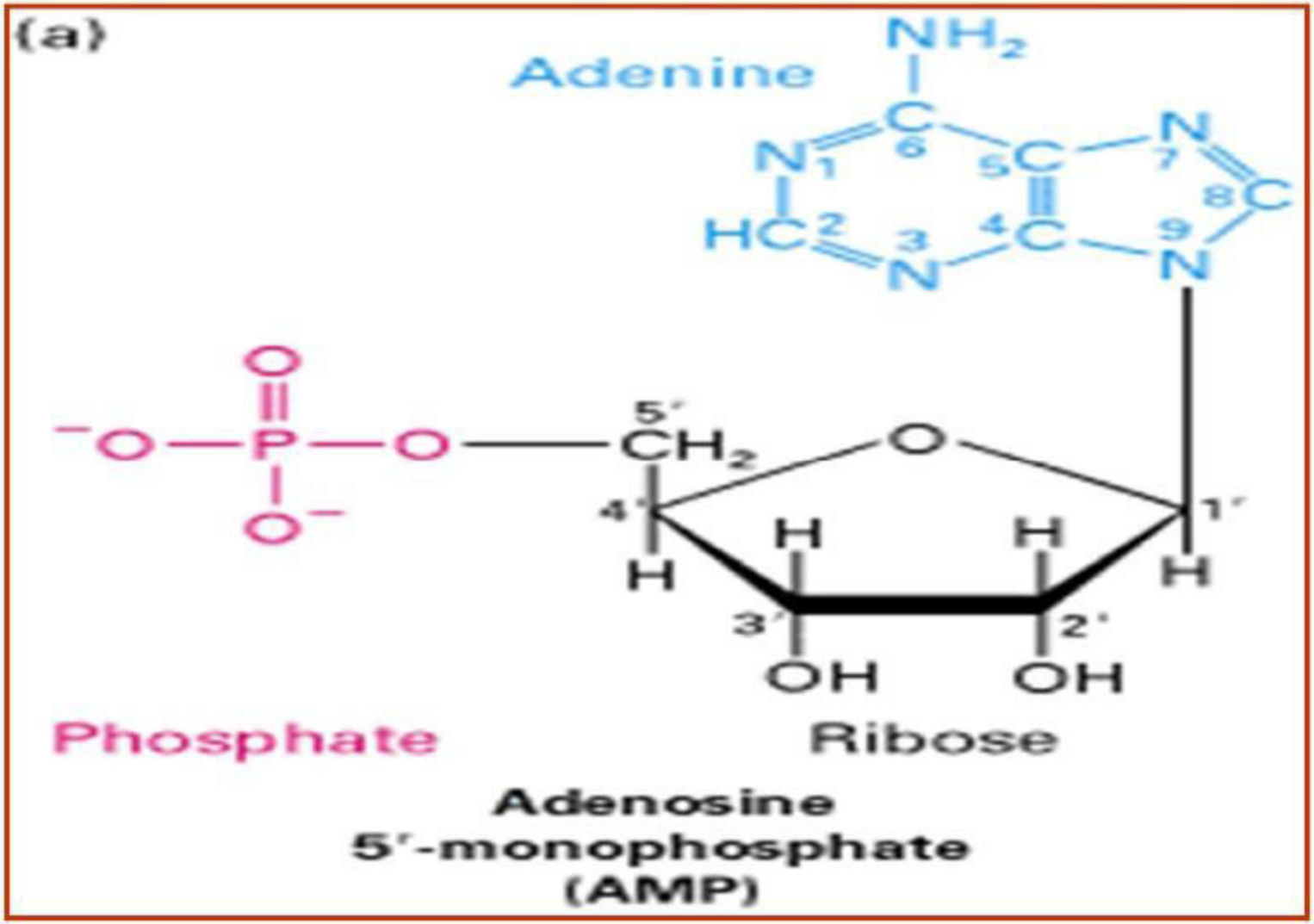
Nucleotides and Nucleic Acids

- In the pentoses of nucleotides and nucleosides the carbon numbers are given a prime (') designation to distinguish them from the numbered atoms of the nitrogenous base.

Nucleotides and Nucleic Acids

- The carbons in the **pentose** are numbered 1' to 5 '.
- Numerals with a prime (e.g., 2' or 3') distinguish atoms of the sugar from those of the heterocycle.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Thus, when the 5'-carbon of a nucleoside (or nucleotide) is referred to, a carbon atom in the pentose, rather than an atom in the base, is being specified.

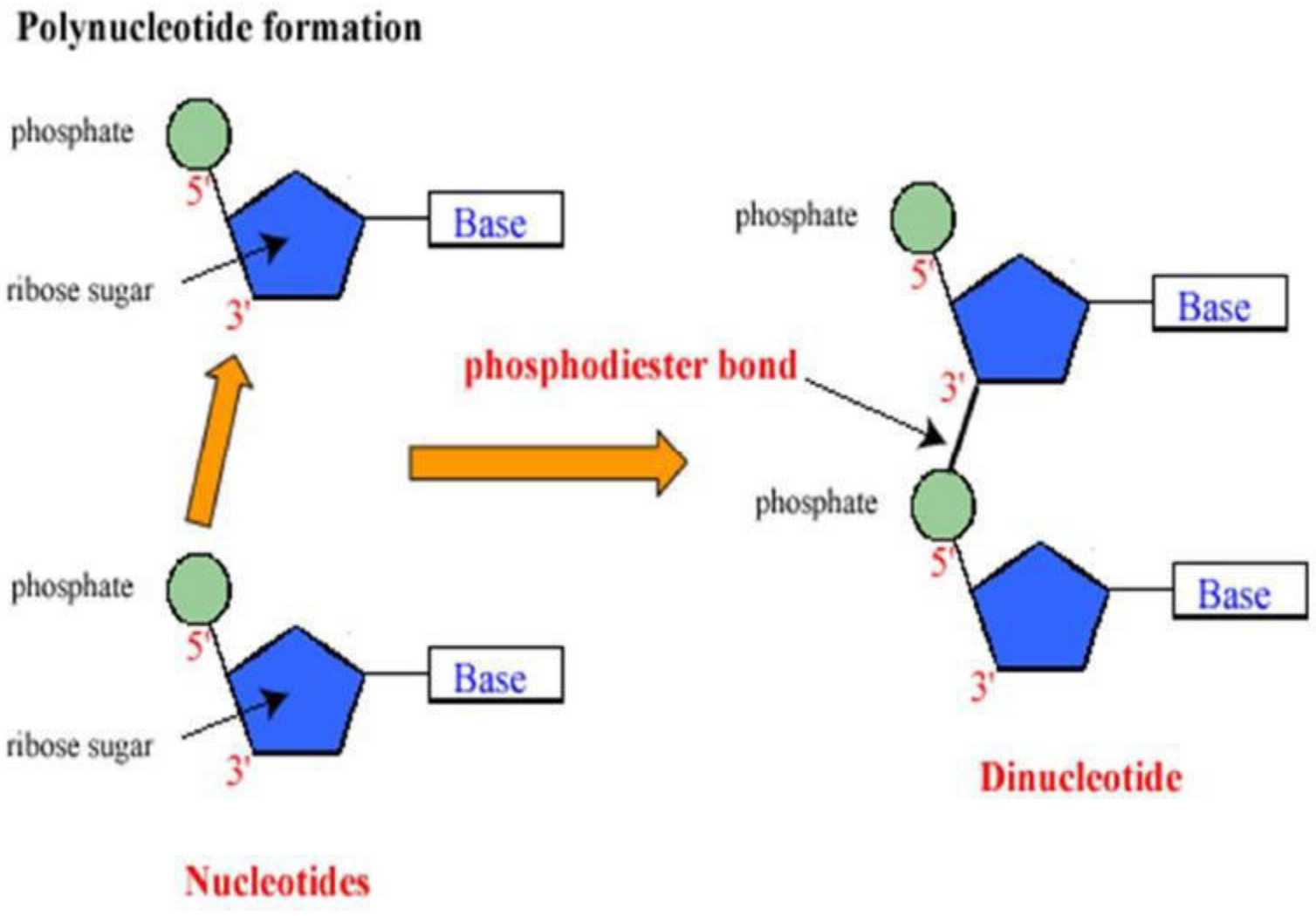
END

Nucleotides and Nucleic Acids

Phosphodiester Bond

- When two or more nucleotides combine together a phosphodiester bond is formed.

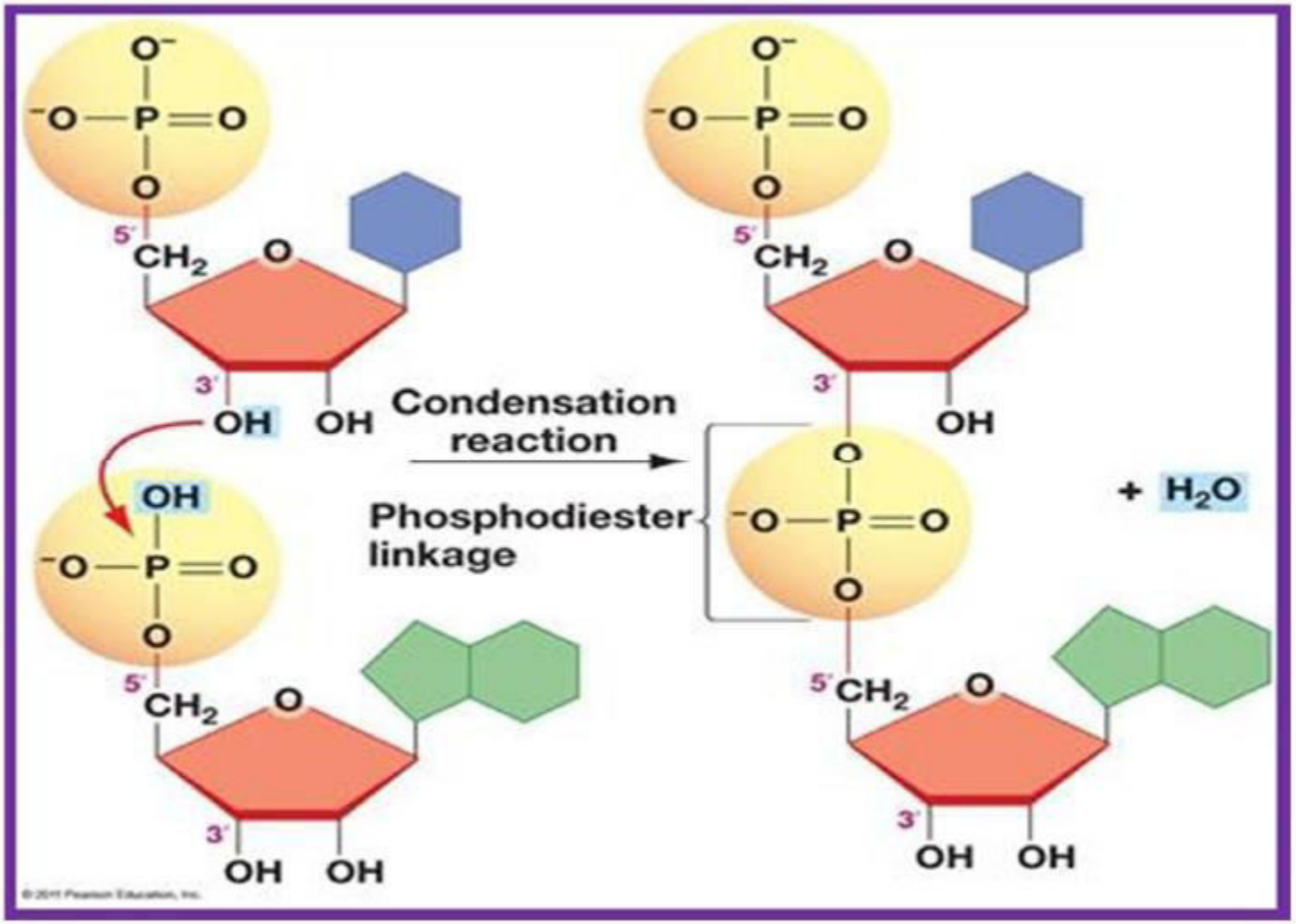
Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- This bond is formed mainly between the 3′ OH group of sugar of one nucleotide and
- 5′ PO_4 group of sugar of another nucleotide creating a phosphodiester linkage.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- By definition, the 5′ end lacks a nucleotide at the 5′ position and the
- 3′ end lacks a nucleotide at the 3′ position.

Nucleotides and Nucleic Acids

- The sugar and phosphate group is called the backbone of the nucleic acid.
- The backbones of both DNA and RNA are hydrophilic.
- The hydroxyl groups of the sugar residues form hydrogen bonds with water.

Nucleotides and Nucleic Acids

- The phosphate groups, are completely ionized and negatively charged at pH 7

Nucleotides and Nucleic Acids

- The negative charges are generally neutralized by ionic interactions with positive charges which are present on proteins, metal ions, and polyamines.

Nucleotides and Nucleic Acids

- Depending on the number of nucleotides
- mono- di-, tri-, oligo- and polynucleotides are formed
- as a result of phosphodiester bond formation.

END

Nucleotides and Nucleic Acids

Cyclic Nucleotides

- There are two important cyclic nucleotides:

- Cyclic AMP cAMP
- Cyclic GMP cGMP

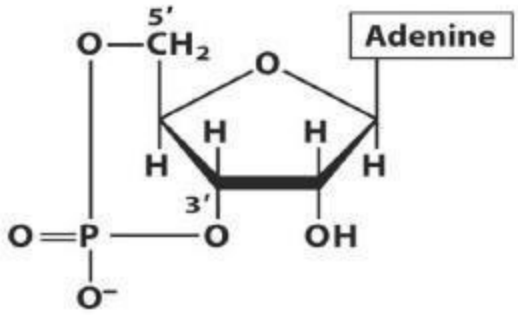
Nucleotides and Nucleic Acids

Cyclic AMP is a cyclic nucleotide

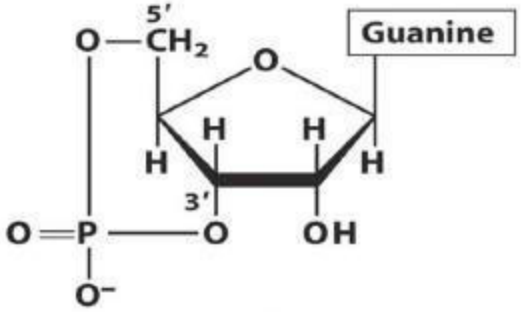
- cAMP is synthesized in tissues from ATP

- chemically it is 3′ -5′ adenosine monopohosphate.

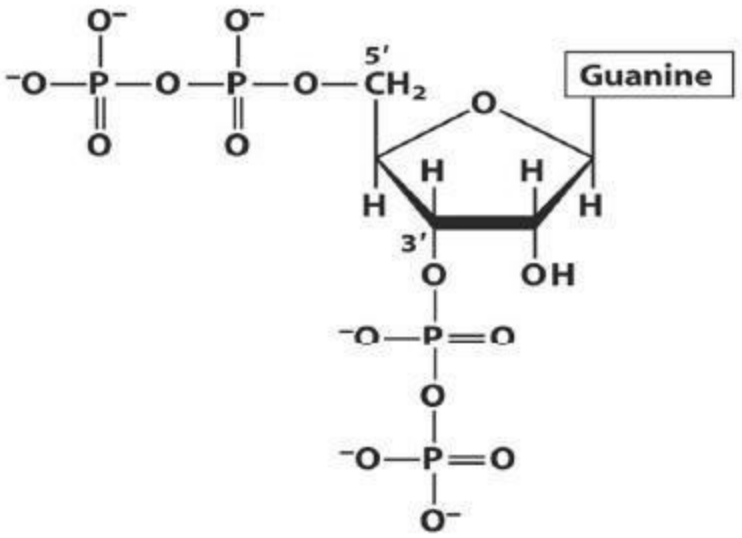
Nucleotides and Nucleic Acids



Adenosine 3′,5′-cyclic monophosphate
(cyclic AMP; cAMP)



Guanosine 3′,5′-cyclic monophosphate
(cyclic GMP; cGMP)



Guanosine 5′-diphosphate, 3′-diphosphate
(guanosine tetraphosphate)
(ppGpp)

Nucleotides and Nucleic Acids

Functions of c-AMP

- Acts as second messenger in the cell
- It has role in glycogen metabolism
- ↑cAMP, ↑glycogenolysis
- ↑cAMP ↑TAG metabolism
- ↑cAMP ↑ lipolysis

Second messengers relay signals received at receptors on the cell surface ,hormones, growth factors, to target

Nucleotides and Nucleic Acids

- It decreases cholesterol synthesis
- It causes activation of protein kinases which in turn;
- activate or deactivate other enzymes.

Nucleotides and Nucleic Acids

- It regulates the cell membrane permeability, by increasing permeability of cell membrane to H_2O, Na^+, K^+& Ca^{+2}

Nucleotides and Nucleic Acids

- Moreover, it regulates
- insulin secretion,
- catecholamine biosynthesis &
- Melatonin synthesis

Nucleotides and Nucleic Acids

Cyclic GMP is synthesized from GTP

- It serves as a second messenger in response to nitric oxide during relaxation of smooth muscle (especially blood vessels) so it has role in smooth muscle relaxation and vasodilatation.

Nucleotides and Nucleic Acids

It also has role in

- Protein phosphorylation
- Neurotransmission
- Insulin action
- Regulation of sodium channels

END

Nucleotides and Nucleic Acids

DNA

- It stands for Doeoxyribonucleic acid
- DNA is present in nuclear chromosomes of eukaryotes
- mitochondria
- chloroplasts and
- plasmids of prokaryotes

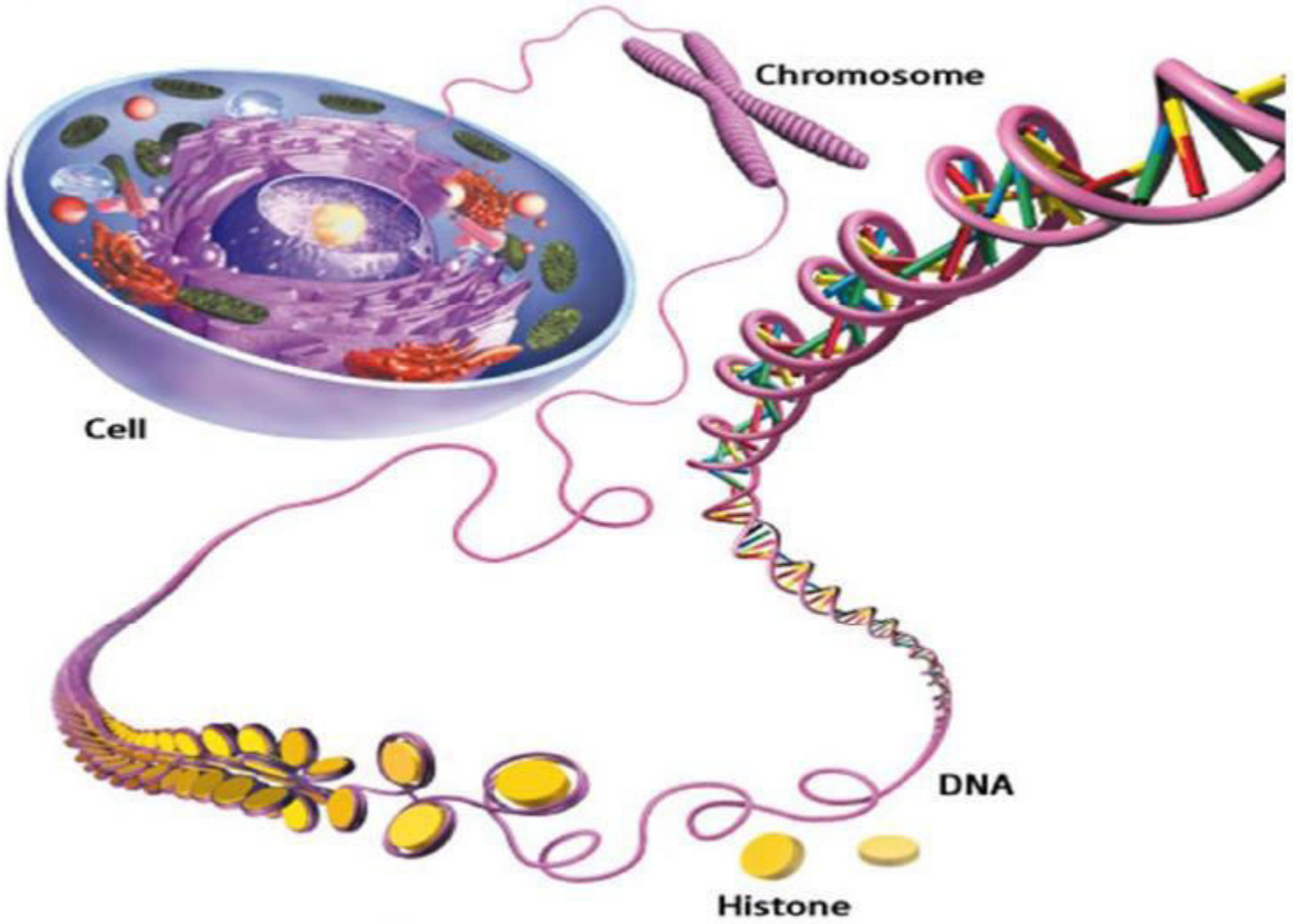
Nucleotides and Nucleic Acids

- DNA is a polymer of deoxyribonucleoside monophosphates
- covalently linked by 3’ 5’-phosphodiester bonds.
- DNA is a repository of genetic information

Nucleotides and Nucleic Acids

- In eukaryotic cells, DNA is present in the chromosomes in the nucleus.
- It is found associated with basic proteins HISTONES and also various other proteins present in nucleus (nucleoproteins)

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Prokaryotic cells lack nuclei, and have a single chromosome.
- The protein-DNA complex is present in a non membrane bound region known as nucleoid.
- It also contain non-chromosomal DNA in the form of plasmids.

Nucleotides and Nucleic Acids

- Nucleic acid structure can be described in terms of hierarchical levels of complexity (primary, secondary, tertiary)

A **hierarchy** (Greek) is an arrangement of items

Nucleotides and Nucleic Acids

- The primary structure of a nucleic acid is its
- covalent structure and
- nucleotide sequence.

Nucleotides and Nucleic Acids

- Any regular, stable structure taken up by some or all of the nucleotides in a nucleic acid can be referred to as secondary structure
- In DNA double **helix**, the two strands of DNA are held together by hydrogen bonds.

Nucleotides and Nucleic Acids

- The nucleotides on one strand base pairs with the nucleotide on the other strand.
- The secondary structure is responsible for the shape that the **nucleic** acid assumes.

Nucleotides and Nucleic Acids

- The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids is generally considered tertiary structure

Nucleotides and Nucleic Acids

DNA Interactions

- The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral pH of the cell.
- Hydrophobic stacking interactions in which bases are positioned with the planes of their rings parallel

Nucleotides and Nucleic Acids

- Hydrophobic stacking are an important interaction between bases in nucleic acids.
- The stacking also involves a combination of van der Waals and dipole-dipole interactions between the bases.

Nucleotides and Nucleic Acids

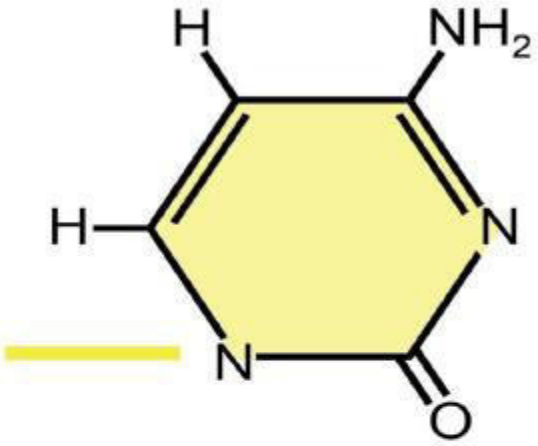
- Base stacking helps to minimize contact of the bases with water.
- Therefore base-stacking interactions are very important in stabilizing the three-dimensional structure of nucleic acids.

Nucleotides and Nucleic Acids

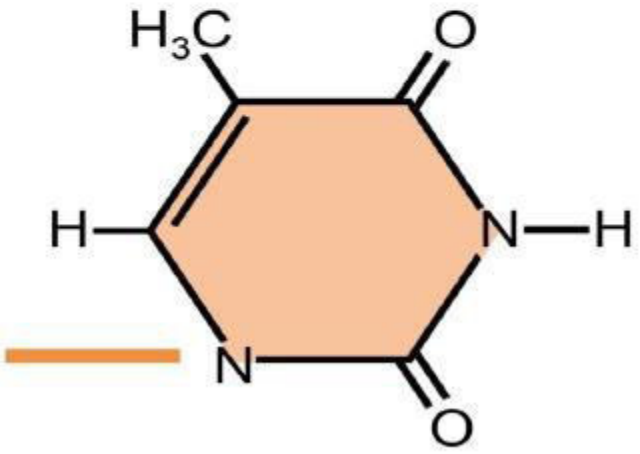
- The most important functional groups of pyrimidines and purines are
- ring nitrogens,
- carbonyl groups, and
- exocyclic amino groups.

Nucleotides and Nucleic Acids

Cytosine



Thymine
(DNA Only)



Blue-H. Red –O, Grey—C, Yellow -N

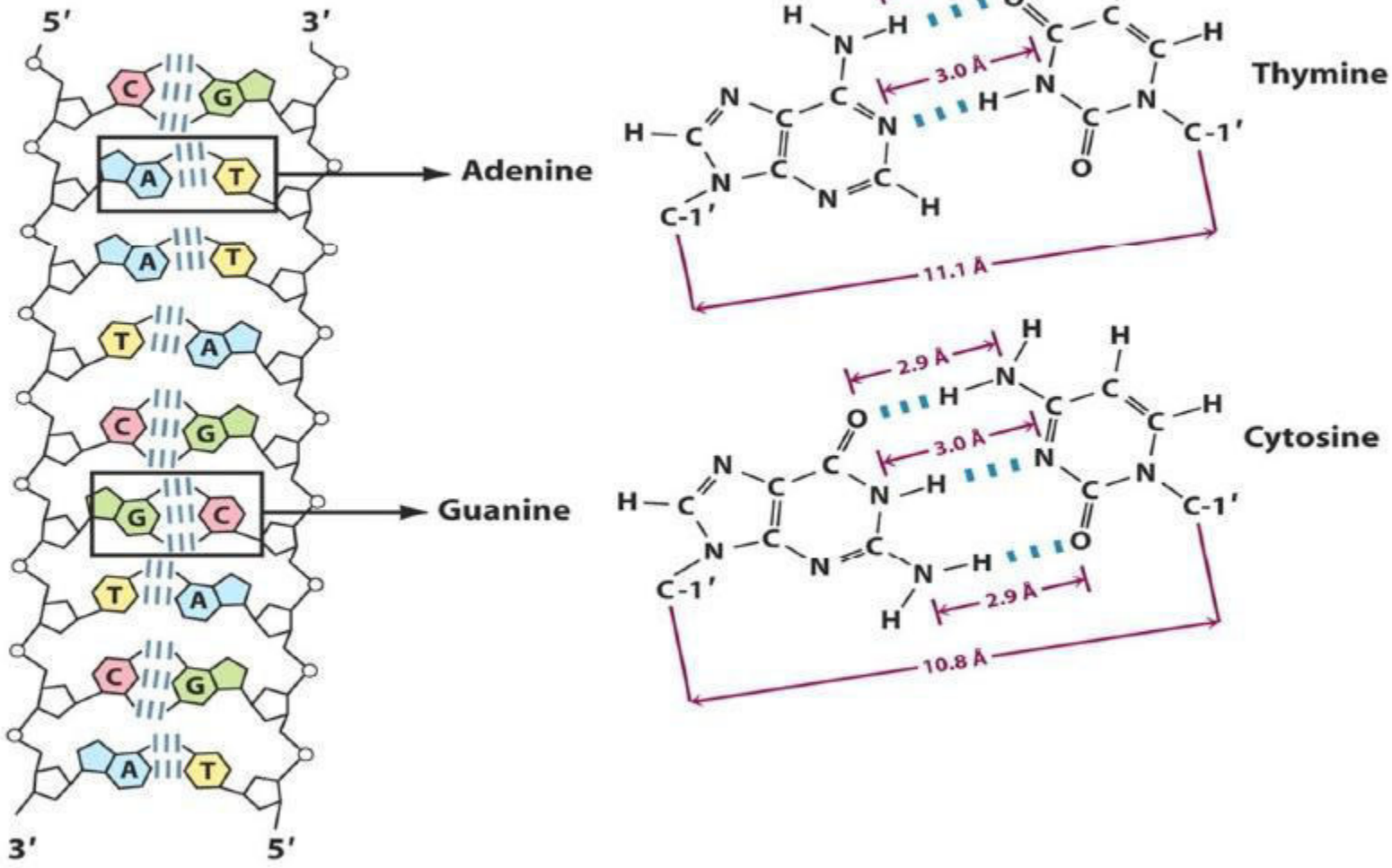
Nucleotides and Nucleic Acids

- Hydrogen bonds involving the amino and carbonyl groups are the second important mode of interaction between bases in nucleic acid molecules.

Nucleotides and Nucleic Acids

- The most important hydrogen-bonding patterns are those defined by James D. Watson and Francis Crick, in which
- A bonds specifically to T (or U) and
- G bonds to C.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- These two types of base pairs predominate in double-stranded DNA and RNA.

Nucleotides and Nucleic Acids

DNA Primary Structure

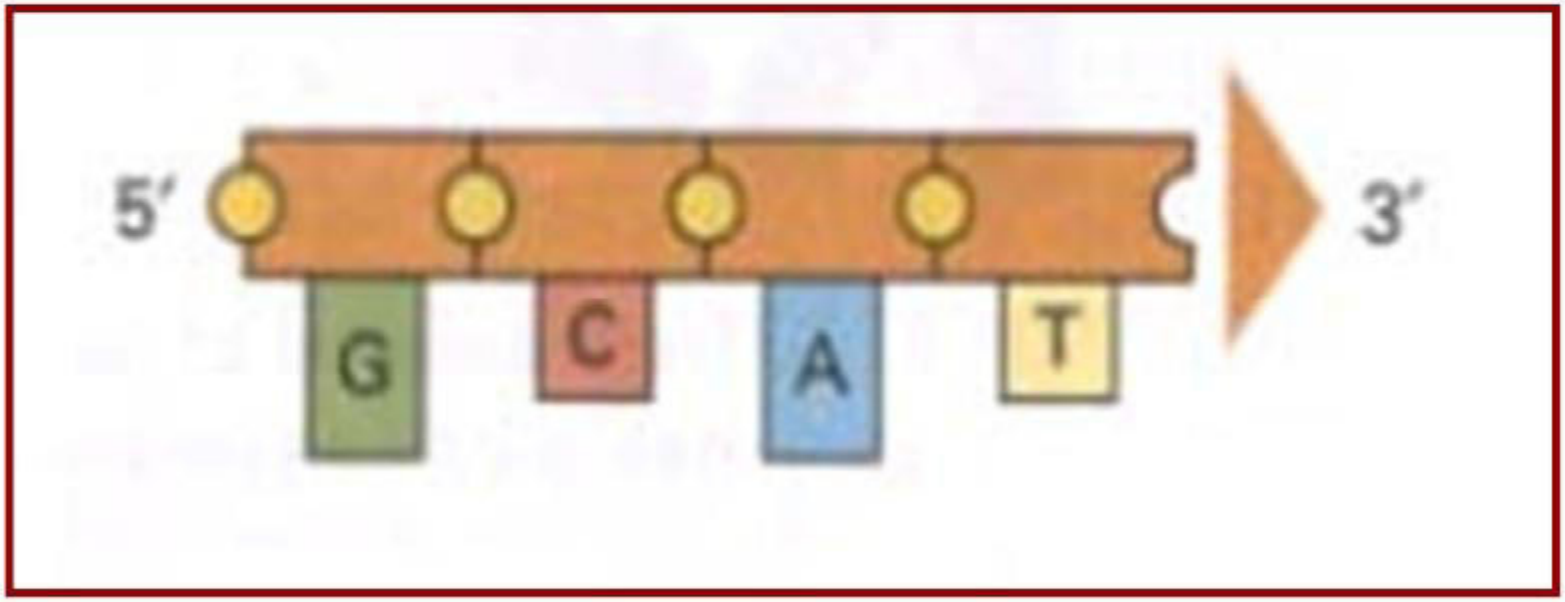
- The primary structure of a nucleic acid is its
- covalent structure and
- nucleotide sequence.

Nucleotides and Nucleic Acids

- The back bone of the primary structure is the linear strand made by sugar phosphate residues, linked together, while the bases project laterally.
- This way a long, un-branched chain is formed.

Nucleotides and Nucleic Acids

Primary structure is a huge linear polymer of dNTPs that are joined to each other by 5'-3' PDE bonds.



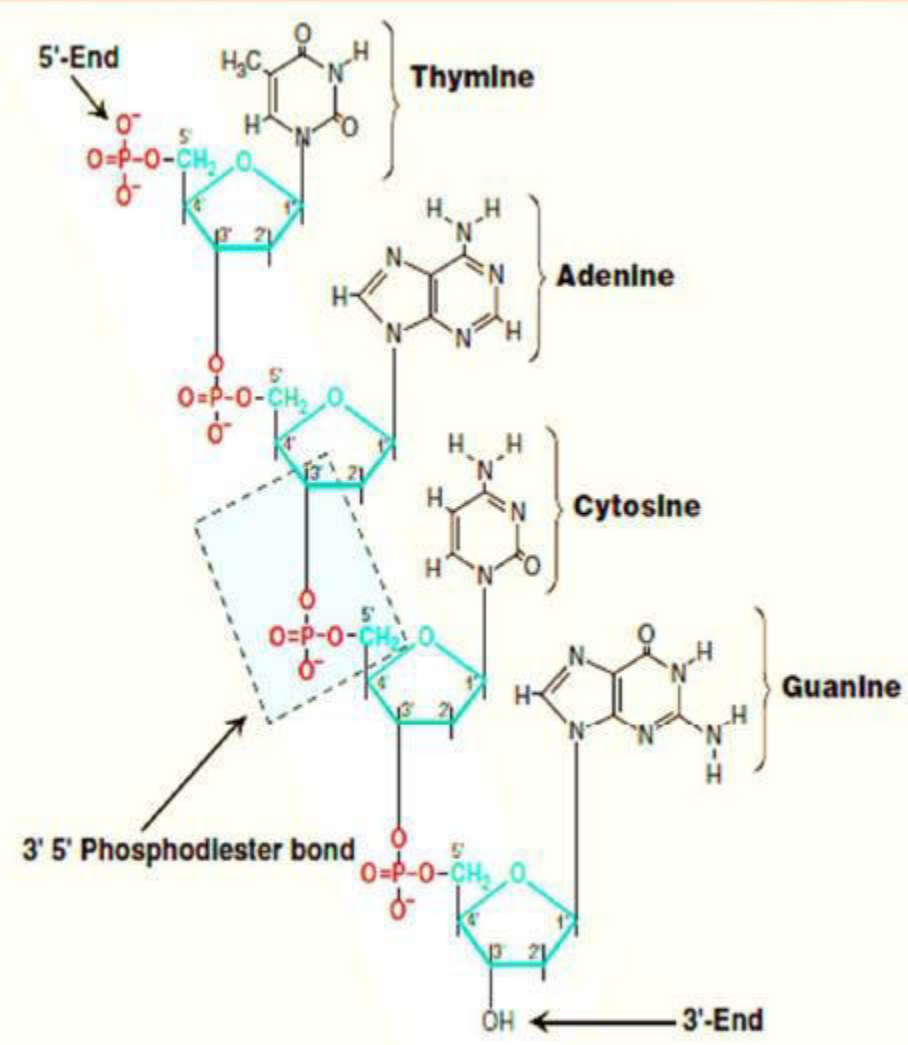
Nucleotides and Nucleic Acids

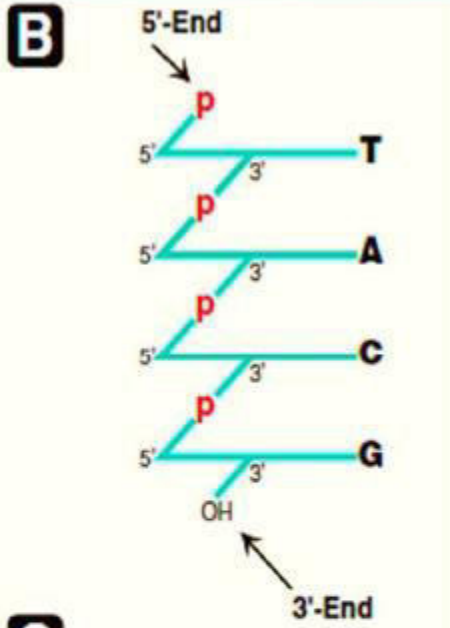
- The resulting long, un-branched chain has polarity.
- Both 5’-end and 3’-end are free.
- at 5’-end there is a free phosphate.
- at 3’-end there is a free OH that are not attached to other nucleotides.

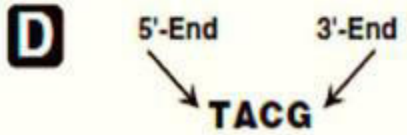
Nucleotides and Nucleic Acids

- Purines and pyrimidines project laterally from the backbone and forms a variable part.
- The variable part is concerned with the expression of genetic information.

Nucleotides and Nucleic Acids







Nucleotides and Nucleic Acids

- By convention, the structure of a single strand of nucleic acid is always written with the 5′ end at the left and the 3′ end at the right
- that is, in the 5′ to 3′ direction.

Nucleotides and Nucleic Acids

- Some simpler representations of this penta deoxy ribonucleotide are
- pA-C-G-T-A OH,
- pApCpGpTpA, and
- finally 5′-ACGTA-3′.

Nucleotides and Nucleic Acids

DNA Secondary Structure

- Any regular, stable structure taken up by some or all of the nucleotides in a nucleic acid can be referred to as secondary structure

Nucleotides and Nucleic Acids

- DNA exist as double stranded molecule (double helix), except few viruses that contain single stranded molecule.
- This model was presented by Watson and Crick in 1953.

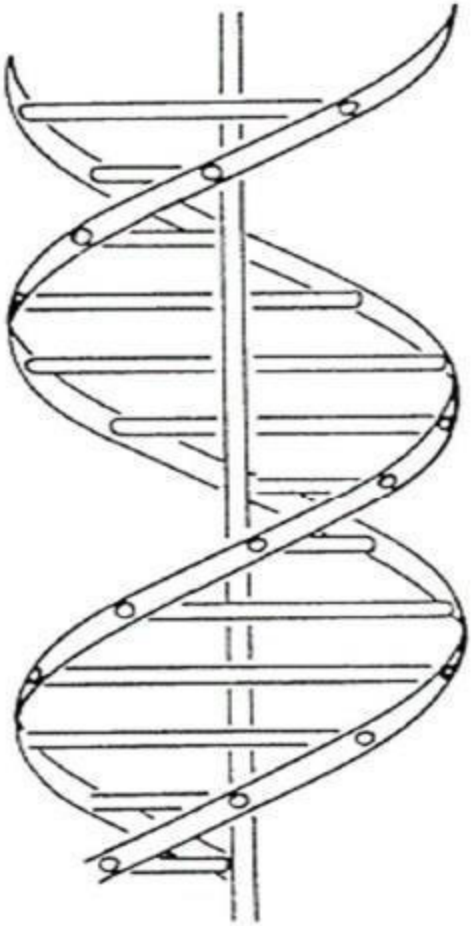
Nucleotides and Nucleic Acids

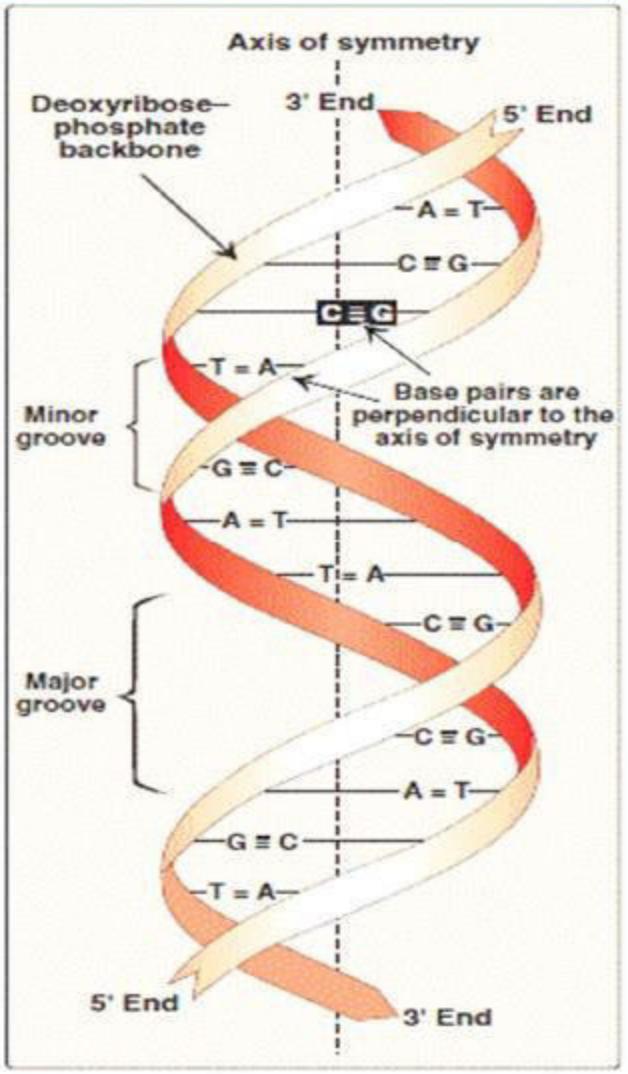
- The two polydeoxyribonucleotide strands are coiled around a common axis called axis of symmetry.
- The chains are paired in anti-parallel manner, i.e., the 5'-end of one strand is paired with the 3'-end of the other strand.

Nucleotides and Nucleic Acids

- In the DNA helix
- Deoxyribose-phosphate backbone is hydrophilic and thus it is on the outside of the molecule.
- The bases are hydrophobic and are thus stacked inside the molecule.

Nucleotides and Nucleic Acids





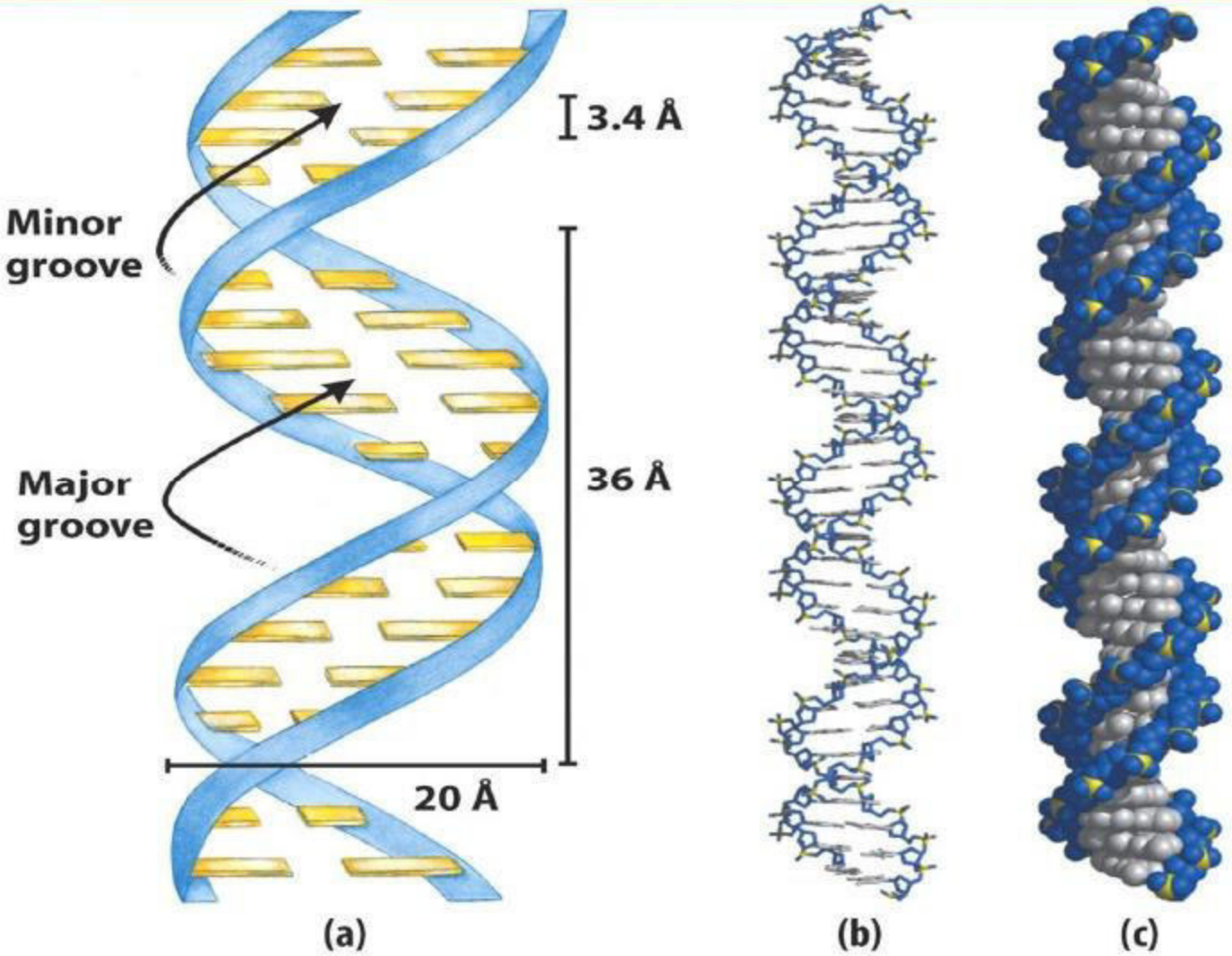
Nucleotides and Nucleic Acids

- The overall structure resembles a twisted ladder.

Nucleotides and Nucleic Acids

- Grooves: The spatial relationship between the two strands in the helix creates
- a major (wide) groove
- a minor (narrow) groove.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- These grooves provide access for the binding of regulatory proteins to their specific recognition sequences along the DNA

Clinical aspect: Certain anticancer drugs, exert their cytotoxic effect by intercalating into the narrow groove of the DNA double helix, thus interfering with RNA and DNA synthesis

Nucleotides and Nucleic Acids

DNA tertiary Structure

- The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids is generally considered tertiary structure

Nucleotides and Nucleic Acids

- Eukaryotic DNA is associated with tightly bound basic proteins, called histones.
- These serve to order the DNA into fundamental structural units, called nucleosomes.

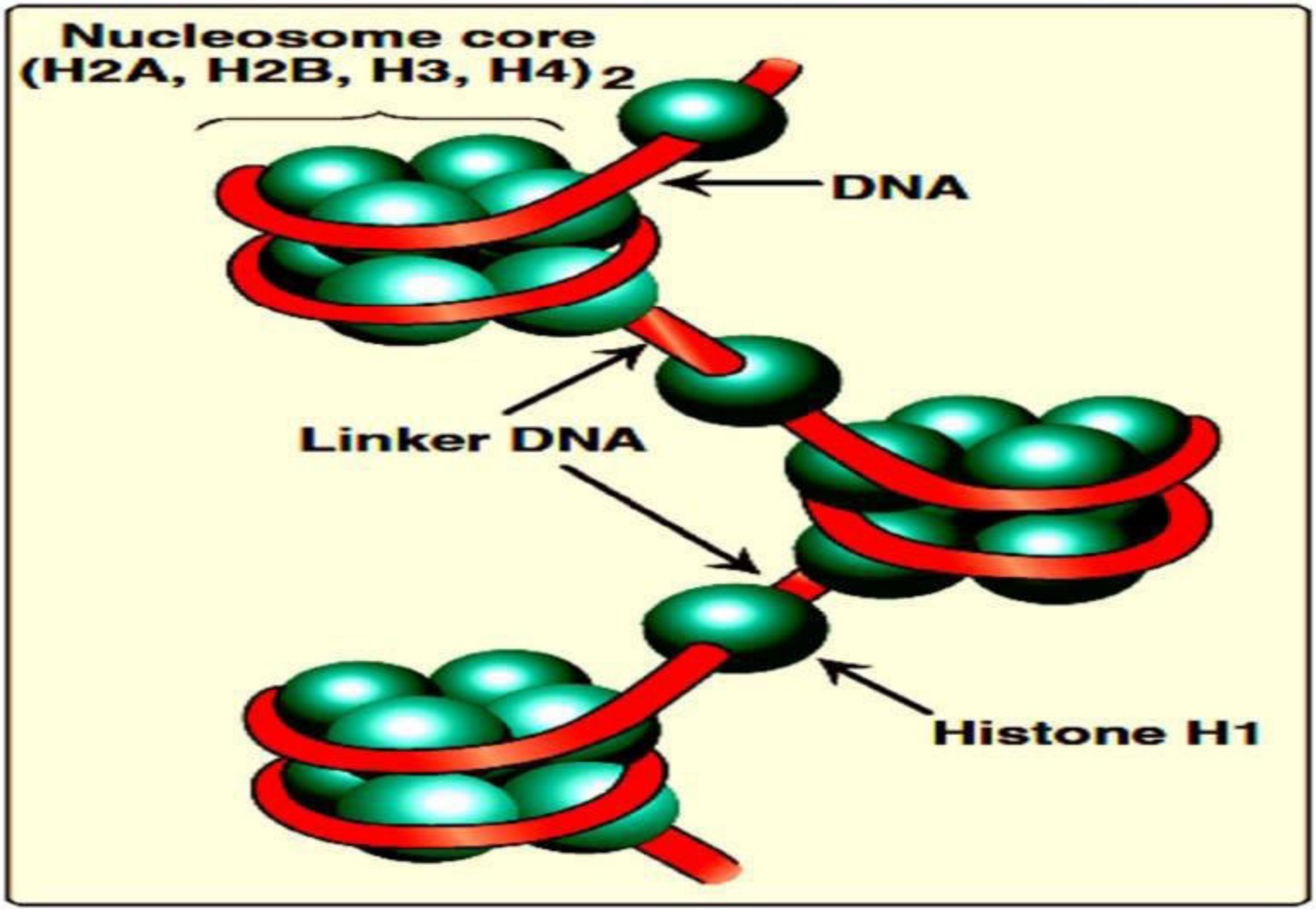
Nucleotides and Nucleic Acids

- There are five classes of histones, designated H1, H2A, H2B, H3, and H4.
- These are positively charged at physiologic pH as a result of their high content of lysine and arginine.

Nucleotides and Nucleic Acids

- Two molecules each of H2A, H2B, H3, and H4 form the structural core of the nucleosome.
- Around this core, a segment of the DNA double helix is wound nearly twice- approximately 140bp

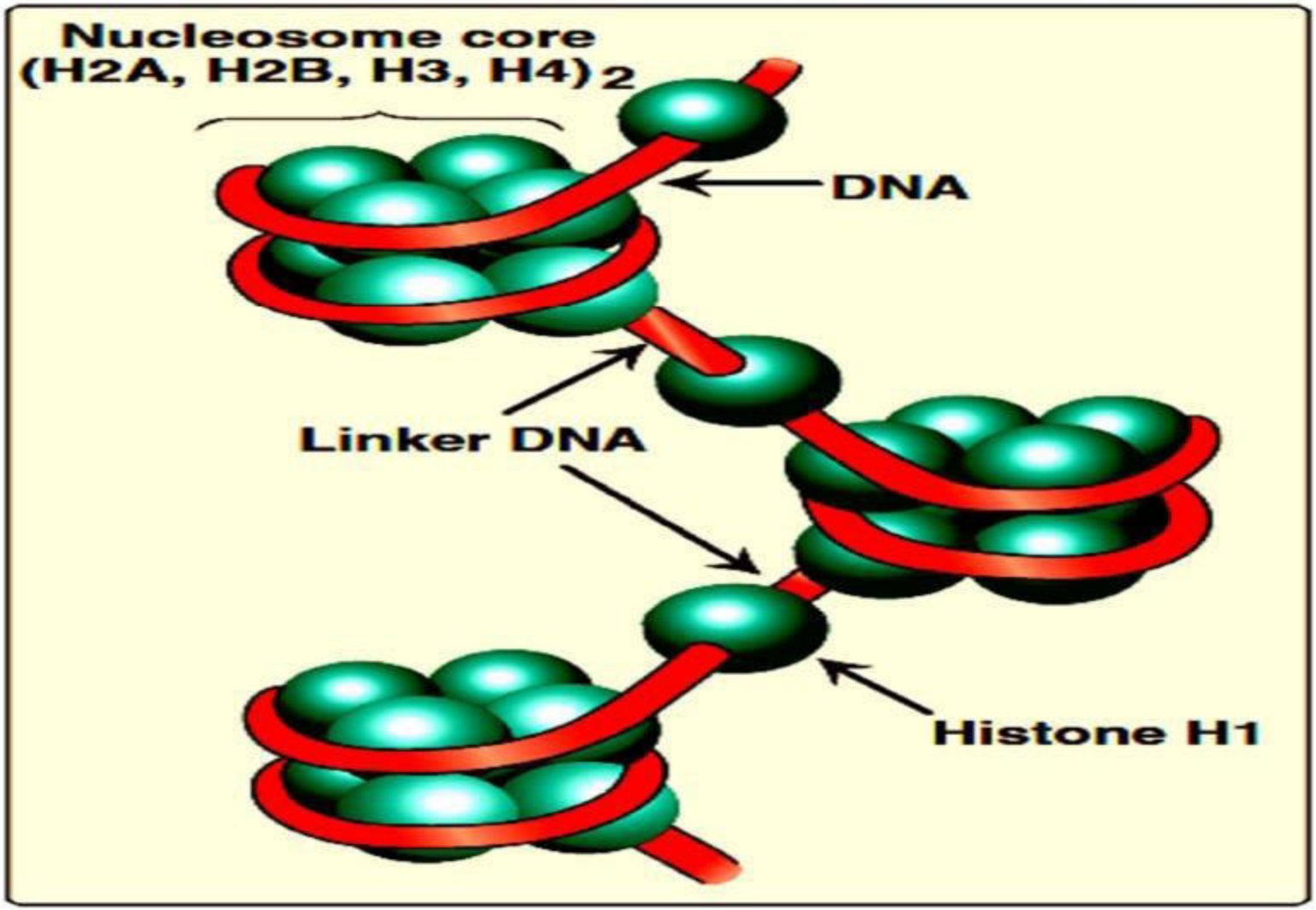
Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- The DNA wrapped around the nucleosome core is continuous and joins one nucleosome core to the next -the linker DNA
- this 50 bp DNA is complexed with the fifth type of histone, H1.

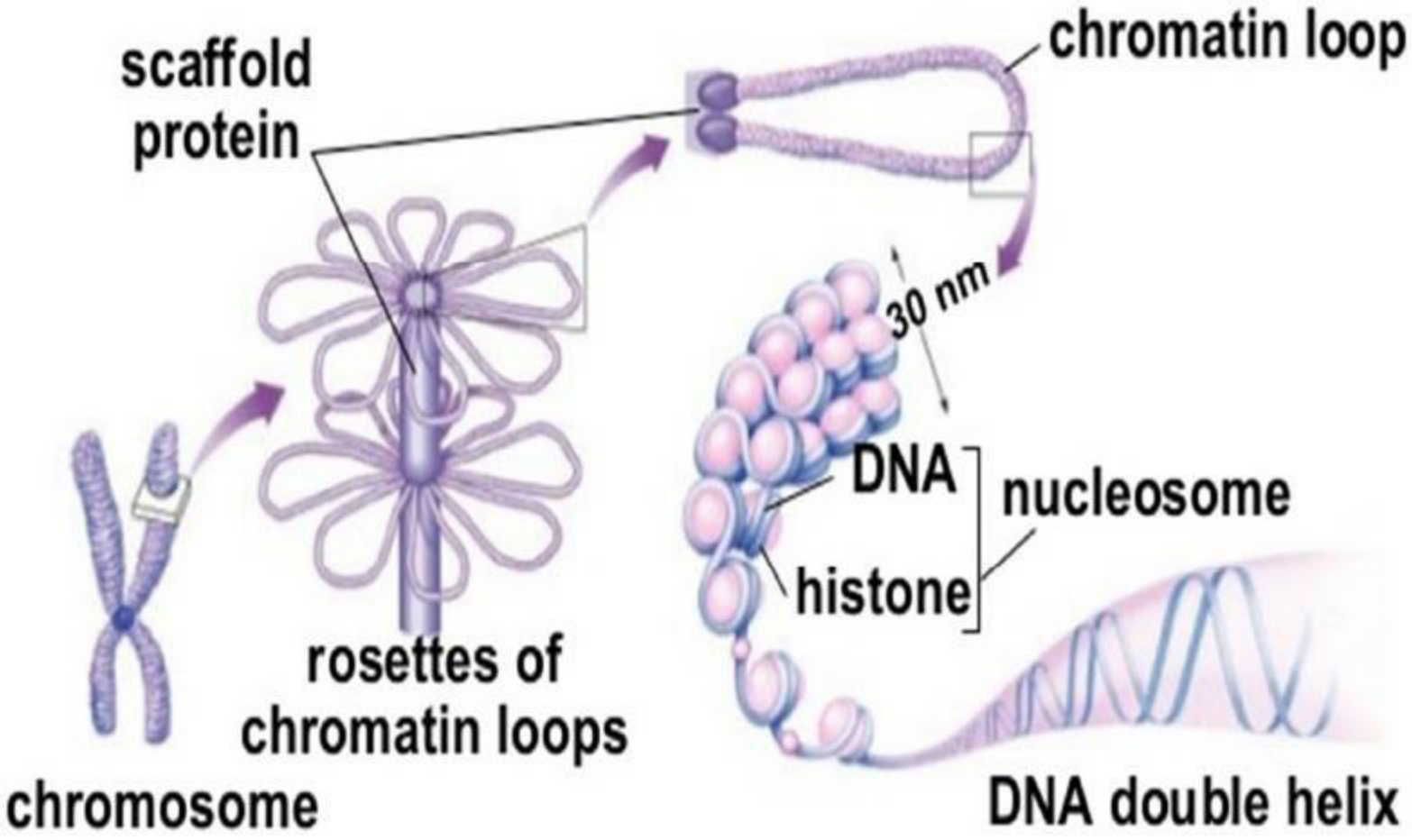
Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Nucleosomes can be packed more tightly to form a polynucleosome also called a nucleofilament or a 30-nm fiber.
- The fiber is organized into loops that are anchored by nuclear scaffolding proteins.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Additional levels of organization lead to the final chromosomal structure

Nucleotides and Nucleic Acids

DNA Properties

Template and Non-template Strands

- The term template strand refers to the sequence of DNA that is copied during the synthesis of mRNA.

Nucleotides and Nucleic Acids

- The opposite strand is called the Non Template or coding strand or the mRNA-like strand
- it has base sequence directly corresponding to the mRNA sequence
- the sequence corresponds to the codons that are translated into protein

Nucleotides and Nucleic Acids

- The 3′ -5′ strand is called Template strand while
- 5′ -3′ strand is called Non Template (coding strand).

Nucleotides and Nucleic Acids

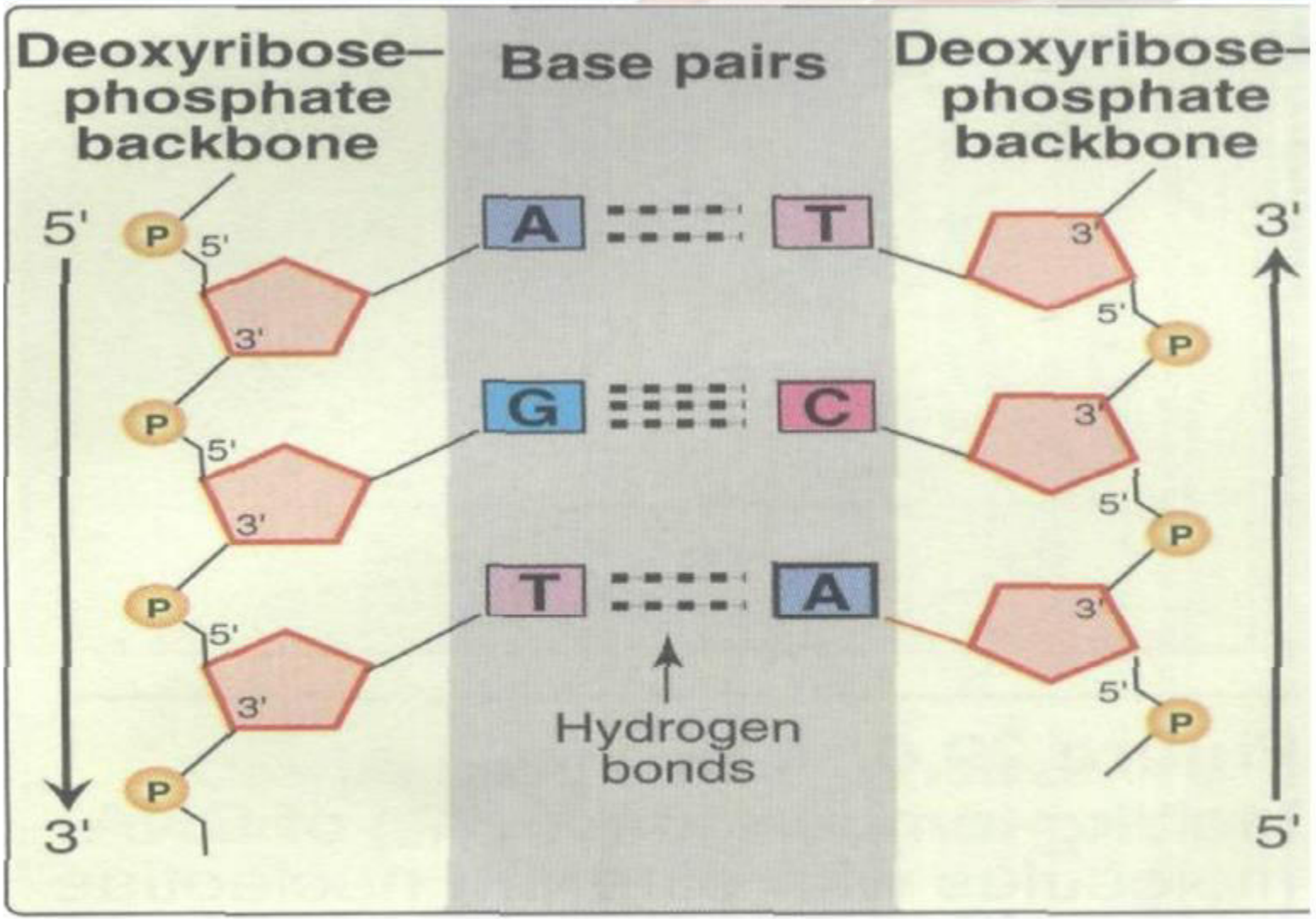
(5′)CGCTATAGCGTTT(3′)	DNA nontemplate (coding) strand
(3′)GCGATATCGCAAA(5′)	DNA template strand
(5′)CGCUAUAGCGUUU(3′)	RNA transcript

Nucleotides and Nucleic Acids

Base Pairing

- The bases of one strand of DNA are paired with the bases of the opposite strand, so that
- Adenine is always paired with thymine
- Cytosine is always paired with guanine.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- One polynucleotide chain of DNA double helix is always complementary to the other
- Given the sequence of bases on one chain, the sequence of bases on the complementary chain can be determined

Nucleotides and Nucleic Acids

DNA Properties (contd.)

Chargaff's Rules

- Due to specific base pairing of DNA i.e A toT and G to C
- In any sample of double-stranded DNA the amount of adenine equals the amount of thymine.

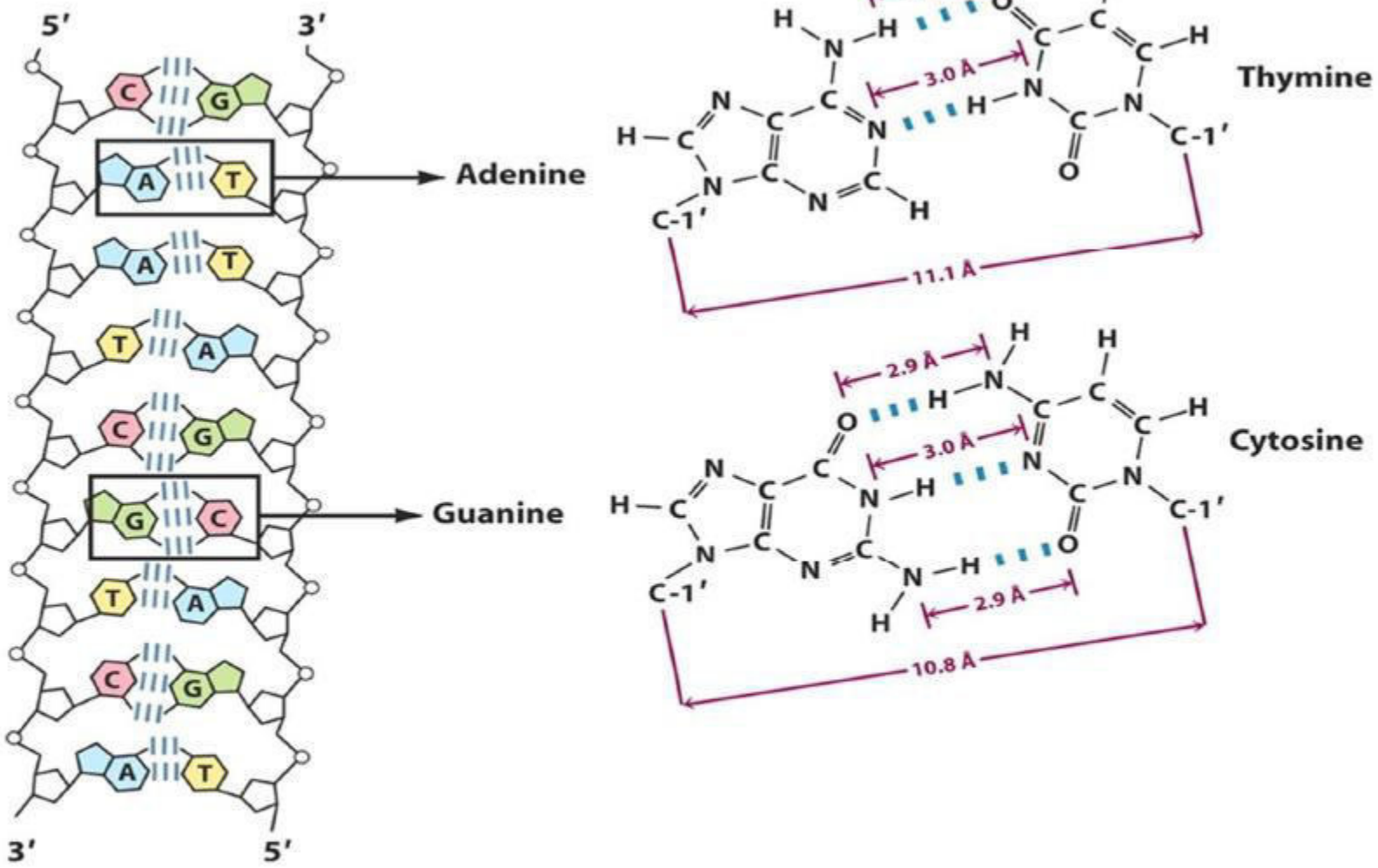
Nucleotides and Nucleic Acids

- The amount of guanine equals the amount of cytosine
- the total amount of purines equals the total amount of pyrimidines ;
- i.e. A + G = T+ C.

Nucleotides and Nucleic Acids

- The base pairs are held together by hydrogen bonds
- Two between A and T
- Three between G and C

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- These hydrogen bonds, plus the van der Waals and hydrophobic interactions between the adjacent stacked bases, stabilize the structure of the double helix.

Nucleotides and Nucleic Acids

- The base composition of DNA varies from one species to another, but is

- same if isolated from different tissues of same species.

Nucleotides and Nucleic Acids

- The base composition does not change with
- age,
- nutritional status and
- environment.

Nucleotides and Nucleic Acids

DNA Denaturation

- Separation of the two strands of the double helix when hydrogen bonds between the paired bases are disrupted.

Nucleotides and Nucleic Acids

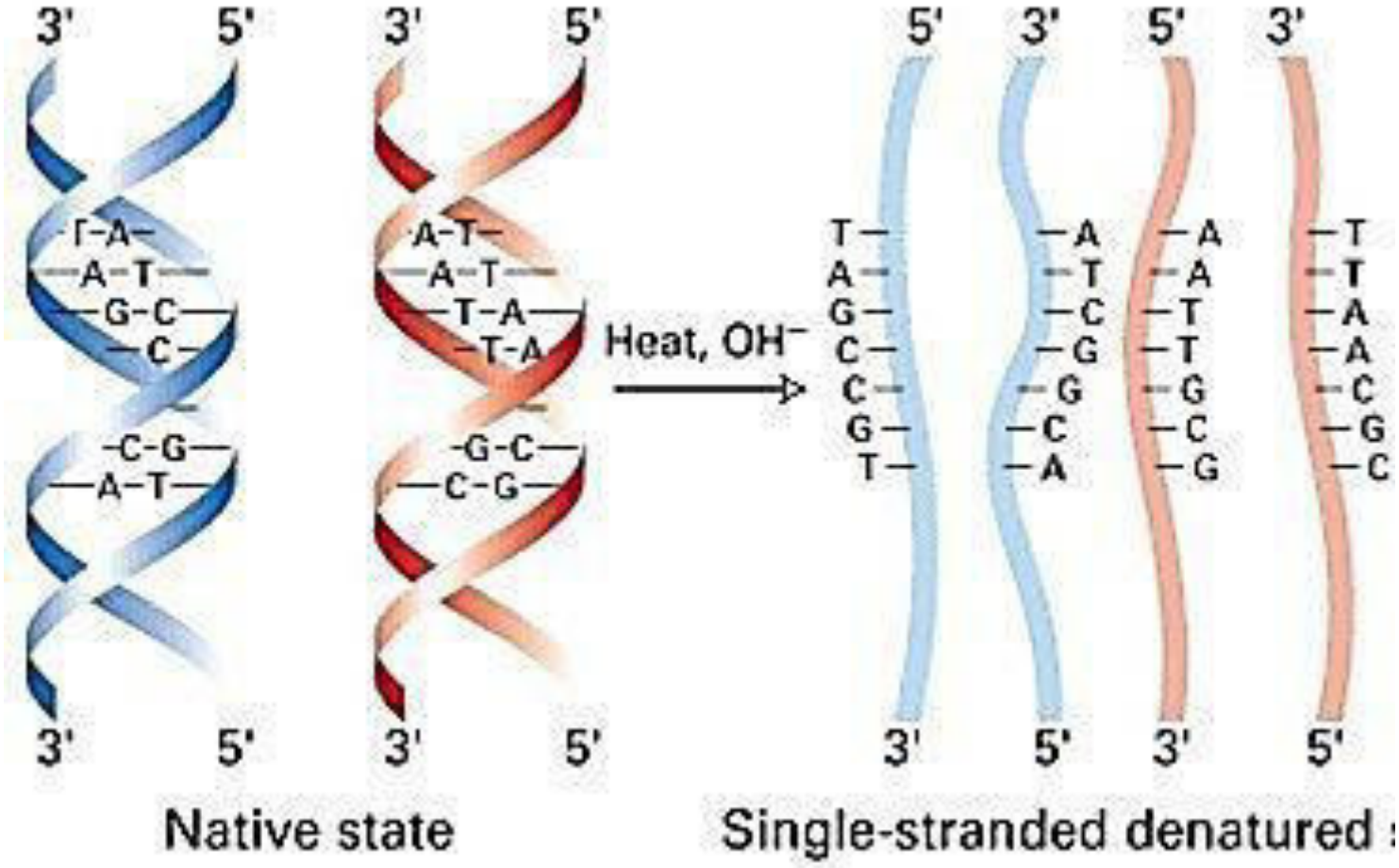
- Disruption can occur in the laboratory if
  - the pH or the salt concentration of the DNA solution is altered
  - if the solution is heated above 80°C

Nucleotides and Nucleic Acids

- Disruption of the hydrogen bonds between paired bases and of base stacking causes unwinding of the double helix to form two single strands complete separation of DNA stands along the entire length.

- No covalent bonds in the DNA are broken

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- RNA duplexes are more stable than DNA duplexes.
- At neutral pH denaturation of a double helical RNA often requires temperatures 20^{o} C or more higher than those required for denaturation of a DNA molecule with a comparable sequence.

Nucleotides and Nucleic Acids

- T_m: When DNA is heated, the temperature at which one half of the helical structure is lost is defined as the melting temperature.

Nucleotides and Nucleic Acids

- A specific sequence of DNA has a characteristic denaturation temperature, or melting point (T_m)

Nucleotides and Nucleic Acids

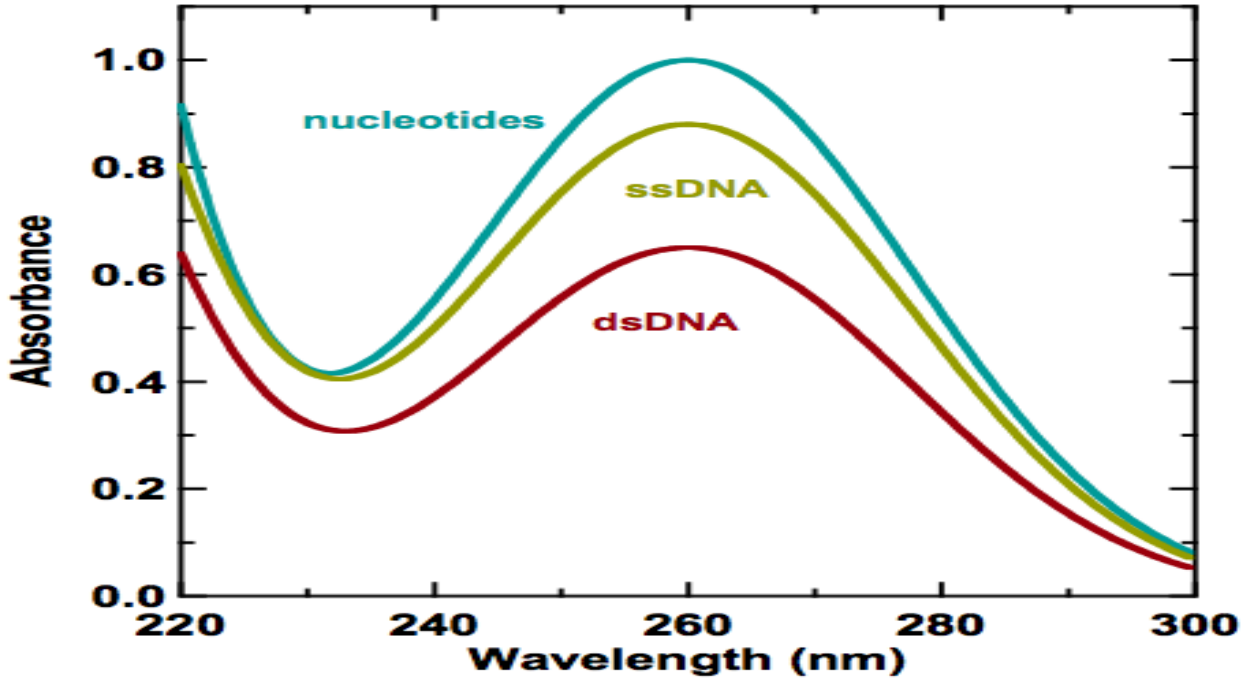
- Careful determination of the melting temperature of a DNA specimen, under fixed conditions of pH and ionic strength, can yield an estimate of a DNA base composition.

END

Nucleotides and Nucleic Acids

- **DNA Denaturation** (Contd.)
- Concomitant with this denaturation of the DNA molecule is an increase in the optical absorbance of the purine and pyrimidine bases—a phenomenon referred to as hyperchromicity of denaturation.

Nucleotides and Nucleic Acids



The absorbance of double-stranded DNA (dsDNA) at 260 nm is less than that of either single-stranded DNA (ssDNA) or the free bases. This is called "hyperchromism."

Nucleotides and Nucleic Acids

- So single-stranded DNA has a higher relative absorbance at this wavelength than does double-stranded DNA.
- Denaturation can be monitored by measuring its absorbance at 260 nm

Nucleotides and Nucleic Acids

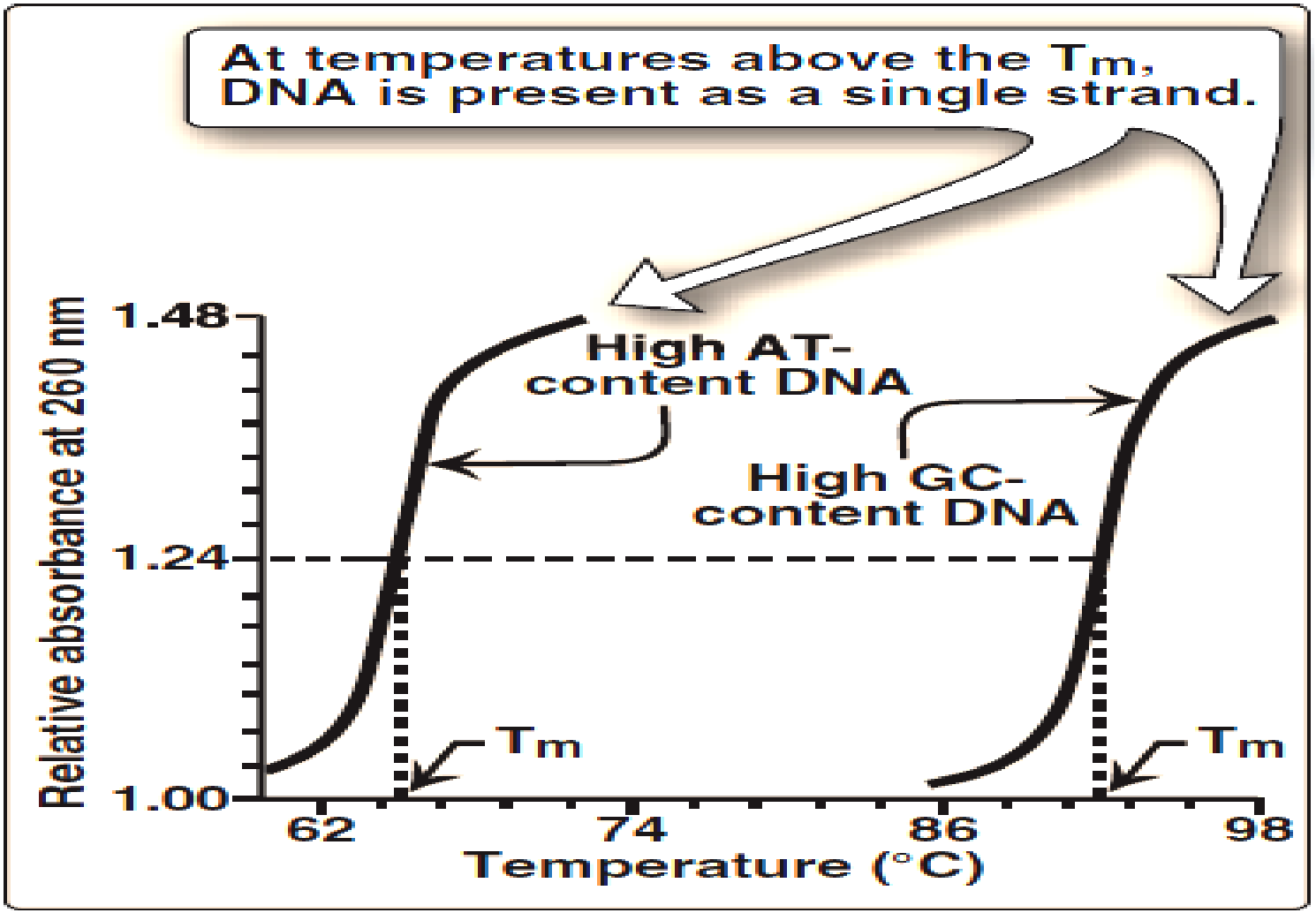
Factors affecting Tm

- The Tm is influenced by
- the base composition of the DNA
- the salt concentration of the solution

Nucleotides and Nucleic Acids

- The higher the content of GC base pairs, the higher the melting point of the DNA.
- This is because GC base pairs, with three hydrogen bonds, require more heat energy to dissociate than AT base pairs.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- An *increase* in salt concentration *increases* and a

- decrease in salt concentration decreases the Tm

END

Nucleotides and Nucleic Acids

DNA Renaturation

- Under appropriate conditions *(temp. & salt concentration)*, separated strands of DNA will renature or reassociate and form the double helix by the process called renaturation (or reannealing).

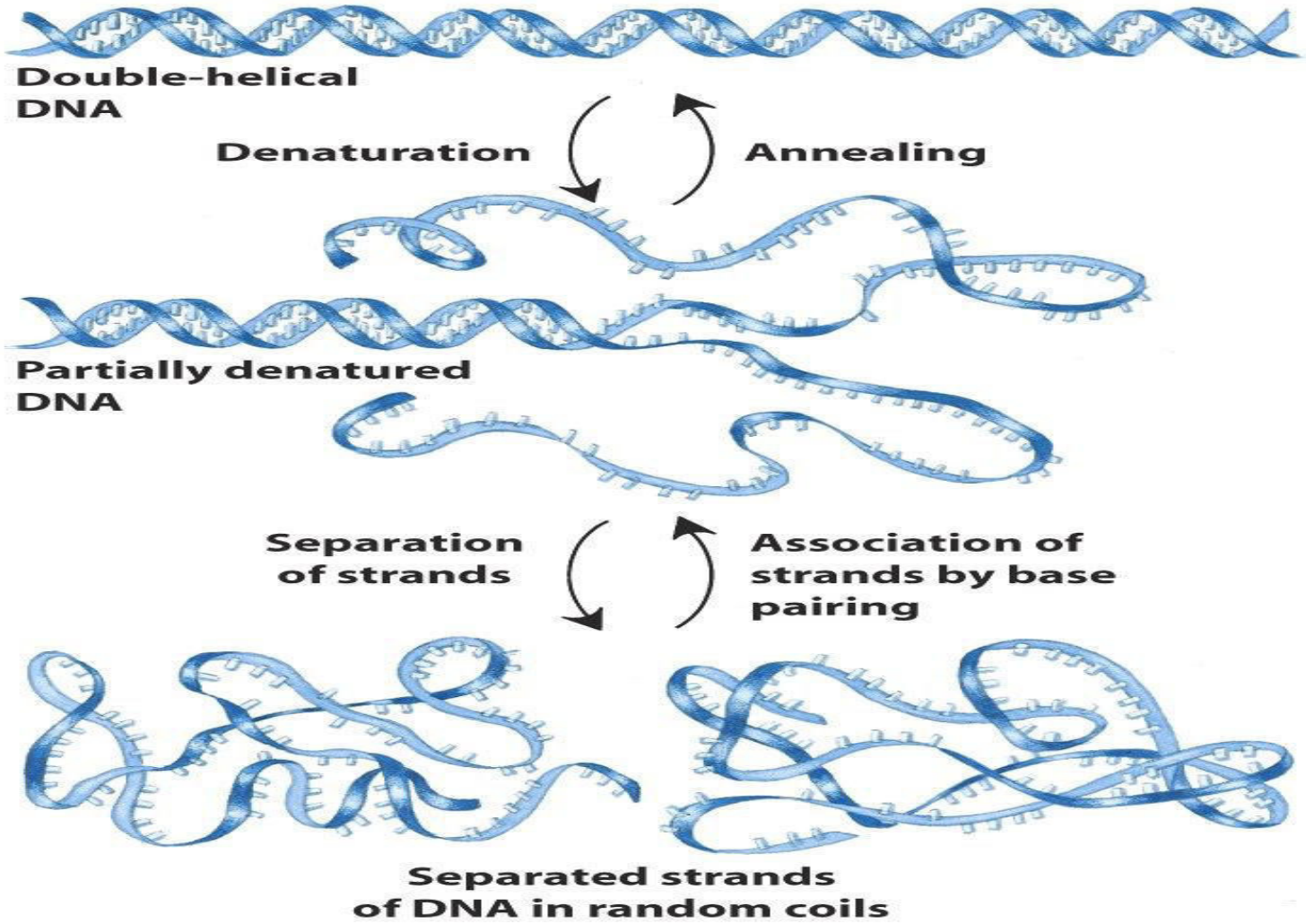
Nucleotides and Nucleic Acids

- This reannealing process is also referred to as hybridization.

Nucleotides and Nucleic Acids

- When the temperature or pH is returned to the range in which most organisms live, the unwound segments of the two strands spontaneously rewind, or anneal, to yield the intact duplex

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Renaturation of a DNA molecule is a rapid one-step process.
- The rate of reassociation depends upon the concentration of the complementary strands.

Nucleotides and Nucleic Acids

- At a given temperature and salt concentration, a particular nucleic acid strand will associate tightly only with a complementary strand
- This property is utilized in analyzing nucleotide sequencing of a given nucleic acid.

Nucleotides and Nucleic Acids

- Under appropriate conditions DNA will form a hybrid with a complementary DNA or with a complementary RNA

Nucleotides and Nucleic Acids

- Hybridization is combined with gel electrophoresis techniques that separate nucleic acids by size,
- coupled with radioactive or fluorescent probe labeling to provide a detection of a nucleotide sequence.

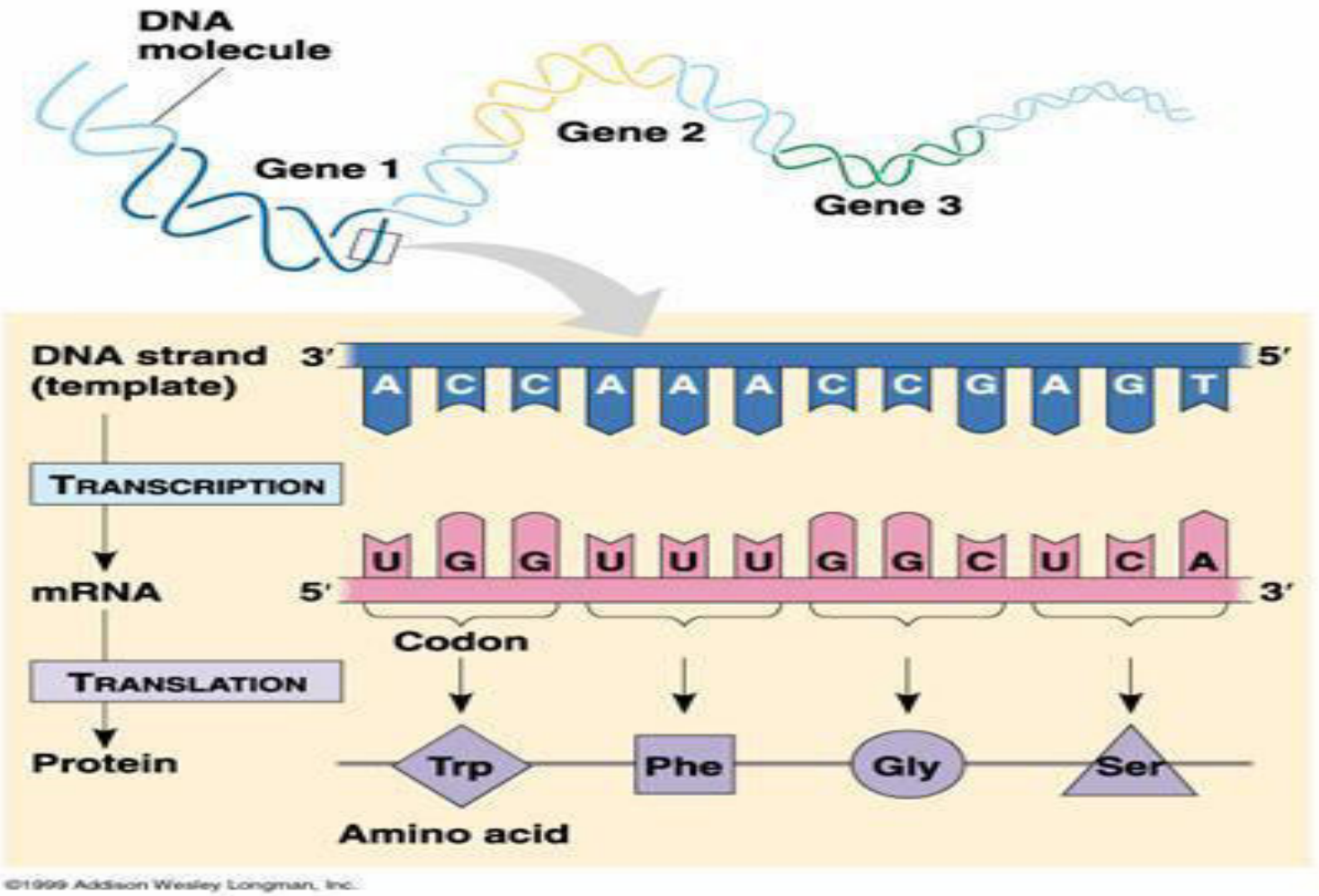
END

Nucleotides and Nucleic Acids

RNA

- The genetic master plan is contained in the nucleotide sequence of DNA.
- It is through the ribonucleic acid (RNA)—the "working copies" of the (DNA) — that the master plan is expressed

Nucleotides and Nucleic Acids



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Nucleotides and Nucleic Acids

- RNA is a polymer of ribonucleotides of Adenine, Uracil, Guanine and Cytosine, joined together by 3´-5´phosphodiester bonds.
- RNA does not contain thymine except in rare cases.

Nucleotides and Nucleic Acids

- The pentose sugar of RNA is D-ribose.
- Location: RNA is found in the nucleolus, ribosomes, mitochondria, and cytoplasm.
- The genetic material for some animal and plant viruses is RNA rather than DNA.

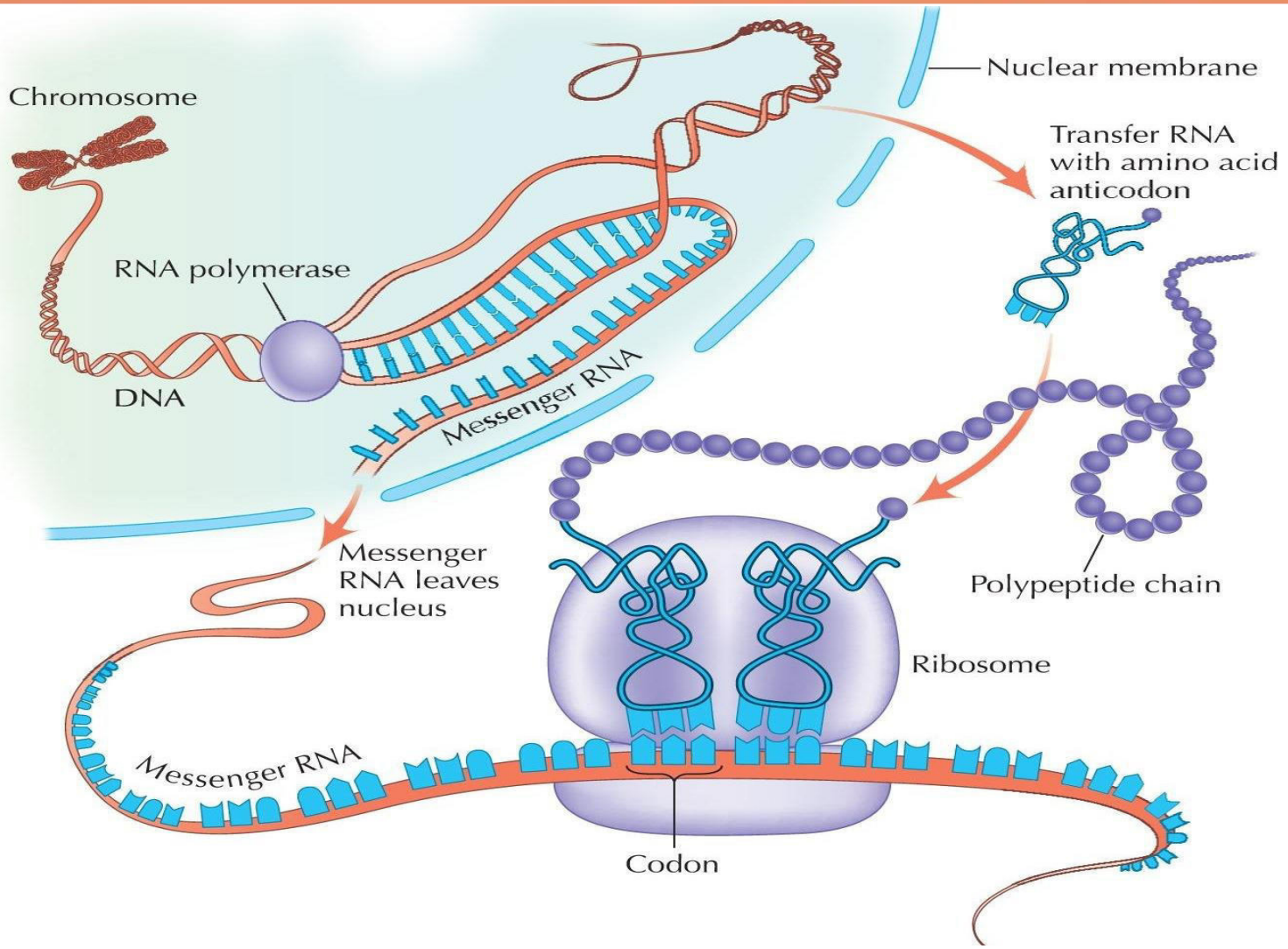
Nucleotides and Nucleic Acids

- There is a wide variety of RNAs,
- messenger RNAs (mRNAs)- transfer genetic information from DNA to the protein-synthesizing machinery.
- ribosomal RNAs (rRNAs)- contribute to the formation and function of ribosomes

Nucleotides and Nucleic Acids

- transfer RNAs (tRNAs)- adapter molecules that carry specific amino acids for protein synthesis

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- small nuclear RNA (snRNA)- play pivotal roles in RNA processing, particularly mRNA processing
- ribozymes — some RNA molecules have intrinsic catalytic activity these RNA enzymes, are called ribozymes

END

Nucleotides and Nucleic Acids

Differences between DNA and RNA

- Although sharing many features with DNA, RNA possesses several specific differences.

Nucleotides and Nucleic Acids

- Size: They are considerably smaller than DNA
- Sugar: In RNA, the sugar moiety to which the phosphates and purine and pyrimidine bases are attached is ribose rather than the 2'-deoxyribose of DNA

Nucleotides and Nucleic Acids

- Pyrimidine: The pyrimidine components of RNA differ from those of DNA.
- Instead of thymine, RNA contains the ribonucleotide of uracil.
- Thymine is present in the rare case of tRNA

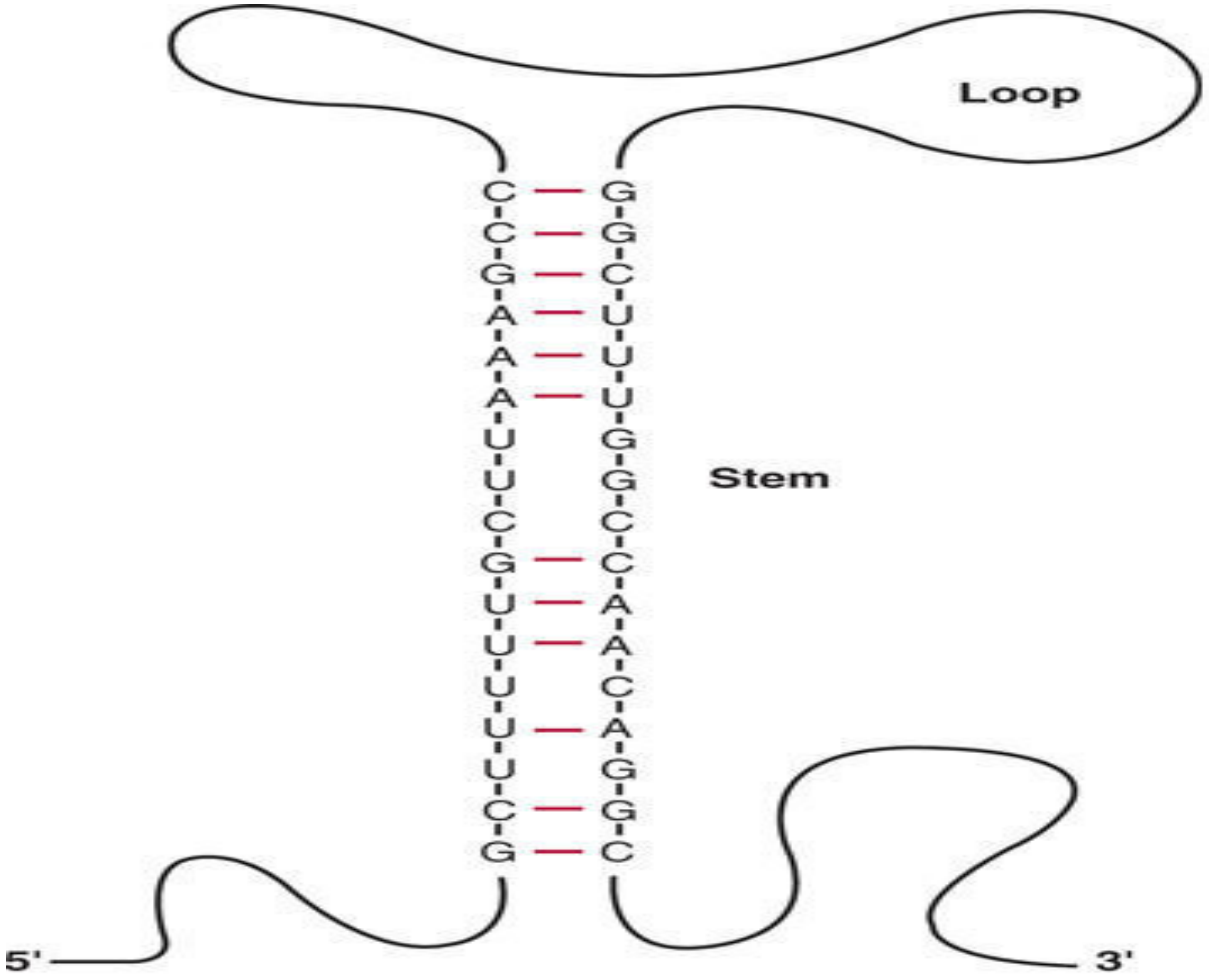
Nucleotides and Nucleic Acids

- Single Strand:
- RNA typically exists as a single strand whereas DNA exists as a double-stranded helical molecule.

Nucleotides and Nucleic Acids

- However, given the proper complementary base sequence with opposite polarity, the single strand of RNA is capable of self- folding like a hairpin and thus acquiring double-stranded characteristics: G pairing with C, and A with U.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- Chargaff's Rules Do Not Apply:
- Since the RNA molecule is a single strand complementary to only one of the two strands of a gene,
- G≠C
- A≠U
- Purines ≠ Pyrimidines

Nucleotides and Nucleic Acids

- Hydrolysis: RNA can be hydrolyzed by alkali to 2',3' cyclic diesters of the mononucleotides, compounds that cannot be formed from alkali-treated DNA because of the absence of a 2'-hydroxyl group.

Nucleotides and Nucleic Acids

- Location: In addition to nucleus, RNA is found in cytoplasm.
- Reverse Transcription: DNA forms RNA by transcription whereas the process by which RNA form DNA is called reverse transcription.

Nucleotides and Nucleic Acids

- **RNA Structural Hierarchy**
- RNA has no simple, structural hierarchy that serves as a reference point, as does the double helix for DNA.
- The three-dimensional structures of many RNAs, like those of proteins, are complex and unique.

Nucleotides and Nucleic Acids

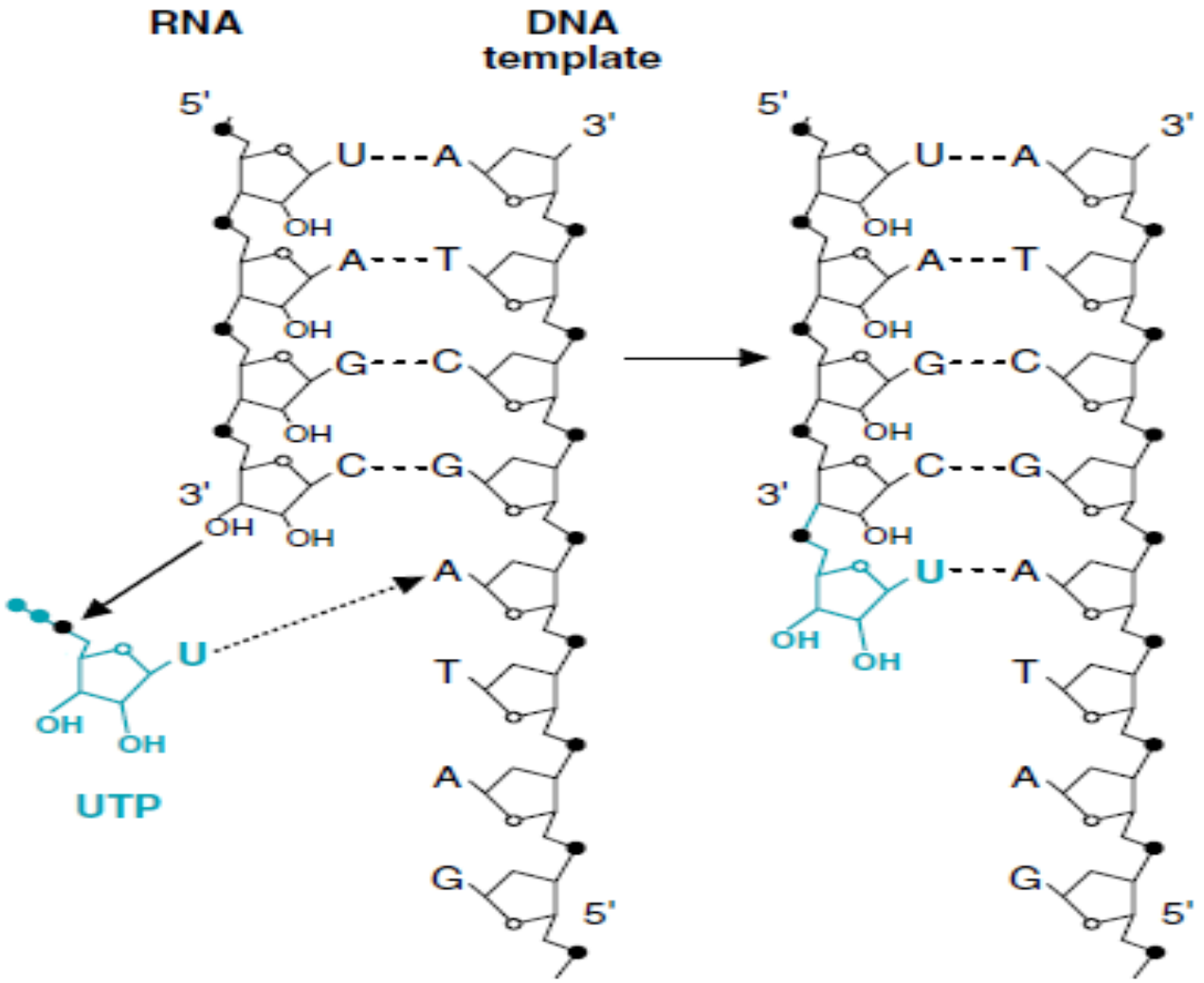
Primary Structure of RNA

- It is defined as the number and sequence of ribonucleotides in the RNA chain.

Nucleotides and Nucleic Acids

- The sequence is complementary to the template strand of the gene from which it was transcribed.

Nucleotides and Nucleic Acids

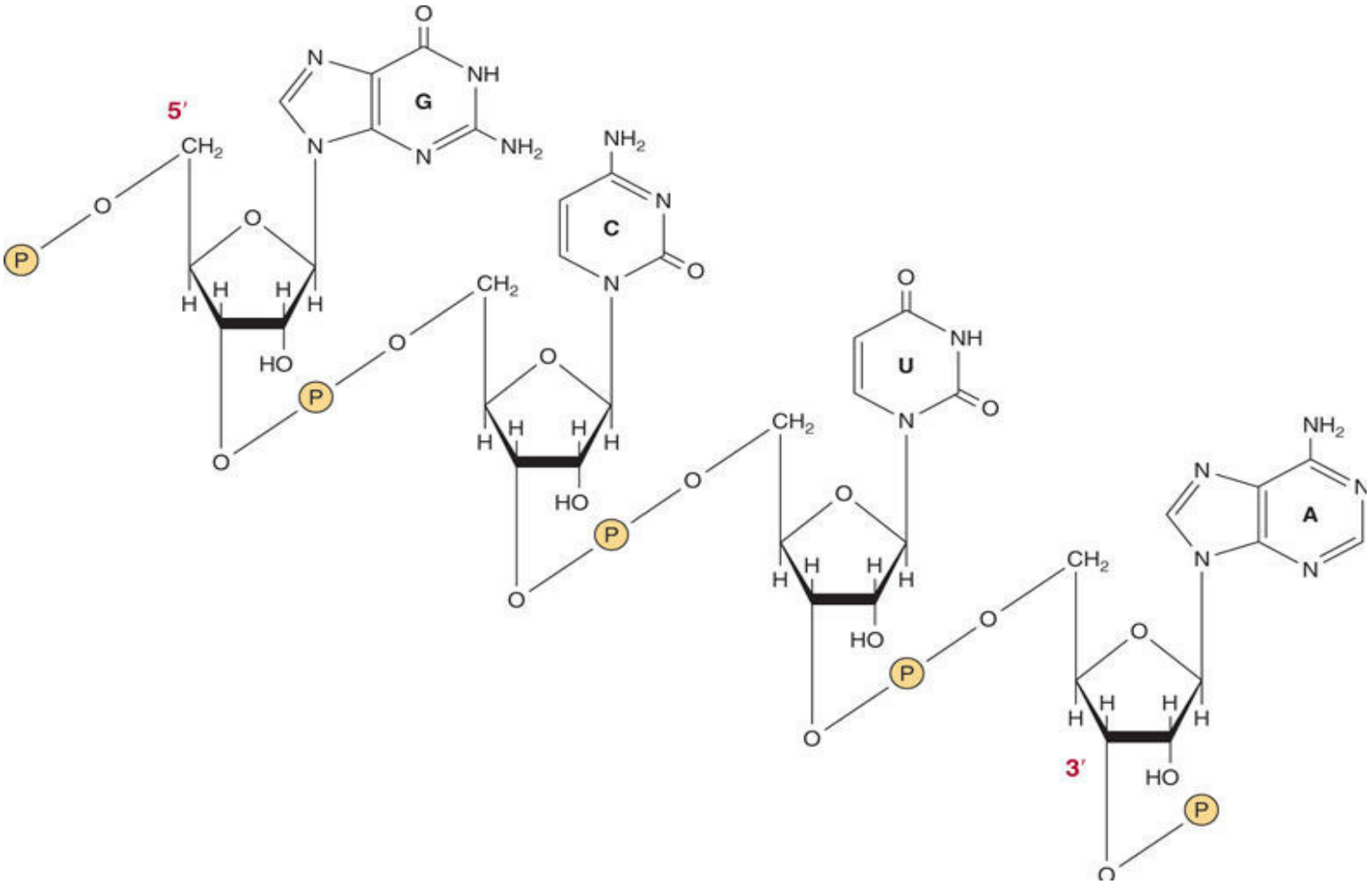


template strand, the sequence of DNA that is copied.Non Template (coding strand).

Nucleotides and Nucleic Acids

- The ribonucleotides are held together by 3´- 5´ phosphodiester bonds.
- 3´-OH group of one nucleotide is bound to 5´-PO_4 of the other nucleotide and form a linear strand.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- The ribosyl moieties are attached to the nucleobases by *N*-glycosidic bonds
- Similar to DNA RNA polymer also has polarity

Nucleotides and Nucleic Acids

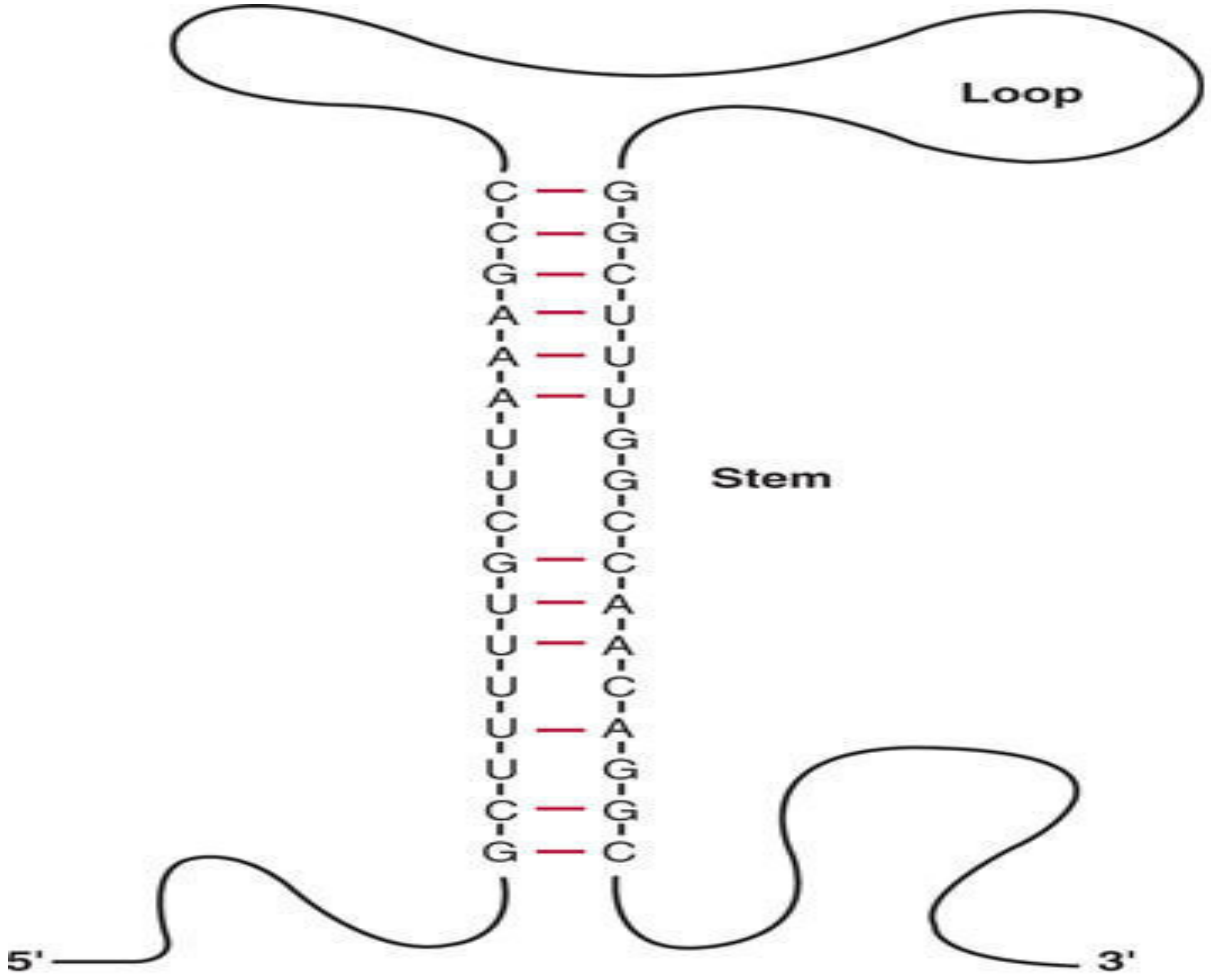
- **Secondary Structure of RNA**

- Secondary structure involves coil formation of the polyribonucleotide chain.

Nucleotides and Nucleic Acids

- The coiled structures are stabilized by
- Hydrophobic interactions between purine and pyrimidine bases.
- Intra-chain hydrogen bonds between G-C and A-U.

Nucleotides and Nucleic Acids



Note: Adenine pairs with Uracil in RNA

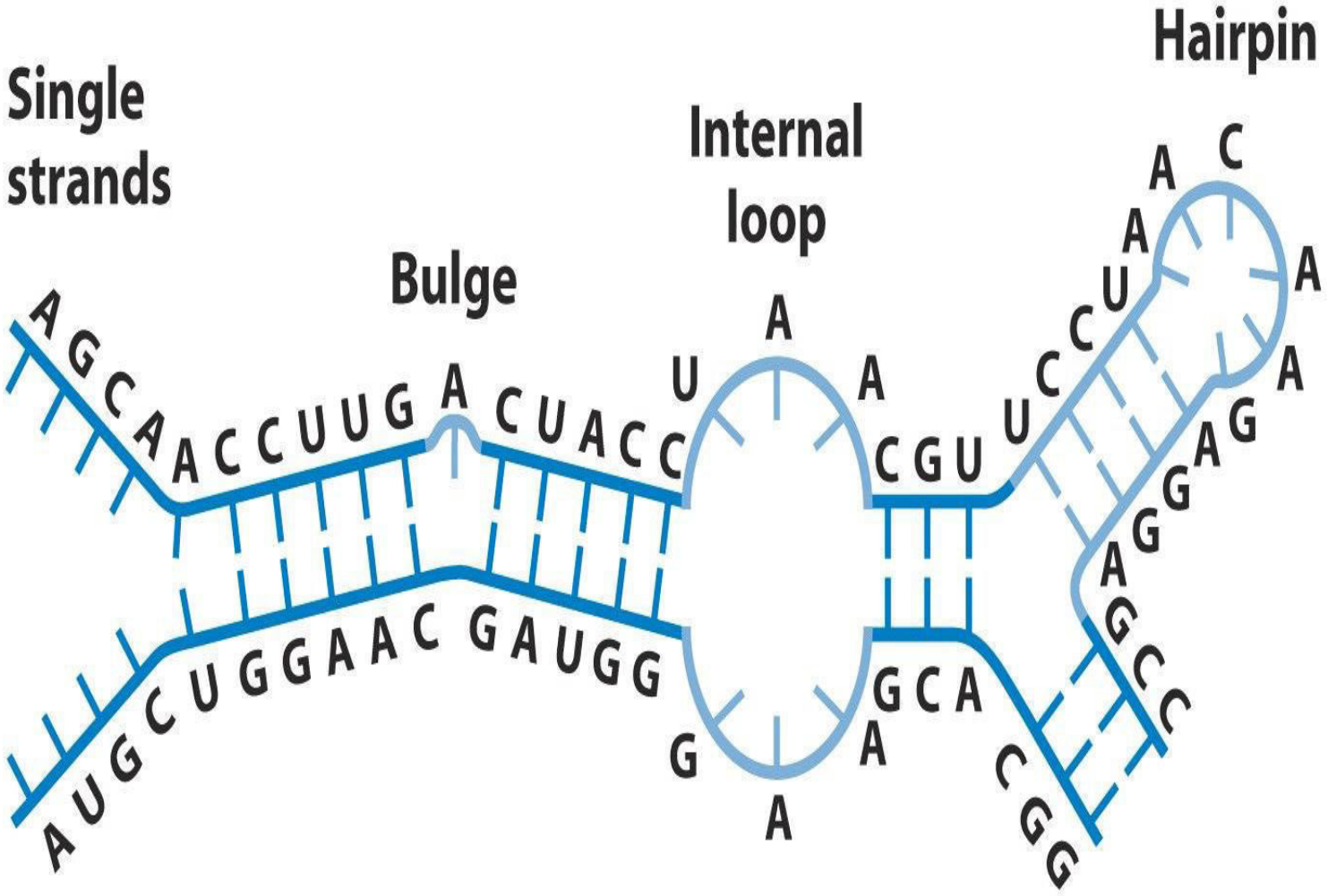
Nucleotides and Nucleic Acids

- Weak interactions, especially base-stacking interactions.
- Where complementary sequences are present, the predominant double-stranded structure is formed.

Nucleotides and Nucleic Acids

- Common RNA secondary structures
- Internal loops: a short series of unpaired bases in a longer paired helix and
- bulges: regions in which one strand of a helix has "extra" inserted bases with no counterparts in the opposite strand

Nucleotides and Nucleic Acids

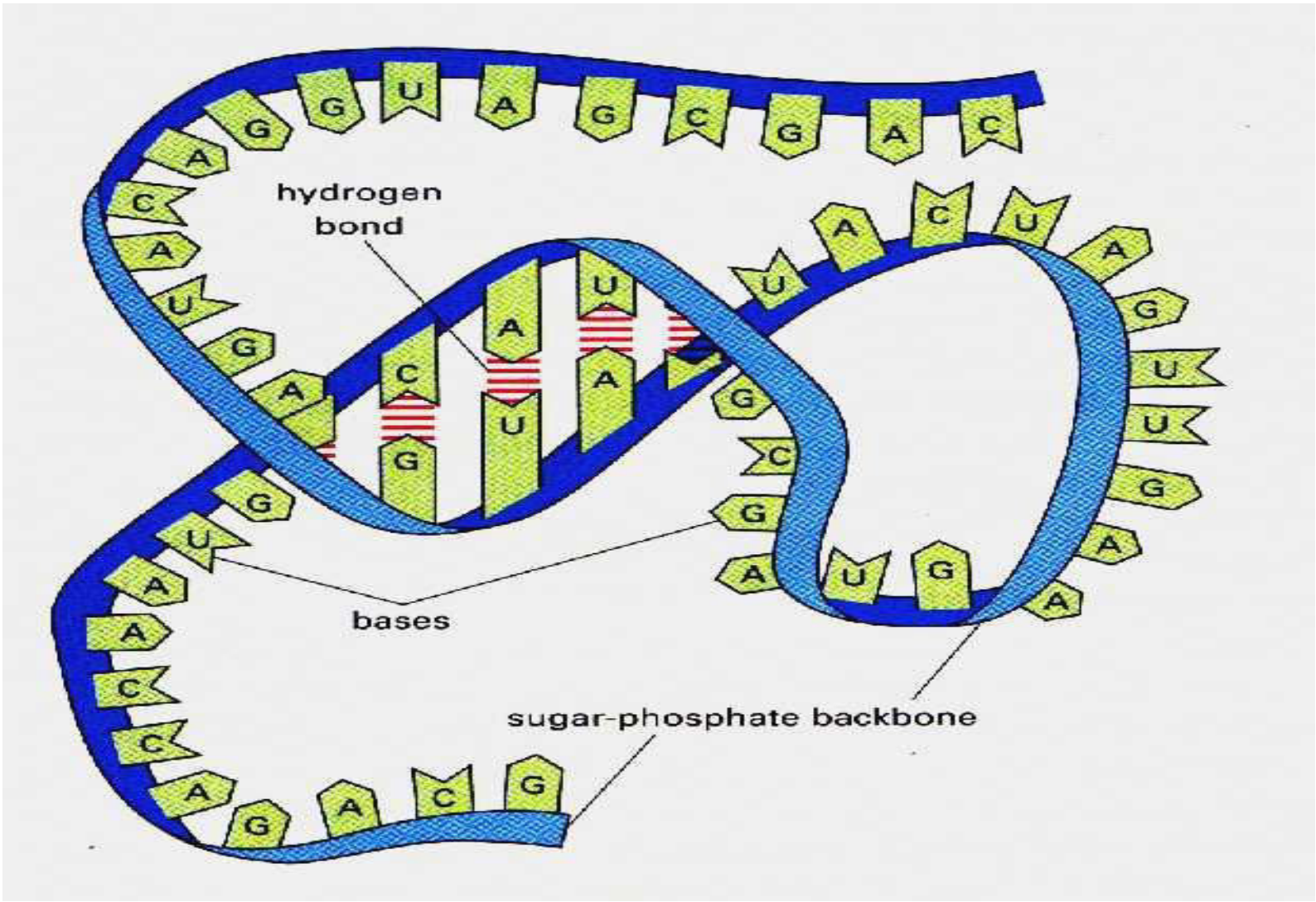


Nucleotides and Nucleic Acids

Tertiary Structure of RNA

- It is the folding of the molecule into three dimensional structure.
- cross-linking at various sites stabilized by hydrophobic and hydrogen bonds produces a compactly coiled globular structure.

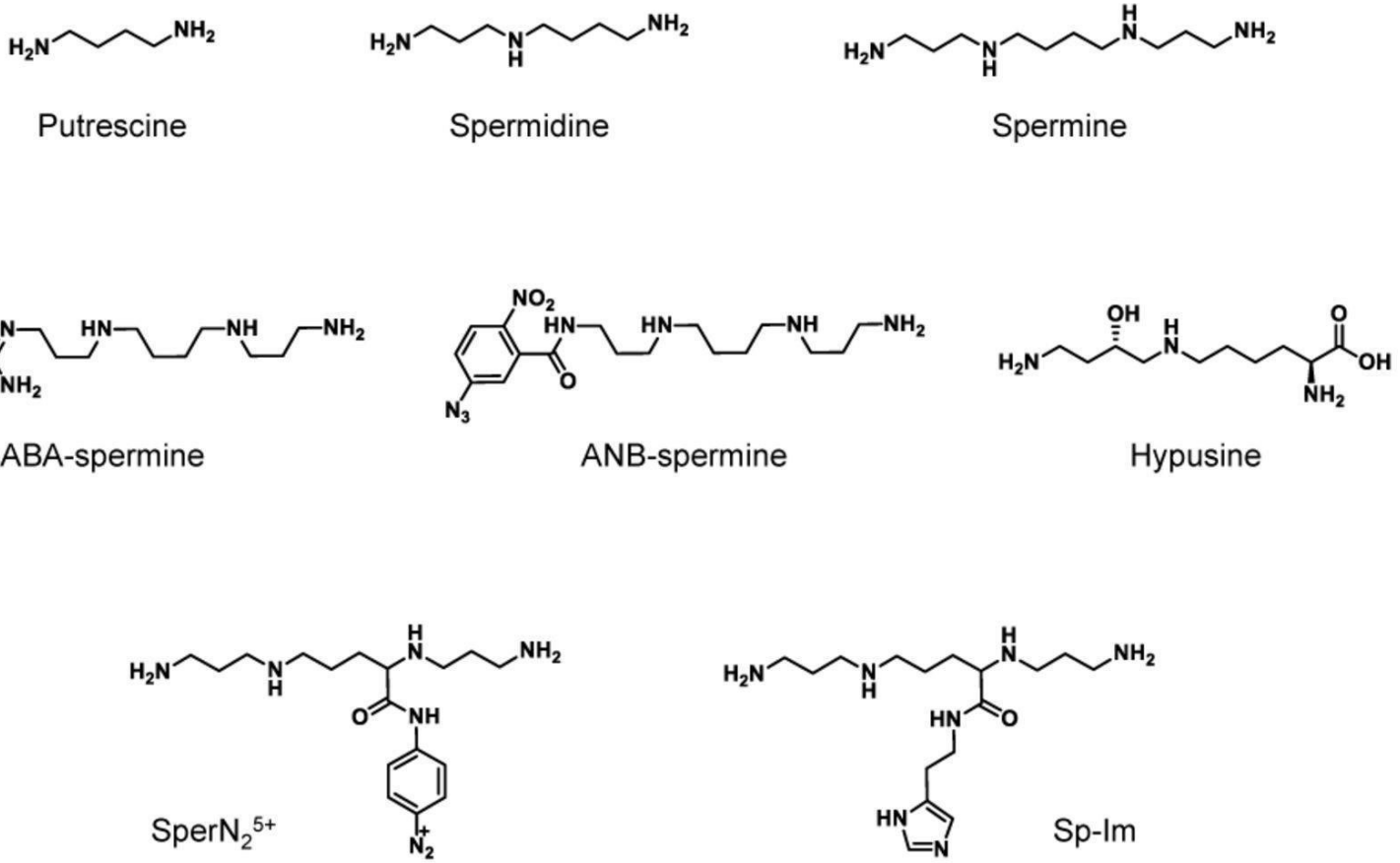
Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- The stacking of helices, together with specific helix–helix contacts or helix–loop interactions, lead to compact tertiary structure of the RNA assemblies, generally in the presence of divalent ions or polyamines

Chemical structures of the polyamines and polyamine analogues discussed throughout this review.



Helen L. Lightfoot, and Jonathan Hall Nucl. Acids Res. 2014;nar.gku837

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Nucleic Acids Research

Nucleotides and Nucleic Acids

Messenger RNA (mRNA)

- This class is the most heterogeneous in
- Abundance
- Size (500-6000 nucleotides)
- base sequence
- Stability

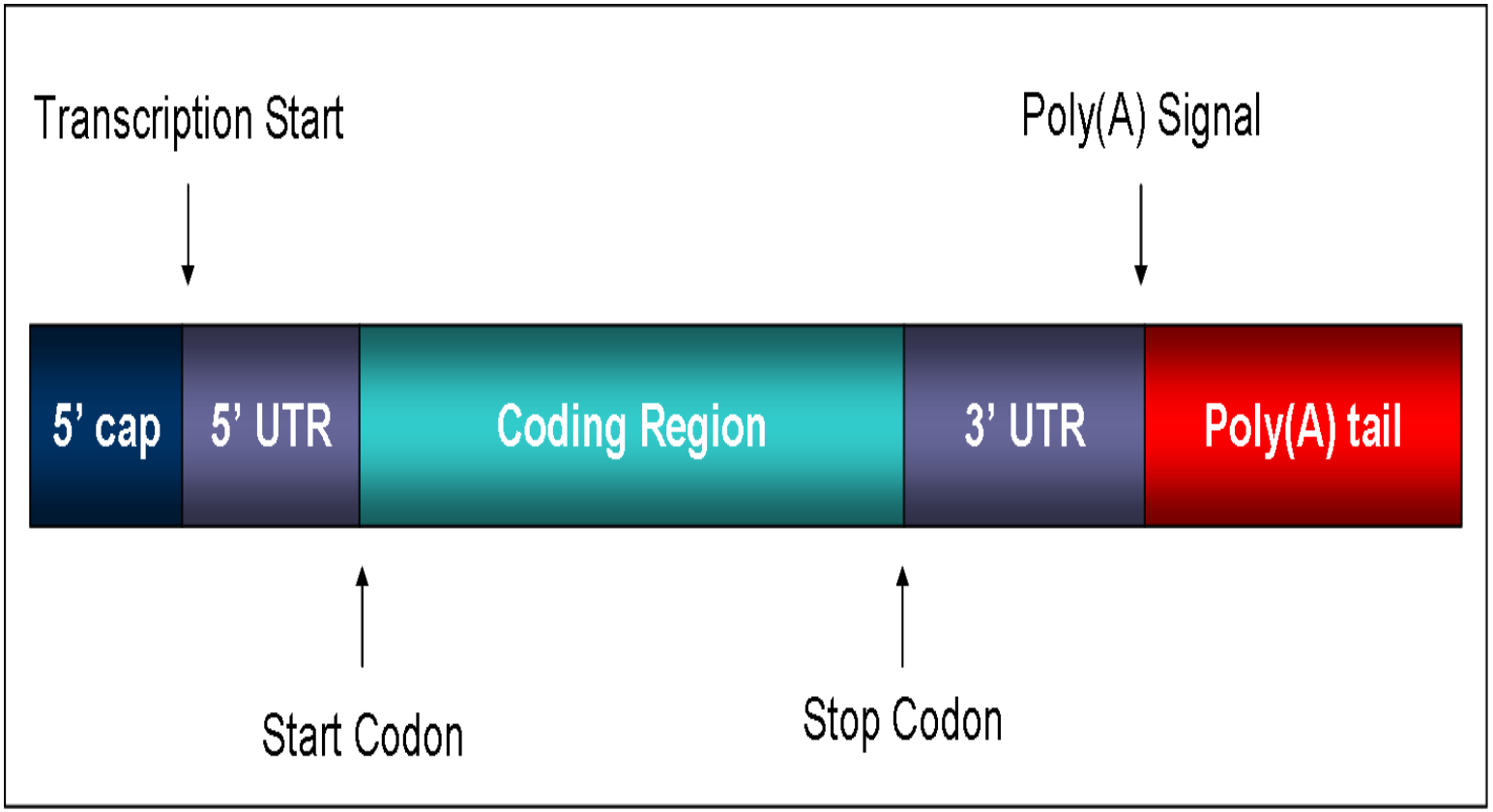
Nucleotides and Nucleic Acids

- mRNA comprise about 2–5% of total cellular RNA
- mRNA molecules are formed with the help of DNA template strand (3´-5´) during the process called transcription.

Nucleotides and Nucleic Acids

- In addition to the protein coding regions in the mature eukaryotic mRNA that can be translated,
- there are untranslated regions at its 5´ and 3´ ends
- Moreover, there is a 5´ cap and
- a poly A tail at 3´ end

Nucleotides and Nucleic Acids

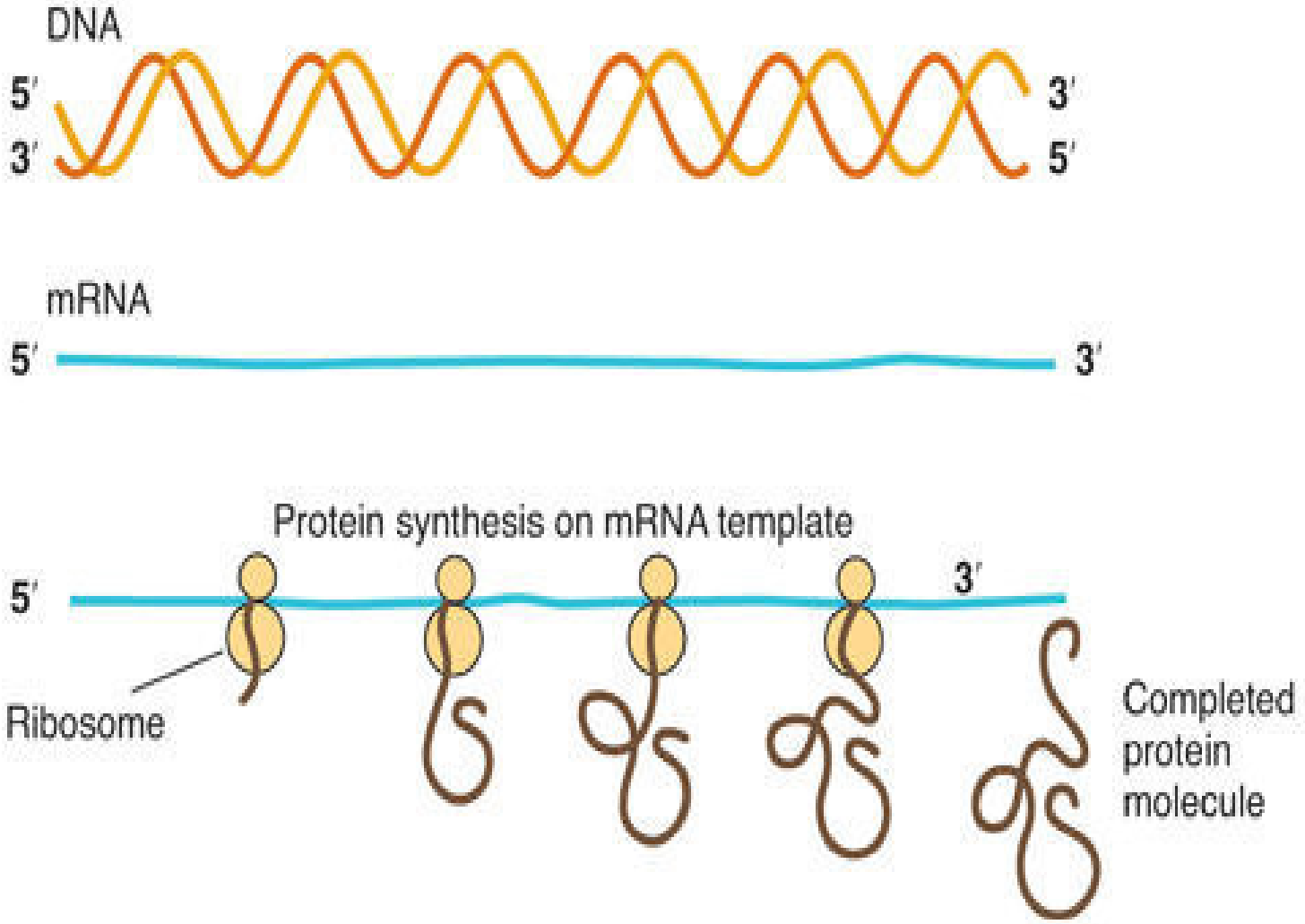


Nucleotides and Nucleic Acids

Function of mRNA

- The members of this class function as messengers to convey the information in a gene to the protein synthesizing machinery.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- The mRNA carries genetic information from the nuclear DNA to the cytosol, where it is used as a template for protein synthesis.

Nucleotides and Nucleic Acids

Transfer RNA (tRNA)

- t RNA is the smallest of the three major species of RNA (4S).
- They are single stranded globular molecules.
- They remain largely in cytoplasm.

Nucleotides and Nucleic Acids

- They are generated by nuclear processing of a precursor molecule.
- tRNAs compose roughly 20% of total cellular RNA
- There are at least 20 species of tRNA molecules in every cell.

Nucleotides and Nucleic Acids

- Although each specific tRNA differs from the others in its sequence of nucleotides, the tRNA molecules as a class have many features in common

Nucleotides and Nucleic Acids

Primary structure

- t RNA molecules consist of 74-95 nucleotides in a particular sequence.
- The t RNA molecules contain not only the usual bases like adenine, guanine, cytosine, uracil but also contain **unusual bases**

Nucleotides and Nucleic Acids

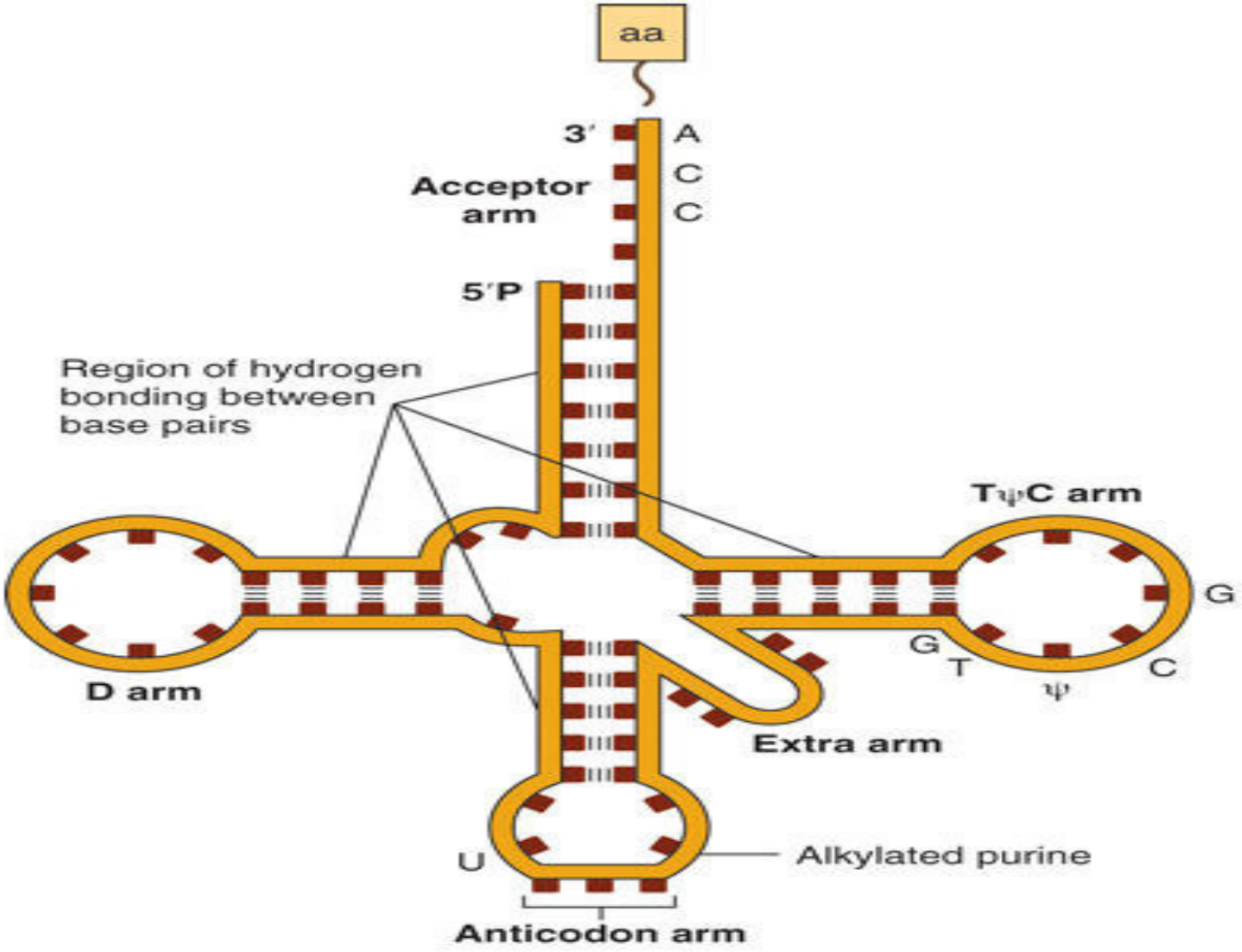
- These unusual bases(also called modified bases) include
- Dihydrouracil
- Pseudouridine
- Thymine.

Nucleotides and Nucleic Acids

Secondary Structure Pseudouridine

- Each single stranded t RNA is folded extensively.
- Extensive intra chain base pairing which leads to a characteristic CLOVER-LEAF structure.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- These folds are stabilized by hydrogen bonds between complementary bases of the same strand.

Nucleotides and Nucleic Acids

Transfer RNA (tRNA) (Countinued)

Arms or loops of tRNA

- All tRNA molecules contain 4 main arms or loops.
- **1-Acceptor arm:** This is made up of unpaired sequences of cytosine-cytosine-adenine (CCA) at the 3´end.

Nucleotides and Nucleic Acids

- The 3´OH group of adenine binds with the carboxylic group of a specific amino acid and carries it to ribosomes for protein synthesis.

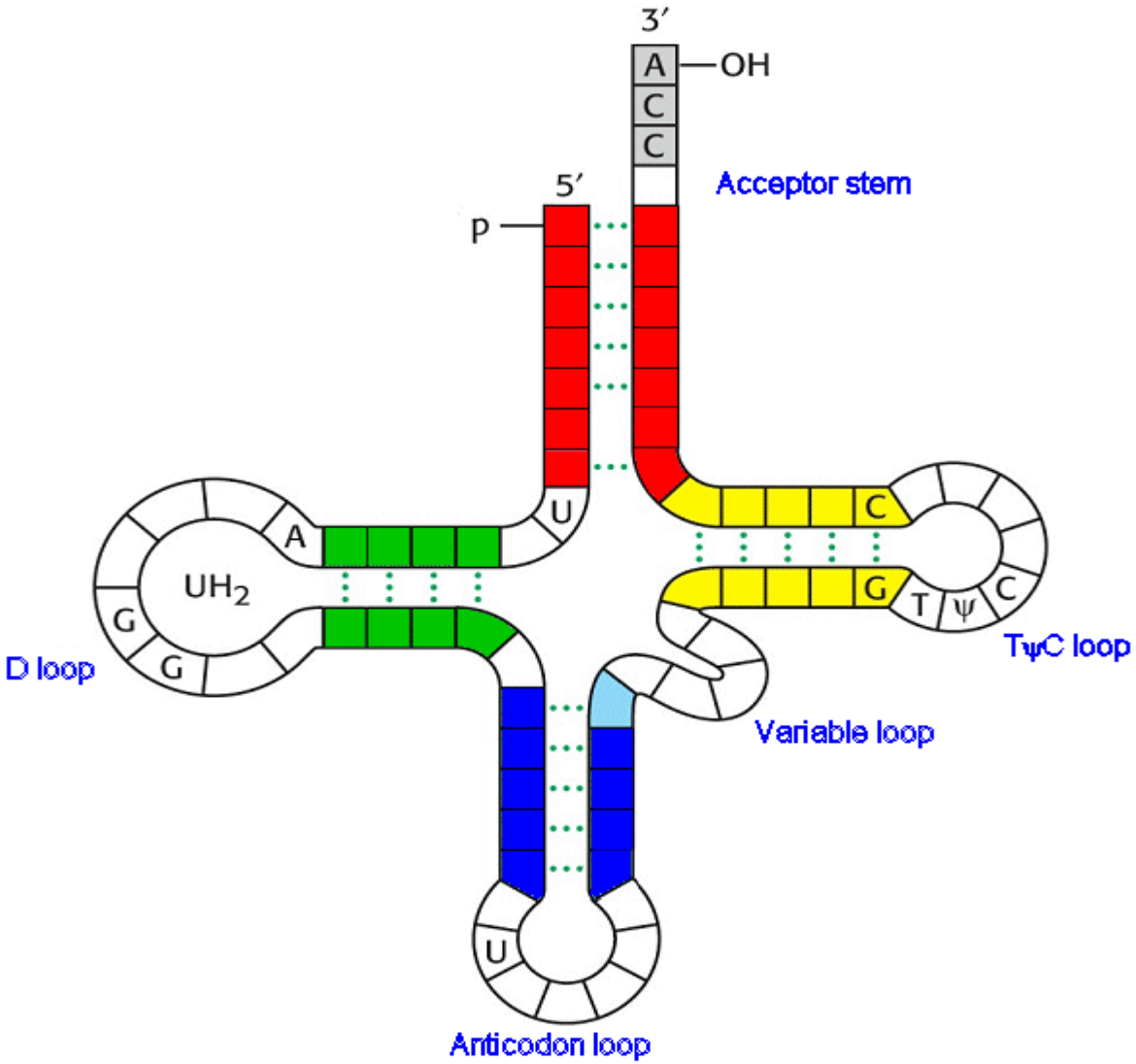
Nucleotides and Nucleic Acids

- **2-Anticodon arm:** It is in the form of a loop and carries specific sequences of three bases which constitute the anticodon.
- The bases of anticodon are bonded with three complementary bases of codon on mRNA.

Nucleotides and Nucleic Acids

- **3-D arm:** It contains the base dihydrouridine.
- **4-TΨC arm:** It contains thymine, pseudouridine and cytosine.
- The extra arm and the TΨC arms help to define a specific tRNA.

Nucleotides and Nucleic Acids



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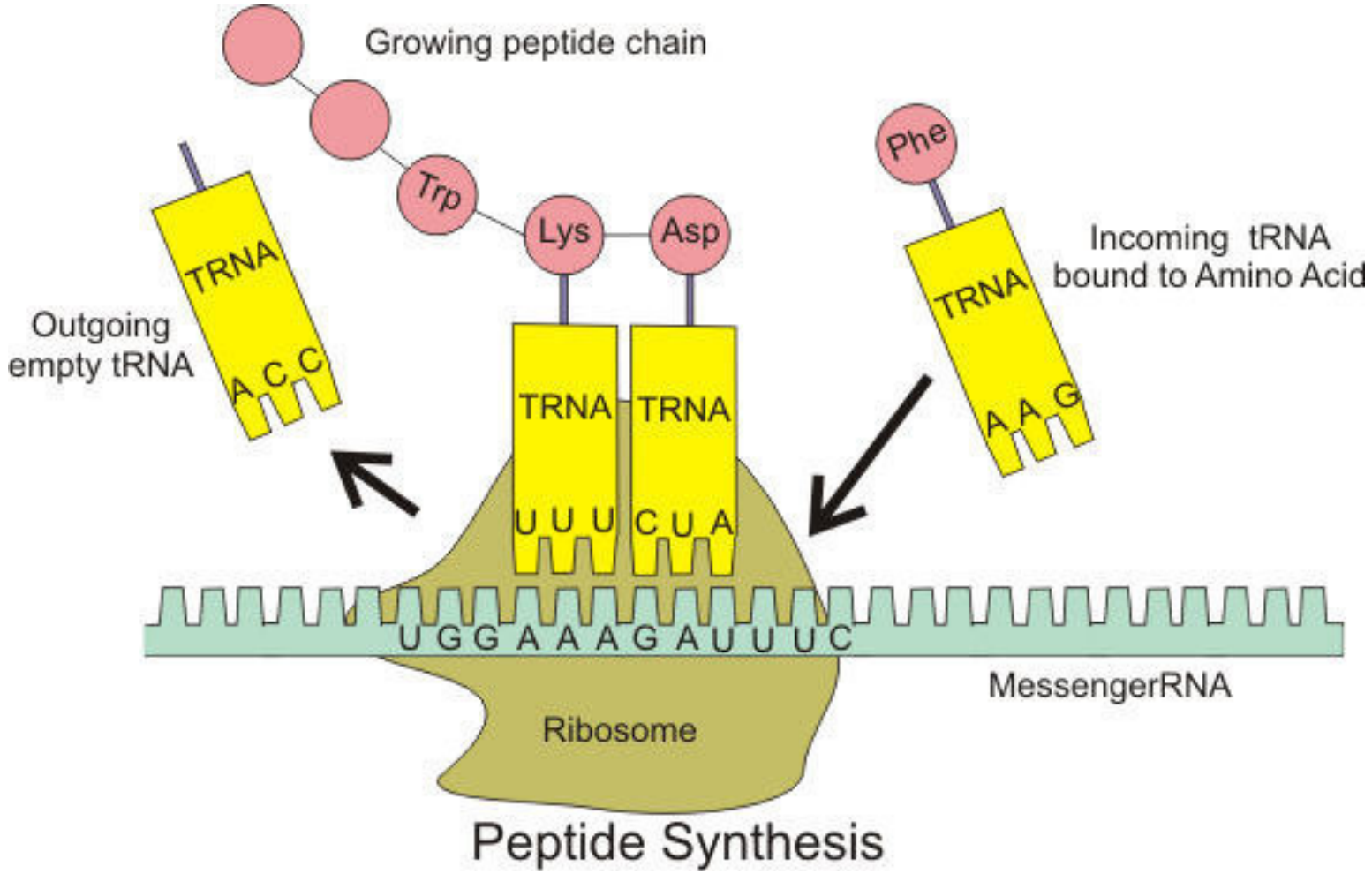
Nucleotides and Nucleic Acids

Function of tRNA

- The t RNA molecules serve as ADAPTERS for the translation of information in the sequence of nucleotides of the mRNA into specific amino acids.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids

- There is at least one (and often several) specific type of tRNA molecule for each of the amino acids commonly found in proteins.
- Each t RNA carries its specific amino acid to the site of protein synthesis.

Nucleotides and Nucleic Acids

- There it recognizes the genetic code word on mRNA (codon) and this specifies the addition of its amino acids to the growing peptide chain.

Nucleotides and Nucleic Acids

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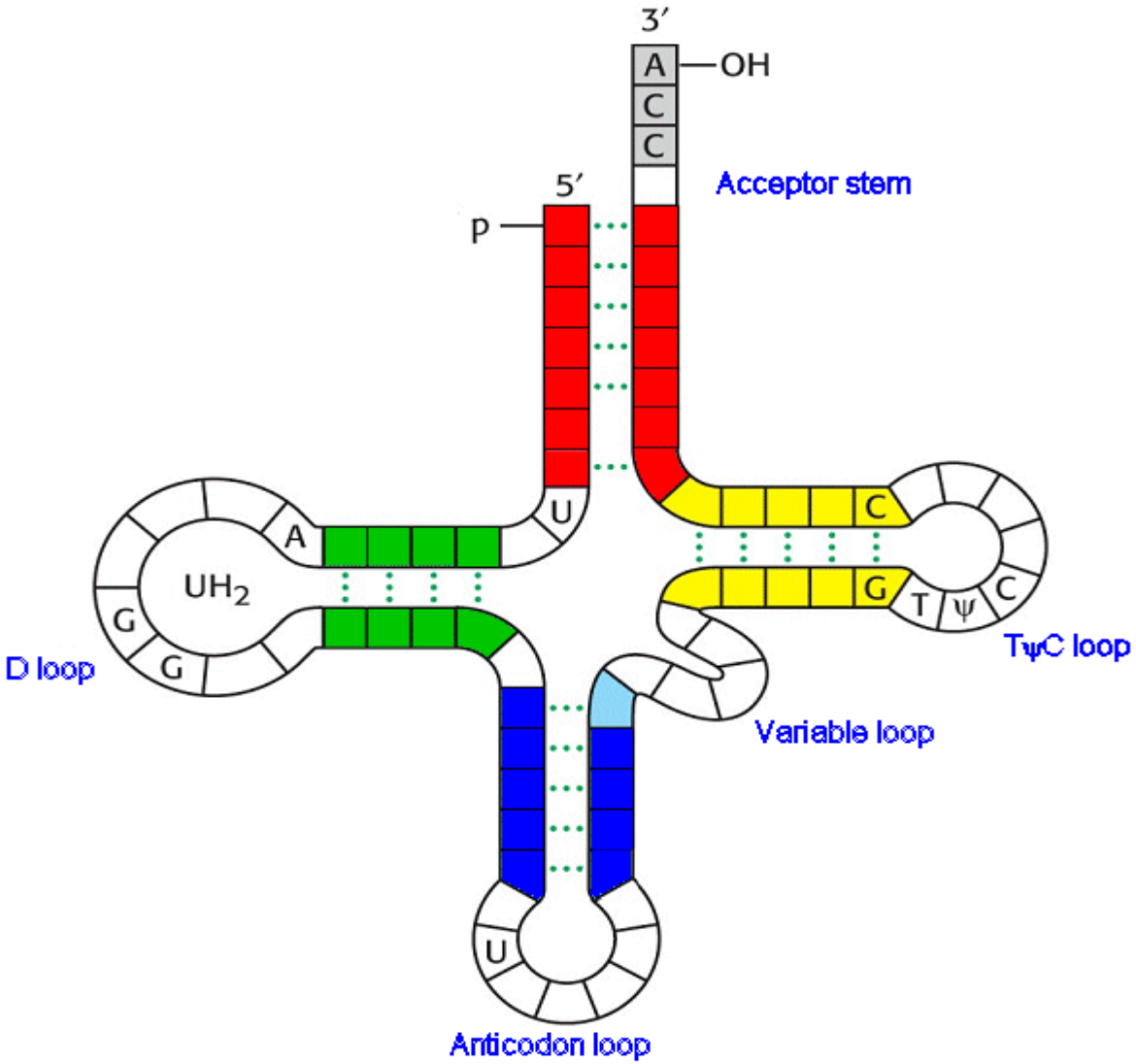

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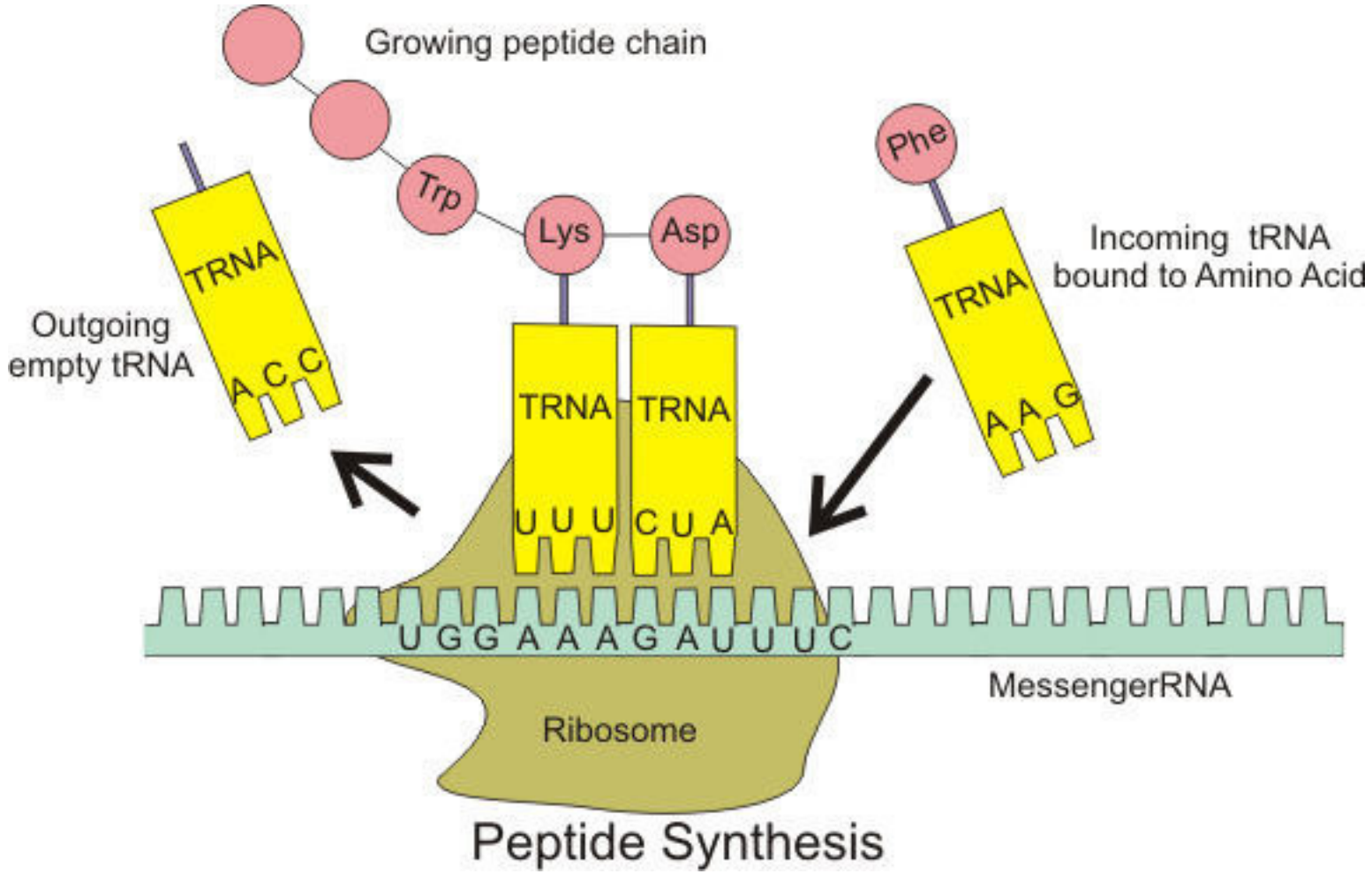

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Nucleotides and Nucleic Acids

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Nucleotides and Nucleic Acids

Ribosomal RNA (rRNA)

- found in association with several proteins as a component of ribosomes--- a cytoplasmic nucleoprotein structure that acts as the machinery for the synthesis of proteins from the mRNA template.

Nucleotides and Nucleic Acids

- RNAs make up 80% of the total RNA in the cell.
- The ribosomal subunits are defined according to their sedimentation velocity in **Svedberg units**.

Nucleotides and Nucleic Acids

- Svedberg unit is related to the molecular weight and shape of the compound.
- The bases in r RNA are mainly adenine, guanine, cytosine and uracil and a few pseudouridine

Nucleotides and Nucleic Acids

Eukaryotic Ribosome

- The mammalian ribosome contains two major nucleoprotein subunits:
- a larger one with 60S
- a smaller one with 40S.

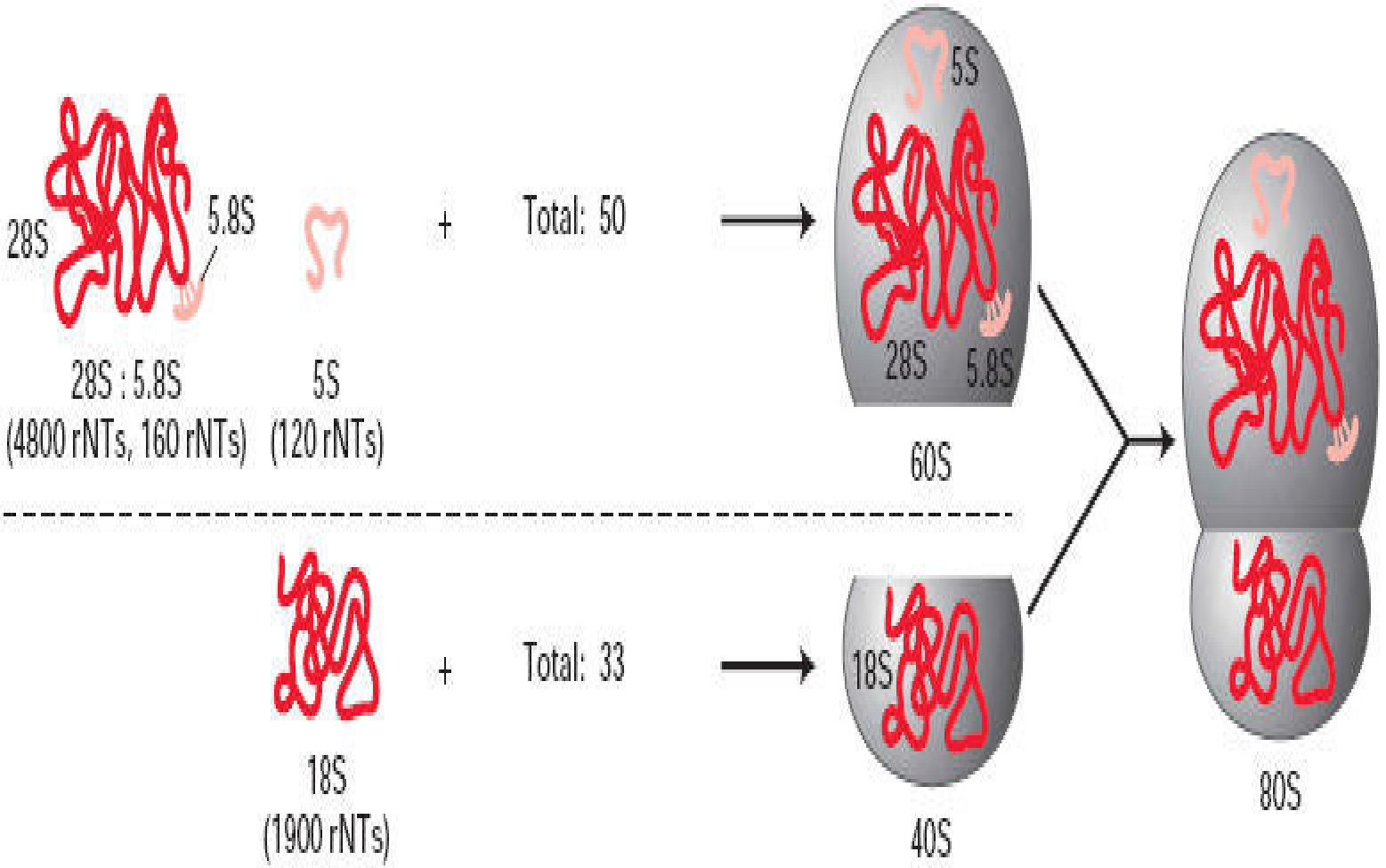
Nucleotides and Nucleic Acids

- The **60S** subunit contains
- a 5S rRNA
- a 5.8S rRNA
- a 28S rRNA
- more than 50 specific polypeptides.

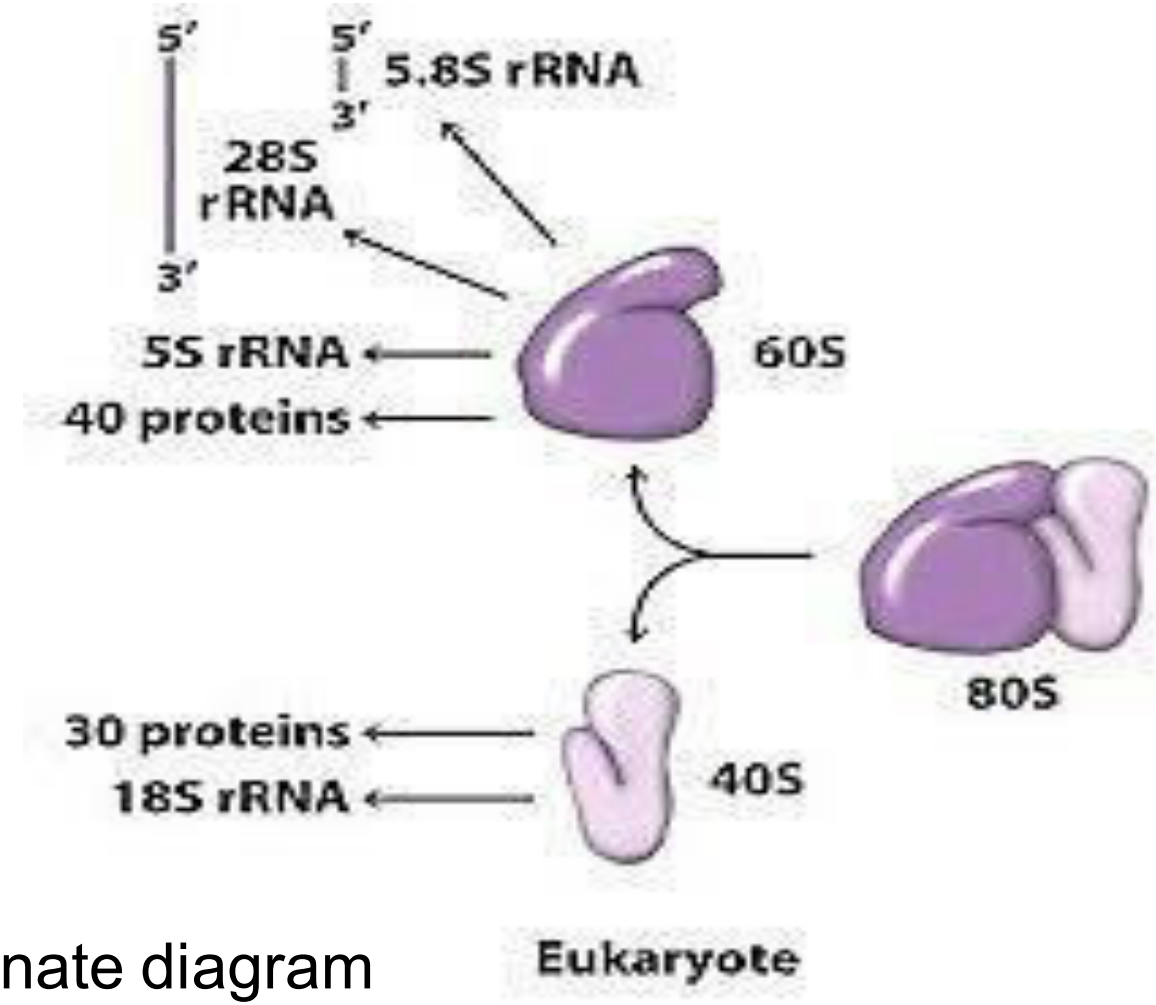
Nucleotides and Nucleic Acids

- The **40S** subunit is smaller and contains
- a single 18S rRNA
- Approx. 30 distinct polypeptide chains.

Nucleotides and Nucleic Acids



Nucleotides and Nucleic Acids



Alternate diagram

Nucleotides and Nucleic Acids

- In eukaryotes, all of the ribosomal RNA molecules except the 5S rRNA, which is independently transcribed, are processed from a single 45S precursor RNA molecule in the nucleolus- packed with the specific ribosomal proteins.

Nucleotides and Nucleic Acids

- The rRNA are necessary for ribosomal assembly and play a key role in the binding of mRNA to ribosomes and its translation
- In the cytoplasm, the ribosomes remain quite stable and capable of many translation cycles

Nucleotides and Nucleic Acids

Other types of RNA

Small nuclear RNA (snRNA)

- Small nuclear RNA (snRNA) are large number of small stable RNA species found in eukaryotic cells.
- Most of them are complexed with proteins to form ribonucleoproteins.

Nucleotides and Nucleic Acids

- They are distributed in the nucleus, in the cytoplasm or in both.
- They are significantly involved in rRNA and mRNA processing and gene regulation.

Nucleotides and Nucleic Acids

- **Large & Small Noncoding Regulatory RNAs**
- One of the most exciting discoveries in the last decade of eukaryotic regulatory biology has been the identification and characterization of regulatory nonprotein coding RNAs (ncRNAs).

Nucleotides and Nucleic Acids

- NcRNAs exist in two general size classes,
- small consisting of microRNA (miRNAs) and silencing (siRNAs) and

- Large consisting of long noncoding RNAs (lncRNAs)

Nucleotides and Nucleic Acids

- The small ncRNAs termed microRNA (miRNAs) and silencing (siRNAs) typically inhibit gene expression at the level of specific protein production by
- targeting mRNAs through one of several distinct mechanisms.

Nucleotides and Nucleic Acids

- Both siRNAs and miRNAs typically hybridize, via the formation of RNA–RNA hybridization to their targeted mRNAs

Nucleotides and Nucleic Acids

- **long noncoding RNAs (lncRNAs).**
- LncRNAs, which as their name implies, do not code for protein (ie, the mRNA encoding genes).
- ncRNAs make up a significant portion of eukaryotic transcription

Nucleotides and Nucleic Acids

- ncRNAs play many roles ranging from contributing to structural aspects of chromatin to regulation of mRNA gene transcription by RNA polymerase II.

END

Enzymes

- Reaction catalysts of biological systems: the enzymes, the most remarkable and highly specialized proteins.
- Enzymes have extraordinary catalytic power, often far greater than that of synthetic or inorganic catalysts.

Enzymes

- With the exception of some catalytic RNA molecules all enzymes are proteins.
- Thus the primary, secondary, tertiary, and quaternary structures of protein enzymes are essential to their catalytic activity.

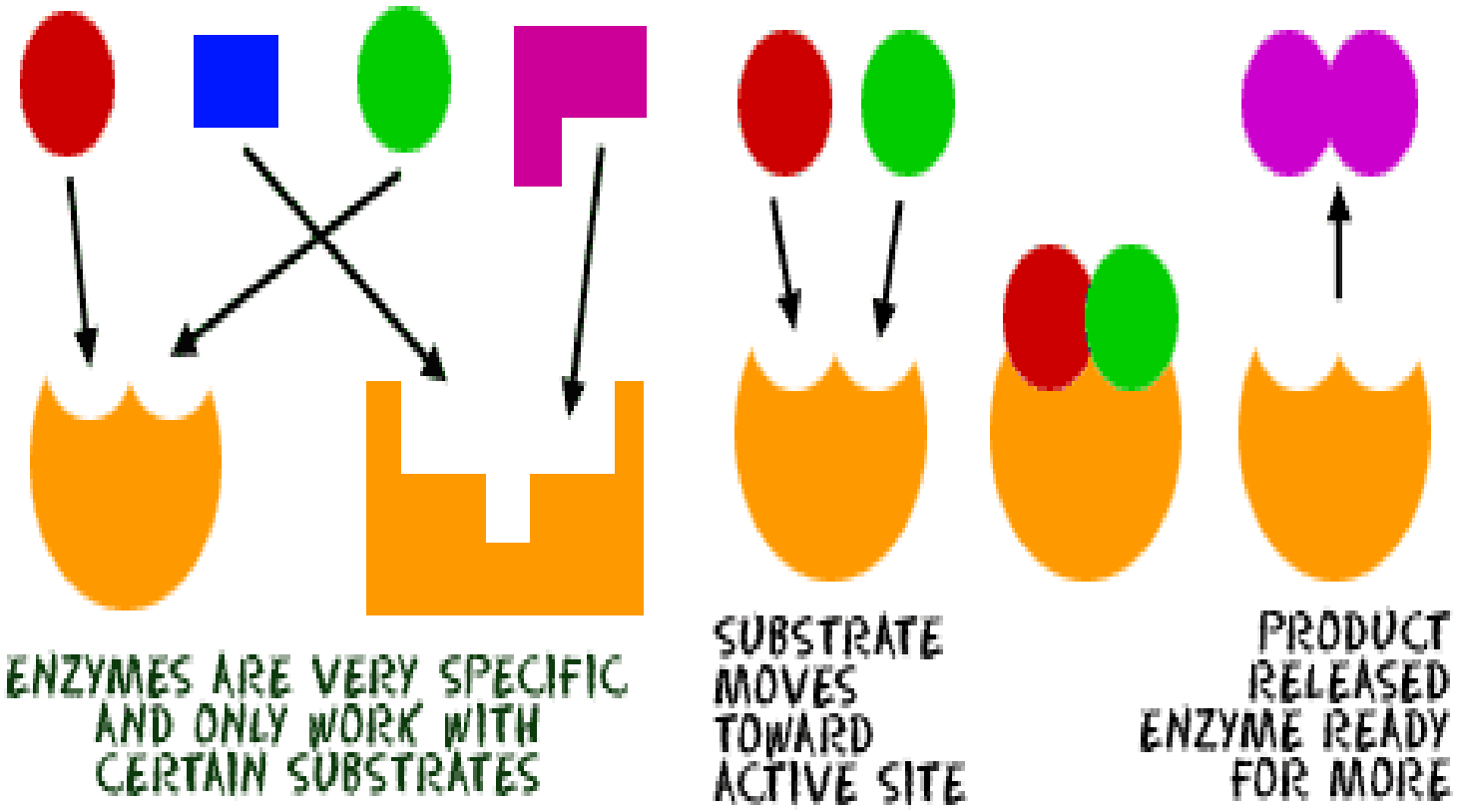
Enzymes

- Enzyme-catalyzed reactions have three basic steps:
- binding of substrate: **$E+S \leftrightarrow ES$**
- conversion of bound substrate to bound product: **$ES \leftrightarrow EP$**
- release of product : **$EP \leftrightarrow E+P$**

Enzymes

- They have a high degree of specificity for their substrates, they function in aqueous solutions under very mild conditions of temperature and pH.
- Enzymes do not invent new reactions; they simply make reactions occur faster.

Enzymes


ENZYMES ARE VERY SPECIFIC AND ONLY WORK WITH CERTAIN SUBSTRATES
SUBSTRATE MOVES TOWARD ACTIVE SITE
PRODUCT RELEASED ENZYME READY FOR MORE

Enzymes

- In addition to increasing the speed of reactions, enzymes provide means for regulating the rate of metabolic pathways in the body.

Enzymes

- The commonly used names for most enzymes describe the type of reaction catalyzed, followed by the suffix ***–ase*** *e.g.*
- **dehydrogen*ases*** remove hydrogen atoms
- **prote*ases*** hydrolyze proteins

Enzymes

- Modifiers may precede the name to indicate, for example,
- the substrate (xanthine oxidase),
- the source of the enzyme (pancreatic ribonuclease),
- its regulation (hormone-sensitive lipase)

Enzymes

- Where needed, alphanumeric designators are added to identify multiple forms of an enzyme e.g,
- RNA polymerase III
- protein kinase C

END

Enzymes

IUB Classification of Enzymes

- International Union of Biochemists (IUB) developed an unambiguous system of enzyme nomenclature in which each enzyme has a
- unique name and
- code number

Unique name and code no identify the type of reaction catalyzed and the substrates involved

Enzymes

- As an example, the formal systematic name of the enzyme (hexokinase) catalyzing the reaction
- ATP + D-glucose→ ADP + D-glucose-6 phosphate is
- ATP:glucose phosphotransferase,

ATP:glucose phosphotransferase

Enzymes

- Its Enzyme Commission (E.C.) number is 2.7.1.1.
- (2) denotes the class name (transferase)
- (7) the subclass phosphotransferase
- (1) denotes a hydroxyl group as acceptor
- (1) D-glucose as the phosphoryl group acceptor.

Enzymes

- In the systematic naming system, enzymes are divided into six major classes each with numerous subgroups

Enzymes

CLASSIFICATION OF ENZYMES

Group of Enzyme	Reaction Catalysed	Examples
1. Oxidoreductases	Transfer of hydrogen and oxygen atoms or electrons from one substrate to another.	Dehydrogenases Oxidases
2. Transferases	Transfer of a specific group (a phosphate or methyl etc.) from one substrate to another.	Transaminase Kinases
3. Hydrolases	Hydrolysis of a substrate.	Estrases Digestive enzymes
4. Isomerases	Change of the molecular form of the substrate.	Phospho hexo Isomerase, Fumarase
5. Lyases	Nonhydrolytic removal of a group or addition of a group to a substrate.	Decarboxylases Aldolases
6. Ligases (Synthetases)	Joining of two molecules by the formation of new bonds.	Citric acid synthetase

Enzymes

CLASSIFICATION OF ENZYMES		
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Enzymes

CLASSIFICATION OF ENZYMES		
Group of Enzyme	**Reaction Catalysed**	**Examples**
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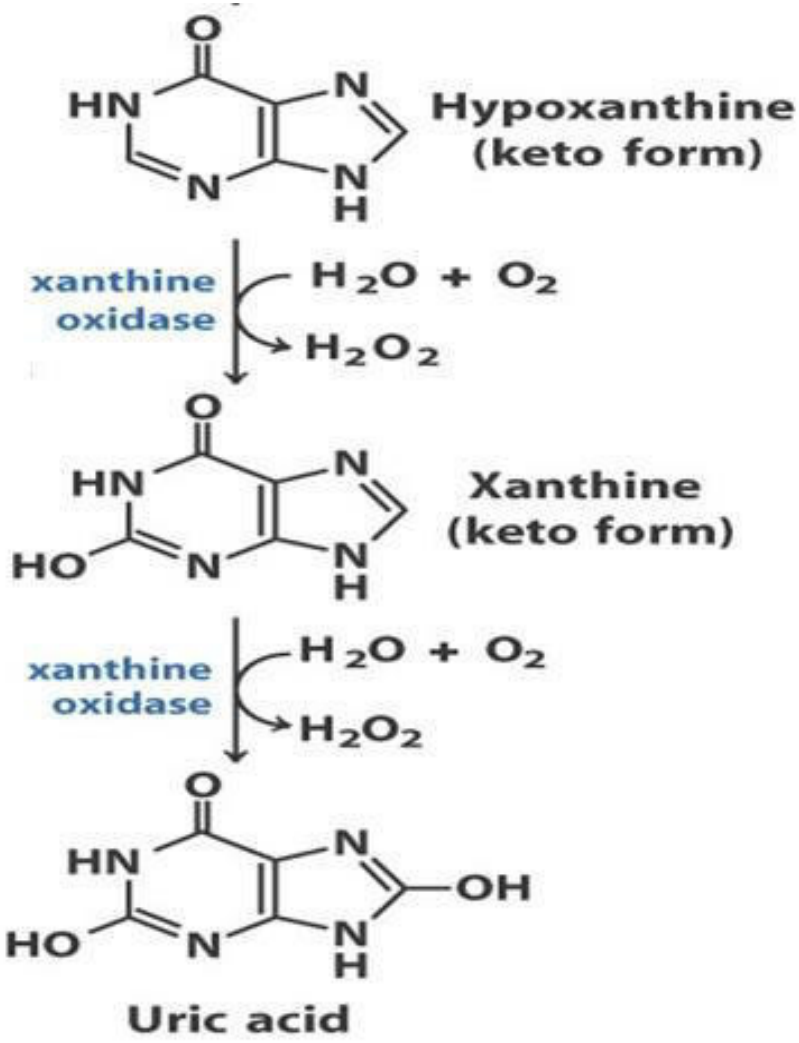
6. Ligases (Synthetases)	**Joining of two molecules by the formation of new bonds.**	**Citric acid Synthetase**

Enzymes

1) Oxidoreductases

- catalyze oxidation reduction reactions
- further divided into four subgroups;
- Oxidase,
- Dehydrogenases,
- Hydroperoxidases
- Oxygenases.

Enzymes



- Two reactions both catalyzed by Xanthine oxidase are given
- Hypoxanthine → xanthine
- Xanthine → Uric acid

Enzymes

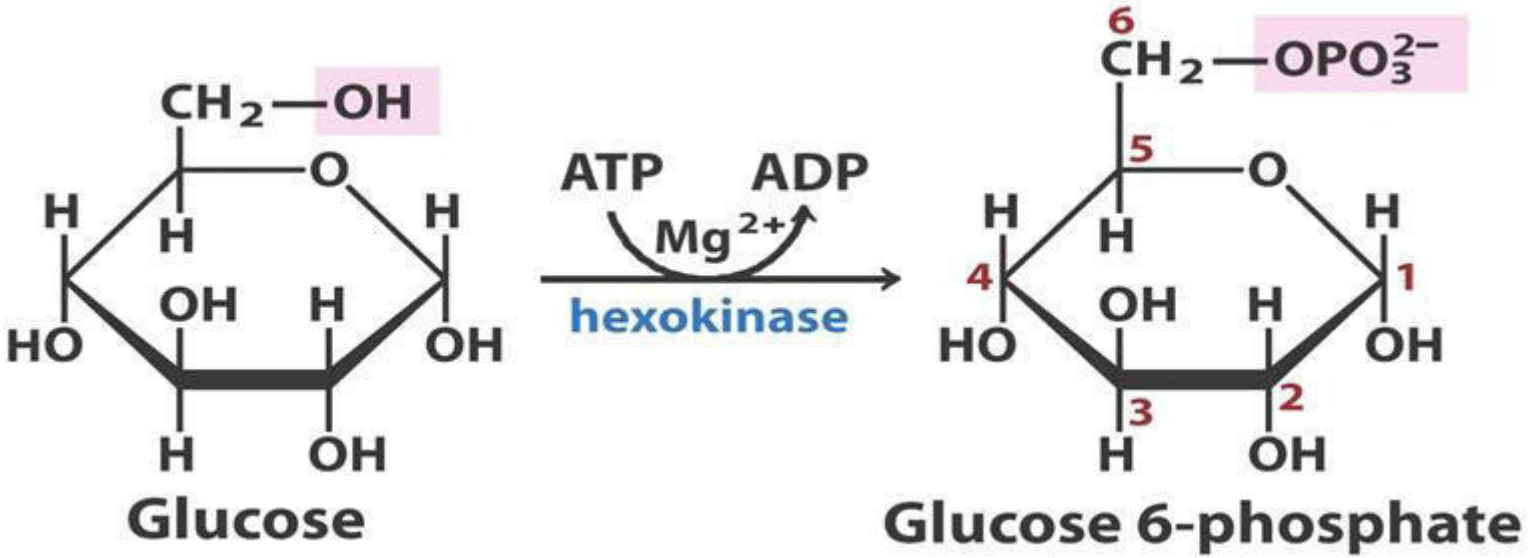
2) Transferases

- These bring about a transfer of functional groups such as
- phosphate and
- amino group
- from one molecule to another molecule called donor and acceptor molecules respectively.

Enzymes

- The common examples of this group are
- Transminases
- Phosphotransfrases (Kinases)

Enzymes



- Hexokinase is a phosphotransfease which catalyze the transfer of phosphate groups.

- Glucose + ATP → Glucose 6-phosphate + ADP.

Enzymes

IUB Classification of Enzymes (Contd.)

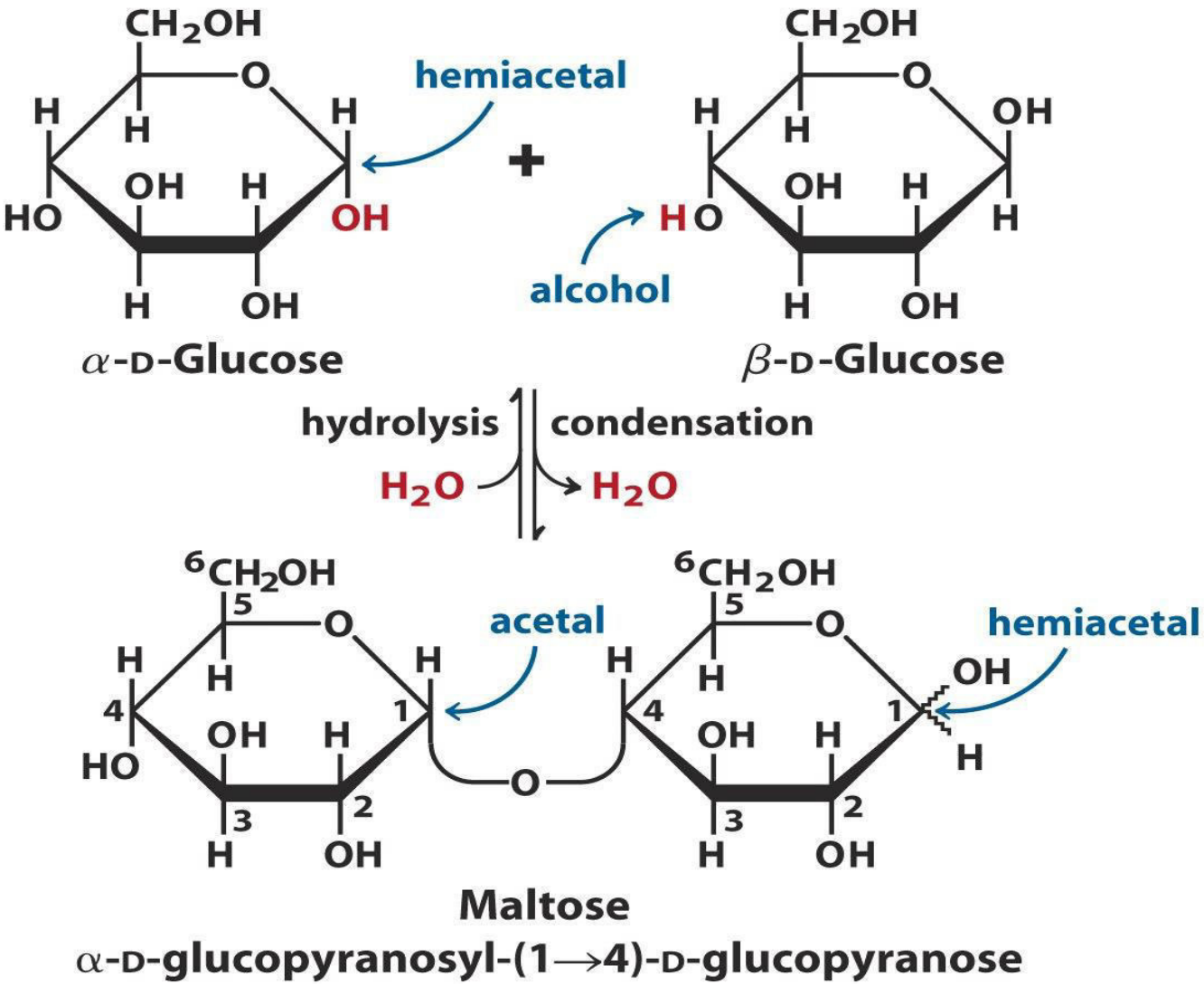
3) Hydrolases

- These enzymes catalyze hydrolysis, i.e.
- add water molecule to the substrate which is simultaneously decomposed; the functional group of substrate is transferred to water.

Enzymes

- Common example of hydrolases are:
- Protein hydrolyzing Enzymes (peptidases).
- Carbohydrases
- Lipid hydrolyzing enzymes e.g. Lipases and
- Phospholipases.

Enzymes

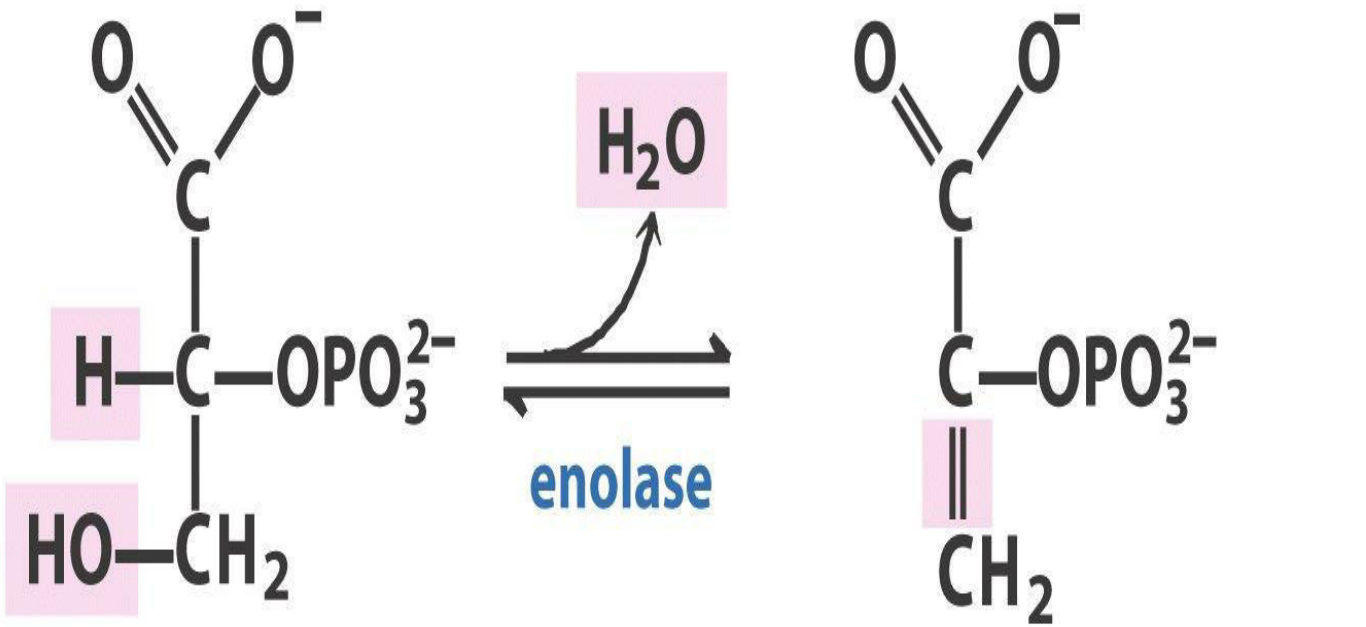

hemiacetal
alcohol
α-D-Glucose
β-D-Glucose
hydrolysis
condensation
H2O
H2O
acetal
hemiacetal
Maltose
α-D-glucopyranosyl-(1→4)-D-glucopyranose

Enzymes

4) Lyases

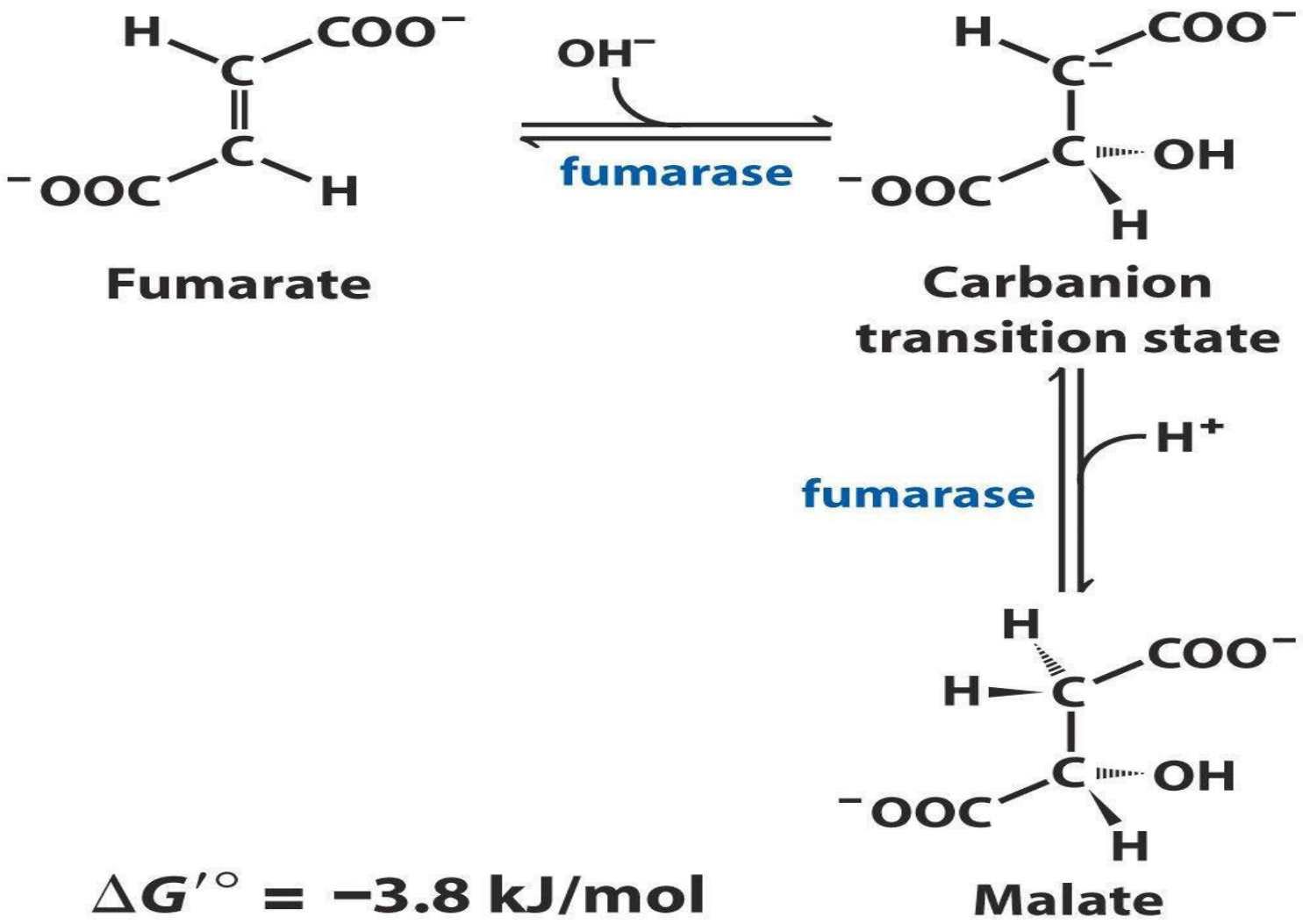
- Theses enzymes catalyze the addition of
- NH_3,
- H_2O or
- CO_2 to double bonds or
- the removal of these groups leaving behind double bonds.

Enzymes



$\Delta G'^{\circ} = 7.5$ kJ/mol

Enzymes


H
COO⁻
C
C
⁻OOC
H
OH⁻
fumarase
H
COO⁻
C⁻
C
OH
⁻OOC
H
Fumarate
Carbanion
transition state
fumarase
H⁺
H
COO⁻
H
C
C
OH
⁻OOC
H
ΔG′° = −3.8 kJ/mol
Malate

Enzymes

- Lyases are included in a separate class because they catalyze these reactions by means other than hydrolysis or oxidation.

END

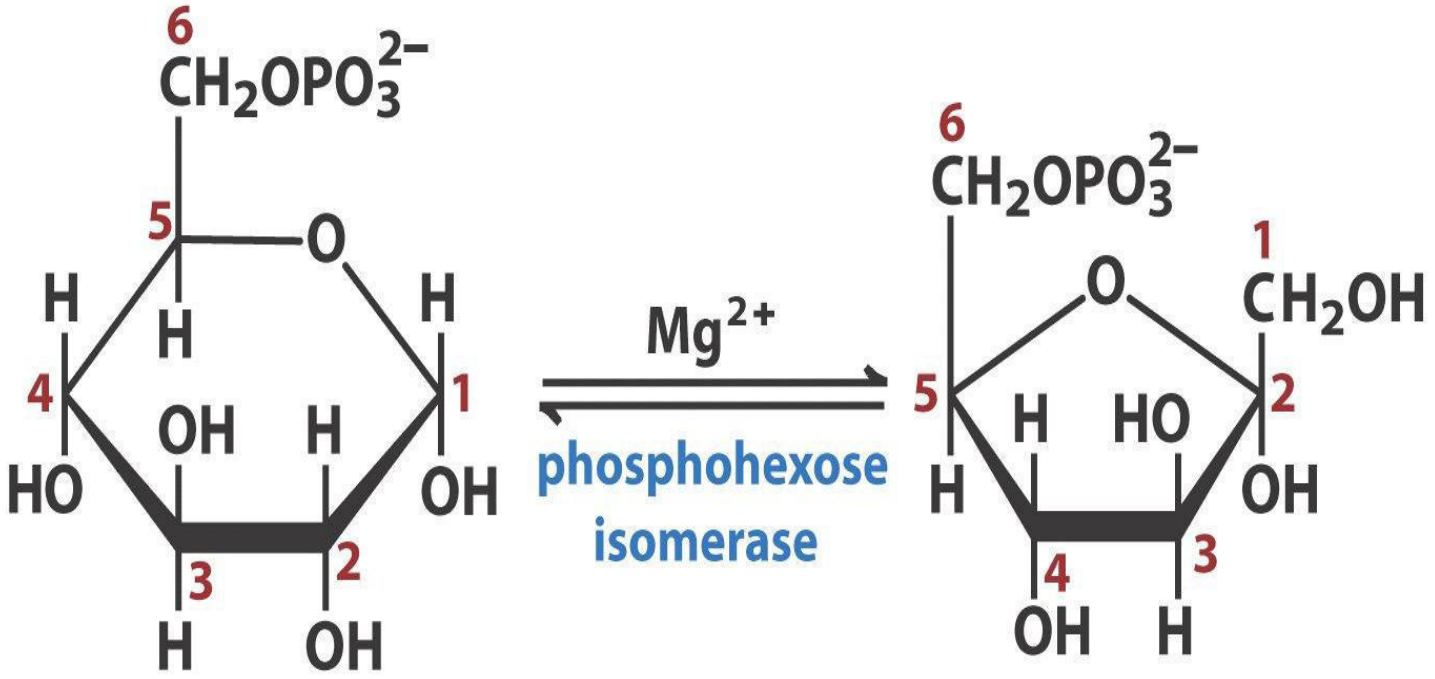
Enzymes

IUB Classification of Enzymes (Contd.)

5) Isomerases

- These enzymes catalyze the structural change within a single molecule by the transfer of groups within it, resulting in the formation of an isomeric form of substrate.

Enzymes

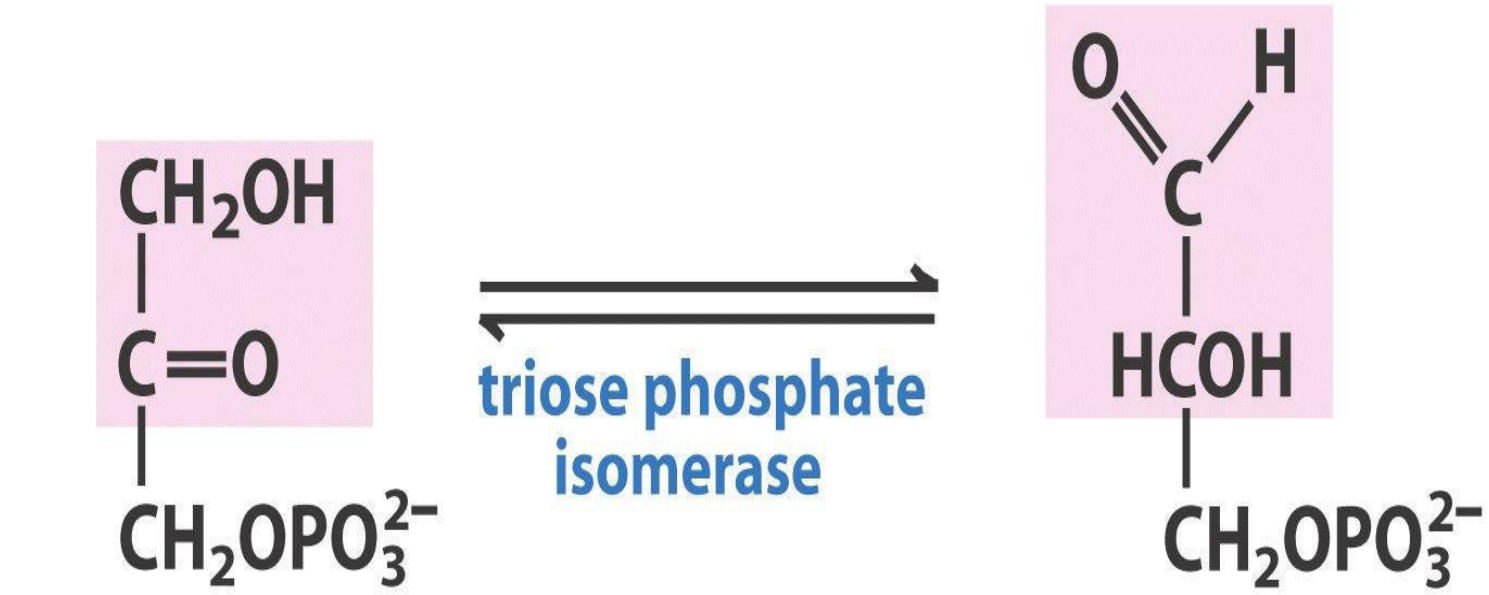


Glucose 6-phosphate

Fructose 6-phosphate

$\Delta G'^{\circ} = 1.7$ kJ/mol

Enzymes



Dihydroxyacetone phosphate

Glyceraldehyde 3-phosphate

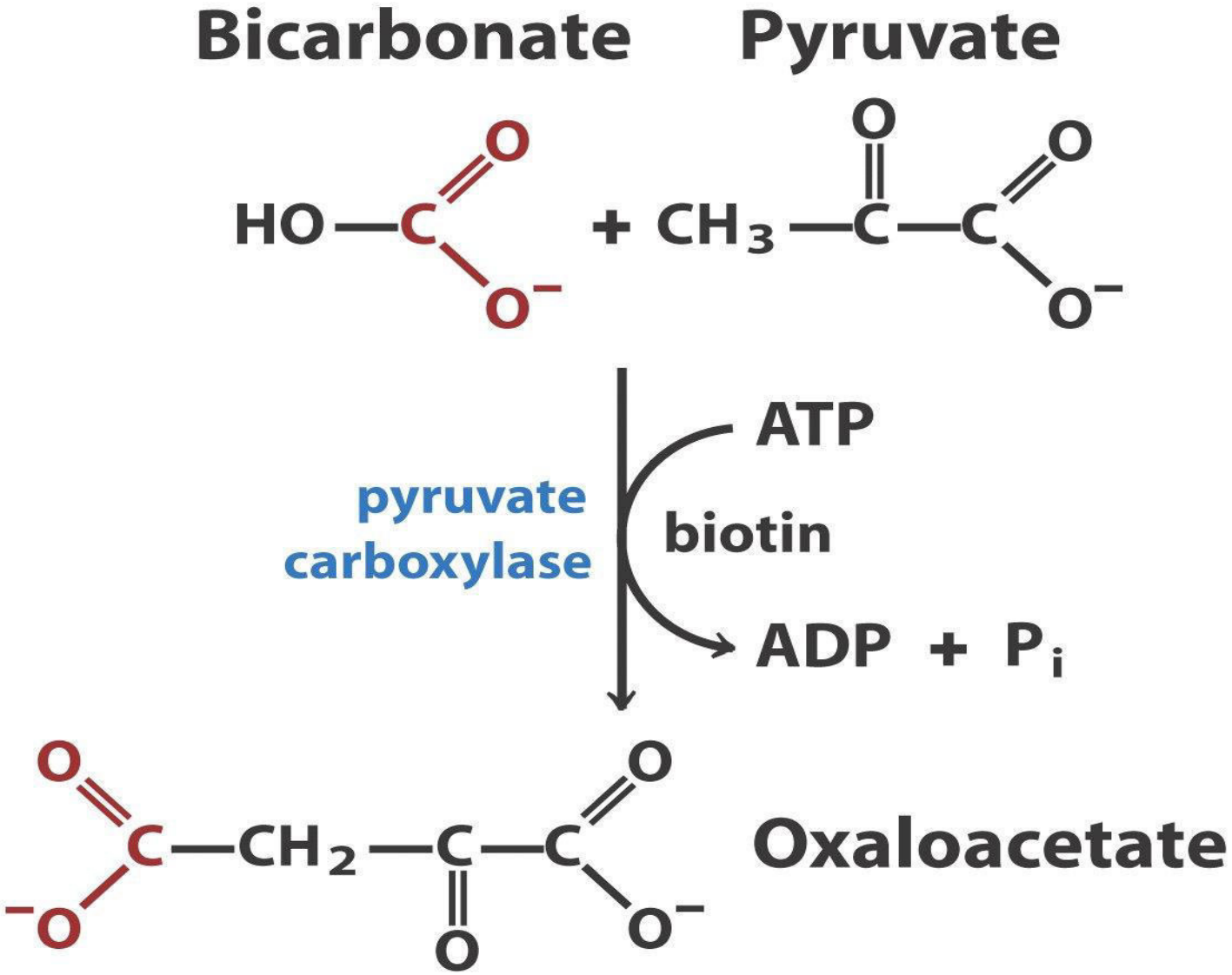
$\Delta G'^{\circ} = 7.5$ kJ/mol

Enzymes

6) Ligases

- These enzymes catalyze condensation reactions joining two molecules by forming
- C-O,
- C-S,
- C-N and
- C-C bonds.

Enzymes


Bicarbonate
Pyruvate
HO
C
O
O⁻
+ CH₃
C
C
O
O
O⁻
pyruvate carboxylase
ATP
biotin
ADP + Pi
O
C
O⁻
CH₂
C
C
O
O
O
O⁻
Oxaloacetate

Enzymes

- The energy for condensation is provided by cleavage of high energy phosphates, e.g. ATP, GTP etc.

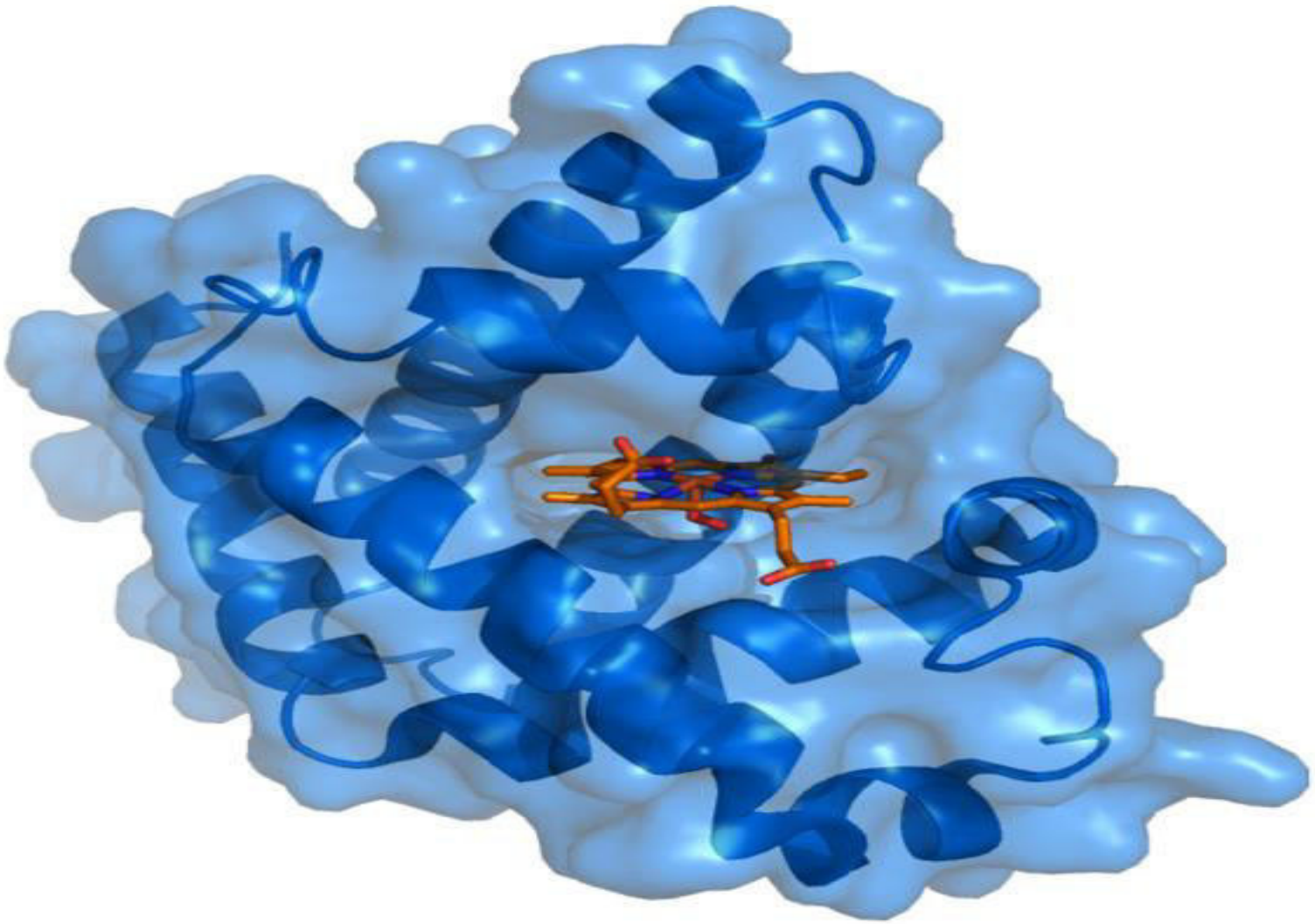
END

Enzymes

Ligand

- In biochemistry and pharmacology, a **ligand** (from the Latin *ligandum, binding*) is a substance (usually a small molecule),
- that forms a complex with a biomolecule to serve a biological purpose.

Enzymes



Enzymes

- In a narrower sense, it is a signal triggering molecule, binding to a site on a target protein
- A molecule bound reversibly by a protein is called a **ligand.**
- Ligands include substrates, inhibitors, activators, and neurotransmitters

Enzymes

- A ligand may be any kind of molecule, including another protein.
- A ligand binds at a site on the protein called the **binding site**,

Enzymes

- **binding site** is complementary to the ligand in
- size,
- shape,
- charge, and
- hydrophobic or
- hydrophilic character.

Enzymes

- Furthermore, the interaction is **specific**: the protein can discriminate among the thousands of different molecules in its environment and selectively bind only one or a few.

Enzymes

- The binding of a protein and ligand is often coupled to a **conformational change** in the protein that makes the binding site more complementary to the ligand, permitting tighter binding called **induced fit.**

Enzymes

- A given protein may have **separate binding sites** for several different ligands.
- In a multi-subunit protein, a conformational change in one subunit often affects the conformation of other subunits.

Enzymes

- Interactions between **ligands** and proteins may be regulated, through interactions with additional ligands.
- These other ligands may cause conformational changes in the protein that affect the binding of the **first ligand**.

END

Enzymes

Mechanism of Enzyme Action

Enzymes bind and chemically transform other molecules— they catalyze reactions.

- The molecules acted upon by enzymes are called reaction **substrates** rather than ligands.

Enzymes

- Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 10^5 to 10^{17}.
- The distinguishing feature of an enzyme-catalyzed reaction is that it takes place within the confines of a pocket on the enzyme called the **active site.**

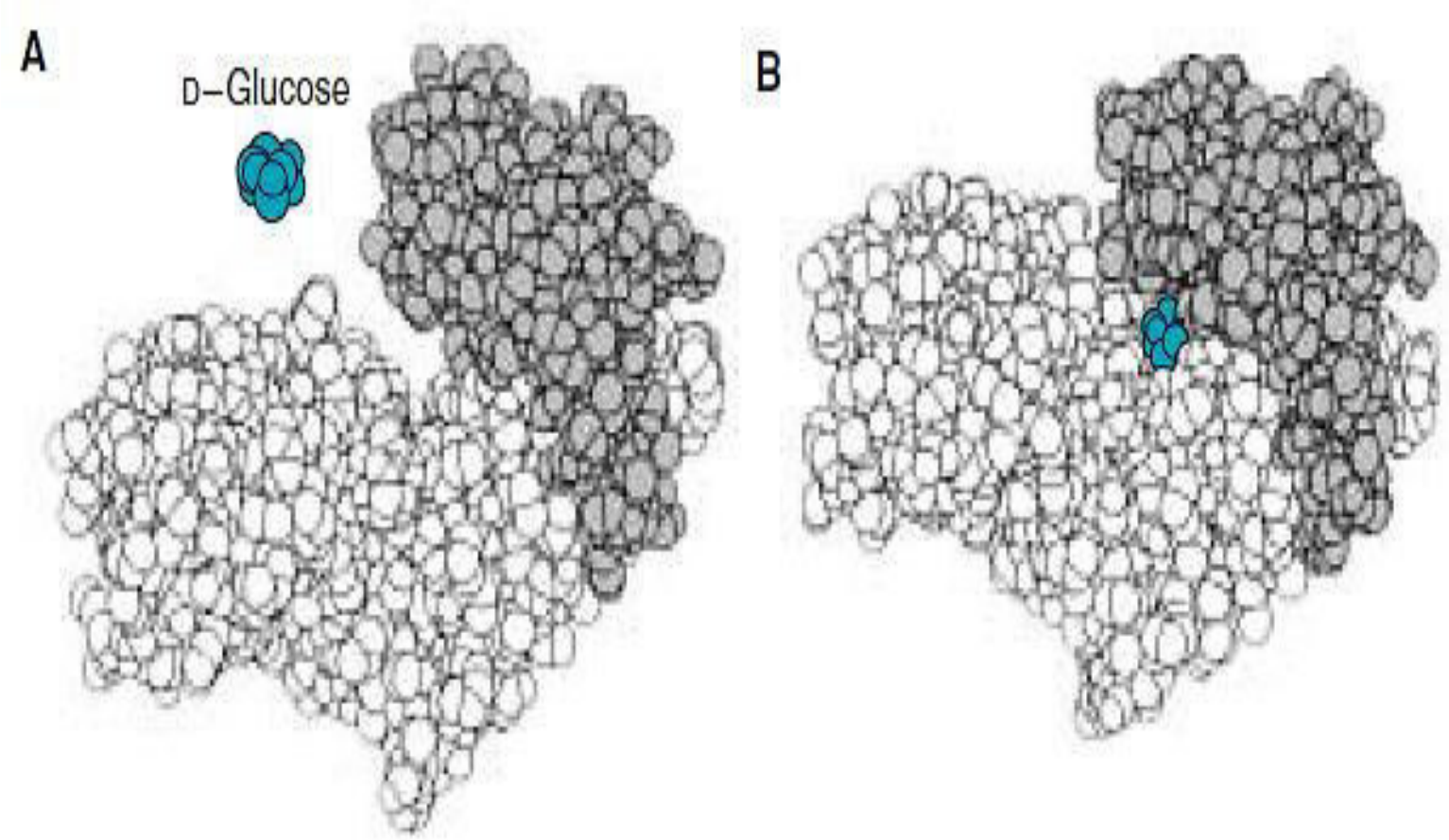
Enzymes

- The surface of the active site is lined with **amino acid** residues with side chains that complement and bind the substrate and catalyze its chemical transformation.

Enzymes

- Often, the active site encloses a substrate, sequestering it completely from solution.
- The enzyme substrate complex, whose existence is central to the action of enzymes.

Enzymes



Enzymes

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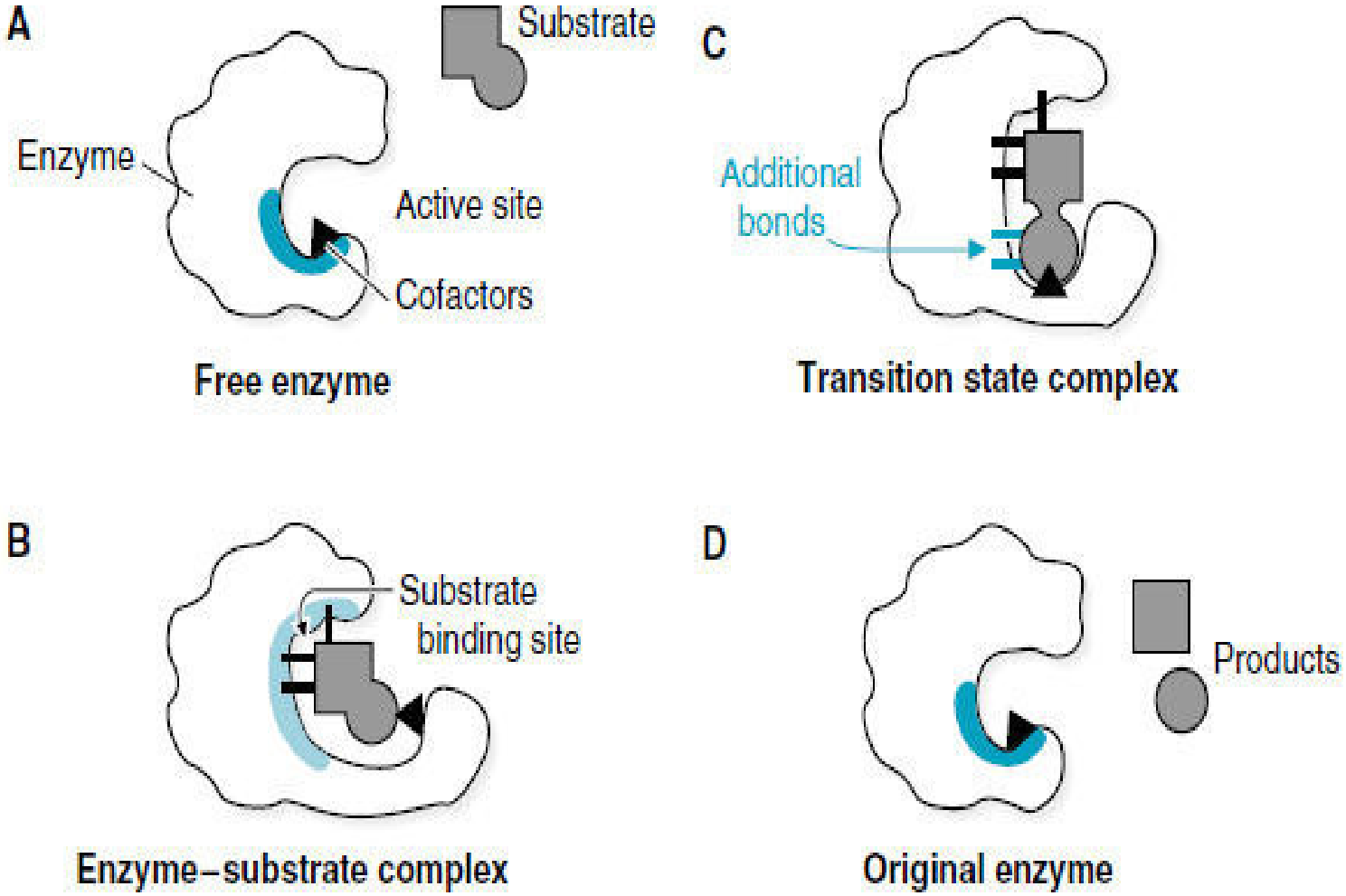
Enzymes


A
Substrate
Enzyme
Active site
Cofactors
Free enzyme
C
Additional
bonds
Transition state complex
B
Substrate
binding site
Enzyme–substrate complex
D
Products
Original enzyme

Enzymes

- To understand catalysis, we must first appreciate the important distinction between
- reaction equilibria and
- reaction rates.

END

Enzymes

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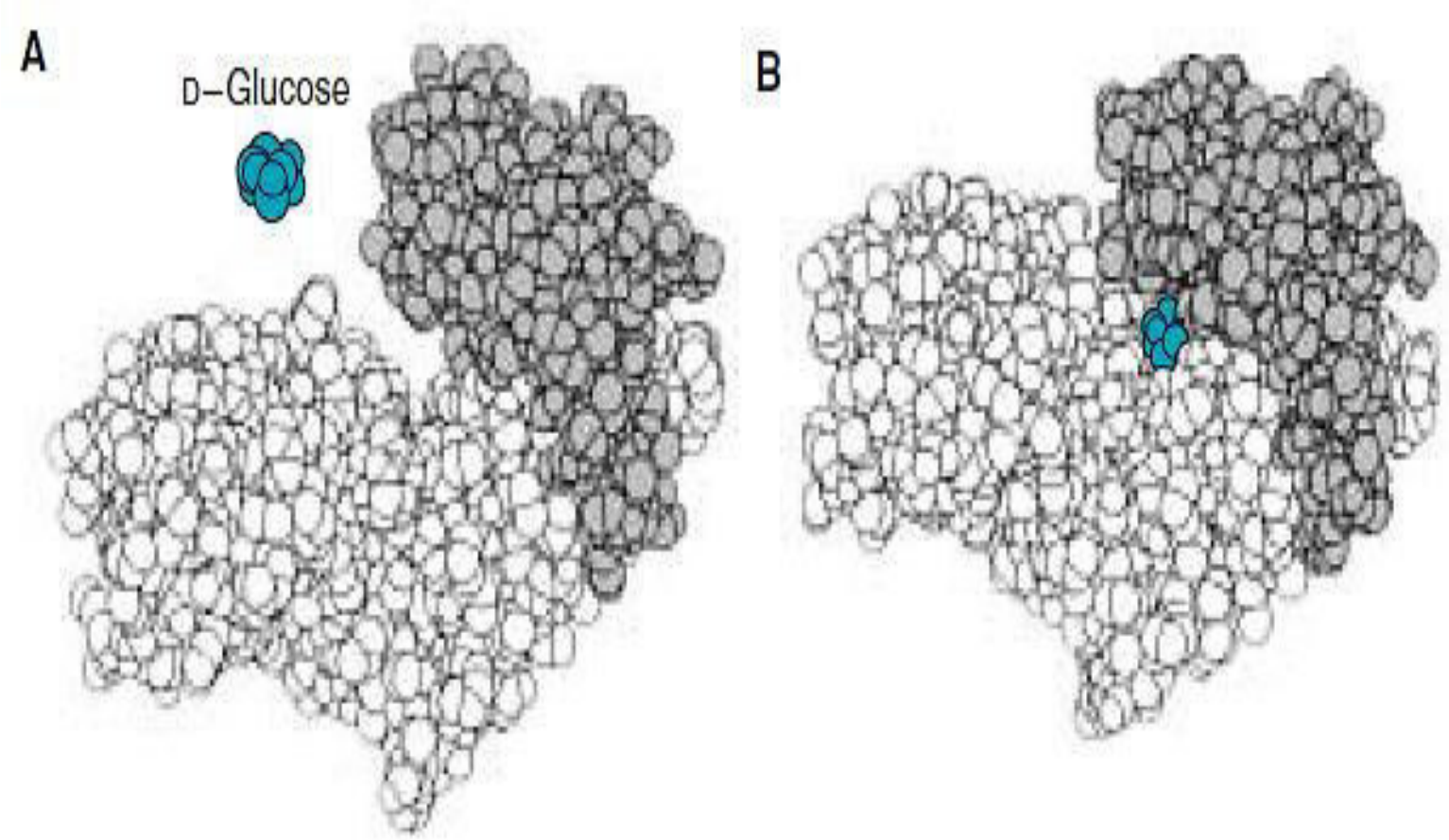

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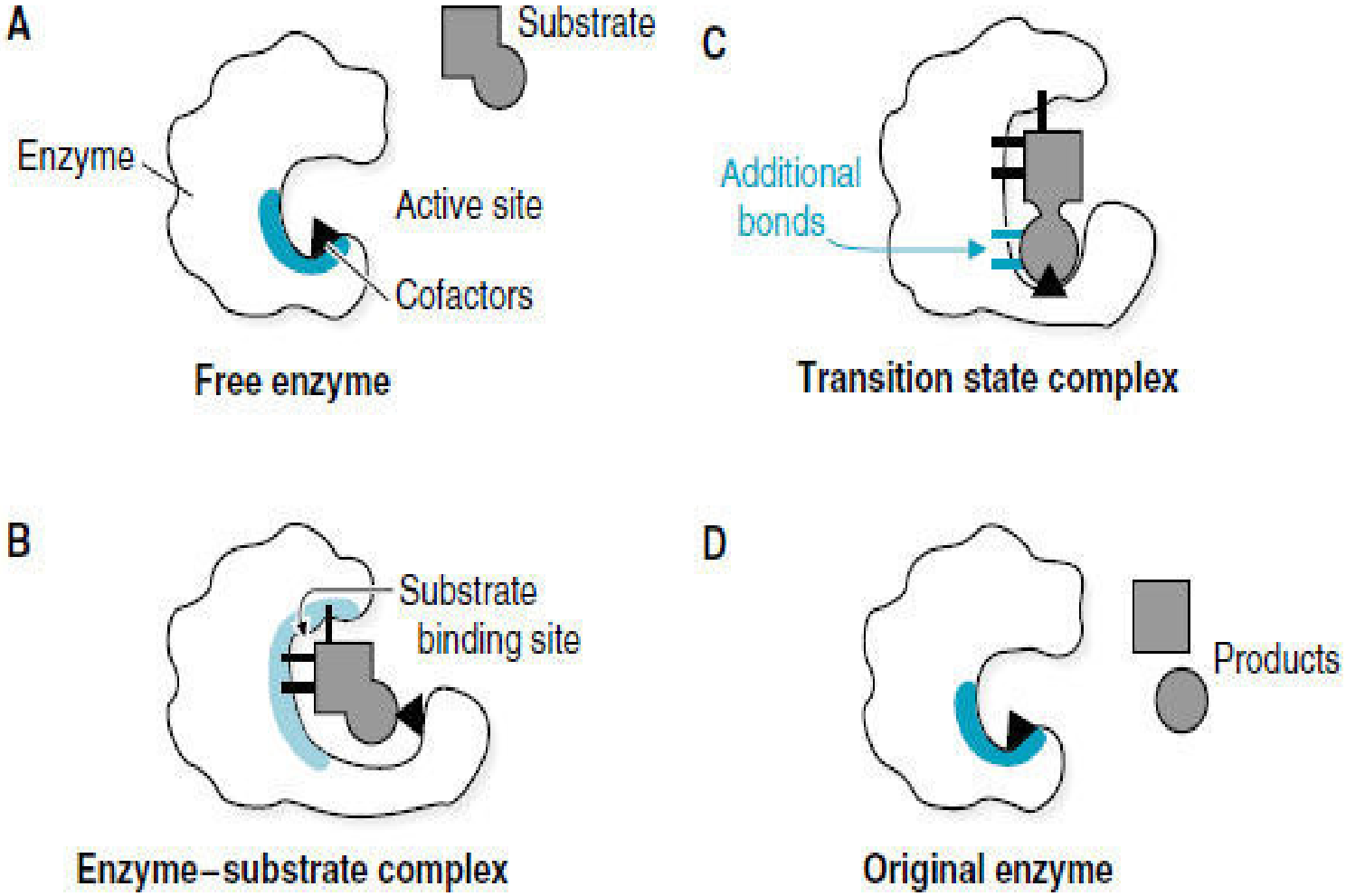
Enzymes


A
Substrate
Enzyme
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Substrate
binding site
Enzyme–substrate complex
D
Products
Original enzyme

Enzymes

- To understand catalysis, we must first appreciate the important distinction between
- reaction equilibria and
- reaction rates.

END

Enzymes

Mechanism of Enzyme Action (Contd.)

Reaction Equilibria

- The function of a catalyst is to increase the rate of a reaction.
- Catalysts *do not affect reaction equilibria*
- Any reaction, such as $S \leftrightarrow P$, can be described by a reaction coordinate diagram

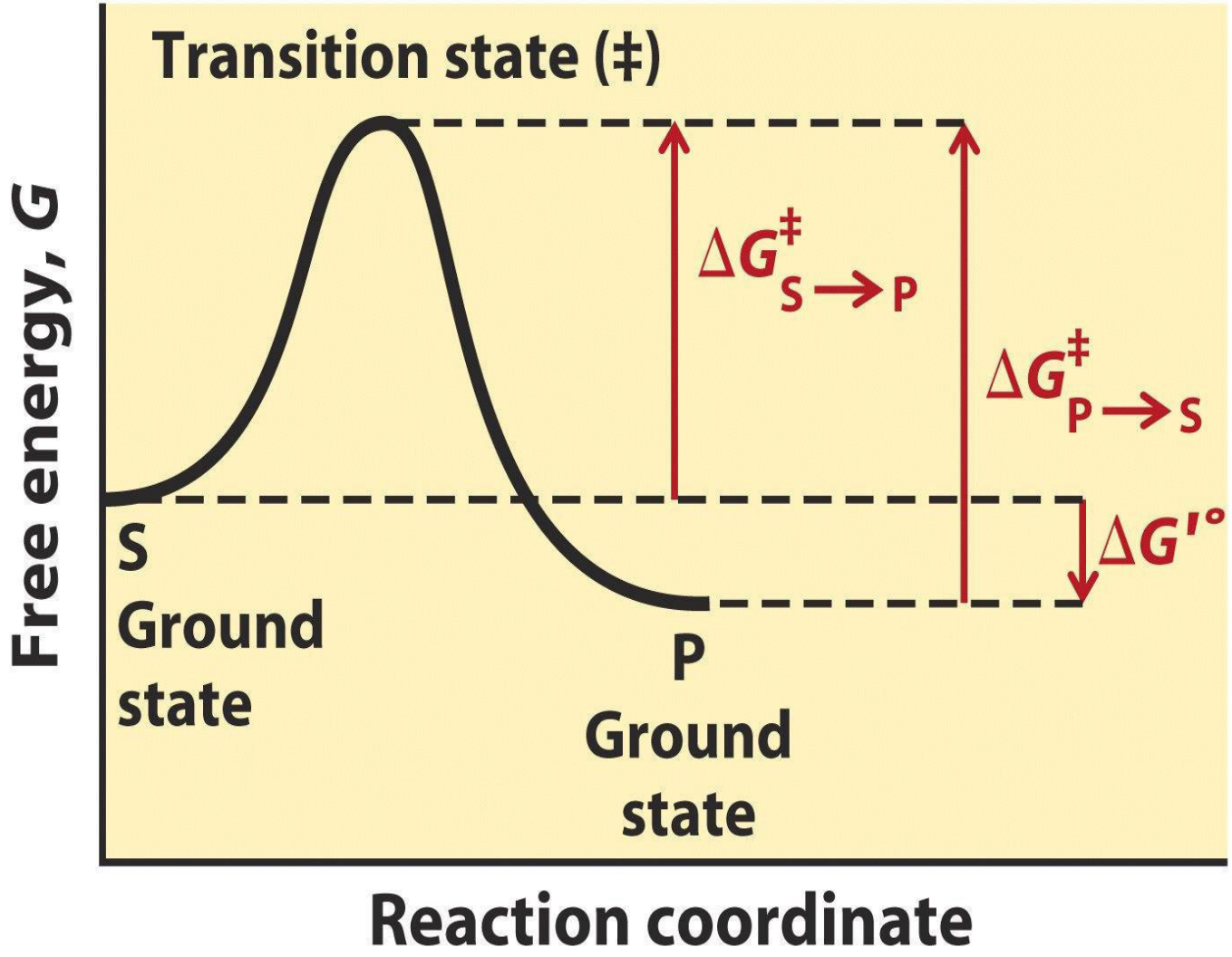
Enzymes

- The free energy of the system is plotted against the progress of the reaction **S** → **P**.
- A diagram of this kind is a description of the energy changes during the reaction.

Enzymes

- The horizontal axis (reaction coordinate) reflects the progressive chemical changes (e.g., bond breakage or formation) as S is converted to P.

Enzymes


Transition state (‡)
$\Delta G^{\ddagger}_{S \rightarrow P}$
$\Delta G^{\ddagger}_{P \rightarrow S}$
$\Delta G'^{\circ}$
S
Ground state
P
Ground state
Free energy, G
Reaction coordinate

Enzymes

Activation energies, *G‡,*

- The activation energies, *G‡, for* the
- **S → P and P → S** reactions are indicated.
- *G′º is the overall standard* free-energy change in the direction **S → P**

Enzymes

Free-energy change ΔG

- When a reacting system is not at equilibrium, the tendency to move toward equilibrium represents a driving force
- the magnitude of which can be expressed as the free-energy change for the reaction, ΔG.

Enzymes

Standard free-energy change, ΔG^{o}

- Under standard conditions (298 K = 25 C)
- when reactants and products are initially present at 1 M concentrations or
- for gases, at partial pressures of 101.3 (kPa), or 1 atm

Enzymes

- the force driving the system toward equilibrium is defined as the standard free-energy change, ΔG^{o}
- However, because the conditions in the body systems are different from standard conditions
- energy in biological systems is described in terms of free energy, G'^{o}.

END

Enzymes

Mechanism of Enzyme Action (Contd.)

Ground State

&

Transition State

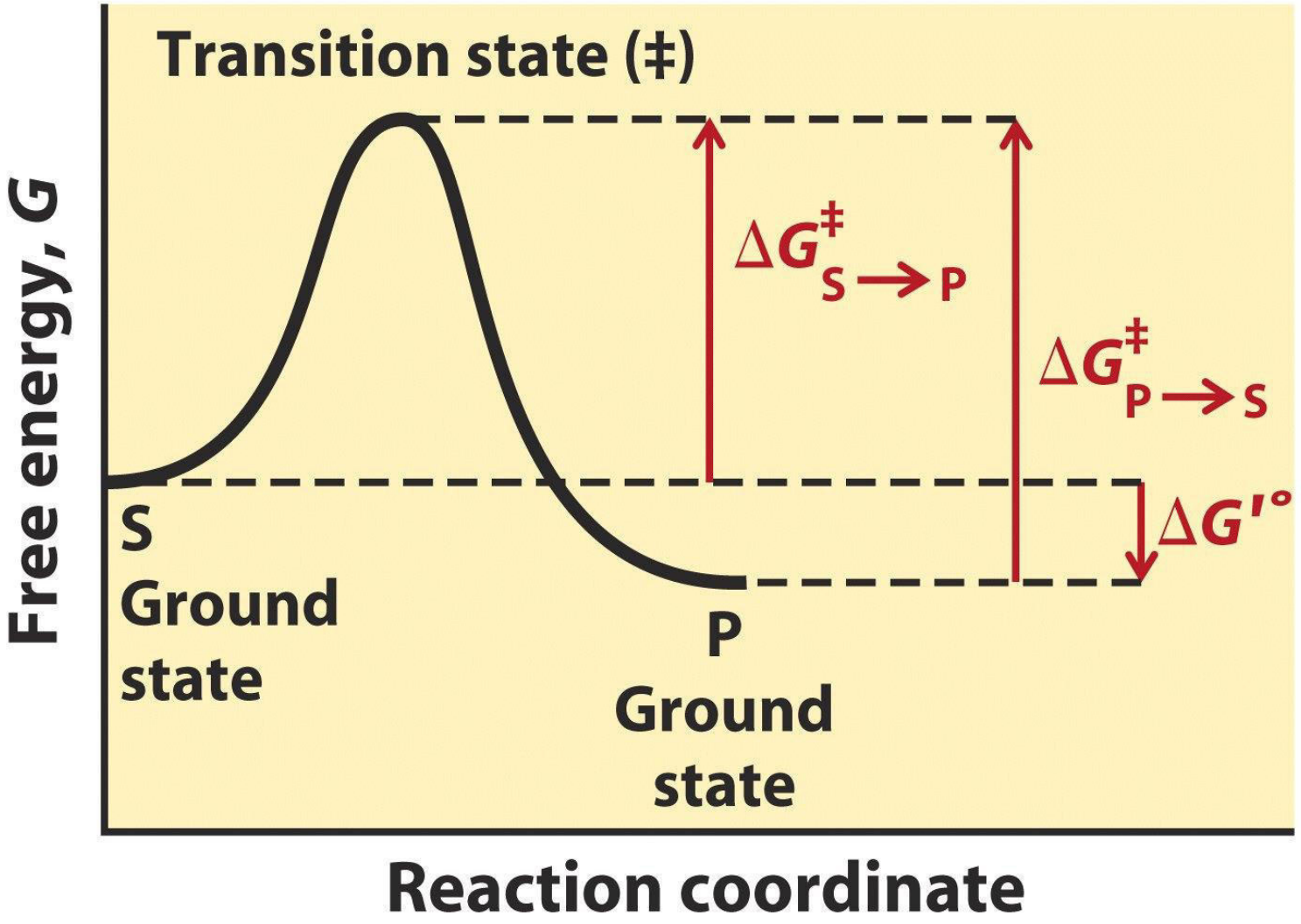
Enzymes

- The starting point for either the forward or the reverse reaction is called the **ground state**
- The equilibrium between S and P reflects the difference in the free energies of their ground states.

Enzymes

- The free energy of the ground state of P is **lower** than that of S
- So G'^{o} *for the* reaction is negative and the equilibrium favors P.
- The position and *direction* of equilibrium are *not affected by* any catalyst.

Enzymes


Transition state (‡)
$\Delta G^{\ddagger}_{S \rightarrow P}$
$\Delta G^{\ddagger}_{P \rightarrow S}$
$\Delta G'^{\circ}$
S
Ground state
P
Ground state
Free energy, G
Reaction coordinate

Enzymes

- But there is an **energy barrier** between S and P: The energy required for
- alignment of reacting groups
- formation of transient unstable charges
- bond rearrangements
- and other transformations

Enzymes

- This is illustrated by the energy “hill”
- To undergo reaction, the molecules must overcome this barrier and therefore must be raised to a higher energy level
- This is called the **transition state.**

Enzymes

- It is simply a fleeting molecular moment in which events such as
- bond breakage,
- bond formation, and
- charge development
- have proceeded to the precise point at which decay to either substrate or product is equally likely.

Enzymes

- The difference between the energy levels of the ground state and the transition state is the **activation energy, *G*‡.**

Enzymes

- At the top of the energy hill is a point at which decay to the S or P state is equally probable (it is downhill either way)

Enzymes

- A substance that modifies the transition state to lower the activation energy is termed a catalyst; a biological catalyst is termed an enzyme.

END

Enzymes

Mechanism of Enzyme Action (Contd.)

Activation energy $\Delta G^{\ddagger}$.

- The difference between the energy levels of the ground state and the transition state is the **activation energy $\Delta G^{\ddagger}$.**

Enzymes

- The rate of a reaction reflects this activation energy: a higher activation energy corresponds to a slower reaction.

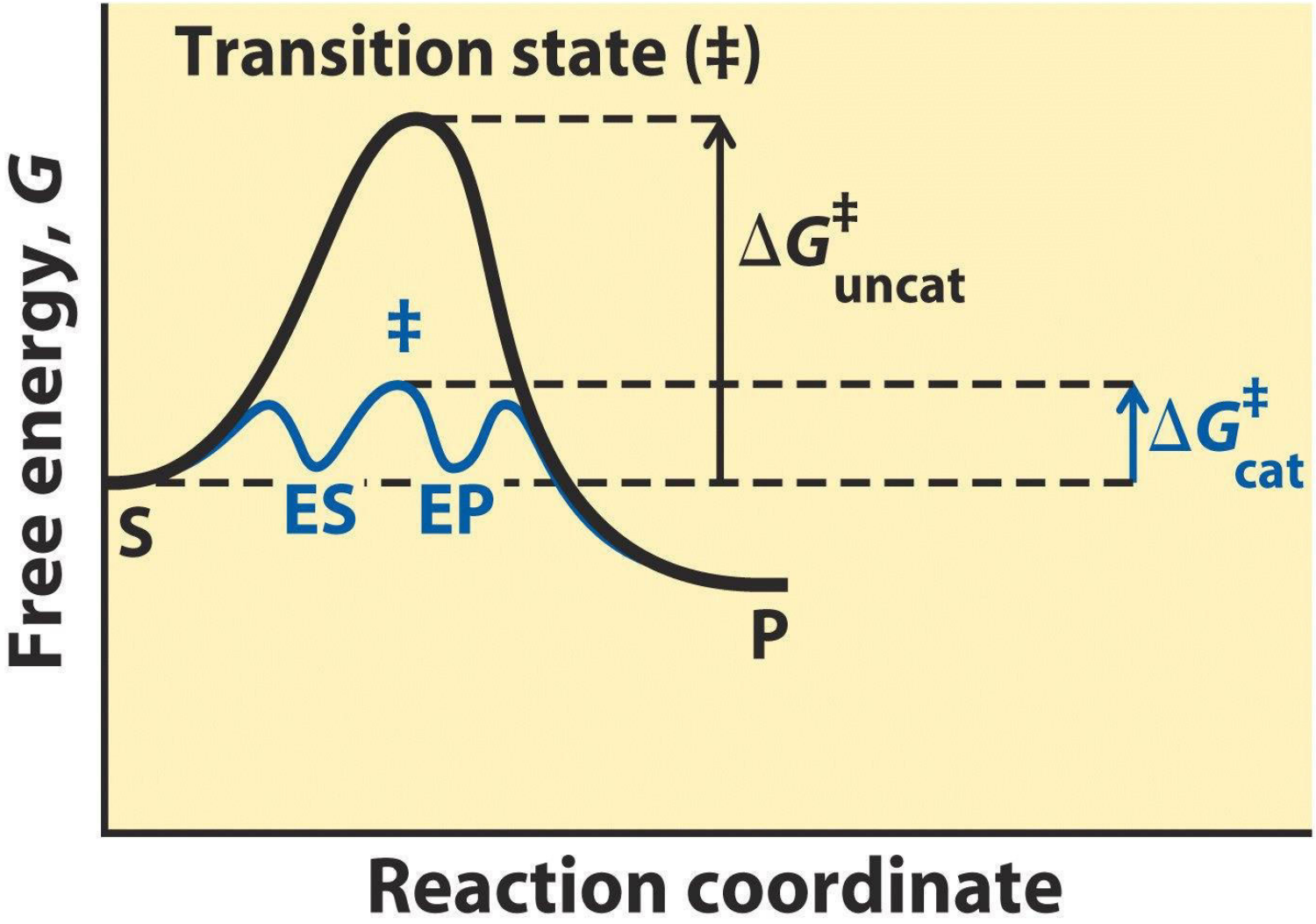
Enzymes

- Reaction rates can be increased by raising the temperature, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier.

Enzymes

- Alternatively, the activation energy can be lowered by adding a catalyst.
- Catalysts enhance reaction rates by lowering activation energies.

Enzymes


Transition state (‡)
$\Delta G^{‡}_{uncat}$
‡
$\Delta G^{‡}_{cat}$
S
ES
EP
P
Free energy, G
Reaction coordinate

Enzymes

- The role of enzymes is to accelerate the inter-conversion of S and P.
- i.e ***enzymes lower the energy of activation, $\Delta G^{\ddagger}$, of a reaction.***
- The enzyme is not used up in the process, and the equilibrium point is unaffected.

Enzymes

- However, the reaction reaches equilibrium much faster when the appropriate enzyme is present,
- because the rate of the reaction is increased.

END

Enzymes

The Induced Fit Hypothesis

- Some proteins can change their shape (conformation)

- When a substrate combines with an enzyme, it induces a change in the enzyme's conformation

Enzymes

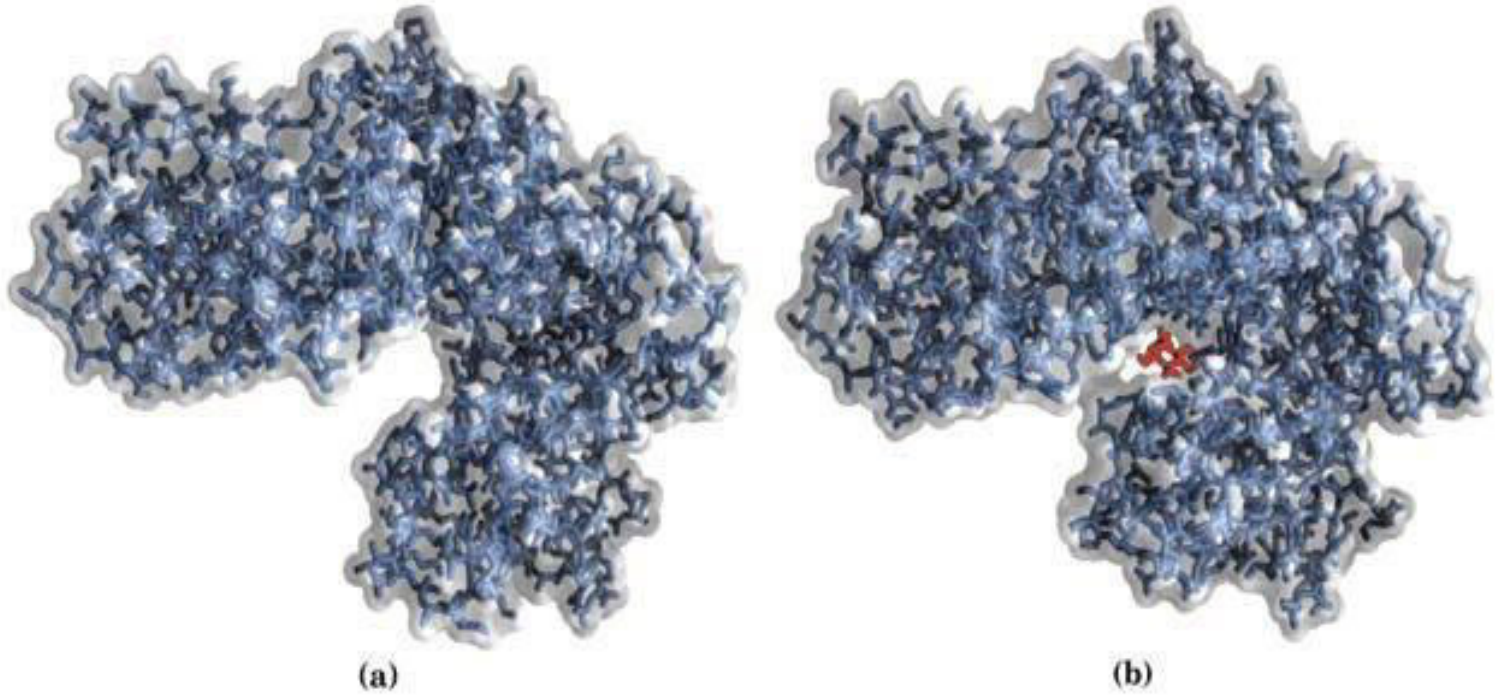
- This change in conformation when the substrate binds is induced by multiple weak interactions with the substrate.
- There may also be rearrangements of covalent bonds during an enzyme-catalyzed reaction.

Enzymes

- This conformational change is referred to as **induced fit.**

Enzymes

The Induced Fit Hypothesis



Hexokinase (a) without (b) with glucose substrate

Enzymes

- Chemical reactions of many types take place between substrates and enzyme's functional groups (specific amino acid side chains, metal ions, and coenzymes

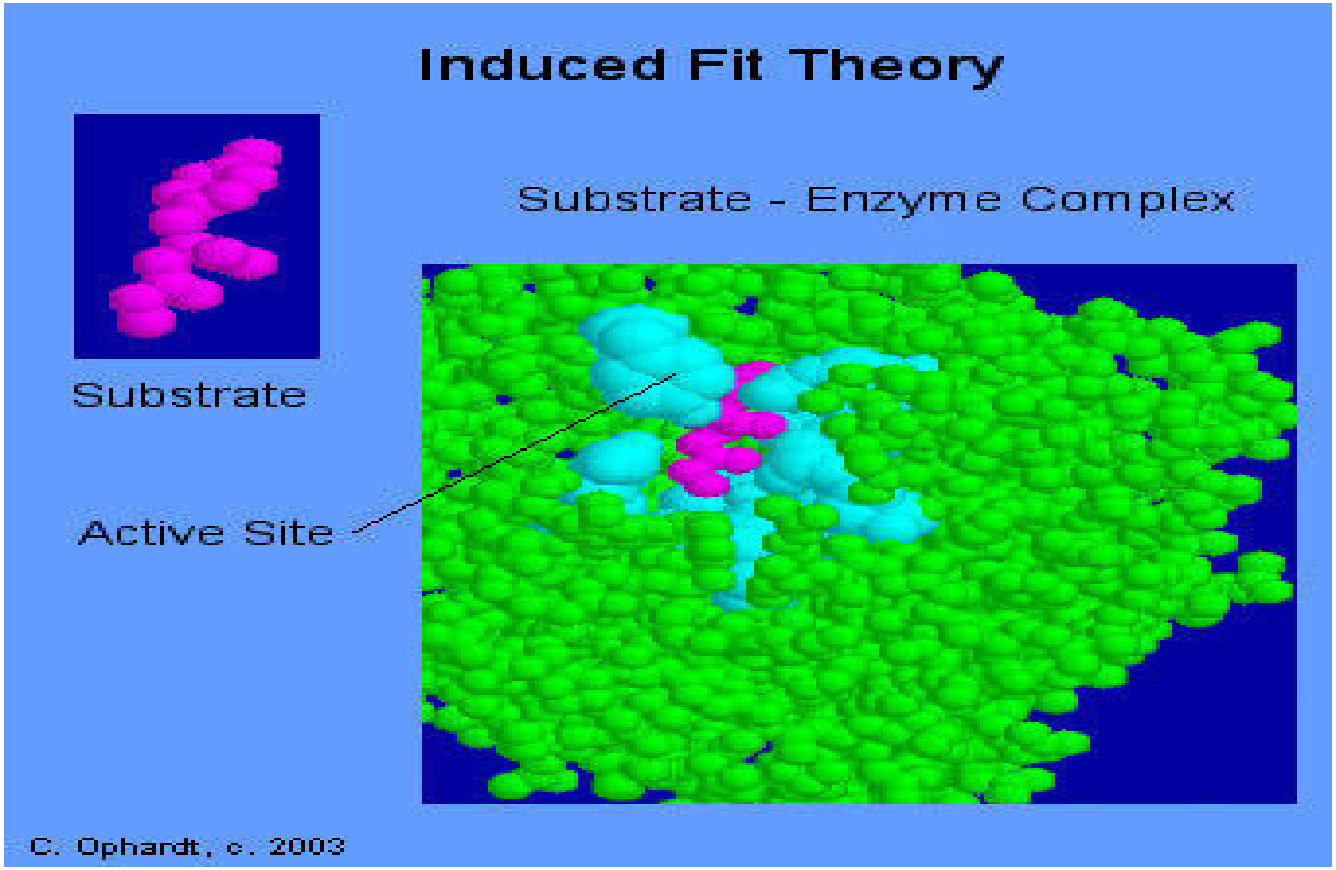
Enzymes

- Induced fit serves to bring specific functional groups on the enzyme into the proper position to catalyze the reaction.

Enzymes

Induced Fit Theory

- Enzyme is not rigid, changes shape with substrate.



Enzymes

- The active site is also moulded into a precise conformation
- Making the chemical environment suitable for the reaction

END

Enzymes

Cofactors, Coenzymes and Prosthetic groups

- Some enzymes require no chemical groups for activity other than their amino acid residues.
- Whereas some enzymes require molecules other than proteins for enzymic activity.

Enzymes

- If the non-protein moiety is a metal ion such as Zn^{2+} or Fe^{2+}, it is called a cofactor.

- If it is a complex organic molecule or metallo-organic compound it is termed a coenzyme.

Enzymes

Some Coenzymes That Serve as Transient Carriers

Coenzyme	*Examples of chemical groups transferred*
Biocytin	CO_2
Coenzyme A	Acyl groups
5′-Deoxyadenosylcobalamin (coenzyme B_{12})	H atoms and alkyl groups
Flavin adenine dinucleotide	Electrons
Lipoate	Electrons and acyl groups
Nicotinamide adenine dinucleotide	Hydride ion ($:H^-$)

Enzymes

Some Inorganic Elements That Serve as Cofactors for Enzymes

Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^{+}	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase

Enzymes

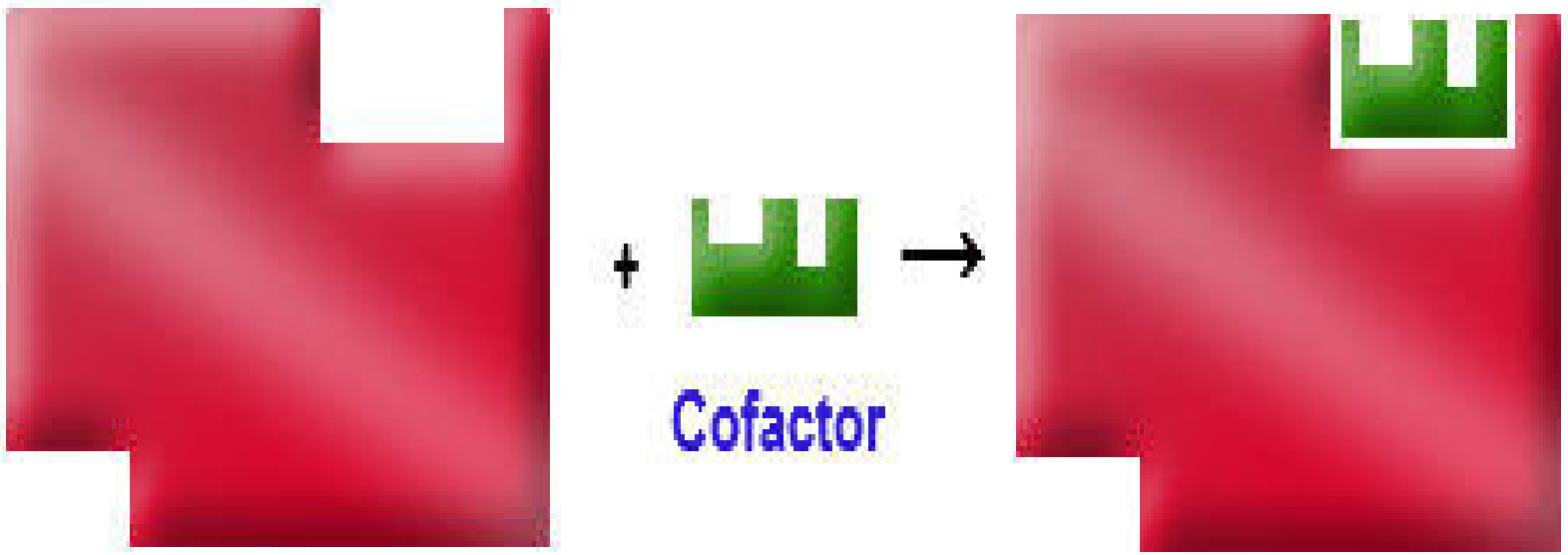
- A coenzyme or metal ion that is very tightly or even covalently bound to the enzyme protein is called a prosthetic group.

Enzymes

- The term holoenzyme refers to the active enzyme with its non-protein component,

- whereas the enzyme without its non-protein moiety is termed an apoenzyme (apoprotein) and is inactive.

Enzymes



Apoenzyme

Holoenzyme

Enzymes

- Coenzymes serve as recyclable shuttles that transport many substrates from one point within the cell to another.
- The function of these shuttles is twofold.

Enzymes

- First, they stabilize species
- such as hydrogen atoms (FADH) or hydride ions (NADH)
- that are too reactive to persist for any significant time in the presence of the water or organic molecules that permeate cells.

Enzymes

- Second, they serve as an adaptor or handle that facilitates the
- recognition and binding of small chemical groups, such as acetate (coenzyme A) or glucose (UDP), by their target enzymes

END

Enzymes

Reaction Rates and Order of Reactions

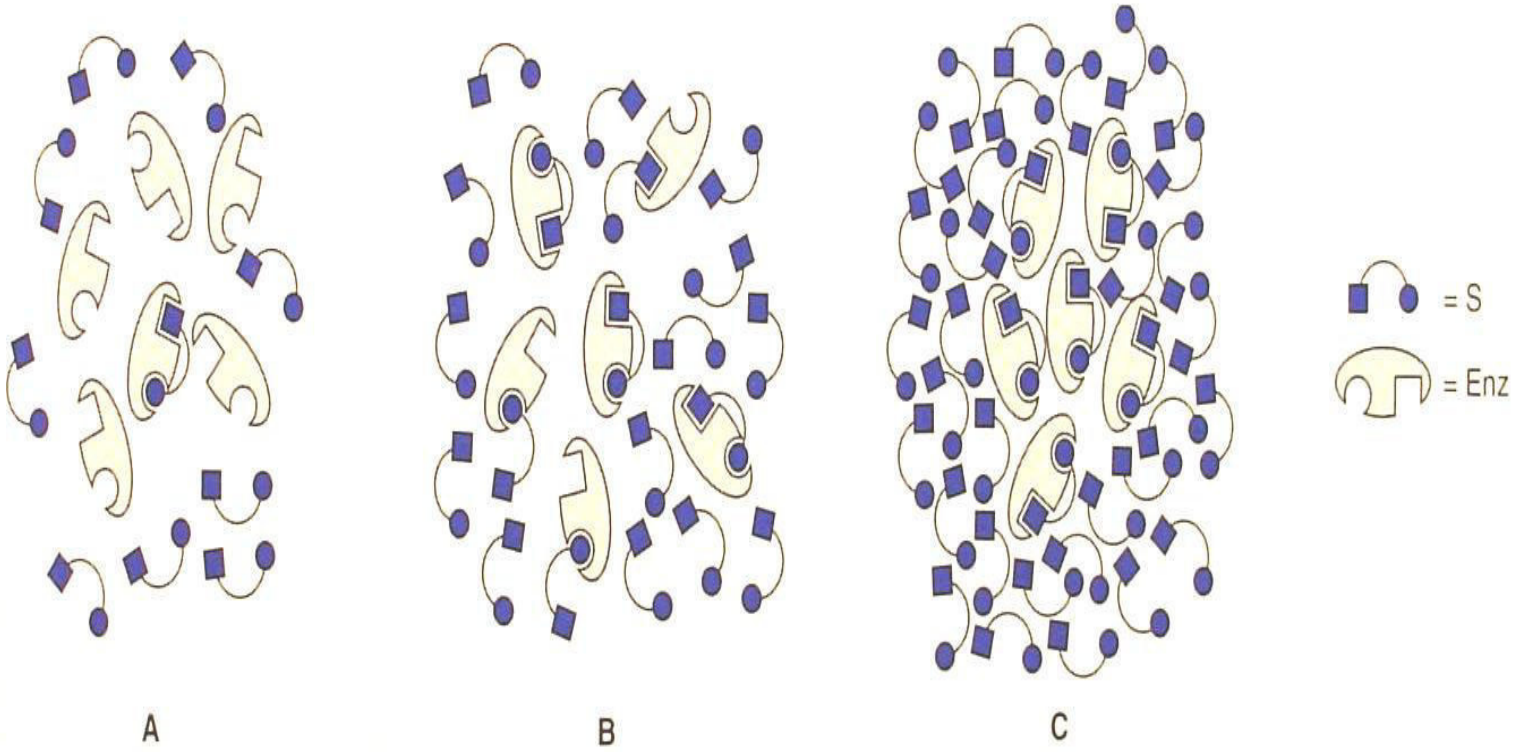
Reaction Velocity (v)

- The rate or velocity of a reaction (v) is the number of substrate molecules converted to product per unit time;
- Velocity is usually expressed as **µmol of product** formed **per minute**.

Enzymes

- The rate of an enzyme-catalyzed reaction increases with substrate concentration until a maximal velocity (Vmax) is reached- reflecting the **saturation** with substrate of **all available binding sites are occupied** on the enzyme molecules present.

Enzymes



Enzymes

- An equilibrium such as S $\leftrightarrow$ P is described by an **equilibrium constant, Keq**, or simply **K.**
- Under the standard conditions used to compare biochemical processes, an equilibrium constant is denoted ***K´eq*** (or ***K´*)*.***

Enzymes

- The rate of any reaction is determined by the concentration of the reactant (or reactants) and by a **rate constant,** usually denoted by **k.**

Enzymes

- For the uni molecular reaction S → P, the rate (or velocity) of the reaction, V—representing the amount of S that reacts per unit time—is expressed by a **rate equation:** V = k[S].

Enzymes

- In this reaction, the rate depends only on the concentration of S.
- This is called a first-order reaction.
- The factor k is a proportionality constant that reflects the probability of reaction under a given set of conditions (pH, temperature, and so forth).

Enzymes

- If a reaction rate depends on the concentration of two different compounds.
- or if the reaction is between two molecules of the same compound, the reaction is second order.

Enzymes

- The rate equation then becomes

- **$V = k[S_1][S_2]$**

END

Enzymes

Factors Affecting Enzymatic Activity

- Enzymes can be isolated from cells, and their properties studied in a test tube (that is, in vitro).

Enzymes

Factors Affecting Enzymatic Activity

- Different enzymes show different responses to changes in;
- substrate concentration
- temperature, and
- pH.

Enzymes

- Enzymic responses to these factors give us valuable clues as to how enzymes function in living cells (that is, in vivo)

Enzymes

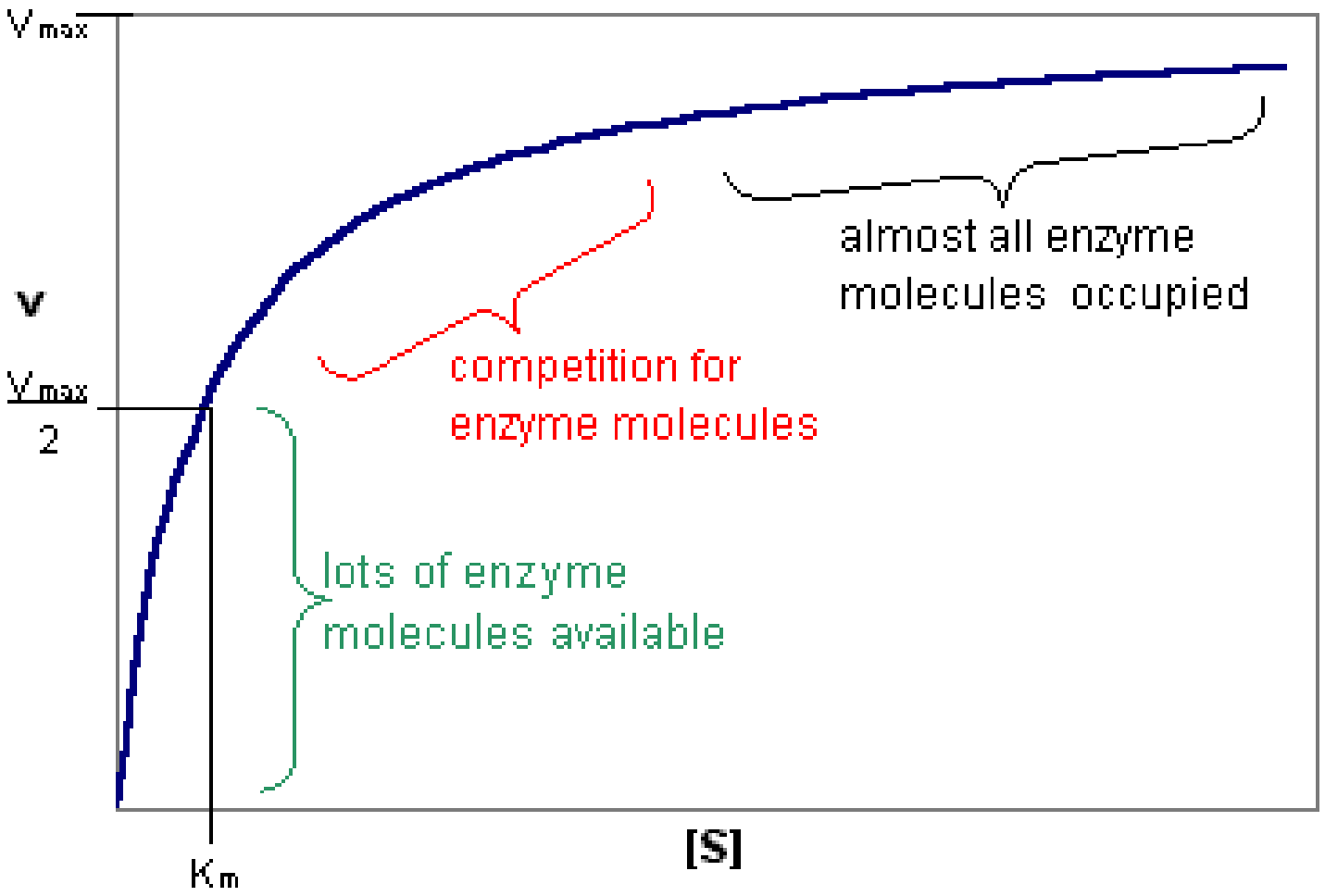
Substrate concentration

- The rate of an enzyme-catalyzed reaction increases with substrate concentration until maximal velocity (Vmax) is reached

Enzymes

- The leveling off of the reaction rate at high substrate concentrations
- reflects the saturation with substrate
- of all available binding sites on the enzyme molecules present.

Enzymes

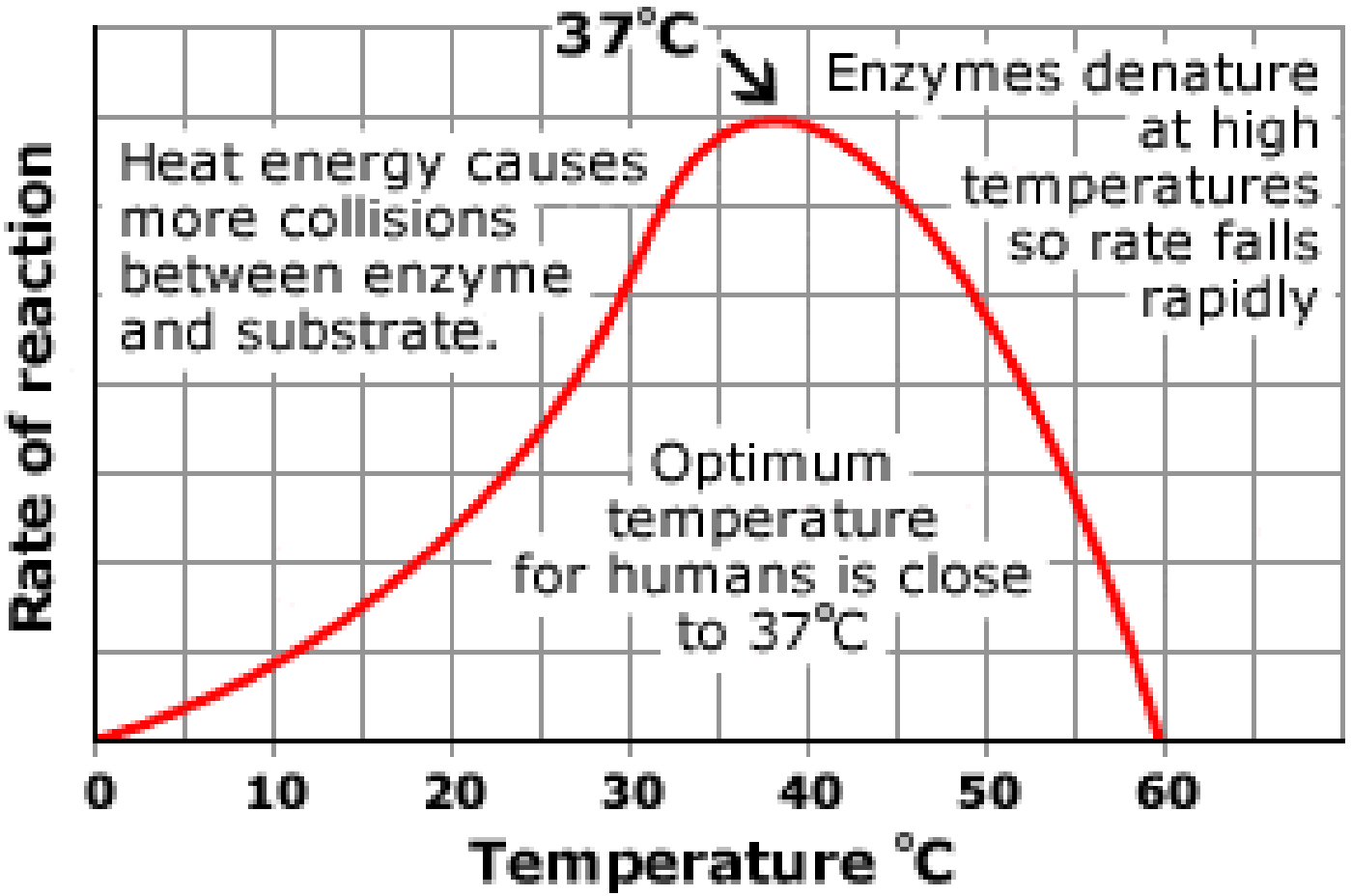

Vmax
V
Vmax
2
Km
[S]
almost all enzyme
molecules occupied
competition for
enzyme molecules
lots of enzyme
molecules available

Enzymes

Temperature

- The reaction velocity increases with temperature until a peak velocity is reached
- This increase is the result of the increased number of molecules having sufficient energy to pass over the energy barrier and form the products.

Enzymes


37°C
Enzymes denature at high temperatures so rate falls rapidly
Heat energy causes more collisions between enzyme and substrate.
Optimum temperature for humans is close to 37°C
Rate of reaction
0
10
20
30
40
50
60
Temperature °C

Enzymes

- Further elevation
- of the temperature results in a decrease in reaction velocity as a
- result of temperature-induced denaturation of the enzyme

END

Enzymes

Factors Affecting Enzymatic Activity (Contd.)

Effect of pH

Enzymes

- **The pH optimum varies for different enzymes:**
- The pH at which maximal enzyme activity is achieved is different for different enzymes, and often reflects the [H+] at which the enzyme functions in the body.

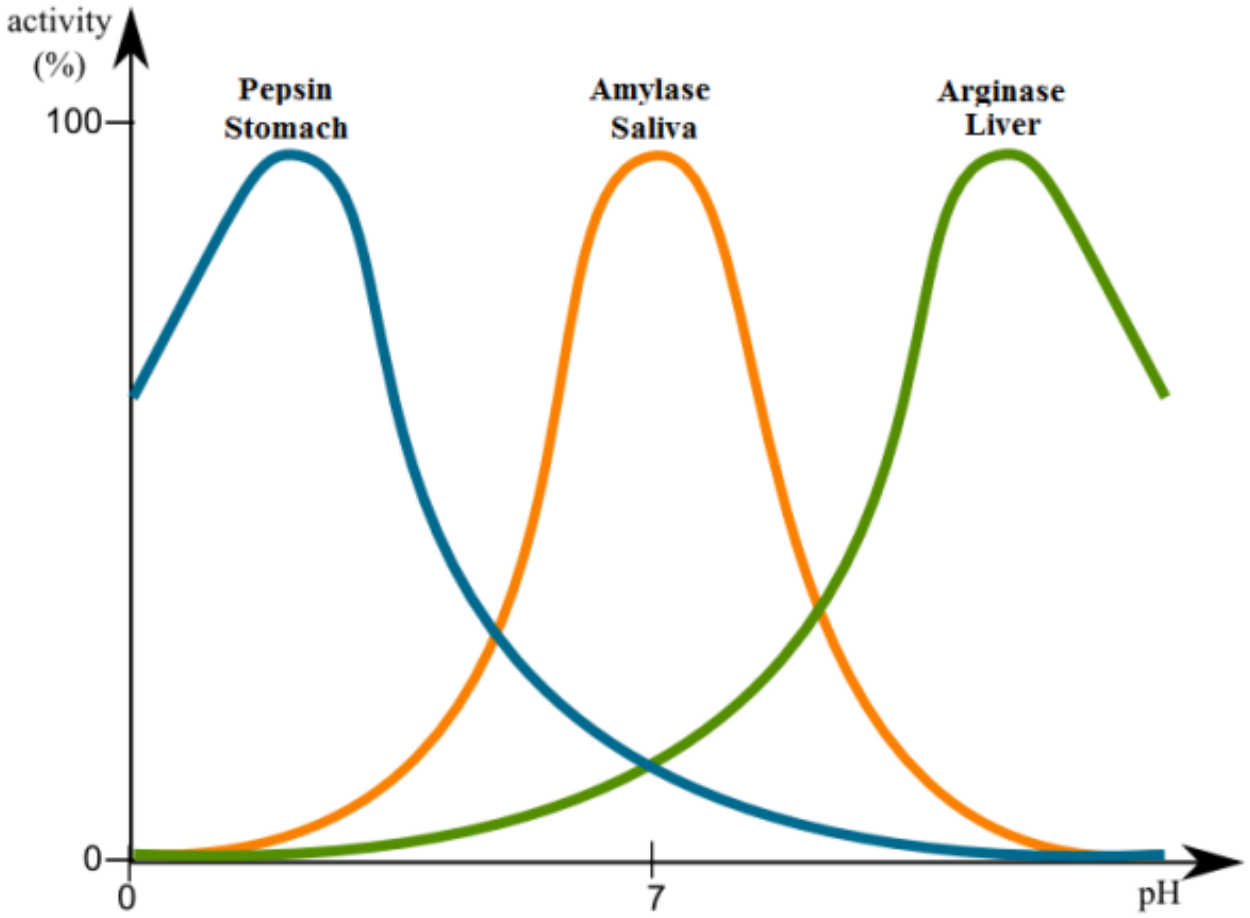
Enzymes

- For example, pepsin, a digestive enzyme in the stomach, is maximally active at pH 2.
- Whereas other enzymes, designed to work at neutral pH, are denatured by such an acidic environment

Enzymes

- Another examples is that there are two types of phosphatases in the body.
- The one that acts in the alkaline pH is called alkaline phosphatase and
- the other which acts at acidic pH is known as acid phosphatase.

Enzymes


activity
(%)
100
0
0
7
pH
Pepsin
Stomach
Amylase
Saliva
Arginase
Liver

Enzymes

- Effect of pH on the ionization of the active site:
- The concentration of H+ affects reaction velocity in several ways.

Enzymes

- First, the catalytic process usually requires that the enzyme and substrate have specific chemical groups in either an ionized or un-ionized state in order to interact.

Enzymes

- For example, catalytic activity may require that an amino group of the enzyme be in the protonated form (–NH3 +).
- At alkaline pH, this group is deprotonated, and the rate of the reaction, therefore, declines.

Enzymes

- Extremes of pH can also lead to denaturation of the enzyme.

END

Enzymes

Enzyme Kinetics

Substrate Concentration Affects the Rate of Enzyme-Catalyzed Reactions

$$E + S \leftrightarrow ES \leftrightarrow EP \leftrightarrow E + P$$

Enzymes

- A key factor affecting the rate of a reaction catalyzed by an enzyme is the concentration of substrate, [S].

Enzymes

- However, studying the effects of substrate concentration is complicated because substrate is converted to product
- and because of reversibility of reactions, e.g. conversion of product back to substrate.

Enzymes

- One simplifying approach in kinetics experiments is to measure the **initial rate (or initial velocity), designated V_0,** when [S] is much greater than the concentration of enzyme, [E].

Enzymes

- The velocity (v) of a reaction is the rate of product formation
- Whereas V_0 is the initial velocity and is measured as soon as the reactants and enzymes are mixed.

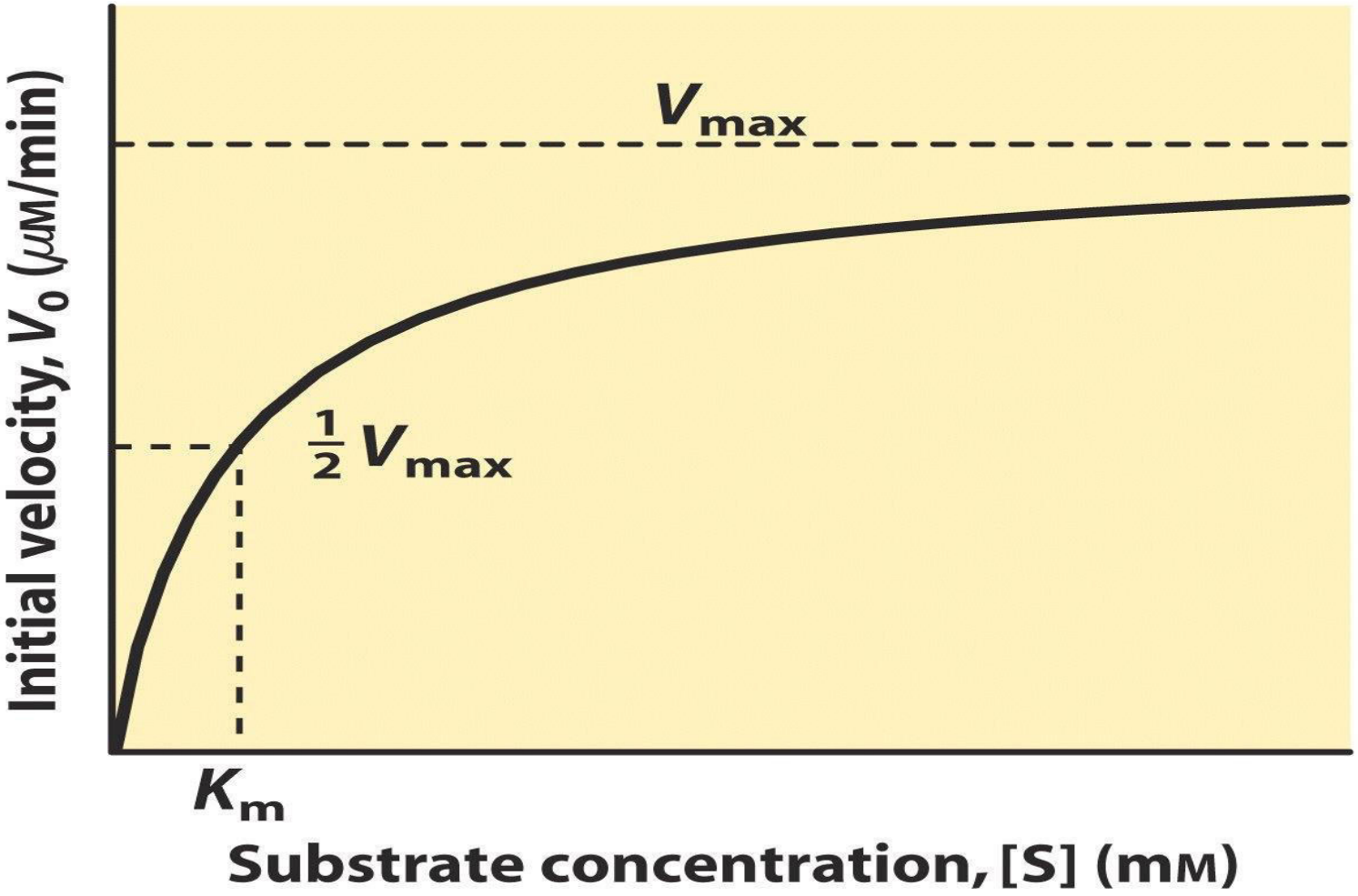
Enzymes

- At the start of a reaction, [S] is in large excess of [P].
- Thus the initial velocity of the reaction will be dependent on substrate concentration

Enzymes

- At that time, the concentration of product is very small and, therefore, the rate of the back reaction from P to S can be ignored.

Enzymes


V_{max}
$\frac{1}{2} V_{max}$
K_m
Initial velocity, V_0 (μM/min)
Substrate concentration, [S] (mM)

Enzymes

- Increase in substrate concentration increases V_0 i.e.
- initial velocity is increased whenever a fixed concentration of enzyme is mixed with an increased concentration of substrate.

END

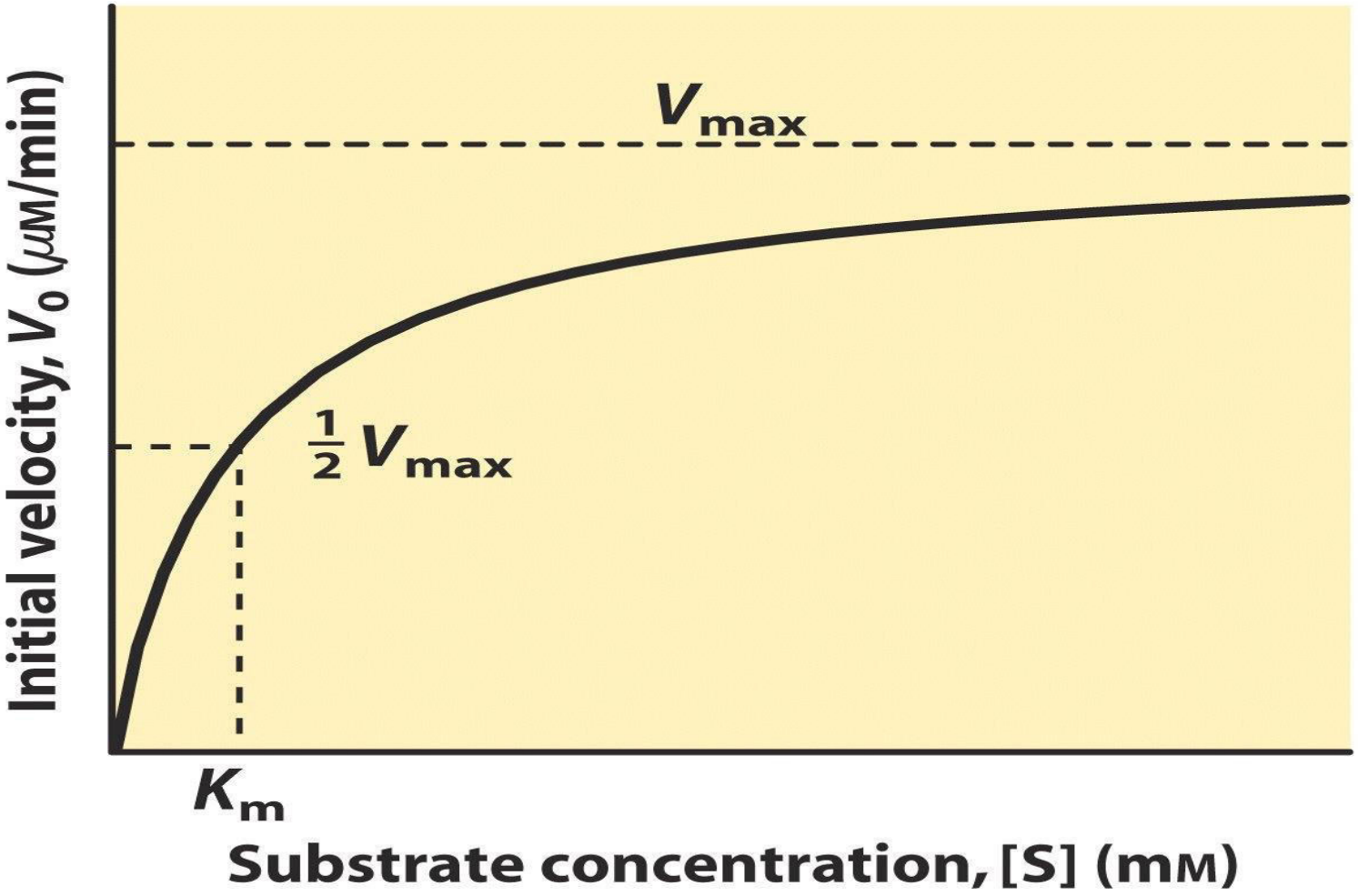
Enzymes

Enzyme Kinetics (Contd.)

Maximal Velocity V_{max}

- When initial velocity **(V_o)** is plotted against [S],
- a **hyperbolic curve** results,
- where V_{max} represents the maximum reaction velocity.

Enzymes


V_{max}
$\frac{1}{2} V_{max}$
K_m
Initial velocity, V_0 (μM/min)
Substrate concentration, [S] (mM)

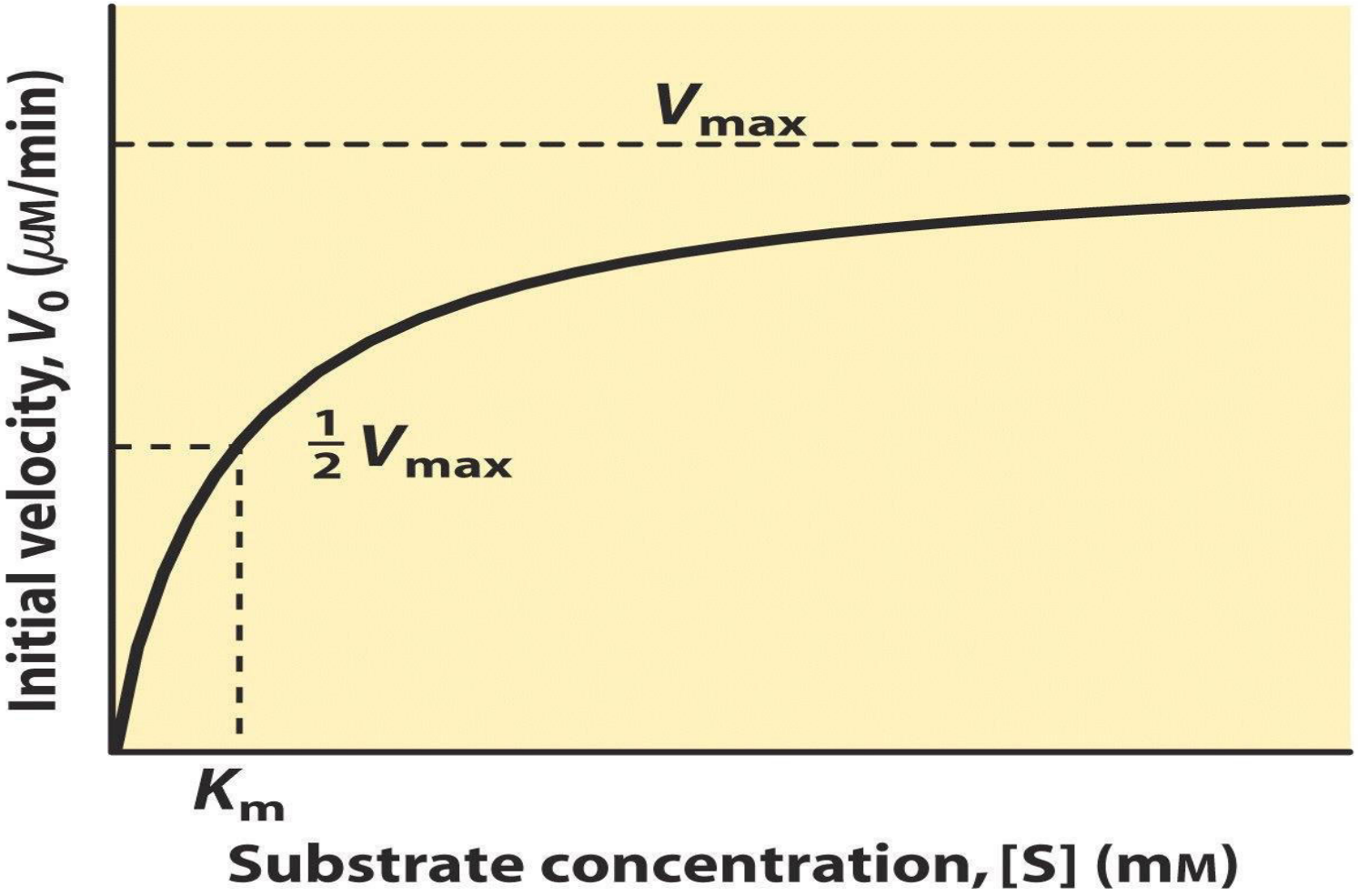
Enzymes

- At relatively low concentrations of substrate, V_o increases almost linearly.
- At higher substrate concentrations, V_o *increases by smaller and smaller amounts in* response to increases in [S].

Enzymes

- This plateau-like V_o *region is close to the* **maximal velocity, V_{max}.**
- V_{max} *is extrapolated from the* plot, because V_o *approaches but never quite reaches* V_{max}.

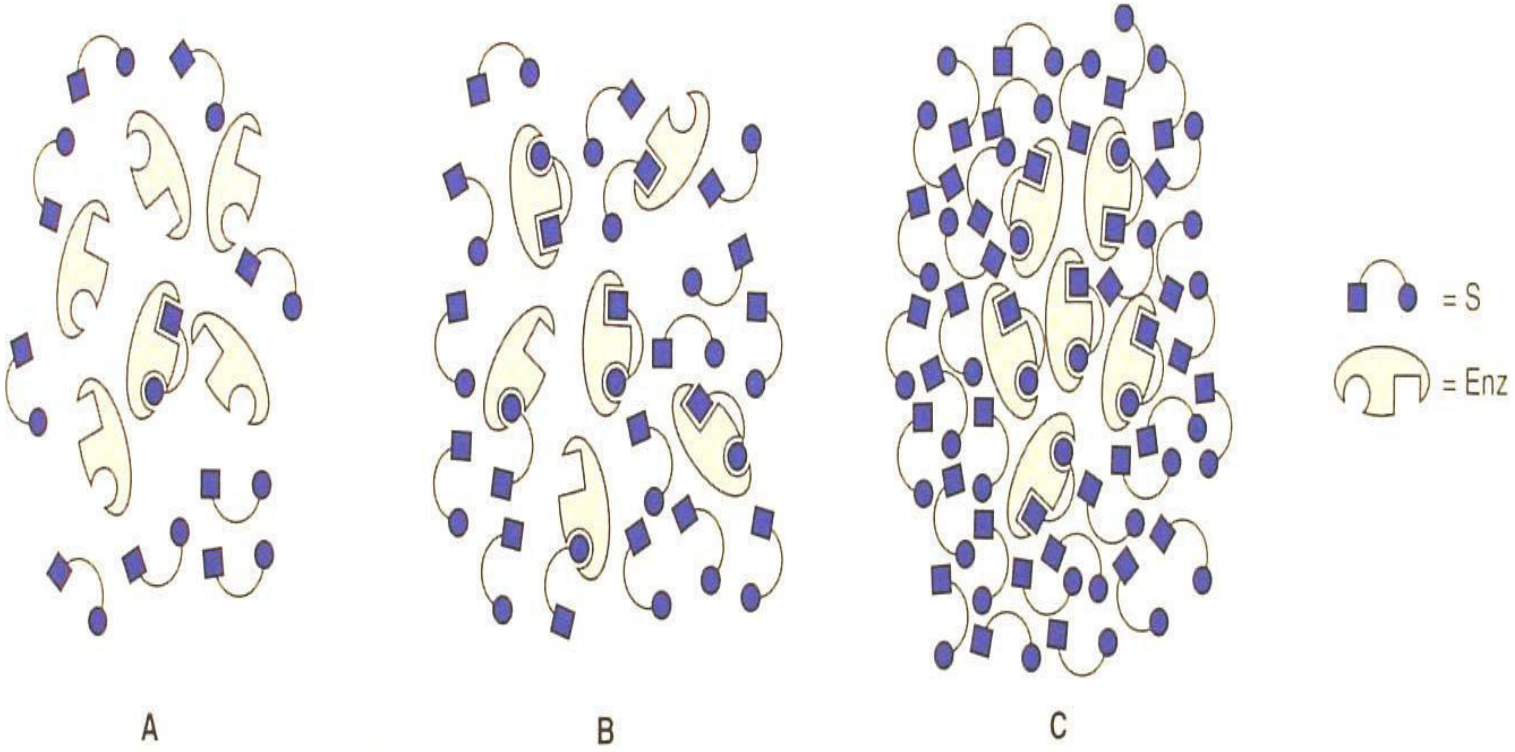
Enzymes


V_{max}
$\frac{1}{2} V_{max}$
K_m
Initial velocity, V_0 (μM/min)
Substrate concentration, [S] (mM)

Enzymes

- At this point in the reaction, if **[S] >> E**, all available enzyme is "saturated" with bound substrate, meaning only the **ES complex** is present.
- This dictates that further increasing the substrate concentration will not result in increased V_0

Enzymes



A. Low [S] B. 50% [S] or K_m C. High, saturating [S]

Enzymes

- The leveling off of the reaction rate at high substrate concentrations
- reflects the saturation with substrate
- of all available binding sites on the enzyme molecules present.

Enzymes

- This finite limit of V_{max} is called saturation kinetics.
- Saturation kinetics is a characteristic property of all rate processes dependent on the binding of a ligand to a protein e.g. membrane transporter proteins.

END

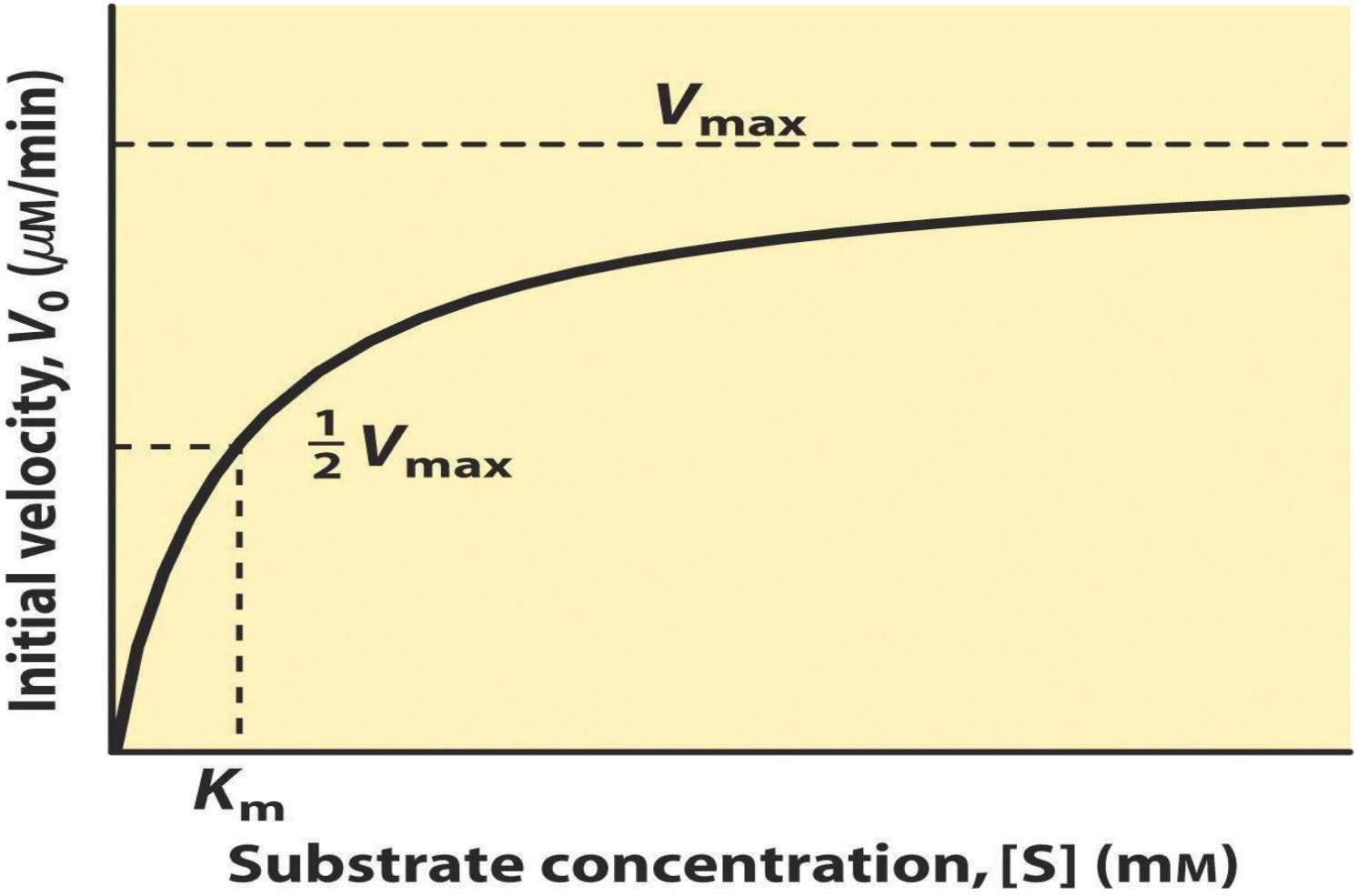
Enzymes

Enzyme Kinetics (Contd.)

Michaelis constant K_m

- The substrate concentration at which V_o is half maximal is K_m, the Michaelis constant.

Enzymes


V_{max}
$\frac{1}{2} V_{max}$
K_m
Initial velocity, V_0 (μM/min)
Substrate concentration, [S] (mM)

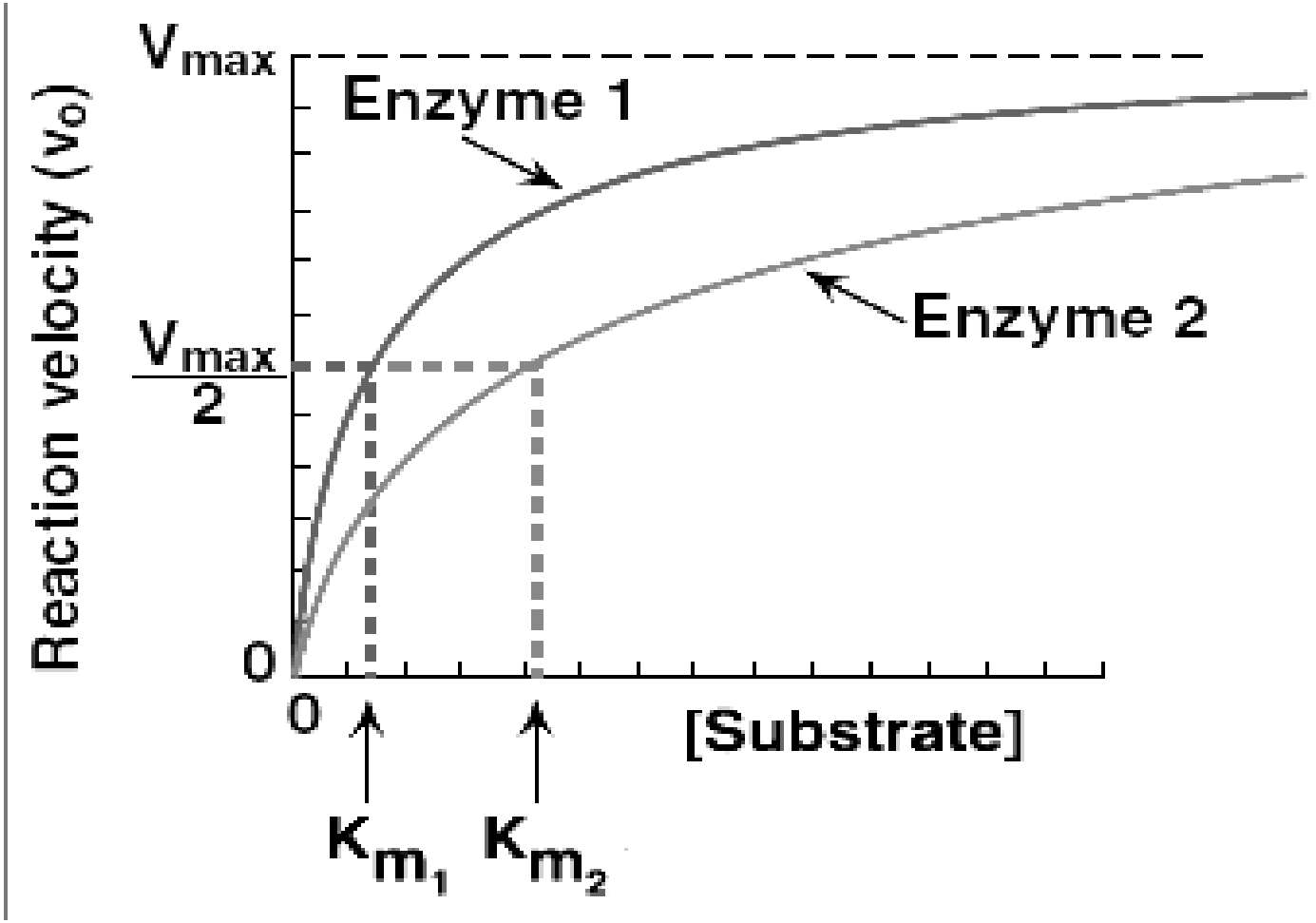
Enzymes

- K_m reflects the affinity of the enzyme for the substrate.
- Small K_m reflects a high affinity of the enzyme for substrate, because a low concentration of substrate is needed to half-saturate the enzyme i.e. to reach a velocity that is $1/2 V_{max}$.

Enzymes

- A numerically large (high) Km reflects a low affinity of enzyme for substrate because
- a high concentration of substrate is needed to half-saturate the enzyme.

Enzymes


Reaction velocity (v_o)
V_{max}
$\frac{V_{max}}{2}$
Enzyme 1
Enzyme 2
0
0
[Substrate]
K_{m_1}
K_{m_2}

Enzymes

a. **Small K_m:** A numerically small (low) K_m reflects a high affinity of the enzyme for substrate, because a low concentration of substrate is needed to half-saturate the enzyme—that is, to reach a velocity that is $1/2\ V_{max}$

Enzymes

b. Large Km: A numerically large (high) Km reflects a low affinity of enzyme for substrate because a high concentration of substrate is needed to half-saturate the enzyme

Enzymes

- The velocity of an enzyme is most sensitive to changes in substrate concentration over a concentration range below its Km.
- At substrate concentrations less than 1⁄10th of the Km, a doubling of substrate concentration nearly doubles the velocity of the reaction

Enzymes

END

- At substrate concentrations 10 times the Km, doubling the substrate concentration has little effect on the velocity.

Enzymes

- A comparison between the isozymes of hexokinase illustrates the significance of the K_m.
- Hexokinase catalyses the first step in glucose metabolism in most cells, the transfer of a phosphate from ATP to glucose to form glucose 6-phosphate.

Enzymes

- Hexokinase I, the isozyme in red blood cells has a low K_m for glucose of approximately 0.05 mM- helpful in utilizing blood glucose even when the blood glucose concentration is very low.

Enzymes

- The isozyme of hexokinase, called glucokinase, which is found in the liver has a much higher K_m of approximately 5 to 6 mM- helpful in storing large amounts of "excess" glucose as glycogen or converting it to fat after a carbohydrate meal.

Enzymes

Michaelis-Menten equation, the rate equation

- Leonor Michaelis and Maud Menten in 1913, proposed a simple model that accounts for most of the features of enzyme-catalyzed reactions.

Enzymes

- They postulated that the enzyme first combines reversibly with its substrate to form an enzyme-substrate complex in a relatively fast reversible step:

$$E + S \underset{K-1}{\overset{k1}{\leftrightarrow}} ES$$

Enzymes

- The ES complex then breaks down in a slower second step to yield the free enzyme (E) and the reaction product (P):

$$\mathbf{ES} \underset{K-2}{\overset{K2}{\leftrightarrow}} \mathbf{E + P}$$

Enzymes

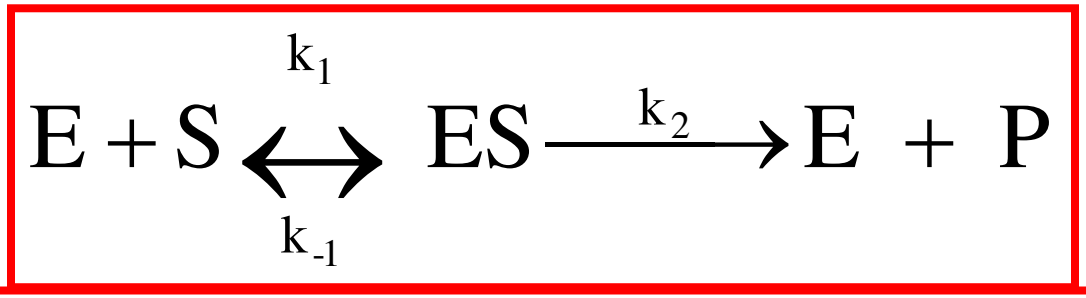
- Early in the reaction, the concentration of the product, **[P], is negligible**, and we make the simplifying assumption that the reverse reaction, P → S (described by *k-2*), *can be ignored*

Enzymes

- This assumption is not critical but it simplifies our task.
- The overall reaction then reduces to

$$E + S \underset{K\text{-}I}{\overset{k1}{\leftrightarrow}} ES \xrightarrow{K2} E + P$$

Enzymes



E = Enzyme S = Substrate P = Product

ES = Enzyme-Substrate complex

k_1 rate constant for the forward reaction

k_{-1} = rate constant for the breakdown of the ES to substrate

k_2 = rate constant for the formation of the products

Enzymes

- Because the slower second reaction must limit the rate of the overall reaction.
- The overall rate must be proportional to the concentration of the species that react in the second step, that is, ES.

Enzymes

- The maximum initial rate of the catalyzed reaction (*Vmax) is observed when virtually all the* enzyme is present as the ES complex and [E] is vanishingly small.

Enzymes

- Under these conditions, the enzyme is “saturated” with its substrate, so that further increases in [S] have no effect on rate.
- This condition exists when [S] is sufficiently high that essentially all the free enzyme has been converted to the ES form.

Enzymes

- After the ES complex breaks down to yield the product P,
- the enzyme is free to catalyze reaction of another molecule of substrate.

END

Enzymes

Michaelis-Menten equation, the rate equation (Contd.)

$$V_o = \frac{Vmax\,[S]}{Km + [S]}$$

- Where;
- V_o = initial reaction velocity

Enzymes

- Vmax = maximal velocity
- Km = Michaelis constant
- = K1+K2 / K-1
- [S] = substrate concentration

Enzymes

- This is the **Michaelis-Menten equation**- statement of the quantitative relationship between the;
- initial velocity V_o,
- *the maximum velocity Vmax, and*
- *the initial substrate concentration [S], all related* through the Michaelis constant *Km*

Enzymes

- An important numerical relationship emerges from the Michaelis-Menten equation in the special case;
- when V_o *is exactly one-half Vmax*

Enzymes

$$V_o = \frac{Vmax\,[S]}{Km + [S]}$$

$$\frac{Vmax}{2} = \frac{Vmax\,[S]}{Km + [S]}$$

- On dividing by *Vmax, we obtain*
- 1/2 = [S] / Km + [S]
- Solving for *Km, we get*
- *Km + [S] = 2[S], or*
- *Km = [S], when* V_o = 1/2 Vmax

Enzymes

- This is a very useful, practical definition of *Km: Km is* equivalent to the substrate concentration at which V_o *is* one-half *Vmax*.

Enzymes

- The equation describes the kinetic behavior of a great many enzymes, and all enzymes that exhibit a hyperbolic dependence of *V_0 on [S] are said to follow* ***Michaelis-Menten kinetics.***

Enzymes

- The practical rule that *Km= [S] when V0 =1⁄2Vmax holds for all enzymes* that follow Michaelis-Menten kinetics.
- The most important exceptions to Michaelis-Menten kinetics are the regulatory enzymes.

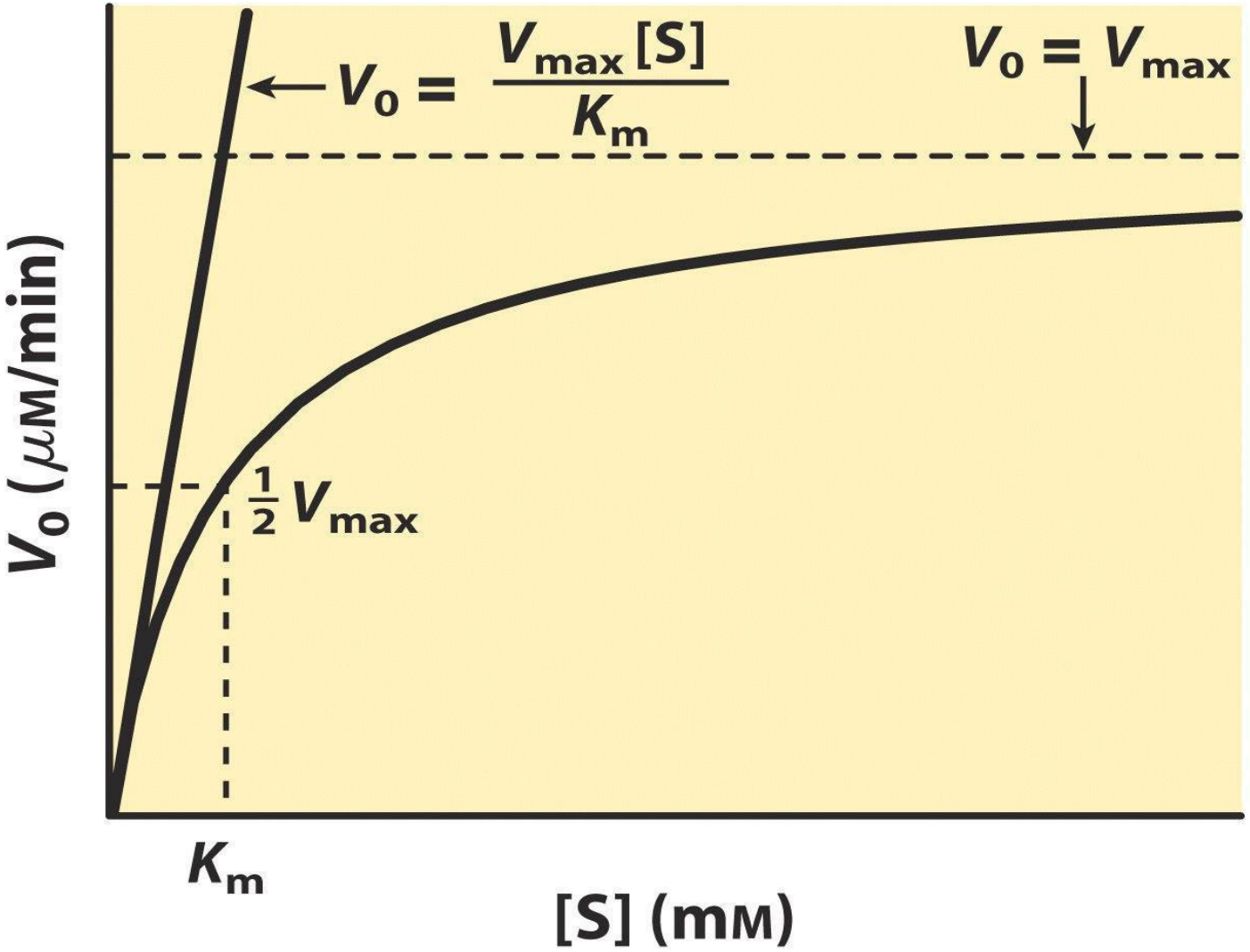
END

Enzymes

The Michaelis-Menton Equation at low [S]

- Interpreting Vmax and Km shows a simple graphical method for obtaining an approximate value for *Km.*
- This graph shows the kinetic parameters that define the limits of the curve at high and low [S].

Enzymes



Enzymes

- At low [S], *Km>>[S]* and the *[S]* term in the denominator of the Michaelis-Menten equation becomes insignificant.

Enzymes

- low [S]; $K_m >> [S]$
- $V_0 = \frac{Vmax\ [S]}{K^m + [S]}$
- $V_0 = \frac{Vmax\ [S]}{K^m}$

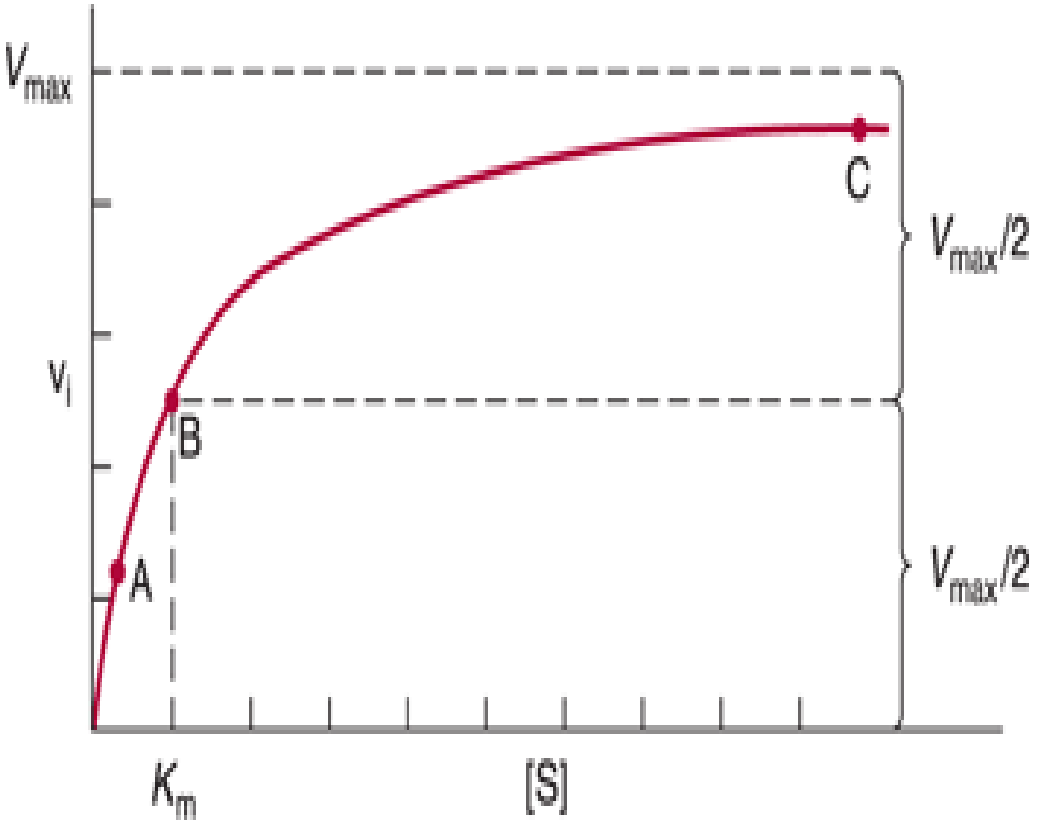
Enzymes

- Since V_{max} and K_m are both constants, their ratio is a constant.
- In other words, when [S] is considerably below K_m,
- V_0 is proportionate to k[S].

Enzymes

- The initial reaction velocity, V_0, therefore is directly proportionate to [S].
- V_0 exhibits a linear dependence on [S], *as observed here.*
- *(First order Reaction)*

Enzymes



Source: Murray RK, Bender DA, Botham KM, Kennelly PJ, Rodwell VW, Weil PA: *Harper's Illustrated Biochemistry, 28th Edition*: http://www.accessmedicine.com

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Enzymes

- Therefore at concentrations below K_m reaction rate is first order i.e. it is directly proportional to the concentration of the substrate.

END

Enzymes

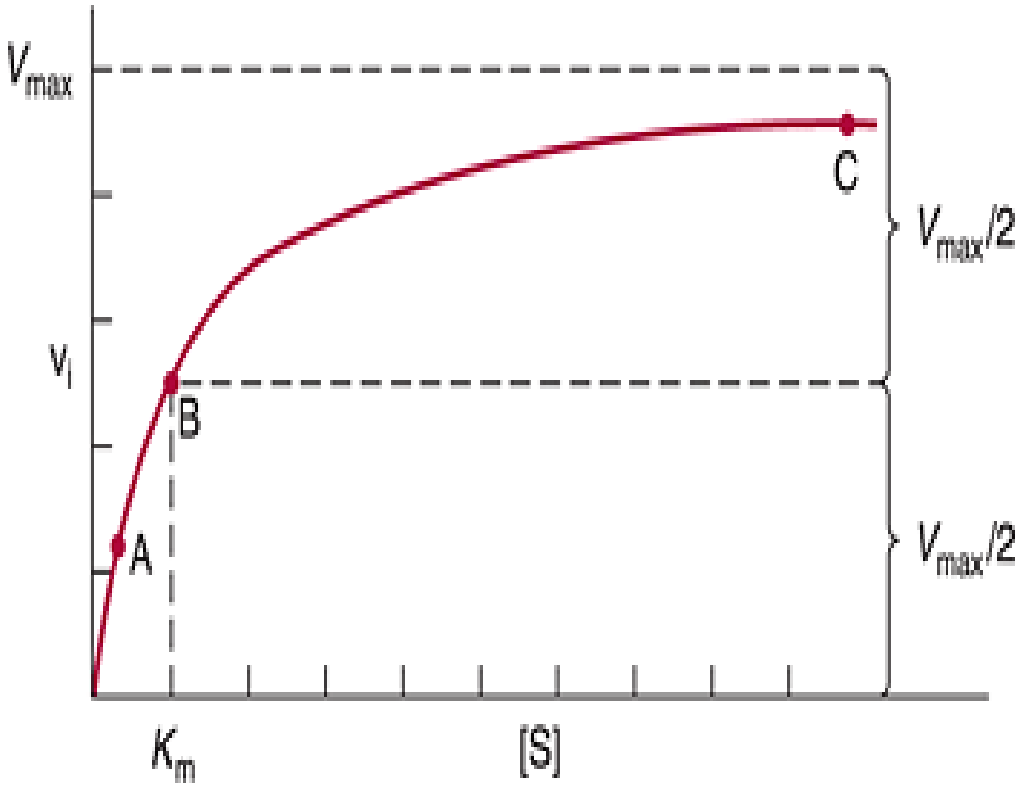
The Michaelis-Menton Equation at high [S]

- At high [S] [S] >> K^m
- The term K_m + [S] is essentially equal to [S].
- *The Km term in the denominator of the Michaelis-* Menten equation becomes insignificant
- Replacing K_m + [S] with [S] reduces equation

Enzymes

- high [S] $[S] >> K^m$
- $V_0 = \frac{Vmax\ [S]}{K_m + [S]}$
- Ignoring K_m
- $V_0 = \frac{Vmax\ [S]}{[S]}$
- V_0 = Vmax.
- *This is consistent with the plateau observed at high [S]. (Zero Order Reaction)*

Enzymes



Source: Murray RK, Bender DA, Botham KM, Kennelly PJ, Rodwell VW, Weil PA: *Harper's Illustrated Biochemistry, 28th Edition*: http://www.accessmedicine.com

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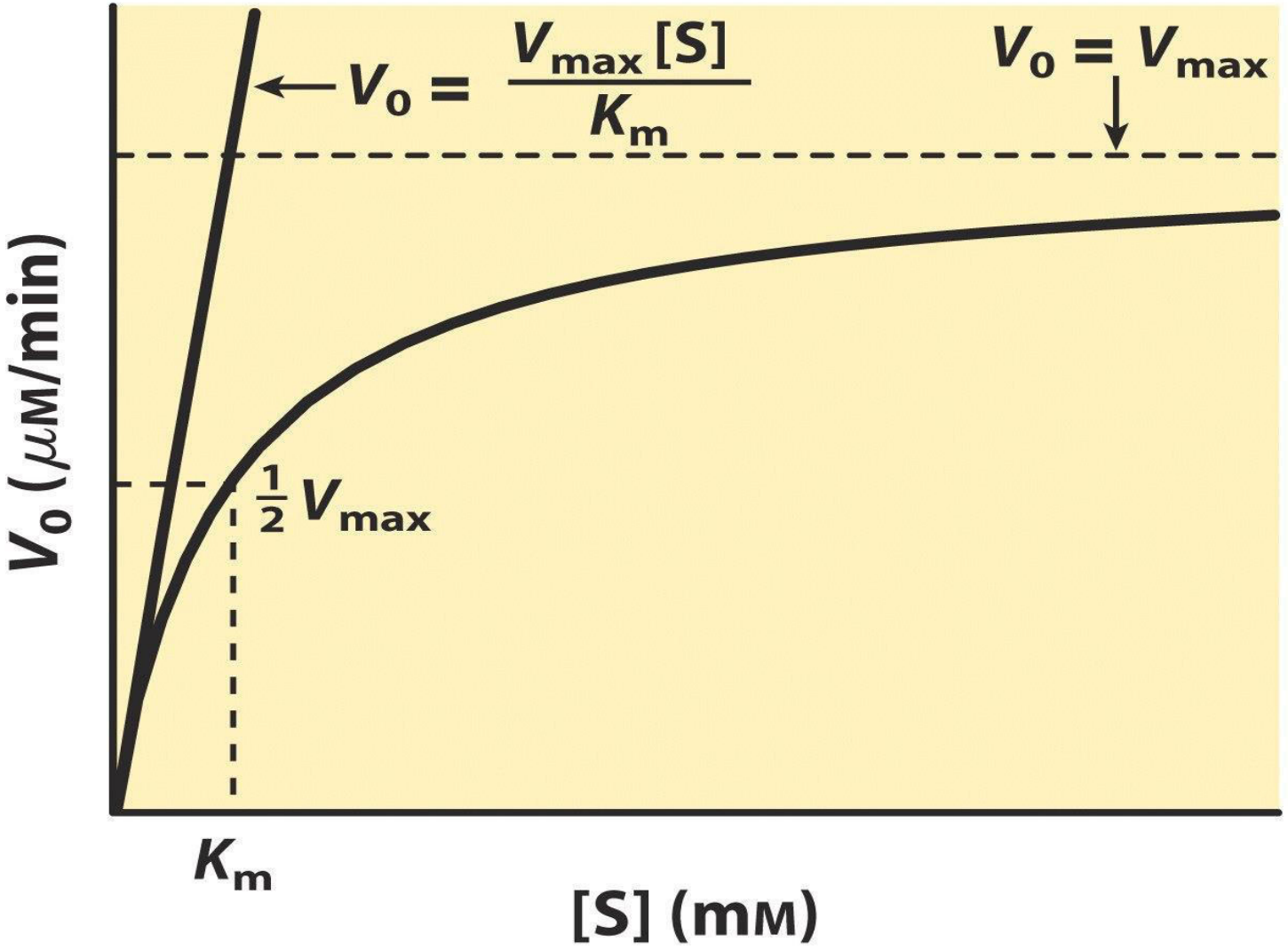
Enzymes

- The rate of reaction is then independent of substrate concentration ***[S]***, and is said to be **zero order** with respect to substrate concentration.

Enzymes



Enzymes

- The Michaelis-Menten equation is therefore consistent with the observed dependence of V_0 on [S], and the shape of the curve is defined by the terms;
- V_{max}/K_m *at low [S] and*
- V_{max} *at high [S].*

END

Enzymes

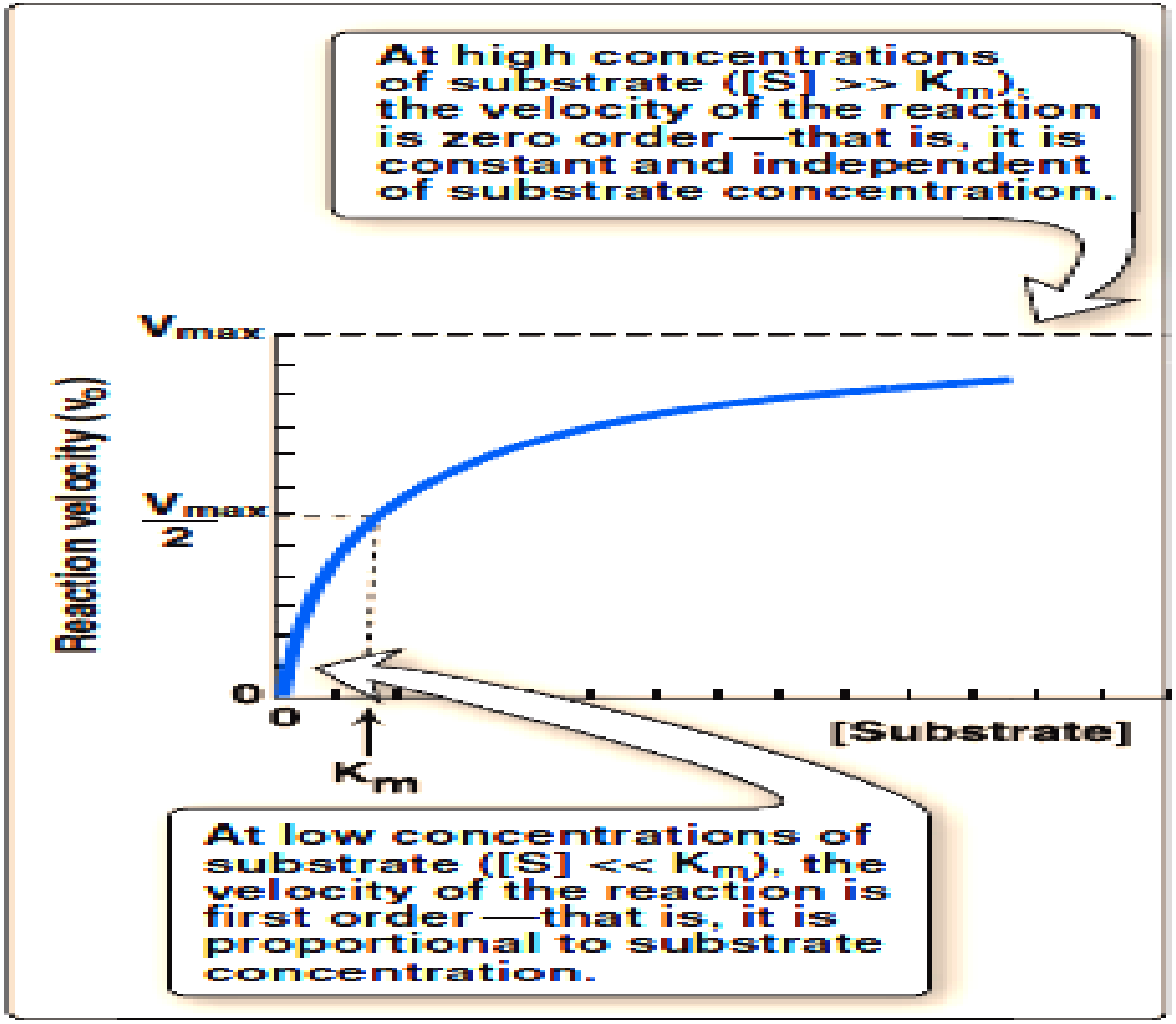
Order of Reaction

- When **[S]** is much less, then the velocity of the reaction is approximately proportional to the substrate concentration.
- The rate of reaction is then said to be first order with respect to substrate.

Enzymes

- When **[S]** is much greater than Km the velocity is constant and equal to V_{max}.
- The rate of reaction is then independent of substrate concentration, and is said to be zero order with respect to substrate concentration

Enzymes


At high concentrations of substrate ([S] >> K_m), the velocity of the reaction is zero order—that is, it is constant and independent of substrate concentration.
V_{max}
$\frac{V_{max}}{2}$
Reaction velocity (V_0)
0
0
[Substrate]
K_m
At low concentrations of substrate ([S] << K_m), the velocity of the reaction is first order—that is, it is proportional to substrate concentration.

Enzymes

Reaction Orders with Respect to Substrate Concentration

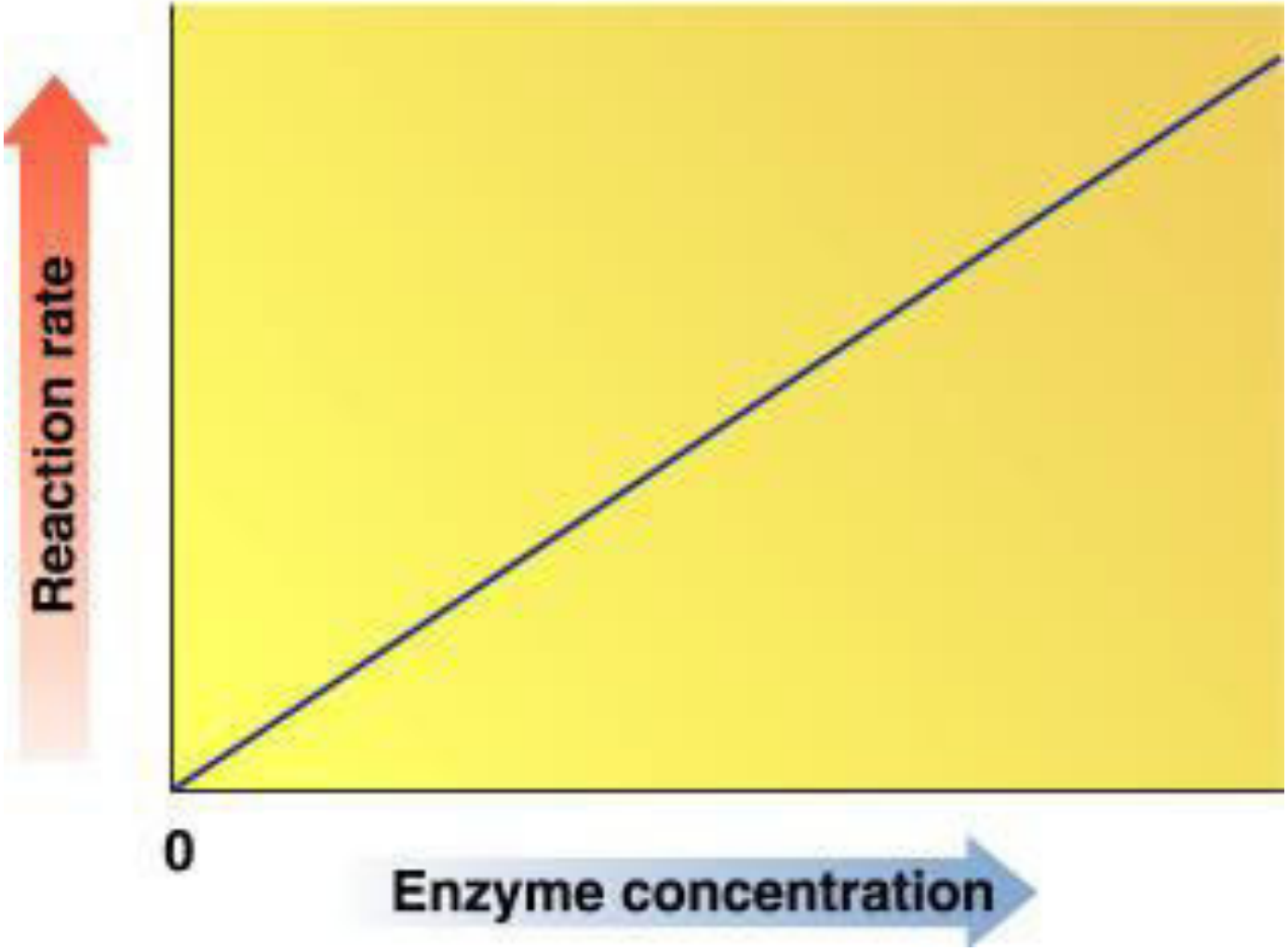
Order	Rate Equation	Comments
Zero	Rate = k	Rate is independent of substrate concentration
First	First rate = k[S]	Rate is proportional to the first power of substrate concentration
Second	Rate = k[S1][S2]	Rate is proportional to the first power of each of two reactants

Enzymes

Relationship of velocity to enzyme concentration

- The rate of the reaction is directly proportional to the enzyme concentration.
- There is a linear relationship between reaction rate and enzyme concentration (at constant substrate concentration)

Enzymes



Enzymes

- For example,
- if the enzyme concentration is halved, the initial rate of the reaction (V_o) as well as that of Vmax are reduced to one half that of the original.

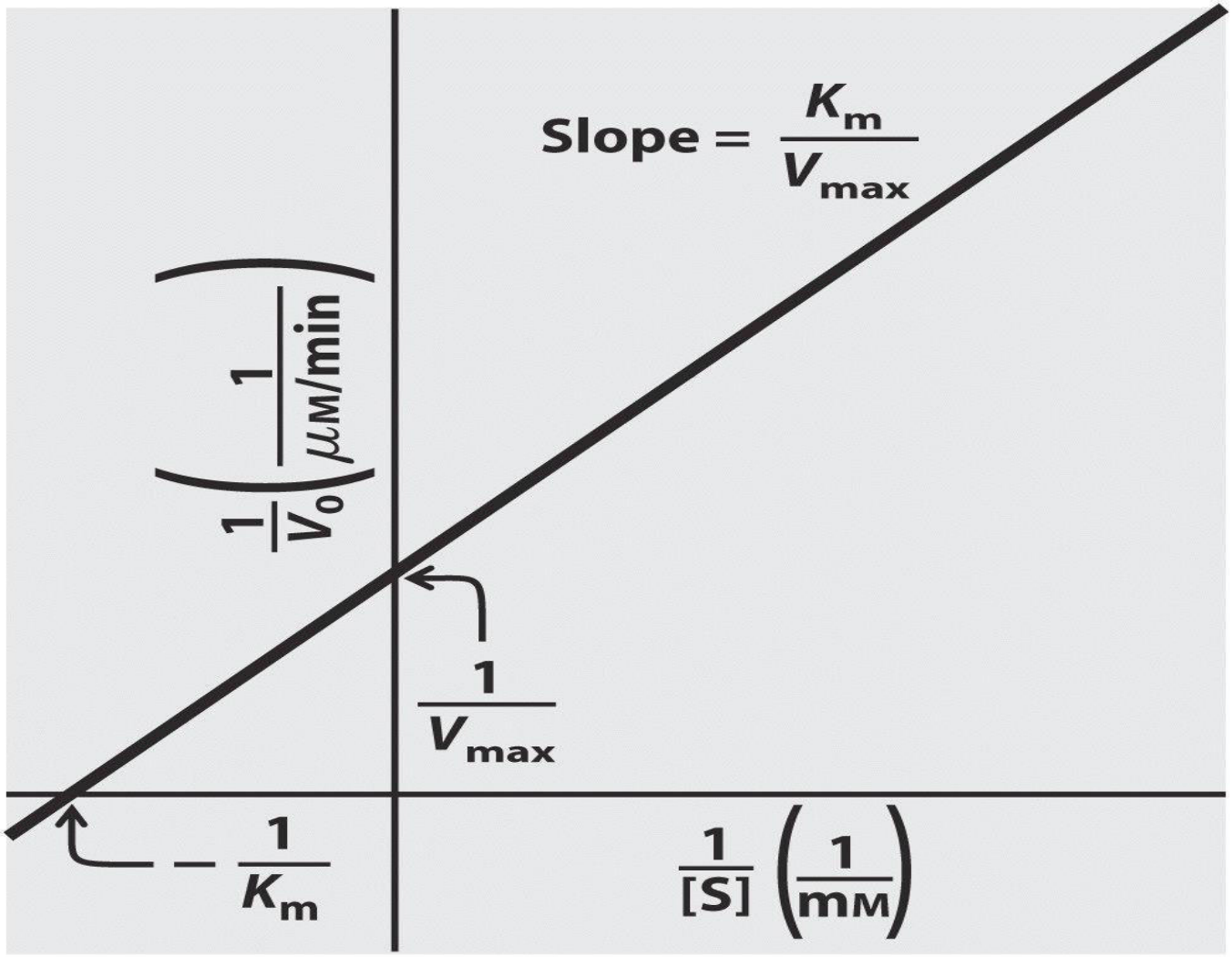
END

Enzymes

Lineweaver-Burke plot

- The Michaelis-Menten equation can be algebraically transformed into **Lineweaver-Burke plot**, a *Double Reciprocal Plot*, that is useful in the practical determination of K_m and $V_{max.}$

Enzymes


Slope = $\frac{K_m}{V_{max}}$
$\frac{1}{V_o}\left(\frac{1}{\mu M/min}\right)$
$\frac{1}{V_{max}}$
$-\frac{1}{K_m}$
$\frac{1}{[S]}\left(\frac{1}{mM}\right)$

Enzymes

- $V_0 = \frac{V_{max}[S]}{K_m + [S]}$
- Lineweaver-Burke transformation is derived simply by taking the reciprocal of both sides of the Michaelis-Menten equation:
- $\frac{1}{V_0} = \frac{K_m + [S]}{V_{max}[S]}$

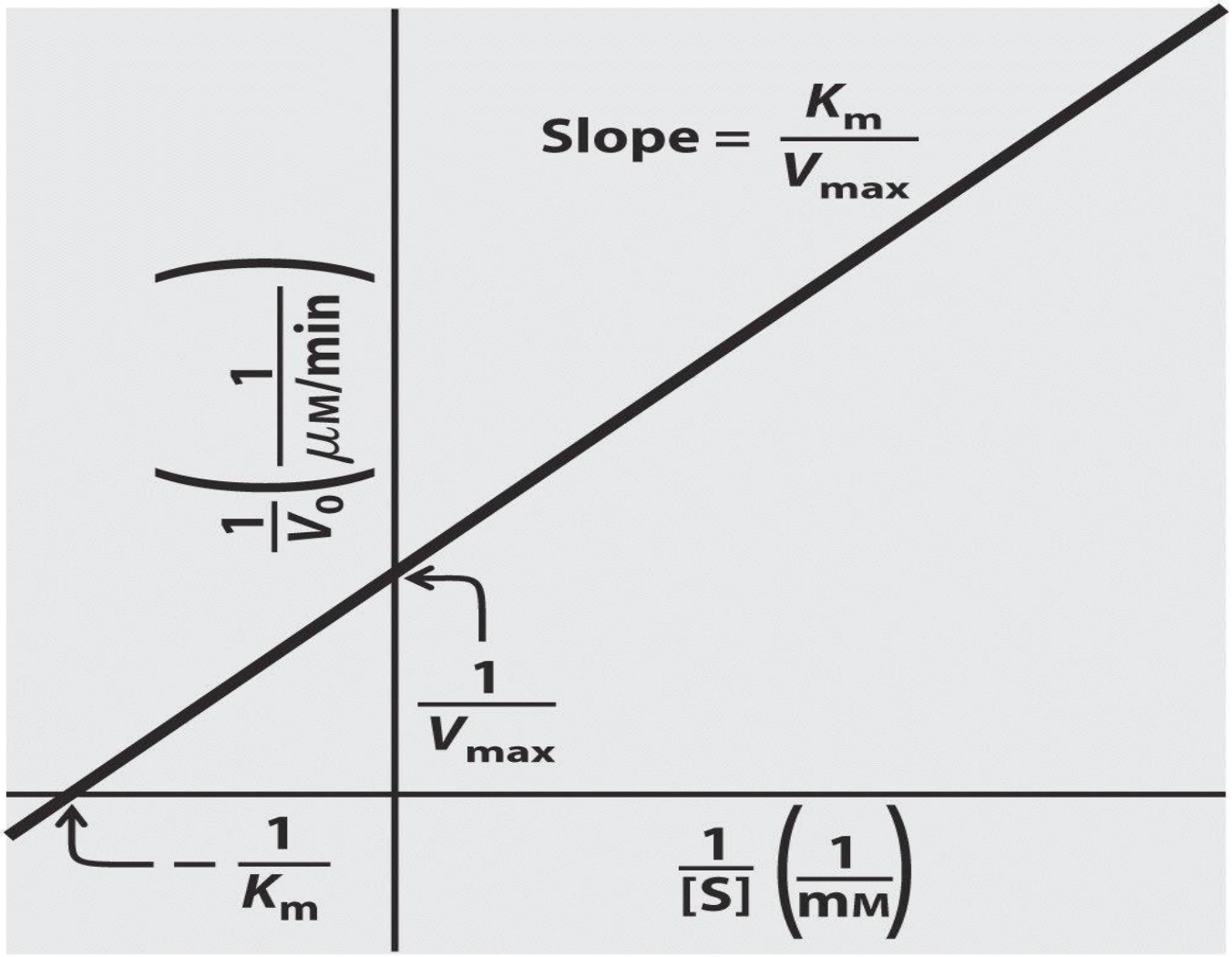
Enzymes

- $\frac{1}{V_0} = \frac{K^m + [S]}{Vmax\,[S]}$

- $\frac{1}{V_0} = \frac{Km}{Vmax[S]} + \frac{[S]}{Vmax[S]}$

- $\frac{1}{V_0} = \frac{K^m}{Vmax[S]} + \frac{1}{Vmax}$

Enzymes


Slope = $\frac{K_m}{V_{max}}$
$\frac{1}{V_o}\left(\frac{1}{\mu M/min}\right)$
$\frac{1}{V_{max}}$
$-\frac{1}{K_m}$
$\frac{1}{[S]}\left(\frac{1}{mM}\right)$

Enzymes

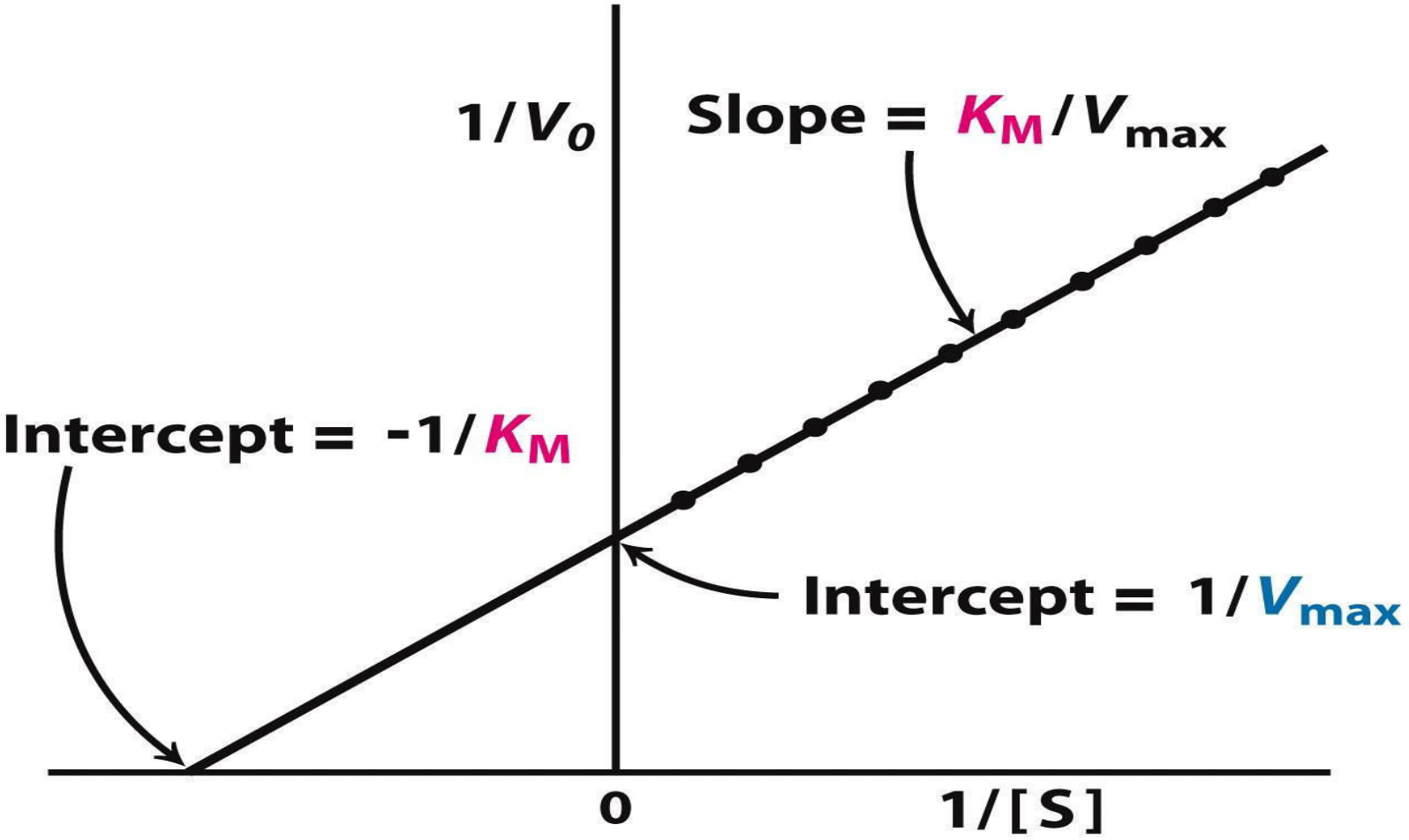


Figure 8.12
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Enzymes

- Lineweaver-Burk plot, has the great advantage of allowing a more accurate determination of V_{max},
- *Which can only be approximated from* a simple plot of V_o *versus [S].*

Enzymes

- The double-reciprocal plot of enzyme reaction rates is very useful in distinguishing between certain types of enzymatic reaction mechanisms and in analyzing enzyme inhibition.

END

Enzymes

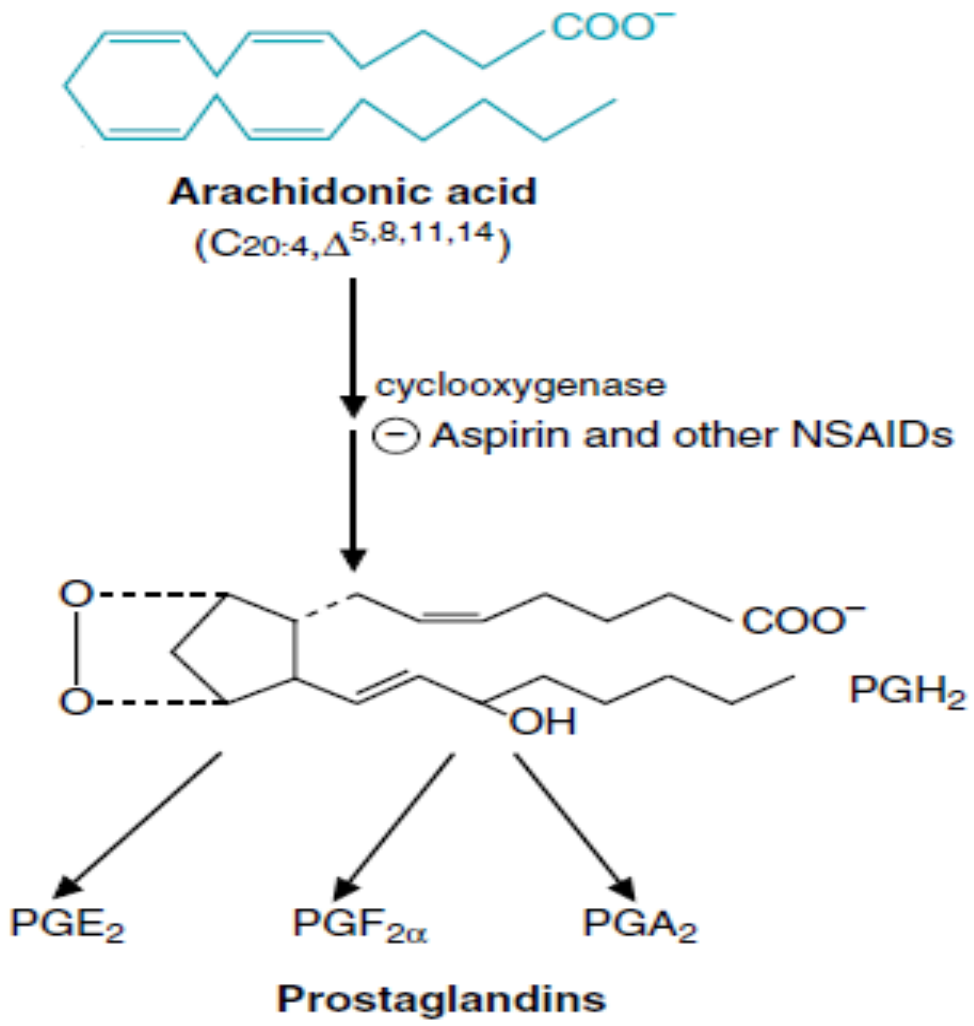
INHIBITION OF ENZYME ACTIVITY

- Enzyme inhibitors are molecular agents that interfere with catalysis, slowing or halting enzymatic reactions.
- Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor.

Enzymes

- Enzymes catalyze virtually all cellular processes, so it should not be surprising that enzyme inhibitors are among the most important pharmaceutical agents known.

Enzymes


COO⁻
Arachidonic acid
(C20:4,Δ5,8,11,14)
cyclooxygenase
⊖ Aspirin and other NSAIDs
O
O
COO⁻
OH
PGH2
PGE2
PGF2α
PGA2
Prostaglandins

Enzymes

- Inhibitors can be classified on the basis of their site of action on the enzyme,
- on whether they chemically modify the enzyme, or on the kinetic parameters they influence.
- Two broad classes of enzyme inhibitors:
- **1. Reversible**
- **2. Irreversible**.

Enzymes

- In general, irreversible inhibitors bind to enzymes through covalent bonds.
- Reversible inhibitors typically bind to enzymes through noncovalent bonds,

Enzymes

- The two most commonly encountered types of reversible inhibition are;
- competitive and
- noncompetitive.

Enzymes

- Competitive inhibitors resemble the substrate and compete for binding to the active site of the enzyme.
- Noncompetitive inhibitors do not bind at the active site. They bind either free enzyme at a site other than active site or the ES complex.

END

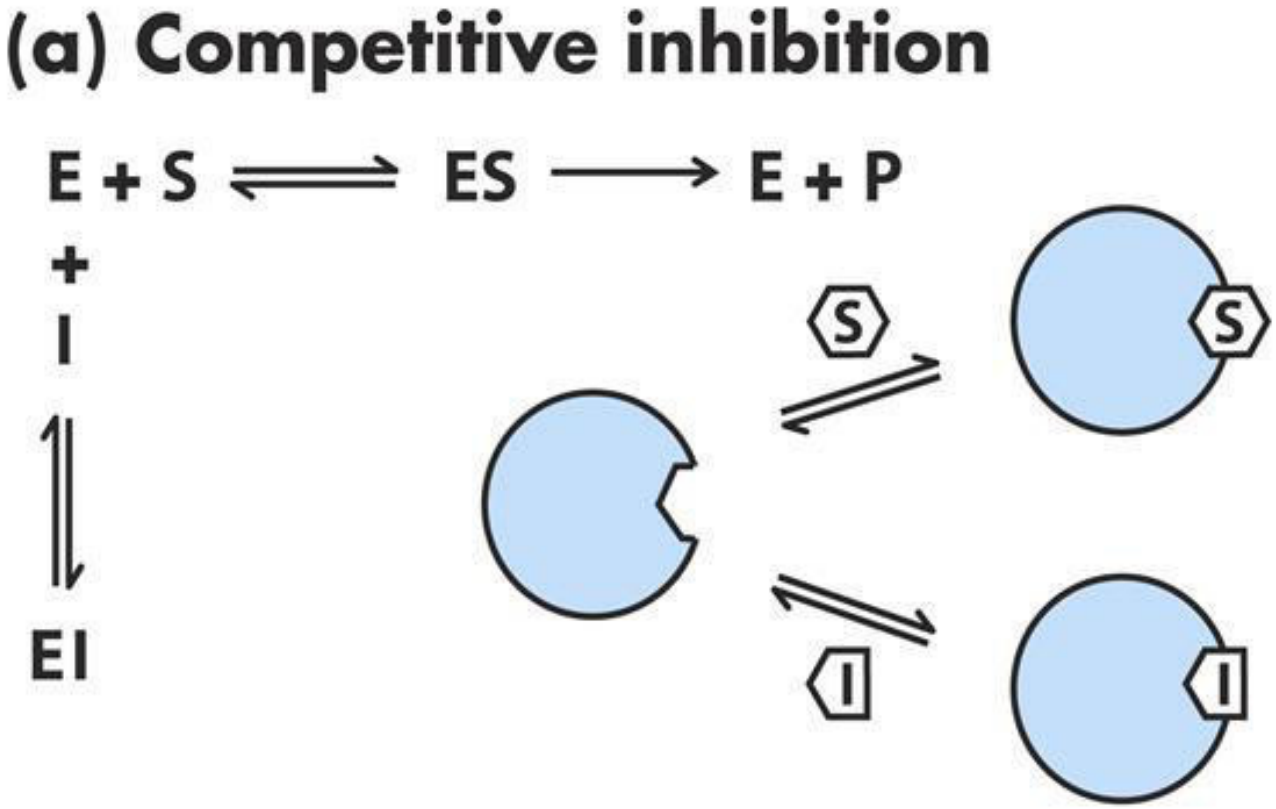
Enzymes

Competitive Inhibition

- This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy i.e. active site and, therefore, competes with the substrate for that site.

Enzymes

- **Competitive inhibitors bind to the enzyme's active site.**



Enzymes

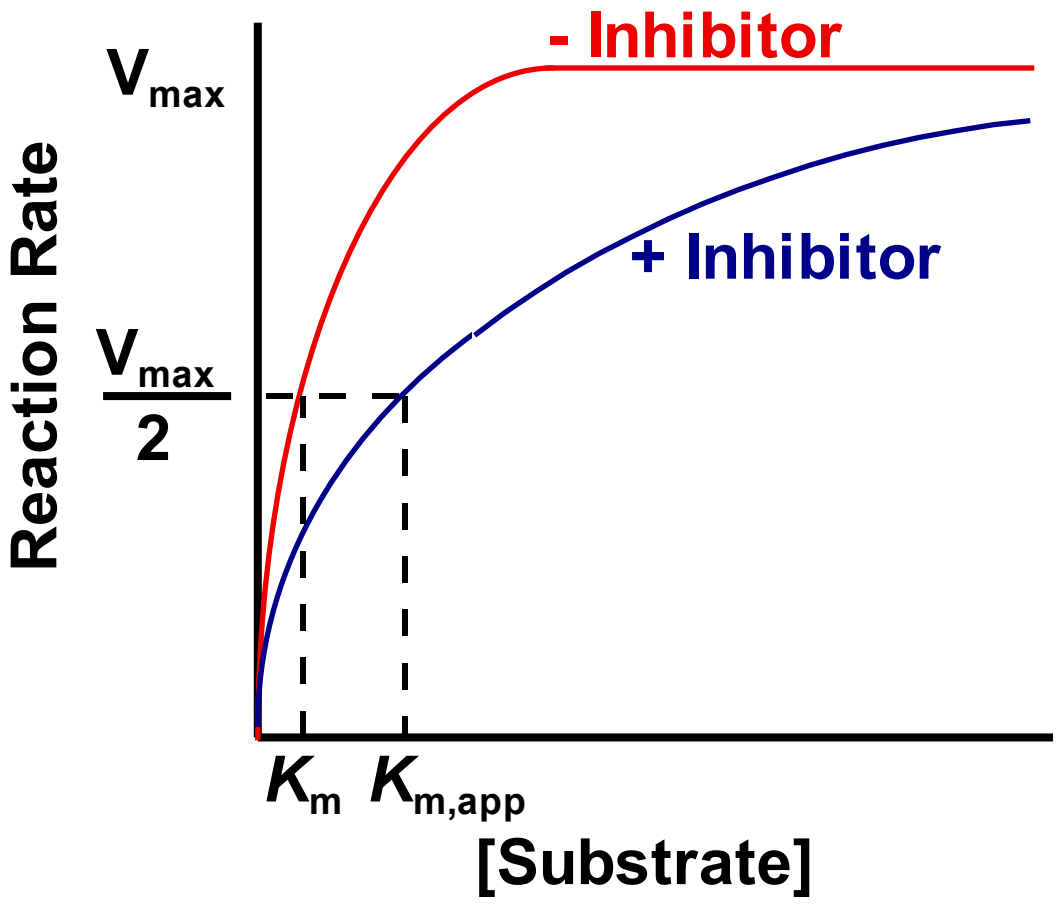
- Reversible inhibitors bind to enzymes through non covalent bonds.
- Dilution of the enzyme-inhibitor complex results in dissociation of the reversibly bound inhibitor, and recovery of enzyme activity.

Enzymes

1. Effect on Vmax

- The effect of a competitive inhibitor is reversed by increasing [S].
- At a sufficiently high substrate concentration, the reaction velocity reaches the V_{max} as observed in the absence of inhibitor.

Enzymes



$V_{max,app} = V_{max}$ $K_{m,app} > K_m$

Enzymes

- When [S] far exceeds [I], the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits a normal V_{max}.
- Therefore, a competitive inhibitor does not decrease V_{max}

END

Enzymes

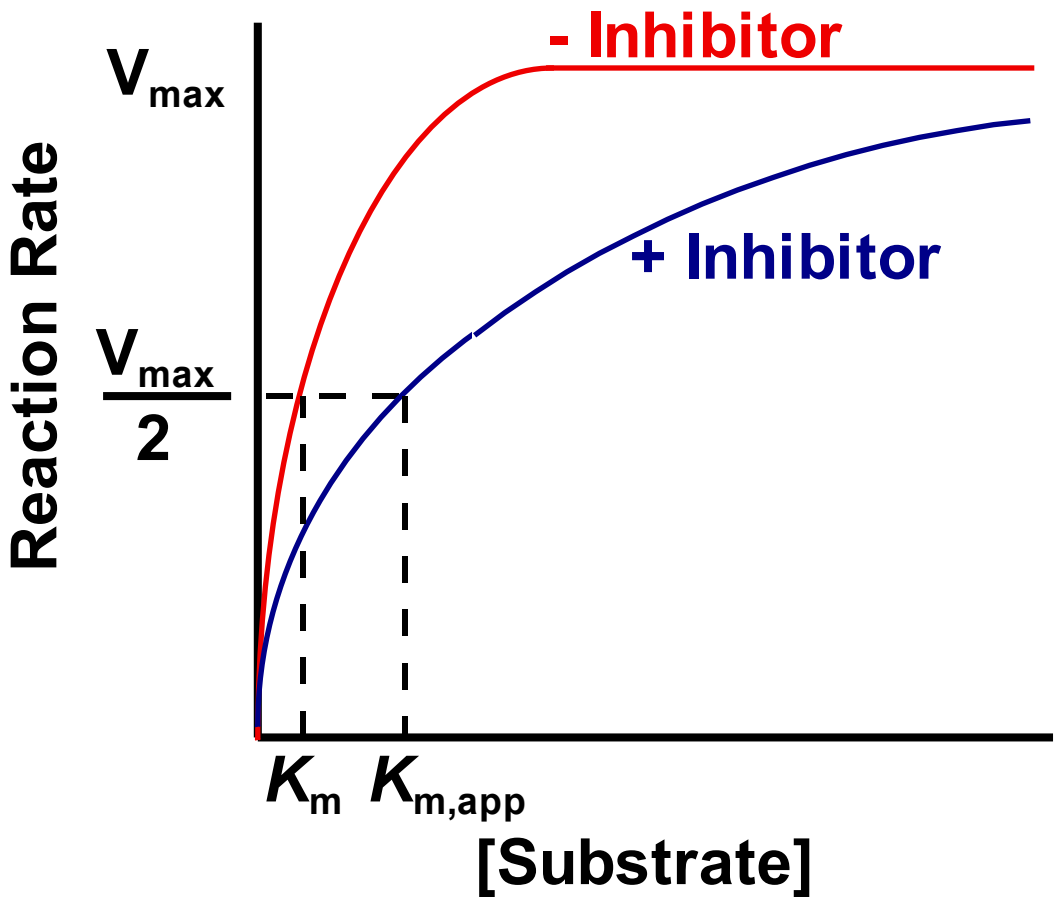
Competitive Inhibition

(Contd.)

2. Effect on Km

- The [S] at which V_0 = 1/2 V_{max}, the apparent Km, increases in the presence of inhibitor.
- A competitive inhibitor increases the apparent K_m (αK_m) for a given substrate.

Enzymes



$V_{max,app} = V_{max}$ $K_{m,app} > K_m$

Enzymes

3. Effect on Lineweaver-Burk plot

- Competitive inhibition shows a characteristic Lineweaver-Burke plot
- the plots of the inhibited and uninhibited reactions intersect at a single point on the y-axis at $1/V_{max.}$
- V_{max} is unchanged

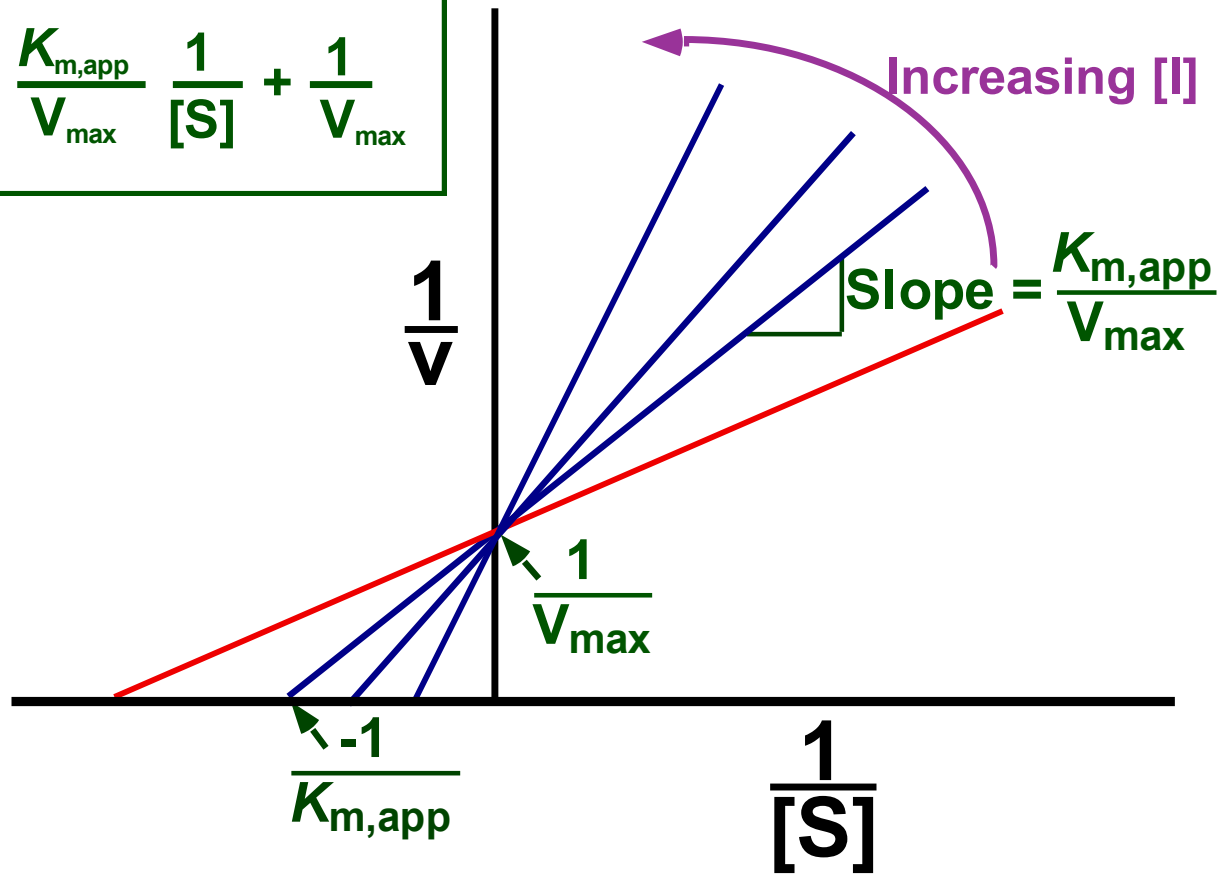
Enzymes

- However, the inhibited and uninhibited reactions show; different x axis intercepts
- Therefore indicating that the apparent K_m is increased in the presence of the competitive inhibitor because $-1/K_m$ moves closer to zero from a negative value.

Enzymes

The Lineweaver-Burk plot is diagnostic for competitive inhibition

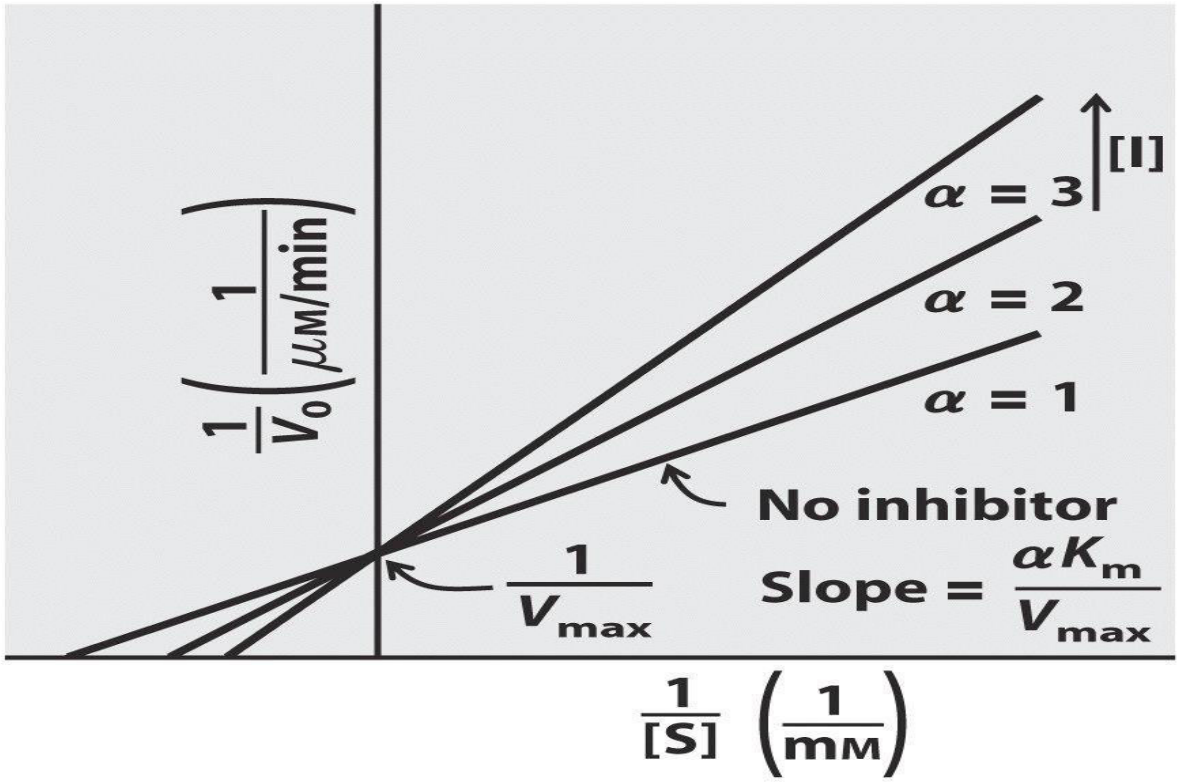
$$\frac{1}{v} = \frac{K_{m,app}}{V_{max}} \frac{1}{[S]} + \frac{1}{V_{max}}$$



Enzymes

Competitive inhibition.

$$\frac{1}{V_0} = \left(\frac{\alpha K_m}{V_{max}}\right)\frac{1}{[S]} + \frac{1}{V_{max}}$$



Enzymes

- We know that V_{max} *is* unchanged by the presence of a competitive inhibitor.
- This is because more substrate means more formation of ES complex vs EI complex.

END

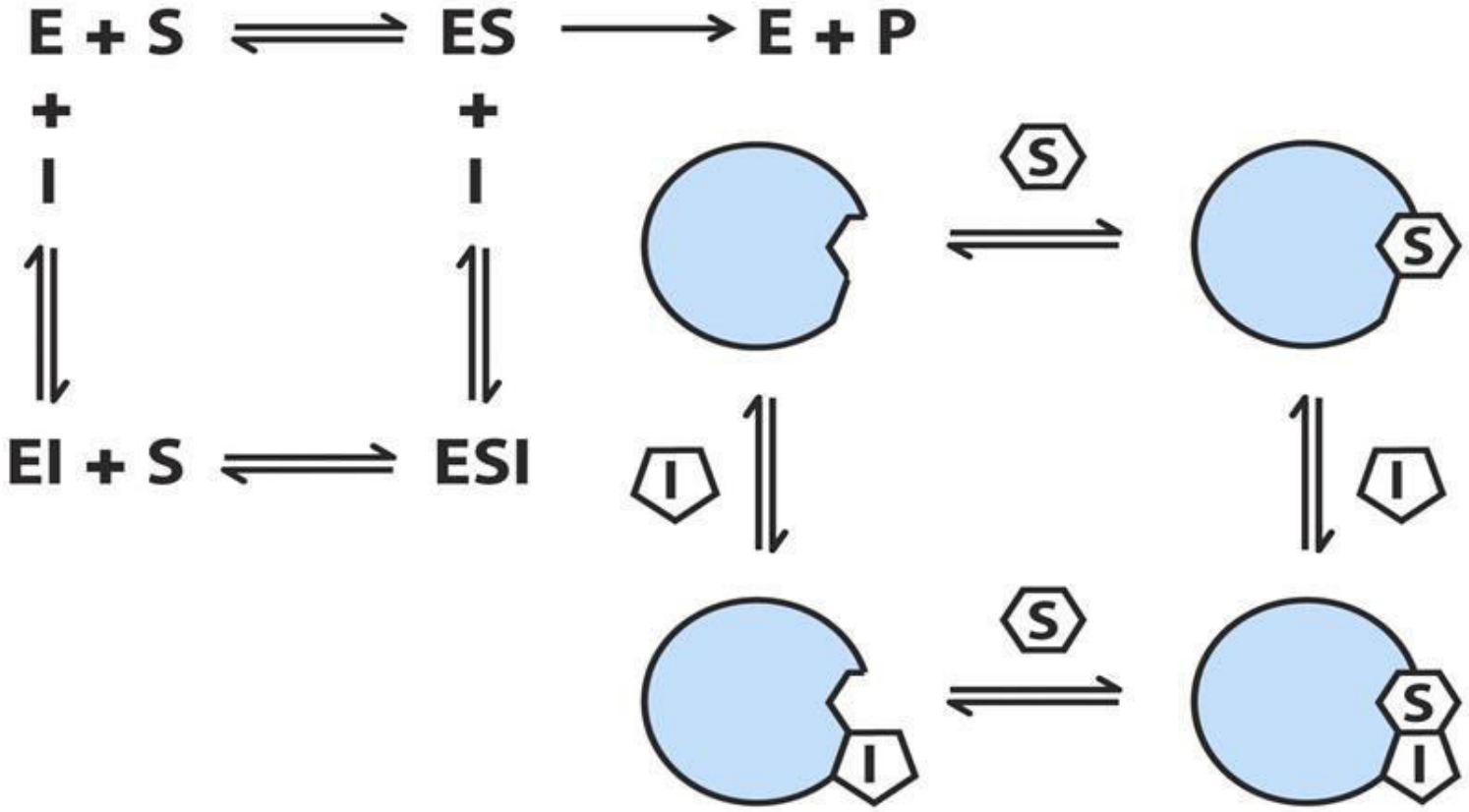
Enzymes

Noncompetitive Inhibition

- Inhibitors bind enzymes at sites distinct from the substrate-binding site and
- generally bear little or no structural resemblance to the substrate.

Enzymes

Noncompetitive inhibitors bind at a separate site, but may bind to either E or ES.



Enzymes

- Noncompetitive inhibition cannot be overcome by increasing the concentration of substrate.

Enzymes

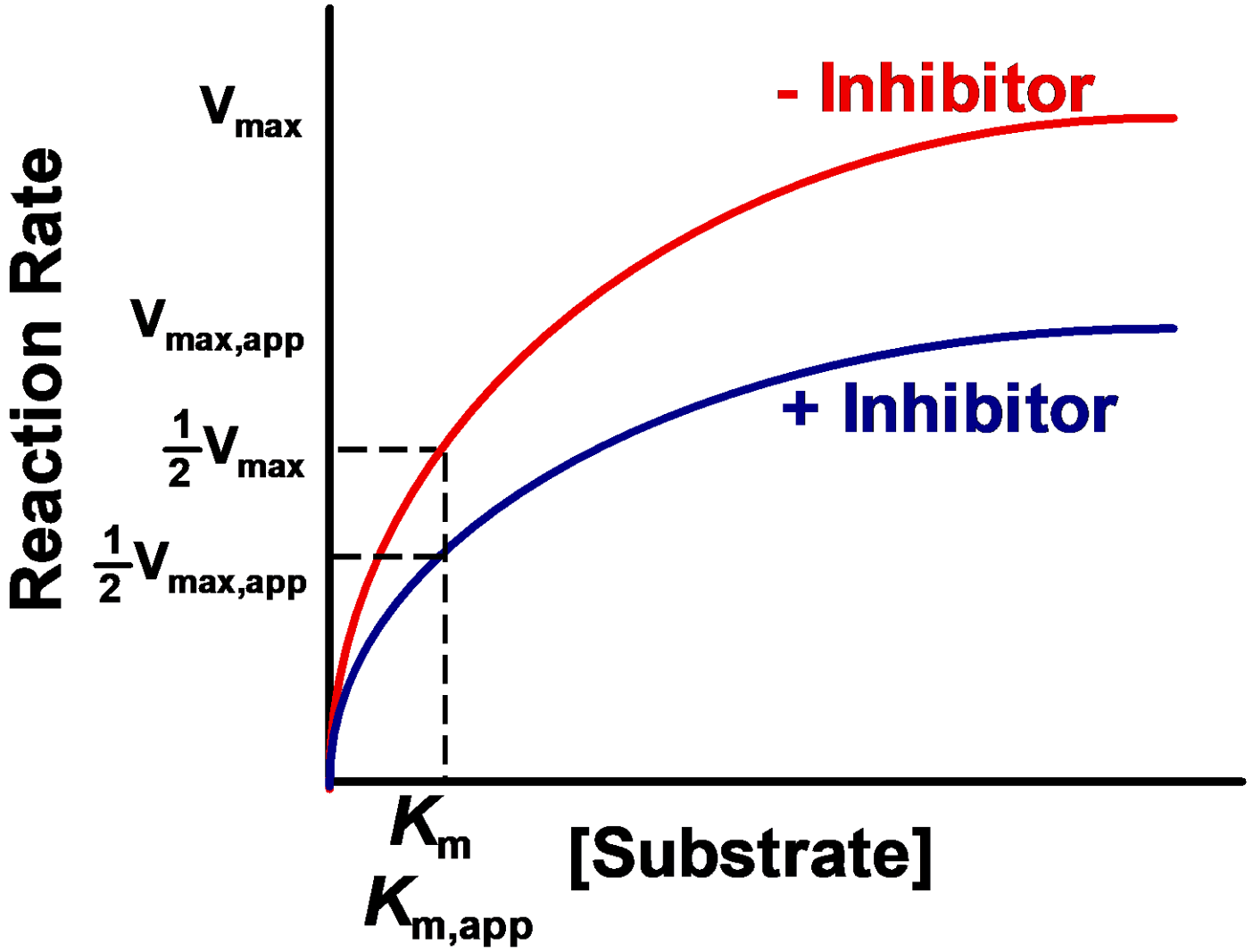
1. Effect on V_{max}

- the apparent V_{max} changes, because the inhibitor is capable of preventing catalysis regardless of whether the substrate is bound to the enzyme.

Enzymes

- Noncompetitive inhibition cannot be overcome by increasing the concentration of substrate.
- Thus, noncompetitive inhibitors decrease the V_{max} of the reaction.

Enzymes


Reaction Rate
V_{max}
$V_{max,app}$
$\frac{1}{2}V_{max}$
$\frac{1}{2}V_{max,app}$
- Inhibitor
+ Inhibitor
K_m
$K_{m,app}$
[Substrate]

Enzymes

- The noncompetitive inhibitor, in effect, lowers the concentration of the active enzyme and therefore decreases the V_{max} of the enzyme.

END

Enzymes

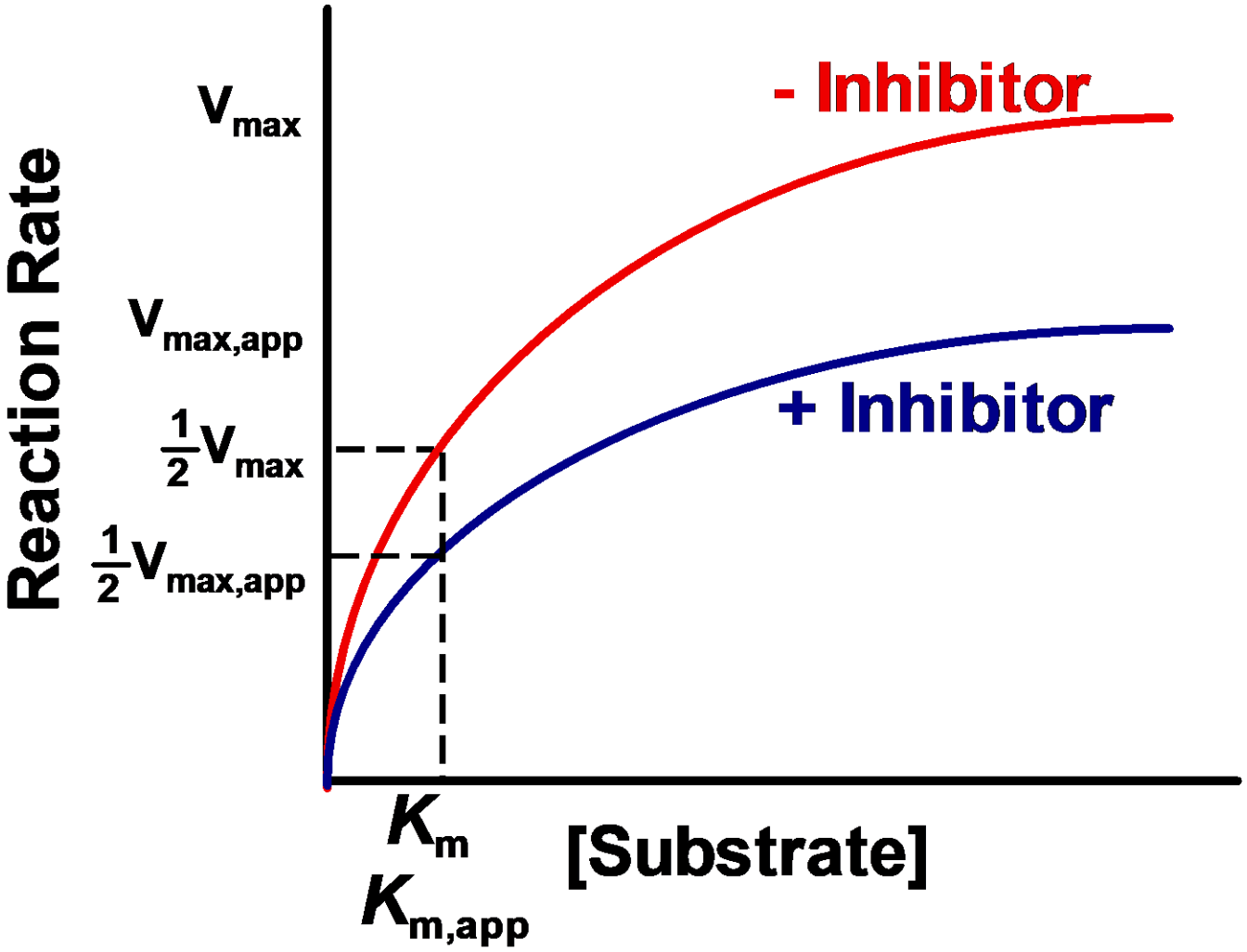
Noncompetitive Inhibition

(Contd.)

2. Effect on Km

- Noncompetitive inhibitors do not interfere with the binding of substrate to enzyme.
- Thus, the enzyme shows the same K_m in the presence or absence of the noncompetitive inhibitor.

Enzymes


Reaction Rate
V_{max}
$V_{max,app}$
$\frac{1}{2}V_{max}$
$\frac{1}{2}V_{max,app}$
- Inhibitor
+ Inhibitor
K_m
$K_{m,app}$
[Substrate]

Enzymes

3. Effect on Lineweaver-Burk plot

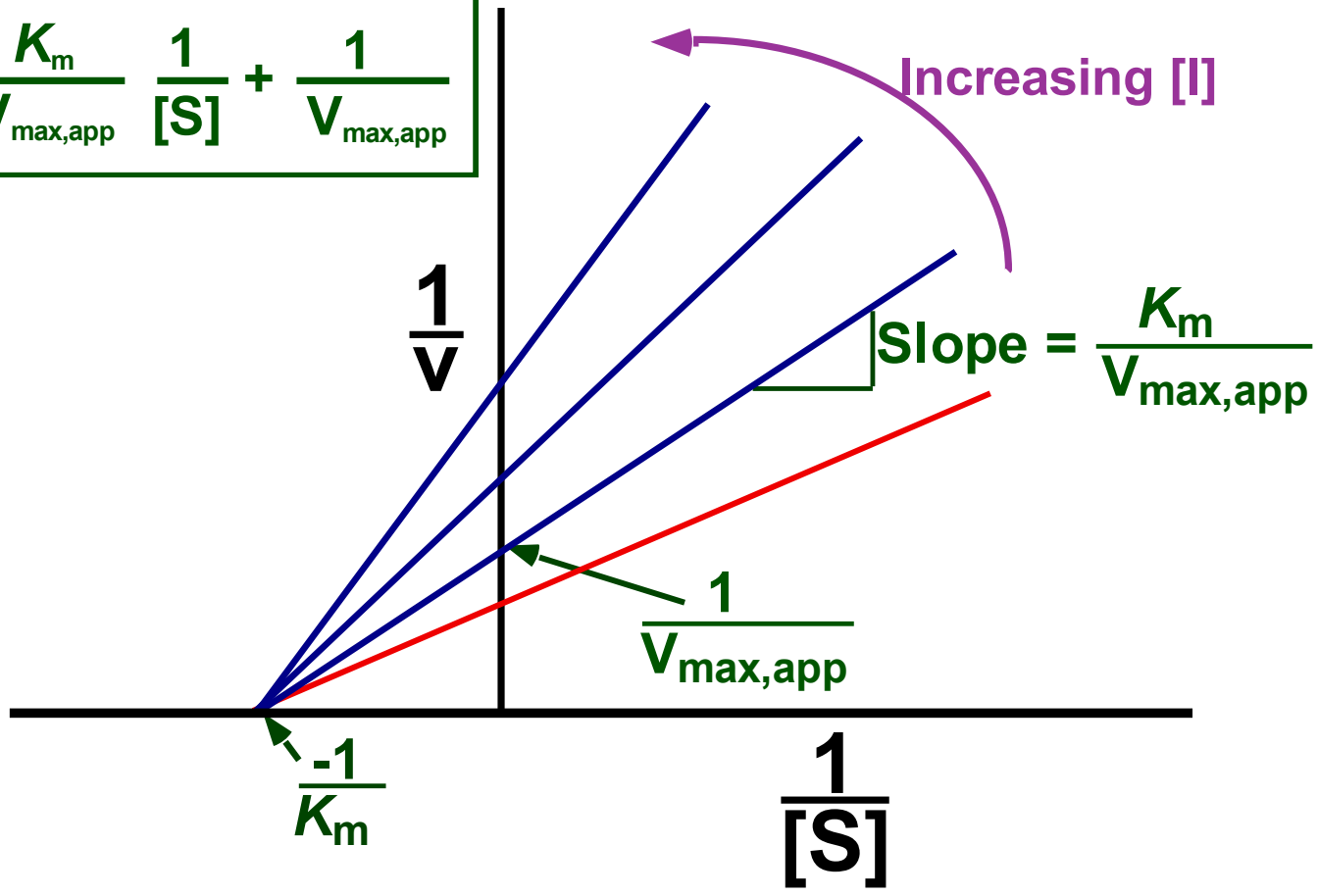
- Noncompetitive inhibition shows a characteristic Lineweaver-Burke plot
- the plots of the inhibited and uninhibited reactions intersect at a single point on the x-axis at K_m
- K_m is unchanged

Enzymes

- However, the inhibited and uninhibited reactions show; different y axis intercepts
- indicating the deccrease in V_{max} in the presence of the competitive inhibitor.

Enzymes

$$\frac{1}{v} = \frac{K_m}{V_{max,app}} \frac{1}{[S]} + \frac{1}{V_{max,app}}$$



Enzymes

- While certain inhibitors exhibit characteristics of a mixture of competitive and noncompetitive inhibition, the evaluation of these inhibitors exceeds the scope of our level of study.

END

Water ,pH & Buffer Systems

Water, pH & buffer systems

Water

- Water is the most abundant substance in living systems, making up 70% or more of the weight of most organisms

Water, pH & buffer systems

- The hydrogen bonding between water molecules and the slight tendency of water to ionize are of crucial importance to the structure and function of biomolecules

Water, pH & buffer systems

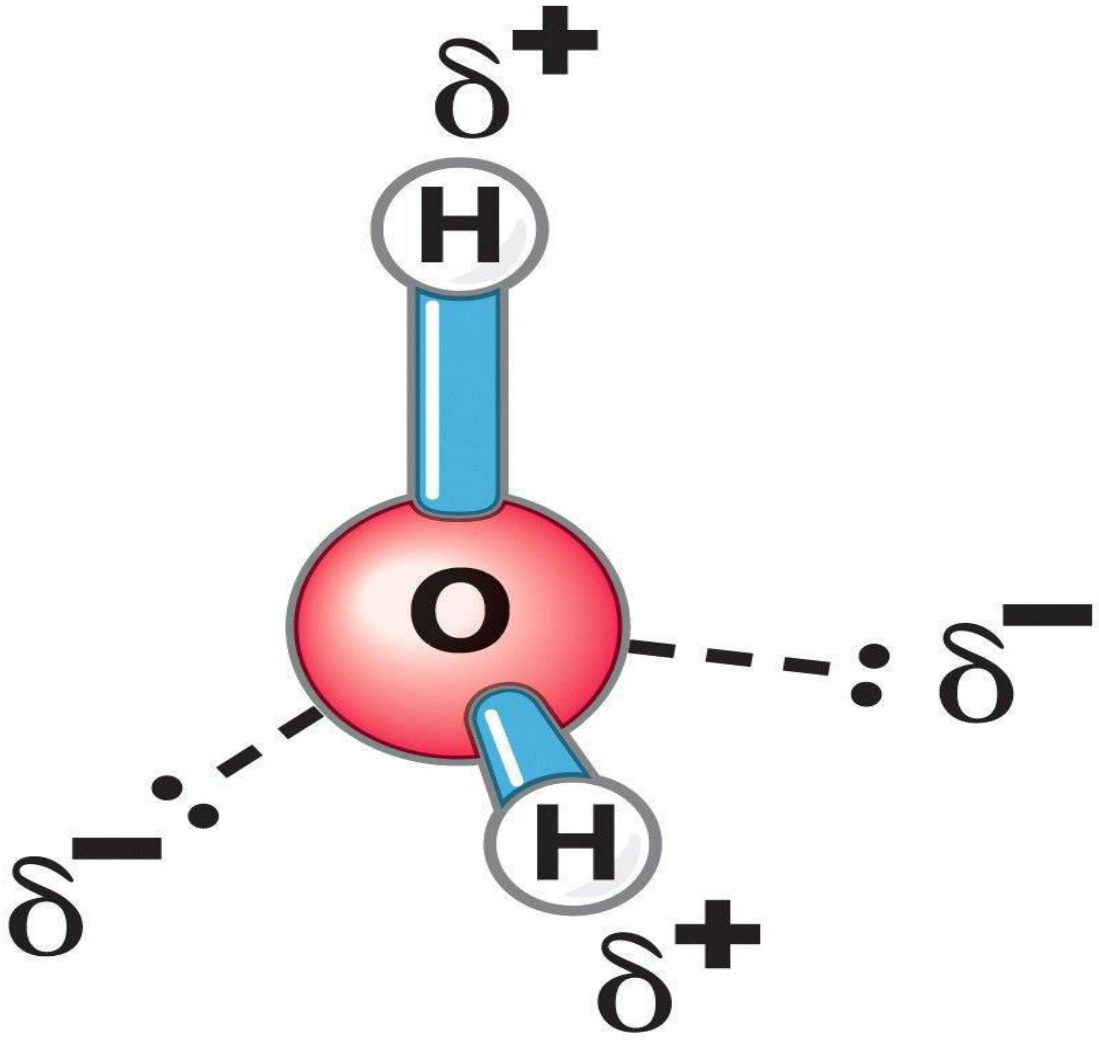
Hydrogen Bonding gives Water its Unusual Properties

- Water has a higher
- melting point,
- boiling point, and
- heat of vaporization than most other common solvents

Water, pH & buffer systems

- Each hydrogen atom of a water molecule shares an electron pair with the central oxygen atom
- The geometry of the molecule is dictated by the shapes of the outer electron orbitals of the oxygen atom

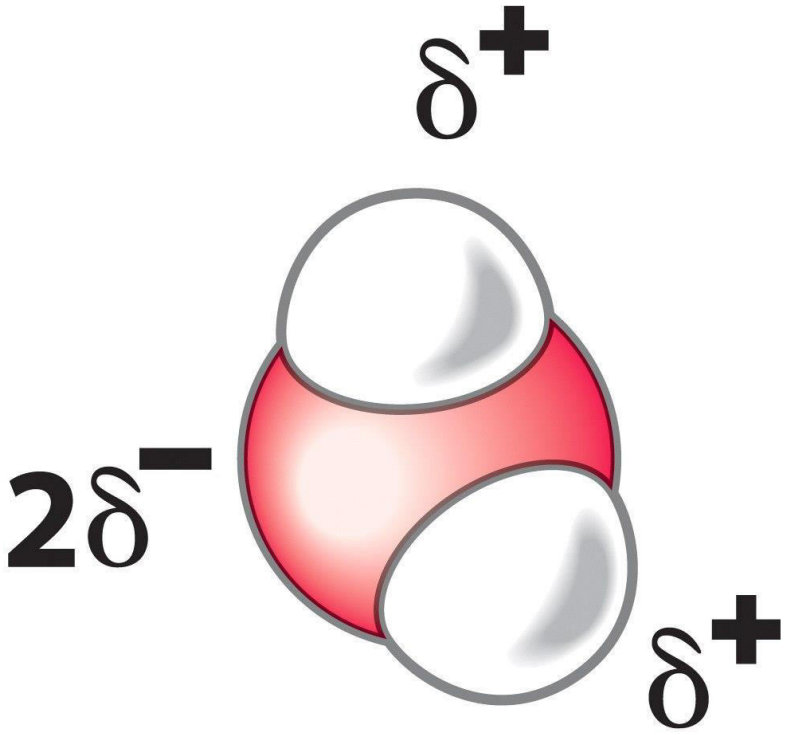
Water, pH & buffer systems



Water, pH & buffer systems

- The result of this unequal electron sharing is two electric dipoles in the water molecule
- Each hydrogen bears a partial positive charge (δ+), and
- the oxygen atom bears a partial negative charge equal to the sum of the two partial positives (2δ-)

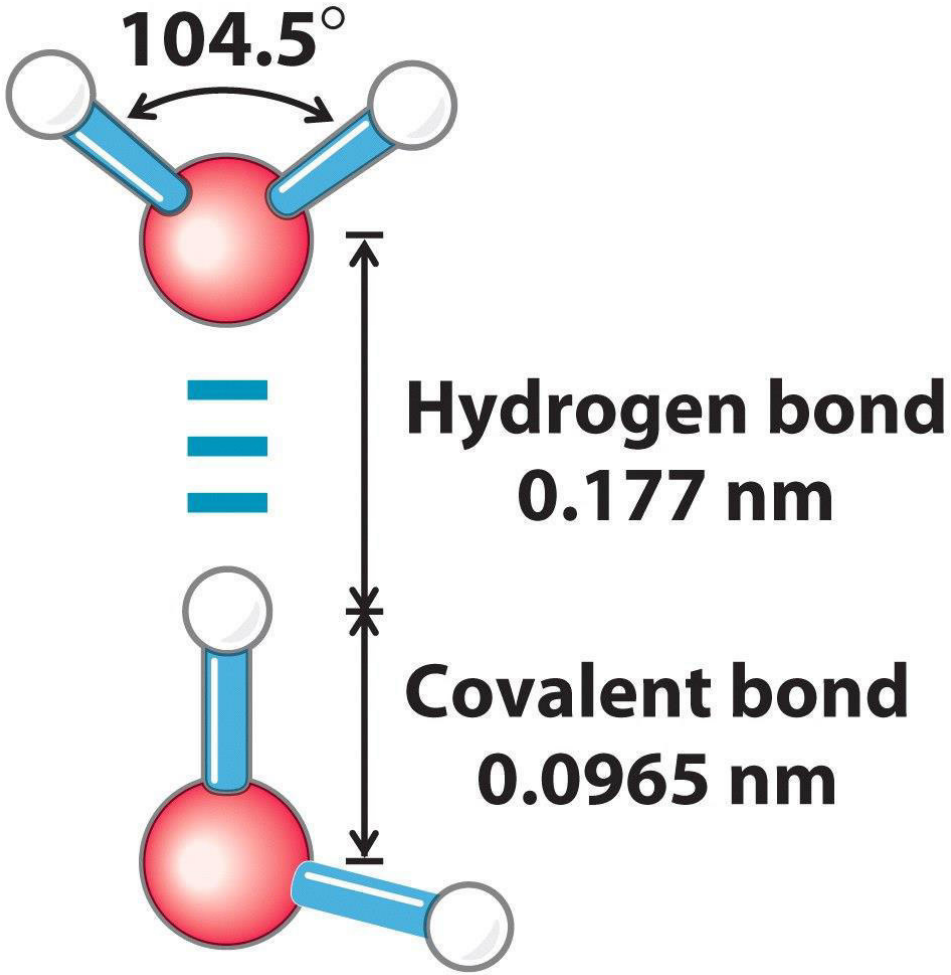
Water, pH & buffer systems



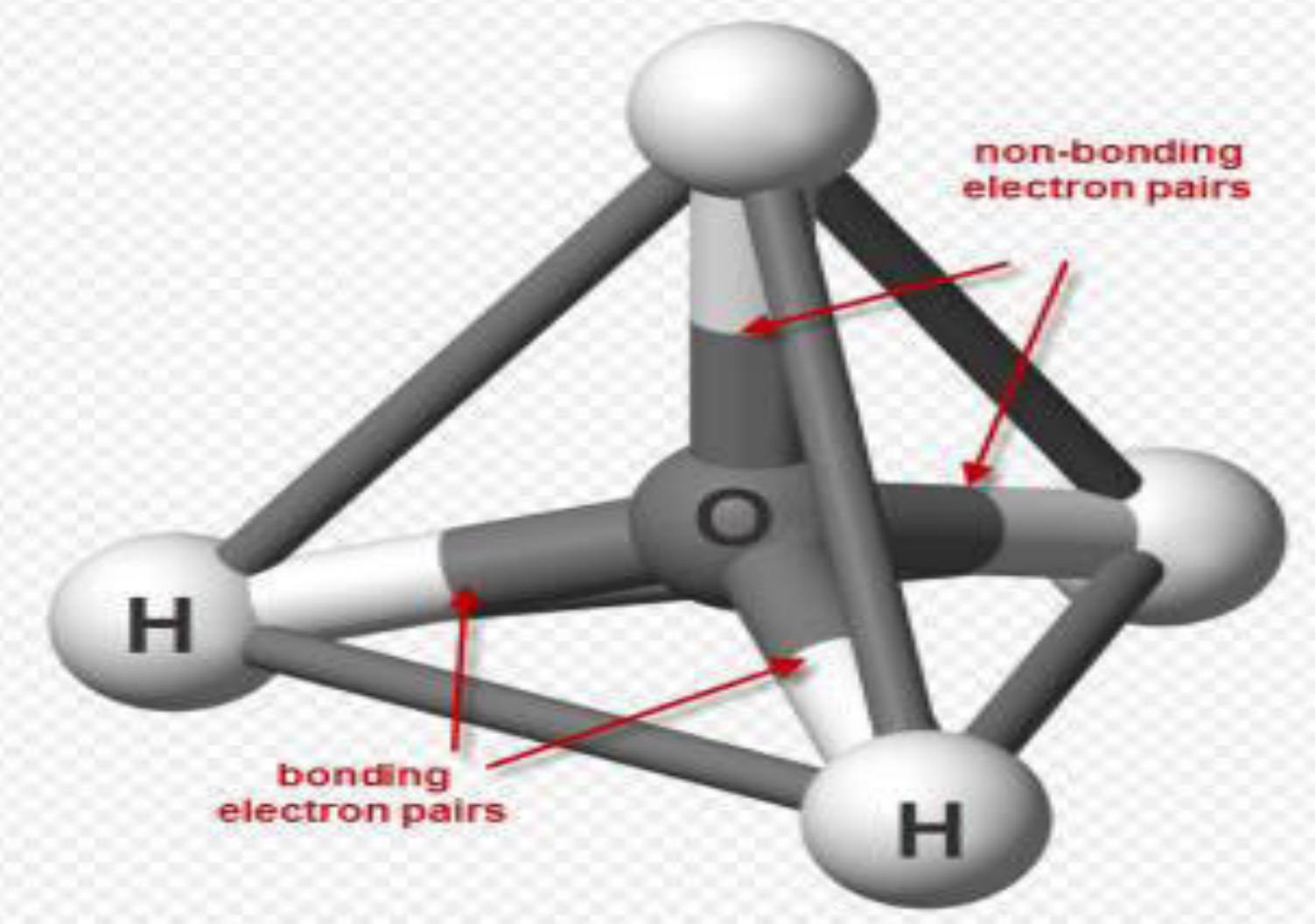
Water, pH & buffer systems

- As a result, there is an electrostatic attraction between the oxygen atom of one water molecule and the hydrogen of another, called a hydrogen bond

Water, pH & buffer systems



Water, pH & buffer systems

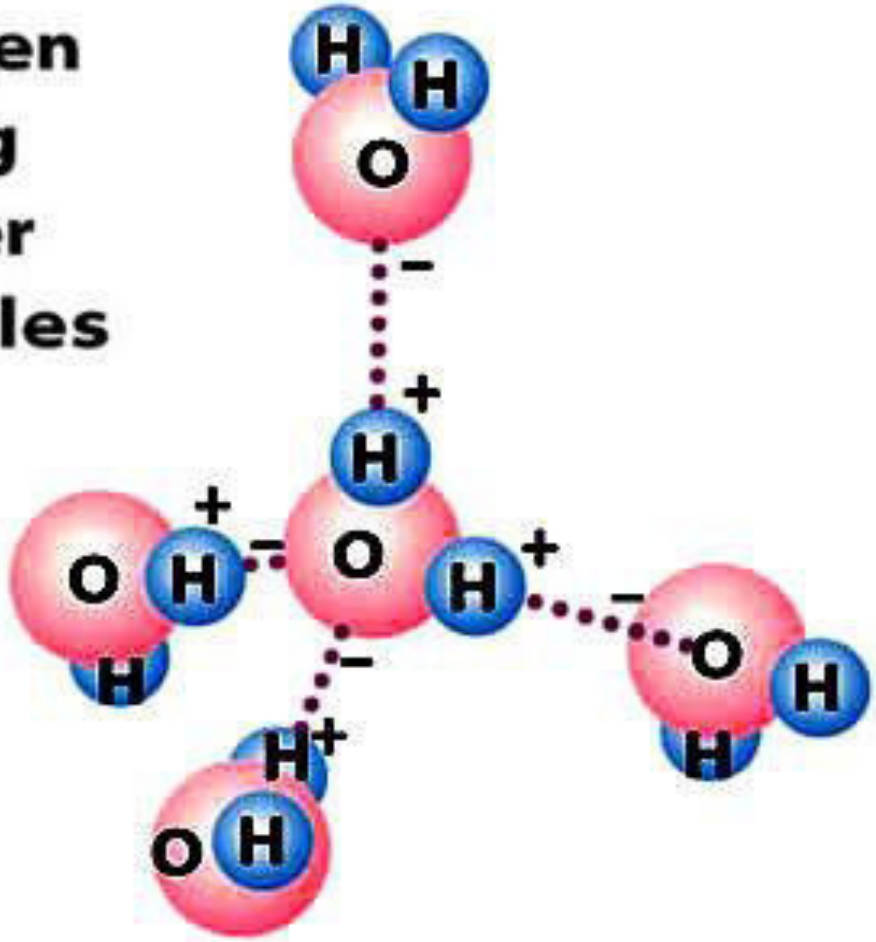


Water, pH & buffer systems

- The nearly tetrahedral arrangement of the orbitals about the oxygen atom allows each water molecule to form hydrogen bonds with as many as four neighboring water molecules

Water, pH & buffer systems

Hydrogen bonding of water molecules



Water, pH & buffer systems

Ionization of Water

- Although many of the solvent properties of water can be explained in terms of the uncharged H_2O molecule, the small degree of ionization of water into hydrogen ions (H^+) and hydroxide ions (OH^-) must also be taken into account

Water, pH & buffer systems

- Pure Water Is Slightly Ionized
- Water molecules have a slight tendency to undergo reversible ionization to yield a hydrogen ion (a proton) and a hydroxide ion, giving the equilibrium reaction

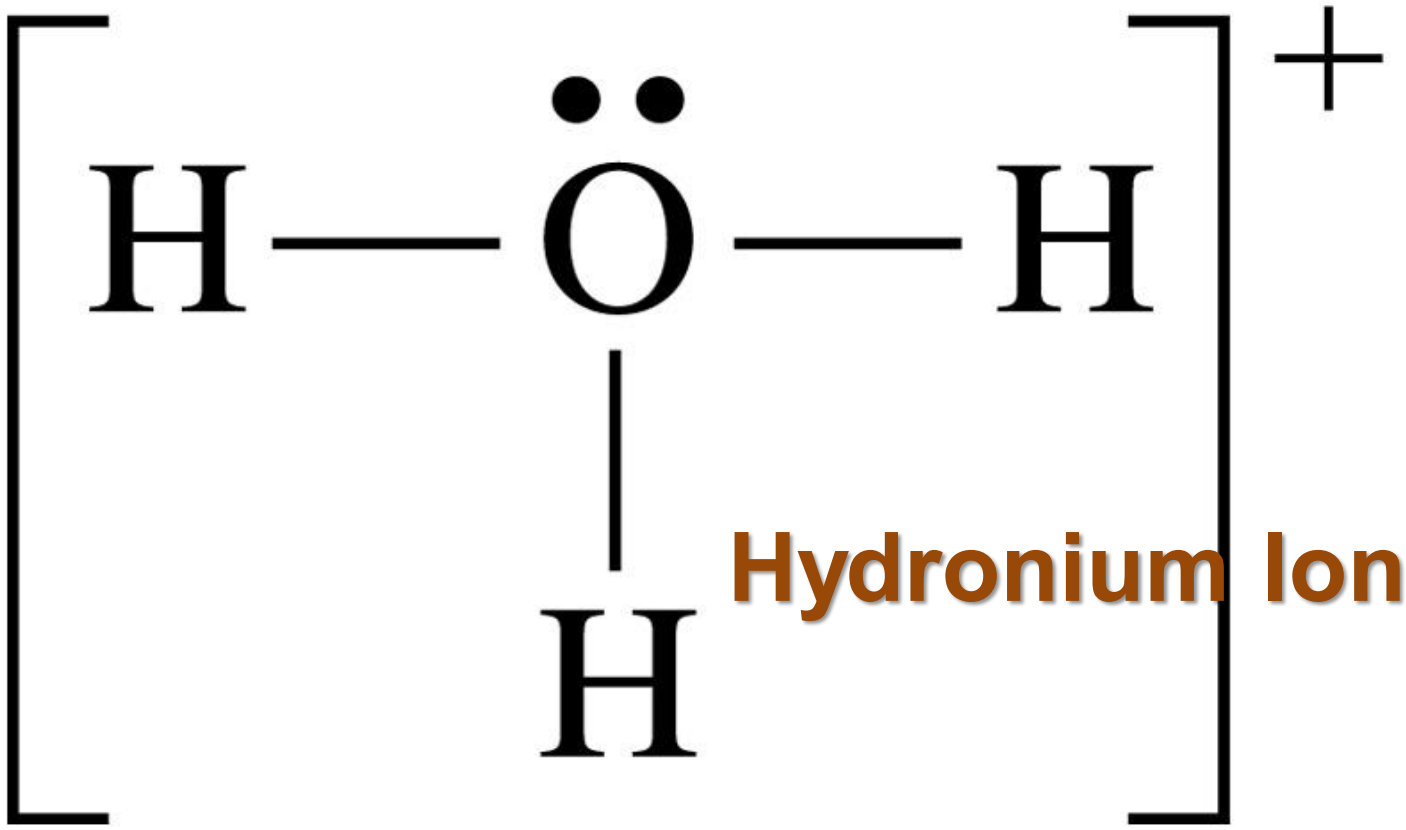
- $H_2O \leftrightarrow H^+ + OH^-$

Water, pH & buffer systems

- Hydrogen ions formed in water are immediately hydrated to hydronium ions ($H3O^+$)
- The ionization of water can be measured by its electrical conductivity
- Pure water carries electrical current as $H3O^+$ migrates toward the cathode and OH^- toward the anode

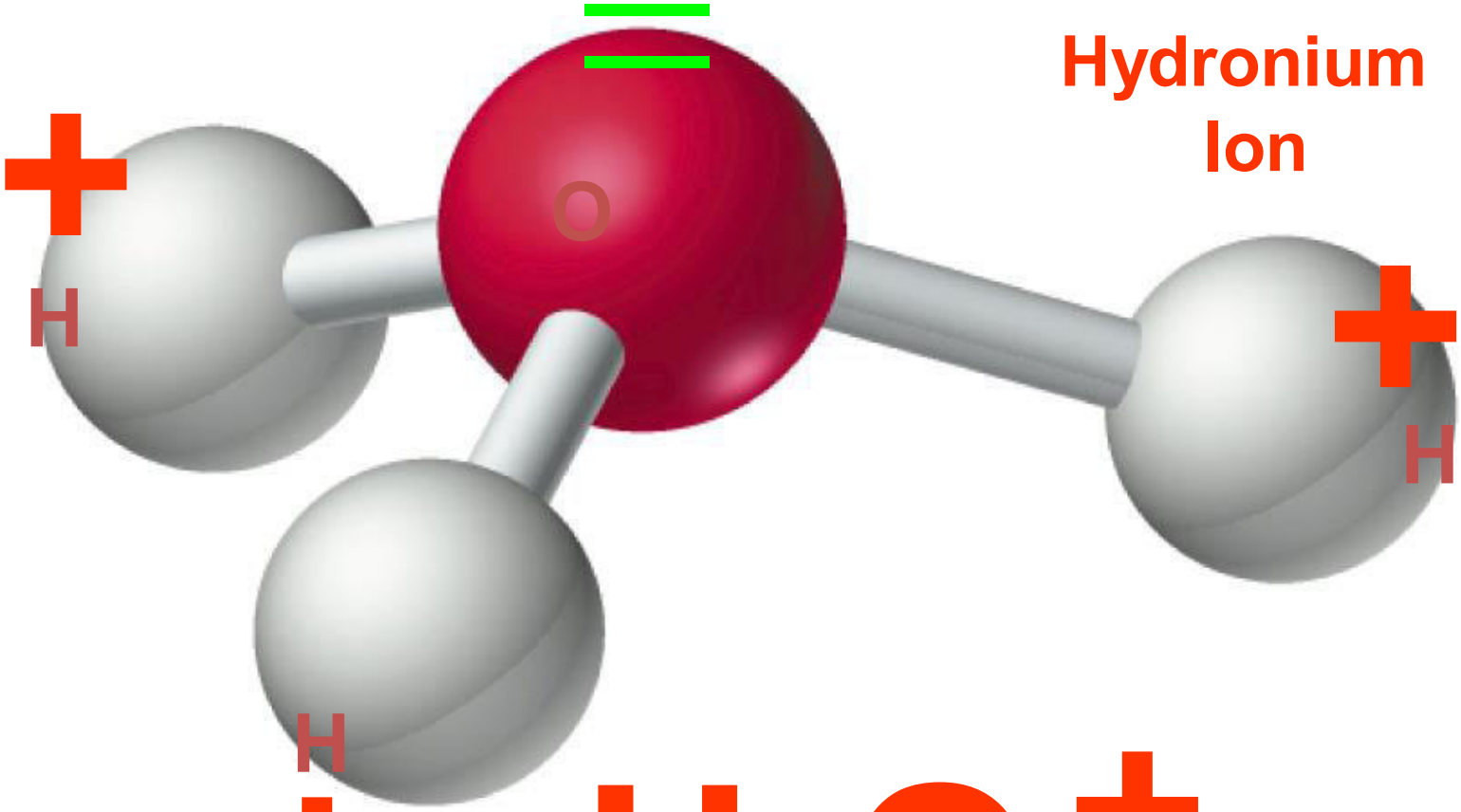
Water, pH & buffer systems

Water, pH & buffer systems



Hydronium Ion

Water, pH & buffer systems



H_3O^+

Water, pH & buffer systems

- The Ionization Of Water Is Expressed By an Equilibrium Constant

- The degree of ionization of water at equilibrium is small
- At 25^0C only about two out of every 10^9 molecules in pure water are ionized at any instant

Water, pH & buffer systems

- The equilibrium constant for the reversible ionization of water is

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]}$$

Water, pH & buffer systems

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]}$$

- Since 1 mole (mol) of water weighs 18 g,
- 1 liter (L) (1000 g) of water contains
- 1000 ÷ 18 = 55.56 mol
- Pure water thus is 55.56 molar(M)
- Accordingly, we can substitute 55.5 M in the equilibrium constant expression to yield
- Keq = $[H^+][OH^-]/55.5$

Water, pH & buffer systems

- $K_{eq} = [H^+][OH^-]/55.5$
- On rearranging, this becomes

  $(K_{eq})(55.5) = [H^+][OH^-] = K_w$

- Where K_w designates the product (55.5M)(K_{eq}), the ion product of water at 25^0C
- The value for K_{eq}, determined by electrical conductivity measurements of pure water, is 1.8×10^{-16} M at 25^0C

Water, pH & buffer systems

- $K_w = (K_{eq})(55.5) = [H^+][OH^-]$
- $K_w = (1.8 \times 10^{-16} M)(55.5M) = [H^+][OH^-]$
- $K_w = 1.0 \times 10^{-14}\ M^2 = [H^+][OH^-]$
- $K_w = [H^+][OH^-] = [H^+]^2 = [OH^-]^2$
- Solving for $[H^+]$ gives:

  $[H^+] = \sqrt{K_w} = \sqrt{10^{-14}\ M^2}$

  $[H^+] = 10^{-7}\ M$

Water, pH & buffer systems

- Thus the product $[H^+][OH^-]$ in aqueous solutions at 25^0C always equals 1×10^{-14} M^2
- When there are exactly equal concentrations of H^+ and OH^- , as in pure water, the solution is said to be at neutral pH

Water, pH & buffer systems

- As the ion product of water is constant, whenever $[H^+]$ is greater than 1×10^{-7} M, $[OH^-]$ must be less than 1×10^{-7} M, and vice versa

END

Water, pH & buffer systems

WORKING EXAMPLES

- From the ion product of water we can calculate $[H^+]$
- if we know $[OH^-]$, and vice versa

Water, pH & buffer systems

- **What is the concentration of H^+ in a solution of 0.1 M NaOH?**

  Solution:

  $Kw = [H^+][OH^-]$

- With $[OH^-] = 0.1$ M, solving for $[H^+]$ gives
- $[H^+] = K_w/[OH^-]$

  $= 1 \times 10^{-14} M^2 / 0.1M$

  $= 10^{-14} M^2 / 0.1M$

  $= 10^{-13} M$

Water, pH & buffer systems

- **What is the concentration of OH^- in a solution with an H^+ concentration of $1.3 \times 10^{-4}M$?**
- **Solution:**
- $K_w = [H^+][OH^-]$
- With $[H+] = 1.3 \times 10^{-4}M$, solving for $[OH^-]$ gives
- $[OH^-] = K_w/[H^+]$
- $= 1 \times 10^{-14}M^2 / 1.3 \times 10^{-4}M$
- $= \mathbf{7.7 \times 10^{-11}}$

Water, pH & buffer systems

The pH Scale

- Designates the H^+ and OH^- Concentrations
- The **pH** of a solution is defined as the logarithm to the base 10 of the reciprocal of the $[H^+]$, i. e, the negative logarithm of the $[H^+]$
- $pH = \log 1/[H^+]$
- $= -\log[H^+]$

Water, pH & buffer systems

- The pH of water at 25°C, in which H^+ and OH^- ions are present in equal numbers, is 7.0
- pH = $-\log[1 \times 10^{-7}] = 7$

- The symbol p denotes "negative logarithm of"

Water, pH & buffer systems

- For each pH unit less than 7.0, the $[H^+]$ is increased tenfold;
- for each pH unit above 7.0, it is decreased tenfold

Water, pH & buffer systems

TABLE 2–6 The pH Scale

$[H^+]$ (M)	pH	$[OH^-]$ (M)	pOH*
10^0 (1)	0	10^{-14}	14
10^{-1}	1	10^{-13}	13
10^{-2}	2	10^{-12}	12
10^{-3}	3	10^{-11}	11
10^{-4}	4	10^{-10}	10
10^{-5}	5	10^{-9}	9
10^{-6}	6	10^{-8}	8
10^{-7}	7	10^{-7}	7
10^{-8}	8	10^{-6}	6
10^{-9}	9	10^{-5}	5
10^{-10}	10	10^{-4}	4
10^{-11}	11	10^{-3}	3
10^{-12}	12	10^{-2}	2
10^{-13}	13	10^{-1}	1
10^{-14}	14	10^0 (1)	0

*The expression pOH is sometimes used to describe the basicity, or OH^- concentration, of a solution; pOH is defined by the expression $pOH = -\log [OH^-]$, which is analogous to the expression for pH. Note that in all cases, $pH + pOH = 14$.

TABLE 2–6 The pH Scale

$[H^+]$ (M)	pH	$[OH^-]$ (M)	pOH*
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10^{-3}	3	10^{-11}	11
10^{-4}	4	10^{-10}	10
10^{-5}	5	10^{-9}	9
10^{-6}	6	10^{-8}	8
10^{-7}	7	10^{-7}	7
10^{-8}	8	10^{-6}	6
10^{-9}	9	10^{-5}	5
10^{-10}	10	10^{-4}	4
10^{-11}	11	10^{-3}	3
10^{-12}	12	10^{-2}	2
10^{-13}	13	10^{-1}	1
10^{-14}	14	10^0 (1)	0

*The expression pOH is sometimes used to describe the basicity, or OH^- concentration, of a solution; pOH is defined by the expression $pOH = -\log [OH^-]$, which is analogous to the expression for pH. Note that in all cases, $pH + pOH = 14$.

Water, pH & buffer systems

- **What will be the pH of 0.1 M HCl?**

- Assuming that being a strong acid HCl is completely dissociated, it's 0.1 M solution will contain 0.1 or 10^{-1} grams H^+ per litre

- $pH = - \log [H^+]$
- $pH = - \log [10^{-1}]$
- $= - [- 1]$
- $= 1$

Water, pH & buffer systems

Weak Acids and Bases

- Each acid has a characteristic tendency to ionize in an aqueous solution
- The stronger the acid, the greater its tendency ionize i.e. to lose its proton
- This tendency is measured by an acid dissociation constant

Water, pH & buffer systems

- Weak Acids and Bases have Characteristic Acid Dissociation Constants
- HCl, H_2SO_4, and HNO_3, commonly called strong acids, are fully ionized in aqueous solutions
- The strong bases NaOH and KOH are also completely ionized

Water, pH & buffer systems

- Of more interest is the behavior of weak acids and bases-those not completely ionized when dissolved in water
- These are ubiquitous in biological systems and play important roles in metabolism and its regulation

Water, pH & buffer systems

- Acids may be defined as proton donors and bases as proton acceptors
- A proton donor and its corresponding proton acceptor make up a conjugate acid-base pair

Water, pH & buffer systems

- Acetic acid (CH_3COOH), a proton donor, and the acetate anion (CH_3COO^-), the corresponding proton acceptor, constitute a conjugate acid-base pair, related by the reversible reaction:

- $CH_3COOH \leftrightarrow CH_3COO^- + H^+$

Water, pH & buffer systems

- The tendency of any acid (HA) to lose a proton and form its conjugate base (A-) is defined by the acid dissociation constant (Ka) for the reversible reaction
- $HA \leftrightarrow H^+ + A^-$;

- $Ka = [H+][A-]/[HA]$

Water, pH & buffer systems

- Stronger acids, have larger dissociation constants (Ka) i.e they ionize completely

- Weaker acids, have smaller dissociation constants (Ka) i.e. they ionize only partially.

Water, pH & buffer systems

END

pKa

- analogous to pH, pKa is defined by the equation
- **pKa = log 1/Ka**
  = -logKa

- The stronger the tendency to dissociate a proton, the stronger is the acid and the lower its pKa

Water, pH & buffer systems

Working with pKa

- Titration is used to determine the amount of an acid in a given solution
- A measured volume of the acid is titrated with a solution of a strong base, usually sodium hydroxide (NaOH), of known concentration

Water, pH & buffer systems

- Consider the titration of a 0.1M solution of acetic acid with 0.1M NaOH at 25^0C
- Two reversible equilibria are involved in the process (here, for simplicity, acetic acid is denoted HAc)

$$H_2O \leftrightarrow H^+ + OH$$

$$HAc \leftrightarrow H^+ + Ac^-$$

Water, pH & buffer systems

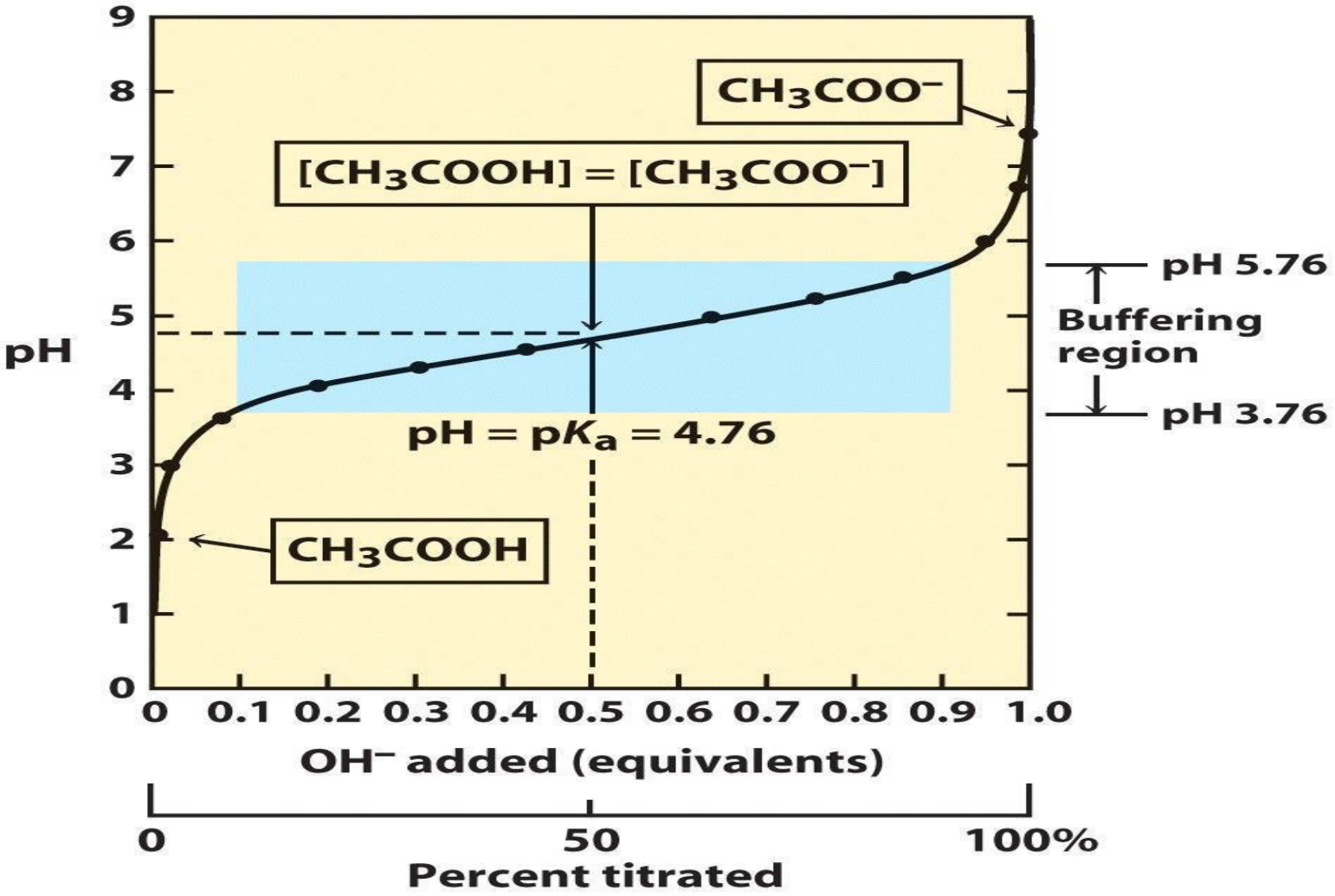
- The equilibria must simultaneously conform to their characteristic equilibrium constants, which are, respectively,
- $K_w = [H^+][OH^-]$
  $= 1 \times 10^{-14}\ M^2$

- $K_a = [H^+][Ac^-]/[HAc]$
  $= 1.74 \times 10^{-5}\ M$

Water, pH & buffer systems

- At the beginning of the titration, the acetic acid is only slightly ionized
- As NaOH is gradually added, OH^- combines with the free H^+ in the solution to form H_2O,
- As free H+ is removed, HAc dissociates further to satisfy its equilibrium constant

Water, pH & buffer systems



Water, pH & buffer systems

- At the midpoint a very important relationship holds:
- The pH of the equimolar solution of acetic acid and acetate is exactly equal to the pKa of acetic acid (pKa = 4.76)

Water, pH & buffer systems

- At pKa of a weak acid, half of the acid is in dissociated form
- whereas other half is un-dissociated.

- Increasing the pH will result in an increased dissociation and vice versa.

END

Water, pH & buffer systems

The Henderson-Hassebalch (HH) Equation

- The HH equation relates pH, pka, and buffer concentration
- This equation is simply a useful way of restating the expression for the ionization constant of an acid

Water, pH & buffer systems

- For the ionization of a weak acid HA, the HH equation can be derived as follows:
- Ka = [H+][A-]/[HA]
- First solve for $[H^+]$:

  $[H^+] = ka[HA]/[A^-]$
- Then take the negative logarithm of both sides:

  $-\log[H^+] = -\log ka - \log[HA]/[A^-]$
- Substitute pH for -log [H+] and pKa for -log Ka

  pH = pka – log[HA]/[A-]
- Now invert -log [HA]/[A-],

  pH = pka + log[A-]/[HA]

Water, pH & buffer systems

- This equation shows why the pKa of a weak acid is equal to the pH of the solution at the midpoint of its titration
- At that point, [HA]= [A-]

- pH = pka + log[A-]/[HA]
- pH = pka + log 1
- pH = pKa + 0
- pH = pka

Water, pH & buffer systems

- **Calculate the pKa of lactic acid, given that when the concentration of lactic acid is 0.01M and the concentration of lactate is 0.087 M, the pH is 4.80**

- pH= pKa + log [lactate]/[lactic acid]
- **pKa** = pH- log [lactate]/[lactic acid]

  = 4.80 - log 0.087/0.01

  = 4.80 - log 8.7

  = 4.80 - 0.94

  = **3.9**

Water, pH & buffer systems

- **Calculate the ratio of the concentrations of acetate and acetic acid required in a buffer system of pH 5.30 (pKa = 4.76)**

- pH= pKa + log [acetate]/[acetic acid]

- log [acetate]/[acetic acid]= pH- pKa
- = 5.30-4.76
- = 0.54
- [acetate]/[acetic acid] = antilog 0.54
- = **3.5**

Water, pH & buffer systems

- In summary,
- when [HA] = [A^-] ; pH=pKa

- when [HA] > [A^-] ; pH<pKa

- when [HA] < [A^-] ; pH>pKa

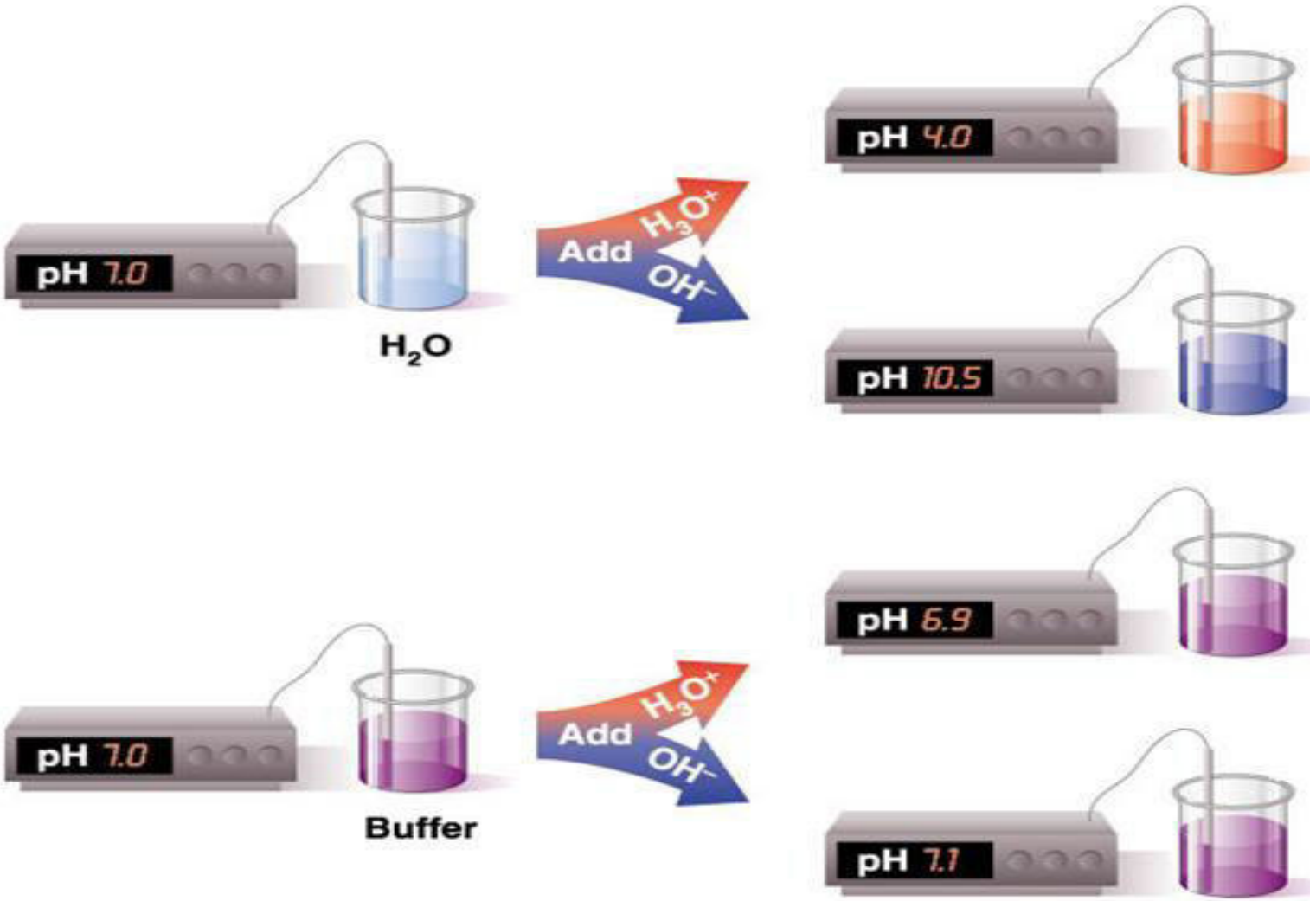
END

Water, pH & buffer systems

Buffer Solutions

- Buffers are aqueous systems that tend to resist changes in pH when small amounts of acid (H^+) or base (OH^-) are added
- A buffer system consists of a weak acid (the proton donor) and its conjugate base (the proton acceptor)

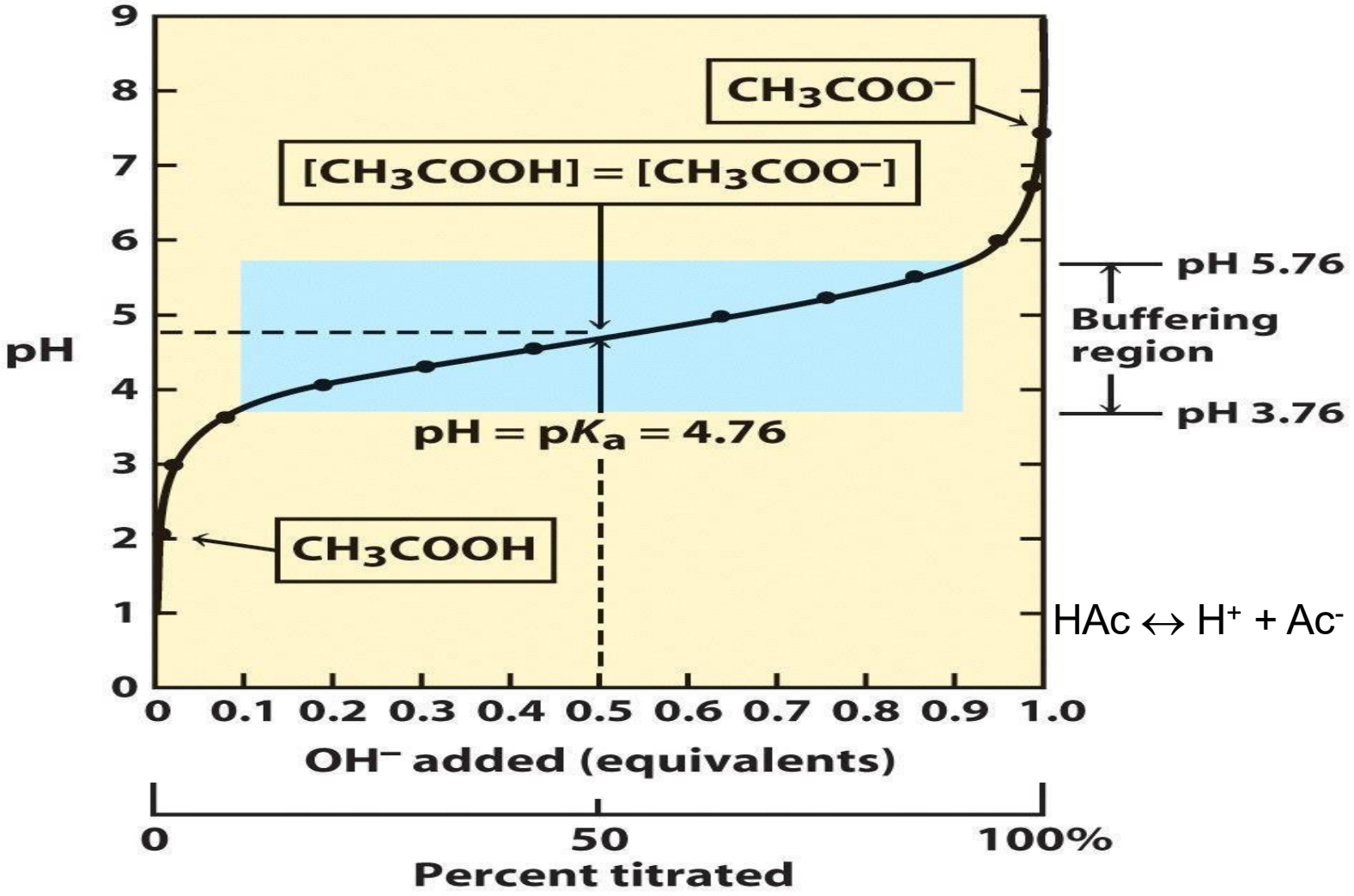
Water, pH & buffer systems



Water, pH & buffer systems

- As an example, a mixture of ;
- acetic acid and
- acetate ion,
- is a buffer system,

Water, pH & buffer systems



Water, pH & buffer systems

- At the midpoint of the buffering region, where the concentration of the proton donor (acetic acid) exactly equals that of the proton acceptor (acetate), the buffering power is maximal;
- The pH at this point in the titration curve of acetic acid is equal to its pKa

Water, pH & buffer systems

- Each conjugate acid-base pair has a characteristic pH zone in which it is an effective buffer
- For example, the $H_2PO_4^-/HPO_4^{2-}$ pair has a pKa of 6.86 and thus can serve as an effective buffer system between approximately pH 5.9 and pH 7.9

END

Water, pH & buffer systems

- The pH of the acetate buffer system does change slightly when a small amount of H^+ or OH^- is added, but this pH change is very small
- compared with the pH change that would result if the same amount of H+ or OH- were added to pure water.

Final Term Syllabus
Lecture 113 to 225

Efforts By

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